Universitext



Carlos S. Kubrusly

The Essentials of Measure Theory

Second Edition

Springer

Universitext

Series Editors

Nathanaël Berestycki, Universität Wien, Vienna, Austria

Carles Casacuberta, Universitat de Barcelona, Barcelona, Spain

John Greenlees, University of Warwick, Coventry, UK

Angus MacIntyre, Queen Mary University of London, London, UK

Claude Sabbah, École Polytechnique, CNRS, Université Paris-Saclay, Palaiseau, France

Endre Süli, University of Oxford, Oxford, UK

Universitext is a series of textbooks that presents material from a wide variety of mathematical disciplines at master's level and beyond. The books, often well class-tested by their author, may have an informal, personal, or even experimental approach to their subject matter. Some of the most successful and established books in the series have evolved through several editions, always following the evolution of teaching curricula, into very polished texts.

Thus as research topics trickle down into graduate-level teaching, first textbooks written for new, cutting-edge courses may find their way into *Universitext*.

Carlos S. Kubrusly

The Essentials of Measure Theory

Second Edition

Springer

Carlos S. Kubrusly
Professor Emeritus
Catholic University of Rio de Janeiro
Rio de Janeiro, Brazil

ISSN 0172-5939 ISSN 2191-6675 (electronic)
Universitext
ISBN 978-3-032-12662-7 ISBN 978-3-032-12663-4 (eBook)
https://doi.org/10.1007/978-3-032-12663-4

Mathematics Subject Classification: 28Axx, 28-01, 28C15

1st edition: © Springer International Publishing Switzerland 2015
2nd edition: © The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

This work is subject to copyright. All rights are solely and exclusively licensed by the Publisher, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, reuse of illustrations, recitation, broadcasting, reproduction on microfilms or in any other physical way, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publisher, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publisher nor the authors or the editors give a warranty, expressed or implied, with respect to the material contained herein or for any errors or omissions that may have been made. The publisher remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

This Springer imprint is published by the registered company Springer Nature Switzerland AG
The registered company address is: Gewerbestrasse 11, 6330 Cham, Switzerland

If disposing of this product, please recycle the paper.

To A & J, ever

'Oh God!' cried Lawson furiously. 'I knew it was that. He wants morality.' He joined his hands and held them towards Heaven in supplication. 'Oh, Christopher Columbus, Christopher Columbus, what did you do when you discovered America?'

W. Somerset Maugham's *Of Human Bondage*
ch. XLI (1915)

Preface to the Second Edition

This is a revised, enlarged, and updated second edition of *Essentials of Measure Theory* (Springer, Cham, 2015).

The preface of the previous edition says: "It is assumed that the reader is familiar with the notions of cardinality, countable and uncountable sets, and also with the following basic results: the set of all rational numbers is countable, a countable union of countable sets is countable, infinite sets (in particular, uncountable sets) have a countably infinite proper subset, and countably infinite families can be reenumerated into a sequence." This familiarity is indeed assumed, and to fix notation and definitions, the following six corresponding basic notions are now added. (See Suggested Readings at the end of Chapter 1 for a short list of references on introductory set theory.)

(1) Cardinality. Two sets have *the same cardinality* if there exists a one-to-one correspondence (i.e., an injective and surjective map) between them. If two sets A and B have the same cardinality, then we write $\#A = \#B$, and say that A and B are *equivalent*. In this sense, $\#$ stands for *cardinality*, and $\#A$ for the cardinality of the set A. *A set A has a cardinality less than or equal to the cardinality of a set B* (notation: $\#A \le \#B$) if there exists an injective mapping of A into B, and $\#A = \#B$ if and only if $\#A \le \#B$ and $\#B \le \#A$. If $\#A \le \#B$ and $\#A \ne \#B$, then we write $\#A < \#B$. Also,

$$A \subseteq B \quad \text{implies} \quad \#A \le \#B.$$

(2) Infinite Sets. Let $\mathbb{N}$ denote the set of all positive integers. A set A is *finite* if it is equivalent to a segment of the positive integers, say, to $\mathbb{I}_n = \{i \in \mathbb{N} \colon i \le n\}$ for some positive integer n. Otherwise, the set A is *infinite*.

A set is infinite if and only if it is equivalent to a proper subset of itself.

A proof of part of this statement requires the Axiom of Choice.

(3) Axiom of Choice. *If* $\{A_\gamma\}_{\gamma\in\Gamma}$ *is an indexed family of nonempty sets, indexed by an arbitrary nonempty index set* Γ*, then there exists an indexed family* $\{a_\gamma\}_{\gamma\in\Gamma}$ *such that* $a_\gamma \in A_\gamma$ *for each* $\gamma \in \Gamma$.

The Axiom of Choice is equivalent to the next statement (Zorn's Lemma) in the sense that each one can be obtained as a consequence of the other.

(4) Zorn's Lemma. *Let* X *be a partially ordered set. If every simply ordered subset of* X (i.e., *if each chain in* X) *has an upper bound in* X*, then* X *has a maximal element.*

(5) Countable Sets. If a set A is finite, then $\#A$ is the number of points in A. The set $\mathbb{N}$ of all positive integers is infinite. A set A is *countably infinite* if $\#A = \#\mathbb{N}$, and A is *countable* if A is either finite or countably infinite. Otherwise, A is *uncountable* (i.e., if $\#\mathbb{N} < \#A$). It can be verified that

$$\#\mathbb{N} = \#\mathbb{Q} < \#(0,1) = \#\mathbb{R} = \#2^{\mathbb{N}},$$

where $\mathbb{Q}$ stands for the set of all rational numbers, $\mathbb{R}$ for the set of all real numbers, and $2^{\mathbb{N}}$ for the collection of all functions from $\mathbb{N}$ to the pair $\{0,1\}$. The cardinality of $\mathbb{N}$, $\#\mathbb{N}$, is usually denoted by $\aleph_0$, and the cardinality of $2^{\mathbb{N}}$ (i.e., $\#2^{\mathbb{N}}$) by $2^{\aleph_0}$. The Continuum Hypothesis is the statement, or assumption, or hypothesis, or postulate, or axiom, that reads as follows.

(6) Continuum Hypothesis. *There is no set whose cardinality is greater than* $\#\mathbb{N}$ *and smaller than* $\#\mathbb{R}$*; equivalently, the cardinality of an uncountable subset of the real line is the same as the cardinality of* $\mathbb{R}$ *itself*.

This new edition differs from the previous edition in a number of places. It surpasses by 30% the previous edition in number of pages, where several new topics have been discussed throughout the text. The attention given to probability measures, some of them scattered among the proposed problems of Part I and all of them summarized in the Appendix to Part I, is an instance of the difference between the two editions, but certainly not the only one. Chapters 7 and 12 include a substantial amount of additional material. A comprehensive discussion on Cantor–Lebesgue measure has been brought forward in Problems 7.15 and 7.16. Rajchman measures have been considered in Problems 7.17 and 7.18. There is a new subsection on Borel regular measures on topological spaces in Section 12.4. The reference list has also increased from 42 to 74 and, as observed in the preface to the previous edition, the claim that it contains only books is still in force up to a single survey paper on Rajchman measures.

All definitions, propositions, lemmas, theorems, and problems of the previous edition have been kept in the present edition. Their numbering remains the same with just three exceptions: a pair of problems in Chapter 2 have had their numbering interchanged, one problem in Chapter 3 has been moved

two steps forward, and the last proposition of the previous edition in Chapter 11 has had its numbering moved one step ahead as well. The quantity of proposed problems closing every chapter of Part I, as well as the additional propositions closing every chapter of Part II, has been increased in all chapters of this second edition.

The sets of problems in each chapter of Part I go beyond conventional exercises and are an integral part of the text. They, together with their hints (which sometimes become sketchy, or even full, proofs), complement the main text of Part I, containing important theorems, some being necessary in forthcoming chapters. Likewise, the additional propositions and their notes extend the theory elaborated in the main text of Part II.

Actually, as in the previous edition, the last section of each chapter of Part II also contains a set of proposed problems, but, unlike Part I, they are, in general, more advanced than those in the Problems sections of Part I. So these sections at the end of every chapter of Part II are referred to as *Additional Propositions*. They are followed by a set of *Notes* in which each proposition is discussed, where proofs or references are provided for all of them.

As said before, in the preface to the previous edition, the book is written for the student, and the aim is teaching, almost from first principles, with detailed explanations and proofs, up to a level allowing them to pursue research in their field, be it in pure mathematics or applications outside of it, such as in statistics, engineering, physics, and economics. Along this line, the book may be a useful source for working mathematicians dealing with measure theory and for scientists wishing to apply it to their field.

I thank again all those people mentioned in the preface to the first edition, who have effectively contributed to improving the book. Thanks are also due to Alan C. Kubrusly, who suggested the inclusion of probability measures as in the Appendix to Part I, and to Jan Stochel, who has called my attention to a systematic misspelling in the previous edition that was inherited from older books: the name of the Polish mathematician Otto M. Nikodym has no accent over the y (which has been corrected throughout this edition). I am grateful to the copyeditor Brian Treadway as well, who did a remarkable job, and to Elizabeth Loew from Springer New York for her ever-pleasant, extremely efficient, and lasting partnership.

In these days of the dawn of the age of AI everywhere, when many doubts still hang in the air, I have chosen to go strictly handmade.

Rio de Janeiro,
October 2025

Carlos Kubrusly

Preface to the First Edition

The book is split into two parts. Part I is a first course in measure theory with integration, consisting of a revised, corrected, enlarged, updated, and thoroughly rewritten text based on the author's 2007 *Measure Theory: A First Course* [42]. Part II is a second course, dealing with measure and integration on topological spaces.

Part I, *Introduction to Measure and Integration*, designed to be a textbook for a first course in measure theory, gives an abstract approach to measure and integration, in which the classical concrete cases of Lebesgue measure and Lebesgue integral are presented as an important particular case of the general theory. Part I contains nine chapters. Chapter 1 considers real-valued (and extended real-valued) measurable functions with respect to a σ-algebra, and Chapter 2 introduces the concepts of measure and signed measure. The integral of nonnegative measurable functions with respect to a given measure is addressed in Chapter 3. The notion of integral is extended to real-valued measurable functions in Chapter 4, and L^p spaces are constructed in Chapter 5. Convergence of sequences of measurable functions is discussed in Chapter 6, where several concepts are compared. Decomposition of measures is investigated in Chapter 7, and extension theorems are treated in Chapter 8, where the Lebesgue measure is built up and discussed in detail. Product measures and integrals with respect to product measures (in particular, iterated integrals) close Part I in Chapter 9.

Part II, *Measures on Topological Spaces*, extends the material of Part I by equipping a nonempty set with a topology and considering σ-algebras of subsets of it containing the topology. This second part investigates measures and integrals on such Borel σ-algebras. It contains four chapters. Chapter 10 is an introduction to Part II, examining fundamental properties of integrals with respect to positive, real, and complex measures. Measures on topologi-

cal spaces are introduced in Chapter 11, whose central theme is the construction of Borel measures on a Borel σ-algebra of subsets of a locally compact Hausdorff space. Several forms of the Riesz Representation Theorem are considered in Chapter 12 after an introduction to continuous functions with compact support and bounded linear functionals. Invariant measures are focused on in Chapter 13, where the main topic is the construction of Haar measure on a Borel σ-algebra of subsets of a locally compact Hausdorff group.

The final section of each chapter in Part I contains *Problems*, and is an integral part of the chapter, not only just a set of routine exercises. The majority of those problems consists of auxiliary results, extensions of the theory, examples, and mainly counterexamples. The reader is encouraged to look at these problems with the same care as expected for a conventional theory section. Indeed, part of the theory is sometimes shifted to the problems section, and when this happens those problems are accompanied by *Hints* (sometimes, by detailed hints). The intention is to motivate readers to take an active part in the development of the theory presented in Part I.

Part II evidently is more advanced than Part I, addressed to more experienced readers. Thus, unlike Part I, the last section of each chapter of Part II consists of *Additional Propositions*, containing auxiliary and complementary results. These are followed by a set of *Notes*, in which each proposition is briefly discussed, and references are provided indicating proofs for all of them. These additional propositions can be viewed as a set of more complex proposed problems, and the respective notes as hints for dealing with them.

Each chapter in both Parts I and II ends with a collection of *Suggested Readings*. This has a triple purpose: to offer a reasonable bibliography including most of the classics as well as some recent texts, to point out where different approaches and proofs can be found, and also to indicate alternate routes towards additional results (so that some of the references are suggested as a second or third reading on the subject).

The material in Part I was devised to be covered in a one-semester beginning graduate course. Although naturally addressed to graduate students, Part I certainly is accessible to advanced undergraduate students as well. In fact, it is self-contained, and the prerequisites are very modest, namely, conventional undergraduate introductory analysis and, just for Chapter 5, linear spaces as usually taught in standard linear algebra courses. No acquaintance with functional analysis is required in Part I, but elementary set theory is obviously required. In particular, it is assumed that the reader is familiar with the notions of cardinality, countable and uncountable sets, and also with the following basic results: the set of all rational numbers is countable, a countable union of countable sets is countable, infinite sets (in particular, uncountable sets) have a countably infinite proper subset, and countably infinite families can be reenumerated into a sequence. Part II

was prepared to be covered in a one-semester graduate course as well, subsequent to a first course based on the material of Part I. The prerequisite for Part II is just Part I.

All in all the resulting text of the whole book is the outcome of attempts to meet the needs of a contemporary course in measure theory for mathematicians that will also be accessible to a wider audience of students in mathematics, statistics, economics, engineering, and physics, bearing a modest prerequisite. I tried to respond to the input from those students by presenting a text that contains complete proofs, including answers to several questions they have raised throughout the years. The logical dependence of the various sections and chapters is roughly linear and reflects approximately the minimum amount of material needed to proceed further.

I have been lecturing on this subject for a long time. Thus I benefited from the help of many friends among students and colleagues and I am truly grateful to all of them, in particular to Renato A.A. da Costa, Sergio Franklin, Leonardo B. Gonçalves, Johnny Kwong, André L. Pulcherio, Luciano R. da Silveira, Alexandre Street, and João Zanni who helped with the quest for typos in Part I, and Lucas Freire and Joaquim D. Garcia who helped with the quest for typos in Part II. Special thanks are due to Jessica Q. Kubrusly from whom I stole the notes for Section 7.3, to Adrian H. Pizzinga who read the entire text of Part I and corrected a number of typos and inaccuracies, to Richard Delaware who (together with his students) has scanned Part I as well and contributed with many corrections, and to an anonymous reviewer who made sensible suggestions to improve the text. Thanks are also due to my friend and colleague Marcelo D. Fragoso who was indeed an accomplice in writing this book. Let me also thank Elizabeth Loew and Ann Kostant from Springer New York for a pleasant and lasting partnership. I am grateful to the Catholic University of Rio de Janeiro for providing the release time that made this project possible.

Rio de Janeiro,
June 2015

Carlos S. Kubrusly

Contents

Preface to the Second Edition vii
Preface to the First Edition xi

PART I Introduction to Measure and Integration

1 Measurable Functions 3
1.1 Measurable Space 3
1.2 Real-Valued Measurable Functions 5
1.3 Extended Real-Valued Measurable Functions 9
1.4 Problems 14

2 Measure on a σ-Algebra 23
2.1 Measure and Measure Space 23
2.2 Signed Measure 29
2.3 Completion of Measure Spaces 32
2.4 Problems 34

3 Integral of Nonnegative Functions 43
3.1 Simple and Nonnegative Functions 43
3.2 The Monotone Convergence Theorem 45
3.3 Monotone Convergence Corollaries 49
3.4 Problems 52

4 Integral of Real-Valued Functions 61
4.1 Integrable Functions 61
4.2 Three Fundamental Lemmas 64
4.3 The Dominated Convergence Theorem 68
4.4 Problems 70

5 Banach Spaces L^p **79**
5.1 Construction of L^1 79
5.2 Spaces L^p and L^∞ 85
5.3 The Riesz–Fischer Completeness Theorem 88
5.4 Problems 91

6 Convergence of Functions **101**
6.1 Four Basic Convergence Notions 101
6.2 Convergence in Measure 106
6.3 Almost Uniform Convergence 111
6.4 Problems 116

7 Decomposition of Measures **121**
7.1 The Jordan Decomposition Theorem 121
7.2 The Radon–Nikodym Theorem 125
7.3 The Lebesgue Decomposition Theorem 131
7.4 Problems 136

8 Extension of Measures **147**
8.1 Measure on an Algebra and Outer Measure 147
8.2 The Carathéodory Extension Theorem 150
8.3 Construction of Lebesgue Measure 156
8.4 Problems 166

9 Product Measures **177**
9.1 Construction of Product Measures 177
9.2 Sections of Sets and Functions 181
9.3 Fubini and Tonelli Theorems 186
9.4 Problems 190

Appendix to Part I

A A Glimpse at Probability Measures **207**
A.1 Introduction 207
A.2 Random Vectors, Sequences, and Functions 209
A.3 Distributions, Characteristic, and Density Functions 212
A.4 Absolute Continuities and Normal Distribution 213
A.5 Joint Distribution and Density Functions 215
A.6 Conditioning 217
A.7 Markov and Martingale Processes 227
A.8 Why Does Normalization Change Everything? 230

PART II Measures on Topological Spaces

10 Remarks on Integrals **235**
10.1 Positive Measures 235
10.2 Real Measures 237
10.3 Complex Measures 240
10.4 Additional Propositions 243

11 Borel Measure **253**
11.1 Topological Spaces 253
11.2 Regular Measures 261
11.3 Construction of Borel Measures 267
11.4 Additional Propositions 270

12 Representation Theorems **277**
12.1 Continuous Functions and Compact Support 277
12.2 Bounded Linear Functionals 282
12.3 The Riesz Representation Theorem 288
12.4 Additional Propositions 298

13 Invariant Measures **305**
13.1 Topological Groups 305
13.2 Haar Measure 311
13.3 Construction of Haar Measures 314
13.4 Additional Propositions 323

References **327**

List of Symbols **331**

Index **337**

Part I

Introduction to Measure and Integration



1

Measurable Functions

1.1 Measurable Space

The *power set* $\wp(X)$ of a given set X is the collection of all subsets of X. We will work with set functions (i.e., functions whose domains are sets of sets) from Chapter 2 onwards. To begin with, we might assume that a natural candidate for the domain of such functions of sets would be the power set $\wp(X)$ of a given set X. This would indeed be an admissible candidate. However, as we will see in the subsequent chapters, there are instances where the power set is too large a set to be the domain of some set functions we wish to consider. This means that some functions may lose essential properties if their domain is too large or, in other words, some functions would not be well-behaved when defined on a domain that is as big as a power set. This will be discussed in Chapter 8 (cf. the remarks that close Section 8.3). Given an arbitrary set X, a collection of subsets of X (i.e., a subcollection of the power set $\wp(X)$) that will be appropriate to our purpose is a σ-*algebra*.

Definition 1.1. An *algebra* $\mathcal{A}$ (or a *field*, or a *Boolean algebra*) of subsets of a set X is a collection of subsets of X (i.e., $\mathcal{A} \subseteq \wp(X)$) fulfilling the axioms:

(a) The whole set X and the empty set $\varnothing$ belong to $\mathcal{A}$.

(b) The complement $X \setminus E$ of a set E in $\mathcal{A}$ belongs to $\mathcal{A}$.

(c) The union of a finite collection of sets in $\mathcal{A}$ belongs to $\mathcal{A}$.

If the restriction of finite union in axiom (c) is relaxed to allow countably infinite unions as well, then the algebra receives a special name and notation.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_1

A *σ-algebra* (or a *σ-field*, or a *Borel field*) of subsets of X (or on X), denoted by $\mathcal{X}$, is an algebra of subsets of X with axiom (c) extended to axiom ($\widehat{\text{c}}$):

($\widehat{\text{c}}$) The union of a countable collection of sets in $\mathcal{X}$ belongs to $\mathcal{X}$.

A pair $(X, \mathcal{X})$ consisting of an arbitrary set X and a σ-algebra $\mathcal{X}$ of subsets of X is called a *measurable space*. Sets in $\mathcal{X}$ are referred to as *measurable sets* (measurable with respect to the σ-algebra $\mathcal{X}$) or as *$\mathcal{X}$-measurable sets*.

Axioms (c) and ($\widehat{\text{c}}$) in the preceding definition of algebra and σ-algebra can be replaced, respectively, by the following axioms.

(c′) The intersection of a finite collection of sets in $\mathcal{A}$ belongs to $\mathcal{A}$.

($\widehat{\text{c}}$′) The intersection of a countable collection of sets in $\mathcal{X}$ belongs to $\mathcal{X}$.

Indeed, (b) and the De Morgan laws (viz., $X\backslash\big(\bigcup_\alpha A_\alpha\big) = \bigcap_\alpha (X\backslash A_\alpha)$ and $X\backslash\big(\bigcap_\alpha A_\alpha\big) = \bigcup_\alpha (X\backslash A_\alpha)$ for any collection $\{A_\alpha\}$ of subsets of X) ensure that (b) and (c) $\big($(b) and ($\widehat{\text{c}}$)$\big)$ is equivalent to (b) and (c′) $\big($to (b) and ($\widehat{\text{c}}$′)$\big)$.

Moreover, as $A\backslash B = A \cap X\backslash B$ for every $A, B \subseteq X$, axioms (a), (b), (c) in Definition 1.1 are equivalent to (a′), (b′), (c), with (a′) and (b′) given by

(a′) X belongs to $\mathcal{A}$.

(b′) The difference $E\backslash E'$ of sets E and E' in $\mathcal{A}$ belongs to $\mathcal{A}$.

A nonempty collection of sets that satisfies axioms (b′) and (c) is called a *ring* (or a *Boolean ring*) of sets. If it satisfies axioms (b′) and ($\widehat{\text{c}}$), then it is a *σ-ring*. Every algebra (every σ-algebra) is a ring (a σ-ring). If a ring (a σ-ring) of subsets of a set X contains X, then it is an algebra (a σ-algebra).

Example 1A. Take a nonempty set X. The power set $\wp(X)$ is a σ-algebra of subsets of X (it is more than that since the union of an arbitrary collection of sets in $\wp(X)$ lies in $\wp(X)$), which is the largest σ-algebra of subsets of X. At the other end, the collection $\{\varnothing, X\}$ is the smallest σ-algebra of subsets of X. (The notions of large and small are defined in terms of the inclusion ordering). A *partition* of a set X is any collection of pairwise disjoint subsets of X that cover X. If $\{A, B\}$ is a partition of X (i.e., if $A \cup B = X$ and $A \cap B = \varnothing$), then $\mathcal{X} = \{\varnothing, A, B, X\}$ is also a σ-algebra of subsets of X.

A subcollection of a σ-algebra $\mathcal{X}$ of subsets of a set X that is a σ-algebra of subsets of a set $E \subseteq X$ (which may be X itself) is a *sub-σ-algebra of $\mathcal{X}$*.

The intersection of an arbitrary collection of σ-algebras of subsets of a set X is again a σ-algebra of subsets of X (by the definition of σ-algebra). *If $\mathcal{C}$ is a nonempty collection of subsets of a set X* (i.e., if $\mathcal{C} \subseteq \wp(X)$), *then there exists a smallest σ-algebra $\mathcal{X}_\mathcal{C}$ of subsets of X that includes $\mathcal{C}$*. Indeed, let $\boldsymbol{X}_\mathcal{C}$ denote the collection of all σ-algebras of subsets of X that include $\mathcal{C}$.

Observe that $\boldsymbol{X}_{\mathcal{C}}$ is nonempty because the power set $\wp(X)$ is an element of $\boldsymbol{X}_{\mathcal{C}}$ (i.e., $\wp(X) \in \boldsymbol{X}_{\mathcal{C}}$). Now set $\mathcal{X}_{\mathcal{C}} = \bigcap \boldsymbol{X}_{\mathcal{C}}$, the intersection of all σ-algebras of subsets of X that include $\mathcal{C}$. Thus $\mathcal{X}_{\mathcal{C}}$ is a σ-algebra of subsets of X that includes $\mathcal{C}$. Moreover, it is included in any σ-algebra of subsets of X that includes $\mathcal{C}$ (i.e., if $\mathcal{X}$ is a σ-algebra of subsets of X such that $\mathcal{C} \subseteq \mathcal{X}$, then $\mathcal{X}_{\mathcal{C}} \subseteq \mathcal{X}$). This smallest σ-algebra $\mathcal{X}_{\mathcal{C}}$ is called *the σ-algebra generated by $\mathcal{C}$*. Alternative notation: $\sigma(\mathcal{C}) = \mathcal{X}_{\mathcal{C}}$. If $\mathcal{C}$ is already a σ-algebra of subsets of a subset of X, then it is clear that $\mathcal{X}_{\mathcal{C}} = \mathcal{C}$.

An important and useful particular case. Let $\mathcal{X}$ be a σ-algebra of subsets of a set X and take an $\mathcal{X}$-measurable set E (i.e., $E \in \mathcal{X} \subseteq \wp(X)$). First note that *nothing prevents E from having nonmeasurable subsets*. Thus set

$$\mathcal{E} = \wp(E) \cap \mathcal{X},$$

the σ-algebra of all $\mathcal{X}$-measurable subsets of the $\mathcal{X}$-measurable set E (intersection of two σ-algebras of subsets of X). So $\mathcal{E}$ is the largest sub-σ-algebra of $\mathcal{X}$ containing only subsets of E. This is *the sub-σ-algebra of $\mathcal{X}$ generated by E*. (The σ-algebra generated by $\mathcal{E}$ is $\mathcal{X}_{\mathcal{E}} = \mathcal{E}$, and the smallest sub-$\sigma$-algebra of $\mathcal{X}$ containing E — the smallest σ-algebra of subsets of E — is $\mathcal{X}_{\{E\}} = \{\varnothing, E\}$.)

Another important and concrete particular case. Let $\mathbb{R}$ stand for the field of all real numbers, the so-called real field. The *Borel σ-algebra of subsets of* $\mathbb{R}$, sometimes also called *Borel algebra of subsets of* $\mathbb{R}$, is the σ-algebra $\mathfrak{R}$ generated by the collection of all open (or half-open, or closed) intervals of the real line $\mathbb{R}$. If $\mathcal{T}$ denotes the collection of all open subsets of $\mathbb{R}$, then $\mathfrak{R}$ coincides with the σ-algebra $\mathcal{R}_{\mathcal{T}}$ of subsets of $\mathbb{R}$ generated by $\mathcal{T}$ (see the remark following Problem 1.14):

$$\mathfrak{R} = \mathcal{R}_{\mathcal{T}}.$$

The elements of $\mathfrak{R}$ (i.e., the $\mathfrak{R}$-measurable sets) are called *Borel sets*. The notion of Borel σ-algebra will be extended in Chapter 11.

1.2 Real-Valued Measurable Functions

Definition 1.2. Consider a measurable space $(X, \mathcal{X})$. A function $f: X \to \mathbb{R}$ is *measurable* with respect to the σ-algebra $\mathcal{X}$, or *$\mathcal{X}$-measurable*, if the inverse image of (α, ∞) under f is a measurable set for any real number α:

$$f^{-1}\big((\alpha, \infty)\big) = \big\{x \in X\colon\ f(x) > \alpha\big\} \ \in\ \mathcal{X} \quad \text{for every } \ \alpha \in \mathbb{R}.$$

The sign $>$ in Definition 1.2 can be replaced with $\geq$, $<$, or $\leq$, yielding equivalent definitions of measurable functions (Problem 1.1). Also, *$f: X \to \mathbb{R}$ is measurable if and only if the inverse images of open sets in $\mathbb{R}$ are measurable sets in $\mathcal{X}$* (see Problem 1.8 — this will be extended in Chapter 11).

Example 1B. Let $(X, \mathcal{X})$ be a measurable space, take any set $E \in \wp(X)$, and consider the *characteristic function* $\chi_E \colon X \to \{0,1\} \subset \mathbb{R}$ of E; that is,

$$\chi_E(x) = \begin{cases} 1, & x \in E, \\ 0, & x \in X \backslash E. \end{cases}$$

The characteristic function $\chi_E : X \to \mathbb{R}$ *of a set* $E \subseteq X$ *is an* $\mathcal{X}$*-measurable function if and only if* E *is an* $\mathcal{X}$*-measurable set* (i.e., if and only if $E \in \mathcal{X}$):

$$\chi_E^{-1}\big((\alpha, \infty)\big) = \big\{x \in X \colon \chi_E(x) > \alpha\big\} = \begin{cases} \varnothing, & 1 \le \alpha, \\ E, & 0 \le \alpha < 1, \\ X, & \alpha < 0. \end{cases}$$

The *unit function* $1 \colon X \to \mathbb{R}$ is defined by $1(x) = 1$ for all $x \in X$. It coincides with the characteristic function χ_X of X. It is clear that γf is measurable for every $\gamma \in \mathbb{R}$ whenever $f \colon X \to \mathbb{R}$ is measurable. So *every real-valued constant function is measurable* (as every constant function is $\gamma \chi_X$ for some $\gamma \in \mathbb{R}$).

Example 1C. Let $\mathbb{N}_e$ and $\mathbb{N}_o$ be subsets of the positive integers $\mathbb{N}$ consisting of all *even positive integers* and of all *odd positive integers*, respectively. So $\{\mathbb{N}_o, \mathbb{N}_e\}$ forms a partition of $\mathbb{N}$. Example 1A says that $\mathcal{N} = \{\varnothing, \mathbb{N}_o, \mathbb{N}_e, \mathbb{N}\}$ is a σ-algebra of subsets of $\mathbb{N}$. Consider the measurable space $(\mathbb{N}, \mathcal{N})$, and take the *identity function* $f \colon \mathbb{N} \to \mathbb{N} \subset \mathbb{R}$ (i.e., $f(n) = n$ for every $n \in \mathbb{N}$). This function $f \colon \mathbb{N} \to \mathbb{R}$ is not measurable (i.e., not $\mathcal{N}$-measurable). In fact,

$$f^{-1}\big((1, \infty)\big) = \big\{n \in \mathbb{N} \colon f(n) > 1\big\} = \big\{2, 3, 4, \ldots\big\} \notin \mathcal{N}.$$

The next proposition says that *restrictions of* $\mathcal{X}$*-measurable functions to* $\mathcal{X}$*-measurable sets are* $\mathcal{E}$*-measurable functions*.

Proposition 1.3. *A restriction* $f|_E \colon E \to \mathbb{R}$ *of an* $\mathcal{X}$*-measurable function* $f \colon X \to \mathbb{R}$ *to an* $\mathcal{X}$*-measurable set* E *is an* $\mathcal{E}$*-measurable function with respect to the sub-*σ*-algebra* $\mathcal{E} = \wp(E) \cap \mathcal{X}$ *of* $\mathcal{X}$.

Proof. Let $(X, \mathcal{X})$ be a measurable space and take an $\mathcal{X}$-measurable function $f \colon X \to \mathbb{R}$. Let E be an $\mathcal{X}$-measurable subset of X and take the restriction $f|_E \colon E \to \mathbb{R}$ of f to E (i.e., $f|_E(x) = f(x)$ for every $x \in E$). Let $\alpha \in \mathbb{R}$ be an arbitrary real number. Note that

$$\big\{x \in E \colon f|_E(x) > \alpha\big\} = \big\{x \in X \colon f(x) > \alpha\big\} \cap E \quad \in \quad \mathcal{E} = \wp(E) \cap \mathcal{X}.$$

In fact, $\{x \in E \colon f|_E(x) > \alpha\}$ lies in $\wp(E)$ because it is a subset of E, and it lies in $\mathcal{X}$ because it is the intersection of two $\mathcal{X}$-measurable sets (since f is an $\mathcal{X}$-measurable function). Thus $f|_E$ is $\mathcal{E}$-measurable. □

Notes: (1) If $\mathcal{X} = \Re$, $E = (1,3) \in \mathcal{X}$, $\mathcal{E}' = \{\varnothing, E\}$, $f = \chi|_{(1,2)}$ ($\mathcal{X}$-measurable), then $\{x \in E \colon f|_E(x) > 0\} = (1,2) \notin \mathcal{E}'$, and so $f|_E$ is not $\mathcal{E}'$-measurable.

(2) We will see, as a particular case of Proposition 1.5 below, that the product $f\chi_E\colon X \to \mathbb{R}$ is $\mathcal{X}$-measurable whenever f and E are $\mathcal{X}$-measurable.

Proposition 1.4. *Let $(X, \mathcal{X})$ be a measurable space, let A and B be $\mathcal{X}$-measurable sets, and consider the sub-σ-algebras generated by A and B,*

$$\mathcal{A} = \wp(A) \cap \mathcal{X} \quad \text{and} \quad \mathcal{B} = \wp(B) \cap \mathcal{X}.$$

Take a function $f\colon X \to \mathbb{R}$ and consider its restrictions to A and B,

$$f|_A\colon A \to \mathbb{R} \quad \text{and} \quad f|_B\colon B \to \mathbb{R}.$$

If $A \cup B = X$, then

$$f \text{ is } \mathcal{X}\text{-measurable} \iff f|_A \text{ is } \mathcal{A}\text{-measurable and } f|_B \text{ is } \mathcal{B}\text{-measurable.}$$

Proof. Proposition 1.3 ensures the "only if" part. To prove the "if" part proceed as follows. Take an arbitrary $\alpha \in \mathbb{R}$ and note that, since $A \cup B = X$,

$$\big\{x \in X\colon\ f(x) > \alpha\big\} = \big\{x \in A\colon\ f|_A(x) > \alpha\big\} \cup \big\{x \in B\colon\ f|_B(x) > \alpha\big\},$$

which is the union of an $\mathcal{A}$-measurable set and a $\mathcal{B}$-measurable set (because $f|_A$ is an $\mathcal{A}$-measurable function and $f|_B$ is a $\mathcal{B}$-measurable function). Since the σ-algebras $\mathcal{A}$ and $\mathcal{B}$ are included in $\mathcal{X}$, both sets lie in $\mathcal{X}$, and so their union $\{x \in X\colon\ f(x) > \alpha\}$ is an $\mathcal{X}$-measurable set. This means that f is an $\mathcal{X}$-measurable function. □

There is a huge supply of measurable functions, as we will see in the rest of this chapter (see also Problem 1.2 for the particular case of real-valued functions on $\mathbb{R}$ with respect to the Borel σ-algebra $\mathfrak{R}$). Let $\mathcal{X}$ be a σ-algebra of subsets of a nonempty set X and take a pair of $\mathcal{X}$-measurable functions, say, $f\colon X \to \mathbb{R}$ and $g\colon X \to \mathbb{R}$. A polynomial $p(f, g)$ of a pair of functions f and g is an arbitrary (finite) linear combination of products of powers of f and g with real coefficients $\gamma_{j,k}$; that is,

$$p(f, g) = \sum_{j,k=0}^{m,n} \gamma_{j,k}\, f^j\, g^k.$$

Proposition 1.5. *If f and g are measurable functions, then so is any $p(f, g)$.*

Proof. Consider a measurable space $(X, \mathcal{X})$. Let f and g be measurable functions and let γ be any real number. We have already seen that γ (a constant function) and γf (a multiple of a measurable function) are measurable functions. Thus it is enough to show that $f + g$ and $f\,g$ are measurable functions in order to ensure that each (finite) linear combination of $\{f^j g^k\}_{j,k\geq 0}$

is a measurable function. Let $\mathbb{Q}$ denote the rational field, take an arbitrary $\rho \in \mathbb{Q}$, an arbitrary $\alpha \in \mathbb{R}$, and note that $f^{-1}((\rho,\infty)) \cap g^{-1}((\alpha-\rho,\infty))$ is a measurable set (it is the intersection of measurable sets since f and g are measurable functions). Let $E_{\alpha,\rho}$ denote such a measurable set, viz.,

$$E_{\alpha,\rho} = \big\{x \in X\colon\ f(x) > \rho \ \text{ and } \ g(x) > \alpha - \rho\big\}.$$

Recall: $(f+g)(x) = f(x) + g(x)$ for every $x \in X$. Thus for each real $\alpha \in \mathbb{R}$, $(f+g)(x) > \alpha$ if and only if $f(x) > \rho$ and $g(x) > \alpha - \rho$ for some rational $\rho \in \mathbb{Q}$. That is, if and only if $x \in E_{\alpha,\rho}$ for some $\rho \in \mathbb{Q}$. So, for every $\alpha \in \mathbb{R}$,

$$(f+g)^{-1}\big((\alpha,\infty)\big) = \big\{x \in X\colon\ (f+g)(x) > \alpha\big\} = \bigcup\nolimits_{\rho} E_{\alpha,\rho}.$$

Since each set $E_{\alpha,\rho}$ is measurable, and since $\mathbb{Q}$ is countable, it follows that $\bigcup_{\rho\in\mathbb{Q}} E_{\alpha,\rho}$ is a measurable set for every $\alpha \in \mathbb{R}$, and so $f+g$ is a measurable function (Definitions 1.1 and 1.2). Next we show that f^2 (given by $f^2(x) = f(x)^2$ for every $x \in X$) is measurable. If $\alpha < 0$, then $\{x \in X\colon f^2(x) > \alpha\} = X$. If $\alpha \ge 0$, then $f(x)^2 > \alpha$ means $f(x) > \sqrt{\alpha}$ or $f(x) < -\sqrt{\alpha}$. Thus the set $\{x \in X\colon f^2(x) > \alpha\}$ is the union of the sets $\{x \in X\colon f(x) > \sqrt{\alpha}\}$ and $\{x \in X\colon f(x) < -\sqrt{\alpha}\}$. So $(f^2)^{-1}((\alpha,\infty)) = \{x \in X\colon f^2(x) > \alpha\}$ is given by

$$(f^2)^{-1}\big((\alpha,\infty)\big) = \begin{cases} X, & \alpha < 0, \\ f^{-1}\big((\sqrt{\alpha},\infty)\big) \cup (-f)^{-1}\big((\sqrt{\alpha},\infty)\big), & \alpha \ge 0, \end{cases}$$

where, for $\alpha \ge 0$, we have a union of measurable sets. This ensures that f^2 is a measurable function. However, as f and g are $\mathbb{R}$-valued functions,

$$fg = \tfrac{1}{4}\big((f+g)^2 - (f-g)^2\big),$$

whose expression involves multiplication by a constant, addition, and squaring of measurable functions. Thus fg is a measurable function. □

Take an arbitrary function $f\colon X \to \mathbb{R}$, consider the sets

$$F^+ = \big\{x \in X\colon\ f(x) \ge 0\big\} \quad \text{ and } \quad F^- = \big\{x \in X\colon\ f(x) \le 0\big\},$$

and let χ_{F^+} and χ_{F^-} be the characteristic functions of F^+ and F^-, respectively. Now consider the following real-valued functions on X:

$$f^+ = f\,\chi_{F^+} \quad \text{ and } \quad f^- = -f\,\chi_{F^-}.$$

These are the *positive part* of f and the *negative part* of f, respectively. Note that $f^+\colon X \to \mathbb{R}$ and $f^-\colon X \to \mathbb{R}$ are both nonnegative functions (i.e., $f^+(x) \ge 0$ and $f^-(x) \ge 0$ for every $x \in X$). Moreover, the functions f and

its *absolute value* $|f|: X \to \mathbb{R}$ (given by $|f|(x) = |f(x)|$ for every $x \in X$) can be expressed in terms of the positive and negative parts of f as follows.

$$f = f^+ - f^- \quad \text{and} \quad |f| = f^+ + f^-.$$

The above identities can be reversed yielding

$$f^+ = \tfrac{1}{2}\big(|f| + f\big) \quad \text{and} \quad f^- = \tfrac{1}{2}\big(|f| - f\big).$$

Proposition 1.6. *The following assertions are pairwise equivalent.*

(a) *f is a measurable function.*

(b) *f^+ and f^- are measurable functions.*

(c) *F^+ and F^- are measurable sets and $|f|$ is a measurable function.*

Proof. Take a measurable space $(X, \mathcal{X})$. Let f be a measurable function. So F^+ and F^- are both measurable sets (Definition 1.2). Hence χ_{F^+} and χ_{F^-} are measurable functions (Example 1B). Then $f^+ = f\chi_{F^+}$ and $f^- = -f\chi_{F^-}$ are measurable functions by Proposition 1.5, and so is $|f| = f^+ + f^-$. Since $f = f^+ - f^-$, we get the converse: if f^+ and f^- are measurable, then f is measurable. Therefore, (a) $\Leftrightarrow$ (b) and (a) $\Rightarrow$ (c). Next we show that (c) $\Rightarrow$ (a). The restrictions of $|f|$ to F^+ and F^- coincide with the restriction of f to F^+ and with the restriction of $-f$ to F^-; that is,

$$f|_{F^+} = |f|\,\big|_{F^+} : F^+ \to \mathbb{R} \quad \text{and} \quad f|_{F^-} = -|f|\,\big|_{F^-} : F^- \to \mathbb{R}.$$

Since $F^+ \cup F^- = X$, it follows by Proposition 1.4 that if F^+ and F^- are $\mathcal{X}$-measurable sets and $|f|$ is an $\mathcal{X}$-measurable function, then $f|_{F^+}$ is $\mathcal{F}^+$-measurable and $-f|_{F^-}$ (and so $f|_{F^-}$) is $\mathcal{F}^-$-measurable, with respect to the σ-algebras $\mathcal{F}^+ = \wp(F^+) \cap \mathcal{X}$ and $\mathcal{F}^- = \wp(F^-) \cap \mathcal{X}$, respectively. Another application of Proposition 1.4 ensures that f is $\mathcal{X}$-measurable. □

Note that "$|f|$ measurable" does not imply "f measurable" (i.e., (c) does not imply (a) without the assumption that F^+ and F^- are measurable). For instance, take a measurable space $(X, \mathcal{X})$ and suppose there exists a partition $\{A, A'\}$ of X, where A and A' are not $\mathcal{X}$-measurable sets, and consider the function $f: X \to \mathbb{R}$ given by $f(x) = 1$ for x in A and $f(x) = -1$ for x in A' so that $F^+ = A$ and $F^- = A'$. It is clear that f is not an $\mathcal{X}$-measurable function (A is not an $\mathcal{X}$-measurable set). But $|f|$ is a real-valued constant function, thus measurable with respect to any σ-algebra (Example 1B).

1.3 Extended Real-Valued Measurable Functions

When considering the notion of length of subsets of the real line $\mathbb{R}$ (as will be done from Chapter 2 onwards), there emerges the need for dealing with

the length of unbounded subsets of $\mathbb{R}$ (such as $\mathbb{R}$ itself), and also with the notion of inf and sup of unbounded subsets of $\mathbb{R}$. This leads us to introduce the *extended real number system*, or the *extended real line*, which is the collection $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$ consisting of the real field $\mathbb{R}$ and a pair of symbols, namely, $-\infty$ and $+\infty$. These symbols are not numbers, certainly not real numbers. We might have chosen any other pair of symbols such as, for instance, (α, ω) or $(\triangleleft, \triangleright)$ instead of the "set-down-eights" $(-\infty, +\infty)$. However, we will stick with the standard notation. Regarding the natural ordering of the real line $\mathbb{R}$, we postulate that $-\infty < x < +\infty$ for all x in $\mathbb{R}$, and extend the natural order of $\mathbb{R}$ to $\overline{\mathbb{R}}$. Note that $\overline{\mathbb{R}}$ is not a field, even though arithmetics with the new symbols are partially defined in the usual fashion, with some exceptions (e.g., $-\infty$ and $+\infty$ cannot be added together in any order; equivalently, the subtraction of $+\infty$ with itself is not defined).

We transfer the definition of measurable function to extended real-valued functions exactly as in Definition 1.2, namely, if $(X, \mathcal{X})$ is a measurable space, then an extended real-valued function $f\colon X \to \overline{\mathbb{R}}$ is *measurable* (with respect to the σ-algebra $\mathcal{X}$) or $\mathcal{X}$-*measurable* if the inverse image of (α, ∞) under f is a measurable set for any real number (i.e., for any $\alpha \in \mathbb{R}$):

$$f^{-1}\big((\alpha, \infty)\big) = \big\{x \in X\colon\ f(x) > \alpha\big\} \in \mathcal{X} \quad \text{for every } \alpha \in \mathbb{R}.$$

As in Definition 2.1, the sign $>$ in the above expression can be replaced with $\geq$, $<$, or $\leq$, yielding equivalent definitions of an extended real-valued measurable function (cf. Problem 1.1). If f is an extended real-valued $\mathcal{X}$-measurable function, then it is easy to verify (cf. Problem 1.4) that

$$F_{+\infty} = \big\{x \in X\colon\ f(x) = +\infty\big\} \quad \text{and} \quad F_{-\infty} = \big\{x \in X\colon\ f(x) = -\infty\big\}$$

are $\mathcal{X}$-measurable sets. Let $\chi_{X\setminus(F_{+\infty}\cup F_{-\infty})}$ be the characteristic function of the complement of $F_{+\infty} \cup F_{-\infty}$ and consider the real-valued function

$$f_{\mathbb{R}} = f\,\chi_{X\setminus(F_{+\infty}\cup F_{-\infty})}\colon X \to \mathbb{R}$$

(i.e., $f_{\mathbb{R}}(x) = f(x)$ if $x \notin (F_{+\infty} \cup F_{-\infty})$ and $f_{\mathbb{R}}(x) = 0$ if $x \in (F_{+\infty} \cup F_{-\infty})$; in other words, $f_{\mathbb{R}}(x) = f(x)$ if $f(x) \neq \pm\infty$, and $f_{\mathbb{R}}(x) = 0$ if $f(x) = \pm\infty$).

Proposition 1.7. *Consider a measurable space $(X, \mathcal{X})$. An extended real-valued function $f\colon X \to \overline{\mathbb{R}}$ is measurable if and only if the real-valued function $f_{\mathbb{R}}\colon X \to \mathbb{R}$ is measurable and both sets $F_{+\infty}$ and $F_{-\infty}$ are measurable.*

Proof. Take a function $f\colon X \to \overline{\mathbb{R}}$, and an arbitrary $\alpha \in \mathbb{R}$. Observe that

$$\big\{x \in X\colon\ f(x) > \alpha\big\} = \begin{cases} \big\{x \in X\colon\ f_{\mathbb{R}}(x) > \alpha\big\} \cup F_{+\infty}, & \alpha \geq 0, \\ \big\{x \in X\colon\ f_{\mathbb{R}}(x) > \alpha\big\} \setminus F_{-\infty}, & \alpha < 0. \end{cases}$$

Then f is a measurable function whenever $f_{\mathbb{R}}$ is a measurable function and $F_{+\infty}$ and $F_{-\infty}$ are measurable sets (recall: $A\backslash B = A \cap X\backslash B$). Since

$$\big\{x \in X\colon\ f_{\mathbb{R}}(x) > \alpha\big\} = \begin{cases} \big\{x \in X\colon\ f(x) > \alpha\big\}\backslash\, F_{+\infty}, & \alpha \geq 0, \\ \big\{x \in X\colon\ f(x) > \alpha\big\} \cup\ F_{-\infty}, & \alpha < 0, \end{cases}$$

and since $F_{+\infty}$ and $F_{-\infty}$ are measurable sets whenever f is a measurable function, it follows that $f_{\mathbb{R}}$ is measurable whenever f is. □

Let S be an arbitrary set. An *S-valued sequence* (or a sequence of elements in S) is just an S-valued function defined on $\mathbb{N}$, the positive integers (or on $\mathbb{N}_0 = \{0\} \cup \mathbb{N}$, the nonnegative integers). Take an arbitrary $\overline{\mathbb{R}}$-valued sequence $\{\alpha_n\}$. Let $\inf_n \alpha_n$ and $\sup_n \alpha_n$ denote the greatest lower bound and the least upper bound of $\{\alpha_n\}$, respectively, which exist and are unique in $\overline{\mathbb{R}}$ (but may not exist in $\mathbb{R}$). The *limit inferior* (notation: $\liminf_n \alpha_n$) and *limit superior* (notation: $\limsup_n \alpha_n$) of $\{\alpha_n\}$ are defined in $\overline{\mathbb{R}}$ by

$$\liminf_n \alpha_n = \sup_n \inf_{n \leq k} \alpha_k \quad \text{and} \quad \limsup_n \alpha_n = \inf_n \sup_{n \leq k} \alpha_k.$$

If $\liminf_n \alpha_n = \limsup_n \alpha_n = \alpha$, then we say that $\{\alpha_n\}$ *converges* to $\alpha \in \overline{\mathbb{R}}$, and write $\lim_n \alpha_n = \alpha$. If $\{\alpha_n\}$ is *increasing* (i.e., $\alpha_n \leq \alpha_{n+1}$), or *decreasing* (i.e., $\alpha_{n+1} \leq \alpha_n$), then it is a *monotone* sequence. An $\overline{\mathbb{R}}$-valued monotone sequence $\{\alpha_n\}$ converges to a limit α in $\overline{\mathbb{R}}$. For $\mathbb{R}$-valued sequences this definition of convergence is equivalent to the standard definition of convergence (restricted to $\mathbb{R}$); viz., a real-valued sequence $\{\alpha_n\}$ *converges* to $\alpha \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists an integer $n_\varepsilon \in \mathbb{N}$ such that $|\alpha_n - \alpha| < \varepsilon$ whenever $n \geq n_\varepsilon$. If a real-valued sequence $\{\alpha_n\}$ is *bounded* (i.e., if $\sup_n |\alpha_n|$ lies in $\mathbb{R}$), then the sequences $\{\inf_{n \leq k} \alpha_k\}$ and $\{\sup_{n \leq k} \alpha_k\}$ converge in $\mathbb{R}$ to $\liminf_n \alpha_n \in \mathbb{R}$ and $\limsup_n \alpha_n \in \mathbb{R}$, respectively. In this case,

$$\liminf_n \alpha_n = \lim_n \inf_{n \leq k} \alpha_k \quad \text{and} \quad \limsup_n \alpha_n = \lim_n \sup_{n \leq k} \alpha_k.$$

An $\mathbb{R}$-valued monotone and bounded sequence $\{\alpha_n\}$ converges to a limit α in $\mathbb{R}$. Let $\{f_n\}$ be a sequence of extended real-valued functions $f_n\colon X \to \overline{\mathbb{R}}$ on X. Since each $f_n(x)$ lies in $\overline{\mathbb{R}}$ for every x in X, set

$$\phi(x) = \inf_n f_n(x), \qquad \Phi(x) = \sup_n f_n(x),$$
$$\underline{f}(x) = \liminf_n f_n(x), \qquad \overline{f}(x) = \limsup_n f_n(x),$$

for every $x \in X$. These values always exist as elements of $\overline{\mathbb{R}}$, and so they define four functions, viz., $\phi\colon X \to \overline{\mathbb{R}}$, $\Phi\colon X \to \overline{\mathbb{R}}$, $\underline{f}\colon X \to \overline{\mathbb{R}}$, and $\overline{f}\colon X \to \overline{\mathbb{R}}$. If $\underline{f}(x) = \overline{f}(x)$ for every $x \in X$, then the $\overline{\mathbb{R}}$-valued sequence $\{f_n(x)\}$ converges

in $\overline{\mathbb{R}}$ for every $x \in X$. The common value $\underline{f}(x) = \overline{f}(x) \in \overline{\mathbb{R}}$, denoted by $f(x)$ for each $x \in X$, is called the *limit* of the $\overline{\mathbb{R}}$-valued sequence $\{f_n(x)\}$. This defines a function $f\colon X \to \overline{\mathbb{R}}$. In this case (i.e., when $\underline{f} = \overline{f}$), we say that the sequence of functions $\{f_n\}$ *converges pointwise* to the *limit function* f and write $f = \lim_n f_n$, which means that for every $x \in X$,

$$f(x) = \lim_n f_n(x).$$

For a sequence $\{f_n\}$ of real-valued functions this coincides with the standard definition of pointwise convergence (restricted to $\mathbb{R}$); viz., a sequence of real-valued functions $f_n\colon X \to \mathbb{R}$ converges pointwise to a real-valued function $f\colon X \to \mathbb{R}$ if $|f_n(x) - f(x)| \to 0$ as $n \to \infty$ for every $x \in X$. In other words, a sequence $\{f_n\}$ of $\mathbb{R}$-valued functions on X converges pointwise if there exists an $\mathbb{R}$-valued function f on X such that, for every $\varepsilon > 0$ and each $x \in X$, there is an integer $n_{\varepsilon,x} \in \mathbb{N}$ such that $|f_n(x) - f(x)| \le \varepsilon$ whenever $n \ge n_{\varepsilon,x}$.

Proposition 1.8. *If each function f_n is measurable, then so are the functions ϕ, Φ, $\underline{f}$ and $\overline{f}$, and so also is the limit $f = \lim_n f_n$ if the limit exists.*

Proof. Recall that arbitrary intersections of closed sets are closed, and arbitrary unions of open sets are open. Take any $\alpha \in \mathbb{R}$ and note that

$$\big\{x \in X\colon\ \phi(x) \ge \alpha\big\} = \bigcap\nolimits_n \big\{x \in X\colon\ f_n(x) \ge \alpha\big\},$$
$$\big\{x \in X\colon\ \Phi(x) > \alpha\big\} = \bigcup\nolimits_n \big\{x \in X\colon\ f_n(x) > \alpha\big\}.$$

Suppose $\{f_n\}$ is a sequence of measurable functions. The preceding sets are measurable (because they consist of countable intersections and countable unions of measurable sets), and hence ϕ and Φ are measurable functions. Set

$$\phi_n(x) = \inf_{n \le k} f_k(x) \quad \text{and} \quad \Phi_n(x) = \sup_{n \le k} f_k(x)$$

for each integer $n \in \mathbb{N}$ and every point $x \in X$. Since each function f_n is measurable, the same argument ensures that the functions ϕ_n and Φ_n are measurable too, and so are the functions $\underline{f}$ and $\overline{f}$. In fact,

$$\underline{f}(x) = \sup_n \inf_{n \le k} f_k(x) = \sup_n \phi_n(x) \quad \text{and} \quad \overline{f}(x) = \inf_n \sup_{n \le k} f_k(x) = \inf_n \Phi_n(x)$$

for every $x \in X$. Recall that f exists if and only if $\underline{f} = \overline{f}$. □

Appropriate versions of Propositions 1.5 and 1.6 still hold for extended real-valued functions, as we will see in Proposition 1.9. Let us first recall some usual conventions. We declare that $0 \cdot (\pm\infty) = 0$ so that if $\gamma = 0$, then $\gamma f(x) = 0$ for all $x \in X$ (i.e., $\gamma f = 0$ if $\gamma = 0$) for every $\overline{\mathbb{R}}$-valued function f on X. If f and g are $\overline{\mathbb{R}}$-valued $\mathcal{X}$-measurable functions on X, then the sets

$$F_{+\infty} \cap G_{-\infty} = \big\{x \in X\colon\ f(x) = +\infty \ \text{ and } \ g(x) = -\infty\big\},$$

$$G_{+\infty} \cap F_{-\infty} = \big\{x \in X\colon\ g(x) = +\infty \ \text{ and } \ f(x) = -\infty\big\},$$

are $\mathcal{X}$-measurable (cf. Problem 1.4). However, the function $f+g$ is not pointwise defined on these sets. That is, $f+g$ is not defined by $(f+g)(x) = f(x)+g(x)$ if x lies in $F_{+\infty} \cap G_{-\infty}$ or in $G_{+\infty} \cap F_{-\infty}$. So we declare that

$$(f+g)(x) = 0 \quad \text{for all} \quad x \in (F_{+\infty} \cap G_{-\infty}) \cup (G_{+\infty} \cap F_{-\infty}).$$

Recall the definition of the $\overline{\mathbb{R}}$-valued functions $f \wedge g$ and $f \vee g$ on X: for each $x \in X$, $(f \wedge g)(x) = \min\{f(x), g(x)\}$ and $(f \vee g)(x) = \max\{f(x), g(x)\}$.

Proposition 1.9. *If f and g are $\overline{\mathbb{R}}$-valued measurable functions on X, then*

$$\gamma f, \quad f+g, \quad f^+, \quad f^-, \quad |f|, \quad f \wedge g, \quad f \vee g, \quad fg$$

are $\overline{\mathbb{R}}$-valued measurable functions on X as well.

Proof. Consider a measurable space $(X, \mathcal{X})$, take a pair of measurable $\overline{\mathbb{R}}$-valued functions f and g on X, and let γ be any *real* number. The previous conventions ensure that the functions γf and $f+g$ are well defined, and they are measurable as well (use the same argument as in the proof of Proposition 1.5). The arguments in the first part of the proof of Proposition 1.6 still hold for $\overline{\mathbb{R}}$-valued functions, so f^+, f^-, and $|f|$ also are well-defined measurable functions. Similarly (cf. Problem 1.3), $f \wedge g$ and $f \vee g$ are well defined and measurable. Note that fg is a well-defined $\overline{\mathbb{R}}$-valued function. To show that it is measurable, proceed as follows. For each pair of integers $m, n \in \mathbb{N}$ take the truncated functions $f_n\colon X \to \mathbb{R}$ and $g_m\colon X \to \mathbb{R}$ given by

$$f_n(x) = \begin{cases} f(x), & |f(x)| \le n, \\ n, & f(x) > n, \\ -n, & f(x) < -n, \end{cases} \qquad g_m(x) = \begin{cases} g(x), & |g(x)| \le m, \\ m, & g(x) > m, \\ -m, & g(x) < -m, \end{cases}$$

for every $x \in X$. Since $\{f_n\}$ and $\{g_m\}$ are sequences of $\mathbb{R}$-valued $\mathcal{X}$-measurable functions (cf. Problem 1.5), it follows that $f_n\, g_m$ are $\mathbb{R}$-valued $\mathcal{X}$-measurable functions for each pair of integers $m, n \in \mathbb{N}$ according to Proposition 1.5. Note that the sequences $\{f_n\}$ and $\{g_m\}$ clearly converge pointwise to f and g, respectively. Since $\{f_n\, g_m\}$ is a sequence of $\mathcal{X}$-measurable functions for each $m \in \mathbb{N}$, and since

$$f g_m = \lim_n f_n\, g_m$$

for each $m \in \mathbb{N}$, it follows that $f g_m$ is $\mathcal{X}$-measurable for each $m \in \mathbb{N}$ by Proposition 1.8. Analogously, since each $f g_m$ is $\mathcal{X}$-measurable, and since

$$fg = \lim_m fg_m,$$

it follows by Proposition 1.8 that fg is $\mathcal{X}$-measurable. □

Observe that the argument in the proof of Proposition 1.5 can be used to show that the function f^2 is well defined and $\mathcal{X}$-measurable, but the above conventions are not enough to ensure, for $\overline{\mathbb{R}}$-valued functions, the identity

$$fg = \tfrac{1}{4}\big((f+g)^2 - (f-g)^2\big).$$

Let $\mathcal{M}(X,\mathcal{X})$ denote the collection of all real-valued functions on X that are measurable with respect to a σ-algebra $\mathcal{X}$ of subsets of X. Write $\mathcal{M}$ for $\mathcal{M}(X,\mathcal{X})$ when the measurable space $(X,\mathcal{X})$ is clear in the context; that is,

$$\mathcal{M} = \mathcal{M}(X,\mathcal{X}) = \big\{f : X \to \mathbb{R} :\ f \text{ is } \mathcal{X}\text{-measurable}\big\}.$$

The collection consisting of all real-valued functions on a set X is usually denoted by $\mathbb{R}^X$. It is well known (and readily verified) that $\mathbb{R}^X$ is a real *linear space*. If $\mathcal{X}$ is a σ-algebra of subsets of X, then Proposition 1.5 ensures that the collection $\mathcal{M}(X,\mathcal{X})$ of all $\mathcal{X}$-measurable real-valued functions on X forms a *linear manifold* of the linear space $\mathbb{R}^X$ (i.e., addition and product by a real scalar of $\mathcal{X}$-measurable real-valued functions are again $\mathcal{X}$-measurable real-valued functions). Thus, regarding *real-valued* functions,

$$\mathcal{M}(X,\mathcal{X}) \text{ is a real linear space.}$$

As usual, the collection of all extended real-valued functions on a set X that are measurable with respect to a σ-algebra $\mathcal{X}$ of subsets of X will also be denoted by $\mathcal{M}(X,\mathcal{X})$ (or simply by $\mathcal{M}$) — the same notation as in the case of real-valued functions. It is worth noticing, however, that in the case of *extended* real-valued functions, $\mathcal{M}(X,\mathcal{X})$ is not a linear space because $\overline{\mathbb{R}}^X$ is not a linear space (since $f-g$ is not well defined in this case).

1.4 Problems

Problem 1.1. Take a measurable space $(X,\mathcal{X})$. If $f : X \to \mathbb{R}$ is a real-valued function on X, then show that the next assertions are pairwise equivalent.

(a) $f^{-1}\big((\alpha,\infty)\big) = \big\{x \in X :\ f(x) > \alpha\big\} \in \mathcal{X}$ for every $\alpha \in \mathbb{R}$,

(b) $f^{-1}\big([\alpha,\infty)\big) = \big\{x \in X :\ f(x) \geq \alpha\big\} \in \mathcal{X}$ for every $\alpha \in \mathbb{R}$,

(c) $f^{-1}\big((-\infty,\alpha)\big) = \big\{x \in X :\ f(x) < \alpha\big\} \in \mathcal{X}$ for every $\alpha \in \mathbb{R}$,

(d) $f^{-1}\big((-\infty,\alpha]\big) = \big\{x \in X :\ f(x) \leq \alpha\big\} \in \mathcal{X}$ for every $\alpha \in \mathbb{R}$.

Moreover, show that these four assertions can be equivalently stated if we replace the assumption "for every $\alpha \in \mathbb{R}$" with "for every $\alpha \in \mathbb{Q}$", where $\mathbb{Q}$ stands for the field of all rational numbers (cf. Proof of Proposition 1.5).

Hint: If $\alpha \in \mathbb{R}$, then $\alpha = \lim_n \alpha_n$ with $\alpha_{n+1} < \alpha_n \in \mathbb{Q}$ for each $n \geq 1$ (decreasing rational sequence) so that $f^{-1}((\alpha, \infty)) = \bigcup_n \{x \in X\colon\ f(x) > \alpha_n\}$.

Thus the preceding eight assertions are equivalent forms of defining a measurable function, and they still hold if $f\colon X \to \overline{\mathbb{R}}$ (i.e., if f takes values in $\overline{\mathbb{R}}$).

Problem 1.2. Consider the real line $\mathbb{R}$ and take the Borel σ-algebra $\mathfrak{R}$. By a measurable real-valued function on $\mathbb{R}$ we mean an $\mathfrak{R}$-*measurable function* (or a *Borel measurable*, or simply a *Borel function*). Prove the assertions below.

(a) Every continuous function $f\colon \mathbb{R} \to \mathbb{R}$ is measurable.

(b) Every monotone function $f\colon \mathbb{R} \to \mathbb{R}$ is measurable.

Now consider the characteristic function $\chi_{\mathbb{Q}}$ of the rational set $\mathbb{Q}$, which is far from being continuous or monotone. This is called the *Dirichlet function*.

(c) The Dirichlet function $\chi_{\mathbb{Q}}\colon \mathbb{R} \to \mathbb{R}$ is measurable.

Problem 1.3. Let $f\colon X \to \mathbb{R}$ and $g\colon X \to \mathbb{R}$ be arbitrary measurable functions (with respect to a σ-algebra $\mathcal{X}$ of subsets of X). Define the functions $e\colon X \to \mathbb{R}$ and $h\colon X \to \mathbb{R}$ as follows. For each $x \in X$,

$$e(x) = \min\big\{f(x), g(x)\big\} \quad \text{and} \quad h(x) = \max\big\{f(x), g(x)\big\},$$

denoted by $e = f \wedge g = \inf\{f, g\}$ and $h = f \vee g = \sup\{f, g\}$. Show that

$$e = \tfrac{1}{2}\big(f + g - |f - g|\big) \quad \text{and} \quad h = \tfrac{1}{2}\big(f + g + |f - g|\big).$$

Also show that e and h are $\mathcal{X}$-measurable functions.

Problem 1.4. Take an extended real-valued function $f\colon X \to \overline{\mathbb{R}}$, and show that if f is measurable with respect to a σ-algebra $\mathcal{X}$ of subsets of X, then

$$F_{+\infty} = \big\{x \in X\colon\ f(x) = +\infty\big\} \quad \text{and} \quad F_{-\infty} = \big\{x \in X\colon\ f(x) = -\infty\big\}$$

are $\mathcal{X}$-measurable sets.

Hint: With n ranging over $\mathbb{N}$,

$$F_{+\infty} = \bigcap\nolimits_n \big\{x \in X\colon\ f(x) > n\big\},$$

$$F_{-\infty} = \bigcap\nolimits_n \big\{x \in X\colon\ f(x) \leq -n\big\} = X \backslash \bigcup\nolimits_n \big\{x \in X\colon\ f(x) > -n\big\};$$

countable intersection and complement of countable union.

Problem 1.5. Consider a measurable space $(X, \mathcal{X})$. Let $f : X \to \overline{\mathbb{R}}$ be an arbitrary extended real-valued measurable function on X. For each *positive real* number β define the *β-truncation* of f as the real-valued function $f_\beta : X \to \mathbb{R}$ on X defined for each $x \in X$ by

$$f_\beta(x) = \begin{cases} f(x), & |f(x)| \le \beta, \\ \beta, & f(x) > \beta, \\ -\beta, & f(x) < -\beta. \end{cases}$$

Show that f_β is a measurable function.

Problem 1.6. Let $\mathcal{X}$ be a σ-algebra of subsets of X, and take a *nonnegative* $\mathcal{X}$-measurable function $f : X \to \overline{\mathbb{R}}$ on X. Show that there exists a sequence $\{\varphi_n\}$ of $\mathcal{X}$-measurable functions $\varphi_n : X \to \overline{\mathbb{R}}$ with the following properties.

(i) Each φ_n is nonnegative (i.e., $0 \le \varphi_n(x)$ for every $x \in X$ for each n).

(ii) $\{\varphi_n\}$ is increasing (i.e., $\varphi_n(x) \le \varphi_{n+1}(x)$ for every $x \in X$ for each n).

(iii) $\{\varphi_n\}$ converges pointwise to f (i.e., $f(x) = \lim_n \varphi_n(x)$ for every $x \in X$).

(iv) Each φ_n has a finite range (i.e., $\{\alpha \in \overline{\mathbb{R}} : \alpha = \varphi_n(x) \text{ for some } x \in X\}$ is a finite set for each n).

Now show that if f is *bounded* (i.e., if $\sup_{x \in X} |f(x)| < \infty$), then $\{\varphi_n\}$ *converges uniformly* to f (i.e., $\sup_{x \in X} |\varphi_n(x) - f(x)| \to 0$ as $n \to \infty$).

Hint: Take an arbitrary $n \in \mathbb{N}$ and, for each integer $0 \le k \le n\,2^n$, set

$$E_{n,k} = \begin{cases} \{x \in X : k\,2^{-n} \le f(x) < (k+1)\,2^{-n}\}, & k < n\,2^n, \\ \{x \in X : f(x) \ge n\}, & k = n\,2^n. \end{cases}$$

Verify that $\{E_{n,k}\}_{0 \le k \le n\,2^n}$ is a partition of X made up of $\mathcal{X}$-measurable sets. Then, for each $n \in \mathbb{N}$, set $\varphi_n(x) = 2^{-n} \sum_{k=0}^{n\,2^n} k\,\chi_{E_{n,k}}(x)$ for every $x \in X$.

Problem 1.7. Let $\mathbb{C}$ stand for the field of all complex numbers, the so-called complex field. Take an arbitrary complex-valued function $f : X \to \mathbb{C}$ on X. Consider its Cartesian decomposition:

$$f = f_1 + i\,f_2,$$

where f_1 and f_2 are real-valued functions on X. These are called the *real* and *imaginary parts* of f, defined by $f_1(x) = \operatorname{Re} f(x)$ and $f_2(x) = \operatorname{Im} f(x)$ for every $x \in X$. We say that a complex-valued function f is *measurable* with respect to the σ-algebra $\mathcal{X}$, or simply *$\mathcal{X}$-measurable*, if its real and imaginary parts f_1 and f_2 are both (real-valued) $\mathcal{X}$-measurable functions. (Compare with Remark 10.2.)

(a) Verify that $f: X \to \mathbb{C}$ is $\mathcal{X}$-measurable if and only if the inverse image of every open rectangle of the complex plane $\mathbb{C}$ is an $\mathcal{X}$-measurable set. That is, the set $\{x \in X: \alpha < f_1(x) < \beta \text{ and } \gamma < f_2(x) < \delta\}$ lies in $\mathcal{X}$ for all real numbers α, β, γ, and δ (with $\alpha < \beta$ and $\gamma < \delta$, of course).

(b) Generalize the above characterization: $f: X \to \mathbb{C}$ is $\mathcal{X}$-measurable if and only if the inverse image of every open set of $\mathbb{C}$ is an $\mathcal{X}$-measurable set.

(c) Also verify that sums and products of complex-valued measurable functions are again measurable, as well as the limit of every (pointwise) convergent sequence of complex-valued measurable functions.

Problem 1.8. Consider two measurable spaces $(X, \mathcal{X})$ and $(Y, \mathcal{Y})$. Let f be a function of X into Y. We say that f is a *measurable transformation* (with respect to the σ-algebras $\mathcal{X}$ and $\mathcal{Y}$) if the inverse image under f of every $\mathcal{Y}$-measurable set is an $\mathcal{X}$-measurable set. That is, $f: X \to Y$ is measurable with respect to the σ-algebras $\mathcal{X}$ and $\mathcal{Y}$ of subsets of X and Y if

$$f^{-1}(E) \in \mathcal{X} \quad \text{for every} \quad E \in \mathcal{Y},$$

where $f^{-1}(E) = \{x \in X: f(x) \in E\}$. Now set $Y = \mathbb{R}$, $\mathcal{Y} = \Re$ (the Borel σ-algebra) and consider a real-valued function $f: X \to \mathbb{R}$. Show that f is $\mathcal{X}$-measurable in the sense of Definition 1.2 if and only if it is measurable in the above sense; that is, $f^{-1}(B) \in \mathcal{X}$ for every Borel set $B \in \Re$. (Note the analogy with the definition of continuous functions between topological spaces — we will discuss this in Problem 1.13 below and also in Theorem 11.4.)

Problem 1.9. Let $f: X \to Y$ be an arbitrary function of a set X into a set Y. Let $\mathcal{Y}$ be a σ-algebra of subsets of Y. Show that the collection of the inverse images under f of $\mathcal{Y}$-measurable sets is a σ-algebra of subsets of X, that is,

$$\mathcal{X}_f = \big\{f^{-1}(E) \subseteq X: E \in \mathcal{Y}\big\} \text{ is a } \sigma\text{-algebra of subsets of } X.$$

Given a function $f: X \to Y$ and a σ-algebra $\mathcal{Y}$ of subsets of Y, also show that this $\mathcal{X}_f$ is the smallest σ-algebra of subsets of X that makes f measurable. This is called *the σ-algebra* (of subsets of X) *inversely induced* (or simply *induced*) *by f* — also called *the σ-algebra generated by f*. (Note the similarity between this concept and that of the topology inversely induced on X by f, which is the weakest topology on X that makes f continuous.)

Hint: Let $f: X \to Y$ be any function X into a set Y, and recall again that the inverse image of any subset B of Y under f is the subset of X given by

$$f^{-1}(B) = \big\{x \in X: f(x) \in B\big\}.$$

Show that $\varnothing = f^{-1}(\varnothing)$, $X = f^{-1}(Y)$, $X \backslash f^{-1}(B) = f^{-1}(Y \backslash B)$ for every $B \subseteq Y$, and that $\bigcup_\gamma f^{-1}(B_\gamma) = f^{-1}\big(\bigcup_\gamma B_\gamma\big)$ for every nonempty collection $\{B_\gamma\}$ of subsets of Y.

Problem 1.10. Consider three measurable spaces, namely, $(X, \mathcal{X})$, $(Y, \mathcal{Y})$, and $(Z, \mathcal{Z})$. Let $f: X \to Y$ and $g: Y \to Z$ be measurable functions (with respect to the σ-algebras $\mathcal{X}$ and $\mathcal{Y}$, and $\mathcal{Y}$ and $\mathcal{Z}$, respectively). Show that

$$\text{the composition } g \circ f: X \to Z \text{ is a measurable function}$$

(with respect to the σ-algebras $\mathcal{X}$ and $\mathcal{Z}$). Compare with Theorem 11.4(d).

Problem 1.11. Let $f: X \to \mathbb{R}$ be a real-valued $\mathcal{X}$-measurable function on a set X, where $\mathcal{X}$ is a σ-algebra of subsets of the set X, and let $g: \mathbb{R} \to \mathbb{R}$ be a continuous function. Show that

$$\text{the composition } g \circ f: X \to \mathbb{R} \text{ is } \mathcal{X}\text{-measurable.}$$

Hint: Problems 1.2 and 1.10.

The result in Problem 1.11 will be extended in Theorem 11.4(e).

Problem 1.12. A collection $\mathcal{T} \subseteq \wp(X)$ of subsets of a set X is a *topology* on X if it satisfies the following conditions, referred to as the *topology axioms*.

(i) The whole set X and the empty set $\varnothing$ belong to $\mathcal{T}$.

(ii) The intersection of a finite collection of sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

(iii) The union of an arbitrary collection of sets in $\mathcal{T}$ belongs to $\mathcal{T}$.

A *topological space*, denoted by $(X, \mathcal{T})$ or simply X (if $\mathcal{T}$ is clear or immaterial), is a set X equipped with a topology $\mathcal{T}$. The sets in $\mathcal{T}$ are called the *open* sets of X with respect to $\mathcal{T}$. Let X be equipped with a topology, and equip the real line $\mathbb{R}$ with its usual topology (the one induced by the usual metric on $\mathbb{R}$). Show that if a real-valued function $f: X \to \mathbb{R}$ is *continuous* (with respect to the above topologies, which means that the inverse image under f of open sets of $\mathbb{R}$ are open sets of X), then f is measurable with respect to the σ-algebra $\mathcal{X}_\mathcal{T}$ generated by a topology $\mathcal{T}$ on X. In the jargon of the forthcoming Section 11.1 this means that

$$\text{every continuous function is Borel measurable.}$$

Problem 1.13. This generalizes Problem 1.12. Let $\mathcal{X}_\mathcal{T}$ be the σ-algebra generated by a topology $\mathcal{T}_X$ on a set X and let $\mathcal{Y}_\mathcal{T}$ be the σ-algebra generated by a topology $\mathcal{T}_Y$ on a set Y. If $f: X \to Y$ is a *continuous mapping*, i.e.,

$$f^{-1}(U) \in \mathcal{T}_X \quad \text{for every} \quad U \in \mathcal{T}_Y,$$

then (again) f is measurable (in the sense of Problem 1.8). Note: "continuity" is with respect to the topologies $\mathcal{T}_X$ and $\mathcal{T}_Y$, and "measurability" is with respect to the σ-algebras $\mathcal{X}_\mathcal{T}$ and $\mathcal{Y}_\mathcal{T}$. The notion of measurable functions will be discussed again in Section 11.1.

Problem 1.14. Take a topological space X with a topology $\mathcal{T}$. A subcollection $\mathcal{B}$ of $\mathcal{T}$ is a *base* (or a *topological base*) for X if it covers every open subset of X (i.e., every $U \in \mathcal{T}$ is the union of some subcollection of $\mathcal{B}$). If X is a metric space equipped with the metric topology $\mathcal{T}$, then it has a base of open balls (every open set is the union of open balls). A topological space is *separable* if it has a countable dense subset. A metric space is separable if and only if it has a countable base of open balls. (We return to these notions in Section 11.1.) Show that

> the σ-algebra $\mathcal{X}_\mathcal{T}$ generated by the metric topology $\mathcal{T}$ on a separable metric space coincides with the σ-algebra $\mathcal{X}_\mathcal{B}$ generated by any countable base $\mathcal{B}$ of open balls.

Remark 1.A (Borel σ-algebra): Recall that the real line $\mathbb{R}$ (equipped with its usual topology) is a separable metric space. (Indeed, the set $\mathbb{Q}$ of all rational numbers is countable and dense in $\mathbb{R}$.) Therefore, the Borel σ-algebra $\Re$ (i.e., the σ-algebra generated by the open intervals of the real line — see the last paragraph of Section 1.1) coincides with the σ-algebra $\mathcal{R}_\mathcal{B}$ generated by any countable base $\mathcal{B}$ of open intervals of the real line, which coincides with the σ-algebra $\mathcal{R}_\mathcal{T}$ generated by the topology $\mathcal{T}$ on $\mathbb{R}$ (i.e., the σ-algebra generated by the open sets). For this reason

> the σ-algebra $\mathcal{X}_\mathcal{T}$ of Problem 1.14 is called the *Borel σ-algebra* of subsets of a set X generated by a topology $\mathcal{T}$ on X.

(Borel σ-algebras will be revisited in Section 11.2.)

Problem 1.15. A nonempty class (i.e., a nonempty collection) $\mathcal{K}$ of subsets of a set X that contains the union of every increasing sequence in $\mathcal{K}$ and the intersection of every decreasing sequence in $\mathcal{K}$ is a *monotone class*. That is,

> $\mathcal{K} \subseteq \wp(X)$ is a *monotone class* if, whenever $\{E_n\}$ is an increasing sequence $(E_n \subseteq E_{n+1})$ of sets in $\mathcal{K}$ and $\{E'_n\}$ is a decreasing sequence $(E'_{n+1} \subseteq E'_n)$ of sets in $\mathcal{K}$, then $\bigcup_n E_n$ and $\bigcap_n E'_n$ are sets in $\mathcal{K}$.

Problems 1.15 to 1.19 are all about monotone classes, having in Problem 1.18 an important result called the Monotone Class Lemma, which will be required in the sequel. To begin with, prove the following assertions (where monotone classes, algebras, and σ-algebras are supposed to consist of subsets of the same fixed set X).

(a) Every σ-algebra is a monotone class.

(b) A monotone class is not necessarily a σ-algebra.

A nonempty collection of subsets of a set X that is both a monotone class and an algebra is called a *monotone algebra*.

(c) Every monotone algebra is a σ-algebra.

Hint: Let $\mathcal{K}$ be a monotone algebra and let $\{E_n\}$ be a sequence of sets in $\mathcal{K}$. Since $\mathcal{K}$ is an algebra, $\{\bigcup_{i=1}^{n} E_i\}$ is an increasing sequence of sets in $\mathcal{K}$. Since $\mathcal{K}$ is a monotone class, $\{\bigcup_{i=1}^{\infty} E_i\}$ lies in $\mathcal{K}$.

Problem 1.16. Let $\mathcal{C} \subseteq \wp(X)$ be a nonempty collection of subsets of a set X. Prove that there exists a smallest (in the inclusion ordering) monotone class $\mathcal{K}_\mathcal{C}$ of subsets of X that includes $\mathcal{C}$. That is, there exists a monotone class $\mathcal{K}_\mathcal{C}$ included in every monotone class that includes $\mathcal{C}$.

Hint: The intersection of any collection of monotone classes of subsets of X is again a monotone class of subsets of X.

This smallest monotone class $\mathcal{K}_\mathcal{C}$ is the *monotone class generated by* $\mathcal{C}$.

Problem 1.17. Let $\mathcal{C} \subseteq \wp(X)$ be an arbitrary nonempty collection of subsets of a given set X. Show that the monotone class $\mathcal{K}_\mathcal{C}$ generated by $\mathcal{C}$ is included in the σ-algebra $\mathcal{X}_\mathcal{C}$ generated by $\mathcal{C}$:

$$\mathcal{C} \subseteq \mathcal{K}_\mathcal{C} \subseteq \mathcal{X}_\mathcal{C}.$$

Give an example where the above inclusions are all proper.

Problem 1.18. Now prove the Monotone Class Lemma, which says that if the preceding collection is an algebra $\mathcal{A}$, then

$$\mathcal{K}_\mathcal{A} = \mathcal{X}_\mathcal{A}.$$

In other words, *the monotone class $\mathcal{K}_\mathcal{A}$ generated by an algebra $\mathcal{A}$ of subsets of a set X coincides with the σ-algebra $\mathcal{X}_\mathcal{A}$ generated by $\mathcal{A}$.* This is the Monotone Class Lemma, which plays a major role in Chapter 9.

Hint: Consider an algebra $\mathcal{A}$ of subsets of a set X, and let $\mathcal{K}_\mathcal{A}$ be the monotone class generated by $\mathcal{A}$. For each $E \in \mathcal{K}_\mathcal{A}$, set

$$\mathcal{E}_\mathcal{A} = \big\{E' \in \mathcal{K}_\mathcal{A} \colon E \backslash E' \in \mathcal{K}_\mathcal{A},\ E \cap E' \in \mathcal{K}_\mathcal{A} \text{ and } E' \backslash E \in \mathcal{K}_\mathcal{A}\big\} \subseteq \mathcal{K}_\mathcal{A}.$$

Show that

(i) $\mathcal{E}_\mathcal{A}$ is a monotone class,

(ii) $E' \in \mathcal{E}_\mathcal{A}$ if and only if $E \in \mathcal{E}'_\mathcal{A}$,

where the definition of $\mathcal{E}'_\mathcal{A}$ is analogous to that of $\mathcal{E}_\mathcal{A}$ (swapping E with E'). Since $\mathcal{A}$ is an algebra and $\mathcal{A} \subseteq \mathcal{K}_\mathcal{A}$ (cf. Problem 1.17), also show that

(iii) $\varnothing$, E, $X \backslash E$, and X all lie in $\mathcal{E}_\mathcal{A}$,

(iv) $E \in \mathcal{A}$ implies $\mathcal{A} \subseteq \mathcal{E}_\mathcal{A}$.

(Recall that $E \cap E' = X \backslash \big((X \backslash E) \cup (X \backslash E')\big)$ and $E \backslash E' = E \cap (X \backslash E')$, and hence $E \backslash E'$, $E \cap E'$, and $E' \backslash E$ all lie in $\mathcal{A}$ if E and E' lie in $\mathcal{A}$.) Now show from (i) and (iv) that (cf. Problem 1.16)

$$\mathcal{E}_{\mathcal{A}} = \mathcal{K}_{\mathcal{A}} \quad \text{for every} \quad E \in \mathcal{A}.$$

Thus, if $E \in \mathcal{A}$ and $E' \in \mathcal{K}_{\mathcal{A}}$, then $E' \in \mathcal{E}_{\mathcal{A}}$, and so $E \in \mathcal{E}'_{\mathcal{A}}$ by (ii). Hence $\mathcal{A} \subseteq \mathcal{E}'_{\mathcal{A}}$ for every $E' \in \mathcal{K}_{\mathcal{A}}$. Then show from (i) that (cf. Problem 1.16)

$$\mathcal{E}'_{\mathcal{A}} = \mathcal{K}_{\mathcal{A}} \quad \text{for every} \quad E' \in \mathcal{K}_{\mathcal{A}}.$$

So, if $E, E' \in \mathcal{K}_{\mathcal{A}}$, then $E \cap E'$, $E \backslash E'$, and $E' \backslash E$ lie in $\mathcal{K}_{\mathcal{A}}$. Also, property (iii) ensures that $\varnothing$ and X lie in $\mathcal{K}_{\mathcal{A}}$. Thus verify that intersections and complements of sets in $\mathcal{K}_{\mathcal{A}}$ remain in $\mathcal{K}_{\mathcal{A}}$, and so a finite union of sets in $\mathcal{K}_{\mathcal{A}}$ remains in $\mathcal{K}_{\mathcal{A}}$. (Recall: $E' \cup E = X \backslash \big((X \backslash E) \cap (X \backslash E')\big)$.) Hence, by Definition 1.1,

$$\mathcal{K}_{\mathcal{A}} \text{ is an algebra.}$$

Finally, use Problem 1.15(c) to show that $\mathcal{K}_{\mathcal{A}}$ is a σ-algebra, and then conclude that $\mathcal{X}_{\mathcal{A}} \subseteq \mathcal{K}_{\mathcal{A}}$, and therefore, by using Problem 1.17, infer that

$$\mathcal{K}_{\mathcal{A}} = \mathcal{X}_{\mathcal{A}}.$$

Problem 1.19. Prove that if a monotone class includes an algebra $\mathcal{A}$, then it also includes the σ-algebra $\mathcal{X}_{\mathcal{A}}$ generated by $\mathcal{A}$:

$$\mathcal{A} \subseteq \mathcal{K} \quad \Longrightarrow \quad \mathcal{X}_{\mathcal{A}} \subseteq \mathcal{K}.$$

Problems 1.15 to 1.19 (and their hints) give a rather complete account of monotone classes, which will be enough for our purposes. In particular, the Monotone Class Lemma, as stated in Problem 1.18, will play an important role in the proof of Lemma 9.7 in Chapter 9. This Lemma 9.7 is crucial for proving the classical versions of the Tonelli and Fubini Theorems (Theorems 9.8 and 9.9) in Chapter 9, which give sufficient conditions for interchanging the order of multiple integrals.

Problem 1.20. A simple example of a nonmeasurable set, and so of a nonmeasurable function, was given in Example 1C. There, the σ-algebra of subsets of $\mathbb{N}$ was $\mathcal{N} = \{\varnothing, \mathbb{N}_o, \mathbb{N}_e, \mathbb{N}\}$, so that $\{2, 3, 4, \ldots\} \notin \mathcal{N}$ trivially. However, we have not seen so far an example of a subset of $\mathbb{R}$ that is not $\mathfrak{R}$-measurable (i.e., not a Borel set). Consequently, we have not seen an example of a function $f : \mathbb{R} \to \mathbb{R}$ that is not $\mathfrak{R}$-measurable.

Are there subsets of $\mathbb{R}$ that are not $\mathfrak{R}$-measurable?

This was one of the deepest questions in the early days of measure theory. Yes, there are, and we will show their existence in Chapter 8 (cf. Property (P_4) at the end of Section 8.3). An example of such sets are the Vitali sets; they exist, as we will see in Problem 8.14, but are not constructible. Let $V \in \wp(\mathbb{R}) \backslash \mathfrak{R}$ be a Vitali set, which is a set of real numbers that is not a Borel set (i.e., a subset of $\mathbb{R}$ that is not $\mathfrak{R}$-measurable).

Note 1: Recall that a *singleton* in a set X is a subset of X containing one and only one point of X (notation: $\{x\} \subseteq X$ is a singleton at $x \in X$).

(a) Show that V is uncountable.

Hint: Singletons in $\mathbb{R}$ are $\mathfrak{R}$-measurable, and so is a countable union of singletons in $\mathbb{R}$, and therefore a countable subset of $\mathbb{R}$ is $\mathfrak{R}$-measurable.

(b) Show that there exists a function $f: \mathbb{R} \to \mathbb{R}$ such that $f^{-1}([0,1]) = V$.

Hint: There exists a one-to-one correspondence $\theta: V \to [0,1]$ because the set $V \subset \mathbb{R}$ is uncountable. Set $f: \mathbb{R} \to \mathbb{R}$ such that $f(x) = \theta(x)$ if $x \in V$ and $f(x) = 2$ if $x \in \mathbb{R} \backslash V$ (clearly, $\varnothing \neq V \neq \mathbb{R}$).

(c) Conclude that $f: \mathbb{R} \to \mathbb{R}$ is not $\mathfrak{R}$-measurable.

Hint: Apply Problem 1.8.

Note 2: Equivalently, take a one-to-one correspondence $\vartheta: V \to (0, \infty)$, set $g(x) = \vartheta(x)$ if $x \in V$ and $g(x) = 0$ if $x \in \mathbb{R} \backslash V$, so that $g^{-1}((0,\infty)) = V$, and hence $g: \mathbb{R} \to \mathbb{R}$ is not $\mathfrak{R}$-measurable according to Definition 1.2.

Note 3: Two distinguished fundamental postulates of set theory are being applied here. When it is assumed that uncountable subsets of $\mathbb{R}$ have the same cardinality as $\mathbb{R}$ itself, the *Continuum Hypothesis* is called into play (as we did in the hint to item (b) and in Note 2 above, regarding the existence of θ or ϑ). Also, the very argument for proving the existence of Vitali sets, as in Problem 8.14 of Chapter 8, uses the *Axiom of Choice*.

Suggested Readings

Bartle [5], Berberian [9], Halmos [31], Royden [62], Rudin [63]. For introductory set theory, see [32], [68], [71] and the first chapter of [14], [21], [31], [37], [41], [43], [51], [62]. For general topology, see [14], [21], [25, Chapter 4], [37], [41, Chapters 2 and 3], [43, Chapter 3], [51, Chapters 7 & 8], [62, Part Two].



2

Measure on a σ-Algebra

2.1 Measure and Measure Space

A function whose domain is a collection of sets is called a *set function*. A *measure* is a nonnegative extended real-valued set function satisfying some further conditions. The domain of a measure is a subcollection of the power set $\wp(X)$ of a given set X. It is advisable to require that the empty set $\varnothing$ and the whole set X itself belong to the domain, and to assign the minimum (zero) for the value of the function at the empty set. It is also convenient to require *additivity* in the following sense. Assume that every finite union of sets in the domain is again a set in the domain. This indicates that the domain might be an algebra. Then assume that the value of the function at any finite union of disjoint sets in the domain equals the sum of the values of the function at each set. Actually, this leads to a possible definition of a concept of measure (measures defined on an algebra will be considered in Chapter 8). However, such an approach lacks an important feature needed to build up a useful theory, viz., *countable additivity*. That is, it is required that the notion of additivity also holds for countably infinite unions of disjoint sets, and so countably infinite unions of sets are supposed to be in the domain of a measure. This compels the domain to be a σ-algebra.

Definition 2.1. A *measure* is an extended real-valued set function μ on a σ-algebra $\mathcal{X}$ of subsets of a set X,

$$\mu\colon \mathcal{X} \to \overline{\mathbb{R}},$$

that fulfills the following axioms (referred to as the *measure axioms*).

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_2

(a) $\mu(\varnothing) = 0$,

(b) $\mu(E) \geq 0$ for every $E \in \mathcal{X}$ (*positiveness*),

(c) $\mu\big(\bigcup_n E_n\big) = \sum_n \mu(E_n)$ (*countable additivity*) for every countable family $\{E_n\}$ of pairwise disjoint ($E_m \cap E_n = \varnothing$ whenever $m \neq n$) sets in $\mathcal{X}$ (i.e., a measure μ is *countably additive*).

A triple $(X, \mathcal{X}, \mu)$ consisting of an arbitrary set X, a σ-algebra $\mathcal{X}$ of subsets of X, and a measure μ on $\mathcal{X}$ is called a *measure space*. A *measure on* $\mathcal{X}$ is sometimes referred to as a *measure over* X (if $\mathcal{X}$ is clear), and if $A \subset X$ is such that $A \in \mathcal{X}$ and $\mu(A) = \mu(X)$, then μ is *concentrated on* A.

Since μ takes values in $\overline{\mathbb{R}}$, it is possible that $\mu(E) = +\infty$ for some $\mathcal{X}$-measurable sets E. If the countable family $\{E_n\}$ in (c) is infinite, and if $\mu(E_n) \in \mathbb{R}$ for every n (i.e., if all the sets E_n have finite measure), then we get an infinite sum of nonnegative real numbers in (c), and therefore either a convergent (in fact, unconditionally convergent), or a divergent series of nonnegative real numbers — in the latter case, $\mu\big(\bigcup_n E_n\big) = +\infty$.

A real-valued measure $\mu\colon \mathcal{X} \to \mathbb{R}$ is referred to as a *finite measure*. This means that $\mu(E) < \infty$ for all $E \in \mathcal{X}$. Equivalently, μ is *finite* if (and only if)

$$\mu(X) < \infty$$

(by Proposition 2.2(a) below). In particular, μ is a *probability measure* if

$$\mu(X) = 1$$

and, in this particular case, $(X, \mathcal{X}, \mu)$ is called a *probability space.* For any measure space $(X, \mathcal{X}, \mu)$, if there is a countable covering of X consisting of $\mathcal{X}$-measurable sets of finite measure, then μ is a *σ-finite measure*. Equivalently, μ is *σ-finite* if there is an $\mathcal{X}$-valued sequence $\{E_n\}$ such that

$$X = \bigcup_n E_n \text{ with } \mu(E_n) < \infty \text{ for every } n.$$

A set E in $\mathcal{X}$ is a *σ-finite set* (with respect to a measure μ) if there is a countable covering of E made up of measurable sets of finite measure. That is,

$$E \subseteq \bigcup_n E_n \text{ with } \mu(E_n) < \infty \text{ for every } n.$$

An *atom* of a measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ is a measurable set A in $\mathcal{X}$ such that $\mu(A) > 0$ and, for every measurable subset E of A, either $\mu(E) = 0$ or $\mu(E) = \mu(A)$. In other words, $A \in \mathcal{X}$ is an *atom* of a measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ if

$$\mu(A) > 0 \text{ and } \mu(E) = \mu(A) \text{ for every } E \in \wp(A) \cap \mathcal{X} \text{ with } \mu(E) \neq 0.$$

A set A in $\mathcal{X}$ is *atomic* (or *purely atomic*) with respect to μ on $\mathcal{X}$ if every measurable subset of it is the union of a set of μ-measure zero and a disjoint collection of atoms. That is, A is *atomic* if, for a collection $\{A_\gamma\}$ of atoms,

$$E \in \wp(A) \cap \mathcal{X} \implies E = N \cup \left(\bigcup_\gamma A_\gamma\right), \quad \mu(N) = 0, \quad A_\alpha \cap A_\beta = \varnothing \text{ for } \alpha \neq \beta.$$

A set in $\mathcal{X}$ is *atom-free* with respect to μ on $\mathcal{X}$ if it has no atom as a subset. A measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ is *atomic* (or *purely atomic*) if every set E in $\mathcal{X}$ with $\mu(E) > 0$ includes an atom. A measure on $\mathcal{X}$ is *atom-free* if every set in $\mathcal{X}$ is atom-free with respect to it.

Example 2A. Let $\mathcal{X}$ be a σ-algebra of subsets of a nonempty set X. Associated with any point $x \in X$, define a set function $\delta_x : \mathcal{X} \to \mathbb{R}$ as follows:

$$\delta_x(E) = \begin{cases} 1, & x \in E, \\ 0, & x \in X \backslash E, \end{cases}$$

for every set $E \in \mathcal{X}$. By Definition 2.1, δ_x is a measure (a probability measure). This δ_x is the *Dirac measure* at x (or the *unit point measure* concentrated at $\{x\}$). Every measurable set containing x is an atom of δ_x, and so δ_x is an atomic measure. If singletons (i.e., sets containing just one element) are measurable, then $\{x\}$ is an atom of δ_x and, for every singleton $\{y\} \subset X$,

$$\delta_x(\{y\}) = \begin{cases} 1, & x = y, \\ 0, & x \neq y. \end{cases}$$

Replacing 1 with a nonnegative $\alpha \in \overline{\mathbb{R}}$ we still get a measure $\delta_x^\alpha : \mathcal{X} \to \overline{\mathbb{R}}$,

$$\delta_x^\alpha(E) = \begin{cases} \alpha, & x \in E, \\ 0, & x \in X \backslash E, \end{cases}$$

for every $E \in \mathcal{X}$ (since $\delta_x^\alpha(\varnothing) = 0$, $\delta_x^\alpha(E) \geq 0$ for $E \in \mathcal{X}$, and $\delta_x^\alpha\left(\bigcup_n E_n\right) = \sum_n \delta_x^\alpha(E_n)$ for countable families $\{E_n\}$ of pairwise disjoint sets in $\mathcal{X}$). But δ_x^α is not a probability measure if $\alpha \neq 1$, and not even σ-finite if $\alpha = +\infty$.

Example 2B. Take the σ-algebra $\wp(\mathbb{N})$ of all subsets of the positive integers $\mathbb{N}$. Let $\#$ stand for cardinality (for a *finite* set S, $\#(S)$ is the number of elements in S). Take the function $\mu\colon \wp(\mathbb{N}) \to \overline{\mathbb{R}}$ defined, for each $E \in \wp(\mathbb{N})$, by

$$\mu(E) = \begin{cases} \#(E), & E \text{ is finite}, \\ +\infty, & E \text{ is infinite}. \end{cases}$$

It is readily verified by Definition 2.1 that μ is a measure, so $(\mathbb{N}, \wp(\mathbb{N}), \mu)$ is a measure space. Set $E_n = \{n\} \in \wp(\mathbb{N})$ for each $n \in \mathbb{N}$ so that $\mathbb{N} = \bigcup_n E_n$ and $\mu(E_n) = 1$. Then the measure μ is σ-finite, but it is not finite (there are infinite sets in $\wp(\mathbb{N})$; e.g., $\mathbb{N}$ itself). This is called the *counting measure* on $\wp(\mathbb{N})$.

Example 2C. Consider the σ-algebra $\mathfrak{R}$ of subsets of $\mathbb{R}$ generated by the collection of all open intervals. That is, consider the Borel σ-algebra $\mathfrak{R}$ (as

defined at the end of in Section 1.1). We will prove in Chapter 8 the existence and uniqueness of a measure $\lambda\colon \mathfrak{R} \to \overline{\mathbb{R}}$ such that

$$\lambda\big((\alpha,\beta)\big) = \beta - \alpha$$

for every open interval (α,β) of $\mathbb{R}$. This is the only measure on $\mathfrak{R}$ with the property of assigning to each open interval its own length. It is the *Lebesgue measure* on $\mathfrak{R}$. (See the introduction to Section 1.1 and the properties that close Section 8.3.) The Lebesgue measure is not finite (e.g., $\lambda(\mathbb{R}) = +\infty$) but is σ-finite. In fact, let $\mathbb{Z}$ be the set of all integers. For an arbitrary $\varepsilon > 0$ and each $k \in \mathbb{Z}$, let $E_k = (q_k - \varepsilon, q_k + \varepsilon)$ be the open interval of radius ε centered at q_k, where $\{q_k\}_{k\in\mathbb{Z}}$ is an enumeration of the rational numbers $\mathbb{Q}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, we get $\mathbb{R} = \bigcup_k E_k$ and $\lambda(E_k) = 2\varepsilon$ for all $k \in \mathbb{Z}$.

Remark 2.A (Not σ-finite and uniformly σ-finite):

(a) The counting measure of Example 2B is well defined on the σ-algebra $\wp(\mathbb{R})$ of all subsets of the real line $\mathbb{R}$. Such a counting measure $\mu\colon \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ on $\wp(\mathbb{R})$ is an example of a measure that is *not σ-finite* (cf. Problem 2.4(b)).

(b) The Lebesgue measure $\lambda\colon \mathfrak{R} \to \overline{\mathbb{R}}$ was shown to be σ-finite in Example 2C by covering $\mathbb{R}$ with a countable collection of measurable sets of Lebesgue measure 2ε. This, however, can be accomplished by covering $\mathbb{R}$ with a countable collection of measurable sets of increasing finite Lebesgue measure (e.g., $B_n = (-n, n) \in \mathfrak{R}$ so that $\lambda(B_n) = 2n$ for $n \in \mathbb{N}$). In other words, σ-finiteness requires a countable covering of sets of finite measure, but does not require a countable covering of sets of uniformly bounded measure. (The Lebesgue measure is, in fact, *uniformly σ-finite* — cf. Problem 2.1).

Example 2D. Scaling the Lebesgue measure λ of Example 2C leads to another measure on $\mathfrak{R}$. Indeed, for any (positive) real number $\gamma > 0$, the function $\lambda_\gamma\colon \mathfrak{R} \to \overline{\mathbb{R}}$ defined by $\lambda_\gamma = \gamma\lambda$, so that

$$\lambda_\gamma\big((\alpha,\beta)\big) = \gamma\lambda\big((\alpha,\beta)\big) = \gamma(\beta - \alpha) = \gamma\beta - \gamma\alpha$$

for every open interval (α,β) in $\mathbb{R}$, is again a measure on $\mathfrak{R}$. This is a homogeneous scaling. Inhomogeneous scaling also yields new measures. For instance, let $F\colon \mathbb{R} \to \mathbb{R}$ be a nondecreasing function (i.e., $F(\alpha) \le F(\beta)$ for $\alpha \le \beta$), which ensures that F has a left and a right limit at each point $\alpha \in \mathbb{R}$, denoted by $F(\alpha^-) = \lim_{\varepsilon\to 0} F(\alpha - |\varepsilon|)$ and $F(\alpha^+) = \lim_{\varepsilon\to 0} F(\alpha + |\varepsilon|)$ — if F is continuous at α, then $F(\alpha^+) = F(\alpha^-) = F(\alpha)$. The same argument used to construct the Lebesgue measure λ in Chapter 8 can be readily modified to show that there is a unique measure $\lambda_F\colon \mathfrak{R} \to \overline{\mathbb{R}}$ such that

$$\lambda_F\big((\alpha,\beta)\big) = F(\beta^-) - F(\alpha^+)$$

for every open interval (α,β) in $\mathbb{R}$. This measure λ_F is the *Borel–Stieltjes measure* generated by F, which again is σ-finite. Two useful particular cases:

(i) if $F = \chi_{[0,\infty)} : \mathbb{R} \to \{0,1\}$ is the characteristic function of $[0,\infty)$, then $\lambda_F = \delta_0 : \Re \to \{0,1\}$ is the Dirac measure at 0 of Example 2A;

(ii) if the nondecreasing function $F : \mathbb{R} \to \mathbb{R}$ is continuously differentiable, then $\lambda_F\big((\alpha,\beta)\big) = F(\beta) - F(\alpha) = \int_\alpha^\beta \frac{dF(\xi)}{d\xi}\, d\xi$.

A sequence $\{A_n\}$ of subsets of a set X is *increasing* (or nondecreasing) if $A_n \subseteq A_{n+1}$, and *decreasing* (or nonincreasing) if $A_{n+1} \subseteq A_n$, for every n. It is *monotone* if it is increasing or decreasing. The next proposition gives basic properties of measures. (More along this line is discussed in Problem 2.8).

Proposition 2.2. *Let $\mathcal{X}$ be a σ-algebra of subsets of a set X. Take arbitrary (measurable) sets A and B in $\mathcal{X}$, and an arbitrary sequence $\{E_n\}$ of sets in $\mathcal{X}$. The following properties hold true for every measure $\mu : \mathcal{X} \to \overline{\mathbb{R}}$.*

(a) $\mu(A) \le \mu(B)$ *if* $A \subseteq B$.

(b) $\mu(B\backslash A) = \mu(B) - \mu(A)$ *if* $A \subseteq B$ *and* $\mu(A) \ne +\infty$.

(c) $\mu\big(\bigcup_n E_n\big) = \lim_n \mu(E_n)$ *if* $\{E_n\}$ *is increasing.*

(d) $\mu\big(\bigcap_n E_n\big) = \lim_n \mu(E_n)$ *if* $\{E_n\}$ *is decreasing and* $\mu(E_1) \ne +\infty$.

Proof. If $A \subseteq B \subseteq X$, then $B = A \cup (B\backslash A)$ and $A \cap (B\backslash A) = \varnothing$. If both A and B are $\mathcal{X}$-measurable, then so is $B\backslash A = B \cap (X\backslash A)$ according to Definition 1.1(b,c)). Then, by Definition 2.1(b,c),

$$\mu(A) \le \mu(A) + \mu(B\backslash A) = \mu(A \cup (B\backslash A)) = \mu(B),$$

proving (a). If, in addition, $\mu(A) \ne +\infty$, then $\mu(B) - \mu(A)$ is in $\overline{\mathbb{R}}$ so that

$$\mu(B) - \mu(A) = \mu(B\backslash A),$$

which proves (b). Next let $\{E_n\}$ be an increasing sequence of $\mathcal{X}$-measurable sets. If one of E_n has an infinite measure, then assertion (c) follows from assertion (a). Thus suppose $\mu(E_n) \in \mathbb{R}$ for every $n \in \mathbb{N}$, and take a sequence $\{E'_n\}$ of $\mathcal{X}$-measurable sets recursively defined as follows: $E'_1 = E_1$ and $E'_{n+1} = E_{n+1}\backslash E_n$ for each $n \in \mathbb{N}$, which is a sequence of pairwise disjoint sets, and so (cf. Definition 2.1(c) and recall that $\bigcup_n E_n = \bigcup_n E'_n$)

$$\mu\Big(\bigcup\nolimits_n E_n\Big) = \mu\Big(\bigcup\nolimits_n E'_n\Big) = \sum\nolimits_n \mu(E'_n) = \mu(E'_1) + \lim_m \sum_{n=1}^m \mu(E'_{n+1}).$$

Then it follows from (b) that for an arbitrary integer $m > 1$,

$$\begin{aligned}\sum_{n=1}^m \mu(E'_{n+1}) &= \sum_{n=1}^m \mu(E_{n+1}\backslash E_n)\\ &= \sum_{n=1}^m \big(\mu(E_{n+1}) - \mu(E_n)\big) = \mu(E_{m+1}) - \mu(E_1),\end{aligned}$$

and hence $\mu\big(\bigcup_n E_n\big) = \lim_m \mu(E_{m+1})$, proving (c). Next suppose $\{E_n\}$ is a decreasing sequence of $\mathcal{X}$-measurable sets and set $E_n'' = E_1 \backslash E_n$ for each n so that $\{E_n''\}$ is an increasing sequence. Thus we can apply (c), but first note that since $E_n \subseteq E_1$ for all n and $\mu(E_1) \neq +\infty$, we get from (a) that $\mu(E_n) \in \mathbb{R}$ for all n (and hence $\mu\big(\bigcap_n E_n\big) \in \mathbb{R}$). Then by (b), for each n,

$$\mu(E_1 \backslash E_n) = \mu(E_1) - \mu(E_n) \quad \text{and} \quad \mu\Big(E_1 \backslash \bigcap\nolimits_n E_n\Big) = \mu(E_1) - \mu\Big(\bigcap\nolimits_n E_n\Big).$$

Recalling the De Morgan laws and applying (c), we get

$$\begin{aligned}\mu(E_1) - \mu\Big(\bigcap\nolimits_n E_n\Big) &= \mu\Big(E_1 \backslash \bigcap\nolimits_n E_n\Big) = \mu\Big(\bigcup\nolimits_n (E_1 \backslash E_n)\Big) = \mu\Big(\bigcup\nolimits_n E_n''\Big) \\ &= \lim_n \mu(E_n'') = \lim_n \mu(E_1 \backslash E_n) = \mu(E_1) - \lim_n \mu(E_n),\end{aligned}$$

completing the proof of (d) since $\mu(E_1) \neq +\infty$. □

Take a measure space $(X, \mathcal{X}, \mu)$. If a statement (or a proposition) $P(x)$ holds for all $x \in X \backslash N$ for some $N \in \mathcal{X}$ such that $\mu(N) = 0$, then we say that

$$P(x) \text{ holds } \mu\text{-}\textit{almost everywhere},$$

or *almost everywhere with respect to* μ, or simply *almost everywhere* if the measure μ is clear in the context. In symbols,

$$P(x) \text{ holds } \mu\text{-a.e.} \quad \text{or} \quad P(x) \text{ holds a.e.}$$

if $P(x)$ is true up to a set of measure zero; that is, there is an $\mathcal{X}$-measurable set N with $\mu(N) = 0$ (which may be $\varnothing$) such that $P(x)$ holds for all $x \in X \backslash N$. To avoid an empty definition of the notion of μ-almost everywhere, suppose

$$\mu\text{-a.e.} \quad \text{implies} \quad \mu(X) > 0.$$

Example 2E. Let $f : X \to Z$ and $g : X \to Z$ be functions of a set X into a set Z. The standard definition of equality between them is pointwise: $f = g$ if $f(x) = g(x)$ for all $x \in X$; that is, $f = g$ *everywhere* in X. Now take a measure $\mu : \mathcal{X} \to \overline{\mathbb{R}}$ on a σ-algebra $\mathcal{X}$ of subsets of X. We say that f and g are *equal almost everywhere with respect to* μ (or *equal* μ*-almost everywhere*), denoted

$$f = g \quad \mu\text{-a.e.},$$

if $f(x) = g(x)$ for every $x \in X \backslash N$ for some $N \in \mathcal{X}$ such that $\mu(N) = 0$. That is, $f(x) = g(x)$ for every x in the complement of a set of measure zero. For instance, take the measure space $(\mathbb{R}, \Re, \lambda)$ of Example 2C, and let f and g be real-valued functions on $\mathbb{R}$. If $f = 0$ (i.e., $f(x) = 0$ for all $x \in \mathbb{R}$), and $g(x) = 0$ for every $x \in \mathbb{R} \backslash \{0\}$ and $g(0) = 1$, then $f = g$ λ-a.e. but $f \neq g$ pointwise. (If f and g are continuous, then $f = g$ λ-a.e. if and only if $f = g$.)

Example 2F. Consider a sequence $\{f_n\}$ of real-valued functions $f_n : X \to \mathbb{R}$ on a set X. The sequence of functions $\{f_n\}$ is said to *converge pointwise* to

a function $f\colon X \to \mathbb{R}$ if the sequence of real numbers $\{f_n(x)\}$ converges to the real number $f(x)$ for every $x \in X$, which means convergence of $\{f_n(x)\}$ *everywhere* in X. Let $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ be a measure on a σ-algebra $\mathcal{X}$ of subsets of X. The sequence of functions $\{f_n\}$ *converges to* f *almost everywhere* (i.e., a.e.) *with respect to* μ (or *converges* μ*-almost everywhere to* f), denoted by

$$f_n \to f \quad \mu\text{-a.e.} \qquad \text{or} \qquad \lim_n f_n = f \quad \mu\text{-a.e.},$$

if $f_n(x) \to f(x)$ for every $x \in X \backslash N$ for some $N \in \mathcal{X}$ such that $\mu(N) = 0$. That is, $f_n(x) \to f(x)$ for every x except perhaps in a set of measure zero.

2.2 Signed Measure

Consider a measurable space $(X, \mathcal{X})$, let μ and η be two measures on $\mathcal{X}$, and let α be a nonnegative real number. It is easy to verify that $\alpha\mu$ and $\mu + \eta$ are again measures on $\mathcal{X}$ (pointwise defined; that is, $(\alpha\mu)(E) = \alpha\mu(E)$ and $(\mu + \eta)(E) = \mu(E) + \eta(E)$ for every $E \in \mathcal{X}$). This is readily extended (cf. Problem 2.1(d)): a linear combination $\sum_{i=1}^{n} \alpha_i \mu_i$ with nonnegative coefficients α_i of measures μ_i on $\mathcal{X}$ is a measure on $\mathcal{X}$. Nonnegative coefficients are imposed to ensure nonnegativeness for the resulting measure. If we ignore this, we may consider real coefficients. For instance, we may consider the set function $\mu - \eta$ on $\mathcal{X}$ (pointwise defined: $(\mu - \eta)(E) = \mu(E) - \eta(E)$ for every $E \in \mathcal{X}$). However, if there exists a set E in $\mathcal{X}$ such that $\mu(E) = \eta(E) = +\infty$, then $\mu(E) - \eta(E)$ may not be well defined. A way to avoid this is to assume that at least one of μ or η does not take on the value $+\infty$. Another (and simpler) way consists of assuming that μ and η are both real-valued measures. (Still another way will be discussed in the remark after Theorem 7.10).

Definition 2.3. A *signed measure* is a real-valued set function ν on a σ-algebra $\mathcal{X}$ of subsets of a set X,

$$\nu\colon \mathcal{X} \to \mathbb{R},$$

fulfilling the next two axioms (the *signed measure axioms*).

(a) $\nu(\varnothing) = 0$,

(b) $\nu\big(\bigcup_n E_n\big) = \sum_n \nu(E_n)$ (*countable additivity*)
for every countable family $\{E_n\}$ of pairwise disjoint sets in $\mathcal{X}$ for which the series $\sum_n \nu(E_n)$ is unconditionally convergent.

A signed measure is a *real-valued set function* on a σ-algebra that may fail to be a measure (a finite measure, at that) because it may not satisfy nonnegativeness in the second axiom of Definition 2.1. We are now dealing with real-valued functions. So the series in Definition 2.3(b) must converge;

otherwise, the left-hand side of (b) is not well defined. In Definition 2.1(c) we had a series of nonnegative terms, where convergence coincides with absolute convergence, which (in $\mathbb{R}$) means unconditional convergence. Now we have a series of real numbers, where unconditional convergence is not a consequence of plain convergence. So the identity in (b) requires unconditional convergence (equivalently, absolute convergence) because the union on the left-hand side is order invariant, and so is the series on the right-hand side.

The properties in Proposition 2.2(b,c,d) still hold for a signed measure (with essentially the same proof). Observe that any (finite) linear combination of signed measures is again a signed measure. If μ and η are finite measures, then the function $\nu = \mu - \eta$ is a signed measure. Such a setup was our motivation for defining signed measures. There are other ways to get signed measures from measures (cf. Lemma 4.6). Another important question is how to get measures from signed measures (cf. Section 7.1). The next example shows that what might seem the obvious way simply fails. The subsequent two examples exhibit measures generated by a signed measure.

Example 2G. Take a measurable space $(X, \mathcal{X})$, let $\nu\colon \mathcal{X} \to \mathbb{R}$ be a signed measure, and consider a set function $\varrho\colon \mathcal{X} \to \mathbb{R}$ defined for each $E \in \mathcal{X}$ by

$$\varrho(E) = |\nu(E)|.$$

This $\varrho\colon \mathcal{X} \to \mathbb{R}$ may *not* be a measure. Actually, if the signed measure ν is not a measure itself, then there is a set $B \in \mathcal{X}$ for which $\nu(B) < 0$. If there is a set $A \in \mathcal{X}$ such that $A \cap B = \varnothing$ and $\nu(A) = -\nu(B)$, then $\nu(A \cup B) = 0$. Thus $A \subset A \cup B$ and $\varrho(A) \not\leq \varrho(A \cup B)$, and so ϱ is not a measure (Proposition 2.2(a) — i.e., ϱ is not increasing). Another way to see this:

$$\varrho(A \cup B) = |\nu(A \cup B)| = 0 < |\nu(A)| + |\nu(B)| = \varrho(A) + \varrho(B).$$

Hence ϱ is not additive, not even finitely additive, and so ϱ is not a measure (Definition 2.1(c) — a concrete example is exhibited in Problem 2.12).

Example 2H. Let $(X, \mathcal{X})$ be a measurable space. Consider the σ-algebra $\mathcal{E} = \wp(E) \cap \mathcal{X}$ for each $\mathcal{X}$-measurable set E. Let $\nu\colon \mathcal{X} \to \mathbb{R}$ be a signed measure. We show that the function $\mu\colon \mathcal{X} \to \mathbb{R}$ defined for each $E \in \mathcal{X}$ by

$$\mu(E) = \sup_{A \in \mathcal{E}} \nu(A)$$

is a finite measure on $\mathcal{X}$. In fact, let $\{E_n\}$ be an arbitrary countable family of pairwise disjoint sets in $\mathcal{X}$, set $E = \bigcup E_n \in \mathcal{X}$, and take any $n \in \mathbb{N}$.

Claim. For each $\varepsilon > 0$ there is an $\mathcal{X}$-measurable set $A_n \subseteq E_n$ such that

$$\nu(A_n) \leq \mu(E_n) \leq \nu(A_n) + \tfrac{\varepsilon}{2^n}\,.$$

Proof of Claim. Observe that $\nu(A_n) \le \mu(E_n)$ according to the definition of μ. Take an arbitrary $\varepsilon > 0$. If $\nu(A) + \varepsilon < \mu(E)$ for every $A \in \mathcal{E}$, then $\mu(E) + \varepsilon = \sup_{A\in\mathcal{E}} \nu(A) + \varepsilon < \mu(E)$, which is a contradiction. Thus, for every $\varepsilon > 0$ and every $E \in \mathcal{X}$ there is an $A_\varepsilon \in \mathcal{E}$ such that $\mu(E) \le \nu(A_\varepsilon) + \varepsilon$. Hence, for any integer $n \in \mathbb{N}$ and any $\varepsilon > 0$, set $\varepsilon_n = \frac{\varepsilon}{2^n}$ so that there exists an $\mathcal{X}$-measurable $A_n \subseteq E_n$ for which $\mu(E_n) \le \nu(A_n) + \frac{\varepsilon}{2^n}$. $\square$

Therefore,

$$\sum\nolimits_n \nu(A_n) \le \sum\nolimits_n \mu(E_n) \le \sum\nolimits_n \nu(A_n) + \varepsilon$$

because $\sum_{n\in\mathbb{N}} \frac{1}{2^n} = 1$. Since $\{A_n\}$ is a disjoint sequence (reason: $\{E_n\}$ is a disjoint sequence) of sets in $\mathcal{X}$, and since ν is a signed measure, we get

$$\nu\Big(\bigcup\nolimits_n A_n\Big) = \sum\nolimits_n \nu(A_n).$$

Set $A = \bigcup_n A_n$ in $\mathcal{X}$ so that $A \subseteq E = \bigcup_n E_n$. Hence $A \in \mathcal{E}$ and

$$\nu(A) \le \sum\nolimits_n \mu(E_n) \le \nu(A) + \varepsilon.$$

This implies that

$$\mu(E) \le \sum\nolimits_n \mu(E_n) \le \mu(E) + \varepsilon$$

for every $\varepsilon > 0$ because $\mu(E) = \sup_{A\in\mathcal{E}} \nu(A)$. Thus, since $\bigcup_n E_n = E$,

$$\mu\Big(\bigcup\nolimits_n E_n\Big) = \mu(E) = \sum\nolimits_n \mu(E_n),$$

so that axiom (c) of Definition 2.1 (countable additivity) is satisfied. Since axioms (a) and (b) are trivially satisfied ($\wp(\varnothing) \cap \mathcal{X} = \{\varnothing\}$ and $\nu(\varnothing) = 0$, so $\mu(\varnothing) = 0$ and $\mu(E) \ge 0$ for every $E \in \mathcal{X}$), μ is a measure on $\mathcal{X}$. Moreover, it is readily verified that μ is indeed a finite measure (actually, it is dominated by the measure in the next example, which will also be shown to be finite).

Example 2I. A *covering* of a subset A of a set X is a collection of subsets of X that cover A (i.e., whose union includes A). A *partition* of A is then a covering of it consisting of disjoint subsets of it (i.e., a disjoint covering). Take a σ-algebra $\mathcal{X}$ of subsets of X and let E be any $\mathcal{X}$-measurable set. A *measurable covering* of E is a covering of it made up of $\mathcal{X}$-measurable sets. A *measurable partition* of E is a disjoint covering of it consisting of sets in $\mathcal{E} = \wp(E) \cap \mathcal{X}$. For each integer $n \in \mathbb{N}$ let $\boldsymbol{E}(n)$ be the collection of all measurable partitions of E containing precisely n sets, so that $\bigcup \boldsymbol{E}(n)$ is the collection of all *finite* measurable partitions of E. Take any signed measure $\nu\colon \mathcal{X} \to \mathbb{R}$ and consider the function $\mu\colon \mathcal{X} \to \mathbb{R}$ defined for each $E \in \mathcal{X}$ by

$$\mu(E) \;=\; \sup_{\{E_i\} \in \cup \boldsymbol{E}(n)} \sum\nolimits_i |\nu(E_i)|.$$

Every finite partition $\{E_i\} \in \bigcup \boldsymbol{E}(n)$ can be written as $\{E_i\} = \{E_j\} \cup \{E_k\}$, where $\{E_j\}$ and $\{E_k\}$ are disjoint collections such that $\nu(E_j) \geq 0$ for each j and $\nu(E_k) \leq 0$ for each k. Set $E^+ = \bigcup_j E_j$ and $E^- = \bigcup_k E_k$ in $\mathcal{E}$ so that

$$\nu(E^+) = \nu\Big(\bigcup\nolimits_j E_j\Big) = \sum\nolimits_j \nu(E_j) \geq 0,$$

$$\nu(E^-) = \nu\Big(\bigcup\nolimits_k E_k\Big) = \sum\nolimits_k \nu(E_k) \leq 0.$$

Therefore, $\sum_i |\nu(E_i)| = \nu(E^+) - \nu(E^-)$, where $\{E^+, E^-\} \in \boldsymbol{E}(2)$. Thus

$$\mu(E) = \sup_{\{E^+,\, E^-\} \in \boldsymbol{E}(2)} \big(\nu(E^+) - \nu(E^-)\big) \quad \text{for every} \quad E \in \mathcal{X},$$

where the supremum is taken over all measurable partitions $\{E^+, E^-\}$ of E consisting of two sets such that $\nu(E^+) \geq 0$ and $\nu(E^-) \leq 0$. This μ is a finite measure on $\mathcal{X}$, called the *variation* of the signed measure ν. Indeed, μ coincides with the "total variation" of ν, which is a finite measure $|\nu|\colon \mathcal{X} \to \mathbb{R}$ that will be discussed in Example 7A of Section 7.1. That is, $\mu = |\nu|$.

2.3 Completion of Measure Spaces

Consider a measure space $(X, \mathcal{X}, \mu)$. It is said to be *complete* if the σ-algebra contains all subsets of sets of measure zero. That is, $(X, \mathcal{X}, \mu)$ is *complete* if

$$N \in \mathcal{X}, \;\; \mu(N) = 0 \;\text{ and }\; A \subseteq N \quad \text{imply} \quad A \in \mathcal{X}.$$

If a measure space $(X, \mathcal{X}, \mu)$ is complete, then $\mathcal{X}$ is said to be a *complete σ-algebra* (with respect to a measure μ) and μ is said to be a *complete measure* on $\mathcal{X}$. Each measure space can be completed by adding enough subsets of measure zero to the σ-algebra, as we will see next. Consider a measure space $(X, \mathcal{X}, \mu)$, and take the collection

$$\mathcal{N} = \big\{N \in \mathcal{X}\colon\; \mu(N) = 0\big\}$$

of all sets in $\mathcal{X}$ of measure zero. Let $\widehat{\mathcal{X}}$ be the collection of all sets of the form $E \cup A$, where E is a set in $\mathcal{X}$ and A is a subset of some set in $\mathcal{N}$:

$$\widehat{\mathcal{X}} = \big\{\widehat{E} \subseteq X\colon\; \widehat{E} = E \cup A \;\text{ with }\; E \in \mathcal{X} \;\text{ and }\; A \subseteq N \;\text{ for some }\; N \in \mathcal{N}\,\big\}.$$

Consider a set function $\widehat{\mu}\colon \widehat{\mathcal{X}} \to \overline{\mathbb{R}}$ defined by

$$\widehat{\mu}(\widehat{E}) = \mu(E)$$

for each set $\widehat{E} \in \widehat{\mathcal{X}}$, where E is any set in $\mathcal{X}$ for which $\widehat{E} = E \cup A$ for some subset A of some N in $\mathcal{N}$. This function $\widehat{\mu}$ is well defined. In fact, take an

arbitrary $\widehat{E} \in \widehat{\mathcal{X}}$ and consider any pair of possible representations of it, say, $\widehat{E} = E_1 \cup A_1 = E_2 \cup A_2$, where E_1, E_2 are sets in $\mathcal{X}$ and A_1, A_2 are subsets of some sets N_1, N_2 in $\mathcal{N}$, respectively. Since $E_1 \subseteq E_1 \cup A_1 = E_2 \cup A_2 \subseteq E_2 \cup N_2$, and since $E_2 \cup N_2$ lies in $\mathcal{X}$ (Definition 1.1(c′)), it follows from Proposition 2.2(a) and Definition 2.1(c) that $\mu(E_1) \le \mu(E_2 \cup N_2) = \mu(E_2)$. Using the same argument, $\mu(E_2) \le \mu(E_1 \cup N_1) = \mu(E_1)$. So $\mu(E_1) = \mu(E_2)$. Thus the function $\widehat{\mu}$ is well defined: it assigns to each $\widehat{E}$ the value $\mu(E)$ for all representations $\widehat{E} = E \cup A$. As will be proved in Proposition 2.4 below,

(i) $\widehat{\mathcal{X}}$ is a σ-algebra of subsets of X that includes the σ-algebra $\mathcal{X}$ of subsets of X (which means that $\mathcal{X}$ is a *sub-σ-algebra* of $\widehat{\mathcal{X}}$), and

(ii) $\widehat{\mu}$ is a measure on $\widehat{\mathcal{X}}$ that agrees with μ on $\mathcal{X}$ (that is, $\widehat{\mu}$ is an *extension of μ over $\widehat{\mathcal{X}}$* or, equivalently, μ is a *restriction of $\widehat{\mu}$ to $\mathcal{X}$*).

Proposition 2.4. *Take an arbitrary measure space $(X, \mathcal{X}, \mu)$.*

(a) *$\widehat{\mathcal{X}}$ is a σ-algebra of subsets of X such that $\mathcal{X} \subseteq \widehat{\mathcal{X}}$.*

(b) *$\widehat{\mu}\colon \widehat{\mathcal{X}} \to \overline{\mathbb{R}}$ is a measure on $\widehat{\mathcal{X}}$ such that $\widehat{\mu}(E) = \mu(E)$ for every $E \in \mathcal{X}$.*

(c) *$(X, \widehat{\mathcal{X}}, \widehat{\mu})$ is a complete measure space.*

Proof. First note that $\mathcal{X} \subseteq \widehat{\mathcal{X}}$, since $E = E \cup \varnothing$ for every $E \in \mathcal{X}$. In particular, the empty set and the whole set trivially lie in $\widehat{\mathcal{X}}$. Take an arbitrary $\widehat{E} \in \widehat{\mathcal{X}}$. Next we show that $X \backslash \widehat{E} \in \widehat{\mathcal{X}}$. Indeed, if $\widehat{E} = E \cup A$, then $X \backslash \widehat{E} = X \backslash (E \cup A) = (X \backslash E) \cap (X \backslash A) = E' \backslash A$. Here $E' = X \backslash E$, which lies in $\mathcal{X}$ because $E \in \mathcal{X}$. Let N be any set in $\mathcal{N} \subseteq \mathcal{X}$ such that $A \subseteq N$. Thus $E' \backslash A = (E' \backslash N) \cup (N \backslash A) = E_1 \cup A_1$, where $E_1 = E' \backslash N$ lies in $\mathcal{X}$ since both E' and N lie in $\mathcal{X}$, and $A_1 = N \backslash A \subseteq N$. Hence

$$X \backslash \widehat{E} = E' \backslash A = E_1 \cup A_1 \quad \text{lies in} \quad \widehat{\mathcal{X}}.$$

Let $\{\widehat{E}_n\}$ be an arbitrary sequence of sets in $\widehat{\mathcal{X}}$. So $\widehat{E}_n = E_n \cup A_n$, where $E_n \in \mathcal{X}$ and $A_n \subseteq N_n$ with $N_n \in \mathcal{N}$, for each n. Set $E = \bigcup_n E_n$ in $\mathcal{X}$ (Definition 1.1($\widehat{\text{c}}$)), and $A = \bigcup_n A_n \subseteq N = \bigcup_n N_n$. Note that $N \in \mathcal{N}$ (Definition 2.1(c)). Then $\widehat{\mathcal{X}}$ is a σ-algebra (Definition 1.1) because

$$\bigcup\nolimits_n \widehat{E}_n = \bigcup\nolimits_n (E_n \cup A_n) = \Big(\bigcup\nolimits_n E_n\Big) \cup \Big(\bigcup\nolimits_n A_n\Big) = E \cup A \quad \text{lies in} \quad \widehat{\mathcal{X}},$$

completing the proof of (a). Note that the set function $\widehat{\mu}$ agrees with the measure μ on $\mathcal{X}$ (i.e., $\widehat{\mu}(E) = \mu(E)$ for every $E \in \mathcal{X}$) by the definition of $\widehat{\mu}$ (because $E = E \cup \varnothing$). In particular, $\widehat{\mu}(\varnothing) = 0$ and $\widehat{\mu}(\widehat{E}) = \mu(E) \ge 0$ for every $\widehat{E} \in \widehat{\mathcal{X}}$. Moreover, suppose the arbitrary sequence $\{\widehat{E}_n\}$ of sets in $\widehat{\mathcal{X}}$ (i.e., $\widehat{E}_n = E_n \cup A_n$, where $E_n \in \mathcal{X}$ and $A_n \subseteq N_n$ with $N_n \in \mathcal{N}$) is made up of pairwise disjoint sets. Thus $\{E_n\}$ is a sequence of pairwise disjoint sets in $\mathcal{X}$, and so it follows from the above displayed identity that

$$\widehat{\mu}\Big(\bigcup\nolimits_n \widehat{E}_n\Big) = \widehat{\mu}\big(E \cup A\big) = \mu(E) = \mu\Big(\bigcup\nolimits_n E_n\Big) = \sum\nolimits_n \mu(E_n) = \sum\nolimits_n \widehat{\mu}(\widehat{E}_n)$$

by the definition of $\widehat{\mu}$, since μ is a measure on $\mathcal{X}$ (Definition 2.1(c)). Then $\widehat{\mu}$ is a measure on $\widehat{\mathcal{X}}$ (Definition 2.1). This proves (b). Finally, set

$$\widehat{\mathcal{N}} = \big\{\widehat{N} \in \widehat{\mathcal{X}}\colon\ \widehat{\mu}(\widehat{N}) = 0\big\}.$$

If $\widehat{N} \in \widehat{\mathcal{N}}$, then $\widehat{N} \in \widehat{\mathcal{X}}$. Thus the set $\widehat{N}$ is of the form $\widehat{N} = N' \cup A$ with $N' \in \mathcal{X}$ and $A \subseteq N$ for some $N \in \mathcal{N}$ (by the definition of $\widehat{\mathcal{X}}$). However, $\mu(N') = \widehat{\mu}(\widehat{N}) = 0$ (cf. definition of $\widehat{\mu}$), and so $N' \in \mathcal{N}$. Outcome:

$$\widehat{N} \in \widehat{\mathcal{N}} \quad \text{implies} \quad \widehat{N} = N' \cup A \ \text{ with } \ A \subseteq N \ \text{ for some } \ N, N' \in \mathcal{N}. \quad \text{(i)}$$

Also, if $A \subseteq N \in \mathcal{N} \subseteq \mathcal{X}$, then $A = \varnothing \cup A$ must lie in $\widehat{\mathcal{X}}$ since $\varnothing \in \mathcal{X}$:

$$A \subseteq N \in \mathcal{N} \quad \text{implies} \quad A \in \widehat{\mathcal{X}}. \qquad \text{(ii)}$$

Hence, if $\widehat{A}$ is any subset of an arbitrary set $\widehat{N}$ in $\widehat{\mathcal{N}}$, then $\widehat{A} \subseteq \widehat{N} = N' \cup A$ with $A \subseteq N$ for some pair $N, N' \in \mathcal{N}$ by (i). Therefore, $\widehat{A} = A' \cup A''$, where $A' \subseteq N'$ and $A'' \subseteq A \subseteq N$. But both A' and A'' lie in $\widehat{\mathcal{X}}$ by (ii), and so $\widehat{A} \in \widehat{\mathcal{X}}$ (Definition 1.1(c′)). That is, $\widehat{\mathcal{X}}$ is a complete σ-algebra with respect to the measure $\widehat{\mu}$, which completes the proof of (c). □

The complete measure space $(X, \widehat{\mathcal{X}}, \widehat{\mu})$ is referred to as the *completion of the measure space* $(X, \mathcal{X}, \mu)$. Accordingly, we say that $\widehat{\mathcal{X}}$ is the *completion of the σ-algebra* $\mathcal{X}$ (with respect to the measure μ) and $\widehat{\mu}$ is the *completion of the measure* μ on $\mathcal{X}$.

Consider the Lebesgue measure λ on the σ-algebra $\mathfrak{R}$ of Borel sets as in Example 2C. The measure space $(\mathbb{R}, \mathfrak{R}, \lambda)$ is not complete. That is, on the Borel σ-algebra $\mathfrak{R}$, the Lebesgue measure λ is not complete. We will see this in Chapter 8, and also how to build its completion $\widehat{\lambda}$. Here $\widehat{\lambda}$ stands for the Lebesgue measure on the *Lebesgue σ-algebra* $\widehat{\mathfrak{R}}$, which in turn stands for the completion of the Borel σ-algebra $\mathfrak{R}$ with respect to Lebesgue measure λ. This makes a complete measure space $(\mathbb{R}, \widehat{\mathfrak{R}}, \widehat{\lambda})$.

2.4 Problems

Problem 2.1. Consider a measurable space $(X, \mathcal{X})$, and let $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ be a measure on the σ-algebra $\mathcal{X}$.

(a) Show that μ is a finite measure if and only if there exists $0 \leq \alpha \in \mathbb{R}$ such that $\mu(E) < \alpha$ for all $E \in \mathcal{X}$.

Consider the definition of a σ-finite measure and suppose there exists an $\mathcal{X}$-valued sequence $\{E_n\}$ and a real number $\alpha > 0$ such that $\mu(E_n) \leq \alpha$ for all n and $X = \bigcup_n E_n$. In this case μ is said to be *uniformly σ-finite*.

(b) Verify that the counting measure of Example 2B and the Lebesgue measure of Example 2C are uniformly σ-finite.

(c) When is the Borel–Stieltjes measure of Example 2D finite? When is it uniformly σ-finite?

(d) Show that a countable sum of measures on the same σ-algebra $\mathcal{X}$ is a measure on $\mathcal{X}$.

Problem 2.2. It is clear that every finite measure is uniformly σ-finite, and every uniformly σ-finite measure is σ-finite. However, the converses fail. Indeed, consider the σ-algebra $\wp(\mathbb{N})$ of all subsets of the natural numbers $\mathbb{N}$. According to the previous problem the counting measure on $\wp(\mathbb{N})$ is uniformly σ-finite, and it is trivially not finite (since the value of it at $\mathbb{N}$ is not finite). Now consider the function $\mu\colon \wp(\mathbb{N}) \to \overline{\mathbb{R}}$ defined by

$$\mu(E) = \sum\nolimits_{k\in E} k$$

for every $E \in \wp(\mathbb{N})$ (by convention, the empty sum is null). Verify that this is a measure on $\wp(\mathbb{N})$, which is σ-finite but not uniformly σ-finite.

Problem 2.3. Prove that a measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ on a σ-algebra $\mathcal{X}$ of subsets of a set X is σ-finite if and only if there exists a countable family $\{E_k\}$ of *disjoint* sets in $\mathcal{X}$ such that $\mu(E_k) < \infty$ for every k and $X = \bigcup_k E_k$.

Hint: Every sequence of sets $\{X_n\}_{n\in\mathbb{N}}$ has a *disjointification* $\{Y_n\}_{n\in\mathbb{N}}$ in the sense that $\{Y_n\}_{n\in\mathbb{N}}$ is a sequence of pairwise disjoint sets and

$$\bigcup\nolimits_n Y_n = \bigcup\nolimits_n X_n.$$

Problem 2.4. Let $\mathcal{X}$ be a σ-algebra of subsets of an arbitrary set X. A measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ is called *semifinite* if every measurable set of infinite measure includes a measurable set of arbitrarily large finite measure.

(a) Prove that every σ-finite measure is semifinite.

Hint: Suppose a measure is not semifinite but σ-finite (Problem 2.3).

Now suppose that X is an uncountable set. Let # stand for cardinality, and take two set functions $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ and $\eta\colon \mathcal{X} \to \overline{\mathbb{R}}$ defined for each $E \in \mathcal{X}$ by

$$\mu(E) = \begin{cases} \#(E), & E \text{ is finite,} \\ +\infty, & E \text{ is infinite,} \end{cases} \qquad \eta(E) = \begin{cases} 0, & E \text{ is countable,} \\ +\infty, & E \text{ is uncountable.} \end{cases}$$

(b) Verify that μ is a measure on $\mathcal{X}$ (the counting measure on an *uncountable* set) that is semifinite but not σ-finite.

(c) Verify that η is a measure on $\mathcal{X}$ that is not semifinite (thus not σ-finite according to (a)).

Hint: Infinite sets have countably infinite (proper) subsets. Countable unions of countable sets are countable.

Problem 2.5. Suppose $\mathcal{X}$ is a σ-algebra of subsets of an infinite set X for which all singletons are measurable sets. Let $\varrho\colon \mathcal{X} \to \overline{\mathbb{R}}$ and $\rho\colon \mathcal{X} \to \overline{\mathbb{R}}$ be set functions defined for each $E \in \mathcal{X}$ by

$$\varrho(E) = \begin{cases} 0, & E \text{ is finite,} \\ +\infty, & E \text{ is infinite,} \end{cases} \qquad \rho(E) = \begin{cases} 0, & E \text{ is finite,} \\ 1, & E \text{ is infinite.} \end{cases}$$

Question: Why are these functions not measures?

Problem 2.6. This is a continuation of the previous problem for an uncountable set. Suppose X is an uncountable set, and let $\mathcal{X}$ be a collection of those subsets of X that either are countable or are the complement of a countable subset of X. First show that $\mathcal{X}$ is a σ-algebra of subsets of X. Next consider the set function $\mu\colon \mathcal{X} \to \mathbb{R}$ defined for each $E \in \mathcal{X}$ by

$$\mu(E) = \begin{cases} 0, & E \text{ is countable,} \\ 1, & X \backslash E \text{ is countable.} \end{cases}$$

Verify that μ is a measure (actually, a probability measure) on $\mathcal{X}$.

Problem 2.7. Consider the Lebesgue measure $\lambda\colon \mathfrak{R} \to \overline{\mathbb{R}}$ on the Borel σ-algebra $\mathfrak{R}$ of subsets of $\mathbb{R}$ (cf. Example 2C). Prove the following assertions.

(a) Every singleton of $\mathbb{R}$ is $\mathfrak{R}$-measurable and has measure zero.

(b) Every countable subset of $\mathbb{R}$ is $\mathfrak{R}$-measurable and has measure zero.

(c) If $\alpha < \beta$, then the intervals (α, β), $[\alpha, \beta)$, $(\alpha, \beta]$, $[\alpha, \beta]$ are $\mathfrak{R}$-measurable and $\lambda\big((\alpha, \beta)\big) = \lambda\big([\alpha, \beta)\big) = \lambda\big((\alpha, \beta]\big) = \lambda\big([\alpha, \beta]\big) = \beta - \alpha$.

(d) Every nonempty open subset U of $\mathbb{R}$ is $\mathfrak{R}$-measurable and $\lambda(U) > 0$.
(*Hint:* $\mathbb{R}$ has a countable base of open intervals — see Problem 1.14.)

(e) Every bounded $\mathfrak{R}$-measurable subset of $\mathbb{R}$ has finite measure.

(f) A closed and bounded subset K of $\mathbb{R}$ is $\mathfrak{R}$-measurable and $\lambda(K) < \infty$.

Also show that λ is uniformly σ-finite by exhibiting a countably infinite family $\{B_k\}$ of *disjoint sets* in $\mathfrak{R}$ such that $\lambda(B_k) = 1$ for all k and $\mathbb{R} = \bigcup_k B_k$.

Problem 2.8. Let $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ be a measure on a σ-algebra $\mathcal{X}$, and let $\{E_n\}$ be a sequence of $\mathcal{X}$-measurable sets. Prove the following propositions.

(a) $\mu\big(\bigcup_n E_n\big) = \lim_m \mu\big(\bigcup_{i=1}^m E_i\big)$.

(b) $\mu\big(\bigcup_n E_n\big) \le \sum_n \mu(E_n)$ (*countable subadditivity*).

(c) *A countable collection of sets of measure zero has measure zero.*

(d) *If μ is σ-finite, a disjoint collection of positive-measure sets is countable.*

Hints: Use Proposition 2.2 to verify (a), (b) and (c). (a) Set $A_m = \bigcup_{i=1}^m E_i$. (b) Set $B_{m+1} = E_{m+1} \setminus \big(\bigcup_{i=1}^m E_i\big)$. So $B_m \subseteq E_m$ and $\{B_m\}$ is pairwise disjoint. (c) Apply (a). For (d) proceed as follows. Take $\{E_\gamma\}_{\gamma\in\Gamma}$ disjoint with $\mu(E_\gamma) > 0$. If $\mu(E) < \infty$ and $\frac{1}{k} \le \mu(E \cap E_{\gamma_k})$, then $\frac{n}{k} \le \sum_{k=1}^n \mu(E \cap E_\gamma) \le \mu(E) < \infty$. So $\{\gamma \in \Gamma\colon \mu(E \cap E_\gamma) \ge \frac{1}{k}\}$ is finite. Thus $\{\gamma \in \Gamma\colon \mu(E \cap E_\gamma) > 0\}$ is countable. If $X = \bigcup_k E'_k$ with $\mu(E'_k) < \infty$, set $\Gamma_k = \{\gamma \in \Gamma\colon \mu(E'_k \cap E_\gamma) > 0\}$, which is countable for each k. So $\Gamma = \bigcup_k \Gamma_k$ is countable.

Problem 2.9. The *Cantor set* is a rather important well-known subset of the interval $[0,1]$ of the real line $\mathbb{R}$, possessing striking properties, which make it a significant source of counterexamples. Six selected references dealing with the construction of the Cantor set and its most important properties are given at the end of this chapter (see Suggested Readings). Roughly speaking, the Cantor set is the intersection of a decreasing sequence $\{C_n\}$,

$$C_{n+1} \subset C_n \subset C_0 = [0,1],$$

of closed subsets of the unit interval, obtained by successive removal of the central open third, where each C_n is the union of 2^n disjoint nondegenerate subintervals of length $\frac{1}{3^n}$. The first three sets are shown below.

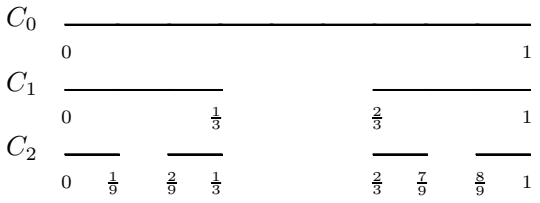


Fig. 2.1. The first three iterations for building the Cantor set.

The *Cantor set* is the set

$$C = \bigcap\nolimits_n C_n \;\subset\; [0,1] \;\subset\; \mathbb{R}.$$

Among the main properties of the Cantor set we point out that C is closed, without isolated points (so a perfect set), and nonempty. It is bounded, has an empty interior, and is totally disconnected. It is also uncountable. Take

the Lebesgue measure $\lambda\colon \Re \to \overline{\mathbb{R}}$ (Example 2C and Problem 2.7). Show that the Cantor set has Lebesgue measure zero. In other words, C lies in $\Re$ and

$$\lambda(C) = 0.$$

Hint: Each set C_n consists of 2^n disjoint intervals of length $\frac{1}{3^n}$.

Problem 2.10. Now we build up a Cantor-like set S whose Lebesgue measure $\lambda(S)$ is not null. Consider the setup and the construction of the previous problem, where each set C_n (for $n \in \mathbb{N}$) is obtained from C_{n-1} by removing 2^{n-1} central open subintervals, each of length $\frac{1}{3^n}$. Now, instead of removing 2^{n-1} central open subintervals of length $\frac{1}{3^n}$ at each iteration, remove 2^{n-1} central open subintervals of length $\frac{1}{4^n}$ at each iteration. Let $\{S_n\}$ be the resulting decreasing sequence of closed subsets of the unit interval $S_0 = [0,1]$. Show that the length of each S_n is $\lambda(S_n) = \frac{1}{2} + \frac{1}{2^{n+1}}$ and conclude that $S = \bigcap_n S_n \subset [0,1] \subset \mathbb{R}$ lies in $\Re$ and

$$\lambda(S) = \tfrac{1}{2}.$$

Hint: Proposition 2.2(d).

Problem 2.11. Take a measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ on a σ-algebra $\mathcal{X}$ of subsets of a set X. Let A be an arbitrary $\mathcal{X}$-measurable set, and consider the σ-algebra $\mathcal{A} = \wp(A) \cap \mathcal{X}$ of subsets of A, so that $\mathcal{A} \subseteq \mathcal{X}$. Define a pair of set functions $\mu_A\colon \mathcal{X} \to \overline{\mathbb{R}}$ and $\mu|_{\mathcal{A}}\colon \mathcal{A} \to \overline{\mathbb{R}}$ (the restriction of μ to $\mathcal{A}$) as follows:

$$\mu_A(E) = \mu(E \cap A) \quad \text{for every} \quad E \in \mathcal{X},$$

$$\mu|_{\mathcal{A}}(E) = \mu(E) \quad \text{for every} \quad E \in \mathcal{A}.$$

Verify that μ_A and $\mu|_{\mathcal{A}}$ are measures on $\mathcal{X}$ and on $\mathcal{A}$, respectively.

Hint: $\big(\bigcup_n E_n\big) \cap A = \bigcup_n (E_n \cap A)$.

The measure $\mu|_{\mathcal{A}}$ is *the restriction of both* μ *and* μ_A *to* $\mathcal{A}$, and so μ and μ_A are (different) *extensions of* $\mu|_{\mathcal{A}}$ *over* $\mathcal{X}$ — all these measures coincide on $\mathcal{A}$. For this reason, it is usual to write $\mu_A = \mu|_{\mathcal{A}}$ for the restriction of μ to $\mathcal{A}$.

Problem 2.12. Consider the Lebesgue measure $\lambda\colon \Re \to \overline{\mathbb{R}}$ on the Borel σ-algebra $\Re$ of subsets of the real line $\mathbb{R}$ (cf. Example 2C and Problem 2.7). Set $A = [1,2]$, $B = [-2,-1]$, and take the measures $\lambda_A\colon \Re \to \overline{\mathbb{R}}$ and $\lambda_B\colon \Re \to \overline{\mathbb{R}}$ defined in Problem 2.11. Show that λ_A and λ_B are finite measures so that their difference $\nu = \lambda_A - \lambda_B\colon \Re \to \mathbb{R}$ is a signed measure. Also show that

$$\nu(A \cup B) = \lambda_A(A \cup B) - \lambda_B(A \cup B) = 0,$$

$$|\nu(A)| + |\nu(B)| = |\lambda_A(A) - \lambda_B(A)| + |\lambda_A(B) - \lambda_B(B)| = 2.$$

Then conclude that the set function $\varrho\colon \Re \to \mathbb{R}$ defined, for each Borel set E in the Borel σ-algebra $\Re$, by $\varrho(E) = |\nu(E)| = |\lambda_A(E) - \lambda_B(E)|$, is not a measure (compare with Example 2G).

Problem 2.13. Consider the measure space $(\mathbb{R}, \Re, \mu)$, where $\mu\colon \Re \to \overline{\mathbb{R}}$ is a measure on the Borel σ-algebra $\Re$ of subsets of $\mathbb{R}$ such that

$$\mu(K) < \infty \text{ for every closed and bounded subset } K \text{ of } \mathbb{R}.$$

This is referred to as a *Borel measure* (recall that all open, and so all closed, sets lie in $\Re$; that is, they are Borel sets). The general notion of Borel measure will be the subject of a whole chapter — Chapter 11. Verify that

(a) every finite measure on $\Re$ is a Borel measure,

(b) the Lebesgue measure of Example 2C is a Borel measure,

(c) every Borel measure is σ-finite.

If μ is a Borel measure, then its *support* is the set

$$[\mu] = \mathbb{R}\backslash U, \text{ where } U \text{ is the union of all open sets of measure zero.}$$

Show that

(d) $[\mu]$ is a closed set (and therefore $[\mu] \in \Re$),

(e) $\mathbb{R}\backslash[\mu]$ is the largest (in the inclusion ordering) open set of measure zero,

(f) a point $\alpha \in \mathbb{R}$ is not in the support of μ if and only if there exists an open subset of measure zero that contains α.

Let A be a closed set with $0 < \mu(A) < \infty$, take the σ-algebra $\mathcal{A} = \wp(A) \cap \Re$, and take the restriction $\eta = \mu|_{\mathcal{A}}\colon \mathcal{A} \to \mathbb{R}$ of μ to $\mathcal{A}$, which is a *finite* measure.

Examples: (i) if μ is finite itself (i.e., a real-valued measure), then A may be any closed subset of $\mathbb{R}$ with $\mu(A) \neq 0$ (e.g., $A = \mathbb{R}$, $\mathcal{A} = \Re$, and $\eta = \mu$); or (ii) if μ is the Lebesgue measure $\lambda\colon \Re \to \overline{\mathbb{R}}$, then A may be any closed and bounded *nondegenerate interval* (i.e., nonempty and not a singleton).

Show that

(g) the support $[\eta]$ of η is the smallest (in the inclusion ordering) closed subset of A such that $\eta([\eta]) = \eta(A)$.

Problem 2.14. Let $(X, \mathcal{X})$ be an arbitrary measurable space. Prove that a (countable) sum of σ-finite measures on $\mathcal{X}$ is a σ-finite measure on $\mathcal{X}$.

Hint: Suppose μ and η are σ-finite measures on $\mathcal{X}$. Let $\{E_n\}$ and $\{E'_n\}$ be sequences in $\mathcal{X}$ such that $\mu(E_n) < \infty$, $\eta(E'_n) < \infty$, and $\bigcup_n E_n = \bigcup_n E'_n = X$. If $\eta(E_i) < \infty$, then take E_i. If $\eta(E_k) = \infty$, then take any $\{E'_{n_j}\}$ such that $E_k = \bigcup_j E'_{n_j} \cap E_k$. Show that the collection of all those E_i and $\{E'_{n_j} \cap E_k\}$ consists of a countable collection, and so conclude that $(\mu + \eta)$ is σ-finite.

Problem 2.15. Take a measure space $(X, \mathcal{X}, \mu)$ and let $(X, \widehat{\mathcal{X}}, \widehat{\mu})$ be a completion of it. Suppose $\widehat{f}\colon X \to \overline{\mathbb{R}}$ is $\widehat{\mathcal{X}}$-measurable. Show that there is an $\mathcal{X}$-measurable function $f\colon X \to \overline{\mathbb{R}}$ such that $f = \widehat{f}$ μ-almost everywhere.

Hint: Let $q \in \mathbb{Q}$ be an arbitrary rational number. Set $\widehat{E}_q = \widehat{f}^{-1}((q, \infty))$ in $\widehat{\mathcal{X}}$, write $\widehat{E}_q = E_q \cup A_q$ with $E_q \in \mathcal{X}$ and $A_q \subseteq N_q \in \mathcal{N}$, and set $N = \bigcup_q N_q$, which is a countable union, so that $N \in \mathcal{N}$ with $A_q \subseteq N$. Consider the function $f\colon X \to \overline{\mathbb{R}}$ such that $f(x) = \widehat{f}(x)$ if $x \in X \backslash N$ and $f(x) = 0$ if $x \in N$. Show that $f^{-1}((q, \infty))$ is either in $\widehat{E}_q \backslash N$ or in $\widehat{E}_q \cup N$, and so $f^{-1}((q, \infty))$ lies in $\mathcal{X}$ for all $q \in \mathbb{Q}$, and hence f is $\mathcal{X}$-measurable (cf. Problem 1.1).

Problem 2.16. Let $\mathcal{X}$ be a σ-algebra of subsets of a set X. A complex-valued set function $\zeta\colon \mathcal{X} \to \mathbb{C}$ satisfying axioms (a) and (b) of Definition 2.3, with absolute convergence on the right-hand side of (b), is a *complex measure.* Show that every complex measure ζ on $\mathcal{X}$ can be expressed as

$$\zeta = \nu_1 + i\,\nu_2,$$

where ν_1 and ν_2 are (real-valued) signed measures on $\mathcal{X}$. Complex measures will be considered again in Chapter 10.

Problem 2.17. (i) Show that for a real-valued measure (equivalently, for a finite measure $\mu\colon \mathcal{X} \to \mathbb{R}$), it is unnecessary to assume condition (a) in Definition 2.1 since this follows from countable additivity in condition (c).

Hint: $\mu(E) = \mu(E \cup \varnothing) = \mu(E) + \mu(\varnothing)$.

(ii) Similarly, since a signed measure was defined as a real-valued set function satisfying Definition 2.3, also show that for a signed measure (and so for a complex measure), it is unnecessary to assume condition (a) in Definition 2.3 since this follows from countable additivity in condition (b).

(iii) Exhibit an extended real-valued measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ for which condition (a) in Definition 2.1 cannot be dismissed. Is there any other example?

Problem 2.18. Consider the definitions of atom, of atomic set and measure, and of atom-free set and measure (given above Example 2A). Take the measure space $(\mathbb{R}, \mathfrak{R}, \delta)$, where (cf. Problem 2.1(d) and Example 2A)

$$\delta = \sum\nolimits_k \delta_{\alpha_k}$$

is an arbitrary countable sum of Dirac measures δ_{α_k} on $\mathfrak{R}$. Verify that

(a) every set in $\mathfrak{R}$ is atomic with respect to δ, and so δ is atomic itself.

Now take the measure space $(\mathbb{R}, \mathfrak{R}, \lambda)$, where λ is the Lebesgue measure on $\mathfrak{R}$ (cf. Example 2C), and show that

(b) every set in $\mathfrak{R}$ is atom-free with respect to λ, and so λ is atom-free.

Problem 2.19. Let $(X, \mathcal{X}, \eta)$ be a measure space and let $\Re$ be the Borel σ-algebra of Borel subsets of $\mathbb{R}$. For an arbitrary $\mathcal{X}$-measurable function $f\colon X \to \overline{\mathbb{R}}$, consider the set function $\mu\colon \Re \to \overline{\mathbb{R}}$ given, for every $B \in \Re$, by

$$\mu(B) = \eta\big(f^{-1}(B)\big) = \eta\big(\{x \in X\colon\ f(x) \in B\}\big). \qquad (\S)$$

(a) Show that μ is a measure on $\Re$.

(See Problem 1.8.) The measure μ is defined by the expression $\mu(B) = \eta(E)$ with $E = f^{-1}(B) \in \mathcal{X}$ for every $B \in \Re$, and so μ depends on f and η. This is

the *measure* μ *on* $\Re$ *induced by the measure* η *on* $\mathcal{X}$ *via* f.

Thus, such a procedure sets up, for each $\mathcal{X}$-measurable function f, a drop from $\mathcal{X}$-measurable subsets of X to Borel subsets of the real line, bringing the measure space $(X, \mathcal{X}, \eta)$ down to the measure space $(\mathbb{R}, \Re, \mu)$. Verify that

(b) $\mu(\mathbb{R}) = \eta(X)$.

Problem 2.20. It is usual to have a special notation, for instance, $\wp$, for a probability measure. So let $(X, \mathcal{X}, \wp)$ be a probability space. In this case, a measurable function (i.e., a real-valued $\mathcal{X}$-measurable function)

$$f\colon X \to \mathbb{R} \qquad \text{is called a} \qquad \textit{random variable.}$$

Denote by $\mu_f(\ )$ the measure on $\Re$ induced by the measure $\wp$ on $\mathcal{X}$ via f,

$$\mu_f(B) = \wp\big(f^{-1}(B)\big) \qquad (\S)$$

for every $B \in \Re$, as defined in Problem 2.19. Note from Problem 2.19(b) that $\mu_f(\mathbb{R}) = 1$ since $\wp(X) = 1$. The probability measure μ_f on $\Re$ induced by the probability measure $\wp$ on $\mathcal{X}$ via a random variable f is referred to as the *probability distribution of the random variable* f, or as the *distribution law of the random variable* f. The function $F\colon \mathbb{R} \to [0,1]$, defined by

$$F(\alpha) = \mu_f\big((-\infty, \alpha]\big) = \wp\big(\{x \in X\colon\ f(x) \le \alpha\}\big)$$

for every $\alpha \in \mathbb{R}$, is the *distribution function of the random variable* f. By a slight abuse of notation, such a pointwise definition of F is often written as

$$F(\alpha) = \mu_f\big((-\infty, \alpha]\big) = \wp(f \le \alpha),$$

and read: "$F(\alpha)$ is the probability of f being less than or equal to α".

(a) Set $X = [0,1]$, $\mathcal{X} = \wp([0,1]) \cap \Re$, $\wp = \lambda$, the Lebesgue measure over $[0,1]$, and $f = \chi_{[0,1]}$, the unit function on $[0,1]$. Show that $f^{-1}(B) = [0,1]$ if $1 \in B$ and $\varnothing$ otherwise, $\mu_f = \delta_1$, the Dirac measure at 1, and $F = \chi_{[1,\infty)}$.

(b) A distribution function F is nondecreasing and right-continuous. Prove.

(c) Verify that $\lim_{\alpha \to -\infty} F(\alpha) = 0$ and $\lim_{\alpha \to +\infty} F(\alpha) = 1$.

Hints: For (b) and (c). *Right continuity* means: $\alpha \le \alpha_n$ and $\alpha_n \to \alpha$ imply $F(\alpha_n) \to F(\alpha)$. See Proposition 2.2(d) and Problem 2.8(a).

Take an arbitrary $\alpha \in \mathbb{R}$. Set (cf. Example 2D)

$$F(\alpha^-) = \lim_{\varepsilon \to 0} F(\alpha - |\varepsilon|) \quad \text{and} \quad F(\alpha^+) = \lim_{\varepsilon \to 0} F(\alpha + |\varepsilon|).$$

Hence, by the right continuity of F in (b),

$$F(\alpha^-) = \mu_f\big((-\infty, \alpha)\big) = \wp(f < \alpha) \quad \text{and} \quad F(\alpha^+) = F(\alpha),$$

so

$$F(\alpha) - F(\alpha^-) = \mu_f(\{\alpha\}) = \wp(f = \alpha).$$

(d) Show that the probability distribution μ_f is the unique Borel–Stieltjes measure λ_F generated by the distribution function F:

$$\mu_f = \lambda_F.$$

Hint: Since F is nondecreasing, the Borel–Stieltjes measure $\lambda_F : \Re \to \mathbb{R}$ generated by F in Example 2D is given by $\lambda_F\big((\alpha, \beta)\big) = F(\beta^-) - F(\alpha^+)$ so that

$$\lambda_F\big((\alpha, \beta)\big) = F(\beta^-) - F(\alpha) = \mu_f\big((-\infty, \beta)\big) - \mu_f\big((-\infty, \alpha]\big) = \mu_f\big((\alpha, \beta)\big).$$

(e) Show that the set of points of discontinuity of F is countable.

Hint: Problem 2.8(d).

Problem 2.21. Prove the following assertions. Suppose $F : \mathbb{R} \to [0, 1]$ is

$$\text{nondecreasing,} \quad \text{right-continuous,} \quad \lim_{\alpha \to -\infty} F(\alpha) = 0, \quad \lim_{\alpha \to +\infty} F(\alpha) = 1.$$

(a) There is a probability space $(X, \mathcal{X}, \wp)$ and a random variable $f : X \to \mathbb{R}$ such that F is a distribution function for f (i.e., $F(\alpha) = \wp(f \le \alpha)$ for $\alpha \in \mathbb{R}$).

(b) If $(X, \mathcal{X})$ is a measurable space and $f : X \to \mathbb{R}$ is $\mathcal{X}$-measurable, then there is a probability measure $\wp : \mathcal{X}_f \to \mathbb{R}$ on *the sub-σ-algebra generated by* f, viz., $\mathcal{X}_f = \{f^{-1}(B) \in \mathcal{X} : B \in \Re\} \subseteq \mathcal{X}$, for which F is a distribution of f.

Hints: (a) Consider Problems 2.19, 2.20. Given F, set $(X, \mathcal{X}, \wp) = (\mathbb{R}, \Re, \lambda_F)$, λ_F as in Problem 2.20(d). Take the identity function $f : \mathbb{R} \to \mathbb{R}$ $(f(\alpha) = \alpha)$. (b) Given X, $\mathcal{X}$, $f : X \to \mathbb{R}$, take $\wp : \mathcal{X}_f \to \mathbb{R}$ defined by $\wp\big(f^{-1}(B)\big) = \lambda_F(B)$ for every $B \in \Re$, with $\lambda_F : \Re \to \mathbb{R}$ as in Example 2D. Show: $\wp$ is a probability measure (and f is an $\mathcal{X}_f$-measurable random variable; Problems 1.8, 1.9).

Suggested Readings

Bartle [5], Berberian [9], Brown and Pearcy [13], Halmos [31], Royden [62]. For a discussion on unconditionally convergent series (as in Definition 2.3) see [43, Section 5.7]. For the construction and properties of the Cantor set (as in Problems 2.9 and 2.10) see, for instance, [1], [6], [14], [43], [58], [64].



3

Integral of Nonnegative Functions

3.1 Simple and Nonnegative Functions

Let X be an arbitrary set. A *simple function* on X is a real-valued function $\varphi\colon X \to \mathbb{R}$ with a finite range (i.e., a function that takes on only a finite number of distinct values). It is clear that φ is a simple function if and only if it can be represented as a linear combination of characteristic functions,

$$\varphi = \sum_{i=1}^{n} \alpha_i \chi_{E_i},$$

where $\{E_i\}_{i=1}^n$ is a finite collection of subsets of X and $\{\alpha_i\}_{i=1}^n$ is a finite set of real numbers. The above *representation* is not unique, but it becomes unique if it is assumed that $\{\alpha_i\}_{i=1}^n$ is a set of *distinct* coefficients and $\{E_i\}_{i=1}^n$ is a partition of X (i.e., a collection of *disjoint* sets that cover X). This is the representation of φ for which the set $\{\alpha_i\}_{i=1}^n$ is the range of φ, and for each index $i = 1, \ldots, n$ the set E_i is the *inverse image* of the singleton $\{\alpha_i\}$, viz., $E_i = \varphi^{-1}(\{\alpha_i\}) = \{x \in X\colon \varphi(x) = \alpha_i\}$. This unique representation is referred to as the *canonical* (or *standard*) *representation* of φ.

Let $\mathcal{X}$ be a σ-algebra of subsets of X. Take any $E \subseteq X$. Its characteristic function $\chi_E\colon X \to \{0, 1\}$ is $\mathcal{X}$-measurable if and only if the set E is $\mathcal{X}$-measurable (Example 1B). Then a simple function $\sum_{i=1}^n \alpha_i \chi_{E_i}\colon X \to \mathbb{R}$ is measurable if $\{E_i\}_{i=1}^n$ is a collection of measurable sets (Proposition 1.5). In general, the converse may fail (e.g., take a partition $\{A, B\}$ of X made up of nonmeasurable sets $A, B \in \wp(X)\backslash\mathcal{X}$ and note that the measurable simple function χ_X can be represented as $\chi_X = \chi_A + \chi_B$). However,

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_3

a simple function is measurable if and only if the partition $\{E_i\}_{i=1}^n$ *of its canonical representation is made up of measurable sets*

or, equivalently, if and only if it has a representation such that all sets E_i are measurable. Recall that the collection of all real-valued measurable (i.e., $\mathcal{X}$-measurable) functions is a (real) linear space, denoted by $\mathcal{M}(X, \mathcal{X})$ (see the last paragraph of Section 1.3). Since sum and (real) scalar multiple of measurable simple functions are measurable simple functions, the collection of all $\mathcal{X}$-measurable simple functions is again a linear manifold of the real linear space $\mathcal{M}(X, \mathcal{X})$ of all $\mathcal{X}$-measurable functions, and so it is itself a linear space. A simple function is nonnegative if and only if all coefficients α_i of any representation are nonnegative numbers.

Definition 3.1. (The integral of a simple function). Consider a measure space $(X, \mathcal{X}, \mu)$. Let $\varphi\colon X \to \mathbb{R}$ be a nonnegative measurable simple function,

$$\varphi = \sum_{i=1}^{n} \alpha_i \chi_{E_i}.$$

The *integral* of φ with respect to μ is the nonnegative extended real number

$$\int \varphi \, d\mu = \sum_{i=1}^{n} \alpha_i \, \mu(E_i).$$

Remark 3.A (Measurability): φ must be measurable because E_i must be measurable; otherwise the definition of $\int \varphi \, d\mu$ (with respect to any measure) would not make sense. In particular (the first identity below is a definition),

$$\int_E d\mu = \int \chi_E \, d\mu = \mu(E) \quad \text{for every} \quad E \in \mathcal{X}.$$

It is readily verified that the integral of a nonnegative measurable simple function is independent of its representation. So the notion of integral of a simple function is unambiguously defined and we may assume the canonical representation of φ without loss of generality. To ensure that the integral of the null function ($\varphi = 0$) is well defined and equal to zero for every measure, including nonfinite measures, we declare again that $0 \cdot +\infty = 0$.

The next proposition considers three fundamental properties, which will survive as long as the notion of integral is extended. The first two point out that the integral is nonnegative homogeneous and additive. The third one shows how the integral with respect to a measure yields a new measure. The reader is invited to prove Proposition 3.2 in Problem 3.2.

Proposition 3.2. *Consider any measure space* $(X, \mathcal{X}, \eta)$. *If* φ *and* ψ *are nonnegative measurable simple functions, and* γ *is a nonnegative real number, then* $\gamma\varphi$ *and* $\varphi + \psi$ *are nonnegative measurable simple functions and*

(a) $\int \gamma\varphi\, d\eta = \gamma \int \varphi\, d\eta$,

(b) $\int (\varphi + \psi)\, d\eta = \int \varphi\, d\eta + \int \psi\, d\eta$,

(c) $\mu(E) = \int \varphi \chi_E\, d\eta$ *for every* $E \in \mathcal{X}$ *defines a measure* $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$.

Let $(X, \mathcal{X})$ be a measurable space. The set of all $\mathcal{X}$-measurable *extended* real-valued functions was also denoted in Chapter 1 by $\mathcal{M}(X, \mathcal{X})$, or simply by $\mathcal{M}$. Accordingly, let $\mathcal{M}(X, \mathcal{X})^+$, or simply $\mathcal{M}^+$, denote the collection of all $\mathcal{X}$-measurable *extended* real-valued nonnegative functions from $\mathcal{M}(X, \mathcal{X})$,

$$\mathcal{M}^+ = \mathcal{M}(X, \mathcal{X})^+ = \big\{f\colon X \to \overline{\mathbb{R}}\colon\ f \text{ is } \mathcal{X}\text{-measurable and } f(x) \geq 0\ \ \forall x \in X\big\}.$$

Extended real-valued functions are allowed in $\mathcal{M}$ and $\mathcal{M}^+$, but note that these collections also contain real-valued functions. In particular, nonnegative $\mathcal{X}$-measurable simple functions lie in $\mathcal{M}(X, \mathcal{X})^+$. Given an arbitrary (extended real-valued) function f in $\mathcal{M}(X, \mathcal{X})^+$, consider the set Φ_f^+ of all simple functions φ in $\mathcal{M}(X, \mathcal{X})^+$ that are dominated by f,

$$\Phi_f^+ = \big\{\varphi \in \mathcal{M}\colon\ \varphi \text{ is simple and } 0 \leq \varphi(x) \leq f(x)\ \ \forall x \in X\big\}\ \subseteq\ \mathcal{M}^+.$$

Definition 3.3. (The integral of a nonnegative measurable function). Consider a measure space $(X, \mathcal{X}, \mu)$. The *integral* of a function $f \in \mathcal{M}(X, \mathcal{X})^+$ with respect to μ is the nonnegative extended real number

$$\int f\, d\mu\ =\ \sup_{\varphi\, \in\, \Phi_f^+} \int \varphi\, d\mu.$$

The *integral* of f *over* a measurable set E with respect to μ is defined by

$$\int_E f\, d\mu\ =\ \int f \chi_E\, d\mu \quad \text{ in } \quad \overline{\mathbb{R}}.$$

Remark 3.B (Measurable functions): the function f in Definition 3.3 must be measurable. Indeed, by the remark that follows Definition 3.1, the simple functions φ in Φ_f^+ must be measurable. As the supremum of measurable functions is measurable, and since $f^{-1}(\alpha, \infty) = \{x \in X\colon \alpha < f(x)\} = \{x \in X\colon \alpha < \sup_{\varphi\, \in\, \Phi_f^+} \varphi(x)\}$, it follows that f must be measurable as well.

3.2 The Monotone Convergence Theorem

The Monotone Convergence Theorem is a fundamental result in the theory of integration, which is due to Beppo Levi. It is a basic tool for almost everything that follows. A sequence $\{f_n\}$ of functions $f_n\colon X \to \overline{\mathbb{R}}$ is *increasing* if $f_n \leq f_{n+1}$ (i.e., $f_n(x) \leq f_{n+1}(x)$ for every $x \in X$) and *decreasing* if

$f_{n+1} \le f_n$ (i.e., $f_{n+1}(x) \le f_n(x)$ for every $x \in X$) for each n. If it is either increasing or decreasing, then it is a *monotone* sequence. Observe, according to Section 1.3, that a monotone sequence of *extended* real-valued functions converges pointwise (to an *extended* real-valued function).

Theorem 3.4. (Monotone Convergence Theorem). *Let $(X, \mathcal{X}, \mu)$ be a measure space. If $\{f_n\}$ is an increasing sequence of functions in $\mathcal{M}(X, \mathcal{X})^+$, then it converges pointwise to a function $f\colon X \to \overline{\mathbb{R}}$ in $\mathcal{M}(X, \mathcal{X})^+$, and*

$$\int f \, d\mu \;=\; \lim_n \int f_n \, d\mu.$$

Proof. As we saw above, $\{f_n\}$ converges pointwise. Let f be its limit. Take any x in X. Since each $f_n(x) \ge 0$, it follows that $f(x) = \lim_n f_n(x) \ge 0$, and so f is in $\mathcal{M}(X, \mathcal{X})^+$ by Proposition 1.8. Since $f_n \le f_{n+1}$, we get $f_n \le f_{n+1} \le f = \lim_n f_n$, and so $\int f_n \, d\mu \le \int f_{n+1} \, d\mu \le \int f \, d\mu$ (cf. Problem 3.3), for every n. Then the extended real-valued increasing sequence $\left\{\int f_n \, d\mu\right\}$ converges and

$$\lim_n \int f_n \, d\mu \;\le\; \int f \, d\mu. \tag{i}$$

To verify the reverse inequality, proceed as follows. Take an arbitrary simple function φ in $\mathcal{M}(X, \mathcal{X})^+$ such that $0 \le \varphi \le f$ (i.e., an arbitrary $\varphi \in \Phi_f^+$), and take an arbitrary $\alpha \in (0, 1)$. Set $\psi = \alpha\varphi$, which is again a simple function in $\mathcal{M}(X, \mathcal{X})^+$ such that $0 \le \psi \le f$. For every positive integer n, set

$$E_n = \big\{x \in X\colon\ \psi(x) \le f_n(x)\big\} = \big\{x \in X\colon\ f_n(x) - \psi(x) \ge 0\big\}.$$

Since f_n and ψ are measurable functions, $f_n - \psi$ is measurable (Proposition 1.9), and so each E_n is a measurable set. Thus, for every n,

$$\alpha \int_{E_n} \varphi \, d\mu \;=\; \int_{E_n} \psi \, d\mu \;\le\; \int_{E_n} f_n \, d\mu \;\le\; \int f_n \, d\mu$$

(Proposition 3.2(a,c)). Moreover, since $\{f_n\}$ is increasing, it follows that $\{E_n\}$ is increasing. Since $f_n \nearrow f$ (i.e., $\{f_n\}$ is increasing and converges to f) and $0 \le \psi(x) \le f(x)$ for each $x \in X$, it follows that for every $x \in X$ there is an m for which $\psi(x) \le f_m(x) \le f(x)$, and so x lies in E_m. Hence $X = \bigcup_n E_n$ so that $\{E_n\}$ is an increasing sequence of sets in $\mathcal{X}$ that covers X. Then

$$\int \varphi \, d\mu \;=\; \eta(X) \;=\; \lim_n \eta(E_n) \;=\; \lim_n \int_{E_n} \varphi \, d\mu$$

by Proposition 2.2(c), where η is the measure given by $\eta(E) = \int_E \varphi \, d\mu$ as in Proposition 3.2(c). So, by the previous two displayed expressions,

$$\alpha \int \varphi \, d\mu \;\le\; \lim_n \int f_n \, d\mu.$$

Since this holds for an arbitrary $\alpha \in (0, 1)$,

$$\int \varphi \, d\mu \;=\; \sup_{\alpha \in (0,1)} \alpha \int \varphi \, d\mu \;\le\; \lim_n \int f_n \, d\mu.$$

Since $\int \varphi \, d\mu \le \lim_n \int f_n \, d\mu$ for an arbitrary $\varphi \in \Phi_f^+$, we get by Definition 3.3

$$\int f \, d\mu \;=\; \sup_{\varphi \in \Phi_f^+} \int \varphi \, d\mu \;\le\; \lim_n \int f_n \, d\mu. \qquad \text{(ii)}$$

Thus (i) and (ii) ensure the claimed identity, viz., $\int f \, d\mu = \lim_n \int f_n \, d\mu$. □

The Monotone Convergence Theorem gives us the first evidence of continuity for the integral transformation. Theorem 3.4 will also give us the first hint of linearity (in Proposition 3.5(a,b) below). Of course, all this would make sense only if the domain $\mathcal{M}^+$ and codomain $\overline{\mathbb{R}}$ of the transformation

$$\int (\ ) \, d\mu \colon \mathcal{M}^+ \to \overline{\mathbb{R}},$$

(that assigns to each function f in $\mathcal{M}^+$ the extended real number $\int f d\mu$) might be equipped with a suitable algebraic and topological structure. We will modify the domain $\mathcal{M}^+$ in order to endow the new domain with the proper algebraic structure in Chapter 4 (that makes the integral transformation a linear one) and with a proper topological structure in Chapter 5 (that makes the integral transformation a continuous one as well).

As a first application of the Monotone Convergence Theorem (among many to come) we extend in Proposition 3.5 below the results of Proposition 3.2, from measurable nonnegative simple functions to arbitrary measurable nonnegative (extended real-valued) functions. In particular, Theorem 3.4 shows (as stated in Proposition 3.5(c) below) how the integral of a *nonnegative* function with respect to a measure yields a new measure, viz., the set function $\mu \colon \mathcal{X} \to \overline{\mathbb{R}}$ given for each set $E \in \mathcal{X}$ by $\mu(E) = \int_E f \, d\eta$.

Proposition 3.5. *Consider a measure space $(X, \mathcal{X}, \eta)$. If $f \colon X \to \overline{\mathbb{R}}$ and $g \colon X \to \overline{\mathbb{R}}$ are functions in $\mathcal{M}(X, \mathcal{X})^+$, and if γ is a nonnegative real number, then the functions γf and $f + g$ lie in $\mathcal{M}(X, \mathcal{X})^+$ and*

(a) $\int \gamma f \, d\eta = \gamma \int f \, d\eta$,

(b) $\int (f + g) \, d\eta = \int f \, d\eta + \int g \, d\eta$,

(c) $\mu(E) = \int_E f \, d\eta$ *for every $E \in \mathcal{X}$ defines a measure $\mu \colon \mathcal{X} \to \overline{\mathbb{R}}$.*

Note: This sets a *measure μ generated by a measure η via a function f*, which is different form the one defined in Problem 2.19.

Proof. If $f, g \in \mathcal{M}(X, \mathcal{X})^+$, then γf and $f + g$ are in $\mathcal{M}(X, \mathcal{X})^+$ by Proposition 1.9. By Problem 1.6(i-iv), there are increasing sequences $\{\varphi_n\}$ and $\{\psi_n\}$ of simple functions in $\mathcal{M}(X, \mathcal{X})^+$ for which $f = \lim_n \varphi_n$ and $g = \lim_n \psi_n$.

(a) Take $\gamma \geq 0$ so that $\{\gamma\varphi_n\}$ is an increasing sequence of simple functions in $\mathcal{M}(X,\mathcal{X})^+$ that converges to γf. Thus Proposition 3.2(a) and Theorem 3.4 (the Monotone Convergence Theorem) ensure that

$$\int \gamma f \, d\eta = \lim_n \int \gamma\varphi_n \, d\eta = \gamma \lim_n \int \varphi_n \, d\eta = \gamma \int f \, d\eta.$$

(b) Note that $\{\varphi_n + \psi_n\}$ is an increasing sequence of simple functions in $\mathcal{M}(X,\mathcal{X})^+$ that converges to $f+g$. Thus, again, Proposition 3.2(b) and the Monotone Convergence Theorem ensure that

$$\begin{aligned}\int (f+g)\, d\eta &= \lim_n \int (\varphi_n + \psi_n)\, d\eta = \lim_n \left(\int \varphi_n \, d\eta + \int \psi_n \, d\eta \right) \\ &= \lim_n \int \varphi_n \, d\eta + \lim_n \int \psi_n \, d\eta = \int f \, d\eta + \int g \, d\eta.\end{aligned}$$

(c) Note that $\mu(\varnothing) = 0$ since $\chi_\varnothing = 0$ so that $\int_\varnothing f \, d\eta = 0$, and $\mu(E) \geq 0$ for all E in $\mathcal{X}$ by the definition of the integral of f in $\mathcal{M}(X,\mathcal{X})^+$. To verify countable additivity of μ (Definition 2.1(c)), take any sequence $\{E_n\}$ of pairwise disjoint sets in $\mathcal{X}$. As $\sum_{n=1}^m f\chi_{E_n} = f\chi_{\bigcup_{n=1}^m E_n}$, set for every integer $m \geq 1$

$$f_m = \sum_{n=1}^m f\chi_{E_n} = f\chi_{\bigcup_{n=1}^m E_n}.$$

Now observe that $\{f_m\}$ is an increasing sequence of functions in $\mathcal{M}(X,\mathcal{X})^+$ (Proposition 1.9) that converges pointwise to the function $f\chi_E$, with $E = \bigcup_n E_n$ in $\mathcal{X}$. Then the Monotone Convergence Theorem ensures that

$$\mu\Big(\bigcup\nolimits_n E_n\Big) = \int_E f \, d\eta = \int f\chi_E \, d\eta = \lim_m \int f_m \, d\eta = \lim_m \int \sum_{n=1}^m f\chi_{E_n} \, d\eta.$$

Thus a trivial induction, using additivity as in item (b), ensures that the integral of a finite sum coincides with the finite sum of each integral, so

$$\lim_m \int \sum_{n=1}^m f\chi_{E_n} \, d\eta = \lim_m \sum_{n=1}^m \int f\chi_{E_n} \, d\eta,$$

which completes the proof of (c). That is, $\mu\big(\bigcup_n E_n\big) = \sum_n \mu(E_n)$. In fact,

$$\lim_m \sum_{n=1}^m \int f\chi_{E_n} \, d\eta = \lim_m \sum_{n=1}^m \int_{E_n} f \, d\eta = \lim_m \sum_{n=1}^m \mu(E_n) = \sum\nolimits_n \mu(E_n). \quad \square$$

Theorem 3.4 deals with functions that are in general extended real-valued and with measures that are not necessarily finite (not even σ-finiteness is assumed). Thus infinite integrals and infinite limits are allowed. For example, if μ is the Lebesgue measure λ on $\Re$ and $f_n = \chi_{[0,n)}$ for each integer

$n \geq 1$, then $\{f_n\}$ is an increasing sequence of functions in $\mathcal{M}(\mathbb{R}, \Re)^+$ with finite integral ($\int f_n \, d\lambda = n$ for each n) converging pointwise to the function $f = \chi_{[0,\infty)}$ that has an infinite integral. However, the real-valued sequence $\{\int f_n \, d\lambda\}$ is unbounded (and so it does not converge in $\mathbb{R}$) but it has the limit $+\infty$ in $\overline{\mathbb{R}}$. Hence, $\int f \, d\lambda = \lim_n \int f_n \, d\lambda = +\infty$. We will see in the next section that the Monotone Convergence Theorem still holds if pointwise convergence is weakened to almost everywhere convergence. Moreover, it holds without monotonicity by assuming convergence from below. This is Corollary 3.10. (Convergence from below can be dismissed if we impose uniform convergence and *finite* measure — see Problems 3.6 and 3.11.)

3.3 Monotone Convergence Corollaries

The next result, Fatou's Lemma, can be viewed as an important consequence of the Monotone Convergence Theorem, which will be applied to prove an extension of the Monotone Convergence Theorem that does not require monotonicity (as in the forthcoming Corollary 3.10). Recall from Chapter 1 that $\underline{f} = \liminf_n f_n$ is a measurable function for any sequence $\{f_n\}$ of extended real-valued measurable functions (Proposition 1.8), and that if $\{f_n\}$ converges pointwise to a function f (i.e., if $f = \lim_n f_n$), then $f = \underline{f}$.

Lemma 3.6. (Fatou's Lemma). *Take a measure space $(X, \mathcal{X}, \mu)$. If $\{f_n\}$ is a sequence of functions in $\mathcal{M}(X, \mathcal{X})^+$, then*

$$\int \underline{f} \, d\mu \leq \liminf_n \int f_n \, d\mu.$$

Proof. Let $\{f_n\}$ be a sequence of functions in $\mathcal{M}(X, \mathcal{X})^+$. For every n set

$$\phi_n = \inf_{n \leq k} f_k,$$

meaning that each $\phi_n \colon X \to \overline{\mathbb{R}}$ is a function defined by $\phi_n(x) = \inf_{n \leq k} f_k(x)$ for every $x \in X$. Proposition 1.8 ensures that each ϕ_n is measurable, and so each ϕ_n lies in $\mathcal{M}(X, \mathcal{X})^+$. By definition, $\{\phi_n\}$ is an increasing sequence, and it converges pointwise to $\underline{f}$. Indeed,

$$\lim_n \phi_n(x) = \lim_n \inf_{n \leq k} f_k(x) = \liminf_n f_n(x) = \underline{f}(x)$$

for each $x \in X$. Then the Monotone Convergence Theorem ensures that

$$\int \underline{f} \, d\mu = \lim_n \int \phi_n \, d\mu.$$

But $\phi_n \leq f_k$ for every $k \geq n$, which implies that $\int \phi_n \, d\mu \leq \int f_k \, d\mu$ for every $k \geq n$, and so $\int \phi_n \, d\mu \leq \inf_{n \leq k} \int f_k \, d\mu$. Hence

$$\lim_n \int \phi_n \, d\mu \le \lim_n \inf_{n \le k} \int f_k \, d\mu = \liminf_n \int f_n \, d\mu. \qquad \square$$

Item (a) in Proposition 3.7 below is an application of Fatou's Lemma. It gives the first hint of what will make the basis for defining a new concept of equality in the spaces L^p (based on the notion of equivalence classes as introduced in Chapter 5), which reads as follows: if $f \in \mathcal{M}(X, \mathcal{X})^+$, then

$$f = 0 \quad \mu\text{-a.e.} \quad \Longleftrightarrow \quad \int f \, d\mu = 0.$$

Proposition 3.7. *Consider a measure space $(X, \mathcal{X}, \eta)$. Let f be a function in $\mathcal{M}(X, \mathcal{X})^+$. Let μ be the measure on $\mathcal{X}$ defined in* Proposition 3.5(c): *$\mu(E) = \int_E f \, d\eta$ for every $E \in \mathcal{X}$. Then the following assertions hold true.*

(a) *$f = 0$ η-almost everywhere if and only if $\int f \, d\eta = 0$.*

(b) *If $\eta(E) = 0$ for some $E \in \mathcal{X}$, then $\mu(E) = 0$.*

Proof. Take a measure space $(X, \mathcal{X}, \eta)$ and let f be a function in $\mathcal{M}(X, \mathcal{X})^+$.

(a) Since f is a measurable function, set, for every positive integer n,

$$E_n = \{x \in X : \tfrac{1}{n} < f(x)\},$$

which forms an increasing sequence $\{E_n\}$ of measurable sets such that $\frac{1}{n} \chi_{E_n} \le f$, and therefore (cf. Proposition 3.5(a) and Problem 3.3(b))

$$0 \le \tfrac{1}{n} \eta(E_n) = \tfrac{1}{n} \int_{E_n} d\eta = \int \tfrac{1}{n} \chi_{E_n} \, d\eta \le \int f \, d\eta$$

for every n. If $\int f \, d\eta = 0$, then $\eta(E_n) = 0$ for all n, and so (since $0 \le f$)

$$\eta\big(\{x \in X : f(x) \neq 0\}\big) = \eta\big(\{x \in X : 0 < f(x)\}\big) = \eta\big(\textstyle\bigcup_n E_n\big) = \lim_n \eta(E_n) = 0$$

by Proposition 2.2(c). Hence $f = 0$ η-almost everywhere. Conversely, set $E = \bigcup_n E_n$ in $\mathcal{X}$ and $f_n = n \, \chi_E$ for each n. It is clear that $\{f_n\}$ is a sequence of functions in $\mathcal{M}(X, \mathcal{X})^+$ converging pointwise to $f_\infty = +\infty \, \chi_E : X \to \overline{\mathbb{R}}$, and so $\underline{f} = \liminf_n f_n = \lim_n f_n = f_\infty$. Also, $0 \le f \le f_\infty = \underline{f}$. Thus

$$0 \le \int f \, d\eta \le \int \underline{f} \, d\eta \le \liminf_n \int f_n \, d\eta$$

by Problem 3.3 and Lemma 3.6 (Fatou's Lemma). If $f = 0$ η-almost everywhere, then $\eta(E) = 0$ so that $\int f_n \, d\eta = n \int \chi_E \, d\eta = n \, \eta(E) = 0$ for all n. Hence $\liminf_n \int f_n \, d\eta = 0$, and so $\int f \, d\eta = 0$ by the preceding inequality.

(b) If $\eta(E) = 0$ for some $E \in \mathcal{X}$, and since $\int \chi_E \, d\eta = \eta(E)$, it follows by item (a) that $\chi_E = 0$, and so $f \chi_E = 0$, η-almost everywhere. Let $\mu : \mathcal{X} \to \overline{\mathbb{R}}$ be the measure of Proposition 3.5(c). Another application of item (a) yields

$$\mu(E) = \int_E f\,d\eta = \int f\chi_E\,d\eta = 0. \qquad \square$$

Remark 3.C (Absolute continuity): The implication $\{\eta(E) = 0 \Rightarrow \mu(E) = 0\}$ in Proposition 3.7(b) is referred to by saying that the measure μ is *absolutely continuous* with respect to the measure η. (Absolute continuity is a property that will be discussed in detail in Chapter 7.) So, by Propositions 3.5(c) and 3.7(b), we can infer that *if η is a measure on $\mathcal{X}$ and f is a function in $\mathcal{M}(X,\mathcal{X})^+$, then the set function μ on $\mathcal{X}$ defined for each $E \in \mathcal{X}$ by*

$$\mu(E) = \int_E f\,d\eta$$

is a measure that is absolutely continuous with respect to η. This is another important consequence of the Monotone Convergence Theorem. A crucial result of Chapter 7 (the Radon–Nikodym Theorem) asserts the converse: *if η and μ are σ-finite measures and μ is absolutely continuous with respect to η, then there exists a function f in $\mathcal{M}(X,\mathcal{X})^+$ such that for each $E \in \mathcal{X}$,*

$$\mu(E) = \int_E f\,d\eta.$$

Corollary 3.8. *Take a measure space $(X,\mathcal{X},\mu)$. If $\{f_n\}$ is an increasing sequence of functions in $\mathcal{M}(X,\mathcal{X})^+$ that converges almost everywhere to a function $f\colon X \to \overline{\mathbb{R}}$ in $\mathcal{M}(X,\mathcal{X})^+$, then*

$$\int f\,d\mu = \lim_n \int f_n\,d\mu.$$

Proof. Suppose $\{f_n\}$ converges μ-almost everywhere to f, so that $\{f_n\}$ converges pointwise to f on $E = X\backslash N$ for some $N \in \mathcal{X}$ with $\mu(N) = 0$. Thus $\{f_n\chi_E\}$ is an increasing sequence of functions in $\mathcal{M}(X,\mathcal{X})^+$ converging pointwise to $f\chi_E \in \mathcal{M}(X,\mathcal{X})^+$. By the Monotone Convergence Theorem,

$$\int f\chi_E\,d\mu = \lim_n \int f_n\chi_E\,d\mu.$$

However, Propositions 3.5(b) and 3.7(a) ensure (see Problem 3.8) that

$$\int f\chi_E\,d\mu = \int_E f\,d\mu = \int f\,d\mu \quad \text{and} \quad \int f_n\chi_E\,d\mu = \int_E f_n\,d\mu = \int f_n\,d\mu. \qquad \square$$

Corollary 3.8 is Theorem 3.4 for almost everywhere convergence, which will also be referred to as the Monotone Convergence Theorem. It leads to the following version of Lemma 3.6 for almost everywhere convergence, again referred to as Fatou's Lemma.

Lemma 3.9. *Take a measure space $(X, \mathcal{X}, \mu)$. If $\{f_n\}$ is a sequence in $\mathcal{M}(X, \mathcal{X})^+$ that converges almost everywhere to $f \in \mathcal{M}(X, \mathcal{X})^+$, then*

$$\int f \, d\mu \leq \liminf_n \int f_n \, d\mu.$$

Proof. Consider the setup of the previous proof, where the sequence $\{f_n \chi_E\}$ converges pointwise to $f\chi_E \in \mathcal{M}(X, \mathcal{X})^+$. By Lemma 3.6 (Fatou's Lemma),

$$\int f\chi_E \, d\mu \leq \liminf_n \int f_n \chi_E \, d\mu.$$

Again, Propositions 3.5(b) and 3.7(a) ensure that (cf. Problem 3.8)

$$\int f\chi_E \, d\mu = \int f \, d\mu \quad \text{and} \quad \int f_n \chi_E \, d\mu = \int f_n \, d\mu. \qquad \square$$

A sequence $\{f_n\}$ *converges from below* to f if $f_n \leq f$ for all n and if it converges to f in some sense. The following extension in Corollary 3.10 yields an ultimate version of the Monotone Convergence Theorem that assumes just almost everywhere convergence from below.

Corollary 3.10. *Let $(X, \mathcal{X}, \mu)$ be a measure space. If a sequence $\{f_n\}$ in $\mathcal{M}(X, \mathcal{X})^+$ converges almost everywhere to $f \in \mathcal{M}(X, \mathcal{X})^+$ from below, then*

$$\int f \, d\mu = \lim_n \int f_n \, d\mu.$$

Proof. Since $f_n \leq f$, we get $\int f_n \, d\mu \leq \int f \, d\mu$ for all n. Thus, by Lemma 3.9,

$$\int f \, d\mu \leq \liminf_n \int f_n \, d\mu \leq \limsup_n \int f_n \, d\mu \leq \int f \, d\mu. \qquad \square$$

3.4 Problems

Problem 3.1. Consider a measurable space $(X, \mathcal{X})$. Thus, in this context, "measurable" means $\mathcal{X}$-measurable. Prove the following assertions.

(a) Sum, scalar multiplication, and product of measurable simple functions are measurable simple functions.

(b) Every polynomial $p(\varphi, \psi)$ of measurable simple functions φ and ψ is a measurable simple function.

(c) $\phi = \varphi \wedge \psi = \min\{\varphi, \psi\}$ and $\Phi = \varphi \vee \psi = \max\{\varphi, \psi\}$ are measurable simple functions whenever φ and ψ are measurable simple functions.

Hint: Proposition 1.5 and Problem 1.3.

Problem 3.2. Show that the definition of integral of a nonnegative measurable simple function does not depend on the representation of the simple function. Then prove Proposition 3.2.

Hint: Verify homogeneity. To prove additivity proceed as follows. Show that

$$\varphi + \psi = \sum\nolimits_i \sum\nolimits_j (\alpha_i + \beta_j)\, \chi_{E_i \cap E'_j},$$

where $\varphi = \sum_i \alpha_i \chi_{E_i}$ and $\psi = \sum_j \beta_j \chi_{E'_j}$ are canonical representations. So

$$\begin{aligned} \int (\varphi + \psi)\, d\eta &= \sum\nolimits_i \sum\nolimits_j (\alpha_i + \beta_j)\, \eta(E_i \cap E'_j) \\ &= \sum\nolimits_i \alpha_i \sum\nolimits_j \eta(E_i \cap E'_j) + \sum\nolimits_j \beta_j \sum\nolimits_i \eta(E_i \cap E'_j) \\ &= \sum\nolimits_i \alpha_i\, \eta(E_i) + \sum\nolimits_j \beta_j\, \eta(E'_j) = \int \varphi\, d\eta + \int \psi\, d\eta \end{aligned}$$

since $\eta(E_i) = \sum_j \eta(E_i \cap E'_j)$ and $\eta(E'_j) = \sum_i \eta(E_i \cap E'_j)$ because $\{E_i\}$ and $\{E'_j\}$ are partitions of X. To prove (c) note that $\varphi \chi_E = \sum_i \alpha_i \chi_{E_i \cap E}$. Hence

$$\mu(E) = \int \varphi \chi_E \, d\eta = \sum\nolimits_i \alpha_i\, \eta(E_i \cap E) = \sum\nolimits_i \alpha_i\, \eta_{E_i}(E) \quad \text{for every} \quad E \in \mathcal{X},$$

a (finite) linear combination with nonnegative coefficients α_i of measures η_{E_i} on $\mathcal{X}$ (Problem 2.11), and so it is itself a measure on $\mathcal{X}$.

Problem 3.3. Let μ and η be measures on a σ-algebra $\mathcal{X}$ of subsets of X, let E and E' be sets in $\mathcal{X}$, and let f and g be functions in $\mathcal{M}(X, \mathcal{X})^+$. Recall that

$$\int_E f\, d\mu = \int f \chi_E \, d\mu, \quad \text{in particular,} \quad \int f\, d\mu = \int f \chi_X \, d\mu = \int_X f\, d\mu$$

(cf. Definition 3.3) and, from the remark following Definition 3.1, verify that

(a) $0 \le \int_E d\mu = \int \chi_E \, d\mu = \mu(E)$.

Prove the next assertions (cf. Definitions 3.1, 3.3, and Proposition 3.7(a)).

(b) $f \le g \quad \mu$-a.e. $\implies \quad 0 \le \int_E f\, d\mu \le \int_E g\, d\mu$.

(c) $E \subseteq E' \quad \implies \quad 0 \le \int_E f\, d\mu \le \int_{E'} f\, d\mu$.

(d) $\mu \le \eta \quad \implies \quad 0 \le \int_E f\, d\mu \le \int_E f\, d\eta$.

(e) $0 < f$ everywhere $\implies \quad 0 < \int f\, d\mu$ (if $\mu(X) > 0$).

Problem 3.4. Let $\mathbb{N}$ be the set of all natural numbers (i.e., of all positive integers), and consider the measurable space $(\mathbb{N}, \wp(\mathbb{N}))$.

(a) Verify that every nonnegative function $f : \mathbb{N} \to \overline{\mathbb{R}}$ lies in $\mathcal{M}(\mathbb{N}, \wp(\mathbb{N}))^+$.

Let μ be the counting measure of Example 2B. Apply Definition 3.3 to show that for every nonnegative function $f\colon \mathbb{N} \to \overline{\mathbb{R}}$,

(b) $\int_E f\,d\mu = \sum_{n\in E} f(n)$ for every $E \in \wp(\mathbb{N})$. Thus $\int f\,d\mu = \sum_{n=1}^{\infty} f(n)$.

Problem 3.5. Take any $\alpha \in \mathbb{R}$ and consider the probability space $(\mathbb{R}, \Re, \delta_\alpha)$, where δ_α is the Dirac measure of Example 2A on the Borel σ-algebra $\Re$. If $f \in \mathcal{M}(\mathbb{R}, \Re)^+$, then use Definition 3.3 to prove the following statements.

(a) $\int_E f\,d\delta_\alpha = f(\alpha)$ if $\alpha \in E \in \Re$.

(b) $\int_E f\,d\delta_\alpha = 0$ if $\alpha \notin E \in \Re$.

(c) $\int f\,d\delta_\alpha = f(\alpha)$.

Now take a countable set $\{\alpha_k\}$ of points in $\mathbb{R}$, consider the Dirac measures δ_{α_k} at each α_k and, as in Problem 2.18, verify that

$$\delta = \sum\nolimits_k \gamma_k \delta_{\alpha_k}$$

is a measure on $\Re$ for an arbitrary similarly indexed set of nonnegative coefficients $\{\gamma_k\}$. Next consider the measure space $(\mathbb{R}, \Re, \delta)$, take any $\Re$-measurable nonnegative function $f \in \mathcal{M}(\mathbb{R}, \Re)^+$, and show that

(d) $\int f\,d\delta = \sum_k \gamma_k \int f\,d\delta_{\alpha_k} = \sum_k \gamma_k f(\alpha_k)$.

We will return to Dirac measures in Problem 7.15(a).

Problem 3.6. A sequence $\{f_n\}$ of real-valued functions on a set X *converges uniformly* to a real-valued function f on X if, for each $\varepsilon > 0$, there is a positive integer n_ε such that $\sup_{x\in X} |f_n(x) - f(x)| < \varepsilon$ for all $n \geq n_\varepsilon$. Use Problem 3.3 and Proposition 3.5 to prove the following assertion.

> If $(X, \mathcal{X}, \mu)$ is a *finite* measure space and $\{f_n\}$ is a sequence of real-valued functions in $\mathcal{M}(X, \mathcal{X})^+$ that converges uniformly to a real-valued function f, then f lies in $\mathcal{M}(X, \mathcal{X})^+$ and
>
> $$\int f\,d\mu = \lim_n \int f_n\,d\mu.$$

Hint: For $k \geq 1$, set $\varepsilon = \frac{1}{k}$, and take $n \geq n_\varepsilon$. Set $E_k = \{x \in X\colon \frac{1}{k} \leq f(x)\}$ in $\mathcal{X}$. Uniform convergence implies $(f - \frac{1}{k})\chi_{E_k} \leq f_n \leq f + \frac{1}{k}$. Hence,

$$\int (f - \tfrac{1}{k})\chi_{E_k} d\mu \leq \liminf_n \int f_n\,d\mu \leq \limsup_n \int f_n\,d\mu \leq \int f\,d\mu + \tfrac{1}{k}\mu(X).$$

Show that $\int (f - \frac{1}{k})\chi_{E_k} d\mu = \int f\chi_{E_k} d\mu - \frac{1}{k}\mu(E_k)$ because $\mu(X) < \infty$. Since $\int f\chi_{E_k} d\mu = \eta(E_k)$ (η is the measure $\eta(E) = \int_E f\,d\mu$; cf. Proposition 3.5(c)),

and $\{E_k\}$ is increasing, verify: $\lim_k \int f\chi_{E_k} d\mu = \eta\big(\bigcup_k E_k\big) = \eta(X) = \int f\, d\mu$ (cf. Proposition 2.2). Since $\mu(X) < \infty$, it follows that $\lim_k \frac{1}{k}\mu(E_k) = 0$.

Problem 3.7. Prove the Beppo Levi Theorem, which says that if $(X, \mathcal{X}, \mu)$ is a measure space and $\{f_n\}$ is a sequence of functions in $\mathcal{M}(X, \mathcal{X})^+$, then

$$\int \sum_{n=1}^{m} f_n \, d\mu = \sum_{n=1}^{m} \int f_n \, d\mu$$

for each positive integer m.

Hint: Use Proposition 3.5(b).

Now use the Monotone Convergence Theorem (Theorem 3.4) to show that

$$\int \sum\nolimits_n f_n \, d\mu = \sum\nolimits_n \int f_n \, d\mu.$$

Problem 3.8. Consider a measure space $(X, \mathcal{X}, \mu)$. Take $f \in \mathcal{M}(X, \mathcal{X})^+$. Let $\{E, E'\}$ be a measurable partition of X. Use Proposition 3.5(c) to show that

(a) $\int f\, d\mu = \int_E f\, d\mu + \int_{E'} f\, d\mu$.

Let N be a set in $\mathcal{X}$. Show that the propositions below hold true.

(b) $\mu(N) = 0$ implies $\int_N f\, d\mu = 0$,

(c) $E = X \backslash N$ and $\mu(N) = 0$ imply $\int_E f\, d\mu = \int f\, d\mu$.

Take, in addition, $g \in \mathcal{M}(X, \mathcal{X})^+$. If the integrals $\int f\, d\mu$ and $\int g\, d\mu$ are finite, apply Propositions 3.5(a,b) and 3.7(a) to show that

(d) $\int_E f\, d\mu = \int_E g\, d\mu$ for every $E \in \mathcal{X}$ implies $f = g$ μ-almost everywhere.

Hints: (b,c) Use Definition 3.3 on $\mathcal{N} = \wp(N) \cap \mathcal{X}$ for $\int_N f d\mu = \int f\chi_N d\mu = \int f' d\mu'$ with $f' = f|_N \in \mathcal{M}(N, \mathcal{N})^+$ and $\mu' = \mu|_N : \mathcal{N} \to \overline{\mathbb{R}}$ (cf. Problem 2.11). (d) Set $A = \{x \in X : f(x) < g(x)\}$, $B = \{x \in X : f(x) > g(x)\}$, and $C = \{x \in X : f(x) = g(x)\}$. Thus $\{A, B, C\}$ is a measurable partition of X. Since $\int_A f\, d\mu = \int_A g\, d\mu$ and $(g - f)\chi_A \in \mathcal{M}(X, \mathcal{X})^+$, $\int_A (g - f)\, d\mu = 0$. So $(g - f) = 0$ μ-a.e. on A. Similarly, $(f - g) = 0$ μ-a.e. on B. And $f = g$ on C trivially.

Problem 3.9. Consider a measure space $(X, \mathcal{X}, \mu)$. Recall the definition of a σ-finite measurable set (cf. paragraph preceding Example 2A). Take an arbitrary $f \in \mathcal{M}(X, \mathcal{X})^+$. Show that if $\int f\, d\mu < \infty$, then

(a) $\mu(\{x \in X : f(x) \geq \varepsilon\}) < \infty$ for each $\varepsilon > 0$,

(b) $\mu(\{x \in X : f(x) = +\infty\}) = 0$,

(c) $\{x \in X : f(x) \neq 0\}$ is a σ-finite set.

Hints: (a) If $F_\varepsilon = \{x \in X\colon \varepsilon \le f(x)\}$, then $\varepsilon \chi_{F_\varepsilon} \le f$. (b) Use Proposition 2.2(d) to $\{F_n\}$. (c) If $E_n = \{x \in X\colon \frac{1}{n} \le f(x)\}$, then $\frac{1}{n}\chi_{E_n} \le f$.

Problem 3.10. Let $(X, \mathcal{X}, \mu)$ be a measure space, take an arbitrary function f in $\mathcal{M}(X, \mathcal{X})^+$, and prove the following assertion.

> If $\int f\, d\mu < \infty$, then for every $\varepsilon > 0$ there exists a set $E_\varepsilon \in \mathcal{X}$ such that $\mu(E_\varepsilon) < \infty$ and $\int f\, d\mu \le \int_{E_\varepsilon} f\, d\mu + \varepsilon$.

Hint: Set $E = \{x \in X\colon f(x) \ne 0\}$. Show that $\int f\, d\mu = \int_E f\, d\mu$ (Problem 3.8(a)). Set $E_n = \{x \in X\colon \frac{1}{n} \le f(x)\}$. Show that $\{E_n\}$ is increasing, that $E = \bigcup_n E_n$, and that $\mu(E_n) = n \int f\, d\mu$ (Problem 3.9(c)). Use Theorem 3.4 to verify that $\lim_n \int f\chi_{E_n} d\mu = \int f\, d\mu$. Conclude: for each $\varepsilon > 0$ there is an n_ε for which, with $E_\varepsilon = E_{n_\varepsilon}$, it follows that $\int f\, d\mu - \int f\chi_{E_\varepsilon} d\mu < \varepsilon$.

Problem 3.11. Monotone (increasing) convergence in Theorem 3.4 was weakened to convergence from below in Corollary 3.10, and such a version of the Monotone Convergence Theorem cannot be improved further (even under the assumption of uniform convergence it will require finite measure — cf. Problem 3.6). In fact, with λ denoting the Lebesgue measure on $\Re$,

$$(\mathbb{R}, \Re, \lambda) \text{ is the } \textit{Lebesgue measure space}.$$

Take the functions $f_n = \frac{1}{n}\chi_{[n,\infty)}$ and $g_n = \frac{1}{n}\chi_{[0,n]}$ for each positive integer n, and set $f = g = 0$, which are all functions in $\mathcal{M}(\mathbb{R}, \Re)^+$. Now show that

(a) $\{f_n\}$ decreases and converges uniformly to f but $\int f_n\, d\lambda = +\infty$ for all n, and so $0 = \int f\, d\lambda \ne \lim_n \int f_n\, d\lambda = +\infty$;

(b) $g \le g_n$ for all n, and $\{g_n\}$, which is not monotone, converges uniformly to g but $\int g_n\, d\lambda = 1$ for all n, and so $0 = \int g\, d\lambda \ne \lim_n \int g_n\, d\lambda = 1$.

Problem 3.12. Let $(X, \mathcal{X}, \mu)$ be a measure space and let $\{f_n\}$ be a sequence of functions in $\mathcal{M}(X, \mathcal{X})^+$ converging pointwise to $f \in \mathcal{M}(X, \mathcal{X})^+$. If $\int f\, d\mu = \lim_n \int f_n\, d\mu < \infty$, then show that, for every measurable set $E \in \mathcal{X}$,

$$\int_E f\, d\mu = \lim_n \int_E f_n\, d\mu,$$

and verify that this may fail without the assumption $\lim_n \int f_n\, d\mu < \infty$.

Problem 3.13. Consider a measure space $(X, \mathcal{X}, \eta)$ and take a function f in $\mathcal{M}(X, \mathcal{X})^+$. Let $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ be the measure generated by η via f as in Proposition 3.5(c), namely, for every $E \in \mathcal{X}$,

$$\mu(E) = \int_E f\, d\eta. \qquad (\star)$$

Show that for every $g \in \mathcal{M}(X, \mathcal{X})^+$ and every $E \in \mathcal{X}$,

$$\int_E g \, d\mu = \int_E g f \, d\eta. \tag{$\diamond$}$$

Hint: By $(\star)$, $\int \chi_E \, d\mu = \int \chi_E f \, d\eta$. Let $\{\phi_n\}$ be a sequence of simple functions converging from below to g (Problem 1.6(i-iv)). Apply Corollary 3.10 to get $\int g \, d\mu = \lim \int \phi_n \, d\mu = \lim \int \phi_n f \, d\eta = \int g f \, d\eta$. Replace g with $\chi_E \, g$ to get $(\diamond)$.

So we get the identity $\int (\ ) \, d\mu = \int (\ ) f \, d\eta$ that is sometimes abbreviated by

$$d\mu = f \, d\eta,$$

where no independent meaning is assigned to the symbols $d\mu$ and $d\eta$. In this case, the function f appearing in $(\star)$ is denoted by

$$f = \frac{d\mu}{d\eta},$$

which again is mere notation (with no independent meaning), so that the expression $(\star)$ can be rewritten as (cf. Problem 3.3(a))

$$\mu(E) = \int_E d\mu = \int_E \tfrac{d\mu}{d\eta} \, d\eta$$

for every $E \in \mathcal{X}$. We will return to this point in Section 7.2.

Problem 3.14. Take a measure space $(X, \mathcal{X}, \eta)$ and an $\mathcal{X}$-measurable function $f \colon X \to \mathbb{R}$. Let $(\mathbb{R}, \Re, \mu)$ be the measure space where $\mu \colon \Re \to \overline{\mathbb{R}}$ is the measure induced by η via f as in Problem 2.19, namely, for every $B \in \Re$,

$$\mu(B) = \eta\big(f^{-1}(B)\big). \tag{§}$$

(a) Prove the following *change of variable* formula. If $\phi \colon \mathbb{R} \to \mathbb{R}$ is a nonnegative $\Re$-measurable function and μ on $\Re$ is the induced measure in (§), then

$$\int_X \phi \circ f \, d\eta = \int_{\mathbb{R}} \phi \, d\mu. \tag{$\dagger$}$$

Hint: Recall: $\phi \circ f : X \to \mathbb{R}$ is $\mathcal{X}$-measurable whenever ϕ is $\Re$-measurable and f is $\mathcal{X}$-measurable (Problem 1.11), and $\phi \circ f$ is nonnegative whenever ϕ is. If $\phi = \chi_B$ is the characteristic function of a Borel set $B \in \Re$, then verify that

$$\chi_B \circ f = \chi_{f^{-1}(B)}. \tag{i}$$

Moreover, in addition to the above auxiliary identity, recall from (§) that

$$\int_{f^{-1}(B)} d\eta = \int_B d\mu.$$

(cf. remark after Definition 3.1). Use (i) to show that (†) holds for $\phi = \chi_B$:

$$\int_X \chi_B \circ f \, d\eta = \int_X \chi_{f^{-1}(B)} \, d\eta = \int_{f^{-1}(B)} d\eta = \int_B d\mu = \int_{\mathbb{R}} \chi_B \, d\mu.$$

Then (†) holds for simple functions $\varphi_n : \mathbb{R} \to \mathbb{R}$; and hence for simple functions such that $0 \le \varphi_n \le \phi$ and $\varphi_n \to \phi$, and therefore for simple functions $\varphi_n \circ f : X \to \mathbb{R}$ such that $0 \le \varphi_n \circ f \le \phi \circ f$ and $\varphi_n \circ f \to \phi \circ f$, pointwise. Now apply the Monotone Convergence Theorem (Theorem 3.10) to get (†).

(b) Show the following definite-integral version of (†). For $B \in \mathfrak{R}$,

$$\int_{f^{-1}(B)} \phi \circ f \, d\eta = \int_B \phi \, d\mu. \tag{‡}$$

Hint: Verify the additional auxiliary identity stated below. For $\psi : \mathbb{R} \to \mathbb{R}$,

$$(\psi \circ f)\,(\phi \circ f) = (\psi\,\phi) \circ f. \tag{ii}$$

Now use (i), (ii), and (†) to get (‡).

Note: The above arguments require that ϕ is nonnegative (beyond being $\mathfrak{R}$-measurable), but it is not required that f is anything but $\mathcal{X}$-measurable. We will see in Problem 4.19 that (†) holds for not necessarily nonnegative functions ϕ. Moreover, it can also be shown, in addition, that (†) can be extended from $(\mathbb{R}, \mathfrak{R}, \mu)$ to an arbitrary measure space $(Y, \mathcal{Y}, \mu)$ with the $\mathcal{X}$-measurable function $f : X \to \mathbb{R}$ replaced by a transformation $T : X \to Y$ measurable with respect to $\mathcal{X}$ and $\mathcal{Y}$ (as in Problem 1.8).

Problem 3.15. Consider the notions introduced in Problem 2.20, namely, a probability space $(X, \mathcal{X}, \wp)$, a real random variable $f : X \to \mathbb{R}$, the probability distribution μ_f of the random variable f (i.e., the probability measure $\mu_f(\ ) = \wp(f^{-1}(\ ))$ on $\mathfrak{R}$ induced by the probability measure $\wp$ on $\mathcal{X}$ via f), and the distribution function $F : \mathbb{R} \to [0, 1]$ of f. Suppose there is a nonnegative $\mathfrak{R}$-measurable function $p : \mathbb{R} \to \mathbb{R}$ such that

$$\mu_f(B) = \int_B p \, d\lambda\,, \quad \text{equivalently,} \quad \mu_f(B) = \int_B d\mu_f = \int_B p \, d\lambda = \int_B p(\alpha)\, d\alpha,$$

for every $B \in \mathfrak{R}$, with $\lambda : \mathfrak{R} \to \overline{\mathbb{R}}$ being the Lebesgue measure on $\mathfrak{R}$. We are writing $d\alpha$ for $d\lambda$, as usual. In Problem 3.13, the above expression was symbolically abbreviated by $d\mu_f = p\, d\lambda$ or (again, setting $d\alpha = d\lambda$),

$$p = \frac{d\mu_f}{d\lambda} = \frac{d\mu_f}{d\alpha}.$$

In this case, where the measure μ_f is the probability distribution of a random variable f, the nonnegative $\mathfrak{R}$-measurable function p, when it exists, is referred to as the *density function of the random variable* f.

Note 1: If p exists, then it depends on μ_f, and so it depends only on $\wp$ and f.

(a) If the density function p of the random variable f exists, then show that it is related to the distribution function F of f by the following relation.

$$F(\beta) = \int_{-\infty}^{\beta} p(\alpha)\, d\alpha \quad \text{for every} \quad \beta \in \mathbb{R}.$$

Hint: Problem 2.20: $\wp\{x \in X\colon f(x) \le \beta\} = \mu_f\big((-\infty, \beta]\big) = \int_{-\infty}^{\beta} d\mu_f$.

Definition: A real-valued function G on $\mathbb{R}$ is *absolutely continuous* if there exists a real-valued $\Re$-measurable function g on $\mathbb{R}$ such that $\int_{[\alpha,\beta]} g\, d\lambda < \infty$ and $G(\beta) - G(\alpha) = \int_{[\alpha,\beta]} g\, d\lambda$ for every closed and bounded interval $[\alpha, \beta]$. Moreover, under these conditions, G is continuous, the derivative G' of G exists λ-a.e. on $\mathbb{R}$, and $G' = g$ λ-a.e. on $\mathbb{R}$.

(b) Prove that if the distribution function F of f is absolutely continuous, then the density function p of f exists and p is the derivative F' of F:

$$p = \frac{dF}{d\alpha}.$$

Hint: Suppose F is absolutely continuous (so that the Fundamental Theorem of Calculus holds with $F(-\infty) = 0$ and $F(+\infty) = 1$). Verify that, for every $[\alpha, \beta]$, $\mu_f\big([\alpha, \gamma]\big) = F(\gamma) - F(\alpha) = \int_{[\alpha,\gamma]} g(\xi)\, d\lambda(\xi)$ whenever $\gamma \in [\alpha, \beta]$. Show that this implies $\mu_f(B) = \int_B g\, d\lambda$ for every $B \in \Re$, and hence $p = g$. (Recall that $\Re$ is the smallest σ-algebra containing all closed intervals.).

Note 2: Absolute continuity implies uniform continuity, which implies continuity, but the reverse implications fail in general.

Note 3: For an example of a continuous distribution function (thus nondecreasing with $\lim_{\alpha \to -\infty} F(\alpha) = 0$ and $\lim_{\alpha \to +\infty} F(\alpha) = 1$) that is not differentiable wherever it is not constant, but is differentiable λ-a.e. with a zero derivative, thus not absolutely continuous, see Problem 7.15(c) in Chapter 7.

(c) Use the definition of p (or item (a)) to verify that, if p exists, then

$$\int_{\mathbb{R}} p(\alpha)\, d\alpha = 1.$$

(d) Show that if p exists, then it is λ-a.e. unique.

Hint: Proposition 3.7(a), since p is nonnegative and $\mu_f(X) = \int_{\mathbb{R}} p\, d\lambda$.

Suggested Readings

Bartle [5], Berberian [9], Halmos [31], Royden [62], Rudin [63]. Also [2], [3], [8], [11], [56, Appendix D], and [17] and [65] for nice historical notes as well.



4

Integral of Real-Valued Functions

4.1 Integrable Functions

A real-valued function $f\colon X \to \mathbb{R}$ on X can be expressed as $f = f^+ - f^-$, where the nonnegative functions $f^+\colon X \to \mathbb{R}$ and $f^-\colon X \to \mathbb{R}$ are the positive and negative parts of f. If f is measurable, then so are f^+ and f^- (Proposition 1.6). Integration of measurable real-valued functions, leading to real-valued integrals, is considered by using the above decomposition.

Consider a measure space $(X, \mathcal{X}, \mu)$, and let $\mathcal{L}(X, \mathcal{X}, \mu)$ — or simply $\mathcal{L}$ if the measure space is clear in the context — be the collection of all *real-valued* $\mathcal{X}$-measurable functions such that *both* positive and negative parts have a *finite* integral with respect to the measure μ. That is,

$$\mathcal{L} = \mathcal{L}(X, \mathcal{X}, \mu) = \big\{f\colon X \to \mathbb{R}\colon\ f \in \mathcal{M}(X, \mathcal{X}),\ \int f^+ d\mu < \infty,\ \int f^- d\mu < \infty\big\}.$$

Definition 4.1. (The *integral* of a real-valued measurable function). Take a measure space $(X, \mathcal{X}, \mu)$. The *integral* of a function $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ with respect to μ is the real number

$$\int f\, d\mu = \int f^+ d\mu - \int f^- d\mu.$$

The *integral* of f *over* a measurable set E with respect to μ is defined by

$$\int_E f\, d\mu = \int_E f^+ d\mu - \int_E f^- d\mu.$$

A function in $\mathcal{L}(X, \mathcal{X}, \mu)$ is called an *integrable* (or *μ-integrable*) *function*.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_4

The integral $\int f d\mu$ of a *real-valued* function f in $\mathcal{M}(X, \mathcal{X})$ is then defined in terms of the integrals $\int f^{\pm} d\mu$ of its positive and negative parts $f^{\pm}$ in $\mathcal{M}(X, \mathcal{X})^+$ if these integrals (as in Definition 3.3) are *finite.* Note that, instead of requiring that both integrals $\int f^+ d\mu$ and $\int f^- d\mu$ be finite, we might require that just one of them be finite to ensure the existence of an integral $\int f d\mu$ as in Definition 4.1, which, in this case, no longer would necessarily be finite. We, however, adopt here the symmetric approach of both $\int f^+ d\mu$ and $\int f^- d\mu$ being finite, so f is integrable in the sense of $\int f d\mu < \infty$.

A special case. Consider the Lebesgue measure space $(\mathbb{R}, \Re, \lambda)$ (or, equivalently, $(\mathbb{R}, \widehat{\Re}, \widehat{\lambda})$). Take a real-valued $\Re$-measurable function f on $\mathbb{R}$.

$$\text{The } \textit{Lebesgue integral } \int f d\lambda \text{ of } f\colon \mathbb{R} \to \mathbb{R}$$

is defined as the integral of f with respect to the Lebesgue measure λ on $\Re$ (or $\widehat{\lambda}$ on $\widehat{\Re}$; see the last paragraph of Section 2.3). If the Lebesgue integral of a real-valued function f exists in $\mathbb{R}$, then f is called *Lebesgue integrable.* Naturally, the same framework applies to the Lebesgue measure space $(B, \mathcal{B}, \lambda|_{\mathcal{B}})$, where $B \in \Re$ is an arbitrary $\Re$-measurable set, $\mathcal{B} = \wp(B) \cap \Re$ is the σ-algebra of all Borel subsets of B, and $\lambda|_{\mathcal{B}}$ is the restriction of the Lebesgue measure λ on $\Re$ to $\mathcal{B}$, as in Problem 2.11. The measure $\lambda|_{\mathcal{B}}$ is referred to as the *Lebesgue measure* on $\mathcal{B}$, which is usually written without the subscript, $\lambda|_{\mathcal{B}} = \lambda$, if there is no risk of ambiguity. So consider the Lebesgue measure space $(B, \mathcal{B}, \lambda)$ and an $\mathbb{R}$-valued $\mathcal{B}$-measurable function g on B (with B being the Borel subset of the real line for which $\mathcal{B} = \wp(B) \cap \Re$).

$$\text{The } \textit{Lebesgue integral } \int g\, d\lambda \text{ of } g\colon B \to \mathbb{R}$$

is again defined as the integral of g with respect to the Lebesgue measure $\lambda = \lambda|_{\mathcal{B}}$. If the Lebesgue integral of g exists in $\mathbb{R}$, then g is called *Lebesgue integrable.* (Observe that $\int_B f d\lambda = \int f|_B \, d\lambda|_{\mathcal{B}}$ — cf. Proposition 4.2(c) below.)

We adopt the simplest notation, $\int f d\mu$, for the integral of f with respect to μ. There are, however, some additional usual notations, such as $\int f(x)\, d\mu$, $\int f(x)\, d\mu(x)$, or $\int f(x)\, \mu(dx)$. In particular, another common notation for the Lebesgue integral, $\int f d\lambda$, in addition to the above ones, is $\int f(x)\, dx$ (coinciding with the common notation for the Riemann integral).

Proposition 4.2. *Let $(X, \mathcal{X}, \mu)$ be a measure space. Let f and g be real-valued functions on X. Suppose f and g lie in $\mathcal{M}(X, \mathcal{X})$.*

(a) *If $f = 0$ μ-a.e., then $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ and $\int f d\mu = 0$.*

(b) *If $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ and $\int_E f d\mu = 0$ for every $E \in \mathcal{X}$, then $f = 0$ μ-a.e.*

(c) *If $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ and $g \in \mathcal{M}(X, \mathcal{X})$ is bounded, then $fg \in \mathcal{L}(X, \mathcal{X}, \mu)$.*

(d) *If $f \in \mathcal{L}(X, \mathcal{X}, \mu)$, then $f|_E \in \mathcal{L}(E, \mathcal{E}, \mu|_{\mathcal{E}})$ and $\int_E f\, d\mu = \int f|_E \, d\mu|_{\mathcal{E}}$ for every $E \in \mathcal{X}$ with $\mathcal{E} = \wp(E) \cap \mathcal{X}$.*

Proof. (a) The quick way. Take $f \in \mathcal{M}(X,\mathcal{X})$ so that $f = f^+ - f^-$ with $f^+ = f\chi_{F^+}$ and $f^- = -f\chi_{F^-}$ (cf. definition before Proposition 1.6), where f^+ and f^- lie in $\mathcal{M}(X,\mathcal{X})^+$ by Proposition 1.6(b). If $f = 0$ μ-a.e., then $f^+ = f^- = 0$ μ-a.e. by the definition of f^+ and f^-. Thus, by Proposition 3.7(a), $\int f^+ d\mu = \int f^- d\mu = 0$. So $f \in \mathcal{L}(X,\mathcal{X},\mu)$ with $\int f d\mu = 0$ by Definition 4.1.

(a) The detailed way. For an arbitrary $f\colon X \to \mathbb{R}$, take the sets

$$F_0 = \big\{x \in X\colon f(x) = 0\big\},$$
$$F_- = \big\{x \in X\colon f(x) < 0\big\}, \quad F_+ = \big\{x \in X\colon f(x) > 0\big\},$$
$$F^- = F_- \cup F_0 = \big\{x \in X\colon f(x) \le 0\big\}, \quad F^+ = F_+ \cup F_0 = \big\{x \in X\colon f(x) \ge 0\big\}.$$

If $f \in \mathcal{M}(X,\mathcal{X})$, then F^+ and F^- lie in $\mathcal{X}$ (Proposition 1.6). Thus F_+ and F_-, and so F_0, also lie in $\mathcal{X}$ (Problem 1.1). If $f = 0$ μ-a.e., then $\mu(F_+) = 0$ (as $\mu(\{x \in X\colon f(x) \ne 0\}) = 0$). Recall: $f^+ = f\chi_{F^+}$, so $f^+ = f^+\chi_{F^+}$. Thus

$$\int f^+ d\mu = \int f^+\chi_{F^+}\, d\mu = \int_{F^+} f^+ d\mu = \int_{F_+} f^+ d\mu + \int_{F_0} f^+ d\mu = \int_{F_0} f^+ d\mu = 0$$

by Problem 3.8(b). Similarly, $\int f^- d\mu = 0$. Hence the real-valued function $f = f^+ - f^-$ lies in $\mathcal{L}(X,\mathcal{X},\mu)$ and $\int f\, d\mu = 0$ according to Definition 4.1.

(b) Take $f = f^+ - f^-$ with $f^+ = f\chi_{F^+}$ and $f^- = -f\chi_{F^-}$ in $\mathcal{M}(X,\mathcal{X})^+$. If $\int_E f d\mu = 0$ for every $E \in \mathcal{X}$, then $\int f^+ d\mu = \int_{F^+} f\, d\mu = \int f^- d\mu = \int_{F^-} f\, d\mu = 0$, so that $f^+ = f^- = 0$ μ-a.e. by Proposition 3.7(a), so that $f = 0$.

(c) Note that $fg = (f^+ - f^-)(g^+ - g^-) = f^+g^+ + f^-g^- - f^+g^- - f^-g^+$. So

$$(fg)^+ = f^+g^+ + f^-g^- \quad \text{and} \quad (fg)^- = f^-g^+ + f^+g^-.$$

If g is bounded, set $\beta = \sup_{x\in X} |g(x)|$ so that $g^+ \le \beta$ and $g^- \le \beta$. Then $f^+g^+ \le \beta f^+$ and so $\int f^+g^+ d\mu \le \beta \int f^+ d\mu < \infty$ by Propositions 1.5, 1.6, 3.5(a) and Problem 3.3(b). Similarly, $\int f^-g^- d\mu < \infty$, $\int f^+g^- d\mu < \infty$, and $\int f^-g^+ d\mu < \infty$. Hence, $\int (fg)^+ d\mu < \infty$ and $\int (fg)^- d\mu < \infty$ according to Proposition 3.5(b), and therefore $fg \in \mathcal{L}(X,\mathcal{X},\mu)$.

(d) Take an arbitrary $E \in \mathcal{X}$ and set $\mathcal{E} = \wp(E) \cap \mathcal{X}$. By Proposition 1.3, $f|_E$ is in $\mathcal{M}(E,\mathcal{E})$ if f is in $\mathcal{M}(X,\mathcal{X})$. Clearly, $f^+|_E = f|_E{}^+$ and $f^-|_E = f|_E{}^-$. Let $\mu|_{\mathcal{E}}$ be the restriction of μ to $\mathcal{E}$ as in Problem 2.11. By Definition 4.1, $f|_E \in \mathcal{L}(E,\mathcal{E},\mu|_{\mathcal{E}})$ whenever $f \in \mathcal{L}(X,\mathcal{X},\mu)$ and $\int_E f d\mu = \int f|_E\, d\mu|_{\mathcal{E}}$. □

It is worth noting that we must refrain from using still unproved, possibly guessable, results when attempting to prove item (c) above. If, in addition to Proposition 1.5, we could apply Lemma 4.4 and Problem 5.1(b) now, then we would be able to show at once that

$$\left|\int f g\, d\mu\right| \le \int |fg|\, d\mu \le \sup|g| \int |f|\, d\mu = \sup|g| \left[\int f^+\, d\mu + \int f^-\, d\mu\right] < \infty.$$

If f_1 and f_2 are real-valued functions in $\mathcal{M}(X,\mathcal{X})^+$ with $f_2 \le f_1$ and $\int f_2\,d\mu < \infty$, then $\int (f_1 - f_2)\,d\mu = \int f_1 d\mu - \int f_2\,d\mu$. In fact, write $f_1 = (f_1 - f_2) + f_2$ and apply Propositions 1.5 and 3.5(b). In the next proposition we replace the assumption $f_2 \le f_1$ by $\int f_1\,d\mu < \infty$.

Proposition 4.3. *Take a measure space $(X,\mathcal{X},\mu)$. If f_1 and f_2 are real-valued functions in $\mathcal{M}(X,\mathcal{X})^+$ with $\int f_1\,d\mu < \infty$ and $\int f_2\,d\mu < \infty$, then*

$$f_1 - f_2 \in \mathcal{L}(X,\mathcal{X},\mu) \quad \textit{and} \quad \int (f_1 - f_2)\,d\mu = \int f_1\,d\mu - \int f_2\,d\mu.$$

Proof. If f_1 and f_2 are real-valued functions in $\mathcal{M}(X,\mathcal{X})^+$, then $f = f_1 - f_2$ lies in $\mathcal{M}(X,\mathcal{X})$ (cf. Proposition 1.5). Since $f_1,\ f_2,\ f^+$, and f^- are functions in $\mathcal{M}(X,\mathcal{X})^+$ and $f^+ - f^- = f = f_1 - f_2$, it follows that $f^+ + f_2 = f_1 + f^-$ is in $\mathcal{M}(X,\mathcal{X})^+$, and so (cf. Proposition 3.5(b))

$$\int f^+ d\mu + \int f_2\,d\mu = \int f_1\,d\mu + \int f^- d\mu.$$

Note that $f^+ \le f_1$ and $f^- \le f_2$. Since $\int f_1\,d\mu < \infty$ and $\int f_2\,d\mu < \infty$, it follows by Problem 3.3(b) that $\int f^+ d\mu < \infty$ and $\int f^- d\mu < \infty$. Therefore $(f_1 - f_2) = f$ lies in $\mathcal{L}(X,\mathcal{X},\mu)$ and, according to Definition 4.1,

$$\int (f_1 - f_2)\,d\mu = \int f\,d\mu = \int f^+ d\mu - \int f^- d\mu = \int f_1\,d\mu - \int f_2\,d\mu \quad \square$$

4.2 Three Fundamental Lemmas

Absolute integrability (of measurable functions), the first property in this section, is of crucial importance. It says that $|f|$ is integrable if and only if f is. This holds for every (measure-theoretic) integral, as given in Definitions 3.3 and 4.1. In particular, absolute integrability holds for the Lebesgue integral (but it does not hold for the Riemann integral).

Remark 4.A (Riemann × Lebesgue): There are many differences between Riemann and Lebesgue integrals. We exhibit below just two of them, which are based on Problem 4.1. (More will be shown in Problems 5.14 and 5.15.)

(1) Problem 4.1(a) ensures that *if a bounded function on a closed and bounded* (i.e., on a compact) *interval of the real line has a Riemann integral, then it is $\Re$-measurable and Lebesgue integrable, and its Riemann and Lebesgue integrals coincide*. However, there are bounded functions defined on closed and bounded intervals that are Lebesgue but not Riemann integrable. For instance, the real-valued measurable function on $[0,1]$

$$f = 2\chi_{\mathbb{Q}} - 1 = 1 - 2\chi_{\mathbb{R}\setminus\mathbb{Q}} : [0,1] \to \{-1,1\} \subset \mathbb{R},$$

that is, $f(x) = 1$ for $x \in \mathbb{Q} \cap [0,1]$ and $f(x) = -1$ for $x \in (\mathbb{R}\backslash\mathbb{Q}) \cap [0,1]$, is such that the Riemann integral does not exist but the Lebesgue integral does exist and is equal to -1. (In fact, $\chi_{\mathbb{Q}} = 1 - \chi_{\mathbb{R}\backslash\mathbb{Q}}$ and, since $\mathbb{Q}$ lies in $\Re$, is dense in $\mathbb{R}$, and has Lebesgue measure zero, it is easy to see that the set of points in $[0,1]$ at which $\chi_{\mathbb{R}\backslash\mathbb{Q}}$ is discontinuous, viz., $(\mathbb{R}\backslash\mathbb{Q}) \cap [0,1]$, lies in $\Re$ and has Lebesgue measure different from zero, and therefore $\chi_{\mathbb{R}\backslash\mathbb{Q}}$ — and so $\chi_{\mathbb{Q}}$ — is not Riemann integrable on $[0,1]$ according to Problem 4.1(b).) But $|f| = 1$. Thus, as a constant function on $[0,1]$, $|f|$ is trivially Riemann integrable (and so it is Lebesgue integrable with the same integral $\int |f|\, dx = 1$).

(2) On the other hand, the situation is different for improper Riemann integrals. If a function either is defined on an unbounded interval or is itself unbounded, then it may have a Riemann integral and not a Lebesgue integral. For instance, the real-valued function g on $[1,\infty)$ given by

$$g(x) = \frac{\sin x}{x} \quad \text{for every} \quad x \in [1,\infty)$$

is Riemann integrable (its improper Riemann integral exists and is finite), but $|g|$ is not integrable (i.e., it has no finite integral in any sense), and so g is not Lebesgue integrable (as we will see in Lemma 4.4(a) below).

Lemma 4.4. *Take a measure space $(X, \mathcal{X}, \mu)$ and a function $f\colon X \to \mathbb{R}$.*

(a) $\qquad f \in \mathcal{L}(X,\mathcal{X},\mu) \quad$ *if and only if* $\quad |f| \in \mathcal{L}(X,\mathcal{X},\mu)$.

If $f \in \mathcal{L}(X,\mathcal{X},\mu)$, then

(b) $$\left| \int f\, d\mu \right| \le \int |f|\, d\mu.$$

(c) $f \in \mathcal{L}(X,\mathcal{X},\mu)$ *if and only if* $f \in \mathcal{M}(X,\mathcal{X})$ *and* $\int |f|\, d\mu < \infty$.

Proof. First note that, for any function $f\colon X \to \mathbb{R}$,

$$|f|^+ = |f| = f^+ + f^- \quad \text{and} \quad |f|^- = 0. \tag{i}$$

Now recall from Proposition 1.6 that, since f is *real*-valued,

$$f \text{ is measurable} \iff f^+ \text{ and } f^- \text{ are measurable} \iff |f| \text{ is measurable.} \tag{ii}$$

Next recall from the definition of $\mathcal{L}(X,\mathcal{X},\mu)$ that

$$f \in \mathcal{L}(X,\mathcal{X},\mu) \iff f \in \mathcal{M}(X,\mathcal{X}) \text{ and } \int f^+ d\mu < \infty \text{ and } \int f^- d\mu < \infty, \tag{iii}$$

so that, by (i), (ii), and Proposition 4.2(a),

$$|f| \in \mathcal{L}(X,\mathcal{X},\mu) \iff f \in \mathcal{M}(X,\mathcal{X}) \text{ and } \int |f|\, d\mu < \infty. \tag{iv}$$

(a) Consider the definition of $\mathcal{L}(X, \mathcal{X}, \mu)$. If $f \in \mathcal{L}(X, \mathcal{X}, \mu)$, then, by (i), (ii), and Proposition 4.2(a), $\int |f|^+ d\mu < \infty$ and $\int |f|^- d\mu = 0$. So $|f| \in \mathcal{L}(X, \mathcal{X}, \mu)$ by (iii). Conversely, if $|f| \in \mathcal{L}(X, \mathcal{X}, \mu)$, then $|f| \in \mathcal{M}(X, \mathcal{X})$ and $\int |f|^+ d\mu < \infty$. But $f^+ \le |f|^+$ and $f^- \le |f|^+$ by (i). Hence $\int f^+ d\mu < \infty$ and $\int f^- d\mu < \infty$ according to Problem 3.3(b), and therefore $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ by (ii).

(b) If $f \in \mathcal{L}(X, \mathcal{X}, \mu)$, then $|f| \in \mathcal{L}(X, \mathcal{X}, \mu)$ by (a). Thus it follows by (i), Proposition 3.5(b), and Definition 4.1 that

$$\left| \int f \, d\mu \right| = \left| \int f^+ d\mu - \int f^- d\mu \right| \le \int f^+ d\mu + \int f^- d\mu = \int |f| \, d\mu.$$

(c) This is just another way to state item (a) — see also item (iv). Indeed, $f \in \mathcal{L} \iff |f| \in \mathcal{L} \iff \int |f| \, d\mu < \infty$ and $f \in \mathcal{M}$. □

As we pointed out in Section 3.2, the Monotone Convergence Theorem gave us in Proposition 3.5(a,b) the first evidences of linearity for the integral transformation. Linearity is definitely accomplished in the next lemma.

Lemma 4.5. *$\mathcal{L}$ is a linear space and $\int : \mathcal{L} \to \mathbb{R}$ is a linear functional.*

Remark 4.B (Linearity): Before proving Lemma 4.5, note that its statement is twofold. First it says that the collection $\mathcal{L}(X, \mathcal{X}, \mu)$ is a (real) *linear space*. Indeed, Proposition 1.5 ensures that $\mathcal{M}(X, \mathcal{X})$ is a real linear space (when consisting of real-valued functions only — see the last paragraph of Section 1.3). Since $\mathcal{L}(X, \mathcal{X}, \mu)$ is a subset of $\mathcal{M}(X, \mathcal{X})$, $\mathcal{L}(X, \mathcal{X}, \mu)$ is a linear space if and only if it is a *linear manifold* of $\mathcal{M}(X, \mathcal{X})$, which means that *if f and g are functions in $\mathcal{L}(X, \mathcal{X}, \mu)$ and γ is any real number, then*

(a) $$\gamma f \text{ and } f + g \text{ lie in } \mathcal{L}(X, \mathcal{X}, \mu).$$

Now consider the transformation $\int : \mathcal{L} \to \mathbb{R}$ that assigns to each function f in $\mathcal{L}(X, \mathcal{X}, \mu)$ the value of its integral $\int f d\mu$ in $\mathbb{R}$. A transformation of a (real or complex) linear space into $\mathbb{R}$ or $\mathbb{C}$ is called a (real or complex) *functional*. If, in addition, this transformation is linear, then it is a *linear functional*. Lemma 4.5 says that $\int : \mathcal{L} \to \mathbb{R}$ is a (real) *linear functional* (i.e., a *homogeneous* and *additive* functional), which means that (i) its domain $\mathcal{L}$ is a linear space, and (ii) if f and g are functions in $\mathcal{L}(X, \mathcal{X}, \mu)$ and γ is a real number, then

(b) $$\int \gamma f \, d\mu = \gamma \int f \, d\mu \quad \text{and} \quad \int (f + g) \, d\mu = \int f \, d\mu + \int g \, d\mu.$$

Thus the proof of Lemma 4.5 is reduced to proving (a) and (b).

Proof of Lemma 4.5. First note that, as a particular case of Proposition 4.2(c), if $\gamma \in \mathbb{R}$ and $f \in \mathcal{L}(X, \mathcal{X}, \mu)$, then $\gamma f \in \mathcal{L}(X, \mathcal{X}, \mu)$. Moreover, since $-f = f^- - f^+$, it follows that

$$\gamma f = (\gamma f)^+ - (\gamma f)^- = \begin{cases} |\gamma| f^+ - |\gamma| f^-, & \gamma \geq 0, \\ |\gamma| f^- - |\gamma| f^+, & \gamma < 0, \end{cases}$$

where the functions $|\gamma| f^+$ and $|\gamma| f^-$ are in $\mathcal{M}(X, \mathcal{X})^+$, with $\int |\gamma| f^+ d\mu = |\gamma| \int f^+ d\mu$ and $\int |\gamma| f^- d\mu = |\gamma| \int f^- d\mu$ (Proposition 3.5(a)), and these integrals are finite since $f \in \mathcal{L}(X, \mathcal{X}, \mu)$. Hence, by Proposition 4.3,

$$\int \left(\pm |\gamma| f^+ - \pm |\gamma| f^- \right) d\mu = |\gamma| \left(\pm \int f^+ d\mu - \pm \int f^- d\mu \right),$$

and therefore,

$$\int \gamma f \, d\mu = \left\{ \begin{array}{ll} |\gamma| \left(\int f^+ d\mu - \int f^- d\mu \right), & \gamma \geq 0, \\ |\gamma| \left(\int f^- d\mu - \int f^+ d\mu \right), & \gamma < 0, \end{array} \right\} = \gamma \int f \, d\mu.$$

Thus homogeneity is proved. Now, to prove additivity, proceed as follows. Let f and g be in $\mathcal{L}(X, \mathcal{X}, \mu)$. Since $|f + g| \leq |f| + |g|$, and since $|f|$, $|g|$, $|f + g|$, and $|f| + |g|$ all lie in $\mathcal{M}(X, \mathcal{X})^+$ (cf. Propositions 1.5 and 1.6), it follows by Problem 3.3(b) and Proposition 3.5(b) that

$$\int |f + g| \, d\mu \leq \int \left(|f| + |g| \right) d\mu = \int |f| \, d\mu + \int |g| \, d\mu,$$

and so $|f + g| \in \mathcal{L}(X, \mathcal{X}, \mu)$. In fact, since f and g are in $\mathcal{L}(X, \mathcal{X}, \mu)$, we get $|f|$ and $|g|$ in $\mathcal{L}(X, \mathcal{X}, \mu)$ by Lemma 4.4, and so $\int |f + g| \, d\mu < \infty$. By Lemma 4.4 again, it follows that $f + g \in \mathcal{L}(X, \mathcal{X}, \mu)$ (since $f + g \in \mathcal{M}(X, \mathcal{X})$ according to Proposition 1.5). Now note that

$$f + g = (f^+ - f^-) + (g^+ - g^-) = (f^+ + g^+) - (f^- + g^-),$$

and also that $(f^+ + g^+)$ and $(f^- + g^-)$ lie in $\mathcal{M}(X, \mathcal{X})^+$ by Propositions 1.5 and 1.6, which have finite integrals by Proposition 3.5(b) — since f^+, f^-, g^+, and g^- have finite integrals because f and g lie in $\mathcal{L}(X, \mathcal{X}, \mu)$. Thus Propositions 4.3 and 3.5(b) and Definition 4.1 ensure that

$$\begin{aligned} \int (f + g) \, d\mu &= \int (f^+ + g^+) \, d\mu - \int (f^- + g^-) \, d\mu \\ &= \int f^+ d\mu - \int f^- d\mu + \int g^+ d\mu - \int g^- d\mu = \int f \, d\mu + \int g \, d\mu. \quad \square \end{aligned}$$

In Proposition 3.5(c) we saw that given a measure, another measure is generated by a nonnegative measurable function. This has a natural extension for general real-valued (not necessarily nonnegative) functions, but this time a signed measure is generated instead. Indeed, the next proposition

shows that given a measure, a signed measure is generated by a real-valued integrable function, called the *indefinite integral* of f with respect to μ.

Lemma 4.6. *Let $(X, \mathcal{X}, \mu)$ be a measure space. If $f \in \mathcal{L}(X, \mathcal{X}, \mu)$, then the real-valued set function $\nu\colon \mathcal{X} \to \mathbb{R}$ defined by*

$$\nu(E) = \int_E f\, d\mu \quad \text{for every} \quad E \in \mathcal{X}$$

is a signed measure.

Proof. If $f = f^+ - f^- \in \mathcal{L}(X, \mathcal{X}, \mu)$, then f^+ and f^- lie in $\mathcal{M}(X, \mathcal{X})^+$ and have finite integrals. Hence ν^+ and ν^- on $\mathcal{X}$ defined for each $E \in \mathcal{X}$ by

$$\nu^+(E) = \int_E f^+ d\mu \quad \text{and} \quad \nu^-(E) = \int_E f^- d\mu$$

are finite measures (cf. Proposition 3.5(c)). Therefore, by Definition 4.1, the set function ν on $\mathcal{X}$ defined for each $E \in \mathcal{X}$ by

$$\nu(E) = \int_E f\, d\mu = \int_E f^+\, d\mu - \int_E f^-\, d\mu = \nu^+(E) - \nu^-(E)$$

is such that $\nu = \nu^+ - \nu^-$. Thus, as a difference of two (positive) finite measures, $\nu\colon \mathcal{X} \to \mathbb{R}$ is a signed measure (cf. first paragraph of Section 2.2). □

4.3 The Dominated Convergence Theorem

A very important convergence theorem for integrable functions is the forthcoming Dominated Convergence Theorem (also referred to as the Lebesgue Dominated Convergence Theorem). It goes along the line of the Monotone Convergence Theorem as in Corollary 3.10, now with no restriction to nonnegative functions, where nonnegativeness and convergence from below are replaced with integrability and dominated convergence. Chronologically, the original version of the Dominated Convergence Theorem was published by Lebesgue in 1904, prior to (and independently of) the original version of the Monotone Convergence Theorem, published by Beppo Levi in 1906 — these refer to Lebesgue measure space $(\mathbb{R}, \Re, \lambda)$. However, the Dominated Convergence Theorem can be easily proved — in a general abstract measure space $(X, \mathcal{X}, \mu)$ — as a consequence of the Monotone Convergence Theorem. In fact, we will prove it by using Fatou's Lemma, which in turn was proved in Lemma 3.6 as consequence of the Monotone Convergence Theorem.

Theorem 4.7. (Dominated Convergence Theorem). *Let $(X, \mathcal{X}, \mu)$ be a measure space. If $\{f_n\}$ is a sequence of real-valued functions in $\mathcal{M}(X, \mathcal{X})$ converging μ-almost everywhere to a real-valued function f in $\mathcal{M}(X, \mathcal{X})$, and*

if there exists a nonnegative function g in $\mathcal{L}(X, \mathcal{X}, \mu)$ such that $|f_n| \le g$ for all n μ-almost everywhere, then each f_n and f lie in $\mathcal{L}(X, \mathcal{X}, \mu)$ and

$$\int f \, d\mu \;=\; \lim_n \int f_n \, d\mu.$$

Proof. Suppose $|f_n| \le g$ for all n μ-a.e. and $f_n \to f$ μ-a.e., so that $|f| \le g$ μ-almost everywhere. Thus each f_n and f are in $\mathcal{L}(X, \mathcal{X}, \mu)$ by Problem 4.4(b) — which in fact is straightforward by Lemma 4.4. Since $0 \le g \pm f_n$ (so that $g \pm f_n \in \mathcal{M}(X, \mathcal{X})^+$), and since $g \pm f_n \to g \pm f$ μ-a.e., use Fatou's Lemma (Lemma 3.9) with the sequence $\{g \pm f_n\}$, and apply Lemma 4.5 as follows:

$$\int g \, d\mu \pm \int f \, d\mu = \int \big(g \pm f\big) \, d\mu \le \liminf_n \int \big(g \pm f_n\big) \, d\mu \le \limsup_n \int \big(g \pm f_n\big) \, d\mu,$$

which implies that

$$\int g \, d\mu + \int f \, d\mu \;\le\; \liminf_n \left(\int g \, d\mu + \int f_n \, d\mu \right) = \int g \, d\mu + \liminf_n \int f_n \, d\mu,$$

$$\int g \, d\mu - \int f \, d\mu \;\le\; \limsup_n \left(\int g \, d\mu - \int f_n \, d\mu \right) = \int g \, d\mu - \limsup_n \int f_n \, d\mu,$$

and hence

$$\limsup_n \int f_n \, d\mu \;\le\; \int f \, d\mu \;\le\; \liminf_n \int f_n \, d\mu. \qquad \square$$

The Dominated Convergence Theorem plays a major part in integration theory. In particular, it is essential in the proof of completeness for the space L^p (Theorem 5.9), which is the main result of the next chapter. Applications of the Dominated Convergence Theorem will be frequent throughout the text from now on. As a special case, the next result, the Bounded Convergence Theorem, is an immediate consequence of Theorem 4.7.

Corollary 4.8. (Bounded Convergence Theorem). *Let $(X, \mathcal{X}, \mu)$ be a finite measure space* (i.e., a measure space equipped with a finite measure μ). *Suppose $\{f_n\}$ is a sequence of real-valued functions in $\mathcal{M}(X, \mathcal{X})$ converging μ-almost everywhere to a real-valued function f in $\mathcal{M}(X, \mathcal{X})$. If $\{f_n\}$ is bounded μ-almost everywhere* (i.e., if there exists a real number $\gamma > 0$ such that $|f_n| \le \gamma$ for all n μ-a.e.), *then each f_n and f lie in $\mathcal{L}(X, \mathcal{X}, \mu)$ and*

$$\int f \, d\mu \;=\; \lim_n \int f_n \, d\mu.$$

Proof. If $(X, \mathcal{X}, \mu)$ is a finite measure space, and if there is a number $\gamma > 0$ such that $|f_n| \le \gamma$ for all n μ-a.e. (i.e., if $\{f_n\}$ is bounded μ-almost everywhere), then the function $g\colon X \to \mathbb{R}$ such that $g(x) = \gamma$ for all $x \in X \backslash N$ for

some $N \in \mathcal{X}$ with $\mu(N) = 0$ (i.e., then the constant function $g = \gamma$ μ-a.e.) lies in $\mathcal{L}(X, \mathcal{X}, \mu)$ since $\int g\, d\mu = \gamma\, \mu(X) < \infty$. Now apply Theorem 4.7. □

4.4 Problems

Problem 4.1. Consider the following well-known fundamental results on the relation between the Riemann integral and the Lebesgue integral.

(a) If the Riemann integral of a real-valued function f defined on a closed and bounded interval of the real line exists, then f is Lebesgue integrable, and the values of the Lebesgue and Riemann integrals coincide.

(b) A real-valued bounded function defined on a closed and bounded interval of the real line is Riemann integrable if and only if the set of points at which it is not continuous has Lebesgue measure zero.

Item (b) is referred to as the Lebesgue Theorem. (References for these results are given in Suggested Readings at the end of this chapter.)

Apply items (a) and (b) to answer the following questions. Let C and S be the Cantor set and the Cantor-like set of Problems 2.9 and 2.10, respectively. Recall that both sets C and S are totally disconnected. Let χ_C and χ_S from $[0, 1]$ to $\{0, 1\}$ be the characteristic functions of C and S, respectively. Why is the function χ_C Riemann integrable? Is the function χ_S Riemann integrable? Are they Lebesgue integrable? What are their integrals?

Problem 4.2. Consider a sequence $\{f_n\}$ of functions in $\mathcal{L}(X, \mathcal{X}, \mu)$. If $\{f_n\}$ converges uniformly to $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ and $\mu(X) < \infty$, then show that

$$\int f\, d\mu \,=\, \lim_n \int f_n\, d\mu,$$

and also show that this identity may fail without the assumption $\mu(X) < \infty$.

Hint: If $f - \varepsilon \le f_n \le f + \varepsilon$, then verify that $f^+ - \varepsilon \le f_n^+ \le f^+ + \varepsilon$ and $f^- - \varepsilon \le f_n^- \le f^- + \varepsilon$ for any $\varepsilon > 0$, and so verify that uniform convergence of $\{f_n\}$ to f implies uniform convergence of $\{f_n^+\}$ and $\{f_n^-\}$ to f^+ and f^-, respectively. Use Problem 3.6 and Definition 4.1 to prove the claimed identity, and use Problem 3.11(b) to verify that it requires $\mu(X) < \infty$.

Problem 4.3. Let f be a function in $\mathcal{L}(X, \mathcal{X}, \mu)$. Show that the integral $\int f\, d\mu$ is unambiguously defined in terms of the integrals of any pair of functions f_1 and f_2 in $\mathcal{M}(X, \mathcal{X})^+ \cap \mathcal{L}(X, \mathcal{X}, \mu)$ — i.e., of any pair of *nonnegative* real-valued measurable functions with finite integrals — such that

$$f = f_1 - f_2.$$

Indeed, show that

$$\int f\,d\mu = \int f^+ d\mu - \int f^- d\mu = \int f_1\,d\mu - \int f_2\,d\mu.$$

Problem 4.4. Take a real-valued function f in $\mathcal{M}(X,\mathcal{X})$, and an arbitrary function $g \in \mathcal{L}(X,\mathcal{X},\mu)$. Show that

(a) $0 \le f \le g$ μ-a.e. implies $f \in \mathcal{L}(X,\mathcal{X},\mu)$ and $\int f\,d\mu \le \int g\,d\mu$,

(b) $|f| \le |g|$ μ-a.e. implies $f \in \mathcal{L}(X,\mathcal{X},\mu)$ and $\int |f|\,d\mu \le \int |g|\,d\mu$,

(c) $f = g$ μ-a.e. implies $f \in \mathcal{L}(X,\mathcal{X},\mu)$ and $\int f\,d\mu = \int g\,d\mu$.

Hints: (a) Use Problems 3.3(b) and 3.8(c) and Definition 4.1. (b) Apply item (a) and Lemma 4.4. (c) Use Proposition 4.2 and Lemma 4.5 to conclude that $\int (f-g)\,d\mu = 0$ and $f = (f-g) + g$ (since the functions are real-valued).

Problem 4.5. Take a pair of functions f and g in $\mathcal{L}(X,\mathcal{X},\mu)$, a real number γ, an arbitrary measurable set $E \in \mathcal{X}$, and prove the following assertions.

(a) $\int_E \gamma f\,d\mu = \gamma \int_E f\,d\mu$ and $\int_E (f+g)\,d\mu = \int_E f\,d\mu + \int_E g\,d\mu$.

(b) $\int_{E'} f\,d\mu \ge 0$ $(= 0)$ for all $E' \in \mathcal{X}$ $\iff$ $f \ge 0$ $(= 0)$ μ-a.e.

(c) $\int_{E'} f\,d\mu = \int_{E'} g\,d\mu$ for every $E' \in \mathcal{X}$ $\iff$ $f = g$ μ–a.e.

Hints: By Proposition 4.2(c), $f\chi_E \in \mathcal{L}(X,\mathcal{X},\mu)$. Use Lemma 4.5 to prove (a) and Propositions 3.7(a), 4.2(b), and Problem 4.4 to prove (b) and (c).

Problem 4.6. Prove that if f and g are functions in $\mathcal{L}(X,\mathcal{X},\mu)$, then so are the functions $f \wedge g$ and $f \vee g$.

Hint: Problem 1.3, Lemmas 4.4 and 4.5.

Problem 4.7. Let $(X,\mathcal{X},\mu)$ be a measure space. A complex-valued function $f\colon X \to \mathbb{C}$ is *measurable* if its real and imaginary parts, f_1 and f_2, are real-valued measurable functions (i.e., if f_1 and f_2 lie in $\mathcal{M}(X,\mathcal{X})$; cf. Problem 1.7). A measurable complex-valued function $f = f_1 + i\,f_2$ is *integrable* if its real and imaginary parts are real-valued integrable functions (i.e., if f_1 and f_2 lie in $\mathcal{L}(X,\mathcal{X},\mu)$). If a complex-valued function is integrable, then the *integral* of f is defined as the complex number

$$\int f\,d\mu = \int f_1\,d\mu + i\int f_2\,d\mu.$$

Prove the complex version of Lemma 4.4. Suppose $f\colon \mathcal{X} \to \mathbb{C}$ is measurable.

(a) f is integrable if and only if $|f|$ is integrable.

(b) f integrable implies $\left|\int f\,d\mu\right| \le \int |f|\,d\mu$.

(c) f is integrable if and only if $|f| \in \mathcal{M}(X,\mathcal{X})$ and $\int |f|\,d\mu < \infty$

(i.e., f is integrable if and only if $|f| \in \mathcal{L}(X, \mathcal{X}, \mu)$ — see Section 10.1).

Now prove the complex version of Problem 4.4(b,c). If f and g are complex-valued functions, f measurable and g integrable, then

(d) $|f| \le |g|$ μ-a.e. implies f integrable and $\int |f|\, d\mu \le \int |g|\, d\mu$,

(e) $f = g$ μ-a.e. implies f integrable and $\int f\, d\mu = \int g\, d\mu$.

Hints: (a) Note that $|f| \le |f_1| + |f_2|$, $|f_1| \le |f|$, and $|f_2| \le |f|$. Use Lemmas 4.4 and 4.5 to show that if f is integrable, then so is $|f_1| + |f_2|$, and so $|f|$ is integrable by Problem 4.4(b) (since $|f| = (|f_1|^2 + |f_2|^2)^{1/2}$ is measurable). Conversely, if f is measurable and $|f|$ is integrable, then use Problem 4.4(b) and Lemma 4.4 to show that f_1 and f_2 are integrable. Note: (a) implies (c).

(b) To prove that $\left|\int f\, d\mu\right| \le \int |f|\, d\mu$ proceed as follows. Write $\int f\, d\mu = \rho\, e^{i\theta}$ and consider the function $g = \mathrm{Re}\,(e^{-i\theta} f) : X \to \mathbb{R}$ so that $|g| \le |e^{-i\theta} f| = |f|$. Use Lemma 4.5 to verify that $\left|\int f\, d\mu\right| = \rho = e^{-i\theta} \int f\, d\mu = \int e^{-i\theta} f\, d\mu = \mathrm{Re} \int e^{-i\theta} f\, d\mu = \int \mathrm{Re}\,(e^{-i\theta} f)\, d\mu = \int g\, d\mu = \left|\int g\, d\mu\right| \le \int |g|\, d\mu \le \int |f|\, d\mu$.

(d) That f measurable, g integrable, and $|f| \le |g|$ μ-a.e. imply f integrable and $\int |f|\, d\mu \le \int |g|\, d\mu$ is again a consequence of Problem 4.4(b).

(e) Use the hint to Problem 4.4(c) to show that $\int f\, d\mu = \int g\, d\mu$ if $f = g$ μ-a.e.

Problem 4.8. Apply Lemma 4.5 to prove its own complex version. That is, show that if f and g are complex-valued integrable functions (with respect to a measure μ) and γ is an arbitrary complex number, then γf and $f + g$ are again integrable complex-valued functions, and

$$\int \gamma f\, d\mu = \gamma \int f\, d\mu \quad \text{and} \quad \int (f + g)\, d\mu = \int f\, d\mu + \int g\, d\mu.$$

Problem 4.9. Prove the complex version of the Dominated Convergence Theorem (Theorem 4.7). If $\{f_n\}$ is a sequence of complex-valued measurable functions that converges pointwise to a complex-valued function f, and if g is a nonnegative integrable function (with respect to a measure μ) such that $|f_n| \le g$ for all n, then f is integrable and

$$\int f\, d\mu = \lim_n \int f_n\, d\mu.$$

Problem 4.10. Show that the indefinite integral of an integrable function, as defined in Lemma 4.6, is countably additive in the following sense. If f is in $\mathcal{L}(X, \mathcal{X}, \mu)$ and $\{E_n\}$ is a countable measurable partition of $E \in \mathcal{X}$, then

$$\int_E f\, d\mu = \sum\nolimits_n \int_{E_n} f\, d\mu.$$

Problem 4.11. Consider a measure space $(X, \mathcal{X}, \mu)$. Suppose a sequence $\{f_n\}$ of real-valued functions in $\mathcal{M}(X, \mathcal{X})$ is such that $\sum_{n=1}^{m} f_n \to f$ almost everywhere to a real-valued function f in $\mathcal{M}(X, \mathcal{X})$. Let g be a nonnegative function in $\mathcal{L}(X, \mathcal{X}, \mu)$. If $\left|\sum_{n=1}^{m} f_n\right| \le g$ for all m almost everywhere, then f and each f_n are integrable (i.e., they lie in $\mathcal{L}(X, \mathcal{X}, \mu)$) and

$$\int f \, d\mu = \sum\nolimits_n \int f_n \, d\mu.$$

Problem 4.12. Take a sequence $\{f_n\}$ of functions in $\mathcal{L}(X, \mathcal{X}, \mu)$. If the series $\sum_{n=1}^{\infty} \int |f_n| \, d\mu$ of positive numbers converges (i.e., $\sum_{n=1}^{\infty} \int |f_n| \, d\mu < \infty$), then show that the sequence $\left\{\sum_{n=1}^{m} f_n\right\}$ of functions in $\mathcal{L}(X, \mathcal{X}, \mu)$ converges μ-almost everywhere as $m \to \infty$ to a function f in $\mathcal{L}(X, \mathcal{X}, \mu)$, and

$$\int f \, d\mu = \sum\nolimits_n \int f_n \, d\mu.$$

Hint: Since the sequence $\left\{\sum_{n=1}^{m} \int |f_n| \, d\mu\right\}$ of positive numbers is bounded, it converges in $\mathbb{R}$. Apply the corollary of the Monotone Convergence Theorem stated in Problem 3.7 (i.e., the Beppo Levi Theorem) to verify that the sequence of positive numbers $\left\{\int \sum_{n=1}^{m} |f_n| \, d\mu\right\}$ converges in $\mathbb{R}$, and that the sequence $\left\{\sum_{n=1}^{m} |f_n|\right\}$ of functions in $\mathcal{M}(X, \mathcal{X})^+$ converges pointwise to a function h in $\mathcal{M}(X, \mathcal{X})^+$ such that $\int h \, d\mu = \sum_{n=1}^{\infty} \int |f_n| \, d\mu$. Set $N = \{x \in X : h(x) = +\infty\}$. Also set $g(x) = h(x)$ if $x \in X \backslash N$, and $g(x) = 0$ if $x \in N$. Apply Problems 3.8 and 3.9 to show that the nonnegative real-valued function g lies in $\mathcal{L}(X, \mathcal{X}, \mu)$, and the sequence $\left\{\sum_{n=1}^{m} |f_n|\right\}$ in $\mathcal{M}(X, \mathcal{X})^+$ converges almost everywhere to g, and so the sequence of real-valued functions $\left\{\sum_{n=1}^{m} f_n\right\}$ in $\mathcal{M}(X, \mathcal{X})$ converges almost everywhere to a function f in $\mathcal{M}(X, \mathcal{X})$ (because it converges absolutely almost everywhere). Show that $\left|\sum_{n=1}^{m} f_n\right| \le g$ and apply the Dominated Convergence Theorem.

Problem 4.13. Let $f \in \mathcal{M}(X, \mathcal{X})$ be a real-valued function. For each positive integer n consider its n-truncation $f_n \in \mathcal{M}(X, \mathcal{X})$ as defined in Problem 1.5. Prove that (i) if $f \in \mathcal{L}(X, \mathcal{X}, \mu)$, then

$$\int f \, d\mu = \lim_n \int f_n \, d\mu.$$

Conversely, prove that (ii) if $\sup_n \int |f_n| \, d\mu < \infty$, then $f \in \mathcal{L}(X, \mathcal{X}, \mu)$.

Hint: Show that $\{f_n\}$ is a sequence of real-valued functions in $\mathcal{M}(X, \mathcal{X})$ that converges pointwise to the real-valued function f, and that $\{|f_n|\}$ is an increasing sequence of real-valued functions in $\mathcal{M}(X, \mathcal{X})^+$ that converges pointwise to $|f|$ (which implies that $|f_n| \le |f|$ for all n). To prove (i) verify that $|f|$ lies in $\mathcal{L}(X, \mathcal{X}, \mu)$ and apply the Dominated Convergence Theorem. To prove the converse in (ii) apply the Monotone Convergence Theorem and conclude that f lies in $\mathcal{L}(X, \mathcal{X}, \mu)$ since $\int |f| \, d\mu < \infty$.

Problem 4.14. If $\{f_n\}$ is a sequence of functions in $\mathcal{L}(X, \mathcal{X}, \mu)$, and if f is a real-valued function in $\mathcal{M}(X, \mathcal{X})$, then prove that

$$\lim_n \int |f_n - f|\, d\mu = 0 \quad \text{implies} \quad f \in \mathcal{L}(X, \mathcal{X}, \mu) \text{ and } \int |f|\, d\mu = \lim_n \int |f_n|\, d\mu.$$

Note: In the jargon of Chapter 5, this means: $f_n \to f$ in $L^1 \Longrightarrow \|f_n\|_1 \to \|f\|_1$.

Hint: Recall: $\big||\alpha| - |\beta|\big| \le |\alpha - \beta|$ for $\alpha, \beta \in \mathbb{R}$. As f and f_n are real-valued, use Problem 3.3(b) and Proposition 3.5(b) to show: $\lim_n \int |f_n - f|\, d\mu = 0$ implies $\limsup_n \int |f_n|\, d\mu \le \int |f|\, d\mu \le \liminf_n \int |f_n|\, d\mu$. Moreover, since $\lim_n \int |f_n - f|\, d\mu = 0$ implies $|f_n - f| \in \mathcal{L}(X, \mathcal{X}, \mu)$, and f_n lies in $\mathcal{L}(X, \mathcal{X}, \mu)$, use Lemmas 4.4 and 4.5 to conclude that f lies in $\mathcal{L}(X, \mathcal{X}, \mu)$.

Problem 4.15. Consider the statement of the Dominated Convergence Theorem. Show that in addition to the results stated there we also have

$$\lim_n \int |f_n - f|\, d\mu = 0.$$

Hint: $|f_n - f| \to 0$ and $|f_n - f| \le 2g$ for all n almost everywhere.

Problem 4.16. Consider again the statement of the Dominated Convergence Theorem. Verify that the dominance assumption (viz., $|f_n| \le g$ for all n almost everywhere for some $g \in \mathcal{L}(X, \mathcal{X}, \mu)$) cannot be dropped from the theorem statement, even under the assumption of uniform convergence (see Problem 4.2). In fact, take the Lebesgue measure space $(\mathbb{R}, \Re, \lambda)$, set $f = h = 0$, set $f_n = n\, \chi_{(0, \frac{1}{n}]}$, and set $h_n = \frac{1}{n}\, \chi_{[0,n]}$ for every integer $n \ge 1$ — all real-valued functions in $\mathcal{L}(\mathbb{R}, \Re, \lambda)$. Show that

(a) $\{f_n\}$ converges pointwise to f but $0 = \int f\, d\lambda \ne \lim_n \int f_n\, d\lambda = 1$,

(b) $\{h_n\}$ converges uniformly to h but $0 = \int h\, d\lambda \ne \lim_n \int h_n\, d\lambda = 1$.

Problem 4.17. Let $(X, \mathcal{X}, \mu)$ be a measure space and consider a function $f\colon X \times [0,1] \to \mathbb{R}$ such that, for each $s \in [0,1]$, the function $f(\ \ , s)\colon X \to \mathbb{R}$ is $\mathcal{X}$-measurable. If, for some $s_0 \in [0,1]$, $f(x, s_0) = \lim_{s \to s_0} f(x, s)$ for every $x \in X$, and if $|f(x,s)| \le g(x)$ for every $x \in X$ and all $s \in [0,1]$, for some g in $\mathcal{L}(X, \mathcal{X}, \mu)$, then use the Dominated Convergence Theorem to show that

$$\int f(x, s_0)\, d\mu \;=\; \lim_{s \to s_0} \int f(x, s)\, d\mu.$$

If, in addition, the function $f(x, \ \ )\colon [0,1] \to \mathbb{R}$ is continuous for each $x \in X$, then show that the function $F\colon [0,1] \to \mathbb{R}$, defined for each $s \in [0,1]$ by

$$F(s) \;=\; \int f(x, s)\, d\mu,$$

is continuous as well. (Continuity is with respect to the usual metric on $\mathbb{R}$.)

Problem 4.18. Consider a measurable space $(X, \mathcal{X}, \mu)$. Let $\{f_n\}$ be a sequence of integrable functions $f_n : X \to \mathbb{F}$, where $\mathbb{F}$ stands for either $\mathbb{R}$ or $\mathbb{C}$, and let $f : X \to \mathbb{F}$ be integrable as well. Show that

$$\lim_n \int |f_n - f|\, d\mu = 0 \quad \text{implies} \quad \lim_n \int f_n\, d\mu = \int f\, d\mu.$$

In the jargon of the forthcoming Chapter 5, this means that the linear functional $\int (\ )\, d\mu$ (cf. Lemma 4.5) is continuous when the linear space of integrable functions (more precisely, the linear space of equivalence classes of integrable functions, as discussed in Section 5.1 of the next chapter) is equipped with the norm $\| \ \|_1 = \int |\ |\, d\mu$, which in turn is equivalent to saying that the integral functional $\int (\ )\, d\mu$ preserves convergence in the sense that if $\lim_n f_n = f$ in L^1, then $\lim_n \int f_n\, d\mu = \int \lim_n f_n\, d\mu = \int f\, d\mu$ in $\mathbb{F}$.

Hint: Verify that (i) $f_n - f$ is integrable, (ii) and so is $|f_n - f|$, and (iii) use Lemma 4.4(b) or Problem 4.7(b).

Now compare the above result with Problem 4.14 and show that the implication in Problem 4.14 still holds if $\mathbb{R}$ is replaced by $\mathbb{C}$; that is, it still holds for complex-valued functions on X.

Problem 4.19. This problem shows that if $\phi \in \mathcal{L}(\mathbb{R}, \Re, \mu)$, rather than $\phi \in \mathcal{M}(\mathbb{R}, \Re)^+$, then the change of variables of Problem 3.14 still holds. Take a measure space $(X, \mathcal{X}, \eta)$ and an $\mathcal{X}$-measurable function $f : X \to \mathbb{R}$. Let $(\mathbb{R}, \Re, \mu)$ be the measure space where $\mu : \Re \to \overline{\mathbb{R}}$ is the measure induced by η via f as in Problem 2.19, namely, for every $B \in \Re$,

$$\mu(B) = \eta\big(f^{-1}(B)\big). \tag{§}$$

(a) As we did in Problem 3.14(a,b), prove the following change of variable formulas. If $\phi : \mathbb{R} \to \mathbb{R}$ is integrable with respect to μ, $\phi \in \mathcal{L}(\mathbb{R}, \Re, \mu)$, then

$$\int_X \phi \circ f\, d\eta = \int_{\mathbb{R}} \phi\, d\mu. \tag{†}$$

Generally, for every Borel set $B \in \Re$,

$$\int_{f^{-1}(B)} \phi \circ f\, d\eta = \int_B \phi\, d\mu. \tag{‡}$$

Hint: $\phi = \phi^+ - \phi^-$ and so (†) in Problem 3.14 holds for ϕ^+ and ϕ^-. According to Definition 4.1, $\phi \in \mathcal{L}$ if and only if ϕ^+ and ϕ^- lie in $\mathcal{L}$. Therefore, $\int_X \phi^+ \circ f\, d\eta = \int_{\mathbb{R}} \phi^+ d\mu < \infty$, $\int_X \phi^- \circ f\, d\eta = \int_{\mathbb{R}} \phi^- d\mu < \infty$. By Lemma 4.5,

$$\begin{aligned}\int_X \phi \circ f\, d\eta &= \int_X (\phi^+ - \phi^-) \circ f\, d\eta = \int_X (\phi^+ \circ f - \phi^- \circ f)\, d\eta \\ &= \int_X \phi^+ \circ f\, d\eta - \int_X \phi^- \circ f\, d\eta = \int_{\mathbb{R}} \phi^+ d\mu - \int_{\mathbb{R}} \phi^- d\mu = \int_{\mathbb{R}} \phi\, d\mu.\end{aligned}$$

(b) In particular, if ϕ is the identity function, and is integrable with respect to μ (i.e., if $\phi(\alpha) = \alpha$ for $\alpha \in \mathbb{R}$ and if $\phi \in \mathcal{L}(\mathbb{R}, \Re, \mu)$), then verify that

$$\int_X f(x)\, d\eta(x) \;=\; \int_{\mathbb{R}} \alpha\, d\mu(\alpha).$$

(c) Now suppose, in addition to the identity function being in $\mathcal{L}(\mathbb{R}, \Re, \mu)$, that there exists a nonnegative $\Re$-measurable function $\psi\colon \mathbb{R} \to \mathbb{R}$ (which must be λ-integrable, i.e., $\psi \in \mathcal{L}(\mathbb{R}, \Re, \lambda)$) such that

$$\mu(B) \;=\; \int_B \psi\, d\lambda$$

for $B \in \Re$ as in Problem 3.13, where $\lambda\colon \Re \to \overline{\mathbb{R}}$ is the Lebesgue measure. Using the notation introduced in Problem 3.13, the above is abbreviated as

$$d\mu = \psi\, d\lambda.$$

(We will see in Theorem 7.8, the Radon–Nikodym Theorem, an instance where this assumption is satisfied; see also the remark after Proposition 3.7.) In this case, show that item (b) is rewritten as

$$\int_X f(x)\, d\eta(x) \;=\; \int_{\mathbb{R}} \alpha\psi(\alpha)\, d\alpha,$$

where we are writing $d\alpha$ for $d\lambda(\alpha)$, λ being the Lebesgue measure.

Problem 4.20. Consider a probability space $(X, \mathcal{X}, \wp)$, where $\wp$ is a probability measure on $\mathcal{X}$, and let $f\colon X \to \mathbb{R}$ be a real random variable (cf. Problem 2.20). The integral of f with respect to $\wp$ is well defined if either f is nonnegative (i.e., if $f \in \mathcal{M}(X, \mathcal{X})^+$, when $\int(\ )\, d\wp$ is an extended real-valued functional) or if f is integrable with respect to $\wp$ (i.e., if $f \in \mathcal{L}(X, \mathcal{X}, \wp)$, not necessarily nonnegative, when $\int(\ )\, d\wp$ is a real-valued functional). In both cases, this integral has a special notation and nomenclature. The *expected value of a random variable* f, also called the *expectation* or *mean* of f, denoted by $\mathrm{Exp}(f)$, is the value of the integral of f with respect to $\wp$;

$$\mathrm{Exp}(f) \;=\; \int_X f\, d\wp.$$

Now let $\mu_f\colon \Re \to \mathbb{R}$ be the probability distribution of the random variable f (i.e., the probability measure $\mu_f(\ ) = \wp(f^{-1}(\ ))$ on $\Re$ induced by $\wp$ via f). Moreover, for such a fixed f, let $\phi\colon \mathbb{R} \to \mathbb{R}$ be an arbitrary function, and take the composition $\phi(f) = \phi \circ f\colon X \to \mathbb{R}$. As observed in Problems 3.14 and 4.19, now with $\eta = \wp$, we get the following assumptions and consequences:

$$(\mathrm{A}_1) \quad f \in \mathcal{M}(X, \mathcal{X})^+ \quad \text{and} \quad \phi \in \mathcal{M}(\mathbb{R}, \Re)^+ \quad \Longrightarrow \quad \phi(f) \in \mathcal{M}(X, \mathcal{X})^+.$$

$$(\mathrm{A}_2) \quad f \in \mathcal{L}(X, \mathcal{X}, \wp) \quad \text{and} \quad \phi \in \mathcal{L}(\mathbb{R}, \Re, \mu_f) \quad \Longrightarrow \quad \phi(f) \in \mathcal{L}(X, \mathcal{X}, \wp).$$

(a) Under these assumptions, use Problem 3.14 or 4.19(a) to show that

$$\mathrm{Exp}(\phi(f)) = \int_{\mathbb{R}} \phi(\alpha)\, d\mu_f(\alpha).$$

Particular case: if integer positive powers $\phi(\ ) = (\ )^n$ are either nonnegative (as even powers are) or integrable with respect to μ_f (e.g., for $n = 1$, if the identity function on $\mathbb{R}$ is integrable with respect to μ_f), then the expectations below are referred to as the nth *moments of the random variable* f:

$$\mathrm{Exp}(f^n) = \int_{\mathbb{R}} \alpha^n d\mu_f(\alpha).$$

Next, as in Problem 3.15, suppose there exists a nonnegative $\Re$-measurable function $p\colon \mathbb{R} \to \mathbb{R}$ such that $\mu_f(B) = \int_B p\, d\lambda$ for every $B \in \Re$, where λ is the Lebesgue measure on $\Re$ (and we sometimes write $d\alpha$ for $d\lambda$ or for $d\lambda(\alpha)$). The identity $\mu_f(B) = \int_B p\, d\lambda$ itself is sometimes expressed by $d\mu_f = p\, d\lambda$ (Problem 3.13). In this probability context, such a nonnegative function p is the density function of the random variable f (Problem 3.15).

(b) Show that the density function p is λ-a.e. unique if it exists.

Hint: Problem 4.5(c).

(c) Use item (a) to verify that, under the above assumption A_2 (in particular, if the identity function on $\mathbb{R}$ is integrable with respect to μ_f),

$$\mathrm{Exp}(f) = \int_{\mathbb{R}} \alpha p(\alpha)\, d\alpha.$$

(d) With the assumptions of items (a) and (c) still in force, verify that

$$\mathrm{Exp}(\phi(f)) = \int_{\mathbb{R}} \phi(\alpha)\, p(\alpha)\, d\alpha; \quad \text{in particular,} \quad \mathrm{Exp}(f^n) = \int_{\mathbb{R}} \alpha^n p(\alpha)\, d\alpha.$$

Suggested Readings

Bartle [5], Berberian [9], Brown and Pearcy [13], Halmos [31], Kingman and Taylor [39], Royden [62], Rudin [63]. For recent introductory texts see [53], [70]. Proofs of the classical results in Problem 4.1 can be found, for instance, in the following well-know books: [41, Theorem 30.4] for item (a), and [1, Theorem 7.6.5] and also [59, p. 3] for item (b).



5

Banach Spaces L^p

5.1 Construction of L^1

A topology to equip the linear space $\mathcal{L}(X, \mathcal{X}, \mu)$, which will turn it into a Banach space, is investigated in this chapter. Section 5.1 summarizes the basics on normed spaces that will be required in Chapter 5 as well as in parts of Chapters 6, 10, and 12. We assume the reader has been introduced to *linear spaces* (or *vector spaces*) before — see, e.g., Lemma 4.5.

Definition 5.1. Let $\mathcal{L}$ be an arbitrary linear space over $\mathbb{F}$, where $\mathbb{F}$ stands either for the real field $\mathbb{R}$ or the complex field $\mathbb{C}$. A real-valued function

$$\| \ \| : \mathcal{L} \to \mathbb{R}$$

is a *norm* on $\mathcal{L}$ if the following conditions (referred to as the *norm axioms*) are satisfied for all vectors f and g in $\mathcal{L}$ and all scalars γ in $\mathbb{F}$.

(i) $\|f\| \geq 0$ (*nonnegativeness*),

(ii) $\|f\| > 0$ if $f \neq 0$ (*positiveness*),

(iii) $\|\gamma f\| = |\gamma| \|f\|$ (*absolute homogeneity*),

(iv) $\|f + g\| \leq \|f\| + \|g\|$ (*subadditivity — triangle inequality*).

Any linear space $\mathcal{L}$ equipped with a norm $\| \ \|$ on it is a *normed space* (or a *normed linear space*, or a *normed vector space*). A linear space is a *real* or a *complex linear space* if $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ and, when equipped with a norm, it is called a *real* or *complex normed space*. Occasionally a subscript may be inserted in the norm $\| \ \|$ on $\mathcal{L}$, such as $\| \ \|_{\mathcal{L}}$, if necessary.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_5

Remark 5.A (Special case I): A functional

$$\langle\ ;\ \rangle\colon \mathcal{L}\times\mathcal{L} \to \mathbb{F}$$

is an *inner product* on a linear space $\mathcal{L}$ if the following conditions (referred to as the *inner product axioms*) are satisfied for all vectors f, g, and h in $\mathcal{L}$ and all scalars γ in $\mathbb{F}$ (with the upper bars $\overline{\langle\ ;\ \rangle}$ and $\overline{\gamma}$ denoting complex conjugates of $\langle\ ;\ \rangle$ and γ, which is innocuous if $\mathbb{F} = \mathbb{R}$).

(i′)	$\langle f+g\,;h\rangle = \langle f\,;h\rangle + \langle g\,;h\rangle$	(*additivity*),
(ii′)	$\langle \gamma f\,;g\rangle = \gamma\langle f\,;g\rangle$	(*homogeneity*),
(iii′)	$\langle f\,;g\rangle = \overline{\langle g\,;f\rangle}$	(*Hermitian symmetry*),
(iv′)	$\langle f\,;f\rangle \geq 0$	(*nonnegativeness*),
(v′)	$\langle f\,;f\rangle = 0$ only if $f = 0$	(*positiveness*).

A linear space equipped with an inner product is an *inner product space*. A real or complex (i.e., $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$) linear space equipped with an inner product is called a *real* or *complex inner product space*. A fundamental property of an inner product is the *Schwarz* (or *Cauchy–Schwarz*) *inequality*:

$$|\langle f\,;g\rangle|^2 \leq \langle f\,;f\rangle\,\langle g\,;g\rangle \quad \text{for every} \quad f, g \in \mathcal{L}.$$

(Indeed, by the first four axioms (i′), (ii′), (iii′), and (iv′),

$$0 \leq \langle f-\gamma g\,;f-\gamma g\rangle = \langle f\,;f\rangle - \overline{\gamma}\langle f\,;g\rangle - \gamma\overline{\langle f\,;g\rangle} + |\gamma|^2\langle g\,;g\rangle$$

for every $\gamma \in \mathbb{F}$. Set $\gamma = \frac{\langle f\,;g\rangle}{\beta}$ for an arbitrary $\beta > 0$ so that

$$0 \leq \langle f\,;f\rangle - \frac{1}{\beta}\left(2 - \frac{\langle g\,;g\rangle}{\beta}\right)|\langle f\,;g\rangle|^2.$$

If $\langle g\,;g\rangle \neq 0$, then set $\beta = \langle g\,;g\rangle$ to get the Schwarz inequality.
If $\langle g\,;g\rangle = 0$, then $g = 0$ by (v′) and we are done by (ii′), (iii′).)

This implies that every inner product induces a norm $\|\ \|$ on $\mathcal{L}$ given by

$$\|f\|^2 = \langle f\,;f\rangle \quad \text{for every} \quad f \in \mathcal{L}.$$

(In fact, the inner product axioms (ii′), (iii′), (iv′), and (v′) imply the norm axioms (i), (ii), and (iii) of Definition 5.1. The Schwarz inequality implies axiom (iv) of Definition 5.1, that is, the triangle inequality:

$$0 \leq \|f+g\|^2 = \|f\|^2 + 2\mathrm{Re}\,\langle f\,;g\rangle + \|g\|^2 \leq \big(\|f\| + \|g\|\big)^2.)$$

Consequently, every inner product space is a normed space, and the norm of an inner product space is always that induced by the inner product.

Elements of any linear space $\mathcal{L}$ are called *vectors*, and elements of a field $\mathbb{F}$ are called *scalars*. If $\mathcal{L}$ is a linear space equipped with a norm $\|\ \|$, then the resulting normed space is denoted by $(\mathcal{L}, \|\ \|)$ or simply by $\mathcal{L}$ if the norm is clear in the context. Observe from axioms (i), (ii), and (iii) in Definition 5.1 that if a function $\|\ \|: \mathcal{L} \to \mathbb{R}$ is a norm, then

$$\|f\| = 0 \quad \text{if and only if} \quad f = 0.$$

If a function $\|\ \|: \mathcal{L} \to \mathbb{R}$ satisfies the three axioms (i), (iii), and (iv) but not necessarily axiom (ii), then it is called a *seminorm* (or a *pseudonorm*). In other words, a seminorm does vanish at the origin (as a norm does) but a seminorm may also vanish at a nonzero vector (as a norm never does). A linear space $\mathcal{L}$ equipped with a seminorm is called a *seminormed space*.

Definition 5.2. A sequence of vectors $\{f_n\}$ in a normed space $(\mathcal{L}, \|\ \|)$ *converges* to a vector f in $\mathcal{L}$ if for each real number $\varepsilon > 0$ there exists a positive integer n_ε such that

$$n \geq n_\varepsilon \quad \text{implies} \quad \|f_n - f\| < \varepsilon.$$

If a sequence of vectors $\{f_n\}$ converges to a vector $f \in \mathcal{L}$, then it is said to be a *convergent sequence* and f is said to be the *limit* of $\{f_n\}$. In order to distinguish it among other convergence modes, we refer to the preceding concept as *norm convergence* (or *convergence in the norm topology*). A sequence $\{f_n\}$ is a *Cauchy sequence* in $(\mathcal{L}, \|\ \|)$ (or satisfies the *Cauchy criterion*) if for each real number $\varepsilon > 0$ there is a positive integer n_ε such that

$$n, m \geq n_\varepsilon \quad \text{implies} \quad \|f_m - f_n\| < \varepsilon.$$

A common notation for the Cauchy criterion is $\lim_{m,n} \|f_m - f_n\| = 0$. A sequence $\{f_n\}$ is a *bounded sequence* in $(\mathcal{L}, \|\ \|)$ if $\sup_n \|f_n\| < \infty$.

Proposition 5.3. *Consider an arbitrary normed space* $(\mathcal{L}, \|\ \|)$.

(a) *Every convergent sequence in* $(\mathcal{L}, \|\ \|)$ *is a Cauchy sequence.*

(b) *Every Cauchy sequence in* $(\mathcal{L}, \|\ \|)$ *is bounded.*

(c) *If a Cauchy sequence in* $(\mathcal{L}, \|\ \|)$ *has a subsequence that converges in* $(\mathcal{L}, \|\ \|)$*, then the sequence itself converges in* $(\mathcal{L}, \|\ \|)$ *and its limit coincides with the limit of that convergent subsequence.*

Proof. Take a normed space $(\mathcal{L}, \|\ \|)$ and a sequence $\{f_n\}$ of vectors in $\mathcal{L}$.

(a) Take any $\varepsilon > 0$. If $\{f_n\}$ converges to $f \in \mathcal{L}$, then there exists $n_\varepsilon \geq 1$ such that $\|f_n - f\| < \frac{\varepsilon}{2}$ for all $n \geq n_\varepsilon$. Since $\|f_m - f_n\| \leq \|f_m - f\| + \|f - f_n\|$ by the triangle inequality, it follows that $\|f_m - f_n\| < \varepsilon$ whenever $m, n \geq n_\varepsilon$.

(b) Suppose $\{f_n\}$ is Cauchy. Then there is an $n_1 > 1$ such that $\|f_m - f_n\| < 1$ for all $m, n \geq n_1$. Let $\beta \in \mathbb{R}$ be the maximum of $\{\|f_m - f_n\| \in \mathbb{R} : m, n < n_1\}$ (a finite set). Thus $\|f_m - f_n\| \leq \|f_m - f_{n_1}\| + \|f_{n_1} - f_n\| \leq 2\max\{1, \beta\}$ for all m, n by the triangle inequality. But $\|f_m\| \leq \|f_m - f_1\| + \|f_1\|$ for all m.

(c) Consider a subsequence $\{f_{n_k}\}$ of a Cauchy sequence $\{f_n\}$ that converges to $f \in \mathcal{L}$ (i.e., $\|f_{n_k} - f\| \to 0$ as $k \to \infty$). Take any $\varepsilon > 0$. Since $\{f_n\}$ is a Cauchy sequence, there is a positive integer n_ε such that $\|f_m - f_n\| < \frac{\varepsilon}{2}$ for all $m, n \geq n_\varepsilon$. Since $\{f_{n_k}\}$ converges to f, there is a positive integer k_ε such that $\|f_{n_k} - f\| < \frac{\varepsilon}{2}$ for all $k \geq k_\varepsilon$. Therefore, if j is any integer for which $j \geq k_\varepsilon$ and $n_j \geq n_\varepsilon$ (for instance, if $j = \max\{n_\varepsilon, k_\varepsilon\}$), then $\|f_n - f\| \leq \|f_n - f_{n_j}\| + \|f_{n_j} - f\| < \varepsilon$ for every $n \geq n_\varepsilon$ by the triangle inequality, which implies that the sequence $\{f_n\}$ converges to f. □

By Proposition 5.3(a), every convergent sequence is a Cauchy sequence. The converse may fail (see Problems 5.13 and 5.15). But there are normed spaces with the crucial property that every Cauchy sequence converges. Normed spaces possessing this property are called *complete*: a normed space $\mathcal{L}$ is complete if every Cauchy sequence in $\mathcal{L}$ is a convergent sequence in $\mathcal{L}$.

A *Banach space* is a complete normed space.

A *Hilbert space* is a complete inner product space;

equivalently, a Hilbert space $\mathcal{L}$ is a Banach space for which the norm $\|\ \|$ is induced by an inner product as $\|f\|^2 = \langle f\,; f\rangle$ for every $f \in \mathcal{L}$.

Proposition 5.4. *Set $\mathcal{L} = \mathcal{L}(X, \mathcal{X}, \mu)$. The function $\|\ \| : \mathcal{L} \to \mathbb{R}$ defined by*

$$\|f\| = \int |f|\, d\mu \quad \textit{for every} \quad f \in \mathcal{L}(X, \mathcal{X}, \mu)$$

is a seminorm on the linear space $\mathcal{L}(X, \mathcal{X}, \mu)$, which is such that

$$\|f\| = 0 \quad \textit{if and only if} \quad f = 0 \ \ \mu\textit{-almost everywhere.}$$

Proof. Recall from Lemma 4.5 that $\mathcal{L}(X, \mathcal{X}, \mu)$ is a real linear space. Set $\mathcal{L} = \mathcal{L}(X, \mathcal{X}, \mu)$. Observe that the function $\|\ \| : \mathcal{L} \to \mathbb{R}$ is well defined since the integral $\int |f|\, d\mu$ exists in $\mathbb{R}$ for every $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ by Lemma 4.4. Let f and g be arbitrary functions in $\mathcal{L}(X, \mathcal{X}, \mu)$, and take any scalar γ in $\mathbb{R}$. Note that $|\gamma f(x)| = |\gamma|\,|f(x)|$ for every $x \in X$, which means $|\gamma f| = |\gamma|\,|f|$. Since the triangle inequality holds in $\mathbb{R}$, that is, since $|\alpha + \beta| \leq |\alpha| + |\beta|$ for every pair $\{\alpha, \beta\}$ of real numbers, it follows that $|f(x) + g(x)| \leq |f(x)| + |g(x)|$ for every $x \in X$, which means $|f + g| \leq |f| + |g|$. Now verify the norm axioms. Axiom (i) in Definition 5.1, $\|f\| \geq 0$, is immediate (Problem 3.3). For axioms (iii) and (iv) recall that $\int : \mathcal{L} \to \mathbb{R}$ is a linear functional (Lemma 4.5). So,

$$\|\gamma f\| = \int |\gamma f|\, d\mu = \int |\gamma|\,|f|\, d\mu = |\gamma| \int |f|\, d\mu = |\gamma|\,\|f\|,$$

$$\|f+g\| = \int |f+g|\, d\mu \le \int \big(|f|+|g|\big)\, d\mu = \int |f|\, d\mu + \int |g|\, d\mu = \|f\| + \|g\|$$

(cf. Problem 3.3(b)). Thus $\|\ \|$ is a seminorm on $\mathcal{L}$. Moreover, Proposition 3.7(a) ensures that $\|f\| = 0$ if and only if $f = 0$ μ-almost everywhere. □

Nevertheless, this seminorm $\|\ \|$ is not a norm on $\mathcal{L}(X, \mathcal{X}, \mu)$. In fact, it does not satisfy axiom (ii) in Definition 5.1: there may be a function f in $\mathcal{L}(X, \mathcal{X}, \mu)$ such that $f \neq 0$ and $\|f\| = 0$ (e.g., take a Lebesgue integrable function f in $\mathcal{L}(\mathbb{R}, \Re, \lambda)$ such that $f(x) = 0$ for all $x \in \mathbb{R}$ except at the origin, where $f(0) = 1$). In order to make this seminorm on $\mathcal{L}(X, \mathcal{X}, \mu)$ into a norm we need to redefine the concept of equality between functions in $\mathcal{L}(X, \mathcal{X}, \mu)$ (other than the usual pointwise definition) so that axiom (ii) is satisfied.

Take a measure space $(X, \mathcal{X}, \mu)$, and let f and g be arbitrary real-valued functions in $\mathcal{M}(X, \mathcal{X})$. We say that f and g are *equivalent* (or *μ-equivalent*), denoted by $f \sim g$, if $f = g$ almost everywhere. Summing up:

$$f \sim g \iff f = g \ \ \mu\text{-a.e.}$$

The relation $\sim$ is in fact an equivalence relation on $\mathcal{M}(X, \mathcal{X})$ (reflexive, transitive, and symmetric). For every real-valued function f in $\mathcal{M}(X, \mathcal{X})$, let $[f]$ be the *equivalence class* of f (with respect to μ),

$$[f] = \big\{ f' \in \mathcal{M}(X, \mathcal{X}) \colon\ f' \sim f \big\}.$$

This $[f]$ is the subset of $\mathcal{M}(X, \mathcal{X})$ consisting of all functions in $\mathcal{M}(X, \mathcal{X})$ that are μ-equivalent to f. The following necessary and sufficient conditions for equality between equivalence classes are readily verified:

$$[f] = [g] \iff f \sim g \iff f = g \ \ \mu\text{-a.e.}$$

In particular, if f and g lie in $\mathcal{L}(X, \mathcal{X}, \mu)$, then (cf. Problem 4.4(c) or 4.5(b)),

$$[f] = [g] \iff f \sim g \iff f = g \ \ \mu\text{-a.e.} \iff \int (f-g)\, d\mu = 0.$$

Therefore, by the above three displayed expressions, if f lies in $\mathcal{L}(X, \mathcal{X}, \mu)$, then every f' in $[f]$ also lies in $\mathcal{L}(X, \mathcal{X}, \mu)$ and $\int f' d\mu = \int f\, d\mu$ (by linearity of the integral as in Lemma 4.5). Moreover, if f lies in $\mathcal{L}(X, \mathcal{X}, \mu)$, then every g' in $[g]$ also lies in $\mathcal{L}(X, \mathcal{X}, \mu)$ whenever $[f] = [g]$, and again in this case we get $\int g' d\mu = \int f\, d\mu$. Then set

$$\begin{aligned} L^1 = L^1(\mu) &= L^1(X, \mu) = L^1(X, \mathcal{X}, \mu) \\ &= \big\{ [f] \subseteq \mathcal{M}(X, \mathcal{X}) \colon\ f \in \mathcal{L}(X, \mathcal{X}, \mu) \big\}, \end{aligned}$$

which is the collection of all equivalence classes of functions in $\mathcal{L}(X, \mathcal{X}, \mu)$. Note that this collection $L^1 = L^1(X, \mathcal{X}, \mu)$ is the *quotient space* of $\mathcal{L}(X, \mathcal{X}, \mu)$ modulo $\sim$, and so it is denoted by $\mathcal{L}(X, \mathcal{X}, \mu)/\sim$ as well. Since $\mathcal{L}(X, \mathcal{X}, \mu)$ is a linear space, it can be shown that $L^1(X, \mathcal{X}, \mu)$ is made into a linear space when scalar multiplication and vector addition are defined by

$$\gamma[f] = [\gamma f] \quad \text{and} \quad [f] + [g] = [f + g]$$

for every $[f]$ and $[g]$ in $L^1(X, \mathcal{X}, \mu)$ and every scalar γ. Observe that the origin $[0]$ of the linear space $L^1(X, \mathcal{X}, \mu)$ is a linear manifold

$$[0] = \big\{ f \in \mathcal{L}(X, \mathcal{X}, \mu) \colon \ f = 0 \ \mu\text{-a.e.} \big\}$$

of the linear space $\mathcal{L}(X, \mathcal{X}, \mu)$, and $L^1(X, \mathcal{X}, \mu)$ can still be viewed as the *quotient space* of $\mathcal{L}(X, \mathcal{X}, \mu)$ modulo $[0]$, also denoted by $\mathcal{L}(X, \mathcal{X}, \mu)/[0]$. The seminorm on $\mathcal{L}(X, \mathcal{X}, \mu)$ given in Proposition 5.4 induces a norm in $L^1(X, \mathcal{X}, \mu)$. In fact, consider the function $\| \ \|_1 \colon L^1 \to \mathbb{R}$ defined by

$$\|[f]\|_1 = \int |f| \, d\mu \quad \text{for every} \quad [f] \in L^1(X, \mathcal{X}, \mu),$$

where $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ is any representative of the equivalence class $[f]$.

Proposition 5.5. *The function* $\| \ \|_1$ *is a norm on the linear space* L^1.

Proof. Consider the seminorm $\| \ \|$ on $\mathcal{L}(X, \mathcal{X}, \mu)$ of Proposition 5.4. So the function $\| \ \|_1 \colon L^1 \to \mathbb{R}$ is well defined. Take an arbitrary equivalence class $[f]$ in $L^1(X, \mathcal{X}, \mu)$, an arbitrary function f in $[f]$, and set

$$\|[f]\|_1 = \|f\|.$$

The above identity makes sense because the value of $\|[f]\|_1$ does not depend on the representative f of the equivalence class $[f]$. Indeed, if f and f' are functions in $\mathcal{L}(X, \mathcal{X}, \mu)$ such that $f = f' \ \mu$-a.e., then $|f'| = |f| \ \mu$-a.e. (since $\big||f| - |f'|\big| \le |f - f'|$), and so $\int |f'| \, d\mu = \int |f| \, d\mu$ (Problem 4.4(c)). Next observe that such a function $\| \ \|_1 \colon L^1 \to \mathbb{R}$ satisfies all the axioms (i), (ii), (iii), and (iv) of Definition 5.1. In fact, take arbitrary classes $[f]$ and $[g]$ in $L^1(X, \mathcal{X}, \mu)$, and an arbitrary scalar $\gamma \in \mathbb{R}$ so that, by Proposition 5.4,

$$\|[f]\|_1 = \|f\| \ge 0,$$

$$\|[f]\|_1 = 0 \iff \|f\| = 0 \iff f = 0 \ \mu\text{-a.e.} \iff [f] = [0],$$

$$\|\gamma[f]\|_1 = \|[\gamma f]\|_1 = \|\gamma f\| = |\gamma| \, \|f\| = |\gamma| \, \|[f]\|_1,$$

$$\|[f] + [g]\|_1 = \|[f + g]\|_1 = \|f + g\| \le \|f\| + \|g\| = \|[f]\|_1 + \|[g]\|_1. \quad \square$$

5.2 Spaces L^p and L^∞

Consider a measure space $(X, \mathcal{X}, \mu)$. In this section we extend the construction of L^1 from the previous section. First we define linear spaces L^p for each $p \geq 1$ and L^∞, and then we equip them with norms $\| \ \|_p$ and $\| \ \|_\infty$. In the next section we show that these normed spaces are actually Banach spaces.

We begin with the spaces L^p. Take an arbitrary real number $p \geq 1$. A *real-valued* function f in $\mathcal{M}(X, \mathcal{X})$ is *p-integrable* if $f^p \in \mathcal{L}(X, \mathcal{X}, \mu)$ or, equivalently, if $\int |f|^p \, d\mu < \infty$ (cf. Lemma 4.4). Along the same line used to construct L^1 (which is the particular case of L^p for $p = 1$), set

$$\begin{aligned} L^p &= L^p(\mu) = L^p(X, \mu) = L^p(X, \mathcal{X}, \mu) \\ &= \big\{[f] \subseteq \mathcal{M}(X, \mathcal{X})\colon \ f^p \in \mathcal{L}(X, \mathcal{X}, \mu)\big\}, \end{aligned}$$

the collection of all equivalence classes of p-integrable functions. In other words, $L^p(X, \mathcal{X}, \mu)$ is the collection of all equivalence classes $[f]$ of real-valued functions f in $\mathcal{M}(X, \mathcal{X})$ for which $\int |f|^p \, d\mu < \infty$ for every representative f of $[f]$. Thus consider the function $\| \ \|_p \colon L^p \to \mathbb{R}$ defined by

$$\|[f]\|_p = \left(\int |f|^p \, d\mu \right)^{\frac{1}{p}} \quad \text{for every} \quad [f] \in L^p(X, \mathcal{X}, \mu),$$

where f is any representative of the equivalence class $[f]$ in $L^p(X, \mathcal{X}, \mu)$.

Next we examine the space L^∞. An extended real-valued function f in $\mathcal{M}(X, \mathcal{X})$ is *essentially bounded* if it is bounded almost everywhere. Roughly speaking, if $\sup_{x \in X} |f(x)| < \infty$ μ-a.e. on X. Precisely, this means that there is a real number $\beta > 0$ such that $|f| \leq \beta$ μ-almost everywhere. If an extended real-valued function f in $\mathcal{M}(X, \mathcal{X})$ is essentially bounded, then set

$$\operatorname{ess\,sup} |f| = \inf \big\{\beta \geq 0\colon \ |f| \leq \beta \ \ \mu\text{-a.e.}\big\} = \inf_{N \in \mathcal{N}} \sup_{x \in X \setminus N} |f(x)| \ \in \mathbb{R},$$

where $\mathcal{N} = \{N \in \mathcal{X}\colon \mu(N) = 0\}$, the set of sets of measure zero. Thus set

$$\begin{aligned} L^\infty &= L^\infty(\mu) = L^\infty(X, \mu) = L^\infty(X, \mathcal{X}, \mu) \\ &= \big\{[f] \subseteq \mathcal{M}(X, \mathcal{X})\colon \ \sup_{x \in X} |f(x)| < \infty \ \ \mu\text{-a.e.}\big\}, \end{aligned}$$

the collection of all equivalence classes of essentially bounded extended real-valued functions. In other words, $L^\infty(X, \mathcal{X}, \mu)$ is the collection of all equivalence classes of extended real-valued functions f in $\mathcal{M}(X, \mathcal{X})$ for which $\operatorname{ess\,sup} |f| < \infty$, whose value does not depend on the representative f of the equivalence class $[f]$. Thus consider the function $\| \ \|_\infty \colon L^\infty \to \mathbb{R}$ defined by

$$\|[f]\|_\infty = \operatorname{ess\,sup} |f| \quad \text{for every} \quad [f] \in L^\infty(X, \mathcal{X}, \mu),$$

where f is any representative of the equivalence class $[f]$ in $L^\infty(X, \mathcal{X}, \mu)$.

We have seen in the previous section that since $\mathcal{L}(X, \mathcal{X}, \mu)$ is a linear space, then $L^1(X, \mathcal{X}, \mu)$ was made into a linear space, with scalar multiplication and vector addition of equivalence classes defined as before (viz., $\gamma[f] = [\gamma f]$ and $[f] + [g] = [f + g]$ for every $[f]$, $[g]$ and scalar γ). This extends immediately to $L^p(X, \mathcal{X}, \mu)$, so that $L^p(X, \mathcal{X}, \mu)$ is made into a linear space as well. Similarly, $L^\infty(X, \mathcal{X}, \mu)$ is also made into a linear space under the same definition of scalar multiplication and vector addition.

Note that the elements of $L^p(X, \mathcal{X}, \mu)$ and $L^\infty(X, \mathcal{X}, \mu)$ are equivalence classes of functions (and not functions themselves). If f is any function of an equivalence class $[f]$, then it is usual and convenient to write f for $[f]$. Since the common usage is simpler, we follow it, and refer to a "function f" in $L^p(X, \mathcal{X}, \mu)$ or in $L^\infty(X, \mathcal{X}, \mu)$ instead of "an equivalence class $[f]$ that contains f". Thus we write $\|f\|_p$ and $\|f\|_\infty$ instead of $\|[f]\|_p$ and $\|[f]\|_\infty$.

We will show that $\| \ \|_p$ and $\| \ \|_\infty$ are norms on the linear spaces L^p and L^∞, respectively. But first we need the fundamental inequalities stated in Propositions 5.6 and 5.7 below. These are the Hölder and Minkowski inequalities. We give detailed hints for their proofs in Problems 5.1 and 5.3. The reader is invited to follow those hints in order to prove them. (References are also given in the Suggested Readings at the end of this chapter.)

Take any real number $p > 1$. Let $q = \frac{p}{p-1} > 1$ be the unique solution to the equation $p + q = p\,q$ or, equivalently, the unique solution to the equation

$$\frac{1}{p} + \frac{1}{q} = 1.$$

In this case, p and q are called *Hölder conjugates* (of each other).

Proposition 5.6. (Hölder inequality). *If $p, q > 1$ are Hölder conjugates, and if $f \in L^p$ and $g \in L^q$, then $fg \in L^1$ and*

(a) $$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

If $f \in L^1$ and $g \in L^\infty$, then $fg \in L^1$ and

(b) $$\|fg\|_1 \leq \|f\|_1 \|g\|_\infty.$$

Remark 5.B (Special case II): If $p = q = 2$, the Hölder inequality leads to the Schwarz inequality (cf. remark after Definition 5.1). In this particular case, where $\mathcal{L} = L^2$, the inner product on the concrete linear space L^2 is

$$\langle f\,;\, g\rangle = \int f\overline{g}\,d\mu = \int fg\,d\mu$$

for every $f, g \in L^2$ (interpreted in the sense of equivalence classes — recall that, so far, we are dealing with real-valued functions only). It is readily verified that, as defined above, the real-valued functional $\langle\ ;\ \rangle\colon L^2 \times L^2 \to \mathbb{R}$ satisfies all the inner product axioms, and hence the real normed space L^2 is a real inner product space. The Schwarz inequality, viz., for $f, g \in L^2$

$$|\langle f\,;\,g\rangle| \le \|f\|_2 \|g\|_2,$$

can also be directly verified by the Hölder inequality for $p = q = 2$, which ensures that $fg \in L^1$, and the Schwarz inequality comes from Lemma 4.4(b):

$$|\langle f\,;\,g\rangle| = \left|\int fg\,d\mu\right| \le \int |fg|\,d\mu = \|fg\|_1 \le \|f\|_2 \|g\|_2.$$

For the special case where $\mu(X) \le 1$ (in particular, if μ is a probability measure, that is, if $\mu(X) = 1$), the unit function 1 (i.e., $1(x) = 1$ for $x \in X$) lies in L^2 with $\|1\|_2 \le 1$ (and with unit norm, $\|1\|_2 = 1$, if $\mu(X) = 1$). Then, by the above inequalities, $|\langle f\,;1\rangle| \le \|f\|_1 \le \|f\|_2$, and so

$$\left(\int f\,d\mu\right)^2 \le \left(\int |f|\,d\mu\right)^2 \le \int f^2\,d\mu \quad \text{whenever} \quad \mu(X) \le 1.$$

We will come back to the inequality $\left(\int f\,d\mu\right)^2 \le \int f^2\,d\mu$ in Problem 5.20.

The hints to Problems 5.1 and 5.3 show how to prove the Hölder inequality and how the Hölder inequality is used to prove the Minkowski inequality stated below, which in fact is the triangle inequality in the spaces L^p and L^∞.

Proposition 5.7. (Minkowski inequality). *Take any real number $p \ge 1$. If $f, g \in L^p$, then $f + g \in L^p$ and*

(a) $$\|f + g\|_p \le \|f\|_p + \|g\|_p.$$

If $f, g \in L^\infty$, then $f + g \in L^\infty$ and

(b) $$\|f + g\|_\infty \le \|f\|_\infty + \|g\|_\infty.$$

Lemma 5.8. *The functions $\|\ \|_p$ and $\|\ \|_\infty$ are norms on L^p and L^∞.*

Proof. Consider the real linear spaces L^p for each $p \ge 1$ and L^∞. By Proposition 5.5, $\|\ \|_1$ is a norm on L^1. The same argument shows that $\|\ \|_p$ is a norm on L^p for each $p > 1$, where the triangle inequality in the linear space L^p is precisely the Minkowski inequality of Proposition 5.7(a). To verify that $\|\ \|_\infty$ is a norm on L^∞, note that properties (i) and (iii) of Definition 5.1 are trivially verified and (ii) follows since $\operatorname{ess\,sup}|f| = 0$ means $|f| = 0$ μ-almost everywhere. As for (iv), the Minkowski inequality of Proposition 5.7(b) is, again, the triangle inequality in the linear space L^∞. □

5.3 The Riesz–Fischer Completeness Theorem

A central result in integration theory is the Riesz Theorem, also called Riesz–Fischer Theorem, or Completeness Theorem (Theorem 5.9 below). It says that the linear spaces L^p and L^∞ equipped with the norms $\| \ \|_p$ and $\| \ \|_\infty$ are *complete* normed spaces. In other words (cf. paragraph before Proposition 5.4 and remark after Proposition 5.6), the next theorem ensures that

L^p and L^∞ are Banach spaces, and therefore, L^2 is a Hilbert space.

Theorem 5.9. *$(L^p, \| \ \|_p)$ for each $p \geq 1$ and $(L^\infty, \| \ \|_\infty)$ are Banach spaces.*

Proof. Take any real number $p \geq 1$. According to Lemma 5.8, consider the normed spaces $(L^p(X, \mathcal{X}, \mu), \| \ \|_p)$ and $(L^\infty(X, \mathcal{X}, \mu), || \ \|_\infty)$. The proof is split into two parts. Part (a) shows that $(L^p, \| \ \|_p)$ is complete, and part (b) shows that $(L^\infty, || \ \|_\infty)$ is also complete.

(a) Let $\{f_n\}$ be an arbitrary Cauchy sequence in L^p (Definition 5.2). Thus for any integer $k \geq 1$ there is another integer $n_k \geq 1$ for which

$$\|f_m - f_n\|_p < \left(\tfrac{1}{2}\right)^k \quad \text{whenever} \quad m, n \geq n_k.$$

Then there exists a subsequence $\{f_{n_k}\}$ of $\{f_n\}$ such that

$$\|f_{n_{k+1}} - f_{n_k}\|_p < \left(\tfrac{1}{2}\right)^k$$

for each $k \geq 1$. Let $g\colon X \to \overline{\mathbb{R}}$ be a function defined for each $x \in X$ by

$$g(x) = |f_{n_1}(x)| + \sum_{k=1}^{\infty} \left|f_{n_{k+1}}(x) - f_{n_k}(x)\right|.$$

First note that g is a well-defined extended real-valued nonnegative $\mathcal{X}$-measurable function on X (i.e., a function in $\mathcal{M}(X, \mathcal{X})^+$ — cf. Proposition 1.8). Thus the Monotone Convergence Theorem as in Corollary 3.10 ensures that

$$\int g^p \, d\mu = \lim_n \int \left(|f_{n_1}| + \sum_{k=1}^{n} \left|f_{n_{k+1}} - f_{n_k}\right|\right)^p d\mu.$$

Recall that each $|f_{n_k}|$ is in L^p, and so $|f_{n_1}| + \sum_{k=1}^{n} |f_{n_{k+1}} - f_{n_k}|$ is in the linear space L^p for each integer $n \geq 1$. The Minkowski inequality of Proposition 5.7 (i.e., the triangle inequality) plus a trivial induction ensures that $\|\sum_{k=1}^{n} g_k\|_p \leq \sum_{k=1}^{n} \|g_k\|_p$ for every $n \geq 1$ if $\{g_k\}$ is a sequence in L^p. Then

$$\left(\int \left(|f_{n_1}| + \sum_{k=1}^{n} \left|f_{n_{k+1}} - f_{n_k}\right|\right)^p d\mu\right)^{\frac{1}{p}} = \left\| |f_{n_1}| + \sum_{k=1}^{n} \left|f_{n_{k+1}} - f_{n_k}\right| \right\|_p$$

$$< \|f_{n_1}\|_p + \sum_{k=1}^{n} \left(\tfrac{1}{2}\right)^k < \|f_{n_1}\|_p + 1$$

for every integer $n \geq 1$. The preceding two expressions ensure that $\int g^p \, d\mu < (\|f_{n_1}\|_p + 1)^p < \infty$, and so the set $E = \{x \in X\colon\ g(x) < \infty\}$ lies in $\mathcal{X}$ with $\mu(X \backslash E) = 0$ by Problem 3.9(b). This implies that $\sum_{k=1}^{\infty}(f_{n_{k+1}}(x) - f_{n_k}(x))$ converges for every $x \in E$ (since it converges absolutely). Thus set

$$f(x) = \begin{cases} f_{n_1}(x) + \sum_{k=1}^{\infty} \big(f_{n_{k+1}}(x) - f_{n_k}(x)\big) = \lim_k f_{n_{k+1}}(x), & x \in E, \\ 0, & x \notin E, \end{cases}$$

defining a real-valued function $f \in \mathcal{M}(X, \mathcal{X})$ in L^p. Indeed, $|f| \leq g$, and so

$$\int |f|^p \, d\mu \leq \int g^p \, d\mu < \big(\|f_{n_1}\|_p + 1\big)^p < \infty$$

(cf. Problem 3.3(b)). Hence $f \in L^p$. Observe that

(i) $\{f_{n_k}\}$ converges almost everywhere to f (i.e., $|f_{n_k} - f| \to 0$ μ-a.e.),

(ii) $|f_{n_k}|^p \leq g^p$ for all k, where g^p is a nonnegative function in $\mathcal{L}(X, \mathcal{X}, \mu)$.

In fact, $f_{n_k}(x) \to f(x)$ for every $x \in E$ and, for all k,

$$|f_{n_k}| \leq \Big|\sum_{j=k}^{\infty} \big(|f_{n_j}| - |f_{n_{j+1}}|\big)\Big| \leq \sum_{j=k}^{\infty} |f_{n_{j+1}} - f_{n_j}| \leq \sum_{j=1}^{\infty} |f_{n_{j+1}} - f_{n_j}| \leq g.$$

Also, since $|f_{n_k} - f|^p \leq (|f_{n_k}| + |f|)^p \leq (g + |f|)^p$ by (ii), and $|f| \leq g$,

(iii) $|f_{n_k} - f|^p \leq (2g)^p$ for all k, where $(2g)^p$ lies in $\mathcal{L}(X, \mathcal{X}, \mu)$.

Therefore, since $|f_{n_k} - f|^p \to 0$ μ-a.e. by (i), and according to (iii), it follows by the Dominated Convergence Theorem (Theorem 4.7) that

$$\|f_{n_k} - f\|_p^p = \int |f_{n_k} - f|^p \, d\mu \to 0 \quad \text{as} \quad k \to \infty,$$

and so the subsequence $\{f_{n_k}\}$ of $\{f_n\}$ converges in $(L^p, \|\ \|_p)$ to $f \in L^p$. Use Proposition 5.3(c) to infer that the arbitrary Cauchy sequence $\{f_n\}$ converges in $(L^p, \|\ \|_p)$, and so the normed space $(L^p, \|\ \|_p)$ is complete.

(b) Let $\{f_n\}$ be any sequence of functions in L^∞. Recall that a countable collection of sets of measure zero is again a set of measure zero (cf. Problem 2.8(c)). Thus, there exists a set $N \in \mathcal{X}$ with $\mu(N) = 0$ such that

$$|f_n(x)| \leq \|f_n\|_\infty \quad \text{and} \quad |f_m(x) - f_n(x)| \leq \|f_m - f_n\|_\infty$$

for all $x \in X \backslash N$, for every $m, n \geq 1$. If $\{f_n\}$ is a Cauchy sequence, then for each $\varepsilon > 0$ there exists a positive integer n_ε such that

$$\|f_m - f_n\|_\infty < \varepsilon \quad \text{whenever} \quad n, m \geq n_\varepsilon,$$

and therefore

$$\sup_{\substack{x \in X\setminus N \\ m,n \geq n_\varepsilon}} \big|f_m(x) - f_n(x)\big| < \varepsilon.$$

Thus the scalar sequence $\{f_n(x)\}$ is a real-valued Cauchy sequence for every x in $X\setminus N$. Since $\mathbb{R}$ (with its usual norm $|\ |$) is a complete normed space, it follows that $\{f_n(x)\}$ converges in $\mathbb{R}$ for every $x \in X\setminus N$. Then set

$$f(x) = \begin{cases} \lim_n f_n(x), & x \in X\setminus N, \\ 0, & x \in N. \end{cases}$$

This defines a real-valued function f in $\mathcal{M}(X, \mathcal{X})$ which, in fact, lies in L^∞. Actually, take x, y arbitrary in $X\setminus N$, and an arbitrary $n \geq n_\varepsilon$. Note that

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)|.$$

Since the function $|\ |\colon \mathbb{R} \to \mathbb{R}$ is continuous, it follows that

$$|f(x) - f_n(x)| = \big|\lim_m f_m(x) - f_n(x)\big| = \lim_m |f_m(x) - f_n(x)| \leq \varepsilon,$$

and since each f_n lies in L^∞, it also follows that

$$|f_n(x) - f_n(y)| \leq |f_n(x)| + |f_n(y)| \leq 2\|f_n\|_\infty.$$

Hence

$$|f(x)| \leq |f(y)| + |f(x) - f(y)| \leq |f(y)| + 2(\varepsilon + \|f_{n_\varepsilon}\|_\infty),$$

which implies that $f \in L^\infty$. Moreover, for all $n \geq n_\varepsilon$,

$$\|f - f_n\|_\infty = \sup_{x\in X\setminus N} |f(x) - f_n(x)| \leq \varepsilon.$$

Outcome: the arbitrary Cauchy sequence $\{f_n\}$ converges in $(L^\infty, \|\ \|_\infty)$ to $f \in L^\infty$, and so the normed space $(L^\infty, \|\ \|_\infty)$ is complete. □

Remark 5.C (Complex-valued functions): Just before Definition 4.1, we defined $\mathcal{L} = \mathcal{L}(X, \mathcal{X}, \mu)$ as the collection of all *real*-valued $\mathcal{X}$-measurable integrable functions on X. This was done such that both the positive and negative parts of those real-valued functions have a finite integral. And their integral was defined in Definition 4.1 as the sum of the integrals of their positive and negative parts. In Problem 4.7, complex-valued measurable functions have been defined together with their integrals: the sum of the integrals of their real-valued real and imaginary parts. In light of Problem 4.7, it is not difficult to verify that the integral theory for real-valued functions developed in Chapter 4 can be naturally adapted to complex-valued functions. In this case, we may consider the collection of all *complex*-valued

$\mathcal{X}$-measurable integrable functions on X, denoted again (with the same notation) by $\mathcal{L} = \mathcal{L}(X, \mathcal{X}, \mu)$, as the collection of all functions

$$f\colon X \to \mathbb{C} \quad \text{such that} \quad |f| \in \mathcal{M}(X, \mathcal{X}) \quad \text{and} \quad \int |f|\, d\mu < \infty,$$

as in Problem 4.7(a). This remains a linear space, but is now a *complex* linear space. Therefore, the same arguments as those applied in the theory developed in this Chapter 5 still hold (almost *ipsis litteris*) for complex-valued integrable functions, including Hölder and Minkowski inequalities and Riesz–Fischer Completeness Theorem. Summing up, the same notation

> $(L^p, \|\ \|_p)$ for $p \geq 1$ and $(L^\infty, \|\ \|_\infty)$, with $\|f\|_p^p = \int |f|^p\, d\mu$ and $\|f\|_\infty = \operatorname{ess\,sup} |f|$, also stand for the Banach spaces of all (equivalence classes of) *complex-valued* functions in $\mathcal{M}(X, \mathcal{X})$ that are p-integrable ($\int |f|^p\, d\mu < \infty$) or essentially bounded ($\operatorname{ess\,sup} |f| < \infty$). In particular, $(L^2, \langle\ ;\ \rangle) = (L^2, \|\ \|_2)$, with $\langle f\,;\, g\rangle = \int f\overline{g}\, d\mu$, also stands for the Hilbert space of all (equivalence classes of) *complex-valued*, square-integrable functions on X.

5.4 Problems

Problem 5.1. Prove the *Hölder inequality*. Let $(X, \mathcal{X}, \mu)$ be a measure space and take a pair of real-valued measurable functions f and g in $\mathcal{M}(X, \mathcal{X})$. If $f^p \in \mathcal{L}(X, \mathcal{X}, \mu)$ and $g^q \in \mathcal{L}(X, \mathcal{X}, \mu)$, where $p > 1$ and $q > 1$ are Hölder conjugates, then show that $fg \in \mathcal{L}(X, \mathcal{X}, \mu)$ and

$$\text{(a)} \qquad \int |fg|\, d\mu \leq \left(\int |f|^p\, d\mu \right)^{\frac{1}{p}} \left(\int |g|^q\, d\mu \right)^{\frac{1}{q}}.$$

Hint: (i) First prove the *Young inequality*, which says that if α and β are positive real numbers, and if p and q are Hölder conjugates, then

$$\alpha\beta \leq \frac{\alpha^p}{p} + \frac{\beta^q}{q}.$$

(Hint on the hint: Suppose $\alpha \leq \beta$. Take the invertible function $x \mapsto x^{p-1}$ of $\mathbb{R}$ onto $\mathbb{R}$ whose inverse is $y \mapsto y^{1/p-1} = y^{q-1}$, since $\frac{1}{p} + \frac{1}{q} = 1$, where both functions are increasing. Consider the area of the rectangle $[0, \alpha] \times [0, \beta]$ in the xy-plane. Verify that $\alpha\beta \leq \int_0^\alpha x^{p-1} dx + \int_0^\beta y^{q-1} dy = \frac{\alpha^p}{p} + \frac{\beta^q}{q}$. Draw a picture.)

(ii) Now if f and g are nonzero, then set $\alpha = \frac{|f|}{\|f\|_p}$ and $\beta = \frac{|g|}{\|g\|_q}$, where $\|f\|_p = (\int |f|^p\, d\mu)^{1/p}$ and $\|g\|_q = (\int |g|^q\, d\mu)^{1/q}$, and integrate.

If $f \in \mathcal{L}(X, \mathcal{X}, \mu)$ and g is essentially bounded, then $fg \in \mathcal{L}(X, \mathcal{X}, \mu)$ and

(b) $$\int |fg|\,d\mu \;\le\; \operatorname{ess\,sup}|g| \int |f|\,d\mu.$$

Hint: Proposition 4.2(c) and Problem 4.4 (or Problem 3.8(c)).

Problem 5.2. Consider the inequality in Problem 5.1(b). If f^p lies in $\mathcal{L}(X,\mathcal{X},\mu)$ for some $p \ge 1$ (or if f is essentially bounded) and g is essentially bounded, then $(fg)^p$ lies in $\mathcal{L}(X,\mathcal{X},\mu)$ (or fg is essentially bounded) and

$$\int |fg|^p\,d\mu \le \operatorname{ess\,sup}|g|^p \int |f|^p\,d\mu \quad \big(\text{or} \quad \operatorname{ess\,sup}|fg| \le \operatorname{ess\,sup}|f| \operatorname{ess\,sup}|g|\,\big).$$

Problem 5.3. Prove the *Minkowski inequality*. Let $(X,\mathcal{X},\mu)$ be a measure space and take p-integrable functions f and g (i.e., real-valued functions in $\mathcal{M}(X,\mathcal{X})$ such that $\int |f|^p\,d\mu < \infty$ and $\int |g|^p\,d\mu < \infty$) for an arbitrary real number $p \ge 1$. Show that $f+g$ is p-integrable (i.e., $\int |f+g|^p\,d\mu < \infty$) and

(a) $$\left(\int |f+g|^p\,d\mu\right)^{\frac{1}{p}} \le \left(\int |f|^p\,d\mu\right)^{\frac{1}{p}} + \left(\int |g|^p\,d\mu\right)^{\frac{1}{p}}.$$

Hint: The special case of $p=1$ was proved in Proposition 5.4. To prove the case of $p>1$ proceed as follows. Take any α and β in $\mathbb{R}$. Since $|\alpha+\beta|^p \le 2^p(|\alpha|^p+|\beta|^p)$, show that $\int |f+g|^p\,d\mu < \infty$. Since $|\alpha+\beta|^p \le (|\alpha|+|\beta|)^p = (|\alpha|+|\beta|)^{p-1}|\alpha| + (|\alpha|+|\beta|)^{p-1}|\beta|$, and recalling that $(p-1)q = p$ if q is the Hölder conjugate of p, use Problem 5.1(a) to prove the claimed inequality.

Moreover, show that if f and g are essentially bounded, then

(b) $$\operatorname{ess\,sup}|f+g| \;\le\; \operatorname{ess\,sup}|f| + \operatorname{ess\,sup}|g|.$$

Problem 5.4. The *Littlewood second principle* says: "every function is nearly simple". That is, if $f \in L^p(X,\mathcal{X},\mu)$ for some $p \ge 1$, then for every $\varepsilon > 0$ there is a measurable simple function φ_ε such that $\|f-\varphi_\varepsilon\|_p < \varepsilon$. Prove it.

Hint: If $0 \le |f| \in L^p$, then Problem 1.6 ensures the existence of an increasing sequence $\{\varphi_n\}$ of measurable simple functions converging pointwise to $|f|$. So $(|f|-\varphi_n)^p \to 0$ pointwise and $0 \le (|f|-\varphi_n)^p \le |f|^p$. Use the Dominated Convergence Theorem (Theorem 4.7) to infer that $\|f-\varphi_n\|_p \to 0$.

Problem 5.5. The notations $x = \{x_n\}$ or $x = \{x_n\}_{n\ge 1}$ for a real sequence are different representations of a real-valued function on $\mathbb{N}$, say $x\colon \mathbb{N} \to \mathbb{R}$, for which we write x_n instead of $x(n)$ when indicating its value in $\mathbb{R}$ for each positive integer $n \in \mathbb{N}$. Now consider the measure space $(\mathbb{N}, \wp(\mathbb{N}), \mu)$, where $\mu\colon \wp(\mathbb{N}) \to \overline{\mathbb{R}}$ is the counting measure of Example 2B. Verify that all functions from $\mathbb{N}$ to $\mathbb{R}$ are $\wp(\mathbb{N})$-measurable. In other words, verify that

$$\mathcal{M}(\mathbb{N}, \wp(\mathbb{N})) = \mathbb{R}^{\mathbb{N}},$$

where $\mathbb{R}^{\mathbb{N}}$ is the linear space of all real-valued functions on $\mathbb{N}$, which is the linear space of all real sequences $\{x_n\}$. Next, in light of Problem 3.4, verify that the integral of a function $x\colon \mathbb{N} \to \mathbb{R}$, that is, of a real sequence $\{x_n\}$, with respect to the counting measure μ, is

$$\int x\,d\mu = \int x(n)\,d\mu(n) = \sum_{n=1}^{\infty} x_n.$$

Thus consider the linear space $\mathcal{L}(\mathbb{N}, \wp(\mathbb{N}), \mu)$ as in Lemma 4.5. Suppose

$$\sum_{k=1}^{\infty} |x_k|^p < \infty \quad \text{or} \quad \sup_{k\geq 1} |x_k| < \infty,$$

and set

$$\ell^p = L^p(\mathbb{N}, \wp(\mathbb{N}), \mu) \quad \text{or} \quad \ell^\infty = L^\infty(\mathbb{N}, \wp(\mathbb{N}), \mu),$$

respectively, with μ being the counting measure on $\wp(\mathbb{N})$. Verify that each equivalence class in $L^p\big(\mathbb{N}, \wp(\mathbb{N}), \mu\big)$ or in $L^\infty(\mathbb{N}, \wp(\mathbb{N}), \mu)$ contains just one element. Therefore, ℓ^p stands for the linear space of all scalar-valued *p-summable sequences*, for each real number $p \geq 1$, and ℓ^∞ stands for the linear space of all scalar-valued *bounded sequences*.

(Here the term scalar-valued sequence means a real-valued or a complex-valued sequence satisfying either $\sum_{k=1}^{\infty} |x_k|^p < \infty$ or $\sup_{k\geq 1} |x_k| < \infty$.)

Problem 5.6. Consider the setup of Problem 5.5. Take the linear spaces ℓ^p for $p \geq 1$ and ℓ^∞. According to Lemma 5.8, verify that

$$\|x\|_p = \Big(\sum_{k=1}^{\infty} |x_k|^p\Big)^{\frac{1}{p}} \quad \text{for every sequence } x = \{x_k\} \in \ell^p,$$

$$\|x\|_\infty = \sup_{k\geq 1} |x_k| \quad \text{for every sequence } x = \{x_k\} \in \ell^\infty,$$

define norms $\|\ \|_p$ and $\|\ \|_\infty$ on ℓ^p and on ℓ^∞. Verify that, in the case of the *sup-norm* $\|x\|_\infty$, we get $\sup_{k\geq 1} |x_k| = \operatorname{ess\,sup}_{k\geq 1} |x_k|$. Then apply Theorem 5.9 to show that $(\ell^p, \|\ \|_p)$ for each $p \geq 1$ and $(\ell^\infty, \|\ \|_\infty)$ are Banach spaces.

Problem 5.7. In particular, the Hölder and Minkowski inequalities (Propositions 5.6 and 5.7) hold for sequences in ℓ^p and ℓ^∞ equipped with the norms of Problem 5.6. Now prove the *Jensen inequality* for sequences, which says that if p and q are real numbers such that $0 < p < q$, then

$$\text{(a)} \qquad \Big(\sum_{k=1}^{\infty} |x_k|^q\Big)^{\frac{1}{q}} \leq \Big(\sum_{k=1}^{\infty} |x_k|^p\Big)^{\frac{1}{p}}$$

for every scalar-valued sequence $x = \{x_k\}$ such that $\sum_{k=1}^{\infty} |x_k|^p < \infty$.

Hint: Since $(\alpha_k \,/\, \sum_{k=1}^{\infty} \alpha_k) \le 1$, first show that, for each real $r \ge 1$,

$$\sum_{k=1}^{\infty} \alpha_k^r \le \Big(\sum_{k=1}^{\infty} \alpha_k \Big)^r$$

if the sequence of *nonnegative* real numbers $\{\alpha_k\}$ is such that $\sum_{k=1}^{\infty} \alpha_k < \infty$. Apply the above inequality for $\sum_{k=1}^{\infty} |x_k|^q = \sum_{k=1}^{\infty} |x_k|^{pr}$ with $r = \frac{q}{p} > 1$.

Moreover, for real numbers q and p show that

(b) $$1 < p < q \quad \text{implies} \quad \ell^1 \subset \ell^p \subset \ell^q \subset \ell^\infty.$$

Hint: Take $\{\frac{1}{k}\} \in \ell^p \backslash \ell^1$ for $p > 1$ to verify that these are proper inclusions.

Problem 5.8. Let p, q, r be real numbers such that $1 \le r < \min\{p, q\}$ and

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{r}.$$

Prove the *generalized Hölder inequality*, which reads as follows. If $f \in L^p$, and $g \in L^q$, then $fg \in L^r$, and

(a) $$\|fg\|_r \le \|f\|_p \|g\|_q.$$

Hint: Show that $\frac{p}{r}$ and $\frac{q}{r}$ are Hölder conjugates.

Also, if $(X, \mathcal{X}, \mu)$ is a *finite* measure space (i.e., $\mu(X) < \infty$), then prove that

(b) $$1 < r < p \quad \text{implies} \quad L^\infty \subset L^p \subset L^r \subset L^1.$$

Hint: $\|f\|_r \le \|f\|_p \, \mu(X)^{\frac{p-r}{pr}}$ by the generalized Hölder inequality.

Problem 5.9. Consider an arbitrary measure space $(X, \mathcal{X}, \mu)$. Prove that the following assertions are pairwise equivalent.

(a) $\mu(X) < \infty$.

(b) $L^\infty \subseteq L^p$ for every $p \ge 1$.

(c) $L^\infty \subseteq L^p$ for some $p \ge 1$.

Problem 5.10. Prove that if $(X, \mathcal{X}, \mu)$ is a *finite* measure space, then

$$\lim_{p \to \infty} \|f\|_p = \|f\|_\infty \quad \text{for every} \quad f \in L^\infty.$$

Hint: Problem 5.9. $|f|^p = |f|^{p-1}|f|$ implies $\int |f|^p \, d\mu \le \|f\|_\infty^{p-1} \int |f| \, d\mu$, and $\|f\|_p \le \|f\|_\infty^{1-(1/p)} \|f\|_1^{1/p} = \|f\|_\infty (\|f\|_1 / \|f\|_\infty)^{1/p}$. So $\limsup \|f\|_p \le \|f\|_\infty$.

For γ in $(0, \|f\|_\infty)$ the set $E = \{x \in X : \gamma < |f(x)|\}$ is such that $\mu(E)^{1/p}\,\gamma \le \big(\int_E |f|^p\,d\mu\big)^{1/p} \le \|f\|_p$. So $\gamma \le \liminf \|f\|_p$ and $\|f\|_\infty = \sup \gamma \le \liminf \|f\|_p$.

Problem 5.11. Let $(X, \mathcal{X}, \mu)$ be an arbitrary measure space. Prove that

$$L^r \cap L^q \subseteq L^p \quad \text{whenever} \quad 1 \le r \le p \le q.$$

Hint: Set $F_1 = \{x \in X : |f(x)| < 1\}$ and $E_1 = X \backslash F_1 = \{x \in X : |f(x)| \ge 1\}$ in $\mathcal{X}$ (Proposition 1.6). Show: $\int_{F_1} |f|^p\,d\mu + \int_{E_1} |f|^p\,d\mu \le \int_{F_1} |f|^r\,d\mu + \int_{E_1} |f|^q\,d\mu$.

Problem 5.12. Let $(\mathbb{R}, \Re, \lambda)$ be the Lebesgue measure space. Take an arbitrary $\Re$-measurable set B and consider the restriction $\lambda|_{\mathcal{B}}$ of the Lebesgue measure λ to the σ-algebra $\mathcal{B} = \wp(B) \cap \Re$ of all Borel subsets of B as in Problem 2.11. Take any real number $p \ge 1$ and consider the *Lebesgue spaces*:

$$L^p(B) = L^p(B, \mathcal{B}, \lambda|_{\mathcal{B}}) \quad \text{and} \quad L^\infty(B) = L^\infty(B, \mathcal{B}, \lambda|_{\mathcal{B}}).$$

(a) For each $p \ge 1$ (or $p = \infty$), a function $f \in \mathcal{M}(\mathbb{R}, \Re)$ is *locally* L^p if $f \in L^p(B)$ for every bounded set $B \in \Re$. Verify that, if $f \in L^p(\mathbb{R}, \Re, \lambda)$ for some p, then f is locally L^p, and show that the converse fails.

Write L^pB for $L^p(B)$ and $L^\infty B$ for $L^\infty(B)$ if B is an interval of $\mathbb{R}$ (e.g., write $L^p(\alpha, \beta]$ instead of $L^p((\alpha, \beta])$) in order to simplify notation.

(b) Set $B = [0, 1]$. Show that the inclusions of Problem 5.8 are proper:

$$L^\infty[0,1] \subset L^p[0,1] \subset L^r[0,1] \subset L^1[0,1]$$

whenever $1 < r < p$, by exhibiting nonnegative functions in

$$L^p[0,1] \backslash L^\infty[0,1] \text{ for any } p \ge 1 \quad \text{and} \quad L^1[0,1] \backslash L^2[0,1],$$

(c) On the other hand, for an unbounded $\Re$-measurable set B we get the opposite. Exhibit nonnegative functions in

$$L^\infty[1,\infty) \backslash L^p[1,\infty) \text{ for any } p \ge 1 \quad \text{and} \quad L^2[1,\infty) \backslash L^1[1,\infty).$$

Hints: $\int \frac{dx}{\sqrt{x}} = 2\sqrt{x}$, $\int \frac{dx}{x} = \log(x)$, and $\int \frac{dx}{x^2} = -\frac{1}{x}$.

Problem 5.13. Consider the set $C[0, 1]$ consisting of all real-valued *continuous* functions $f : [0, 1] \to \mathbb{R}$ defined on the closed and bounded interval $[0, 1]$. First verify that $C[0, 1]$ is a linear space when vector addition and scalar multiplication are pointwise defined. Now show that the functions $\|\ \|_p : C[0,1] \to \mathbb{R}$ for each real $p \ge 1$ and $\|\ \|_\infty : C[0,1] \to \mathbb{R}$, given by

$$\|f\|_p = \left(\int_{[0,1]} |f(x)|^p\,dx\right)^{\frac{1}{p}} \quad \text{and} \quad \|f\|_\infty = \max_{x \in [0,1]} |f(x)|,$$

are well defined for every $f \in C[0,1]$, and are norms on $C[0,1]$ (cf. Minkowski inequality). Also show that it makes no difference whether the preceding integral is Riemann or Lebesgue. Next prove the following assertions.

(a) The normed space $(C[0,1], \| \ \|_p)$ is not complete for any $p \geq 1$.

Hint: Take the sequence $\{f_n\}$ of functions in $C[0,1]$ given by

$$f_n(x) = \begin{cases} 1, & x \in [0, \frac{1}{2}], \\ n+1-2nx, & x \in [\frac{1}{2}, \frac{n+1}{2n}], \\ 0, & x \in [\frac{n+1}{2n}, 1]. \end{cases}$$

Show that $\{f_n\}$ is a Cauchy sequence in $(C[0,1], \| \ \|_p)$ but does not converge in $(C[0,1], \| \ \|_p)$ to any (continuous) function in $C[0,1]$.

(b) However, $(C[0,1], \| \ \|_\infty)$ is a Banach space.

Hint: Proof of Theorem 5.9.

Problem 5.14. Take an arbitrary $p \geq 1$. Now consider the Lebesgue space $L^p[0,1]$ of all p-integrable functions on $[0,1]$ (defined in Problem 5.12). Let $R^p[0,1]$ denote the subset of $L^p[0,1]$ consisting of all (equivalence classes of) Riemann integrable functions f for which $|f|^p$ has a finite Riemann integral. $R^p[0,1]$ is a linear manifold of $L^p[0,1]$, and so it is a linear space. Equip it with the norm $\| \ \|_p$ and consider the normed spaces $(R^p[0,1], \| \ \|_p)$. Let $\{f_n\}$ be a sequence of real-valued functions on $[0,1]$ defined by

$$f_n(x) = \begin{cases} 1, & x = \frac{k}{n!} \in [0,1] \text{ for some integer } k \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Verify that each f_n lies in $R^p[0,1]$.

Hint: See Problem 4.1 and the remark that precedes Lemma 4.4.

Apply the same argument to show that the Dirichlet function f on $[0,1]$,

$$f(x) = \chi_{[0,1]\cap\mathbb{Q}}(x) = \begin{cases} 1, & x \in [0,1] \cap \mathbb{Q}, \\ 0, & x \in [0,1] \backslash \mathbb{Q}, \end{cases}$$

lies in $L^p[0,1] \backslash R^p[0,1]$. Then show that

$$f_n \to f \quad \text{pointwise.}$$

Hints: (i) $[0,1]\backslash\mathbb{Q}$ is totally disconnected of measure 1 (Problems 2.7(b), 4.1).
(ii) If $f_n(x) = 1$, then $x = \frac{k}{n!}$, so $x = \frac{k(n+1)}{(n+1)!}$ and hence $f_{n+1}(x) = 1$.

Can we infer from this problem that $(R^p[0,1], \| \ \|_p)$ is not complete?

Problem 5.15. Consider the setup of Problems 5.12 and 5.14, and take the linear manifold $(R^p[0,1], \|\ \|_p)$ of the Banach space $(L^p[0,1], \|\ \|_p)$. We show that the normed space $(R^p[0,1], \|\ \|_p)$ is not complete for any $p \geq 1$. To begin with, take the decreasing collection $\{S_n\}$ of closed subsets of $S_0 = [0,1]$ used to build up the Cantor-like set $S = \bigcap_n S_n$ of Problem 2.10. Observe that

$$\lambda(S_n) = 1 - \sum_{i=0}^{n-1} \frac{2^i}{4^{i+1}} = \frac{1}{2} + \frac{1}{2^{n+1}}$$

is the length of S_n (and so the Lebesgue measure of S_n) for each $n \geq 1$. Take the sequence $\{f_n\}$ of characteristic functions of S_n for every $n \geq 1$,

$$f_n(x) = \chi_{S_n}(x) = \begin{cases} 1, & x \in S_n, \\ 0, & x \in S_0 \backslash S_n, \end{cases}$$

and verify that each f_n belongs to $R^p[0,1]$ for every $p \geq 1$ (see Problem 4.1). Let f be the characteristic function of S; that is,

$$f(x) = \chi_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \in S_0 \backslash S. \end{cases}$$

(a) Show that $\{f_n\}$ is a Cauchy sequence in $(R^p[0,1], \|\ \|_p)$.

Hint: Verify that $\|f_m - f_n\|_p^p \leq \frac{1}{2^{m+1}}$ whenever $m \leq n$.

(b) Show that $f \in L^p[0,1]$ and $\{f_n\}$ converges in $(L^p[0,1], \|\ \|_p)$ to f.

Hint: Verify that $\|f_n - f\|_p^p = \lambda(S_n \backslash S) = \frac{1}{2^{n+1}}$ since $f_n - f = \chi_{S_n \backslash S}$.

(c) Show that $f \notin R^p[0,1]$.

Hint: Problems 2.10 and 4.1.

Use (a), (b), (c) to infer that there is a Cauchy sequence $\{f_n\}$ of functions in $(R^p[0,1], \|\ \|_p)$ that does not converge in $(R^p[0,1], \|\ \|_p)$. Thus conclude that for any $p \geq 1$ the normed space $(R^p[0,1], \|\ \|_p)$ is not a Banach space:

$$(R^p[0,1], \|\ \|_p) \text{ is an incomplete normed space.}$$

Remark 5.D (Continuity): We have promised in Chapter 3, after proving the Monotone Convergence Theorem (Theorem 3.4), to show that the integral

$$\int (\ )\, d\mu \colon L^1 \to \mathbb{R}$$

is a *continuous linear functional*. In fact, Lemma 4.5 ensures linearity (since linearity of $\mathcal{L}$ was extended to L^1 in Proposition 5.5, and so linearity of the integral on L^1 follows by the same argument as was used in Lemma 4.5). As for continuity, our first steps were taken in Problem 4.18. Indeed, recall that

a map between normed spaces (in general, between metric spaces) is continuous if and only if it preserves convergence.

Thus the integral $\int(\ )\,d\mu\colon L^1 \to \mathbb{R}$ is a continuous functional if and only if, whenever a sequence $\{f_n\}$ of functions in L^1 converges in L^1 to $f \in L^1$, then the sequence $\left\{\int f_n\,d\mu\right\}$ of real numbers converges in $\mathbb{R}$ to $\int f_n\,d\mu \in \mathbb{R}$. So the continuity of the integral functional on L^p follows from Problem 5.16(a) below. (This will be extended in Proposition 10.G of Chapter 10.)

Problem 5.16. Let $(X, \mathcal{X}, \mu)$ be a measure space and take any $p \geq 1$.

(a) If a sequence $\{f_n\}$ of functions in L^1 converges in L^1 to $f \in L^1$, then

$$\int f\,d\mu \;=\; \lim_n \int f_n\,d\mu.$$

Hint: $\left|\int f_n\,d\mu - \int f\,d\mu\right| \leq \int |f_n - f|\,d\mu = \|f_n - f\|_1$.

(b) If a sequence $\{f_n\}$ of functions in L^p converges in L^p to $f \in L^p$, then

$$\int |f|^p\,d\mu \;=\; \lim_n \int |f_n|^p\,d\mu.$$

Hint: Verify that, by the triangle inequality, $\left|\|f_n\|_p - \|f\|_p\right| \leq \|f_n - f\|_p$.

(c) If, in addition, $\mu(X) < \infty$, then

$$\int f\,d\mu \;=\; \lim_n \int f_n\,d\mu.$$

Hint: Hölder inequality with a constant function g, and the hint to (a).

Problem 5.17. Prove the Riesz Theorem, which reads as follows.

Consider a measure space $(X, \mathcal{X}, \mu)$, take an arbitrary $p \geq 1$, and suppose a sequence $\{f_n\}$ of functions in L^p converges μ-almost everywhere to $f \in L^p$. Then

$$\int |f|^p\,d\mu \;=\; \lim_n \int |f_n|^p\,d\mu$$

if and only if $\{f_n\}$ converges in L^p to f. In other words,

$$f_n \in L^p \to f \in L^p \;\; \mu\text{-a.e.} \;\;\Longrightarrow\;\; \left\{\, \|f_n\|_p \to \|f\|_p \;\Leftrightarrow\; \|f_n - f\|_p \to 0 \,\right\}.$$

(Recall: $\|f_n - f\| \to 0 \Longrightarrow \|f_n\| \to \|f\|$ for every norm on every normed space because $|\,\|f_n\| - \|f\|\,| \leq \|f_n - f\|$; what is new here is the converse.)

Hint: $0 \leq h_n = 2^p(|f_n|^p + |f|^p) - |f_n - f|^p \in L^p \to 2^{p+1}|f|^p \in L^p$ a.e. (the hint to Problem 5.3 ensures $0 \leq h_n$, $f_n \to f$ a.e. implies a.e. convergence).

If $\|f_n\|_p \to \|f\|_p$ then, by Lemma 3.9, $2^{p+1}\int|f|^p\,d\mu \le \liminf_n \int h_n\,d\mu = 2^{p+1}\int|f|^p\,d\mu - \limsup_n \int|f_n - f|^p\,d\mu$. This implies $\lim_n \int|f_n - f|^p\,d\mu = 0$.

See Problem 4.14. Now give another solution to Problem 4.15; that is, under the assumption of the Dominated Convergence Theorem, $f_n \to f$ in L^1.

Note: Recall that a subset A of a metric space X is dense in X if $\overline{A} = X$, with $\overline{A}$ denoting the closure of A. Equivalently, A is dense in a metric space X if every point of X is the limit of an A-valued sequence.

Warning: The notation $\overline{\mathbb{R}}$ for $\mathbb{R} \cup \{-\infty\} \cup \{+\infty\}$, the extended real number system introduced in Section 1.3, may be tricky; although it is the same notation as that for closure, $\overline{\mathbb{R}}$ does not stand for the closure of anything.

Problem 5.18. Consider the setup of Problem 5.5, the results of Problem 5.6, and take the Banach spaces $(\ell^p, \|\ \|_p)$ for each real $p \ge 1$. Let ℓ^0 stand for the linear space of all scalar-valued (real or complex) sequences $x = \{x_k\}$ with a finite number of nonzero entries. Check that $\ell^0 \subset \ell^p$ (proper inclusion) trivially, and therefore (cf. Problem 5.7(b)),

$$1 \le p < q \quad \text{implies} \quad \ell^0 \subset \ell^p \subset \ell^q.$$

Show that, for every $1 \le p \le q$,

$$\ell^0 \text{ is dense in } (\ell^q, \|\ \|_q) \quad \text{and hence} \quad \ell^p \text{ is dense in } (\ell^q, \|\ \|_q).$$

Hint: Take $x = \{x_k\} \in \ell^q$, set $x_n = \{x_1, x_2, \ldots, x_n, 0, 0, \ldots\} \in \ell^0$, show that $\|x - x_n\|_p \to 0$, and hence $x_n \to x$ in ℓ^q.

Problem 5.19. Let $(X, \mathcal{X}, \mu)$ be a *finite* measure space and consider the inclusions in Problem 5.8(b), namely,

$$1 \le p < q \quad \text{implies} \quad L^\infty \subset L^q \subset L^p.$$

Show that, for every $1 \le p \le q$,

$$L^\infty \text{ is dense in } (L^p, \|\ \|_p) \quad \text{and hence} \quad L^q \text{ is dense in } (L^p, \|\ \|_p).$$

Hint: If μ is a finite measure, then $L^\infty \subset L^p$. Thus take any $f \in L^p \backslash L^\infty$. Set $F_n = \{x \in X : |f(x)| < n\}$ and $E_n = X \backslash F_n = \{x \in X : |f(x)| \ge n\}$ in $\mathcal{X}$. So $\mu(E_n) = \int_{E_n} d\mu = \frac{1}{n^p}\int_{E_n} n^p d\mu \le \frac{1}{n^p}\int_{E_n}|f|^p d\mu \le \frac{1}{n^p}\int_X |f|^p d\mu = \big(\|f\|_p/n\big)^p \to 0$ (independently of μ being finite since $f \in L^p$ — this can also be verified by Proposition 2.2(d): $\lim_n \mu(E_n) = \mu(\bigcap_n E_n) = \mu(\varnothing) = 0$ as $\{E_n\}$ is decreasing and $\mu(E_1) < \infty$). Now set $f_n = f\chi_{F_n}$ so that $f_n \in L^\infty$ (with $\|f\|_\infty \le n$), and conclude: $\|f - f_n\|_p^p = \int_{E_n}|f|^p d\mu$. However, as we will see in Problem 6.11, $\int_{E_n}|f|^p d\mu \to 0$. Thus $\|f - f_n\|_p^p \to 0$ and hence $f_n \to f$ in L^p.

Problem 5.20. Let $(X, \mathcal{X}, \wp)$ be a probability space and let $f\colon X \to \mathbb{R}$ be a real random variable (as in Problem 2.20). Take the expected value of f. That is, the integral of f with respect to $\wp$, as defined in Problem 4.20:

$$\mathrm{Exp}(f) = \int_X f\, d\wp = \int_{\mathbb{R}} \alpha\, d\mu_f(\alpha),$$

where the probability measure μ_f on $\Re$ is the probability distribution of f.

Note: If μ_f is such that there exists a density function p of f (Problem 3.15), then we may write $d\mu_f(\alpha) = p(\alpha)\, d\lambda(\alpha)$, equivalently $d\mu_f = p\, d\alpha$ (e.g., this holds if μ_f on $\Re$ is absolutely continuous with respect to Lebesgue measure λ on $\Re$, as we will see in Theorem 7.8 — the Radon–Nikodym Theorem).

The *second moment* of f is

$$\mathrm{Exp}(f^2) = \int_X f^2 d\wp = \int_{\mathbb{R}} \alpha^2 d\mu_f(\alpha)$$

(as in Problem 4.20). Show that

$$\mathrm{Exp}(f)^2 \le \mathrm{Exp}(f^2).$$

Hint: See the remark after Proposition 5.6, and observe that the requirement of probability measure (i.e., $\wp(X) = 1$ as a particular case of $\wp(X) \le 1$) is crucial here; just finite measure would not be enough.

Their nonnegative difference is the *variance*, $\mathrm{Var}(f)$, of f. Verify that

$$\mathrm{Var}(f) = \mathrm{Exp}(f^2) - \mathrm{Exp}(f)^2 = \mathrm{Exp}\big[\big(f - \mathrm{Exp}(f)\big)^2\big],$$

and so $\mathrm{Var}(f)$ is also called the *second central moment* of f. If $\mathrm{Exp}(f^2) < \infty$, then f is called a *second-order random variable*, which is a vector in the real Hilbert space $L^2(X, \mathcal{X}, \wp)$ for which

$$\|f\|^2 = \mathrm{Exp}(f^2) = \int_X f^2\, d\wp.$$

Suggested Readings

Bartle [5], Bauer [8], Brown and Pearcy [13], Halmos [31], Royden [62], Rudin [63]. See also [9], [22], [25], [29], [72], [73] and, exclusively for Banach space results, [52] (also, [43, Chapter 4]). For detailed proofs of Hölder and Minkowski inequalities of Problems 5.1, 5.3, see, e.g., [22, Lemmas III.3.2,3].



6

Convergence of Functions

6.1 Four Basic Convergence Notions

Major convergence concepts for sequences of *real-valued* functions will be considered in this chapter. We have already met four convergence concepts so far, viz., pointwise, uniform, almost everywhere, and convergence in L^p. These are reviewed and compared in this section. Further concepts, namely, convergence in measure, uniform almost everywhere, and almost uniform convergence, will be discussed and compared in subsequent sections.

Definition 6.1. Let $\{f_n\}$ be a sequence of real-valued functions $f_n : X \to \mathbb{R}$ acting on a (nonempty) set X. The sequence $\{f_n\}$ *converges pointwise* to a function $f : X \to \mathbb{R}$ if the real-valued sequence $\{f_n(x)\}$ converges in $\mathbb{R}$ to the real number $f(x)$ for every $x \in X$. That is, if $|f_n(x) - f(x)| \to 0$ as $n \to \infty$ for every $x \in X$. In other words, $\{f_n\}$ converges pointwise to f if for every $\varepsilon > 0$ and each $x \in X$ there is a positive integer $n_{\varepsilon,x}$ such that

$$n \geq n_{\varepsilon,x} \quad \text{implies} \quad |f_n(x) - f(x)| < \varepsilon.$$

In this case we write $f_n \to f$ pointwise. If the integer $n_{\varepsilon,x}$ does not depend on x, then $\{f_n\}$ *converges uniformly* to f, and we write $f_n \to f$ uniformly. Thus the sequence of functions $\{f_n\}$ converges uniformly to $f : X \to \mathbb{R}$ if $\sup_{x \in X} |f_n(x) - f(x)| \to 0$ as $n \to \infty$ or, equivalently, if for every $\varepsilon > 0$ there is a positive integer n_ε such that

$$n \geq n_\varepsilon \quad \text{implies} \quad \sup_{x \in X} |f_n(x) - f(x)| < \varepsilon.$$

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_6

Pointwise convergence is convergence in the normed space $(\mathbb{R}, |\ |)$. Since $(\mathbb{R}, |\ |)$ is a Banach space, the real sequence $\{f_n(x)\}$ converges to $f(x)$ in $\mathbb{R}$ for each $x \in X$ if and only if $\{f_n(x)\}$ is a Cauchy sequence in $(\mathbb{R}, |\ |)$ for each $x \in X$. So $\{f_n\}$ is pointwise convergent if and only if it is a *Cauchy sequence*: for every $\varepsilon > 0$ and each $x \in X$ there is a positive integer $n_{\varepsilon,x}$ such that

$$m, n \geq n_{\varepsilon,x} \quad \text{implies} \quad |f_m(x) - f_n(x)| < \varepsilon,$$

which is denoted as $\lim_{m,n} |f_m(x) - f_n(x)| = 0$ for every $x \in X$. If such an $n_{\varepsilon,x}$ does not depend on x, then $\{f_n\}$ is a *uniform Cauchy sequence*. This means that for every $\varepsilon > 0$ there is a positive integer n_ε such that

$$m, n \geq n_\varepsilon \quad \text{implies} \quad \sup_{x \in X} |f_m(x) - f_n(x)| < \varepsilon,$$

which is denoted as $\lim_{m,n} \sup_{x \in X} |f_m(x) - f_n(x)| = 0$. It is clear that if $f_n \to f$ uniformly, then $\{f_n\}$ is a uniform Cauchy sequence. Actually, by the triangle inequality, for every m and n,

$$\sup_{x \in X} |f_m(x) - f_n(x)| \leq \sup_{x \in X} |f_m(x) - f(x)| + \sup_{x \in X} |f(x) - f_n(x)|.$$

Conversely, if $\{f_n\}$ is a uniform Cauchy sequence, then $\{f_n(x)\}$ is a real Cauchy sequence, and so it converges in $\mathbb{R}$ to a real number, say $f(x)$, for each $x \in X$. This defines a function $f\colon X \to \mathbb{R}$ such that $f_n \to f$ pointwise. Since $\{f_n\}$ is a uniform Cauchy sequence, it follows that $f_n \to f$ uniformly. In fact, for arbitrary $x \in X$ and $\varepsilon > 0$ there is a positive integer n_ε for which

$$m \geq n_\varepsilon \quad \text{implies} \quad |f_m(x) - f(x)| = \lim_n |f_m(x) - f_n(x)| < \varepsilon.$$

Examples throughout the chapter will compare the several convergence notions. Problems 6.6 and 6.7 summarize all possible implications among some convergence notions that are based on measure-theoretical concepts.

Example 6A. $f_n \to f$ uniformly $\implies\!\!\!\!\!\!\not\quad$ $f_n \to f$ pointwise.

It is clear that uniform convergence implies pointwise convergence to the *same* and unique limit. It is readily verified that the converse fails. For instance, for each integer $n \geq 1$ let $f_n\colon [0,1] \to \mathbb{R}$ be given by

$$f_n(x) = \begin{cases} x^n, & x \in [0,1), \\ 0, & x = 1. \end{cases}$$

The sequence $\{f_n\}$ converges pointwise to the null function 0 (i.e., to the function $0\colon [0,1] \to \mathbb{R}$ such that $0(x) = 0$ for all $x \in [0,1]$) but it does not converge to the function 0 uniformly because $\sup_{x \in [0,1]} |f_n(x)| = 1$ for all $n \geq 1$. Hence it does not converge uniformly to any function.

The previous convergence notions are all measure-free. The next notions require a measure space. So, from now on, take a measure space $(X, \mathcal{X}, \mu)$, and by a measurable function we mean an $\mathcal{X}$-measurable function on X.

Definition 6.2. A sequence $\{f_n\}$ of real-valued functions $f_n\colon X \to \mathbb{R}$ *converges almost everywhere* with respect to the measure $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ (or *μ-almost everywhere*) to a real-valued function $f\colon X \to \mathbb{R}$ if the real-valued sequence $\{f_n(x)\}$ converges in $\mathbb{R}$ to the real number $f(x)$ for every x except in a set of measure zero. In this case we write $f_n \to f$ μ-a.e. (or $f_n \to f$ a.e. if the measure μ is clear or is not relevant in the context). That is, $\{f_n\}$ converges almost everywhere to f if there exists a set $N \in \mathcal{X}$ with $\mu(N) = 0$ such that for every $\varepsilon > 0$ and each $x \in X \backslash N$ there is a positive integer $n_{\varepsilon,x}$ such that

$$n \geq n_{\varepsilon,x} \quad \text{implies} \quad |f_n(x) - f(x)| < \varepsilon.$$

A sequence $\{f_n\}$ of real-valued functions on a set X *converges in* $L^p(X, \mathcal{X}, \mu)$ for some $p \geq 1$ if it converges in the Banach space $(L^p, \|\ \|_p)$. In this case we write $f_n \to f$ in L^p. In other words, $\{f_n\}$ converges in L^p if $\|f_n - f\|_p \to 0$ as $n \to \infty$ for some real-valued function $f \in L^p$. That is, for each $\varepsilon > 0$ there is a positive integer n_ε such that

$$n \geq n_\varepsilon \quad \text{implies} \quad \|f_n - f\|_p < \varepsilon.$$

Example 6B. $\{f_n\}$ converges pointwise $\implies$ $\not\Longleftarrow$ $\{f_n\}$ converges a.e.,

where, for any measure space $(X, \mathcal{X}, \mu)$, the implication

$$f_n \to f \text{ pointwise} \implies f_n \to f \text{ a.e.}$$

holds trivially, since $\mu(\varnothing) = 0$ for every measure μ on any σ-algebra $\mathcal{X}$. We will now see *how* the converse fails. Recall that the limit of a sequence that converges pointwise or uniformly is unique, and so is the limit of a sequence that converges in L^p, where in this case uniqueness is understood almost everywhere (the class of equivalence $[f]$ containing f is unique). But unlike pointwise, uniform, and convergence in L^p, the notion of almost everywhere convergence does not imply uniqueness of the almost everywhere limit. For instance, let $f\colon \mathbb{R} \to \mathbb{R}$ be the characteristic function of $\{0\}$, that is,

$$f(x) = \chi_{\{0\}}(x) = \begin{cases} 1, & x = 0, \\ 0, & x \in \mathbb{R}\backslash\{0\}, \end{cases}$$

let $\{f_n\}$ be a sequence constantly equal to f (i.e., $f_n = f$ for all $n \geq 1$), and consider the Lebesgue measure space $(\mathbb{R}, \Re, \lambda)$. The sequence $\{f_n\}$ trivially converges pointwise to f, which is its unique pointwise limit of $\{f_n\}$, and $\{f_n\}$ also converges a.e. to every function $f'\colon \mathbb{R} \to \mathbb{R}$ for which $f' = f$ a.e.

(i.e., $f'(x) = f(x)$ for every $x \in \mathbb{R}\backslash N$ for some $N \in \Re$ such that $\lambda(N) = 0$). In particular, it converges almost everywhere to the null function $0\colon \mathbb{R} \to \mathbb{R}$ (since $\lambda(\{0\}) = 0$ and $f(x) = 0$ for all $x \in \mathbb{R}\backslash\{0\}$), and also to f itself (since $\lambda(\varnothing) = 0$). Thus, $f_n \to 0$ λ-a.e. and $f_n \to f \neq 0$ pointwise (and so λ-a.e.). Now note that we might argue that such an example would be meaningless had we decided to work with equivalence classes of functions,

$$[f] = \big\{f'\colon \mathbb{R} \to \mathbb{R}\colon\ f' = f\ \lambda\text{-a.e.}\big\},$$

rather than with single functions. In fact, this would yield uniqueness for the almost everywhere limit in this particular example, but would not be enough to avoid the failure of the converse in the preceding implication; the converse in the preceding implication fails anyway. Indeed, if

$$g_n = (-1)^n f,$$

then $g_n \to 0$ λ-a.e. and $\{g_n\}$ does not converge pointwise (to any function).

In general, the notions of uniform convergence and convergence in L^p are not related (see Problem 6.1):

$$\{f_n\} \text{ converges uniformly} \quad \begin{matrix}\not\Rightarrow \\ \not\Leftarrow\end{matrix} \quad \{f_n\} \text{ converges in } L^p.$$

However, if $\{f_n\}$ converges both uniformly and in L^p, then the limits coincide almost everywhere. Moreover, in a *finite* measure space uniform convergence implies convergence in L^p to the same (a.e.) limit.

Proposition 6.3. *Take any $p \geq 1$. Consider a sequence $\{f_n\}$ of functions in L^p, and let f, f', f'' be real-valued measurable functions.*

(a) *If $f_n \to f'$ uniformly and $f_n \to f''$ in L^p, then $f' = f''$ almost everywhere.*

(b) *If $\mu(X) < \infty$ and $f_n \to f$ uniformly, then $f \in L^p$ and $f_n \to f$ in L^p.*

Proof. Take real-valued $\mathcal{X}$-measurable functions f_n, f, f', and f'' on X.

(a) $|f' - f''| \leq |f' - f_n| + |f_n - f''| \leq \sup_{x\in X} |f'(x) - f_n(x)| + |f_n - f''|$ for each n. Since $f_n \to f'$ uniformly, we get $|f' - f''| \leq \limsup_n |f_n - f''|$. Since $f_n \to f''$ in L^p, it follows (cf. Problem 3.3(b) and Theorem 4.7) that

$$0 \leq \int |f' - f''|^p \, d\mu \leq \limsup_n \int |f_n - f''|^p \, d\mu = \lim_n \|f_n - f''\|_p^p = 0,$$

and so $f' = f''$ a.e. (cf. Propositions 1.5, 1.6, and 3.7(a)).

(b) If each f_n lies in L^p for an arbitrary $p \geq 1$, then

$$\|f_m - f_n\|_p^p = \int |f_m - f_n|^p \, d\mu \leq \sup_{x\in X} |f_m(x) - f_n(x)|^p \, \mu(X)$$

for every pair of positive integers m and n. Since $\{f_n\}$ is uniformly Cauchy (because $f_n \to f$ uniformly) and $\mu(X) < \infty$, it follows that $\{f_n\}$ is a Cauchy sequence in the Banach space $(L^p, \| \ \|_p)$, and so it converges in L^p; and the L^p-limit coincides a.e. with the uniform limit f (i.e., it is in $[f]$) by (a). □

According to Problem 6.2(a), observe that even if $\mu(X) < \infty$,

$$\{f_n\} \text{ converges pointwise } \not\Longrightarrow \ \{f_n\} \text{ converges in } L^p.$$

But under the assumption of *dominated convergence*, just convergence almost everywhere is enough to ensure convergence in L^p.

Proposition 6.4. *Take any $p \geq 1$. Consider a sequence $\{f_n\}$ of functions in L^p, let f be a real-valued measurable function, and take g in L^p.*

$$|f_n| \leq g \text{ for all } n \text{ and } f_n \to f \text{ a.e.} \implies f \in L^p \text{ and } f_n \to f \text{ in } L^p.$$

Proof. The dominance assumption, namely, $|f_n| \leq g$ for all n, is equivalent to almost everywhere dominance; that is, $|f_n| \leq g$ for all n a.e., since the functions f_n and g are in L^p, where inequalities (and equalities) are understood in the sense of equivalence classes, and so they are always interpreted almost everywhere. Since $|f_n|^p \leq g^p$ for all n a.e. and $f_n \to f$ a.e., it follows that $|f|^p \leq g^p$ almost everywhere. Thus, if $g \in L^p$, then f^p is integrable (cf. Problem 4.4(b)); that is, $f \in L^p$. Moreover,

$$|f_n - f|^p \leq \big(|f_n| + |f|\big)^p \leq (2g)^p \in L^1 \quad \text{and} \quad |f_n - f|^p \to 0 \ \ \mu\text{-a.e.},$$

so $f_n \to f$ in L^p by the Dominated Convergence Theorem (Theorem 4.7):

$$\lim_n \|f_n - f\|_p^p = \lim_n \int |f_n - f|^p \, d\mu = 0. \qquad \square$$

Note that all constant functions lie in every L^p if the measure is *finite*. This yields the following *uniformly bounded* version of Proposition 6.4.

$$\sup_n |f_n| < \infty, \ \ \mu(X) < \infty \ \text{ and } \ f_n \to f \ \mu\text{-a.e.} \implies f_n \to f \in L^p \text{ in } L^p.$$

Convergence in L^p is of crucial importance since it is convergence in the norm topology of the Banach space $(L^p, \| \ \|_p)$. Generally, it does not imply uniform convergence, nor is it implied by uniform convergence (cf. Problem 6.1), although in a finite measure space it is weaker than uniform convergence (Proposition 6.3). Even in a finite measure space it is not implied by almost everywhere convergence, and not even by pointwise convergence (cf. Problem 6.2(a)). But, under the dominance hypothesis it becomes weaker than almost everywhere convergence (Proposition 6.4). However,

$$\{f_n\} \text{ converges in } L^p \ \not\Longrightarrow \ \{f_n\} \text{ converges a.e.},$$

even under finite measure and dominance condition (cf. Problem 6.2(b)). On the other hand, convergence in measure (defined in the next section) is weaker than L^p-convergence.

6.2 Convergence in Measure

Throughout this section, let $(X, \mathcal{X}, \mu)$ be a measure space and let $\{f_n\}$ be a sequence of real-valued $\mathcal{X}$-measurable functions (which means that each function $f_n : X \to \mathbb{R}$ lies in $\mathcal{M}(X, \mathcal{X})$).

Definition 6.5. A sequence $\{f_n\}$ of real-valued functions in $\mathcal{M}(X, \mathcal{X})$ *converges in measure* to a real-valued function f in $\mathcal{M}(X, \mathcal{X})$ if

$$\lim_n \mu\big(\{x \in X : \ |f_n(x) - f(x)| \geq \alpha\}\big) = 0$$

for every $\alpha > 0$. In this case we write $f_n \to f$ in measure. The sequence $\{f_n\}$ is *Cauchy in measure* if for every real $\alpha > 0$,

$$\lim_{m,n} \mu\big(\{x \in X : \ |f_m(x) - f_n(x)| \geq \alpha\}\big) = 0.$$

Since f_n and f lie in $\mathcal{M}(X, \mathcal{X})$, for each integer $n \geq 1$ the set

$$F_n(\alpha) = \{x \in X : \ |f_n(x) - f(x)| \geq \alpha\}$$

lies in $\mathcal{X}$, for every $\alpha > 0$. So convergence in measure means

$$\lim_n \mu(F_n(\alpha)) = 0 \quad \text{for every} \quad \alpha > 0.$$

Then the sequence $\{\mu(F_n(\alpha))\}$ is eventually real-valued (i.e., eventually, it does not take on the value $+\infty$). Thus $f_n \to f$ in measure if and only if for every $\varepsilon > 0$ and every $\alpha > 0$ there is a positive integer $n_{\varepsilon,\alpha}$ such that

$$n \geq n_{\varepsilon,\alpha} \quad \text{implies} \quad \mu\big(\{x \in X : \ |f_n(x) - f(x)| \geq \alpha\}\big) < \varepsilon.$$

Similarly, the sequence $\{f_n\}$ is Cauchy in measure if and only if for every $\varepsilon > 0$ and every $\alpha > 0$ there exists a positive integer $n_{\varepsilon,\alpha}$ such that

$$m, n \geq n_{\varepsilon,\alpha} \quad \text{implies} \quad \mu\big(\{x \in X : \ |f_m(x) - f_n(x)| \geq \alpha\}\big) < \varepsilon.$$

Example 6C. $f_n \to f$ in L^p $\quad \Longrightarrow \quad \not\Longleftarrow \quad$ $f_n \to f$ in measure.

Convergence in L^p implies convergence in measure to the same limit. In fact, for every $\alpha > 0$ and every integer $n \geq 1$,

$$\alpha^p \mu\big(F_n(\alpha)\big) = \int_{F_n(\alpha)} \alpha^p \, d\mu \leq \int_{F_n(\alpha)} |f_n - f|^p \, d\mu \leq \int |f_n - f|^p \, d\mu = \|f_n - f\|_p^p.$$

The converse, however, fails even under the assumption that $\mu(X) < \infty$. Actually, the sequence $\{f_n\}$ of Problem 6.2(a) does not converge in L^p but it is readily verified that it converges in measure to the null function.

Example 6D. $f_n \to f$ uniformly $\implies\!\!\!\!\!\!\not\quad$ $f_n \to f$ in measure.

Indeed, if $f_n \to f$ uniformly, then for any $\varepsilon > 0$ there is an n_ε such that

$$n \geq n_\varepsilon \implies \sup_{x \in X} |f_n(x) - f(x)| < \varepsilon \implies F_n(\varepsilon) = \varnothing \implies \mu\big(F_n(\varepsilon)\big) = 0.$$

This means that the sequence $\{\mu(F_n(\alpha))\}$ not only converges to zero but is eventually null for every $\alpha > 0$. Again, the converse fails. For example, the sequence $\{f_n\}$ of Problem 6.1(b) converges in L^p, and so it converges in measure, but it does not converge uniformly. Observe that the sequence $\{f_n\}$ of Problem 6.1(a) also yields another example of a sequence that converges in measure (since it converges uniformly) but not in L^p.

Proposition 6.6. *Let $\{f_n\}$ be a sequence of real-valued measurable functions.*

(a) *If $\{f_n\}$ converges in measure, then it is Cauchy in measure.*

(b) *If $\{f_n\}$ converges in measure, then the limit is unique almost everywhere.*

(c) *If $\{f_n\}$ is Cauchy in measure and has a subsequence that converges in measure, then it converges in measure itself and its limit coincides with the limit of that subsequence.*

Proof. Take real-valued measurable functions f_n, f, and f'. Set $F_n(\alpha) = \{x \in X : |f_n(x) - f(x)| \geq \alpha\}$ and $F_n'(\alpha) = \{x \in X : |f_n(x) - f'(x)| \geq \alpha\}$.

(a) Since $|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f_n(x) - f(x)|$, by the triangle inequality, we get $\{x \in X : |f_m(x) - f_n(x)| \geq \alpha\} \subseteq F_m(\frac{\alpha}{2}) \cup F_n(\frac{\alpha}{2})$, and so

$$\mu\big(\{x \in X : |f_m(x) - f_n(x)| \geq \alpha\}\big) \leq \mu\big(F_m(\tfrac{\alpha}{2})\big) + \mu\big(F_n(\tfrac{\alpha}{2})\big).$$

Suppose $f_n \to f$ in measure. Then, by definition, $\lim_n \mu(F_n(\frac{\alpha}{2})) = 0$, and hence $\lim_{m,n} \mu(\{x \in X : |f_m(x) - f_n(x)| \geq \alpha\}) = 0$, for every $\alpha > 0$.

(b) Since $|f(x) - f'(x)| \leq |f_n(x) - f(x)| + |f_n(x) - f'(x)|$ (triangle inequality again), we get $\{x \in X : |f(x) - f'(x)| \geq \alpha\} \subseteq F_n(\frac{\alpha}{2}) \cup F_n'(\frac{\alpha}{2})$, and so

$$\mu\big(\{x \in X : |f(x) - f'(x)| \geq \alpha\}\big) \leq \mu\big(F_n(\tfrac{\alpha}{2})\big) + \mu\big(F_n'(\tfrac{\alpha}{2})\big).$$

Suppose $\{f_n\}$ converges in measure to both f and f', then $\lim_n \mu(F_n(\frac{\alpha}{2})) = \lim_n \mu(F_n'(\frac{\alpha}{2})) = 0$. This implies that $\mu(\{x \in X : |f(x) - f'(x)| \geq \alpha\}) = 0$ for every $\alpha > 0$. Thus $f' = f$ a.e., which is a consequence of the inclusion $\{x \in X : |f(x) - f'(x)| > 0\} \subseteq \bigcup_{k=1}^{\infty} \{x \in X : |f(x) - f'(x)| \geq \frac{1}{k}\}$.

(c) Take a subsequence $\{f_{n_k}\}$ of a sequence $\{f_n\}$. Let $\alpha > 0$ and $\varepsilon > 0$ be arbitrary positive numbers. Suppose $\{f_n\}$ is Cauchy in measure. Thus there is an integer $n_{\varepsilon,\alpha}$ for which $\mu\{x \in X\colon |f_m(x) - f_n(x)| \geq \frac{\alpha}{2}\} < \frac{\varepsilon}{2}$ whenever $m, n \geq n_{\varepsilon,\alpha}$. If $\{f_{n_k}\}$ converges in measure to f, then there is an integer $k_{\varepsilon,\alpha}$ such that $\mu\{x \in X\colon |f_{n_k}(x) - f(x)| \geq \frac{\alpha}{2}\} < \frac{\varepsilon}{2}$ for every $k \geq k_{\varepsilon,\alpha}$. Hence, if j is an integer such that $j \geq k_{\varepsilon,\alpha}$ and $n_j \geq n_{\varepsilon,\alpha}$, then

$$\mu\big\{x \in X\colon\ |f_n(x) - f_{n_j}(x)| \geq \tfrac{\alpha}{2}\big\} < \tfrac{\varepsilon}{2},$$

$$\mu\big\{x \in X\colon\ |f_{n_j}(x) - f(x)| \geq \tfrac{\alpha}{2}\big\} < \tfrac{\varepsilon}{2},$$

for $n \geq n_{\varepsilon,\alpha}$. Since $|f_n(x) - f(x)| \leq |f_n(x) - f_{n_j}(x)| + |f_{n_j}(x) - f(x)|$,

$$\begin{aligned}&\big\{x \in X\colon\ |f_n(x) - f(x)| \geq \alpha\big\}\\ &\subseteq\ \big\{x \in X\colon\ |f_n(x) - f_{n_j}(x)| \geq \tfrac{\alpha}{2}\big\} \cup \big\{x \in X\colon\ |f_{n_j}(x) - f(x)| \geq \tfrac{\alpha}{2}\big\},\end{aligned}$$

so that

$$\begin{aligned}&\mu\big(\big\{x \in X\colon |f_n(x) - f(x)| \geq \alpha\big\}\big)\\ &\leq\ \mu\big(\big\{x \in X\colon |f_n(x) - f_{n_j}(x)| \geq \tfrac{\alpha}{2}\big\}\big) + \mu\big(\big\{x \in X\colon |f_{n_j}(x) - f(x)| \geq \tfrac{\alpha}{2}\big\}\big),\end{aligned}$$

and hence $\mu(\{x \in X\colon |f_n(x) - f(x)| \geq \alpha\}) < \varepsilon$ for all $n \geq n_{\varepsilon,\alpha}$. Therefore, $\lim_n \mu(F_n(\alpha)) = 0$, and so $\{f_n\}$ converges in measure to f. □

Convergences in measure and almost everywhere are not related.

$$\{f_n\} \text{ converges in measure} \quad \not\Longrightarrow \quad \{f_n\} \text{ converges a.e.},$$

even in the case of $\mu(X) < \infty$. Indeed, the sequence $\{f_n\}$ of Problem 6.2(b) acts on a finite measure space, converges in measure (since it converges in L^p) to the null function, but $\{f_n(x)\}$ fails to converge for every x, and so $\{f_n\}$ does not converge a.e. (thus it does not converge pointwise). Even though uniform convergence implies convergence in measure (Example 6D),

$$\{f_n\} \text{ converges pointwise} \quad \not\Longrightarrow \quad \{f_n\} \text{ converges in measure}$$

and, consequently, convergence a.e. does not imply convergence in measure. In fact, the sequence $\{f_n\}$ of Problem 6.3(a,b) converges pointwise (i.e., everywhere in $\mathbb{R}$, and so a.e.) to the null function, but it is not Cauchy in measure and so it does not converge in measure by Proposition 6.6(a). Observe that if a sequence does not converge in measure, then it does not converge both in L^p and uniformly (see Examples 6C and 6D). But convergence almost everywhere implies convergence in measure whenever $\mu(X) < \infty$, as will be verified in the forthcoming Propositions 6.12 and 6.13.

Proposition 6.7. *If a sequence of real-valued measurable functions is Cauchy in measure, then it has a subsequence that converges both in measure and almost everywhere.*

Proof. Let $(X, \mathcal{X}, \mu)$ be a measure space, and take a sequence $\{f_n\}$ of real-valued functions in $\mathcal{M}(X, \mathcal{X})$. Suppose $\{f_n\}$ is Cauchy in measure. Take an arbitrary integer $k \geq 1$. Thus there is another integer $n_k \geq 1$ such that

$$\mu\big(\big\{x \in X\colon\ |f_m(x) - f_n(x)| \geq \big(\tfrac{1}{2}\big)^k\big\}\big) < \big(\tfrac{1}{2}\big)^k \quad \text{whenever} \quad m, n \geq n_k.$$

This implies that there is a subsequence $\{f_{n_k}\}$ of $\{f_n\}$ for which

$$\mu\big(\big\{x \in X\colon\ |f_{n_{k+1}}(x) - f_{n_k}(x)| \geq \big(\tfrac{1}{2}\big)^k\big\}\big) < \big(\tfrac{1}{2}\big)^k.$$

Now, for each integer $k \geq 1$, consider the set $E_k \in \mathcal{X}$ given by

$$E_k = \bigcup_{j=k}^{\infty} \big\{x \in X\colon\ |f_{n_{j+1}}(x) - f_{n_j}(x)| \geq \big(\tfrac{1}{2}\big)^j\big\},$$

so that $\mu(E_k) < \sum_{j=k}^{\infty} \big(\frac{1}{2}\big)^j = \big(\frac{1}{2}\big)^{k-1}$. Set $N = \bigcap_{k=1}^{\infty} E_k \in \mathcal{X}$, which is such that $\mu(N) = 0$ because $N \subseteq E_k$ and so $\mu(N) \leq \mu(E_k)$, for all $k \geq 1$. Since

$$f_{n_i}(x) - f_{n_j}(x) = \sum_{\ell=1}^{i-j} f_{n_{i-\ell+1}}(x) - f_{n_{i-\ell}}(x) = \sum_{\ell=j}^{i-1} f_{n_{\ell+1}}(x) - f_{n_\ell}(x)$$

for every $1 \leq j < i$, it follows that if $x \in X\backslash E_k$ and $k \leq j < i$, then

$$|f_{n_i}(x) - f_{n_j}(x)| \leq \sum_{\ell=j}^{i-1} |f_{n_{\ell+1}}(x) - f_{n_\ell}(x)| < \sum_{\ell=j}^{i-1} \big(\tfrac{1}{2}\big)^\ell < \big(\tfrac{1}{2}\big)^{j-1}.$$

Thus, if x lies in $X\backslash N = X\backslash \bigcap_{k=1}^{\infty} E_k = \bigcup_{k=1}^{\infty}(X\backslash E_k)$ or, equivalently, if x lies $X\backslash E_k$ for some $k \geq 1$, then the above inequality holds for every pair of distinct integers $i, j \geq k$. This leads to two results, as stated below. The result in (i) ensures that $\{f_{n_k}\}$ converges almost everywhere, and the result in (ii) ensures that $\{f_{n_k}\}$ converges in measure to the same limit f.

(i) By the above inequality $\{f_{n_k}(x)\}$ is a Cauchy sequence in $\mathbb{R}$, and so it converges in $\mathbb{R}$ for every x in $X\backslash N$. Since $\mu(N) = 0$, we get

$$f_{n_k} \to f \ \ \mu\text{-a.e.}, \quad \text{where} \quad f(x) = \begin{cases} \lim_k f_{n_k}(x), & x \in X\backslash N, \\ 0, & x \in N, \end{cases}$$

defining the real-valued measurable function f on X.

(ii) Take any $k \geq 1$. The above convergence ensures that $f_{n_j}(x) \to f(x)$ for every x in $X\backslash E_k$, and this implies that for every $j \geq k$

$$|f_{n_j}(x) - f(x)| = \lim_i |f_{n_i}(x) - f_{n_j}(x)| \leq \big(\tfrac{1}{2}\big)^{j-1} \leq \big(\tfrac{1}{2}\big)^{k-1}.$$

Moreover, since $\mu(E_k) < \left(\frac{1}{2}\right)^{k-1}$ for each $k \geq 1$, it follows that for every $\varepsilon > 0$ and $\alpha > 0$ there exists an integer $k' = k_{\varepsilon,\alpha} \geq 1$ such that

$$\mu(E_{k'}) < \left(\tfrac{1}{2}\right)^{k'-1} < \min\{\varepsilon, \alpha\}.$$

However (since $|f_{n_j}(x) - f(x)| \leq \left(\frac{1}{2}\right)^{k'-1}$ for every $j \geq k'$), if $j \geq k'$, then

$$\{x \in X\colon |f_{n_j}(x) - f(x)| \geq \alpha\} \subseteq \{x \in X\colon |f_{n_j}(x) - f(x)| \geq \left(\tfrac{1}{2}\right)^{k'}\} \subseteq E_{k'}.$$

Therefore,

$$\mu\big(\{x \in X\colon |f_{n_j}(x) - f(x)| \geq \alpha\}\big) \leq \mu(E_{k'}) < \varepsilon$$

for all $j \geq k' = k_{\varepsilon,\alpha}$, which means that $f_{n_j} \to f$ in measure. □

The following theorem is an important consequence of Proposition 6.7, which is referred to as the Riesz–Weyl Theorem. It says that *a sequence converges in measure if and only if it is Cauchy in measure.*

Theorem 6.8. *If a sequence of real-valued measurable functions is Cauchy in measure, then it converges in measure.*

Proof. Propositions 6.6(c) and 6.7. □

We have already seen that convergence in L^p implies convergence in measure, but the converse fails even in a finite measure space (see Example 6C). However, dominated convergence in measure implies convergence in L^p, as a consequence of another application of Proposition 6.7.

Proposition 6.9. *Take any $p \geq 1$. If $\{f_n\}$ is a sequence of functions in L^p, if f is a real-valued measurable function, and if g lies in L^p, then*

$$|f_n| \leq g \text{ for all } n \text{ and } f_n \to f \text{ in measure} \implies f \in L^p \text{ and } f_n \to f \text{ in } L^p.$$

Proof. Consider the dominance assumption, and note that in this context plain dominance (i.e., $|f_n| \leq g$ for all n) is equivalent to almost everywhere dominance (i.e., $|f_n| \leq g$ for all n a.e.) — cf. proof of Proposition 6.4. We carry on a proof by contradiction. If $\{f_n\}$ does not converge in L^p to f, then there is a subsequence $\{h_k\}$ of $\{f_n\}$ and a real $\varepsilon > 0$ such that

$$\|h_k - f\|_p \geq \varepsilon \quad \text{for every} \quad k \geq 1. \tag{i}$$

Suppose $\{f_n\}$ converges in measure to f. Thus every subsequence of $\{f_n\}$ converges in measure to f. In particular, $\{h_k\}$ converges in measure to f. Propositions 6.6(a) and 6.7 ensure that $\{h_k\}$ has a subsequence $\{h_{k_j}\}$ that converges both in measure and almost everywhere. Since $\{h_{k_j}\}$ converges

in measure, Proposition 6.6 ensures that it must converge to f. Since $\{h_{k_j}\}$ also converges almost everywhere to f, and since $|h_{k_j}| \le g \in L^p$ for all j, it follows by Proposition 6.4 that f lies in L^p and $\{h_{k_j}\}$ converges in L^p to f, which contradicts the assertion in (i). Then $\{f_n\}$ converges in L^p to f. □

6.3 Almost Uniform Convergence

Consider a measure space $(X, \mathcal{X}, \mu)$. Let $\{f_n\}$ be a sequence of real-valued functions on X, and let f be a real-valued function on X. We say that $\{f_n\}$ *converges uniformly almost everywhere* to f if $\{f_n\}$ converges uniformly to f on $X\backslash N$ (i.e., $\lim_n \sup_{x\in X\backslash N} |f_n(x) - f(x)| = 0$) for some set N in $\mathcal{X}$ with $\mu(N) = 0$. Equivalently, if there exists a set N in $\mathcal{X}$ with $\mu(N) = 0$ such that for every $\varepsilon > 0$ there is a positive integer n_ε for which

$$n \ge n_\varepsilon \quad \text{implies} \quad \sup_{x\in X\backslash N} |f_n(x) - f(x)| < \varepsilon.$$

In other words, if the sequence converges uniformly on the complement of a set of measure zero. However, we will be dealing in this section with a weaker notion of convergence, which requires uniform convergence on the complement of sets that have arbitrarily small measure. Actually, we have already met this notion in the proof of Proposition 6.7, part (ii).

Definition 6.10. A sequence $\{f_n\}$ of real-valued functions f_n on X *converges almost uniformly* (with respect to μ) to a real-valued function f on X if for each $\delta > 0$ there is a set E_δ in $\mathcal{X}$ with $\mu(E_\delta) < \delta$ such that $\{f_n\}$ converges uniformly to f on $X\backslash E_\delta$ (i.e., $\lim_n \sup_{x\in X\backslash E_\delta} |f_n(x) - f(x)| = 0$). Equivalently, if for every $\delta > 0$ and every $\varepsilon > 0$ there is a set E_δ in $\mathcal{X}$ with $\mu(E_\delta) < \delta$ and a positive integer $n_{\varepsilon,\delta}$ such that

$$n \ge n_{\varepsilon,\delta} \quad \text{implies} \quad \sup_{x\in X\backslash E_\delta} |f_n(x) - f(x)| < \varepsilon.$$

In this case, write $f_n \to f$ a.u. (or $f_n \to f$ μ-a.u.). A sequence $\{f_n\}$ is *almost uniformly Cauchy* if for each $\delta > 0$ there is a set E_δ in $\mathcal{X}$ with $\mu(E_\delta) < \delta$ such that $\{f_n\}$ is a uniform Cauchy sequence on $X\backslash E_\delta$ (which means that $\lim_{m,n} \sup_{x\in X\backslash E_\delta} |f_m(x) - f_n(x)| = 0$). Equivalently, if for every $\delta > 0$ and every $\varepsilon > 0$ there is an $E_\delta \in \mathcal{X}$ with $\mu(E_\delta) < \delta$ and an $n_{\varepsilon,\delta} \ge 1$ such that

$$m, n \ge n_{\varepsilon,\delta} \quad \text{implies} \quad \sup_{x\in X\backslash E_\delta} |f_m(x) - f_n(x)| < \varepsilon.$$

Example 6E. Consider the string of implications,

$$f_n \to f \text{ uniformly} \quad \Longrightarrow \not\Longleftarrow \quad f_n \to f \text{ uniformly a.e.} \quad \Longrightarrow \not\Longleftarrow \quad f_n \to f \text{ a.u.},$$

where uniform convergence trivially implies uniform almost everywhere convergence (set $N = \varnothing$), which in turn trivially implies almost uniform convergence (set $E_\delta = N$). The converses, however, fail even if $\mu(X) < \infty$. Actually, take the *finite* Lebesgue measure space $\big([0,1], \wp([0,1]) \cap \Re, \lambda\big)$ and let $\{f_n\}$ be a sequence of real-valued functions on $[0,1]$ such that, for each $n \geq 1$,

$$f_n(x) = \begin{cases} (-1)^n, & x = 0, \\ 0, & x \in (0,1]. \end{cases}$$

It is clear that $\{f_n\}$ does not converge pointwise, and so it does not converge uniformly, but it converges uniformly almost everywhere. It is also clear that the sequence $\{f_n\}$ of Example 6A converges almost uniformly to the null function 0, but it does not converge to 0 uniformly almost everywhere since $\sup_{[0,1]\setminus N} |f_n(x)| = 1$ for any Borel set $N \subseteq [0,1]$ with $\lambda(N) = 0$, for all n.

Note that if $f_n \to f$ almost uniformly, then $\{f_n\}$ is an almost uniform Cauchy sequence. In fact, for each m and n,

$$\sup_{x \in X \setminus E_\delta} |f_m(x) - f_n(x)| \leq \sup_{x \in X \setminus E_\delta} |f_m(x) - f(x)| + \sup_{x \in X \setminus E_\delta} |f(x) - f_n(x)|.$$

The next result ensures the converse, and therefore *a sequence converges almost uniformly if and only if it is an almost uniform Cauchy sequence.*

Proposition 6.11. *Let $\{f_n\}$ be a sequence of real-valued functions.*

(a) *If $\{f_n\}$ is almost uniformly Cauchy, then it converges almost uniformly, and it also converges almost everywhere to the same real-valued limit f.*

(b) *If each function f_n is measurable, then so is the limit function.*

Proof. (a) Take a measure space $(X, \mathcal{X}, \mu)$. Suppose a sequence $\{f_n\}$ of real-valued functions on X is almost uniformly Cauchy (with respect to μ). Then for each integer $k \geq 1$ there is a set E_k in $\mathcal{X}$ with $\mu(E_k) < \frac{1}{k}$ such that

$$\lim_{m,n} \sup_{x \in X \setminus E_k} |f_m(x) - f_n(x)| = 0.$$

Set $N = \bigcap_{k=1}^\infty E_k$ in $\mathcal{X}$ so that $\mu(N) \leq \mu(E_k) < \frac{1}{k}$ for every $k \geq 1$, and so $\mu(N) = 0$. If $x \in X \setminus N = X \setminus \bigcap_{k=1}^\infty E_k = \bigcup_{k=1}^\infty (X \setminus E_k)$, then the real-valued sequence $\{f_n(x)\}$ is Cauchy in $\mathbb{R}$, and hence it converges in $\mathbb{R}$. Thus,

$$f_n \to f \text{ a.e.}, \quad \text{where} \quad f(x) = \begin{cases} \lim_n f_n(x), & x \in X \setminus N, \\ 0, & x \in N. \end{cases}$$

This convergence defines a real-valued function $f \colon X \to \mathbb{R}$ on X,

$$f = \lim_n \big(f_n \chi_{X \setminus N}\big).$$

Since $\{f_n\}$ is almost uniformly Cauchy, and since $f_n(x) \to f(x)$ for every x on each $X\backslash E_k \subseteq X\backslash N$, it follows that $f_n(x) \to f(x)$ uniformly on $X\backslash E_k$. In fact, for $k \geq 1$ and $\varepsilon > 0$ arbitrary there exists $n_{\varepsilon,k} \geq 1$ for which

$$m, n \geq n_{\varepsilon,k} \quad \text{implies} \quad \sup_{x \in X\backslash E_k} |f_m(x) - f_n(x)| < \varepsilon,$$

and so, for every $x \in X\backslash E_k$,

$$m \geq n_{\varepsilon,k} \quad \text{implies} \quad |f_m(x) - f(x)| = \lim_n |f_m(x) - f_n(x)| < \varepsilon,$$

which implies $\sup_{x \in X\backslash E_k} |f_m(x) - f(x)| < \varepsilon$. This ensures that

$$f_n \to f \text{ a.u.}$$

Indeed, for each $\delta > 0$ take k large enough so that $\frac{1}{k} \leq \delta$, and set $E_\delta = E_k$ in $\mathcal{X}$, which implies that $\mu(E_\delta) < \delta$, and $\{f_n\}$ converges uniformly on $X\backslash E_\delta$.

(b) Since $f = \lim_n f_n$ μ-a.e., it follows by Propositions 1.8 and 1.9 that f is $\mathcal{X}$-measurable whenever each f_n is $\mathcal{X}$-measurable. □

So almost uniform convergence implies almost everywhere convergence,

$$f_n \to f \text{ a.u.} \quad \begin{array}{c} \Longrightarrow \\ \not\Longleftarrow \end{array} \quad f_n \to f \text{ a.e.},$$

but the converse fails in general. Indeed, even pointwise convergence does not imply almost uniform convergence (cf. Problem 6.3). However, the converse holds in a finite measure space, as will be seen in Proposition 6.13.

Proposition 6.12. *Consider a sequence $\{f_n\}$ of measurable functions. If $f_n \to f$ almost uniformly, then $f_n \to f$ in measure.*

Proof. Let $(X, \mathcal{X}, \mu)$ be a measure space. For each $\alpha > 0$, consider again the set $F_n(\alpha) = \{x \in X : |f_n(x) - f(x)| \geq \alpha\}$ in $\mathcal{X}$. If $f_n \to f$ a.u., then for each $\delta > 0$ there is a set $E_\delta \in \mathcal{X}$ with $\mu(E_\delta) < \delta$ and an integer $n_{\alpha,\delta} \geq 1$ such that

$$n \geq n_{\alpha,\delta} \quad \text{implies} \quad \sup_{x \in X\backslash E_\delta} |f_n(x) - f(x)| < \alpha.$$

Thus $F_n(\alpha) \subseteq E_\delta$, and hence $\mu(F_n(\alpha)) < \mu(E_\delta) < \delta$ for every $n \geq n_{\alpha,\delta}$. Therefore $\lim_n \mu(F_n(\alpha)) = 0$, which means $f_n \to f$ in measure. □

So almost uniform convergence also implies convergence in measure,

$$f_n \to f \text{ a.u.} \quad \begin{array}{c} \Longrightarrow \\ \not\Longleftarrow \end{array} \quad f_n \to f \text{ in measure},$$

but the converse fails even if $\mu(X) < \infty$. In fact, the sequence of Problem 6.2(b), which acts on a finite measure space, converges in measure (since it converges in L^p), but does not converge a.u. (since it does not converge a.e.). Furthermore, convergences almost uniform and in L^p are not related,

$$\{f_n\} \text{ converges a.u.} \quad \begin{matrix} \Longrightarrow \\ \not\Longleftarrow \end{matrix} \quad \{f_n\} \text{ converges in } L^p,$$

even if $\mu(X) < \infty$. Take the finite measure space of Problem 6.2(a). If

$$f_n = (n+1)\,\chi_{[1/(n+1),\,2/(n+1)]} = \begin{cases} (n+1), & x \in [\frac{1}{n+1}, \frac{2}{n+1}], \\ 0, & x \in [0,1] \setminus [\frac{1}{n+1}, \frac{2}{n+1}], \end{cases}$$

for each $n \geq 1$, then $\{f_n\}$ converges almost uniformly to the null function but it does not converge in L^p (Problem 6.2(a)). Conversely, we saw above that convergence in L^p (in a finite measure space) does not imply almost uniform convergence. However, almost uniform dominated convergence implies convergence in L^p by Propositions 6.9 and 6.12.

By Proposition 6.11, almost uniform convergence implies almost everywhere convergence. We close this chapter by showing that the converse holds in a finite measure space. This is referred to as the Egoroff Theorem (Proposition 6.13 below). The finite measure assumption in the Egoroff Theorem can be replaced with dominated convergence (Corollary 6.14).

Proposition 6.13. *Consider a sequence $\{f_n\}$ of measurable functions. If $\mu(X) < \infty$ and $f_n \to f$ almost everywhere, then $f_n \to f$ almost uniformly.*

Proof. Let $(X, \mathcal{X}, \mu)$ be a measure space. Suppose $\{f_n\}$ converges almost everywhere to f. That is, $f_n(x) \to f(x)$ for every x in $X \backslash N \in \mathcal{X}$ for some $N \in \mathcal{X}$ with $\mu(N) = 0$, which means $f_n \to f$ pointwise on the complement $X' = X \backslash N$ of a set of measure zero. Now set $f'_n = f_n \chi_{X'}$ for each positive integer n and $f' = f\chi_{X'}$. Therefore $f'_n \to f'$ pointwise on X. In other words, $f'_n(x) \to f'(x)$ for every $x \in X$. Next take an arbitrary integer $m > 0$ and, for each $n \geq 1$, consider the set

$$F'_n(m) = \big\{x \in X' \colon |f_n(x) - f(x)| \geq \tfrac{1}{m}\big\} = \big\{x \in X \colon |f'_n(x) - f'(x)| \geq \tfrac{1}{m}\big\}.$$

Recall that f' and f'_n are measurable (since f and f_n are). Thus $|f'_n - f'|$ is a measurable function. Then each $F'_n(m)$ is a measurable set, and so

$$E'_n(m) = \bigcup_{k=n}^{\infty} F'_k(m)$$

is a measurable set for each $n \geq 1$. Hence $\{E'_n(m)\}$ is a decreasing sequence of sets in $\mathcal{X}$ (i.e., $E'_{n+1}(m) \subseteq E'_n(m) \in \mathcal{X}$). Take an arbitrary x in X. Since $f'_n(x) \to f'(x)$, it follows that there exists an integer $n_{m,x} \geq 1$ such that

$$k \geq n_{m,x} \implies |f'_k(x) - f'(x)| < \tfrac{1}{m} \implies x \notin F'_k(m) \implies x \notin E'_{n_{m,x}}(m),$$

and so $x \notin \bigcap_{n=1}^{\infty} E'_n(m)$. Thus

$$\bigcap_{n=1}^{\infty} E'_n(m) = \varnothing, \quad \text{which implies} \quad \mu\Big(\bigcap_{n=1}^{\infty} E'_n(m)\Big) = 0.$$

From now on suppose μ is a finite measure. Proposition 2.2(d) ensures that

$$\mu(X) < \infty \implies \mu\big(E'_1(m)\big) < \infty \implies \lim_n \mu\big(E'_n(m)\big) = 0.$$

Then for every $\varepsilon > 0$ there is an integer $n_{\varepsilon,m} \geq 1$ such that $\mu(E'_n(m)) < \frac{\varepsilon}{2^m}$ whenever $n \geq n_{\varepsilon,m}$. Since this happens for an arbitrary integer $m \geq 1$, set

$$E'_\varepsilon = \bigcup_{m=1}^{\infty} E'_{n_{\varepsilon,m}}(m)$$

in $\mathcal{X}$ so that

$$\mu(E'_\varepsilon) = \mu\Big(\bigcup_{m=1}^{\infty} E'_{n_{\varepsilon,m}}(m)\Big) \leq \sum_{m=1}^{\infty} \mu\big(E'_{n_{\varepsilon,m}}(m)\big) < \sum_{m=1}^{\infty} \frac{\varepsilon}{2^m} = \varepsilon.$$

Suppose $x \in X\backslash E'_\varepsilon$, so that $x \notin E'_{n_{\varepsilon,m}}(m)$. Hence

$$n \geq n_{\varepsilon,m} \implies x \notin F'_n(m) \implies |f'_n(x) - f'(x)| < \tfrac{1}{m}.$$

Thus $\{f'_n\}$ converges uniformly to f' on $X\backslash E'_\varepsilon$. Take E_ε in $\mathcal{X}$ given by

$$E_\varepsilon = E'_\varepsilon \cup N.$$

Since $X\backslash E_\varepsilon = X\backslash(E'_\varepsilon \cup N) = (X\backslash E'_\varepsilon) \cap (X\backslash N) = (X\backslash E'_\varepsilon) \cap X'$, we get

$$\sup_{x \in X\backslash E_\varepsilon} |f_n(x) - f(x)| = \sup_{x \in X\backslash E'_\varepsilon} |f'_n(x) - f'(x)|.$$

Then $\{f_n\}$ converges uniformly to f on $X\backslash E_\varepsilon$ because $\{f'_n\}$ converges uniformly to f'on $X\backslash E'_\varepsilon$. Since $\mu(E_\varepsilon) \leq \mu(E'_\varepsilon) + \mu(N) = \mu(E'_\varepsilon) < \varepsilon$, it follows that $\{f_n\}$ converges almost uniformly. □

Corollary 6.14. *Take a sequence $\{f_n\}$ of measurable functions. Suppose $|f_n| \leq g \in L^p$ and $f_n \to f$ almost everywhere. Then $f_n \to f$ almost uniformly.*

Proof. Consider the proof of Proposition 6.13. Observe that the assumption $\mu(X) < \infty$ was used there only to ensure that $\mu(E'_1(m)) < \infty$. We now show that the condition $\mu(E'_1(m)) < \infty$ still holds if we assume dominated convergence instead, and so we are done, as the argument reduces to the previous proof. If $|f_n| \leq g \in L^p$ and $f_n \to f$ a.e., then $|f| \leq g$ (so that $f_n, f \in L^p$) and $|f_n - f| \leq |f_n| + |f| \leq 2g$ (see Proposition 6.4). By setting $G'(m) = \{x \in X' \colon 2g(x) \geq \frac{1}{m}\}$ in $\mathcal{X}$, we get

$$F'_n(m) = \big\{x \in X' \colon \tfrac{1}{m} \leq |f_n(x) - f(x)|\big\} \subseteq \big\{x \in X' \colon \tfrac{1}{m} \leq 2g(x)\big\} = G'(m)$$

for all n, and hence

$$E_1'(m) = \bigcup_{n=1}^{\infty} F_n'(m) \subseteq G'(m).$$

Since $\int g^p \, d\mu < \infty$, it follows that $\mu(\{x \in X\colon g^p(x) \geq \varepsilon\}) < \infty$ for every $\varepsilon > 0$ (Problem 3.9), and so $\mu(G'(m)) < \infty$. Thus $\mu(E_1'(m)) < \infty$. □

6.4 Problems

Problem 6.1. Let $(\mathbb{R}, \Re, \lambda)$ be the Lebesgue measure space.

(a) If $f_n = n^{-1/p}\chi_{[0,\,n]}$ for each $n \geq 1$, then $\{f_n\}$ converges uniformly to the null function but does not converge in L^p for any $p \geq 1$.

Hint: If $n = 2m$, then show that $\|f_n - f_m\|_p^p \geq \frac{n-m}{n} = \frac{1}{2}$.

(b) If $f_n = n^{1/p}\chi_{[n,\,n+(1/n^2)]}$ for each $n \geq 1$, then $\{f_n\}$ converges in L^p to the null function 0 for every $p \geq 1$ but does not converge uniformly.

Hint: Show that $\{f_n\}$ converges pointwise to 0 but $\sup_{x\in\mathbb{R}} |f_n(x)| = n^{1/p}$.

Problem 6.2. Take the *finite* measure space $\big([0,1],\, \wp([0,1]) \cap \Re,\, \lambda\big)$, where λ is the restriction of Lebesgue measure to the σ-algebra $\wp([0,1]) \cap \Re$ (cf. Problem 2.11). In other words, let λ be the Lebesgue measure acting on the Borel subsets of $[0,1]$, which is a probability measure.

(a) If $f_n = (n+1)\chi_{[1/(n+1),\,2/(n+1)]}$ for $n \geq 1$, then $\{f_n\}$ converges pointwise to the null function 0 but does not converge in L^p for any $p \geq 1$.

Hint: If $n \geq 2m+1$, then $\|f_n - f_m\|_p^p \geq (2^{p-1}+1)(m+1)^{p-1}$.

Note that the sequence $\{f_n\}$ does not converge uniformly because it converges pointwise to 0 but $\sup_{x\in[0,1]} |f_n(x)| = (n+1)$, which is a consequence of Proposition 6.3(b) since $\{f_n\}$ does not converge in L^p.

(b) Consider the intervals $I_{k,j} = \big[\frac{j-1}{k}, \frac{j}{k}\big]$ for each pair of integers j and k such that $1 \leq j \leq k$. For each $k \geq 1$ take the finite sequence $\{I_{k,j}\}_{1\leq j\leq k}$. Stack these finite sequences to get the infinite sequence of intervals

$$\begin{aligned}
&\{I_n\}_{n\geq 1} = \big\{\{I_{k,j}\}_{1\leq j\leq k}\big\}_{k\geq 1}\\
&= \big\{\{I_{1,1}\}, \{I_{2,1}, I_{2,2}\}, \{I_{3,1}, I_{3,2}, I_{3,3}\}, \{I_{4,1}, I_{4,2}, I_{4,3}, I_{4,4}\}, \ldots\big\}\\
&= \big\{[0,1], [0, \tfrac{1}{2}], [\tfrac{1}{2}, 1], [0, \tfrac{1}{3}], [\tfrac{1}{3}, \tfrac{2}{3}], [\tfrac{2}{3}, 1], [0, \tfrac{1}{4}], [\tfrac{1}{4}, \tfrac{2}{4}], [\tfrac{2}{4}, \tfrac{3}{4}], [\tfrac{3}{4}, 1], \ldots\big\}.
\end{aligned}$$

Show that if $f_n = \chi_{I_n}$ for each $n \geq 1$, then $\{f_n\}$ converges in L^p to the null function for every $p \geq 1$, but the real-valued sequence $\{f_n(x)\}$ does

not converge for every x in $[0,1]$ (i.e., $\{f_n\}$ does not converge pointwise everywhere, and so it does not converge almost everywhere).

Hint: First note that the real-valued sequence $\{\lambda(I_n)\}$ is bounded and decreasing, thus convergent. For every $m \geq 1$ there is an n_m such that $\lambda(I_{n_m}) \leq \frac{1}{m}$. Hence $\|f_n\|_p \to 0$. Next take an arbitrary x in $[0,1]$. The real-valued sequence $\{f_n(x)\}$ has a subsequence constantly equal to 1 and another constantly equal to 0. Thus $\{f_n(x)\}$ does not converge.

Problem 6.3. Consider the Lebesgue measure space $(\mathbb{R}, \Re, \lambda)$ and, for each $n \geq 1$, take the characteristic function $f_n = \chi_{[n,\, n+1]}$. Prove the assertions.

(a) $\{f_n\}$ converges pointwise (and so a.e.) to the null function.

(b) $\{f_n\}$ does not converge in measure (so not uniformly and not in L^p).

Hint: Verify that $\lambda(\{x \in X : |f_m(x) - f_n(x)| \geq \frac{1}{2}\}) = 2$ for every $m \neq n$. Thus conclude that $\{f_n\}$ is not Cauchy in measure (Proposition 6.6(a)).

(c) $\{f_n\}$ does not converge almost uniformly (and so not uniformly a.e.).

Problem 6.4. Let $g : \mathbb{R} \to \mathbb{R}$ and $f_n : \mathbb{R} \to \mathbb{R}$ for each positive integer n be real-valued functions on $\mathbb{R}$ given by

$$g(x) = \begin{cases} 0, & x \leq 0, \\ \frac{1}{\sqrt{x}}, & x \in (0,1], \\ \frac{1}{x^2}, & x \in [1,\infty), \end{cases} \qquad f_n(x) = f_{g,n}(x) = \begin{cases} n, & g(x) \geq n, \\ 0, & \text{otherwise.} \end{cases}$$

Consider the Lebesgue measure space $(\mathbb{R}, \Re, \lambda)$. Show that $g \in L^1$, and

(a) $\{f_n\}$ is dominated by g and converges pointwise to the null function 0,

(b) $\{f_n\}$ converges to 0 almost uniformly but does not converge uniformly almost everywhere.

Hint: $\sup_{x \in X \setminus N} |f_n(x)| \to \infty$ for every set N of Lebesgue measure zero.

Problem 6.5. The *symmetric difference* of two sets A and B is the set

$$A \triangledown B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B).$$

Consider a measure space $(X, \mathcal{X}, \mu)$ and let E and F be arbitrary sets in $\mathcal{X}$. We say that the sets E and F are *equivalent* (or *μ-equivalent*), denoted by $E \sim F$, if $\mu(E \triangledown F) = 0$. The relation $\sim$ is in fact an equivalence relation on $\mathcal{X}$. Define the function $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ by $d(E,F) = \mu(E \triangledown F)$ for every E, F in $\mathcal{X}$. Verify that $d(E,F) \geq 0$, $d(E,F) = d(F,E)$, and (triangle inequality) $d(E,F) \leq d(E,G) + d(G,F)$ for every E, F, G in $\mathcal{X}$. (Observe that d is a

pseudometric on $\mathcal{X}$ — cf. Section 11.1 — and so it induces a *metric* on the quotient space $\mathcal{X}/\!\sim$.) Take a sequence $\{E_n\}$ of sets in $\mathcal{X}$ and, for each $n \geq 1$, set $f_n = \chi_{E_n} : X \to \mathbb{R}$, the characteristic function of E_n. Show that $\{\chi_{E_n}\}$ is Cauchy in measure if and only if $\lim_{m,n} \mu(E_m \triangledown E_n) = 0$. That is,

$$\{f_n\} \text{ is Cauchy in measure if and only if } \lim_{m,n} d(E_m, E_n) = 0.$$

Problem 6.6. The following diagrams show the relationships among almost everywhere convergence, almost uniform convergence, convergence in L^p, and convergence in measure. These will be denoted by (a.e.), (a.u.), (L^p), and (μ), respectively. The first diagram considers the general case (with no additional assumption), the second one considers the case of finite measure (i.e., if $\mu(X) < \infty$), and the third diagram considers the case of dominated convergence (i.e., when $|f_n| \leq g \in L^p$).

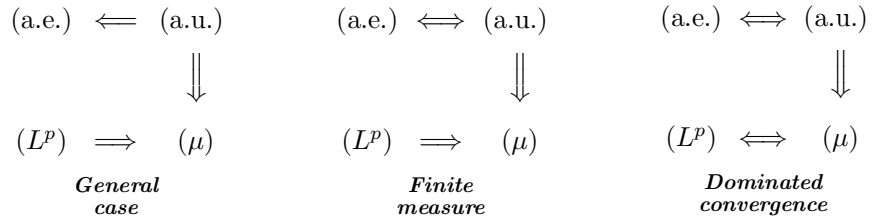


Fig. 6.1. Relationships among convergence modes in three cases.

As usual, the arrows mean implication. Show that the above diagrams are correct and complete in the sense that all arrows are true and no arrow can be added except for the trivial ones (i.e., up to *modus ponens* — for example, it is obvious from these diagrams that (a.e.) implies (μ) in a finite measure space and also that (a.e.) implies (L^p) under the dominance assumption).

Problem 6.7. Now let (u.a.e.) denote uniform convergence almost everywhere. Show that the implications

$$\text{(u.a.e.)} \implies \text{(a.e.)} \qquad \text{and} \qquad \text{(u.a.e.)} \implies \text{(a.u.)}$$

hold true, and their converses fail even under finite measure and dominance assumptions. Also, (u.a.e.) implies (L^p) under finite measure or dominance but not in general, and the converse fails even under both assumptions.

Problem 6.8. Show that the Dominated Convergence Theorem holds if almost everywhere convergence is replaced with convergence in measure.

Hint: Consider the Dominated Convergence Theorem (Theorem 4.7). Use Problem 6.6 (specifically, Proposition 6.9) and Problem 5.16 for $p = 1$.

Problem 6.9. Prove the Vitali Convergence Theorem, which reads as follows. Take an arbitrary $p \geq 1$. If $\{f_n\}$ is a sequence of real-valued functions in $L^p(X, \mathcal{X}, \mu)$, then $f_n \to f$ in L^p for some $f \in L^p$ if and only if the following three assumptions hold true.

(a) $f_n \to f$ in measure.

(b) For each $\varepsilon > 0$ there is an $E_\varepsilon \in \mathcal{X}$ with $\mu(E_\varepsilon) < \infty$ such that for all $n \geq 1$

$$\int_E |f_n|^p \, d\mu < \varepsilon^p \quad \text{for every} \quad E \in \mathcal{X} \text{ for which } E \cap E_\varepsilon = \varnothing.$$

(c) For each $\varepsilon > 0$ there is a $\delta_\varepsilon > 0$ such that for all $n \geq 1$

$$\int_E |f_n|^p \, d\mu < \varepsilon^p \quad \text{for every} \quad E \in \mathcal{X} \text{ with } \mu(E) < \delta_\varepsilon.$$

Hint: To prove that assumptions (a), (b), (c) imply $f_n \to f$ in L^p proceed as follows. Use the Minkowski inequality to show that assumption (b) implies $\|f_m - f_n\|_p \leq \left(\int_{E_\varepsilon} |f_n - f_m|^p\right)^{1/p} + 2\varepsilon$ for every $m, n \geq 1$. By the Minkowski inequality also show that (a) and (c) imply $\left(\int_{E_\varepsilon} |f_n - f_m|^p\right)^{1/p} \leq 3\varepsilon$ for every $m, n \geq n_\varepsilon$ for some $n_\varepsilon \geq 1$. Thus conclude that $\{f_n\}$ is Cauchy in L^p. Apply Proposition 6.6 for uniqueness of the limit f.

Problem 6.10. Take an L^p-valued sequence $\{f_n\}$ for any $p \geq 1$. Prove:

$$f_n \to f \in L^p \text{ a.e. and } \|f_n\|_p \to \|f\|_p \quad \text{implies} \quad f_n \to f \in L^p.$$

Hint: Use Proposition 6.4. Compare with Problems 4.14 and 5.17. Note that Problem 6.10 is the nontrivial converse of Problem 5.17.

Problem 6.11. For any $p \geq 1$, take $f \in L^p(X, \mathcal{X}, \mu)$. For each integer $n \geq 0$, consider the set $E_n = \{x \in X : |f(x)| \geq n\} \in \mathcal{X}$. Show that, as $n \to \infty$,

$$\int_{E_n} |f|^p d\mu \to 0.$$

Hint: Recall that $\lim_n \mu(E_n) = 0$ (cf. hint to Problem 5.19). Set $g_n = \chi_{E_n}|f|$. So $\|g_n\|^p = \int_{E_n} |f|^p d\mu$. Verify that $E_n = \{x \in X : g_n \geq n\}$. Take $\alpha > 0$. Set $E_\alpha(g) = \{x \in X : g \geq \alpha\}$ for an arbitrary measurable function g. Therefore, $E_\alpha(g_n) = \{x \in X : g_n \geq \alpha\} = \{x \in X : |\chi_{E_n} f(x)| > \alpha\} = \{x \in E_n : |f(x)| > \alpha\}$ $= E_n \cap E_\alpha(f)$. Hence $\mu(E_\alpha(g_n)) \leq \mu(E_n)$. Show that $g_n \to 0$ in measure. Since $g_n \leq |f| \in L^p$, Proposition 6.9 ensures that $g_n \to 0$ in L^p.

Suggested Readings

Bartle [5], Berberian [9], Halmos [31], Kingman and Taylor [39], Munroe [54].



7

Decomposition of Measures

7.1 The Jordan Decomposition Theorem

Take a signed measure $\nu\colon \mathcal{X} \to \mathbb{R}$ on a σ-algebra $\mathcal{X}$ of subsets of a set X (cf. Definition 2.3). We saw in Section 2.2 that if μ and η are finite measures, then $\nu = \mu - \eta$ is a signed measure. In this section we show that all signed measures ν admit a decomposition into a difference of two finite measures.

Definition 7.1. Consider a signed measure ν on a σ-algebra $\mathcal{X}$. A set A^+ in $\mathcal{X}$ is *positive* with respect to ν if $\nu(A^+ \cap E) \geq 0$ for all E in $\mathcal{X}$. A set A^- in $\mathcal{X}$ is *negative* with respect to ν if $\nu(A^- \cap E) \leq 0$ for all E in $\mathcal{X}$. A set N in $\mathcal{X}$ is *null* with respect to ν if $\nu(N \cap E) = 0$ for all E.

In other words, let ν be an arbitrary signed measure on an arbitrary σ-algebra $\mathcal{X}$ of subsets of an arbitrary set X. A measurable set (i.e., a set A in $\mathcal{X}$) is *positive*, *negative*, or *null* with respect to ν if each measurable subset of it (i.e., if each set in the sub-σ-algebra $\wp(A) \cap \mathcal{X}$) has nonnegative, nonpositive, or null measure, respectively. The set X always has a measurable partition consisting of a positive and a negative set with respect to ν.

Theorem 7.2. (Hahn Decomposition Theorem). Let $\mathcal{X}$ be a σ-algebra of subsets of a set X. *If ν is a signed measure on $\mathcal{X}$, then there exists a measurable partition $\{A^+, A^-\}$ of X such that A^+ is positive and A^- is negative with respect to ν.*

Proof. Consider a signed measure ν on $\mathcal{X}$. We show that there exists a pair of sets A^+ and A^- in $\mathcal{X}$ such that $A^+ \cup A^- = X$, $A^+ \cap A^- = \varnothing$, A^+ is

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_7

positive, and A^- is negative. Let $\mathcal{A}^+ \subseteq \mathcal{X}$ be the collection of all positive sets with respect to ν, which is not empty (since it contains the empty set). Set $\alpha = \sup_{A\in\mathcal{A}^+} \nu(A)$ and take a sequence $\{A_n\}$ of sets in $\mathcal{A}^+$ such that $\sup_n \nu(A_n) = \alpha$. Thus $A^+ = \bigcup_n A_n$ is a positive set (Problem 7.2(b)) with $0 \le \nu(A^+) = \alpha < \infty$ (since $\nu(A_n) \le \nu(A^+) \le \alpha$ for all n — Problem 7.3). Take its complement $A^- = X\backslash A^+$. If A^- is a negative set, then we are done.

Claim. $A^- = X\backslash A^+$ is a negative set.

Proof of Claim. Suppose A^- is not negative. Then it has a measurable subset E_0 such that $\nu(E_0) > 0$. If E_0 is a positive set, then $\nu(A^+ \cup E_0) > \alpha$ (because $A^+ \cap E_0 = \varnothing$), which is a contradiction ($\alpha = \sup_{A\in\mathcal{A}} \nu(A)$). Thus E_0 is not positive, so it has measurable subsets of negative measure. Let n_0 be the smallest positive integer such that E_0 has a measurable subset of measure not greater than $-\frac{1}{n_0}$, say E_1 with $\nu(E_1) \le -\frac{1}{n_0}$. Observe that

$$\nu(E_0\backslash E_1) \;=\; \nu(E_0) - \nu(E_1) \;>\; \nu(E_0) \;>\; 0$$

(cf. Problem 7.1(a)). If $E_0\backslash E_1$ is a positive set, then $\nu(A^+ \cup (E_0\backslash E_1)) > \alpha$ (because $A^+ \cap (E_0\backslash E_1) = \varnothing$), which is again a contradiction. Thus $E_0\backslash E_1$ is not positive, so it has measurable subsets of negative measure. Let n_1 be the smallest positive integer such that $E_0\backslash E_1$ has a measurable subset of measure not greater than $-\frac{1}{n_1}$, say E_2 with $\nu(E_2) \le -\frac{1}{n_1}$. Again, note that

$$\nu\big(E_0\backslash(E_1\cup E_2)\big) = \nu(E_0) - \nu(E_1\cup E_2) = \nu(E_0) - \big(\nu(E_1)+\nu(E_2)\big) > \nu(E_0) > 0$$

(because $E_1 \cap E_2 = \varnothing$). As before, $E_0\backslash(E_1 \cup E_2)$ is not positive, so it has measurable subsets of negative measure. Let n_2 be the smallest positive integer such that $E_0\backslash(E_1 \cup E_2)$ has a measurable subset of measure not greater than $-\frac{1}{n_2}$, say E_3 with $\nu(E_3) \le -\frac{1}{n_2}$. This leads to the inductive construction of a sequence $\{E_k\}_{k=1}^{\infty}$ of pairwise disjoint measurable sets and a sequence $\{n_k\}_{k=1}^{\infty}$ of integers with each n_k being the smallest positive integer for which $E_0\backslash\bigcup_{i=1}^{k} E_i$ has a measurable subset of measure not greater than $-\frac{1}{n_k}$. Moreover, $\nu(E_{k+1}) \le -\frac{1}{n_k}$ for every $k \ge 0$, and so $\sum_{k=0}^{\infty} \frac{1}{n_k} < \infty$. Actually, by setting $E = \bigcup_{k=1}^{\infty} E_k$ in $\mathcal{X}$ we get

$$-\infty \;<\; \nu(E) \;=\; \sum_{k=1}^{\infty} \nu(E_k) \;\le\; -\sum_{k=0}^{\infty} \tfrac{1}{n_k} \;<\; 0,$$

since $\{E_k\}_{k=1}^{\infty}$ consists of disjoint sets. Thus $\frac{1}{n_k} \to 0$ as $k \to \infty$. Note that

$$\nu(E_0\backslash E) = \nu(E_0) - \nu(E) > \nu(E_0) > 0.$$

The set $E_0\backslash E$ is indeed positive. In fact, suppose $E_0\backslash E$ has a measurable subset of negative measure, say E' with $\nu(E') < 0$. Since $n_k \to \infty$ as $k \to \infty$,

take k large enough so that $\frac{1}{n_k-1} < -\nu(E')$; that is, $\nu(E') < -\frac{1}{n_k-1}$. But $E' \subseteq E_0\backslash E \subseteq E_0\backslash \bigcup_{i=1}^{k} E_i$, so $E_0\backslash \bigcup_{i=1}^{k} E_i$ has a measurable subset of measure less than $-\frac{1}{n_k-1}$, which contradicts the fact that n_k is the smallest positive integer for which $E_0\backslash \bigcup_{i=1}^{k} E_i$ has a measurable subset of measure not greater than $-\frac{1}{n_k}$. Thus every measurable subset of $E_0\backslash E$ has a nonnegative measure, and so $E_0\backslash E$ is a positive set. Therefore, since $A^+ \cap (E_0\backslash E) = \varnothing$ (because $E_0 \subseteq A^-$) and $\nu(E_0\backslash E) > 0$, it follows that $\nu(A^+ \cup (E_0\backslash E)) > \alpha$, which is again a contradiction. Outcome: A^- must be a negative set. □

Let $(X, \mathcal{X})$ be a measurable space. A measurable partition $\{A^+, A^-\}$ of X, where $A^+ \in \mathcal{X}$ is positive and $A^- \in \mathcal{X}$ is negative with respect to a signed measure ν on $\mathcal{X}$, is called a *Hahn decomposition* of X with respect to ν. Given a signed measure ν on $\mathcal{X}$, a Hahn decomposition of X is not unique (if there exists a nonempty null set with respect to ν). In fact, if $\{A^+, A^-\}$ is a Hahn decomposition of X and N is a null set, then $\{A^+ \cup N, A^-\backslash N\}$ and $\{A^+\backslash N, A^- \cup N\}$ are also Hahn decompositions of X (all with respect to ν). However, this lack of uniqueness is indistinguishable for the signed measure ν, and so it is not a disadvantage of the Hahn decomposition.

Proposition 7.3. *Suppose* $\{A_1^+, A_1^-\}$ *and* $\{A_2^+, A_2^-\}$ *are Hahn decompositions of* X *with respect to a signed measure* ν *on* $\mathcal{X}$*. Then, for every* $E \in \mathcal{X}$,

$$\nu(A_1^+ \cap E) = \nu(A_2^+ \cap E) \quad \textit{and} \quad \nu(A_1^- \cap E) = \nu(A_2^- \cap E).$$

Proof. If A, B, C are arbitrary sets, then $\{A\cap(B\backslash C), A\cap B\cap C\}$ is a partition of $A \cap B$. In $\mathcal{X}$, if $\nu(A \cap (B\backslash C)) = 0$, then $\nu(A \cap B) = \nu(A \cap B \cap C)$. Since $E \cap (A_1^+\backslash A_2^+) \subseteq A_1^+ \cap A_2^-$ and $E \cap (A_2^+\backslash A_1^+) \subseteq A_2^+ \cap A_1^-$, it follows that

$$\nu(E \cap (A_1^+\backslash A_2^+)) = 0 \quad \text{and} \quad \nu(E \cap (A_2^+\backslash A_1^+)) = 0.$$

Thus

$$\nu(E \cap A_1^+) = \nu(E \cap A_1^+ \cap A_2^+) \quad \text{and} \quad \nu(E \cap A_2^+) = \nu(E \cap A_2^+ \cap A_1^+),$$

and so

$$\nu(E \cap A_1^+) = \nu(E \cap A_2^+).$$

Analogously, replacing A_1^+ with A_1^-, and A_2^+ with A_2^-, we get

$$\nu(E \cap A_1^-) = \nu(E \cap A_2^-).$$ □

Consider a Hahn decomposition $\{A^+, A^-\}$ of X with respect to a signed measure ν on $\mathcal{X}$. Problem 2.11 ensures that the set functions $\nu^+ : \mathcal{X} \to \mathbb{R}$ and $\nu^- : \mathcal{X} \to \mathbb{R}$ defined for every set E in $\mathcal{X}$ by

$$\nu^+(E) = \nu(A^+ \cap E) \quad \text{and} \quad \nu^-(E) = -\nu(A^- \cap E)$$

are finite measures on $\mathcal{X}$. The measures ν^+ and ν^- are called the *positive variation* and *negative variation* of ν, respectively. Note: (i) ν^+ and ν^- are unambiguously defined since their definitions do not depend on the Hahn decomposition $\{A^+, A^-\}$ (by Proposition 7.3), and (ii) $\nu^+(A^-) = \nu^-(A^+) = 0$, signifying that ν^+ and ν^- are *singular* (as it will be defined in Definition 7.9).

Theorem 7.4. (Jordan Decomposition Theorem). *Let $(X, \mathcal{X})$ be a measurable space. Suppose ν is a signed measure on $\mathcal{X}$. Then*

$$\nu = \nu^+ - \nu^-,$$

where ν^+ and ν^- are the positive and negative variations of ν. If

$$\nu = \mu - \eta,$$

where μ and η are finite measures on $\mathcal{X}$, then

$$\nu^+ \le \mu \quad \text{and} \quad \nu^- \le \eta.$$

Proof. Take an arbitrary Hahn decomposition $\{A^+, A^-\}$ of X with respect to a signed measure ν on $\mathcal{X}$. Since $\{A^+, A^-\}$ is a partition of X, it follows that $(A^+ \cap E) \cup (A^- \cap E) = E$ and $(A^+ \cap E) \cap (A^- \cap E) = \varnothing$, and hence

$$\nu(E) = \nu(A^+ \cap E) + \nu(A^- \cap E) = \nu^+(E) - \nu^-(E),$$

for every $E \in \mathcal{X}$. If μ and η are finite measures on $\mathcal{X}$ (so that $0 \le \mu(E) < \infty$ and $0 \le \eta(E) < \infty$ for every $E \in \mathcal{X}$) such that $\nu = \mu - \eta$, then

$$\nu^+(E) = \nu(A^+ \cap E) = \mu(A^+ \cap E) - \eta(A^+ \cap E) \le \mu(A^+ \cap E) \le \mu(E),$$

$$\nu^-(E) = -\nu(A^- \cap E) = -\mu(A^- \cap E) + \eta(A^- \cap E) \le \eta(A^- \cap E) \le \eta(E),$$

for every $E \in \mathcal{X}$ (Proposition 2.2(a)). Therefore, $\nu^+ \le \mu$ and $\nu^- \le \eta$. □

The sum of finite measures is again a finite measure. The *total variation* of a signed measure $\nu\colon \mathcal{X} \to \mathbb{R}$ is the finite measure $|\nu|\colon \mathcal{X} \to \mathbb{R}$ defined by

$$|\nu| = \nu^+ + \nu^-.$$

Example 7A. Let ν be a signed measure on $\mathcal{X}$. The total variation $|\nu|$ coincides with the (ordinary) variation μ discussed in Example 2I. In fact, as in Example 2I, the (ordinary) variation μ is the measure defined by

$$\mu(E) = \sup_{\{E^+, E^-\} \in \boldsymbol{E}(2)} \big(\nu(E^+) - \nu(E^-)\big) \quad \text{for every} \quad E \in \mathcal{X},$$

where the supremum is taken over all measurable partitions $\{E^+, E^-\}$ of E consisting of two sets such that $\nu(E^+) \ge 0$ and $\nu(E^-) \le 0$. If $\{A^+, A^-\}$ is any Hahn decomposition of X with respect to ν and $\mathcal{E} = \wp(E) \cap \mathcal{X}$, then

$$\begin{aligned}\nu^+(E)+\nu^-(E) &= \nu(A^+\cap E)-\nu(A^-\cap E)\le \mu(E)\\ &\le \sup_{E'\in\mathcal{E}}\nu(E')-\inf_{E'\in\mathcal{E}}\nu(E') = \nu^+(E)+\nu^-(E),\end{aligned}$$

the last identity follows from Theorem 7.4 (via Problem 7.5). So $\mu=|\nu|$,

$$\mu(E)=\nu^+(E)+\nu^-(E)=|\nu|(E) \quad \text{for every} \quad E\in\mathcal{X}.$$

Proposition 7.5. *Consider a measure space $(X,\mathcal{X},\mu)$, and take a function $f\in\mathcal{L}(X,\mathcal{X},\mu)$. If $\nu\colon\mathcal{X}\to\mathbb{R}$ is the signed measure defined in Lemma 4.6,*

$$\nu(E)=\int_E f\,d\mu \quad \textit{for every} \quad E\in\mathcal{X},$$

then the measures ν^+, ν^-, and $|\nu|$ are given for each $E\in\mathcal{X}$ by

$$\nu^+(E)=\int_E f^+d\mu, \qquad \nu^-(E)=\int_E f^-d\mu, \qquad \textit{and} \qquad |\nu|(E)=\int_E |f|\,d\mu.$$

Proof. For each $f\in\mathcal{L}(X,\mathcal{X},\mu)$, consider the sets

$$F_+=\{x\in X\colon\ f(x)>0\}, \qquad F_-=\{x\in X\colon\ f(x)<0\},$$
$$F_0=\{x\in X\colon\ f(x)=0\},$$
$$F^+=F_+\cup F_0=\big\{x\in X\colon\ f\ge 0\big\}, \qquad F^-=F_-\cup F_0=\big\{x\in X\colon\ f\le 0\big\}.$$

Thus $\{F^+,F_-\}$ is a measurable partition of X such that $\nu(F^+\cap E)\ge 0$ and $\nu(F_-\cap E)\le 0$ for each $E\in\mathcal{X}$, and so $\{F^+,F_-\}$ is a Hahn decomposition of X with respect to ν. Since $\int_{F_-\cap E} f\,d\mu=\int_{F^-\cap E} f\,d\mu$ (cf. Proposition 3.7(a), Problem 3.8(a), and Definition 4.1), it follows for every $E\in\mathcal{X}$ that

$$\nu^+(E) = \nu(F^+\cap E) = \int_{F^+\cap E} f\,d\mu = \int_E f\chi_{F^+}\,d\mu = \int_E f^+d\mu,$$

$$\nu^-(E) = -\nu(F_-\cap E) = -\int_{F_-\cap E} f\,d\mu = \int_E -f\chi_{F^-}\,d\mu = \int_E f^-d\mu,$$

$$|\nu|(E) = \nu^+(E)+\nu^-(E) = \int_E (f^++f^-)\,d\mu = \int_E |f|\,d\mu. \qquad \square$$

7.2 The Radon–Nikodym Theorem

Definition 7.6. Take a measurable space $(X,\mathcal{X})$ and let $E\in\mathcal{X}$ be an arbitrary measurable set. A measure μ on $\mathcal{X}$ is *absolutely continuous* with respect to a measure η on $\mathcal{X}$ if

$$\eta(E)=0 \quad \text{implies} \quad \mu(E)=0$$

(i.e., $\mu(E)=0$ for every $E\in\mathcal{X}$ such that $\eta(E)=0$). Notation: $\mu\ll\eta$.

Let μ and η be measures on $\mathcal{X}$, let E be an arbitrary measurable set in $\mathcal{X}$, and consider the following statements.

(a) For every $\varepsilon > 0$ there is a $\delta_\varepsilon > 0$ (that does not depend on E) for which

$$\eta(E) < \delta_\varepsilon \quad \text{implies} \quad \mu(E) < \varepsilon.$$

(b) The measure μ is absolutely continuous with respect to η (i.e., $\mu \ll \eta$).

These are equivalent if μ is a finite measure, as will be shown in Proposition 7.7 below, thus justifying the terminology "absolute continuity".

Proposition 7.7. *Assertions* (a) *and* (b) *above are related as follows*:

(a) *implies* (b) *and* (b) *implies* (a) *if* μ *is finite.*

Proof. Suppose (a) holds. If $\eta(E) = 0$ for some $E \in \mathcal{X}$, then $\mu(E) < \varepsilon$ for all $\varepsilon > 0$, which means that $\mu(E) = 0$. So (a) implies (b). Conversely, suppose (a) fails. Thus there is an $\varepsilon > 0$ such that for every $\delta > 0$ there is an $E_\delta \in \mathcal{X}$ for which $\eta(E_\delta) < \delta$ and $\mu(E_\delta) \geq \varepsilon$. In particular, for every $n \geq 1$ there is an $E_n \in \mathcal{X}$ such that $\eta(E_n) < \frac{1}{2^n}$ and $\mu(E_n) \geq \varepsilon$. Set $E'_n = \bigcup_{k=n}^{\infty} E_k$ in $\mathcal{X}$. So

$$\eta(E'_n) \leq \sum_{k=n}^{\infty} \eta(E_k) < \sum_{k=n}^{\infty} \frac{1}{2^k} = \frac{1}{2^{n-1}} \quad \text{and} \quad \mu(E'_n) \geq \mu(E_n) \geq \varepsilon,$$

for every $n \geq 1$. Now set $E' = \bigcap_{n=1}^{\infty} E'_n$ in $\mathcal{X}$. Since $\{E'_n\}$ is a decreasing sequence of sets in $\mathcal{X}$, and since $\eta(E'_1) \leq 1$ and $\mu(E'_1) < \infty$ if μ is a finite measure, it follows by Proposition 2.2(d) that

$$\eta(E') = \lim_n \eta(E'_n) = 0 \quad \text{and} \quad \mu(E') = \lim_n \mu(E'_n) \geq \varepsilon.$$

Hence (b) fails. Equivalently, (b) implies (a) if μ is a finite measure. □

Propositions 3.5(c) and 3.7(b) ensure that *if* η *is a measure on* $\mathcal{X}$ *and* f *is a function in* $\mathcal{M}(X, \mathcal{X})^+$ (a nonnegative extended real-valued measurable function), *then the set function* μ *on* $\mathcal{X}$ *defined by*

$$\mu(E) = \int_E f \, d\eta \quad \textit{for every} \quad E \in \mathcal{X}$$

is a measure (*a measure* μ *generated by a measure* η *via a function* f) *that is absolutely continuous with respect to* η. What comes as a nice and perhaps unexpected result is that the converse holds if the measures μ and η are σ-finite. That is, in this case, there exists a function f in $\mathcal{M}(X, \mathcal{X})^+$ such that μ is expressed as an integral of f with respect to η. Moreover, the function f is unique η-a.e. (i.e., if g in $\mathcal{M}(X, \mathcal{X})^+$ is such that $\mu(E) = \int_E g \, d\eta$ for every $E \in \mathcal{X}$, then $g = f$ η-almost everywhere). Such a converse is a fundamental result in measure theory. We prove it next.

Theorem 7.8. (Radon–Nikodym Theorem). *Take an arbitrary measurable space $(X, \mathcal{X})$. If μ and η are σ-finite measures on $\mathcal{X}$, and if μ is absolutely continuous with respect to η, then there exists a unique (η-almost everywhere unique) real-valued function f in $\mathcal{M}(X, \mathcal{X})^+$ such that*

$$\mu(E) = \int_E f \, d\eta \quad \text{for every} \quad E \in \mathcal{X}.$$

Proof. Let μ and η be measures on $\mathcal{X}$ such that $\mu \ll \eta$. We split the proof into two parts. Part (a) proves the theorem for finite measures. Part (b) extends the proof for σ-finite measures.

(a) Take an arbitrary real number $\alpha > 0$. Suppose μ and η are finite measures. Thus $\nu_\alpha = \mu - \alpha\eta$ is a (real-valued) signed measure. Let $\{A_\alpha^+, A_\alpha^-\}$ be a Hahn decomposition for X with respect to the signed measure ν_α. Consider a sequence $\{E_k\}_{k\geq 1}$ of sets in $\mathcal{X}$ recursively defined by

$$E_{k+1} = A_{(k+1)\alpha}^- \Big\backslash \bigcup_{j=1}^{k} E_j \quad \text{with} \quad E_1 = A_\alpha^-.$$

It is immediately verified by induction that

(i) $\{E_k\}_{k\geq 1}$ is a sequence of disjoint sets,

(ii) $$\bigcup_{j=1}^{k} E_j = \bigcup_{j=1}^{k} A_{j\alpha}^- \quad \text{for every } k \geq 1,$$

and hence

$$E_k = A_{k\alpha}^- \Big\backslash \bigcup_{j=1}^{k-1} A_{j\alpha}^- = A_{k\alpha}^- \cap \bigcap_{j=1}^{k-1} A_{j\alpha}^+ \quad \text{for every } k \geq 2.$$

Then $E_k \subseteq (A_{k\alpha}^- \cap A_{(k-1)\alpha}^+)$, which implies that for each $E \in (\wp(E_k) \cap \mathcal{X})$, $\mu(E) - k\,\alpha\,\eta(E) \leq 0$ and $\mu(E) - (k-1)\,\alpha\,\eta(E) \geq 0$, and therefore

(iii) $$(k-1)\,\alpha\,\eta(E) \leq \mu(E) \leq k\alpha\,\eta(E),$$

for every $E \in (\wp(E_k) \cap \mathcal{X})$ and every $k \geq 2$. Set (cf. property (ii))

$$E' = X \Big\backslash \bigcup_{j=1}^{\infty} E_j = X \Big\backslash \bigcup_{j=1}^{\infty} A_{j\alpha}^- = \bigcap_{j=1}^{\infty} A_{j\alpha}^+ \subseteq A_{k\alpha}^+ \quad \text{for all } k \geq 1.$$

So $\mu(E') - k\,\alpha\,\eta(E') \geq 0$, equivalently, $0 \leq k\,\alpha\,\eta(E') \leq \mu(E')$, for all $k \geq 1$. Since μ is a finite measure, it follows that $\eta(E') = 0$. Since $\mu \ll \eta$, we get

(iv) $$\mu(E') = 0.$$

Take the nonnegative real-valued function f_α in $\mathcal{M}^+ = \mathcal{M}(X, \mathcal{X})^+$ given by

$$f_\alpha(x) = \begin{cases} (k-1)\,\alpha, & x \in E_k \text{ for some } k \geq 1, \\ 0, & x \in E' = X \setminus \bigcup_{k=1}^{\infty} E_k. \end{cases}$$

Observe that $\{E', E_k;\, k \geq 1\}$ is a measurable partition of X by property (i). Take an arbitrary $E \in \mathcal{X}$ so that $\{E \cap E', E \cap E_k;\, k \geq 1\}$ is a measurable partition of E. Thus (cf. Problems 3.3(a) and 4.10)

$$\begin{aligned} \int_E f_\alpha \, d\eta &= \int_{\bigcup_{k=1}^{\infty}(E \cap E_k)} f_\alpha \, d\eta = \sum_{k=1}^{\infty} \int_{E \cap E_k} f_\alpha \, d\eta \\ &= \sum_{k=1}^{\infty} (k-1)\,\alpha \int_{E \cap E_k} d\eta = \sum_{k=2}^{\infty} (k-1)\,\alpha\,\eta(E \cap E_k) \\ &\leq \sum_{k=2}^{\infty} \mu(E \cap E_k) = \mu\big(\textstyle\bigcup_{k=2}^{\infty}(E \cap E_n)\big) \leq \mu(E) \end{aligned}$$

by properties (iii) and (i). Similarly, by properties (i), (iii), and (iv) we get

$$\begin{aligned} \mu(E) &= \int_E d\mu = \int_{\bigcup_{k=1}^{\infty}(E \cap E_k)} d\mu = \mu\big(\textstyle\bigcup_{k=1}^{\infty}(E \cap E_k)\big) = \sum_{k=1}^{\infty} \mu(E \cap E_k) \\ &\leq \sum_{k=1}^{\infty} k\,\alpha\,\eta(E \cap E_k) = \sum_{k=1}^{\infty} \int_{E \cap E_k} k\,\alpha \, d\eta = \sum_{k=1}^{\infty} \int_{E \cap E_k} (f_\alpha + \alpha)\, d\eta \\ &= \int_{\bigcup_{k=1}^{\infty}(E \cap E_k)} (f_\alpha + \alpha)\, d\eta \leq \int_E (f_\alpha + \alpha)\, d\eta = \int_E f_\alpha \, d\eta + \alpha\,\eta(E). \end{aligned}$$

Take an arbitrary integer $n \geq 1$, set $\alpha = \left(\frac{1}{2}\right)^n$ and $f_n = f_{(1/2)^n}$. The previously displayed inequalities ensure that

$$\text{(v)} \qquad \int_E f_n \, d\eta \leq \mu(E) \leq \int_E f_n \, d\eta + \left(\tfrac{1}{2}\right)^n \eta(X)$$

for all $n \geq 1$. Hence, for an arbitrary pair of positive integers m and n,

$$\int_E f_n \, d\eta \leq \mu(E) \leq \int_E f_m \, d\eta + \left(\tfrac{1}{2}\right)^m \eta(X),$$

$$\int_E f_m \, d\eta \leq \mu(E) \leq \int_E f_n \, d\eta + \left(\tfrac{1}{2}\right)^n \eta(X),$$

so that

$$\int_E f_n \, d\eta - \int_E f_m \, d\eta \leq \mu(E) - \int_E f_m \, d\eta \leq \left(\tfrac{1}{2}\right)^m \eta(X),$$

$$\int_E f_m \, d\eta - \int_E f_n \, d\eta \leq \mu(E) - \int_E f_n \, d\eta \leq \left(\tfrac{1}{2}\right)^n \eta(X),$$

which implies

$$\left|\int_E (f_m - f_n)\,d\eta\right| \le \left(\tfrac{1}{2}\right)^m \eta(X)$$

whenever $m \le n$. Since this holds for all $E \in \mathcal{X}$, we get

$$\begin{aligned}\int |f_m - f_n|\,d\eta &= \int (f_m - f_n)^+\,d\eta + \int (f_m - f_n)^-\,d\eta \\ &= \int_{F^+_{m,n}} (f_m - f_n)\,d\eta - \int_{F^-_{m,n}} (f_m - f_n)\,d\eta \\ &= \left|\int_{F^+_{m,n}} (f_m - f_n)\,d\eta\right| + \left|\int_{F^-_{m,n}} (f_m - f_n)\,d\eta\right| \le 2\left(\tfrac{1}{2}\right)^m \eta(X)\end{aligned}$$

for each m, n such that $m \le n$, with $F^+_{m,n} = \{x \in X\colon (f_m - f_n)(x) \ge 0\}$ and $F^-_{m,n} = \{x \in X\colon (f_m - f_n)(x) \le 0\}$. Since property (v) holds for all E in $\mathcal{X}$, μ is a finite measure, and f_n is a real-valued function in $\mathcal{M}^+$, it follows that each function f_n lies in $L^1(\eta) = L^1(X, \mathcal{X}, \eta)$. Thus, by the above inequality,

$$\|f_m - f_n\|_1 \le \left(\tfrac{1}{2}\right)^{m-1} \eta(X)$$

if $1 \le m \le n$. Since η is a finite measure, the above inequality says that $\{f_n\}$ is a Cauchy sequence in the Banach space $L^1(\eta)$, and so it converges in $L^1(\eta)$ to, say, $f \in L^1(\eta)$. Then the real-valued sequence $\left\{\int_E f_n\,d\eta\right\}$ converges in $\mathbb{R}$ to $\int_E f\,d\eta$ for every $E \in \mathcal{X}$. In fact, by continuity of the integral (cf. Problem 5.16 and remark preceding it),

$$\left|\int_E f_n\,d\eta - \int_E f\,d\eta\right| \le \int_E |f_n - f|\,d\eta \le \int |f_n - f|\,d\eta = \|f_n - f\|_1 \to 0.$$

Recall that $f = \lim_n f_n \in L^1(\eta)$ is real-valued. Therefore, by property (v),

$$\mu(E) = \lim_n \int_E f_n\,d\eta = \int_E f\,d\eta \quad \text{for every } E \in \mathcal{X}.$$

Note that we may take a nonnegative function f in the equivalence class $[f]$, that is, we may take $f \in \mathcal{M}^+$. Indeed, since $0 \le \mu(E) = \int_E f\,d\eta$ for every $E \in \mathcal{X}$, it follows by Problem 4.5(b) that $f \ge 0$ η-a.e. (and also μ-a.e. because $\mu \ll \eta$) for every $f \in [f]$. Moreover, such an f is η-a.e. unique. In fact, if $g \in L^1(\eta)$ (and $g \in \mathcal{M}^+$) is such that $\mu(E) = \int_E f\,d\eta = \int_E g\,d\eta$ for every $E \in \mathcal{X}$, then $f = g$ η-a.e. by Problem 3.8(d) (or Problem 4.5(c)).

(b) Next assume that the measures μ and η are σ-finite. Thus there are two sequences of $\mathcal{X}$-measurable sets, say $\{A_n\}$ and $\{B_n\}$, such that $\mu(A_n) < \infty$ and $\eta(B_n) < \infty$ for every n, both covering X. Actually, by considering successive unions of them if necessary, we may assume that these sequences are increasing. Set $X_n = A_n \cap B_n$ so that $\{X_n\}$ is an increasing sequence of $\mathcal{X}$-measurable sets such that $\bigcup X_n = \bigcup A_n \cap \bigcup B_n = X$ and, for each n,

$$\mu(X_n) < \infty \quad \text{and} \quad \eta(X_n) < \infty.$$

Take an arbitrary index n. Consider the σ-algebra $\mathcal{X}_n = \wp(X_n) \cap \mathcal{X}$. By part (a) there is a function $g_n \in \mathcal{M}(X_n, \mathcal{X}_n)^+$ such that $\mu(E') = \int_{E'} g_n \, d\eta|_{\mathcal{X}_n}$ for every $E' \in \mathcal{X}_n$ (cf. Problem 2.11). Let f_n be a function on X defined as follows: $f_n(x) = g_n(x)$ if $x \in X_n$, and $f_n(x) = 0$ if $x \in X \backslash X_n$. This function f_n lies in $\mathcal{M}^+ = \mathcal{M}(X, \mathcal{X})^+$ and, for every $E' \in \mathcal{X}_n \subseteq \mathcal{X}$,

$$\mu(E') = \int_{E'} f_n \, d\eta.$$

Recall that $X_n \subseteq X_k$ for every $n \leq k$. Then $E' \in \mathcal{X}_k$ for every $k \geq n$ whenever $E' \in \mathcal{X}_n$. So the above identity holds for all $k \geq n$. That is, the sequence $\{f_n\}$ of functions in $\mathcal{M}^+$ is such that if $E' \in \mathcal{X}_n$ for some $n \geq 1$, then

$$\mu(E') = \int_{E'} f_n \, d\eta = \int_{E'} f_k \, d\eta$$

for every $k \geq n$. Again from part (a), g_n is unique η-a.e., and then so is f_n, for each n. Thus the previous identity ensures that $f_k = f_n$ η-a.e. on X_n for all $k \geq n$. (Same uniqueness argument: see Problem 3.8(d) or Problem 4.5(a) — nonnegative functions in $L^1(\eta)$.) Hence, since each f_n vanishes outside X_n, and since $\{X_n\}$ is an increasing sequence of sets, it follows that $\{f_n\}$ is an increasing sequence of functions in $\mathcal{M}^+$ such that

$$\mu(E \cap X_n) = \int_{E \cap X_n} f_n \, d\eta = \int_E f_n \chi_{X_n} \, d\eta = \int_E f_n \, d\eta$$

for each $E \in \mathcal{X}$ (since $E \cap X_n \in \mathcal{X}_n$ and $f_n = f_n \chi_{X_n}$) and for each $n \geq 1$. Take an arbitrary set E in $\mathcal{X}$ and note that $\{E \cap X_n\}$ is an increasing sequence of sets in $\mathcal{X}$ that covers E (because $\{X_n\}$ is an increasing sequence of sets in $\mathcal{X}$ that covers X). Thus, by Proposition 2.2(c),

$$\mu(E) = \mu\Big(\bigcup\nolimits_n (E \cap X_n)\Big) = \lim_n \mu(E \cap X_n) = \lim_n \int_E f_n \, d\eta.$$

The Monotone Convergence Theorem completes the existence proof. Indeed, since $\{f_n\}$ is an increasing sequence of functions in $\mathcal{M}^+$, it converges pointwise to a function $\tilde{f}$ in $\mathcal{M}^+$, and so it follows by Theorem 3.4 that

$$\mu(E) = \lim_n \int_E f_n \, d\eta = \lim_n \int f_n \chi_E \, d\eta = \int \lim_n f_n \chi_E \, d\eta = \int \tilde{f} \chi_E \, d\eta = \int_E \tilde{f} \, d\eta.$$

The function $\tilde{f} \in \mathcal{M}^+$ is the pointwise limit of an increasing sequence of functions in $\mathcal{M}^+$. Thus it may possibly be extended real-valued; but, it is η-a.e. real-valued. Equivalently, the set $F_{+\infty} = \{x \in X\colon \tilde{f}(x) = +\infty\}$ has measure zero (i.e., $\eta(F_{+\infty}) = 0$). Indeed, recall that $\{X_n\}$ is an increasing sequence of sets that covers X, each f_n is null outside X_n, and $f_k = f_n$ η-a.e. on X_n for all $k \geq n$. Thus, since $\tilde{f}(x) = \lim_n f_n(x)$ for every $x \in X$, it follows

that $f_n = \tilde{f}\chi_{X_n}$ η-a.e., and so $\eta(F_{+\infty} \cap X_n) = \eta(\{x \in X_n\colon\ f_n(x) = +\infty\})$, for each n. However, $f_n \in L^1(X, \mathcal{X}, \eta)$ because $g_n = f_n \chi_{X_n} \in L^1(X_n, \mathcal{X}_n, \eta)$ and $f_n = 0$ on $X \backslash X_n$. Hence $\eta(F_{+\infty} \cap X_n) = 0$ for all n by Problem 3.9(b). Since $F_{+\infty} = F_{+\infty} \cap X = F_{+\infty} \cap \bigcup_n X_n = \bigcup_n (F_{+\infty} \cap X_n)$, it follows that $\eta(F_{+\infty}) \leq \sum_n \eta(F_{+\infty} \cap X_n) = 0$ (Problem 2.8(b)), and so $\eta(F_{+\infty}) = 0$. Set $f = \tilde{f}\chi_{X \backslash F_{+\infty}}$, a real-valued function in $\mathcal{M}^+$ such that (see Problem 3.8)

$$\mu(E) = \int_E \tilde{f}\, d\eta = \int_{E \cap (X \backslash F_{+\infty})} \tilde{f}\, d\eta + \int_{E \cap F_{+\infty}} \tilde{f}\, d\eta = \int_E \tilde{f}\chi_{X \backslash F_{+\infty}}\, d\eta = \int_E f\, d\eta$$

for every $E \in \mathcal{X}$. Such an $f \in \mathcal{M}^+$ is unique η-a.e. by Problem 3.8(d). □

The real-valued function $f \in \mathcal{M}(X, \mathcal{X})^+$ in the statement of the Radon–Nikodym Theorem was not claimed to be integrable. Actually, f is η-integrable (i.e., f lies in $\mathcal{L}(X, \mathcal{X}, \eta)$) if and only if μ is a finite measure. This function f is called the *Radon–Nikodym derivative* of μ with respect to η, which is often written as $f = \frac{d\mu}{d\eta}$ (or $d\mu = f\, d\eta$). As noted in Problem 3.13, no independent meaning is assigned to the symbols $d\mu$ and $d\eta$. So, if μ and η are σ-finite measures such that $\mu \ll \eta$, then there is a unique (η-a.e.) real-valued function $\frac{d\mu}{d\eta}$ in $\mathcal{M}(X, \mathcal{X})^+$ such that for every E in $\mathcal{X}$

$$\mu(E) = \int_E \frac{d\mu}{d\eta}\, d\eta.$$

Remark 7.A (Riesz Representation Theorems): A major application of the Radon–Nikodym Theorem is the Riesz Representation Theorem. One of the versions of it says that *if* $\Phi\colon L^p(\mu) \to \mathbb{R}$ *is a bounded linear functional on the Banach space* $L^p(\mu)$*, then there exits a unique* $g \in L^q(\mu)$ *such that* $\Phi(f) = \int fg\, d\mu$ *for every* $f \in L^p(\mu)$ *and* $\|\Phi\| = \|g\|_q$ (where q is the Hölder conjugate of p; if $p = 1$ so that $q = \infty$, then μ is supposed to be σ-finite). See Proposition 12.A. The Riesz Representation Theorem holds in every Hilbert space (cf. Proposition 12.B), and so it can be proved for $p = 2$ without using the Radon–Nikodym Theorem and, perhaps surprisingly, this can be used to prove the Radon–Nikodym Theorem itself. Chapter 12 is entirely dedicated to Riesz Representation Theorems.

7.3 The Lebesgue Decomposition Theorem

If μ is absolutely continuous with respect to η, the actions of the two measures are synchronized in the sense that sets of small η-measure have small μ-measure (Proposition 7.7). At the opposite end there are measures μ and η whose actions are complementary in the sense that sets of small η-measure may have large μ-measure.

Definition 7.9. Take a measurable space $(X, \mathcal{X})$. A measure μ on $\mathcal{X}$ is *singular* with respect to a measure η on $\mathcal{X}$ (notation: $\mu \perp \eta$) if there exists a measurable partition $\{A, B\}$ of X (i.e., if there exist sets $A, B \in \mathcal{X}$ such that $A \cap B = \varnothing$ and $A \cup B = X$) for which

$$\mu(A) = \eta(B) = 0.$$

The notion of singular measures is implied by the notion of *disjoint supports* (see Problem 2.13), which is also referred to by saying that μ is *concentrated on a set of η-measure zero*. It is clear that $\perp$ is a symmetric relation on the collection of all measures on $\mathcal{X}$ (i.e., $\mu \perp \eta$ if and only if $\eta \perp \mu$). Thus we say that μ and η are *mutually singular*, or simply *singular*, instead of μ is singular with respect to η (or vice versa). The above definition can be easily shown to be equivalently restated as follows: $\mu \perp \eta$ if there exists a partition $\{A, B\}$ of X such that $A \cap E$ and $B \cap E$ lie in $\mathcal{X}$ for every $E \in \mathcal{X}$ and

$$\mu(A \cap E) = \eta(B \cap E) = 0.$$

The partition $\{A, B\}$ is, in fact, a *measurable* partition because $A = A \cap X$ and $B = B \cap X$ lie in $\mathcal{X}$ once X lies in $\mathcal{X}$. The next result is another important consequence of the Radon–Nikodym Theorem.

Theorem 7.10. (Lebesgue Decomposition Theorem). *Consider a measurable space $(X, \mathcal{X})$. If μ and η are σ-finite measures on $\mathcal{X}$, then there exists a unique pair of measures μ_a and μ_s on $\mathcal{X}$ such that $\mu_a \ll \eta$, $\mu_s \perp \eta$, and*

$$\mu = \mu_a + \mu_s.$$

Proof. Let μ and η be σ-finite measures on $\mathcal{X}$. Set $\nu = \eta + \mu$, which is again a σ-finite measure on $\mathcal{X}$ (Problem 2.14). Both η and μ are absolutely continuous with respect to ν (i.e., $\nu(E) = 0$ implies $\eta(E) = \mu(E) = 0$, and so $\eta \ll \nu$ and $\mu \ll \nu$). Then the Radon–Nikodym Theorem says that there are real-valued functions f and g in $\mathcal{M}(X, \mathcal{X})^+$ such that for every $E \in \mathcal{X}$,

$$\eta(E) = \int_E f \, d\nu \quad \text{and} \quad \mu(E) = \int_E g \, d\nu.$$

Take the measurable partition $\{F_0, F_+\}$ of X such that

$$F_0 = \{x \in X\colon\ f(x) = 0\} \quad \text{and} \quad F_+ = \{x \in X\colon\ f(x) > 0\}.$$

Set $\mu_s = \mu_{F_0}$ and $\mu_a = \mu_{F_+}$, where μ_{F_0} and μ_{F_+} are the measures on $\mathcal{X}$ given in Problem 2.11. That is, let μ_s and μ_a be defined, for every $E \in \mathcal{X}$, by

$$\mu_s(E) = \mu(E \cap F_0) \quad \text{and} \quad \mu_a(E) = \mu(E \cap F_+).$$

Since $\eta(F_0) = \int_{F_0} f \, d\nu = 0$ and $\mu_s(F_+) = \mu(F_+ \cap F_0) = \mu(\varnothing) = 0$, we get

$$\mu_s \perp \eta.$$

If $E \in \mathcal{X}$ is such that $\eta(E) = 0$, then $\int f\chi_E \, d\nu = \int_E f \, d\nu = \eta(E) = 0$, and hence $f\chi_E = 0$ ν-a.e. (cf. Proposition 3.7(a)). That is, $f = 0$ ν-a.e. on E, and so $\nu(E \cap F_+) = 0$. As $\mu \ll \nu$, this implies $\mu_a(E) = \mu(E \cap F_+) = 0$. Thus

$$\mu_a \ll \eta.$$

Now note that $\mu(E) = \mu((E \cap F_0) \cup (E \cap F_+)) = \mu(E \cap F_0) + \mu(E \cap F_+) = \mu_s(E) + \mu_a(E)$ for every $E \in \mathcal{X}$, and therefore

$$\mu = \mu_s + \mu_a.$$

To prove uniqueness, suppose $\mu = \mu_1 + \mu_2$, where μ_1 and μ_2 are (σ-finite) measures on $\mathcal{X}$ such that $\mu_1 \perp \eta$ and $\mu_2 \ll \eta$. Take the signed measures $\mu_s - \mu_1$ and $\mu_a - \mu_2$ on $\mathcal{X}$ so that $\mu_s - \mu_1 \perp \eta$ and $\mu_a - \mu_2 \ll \eta$ (cf. Problems 7.10 and 7.11). Since $\mu_s + \mu_a = \mu_1 + \mu_2$, it follows by Problem 7.12 that $\mu_s - \mu_1 = \mu_a - \mu_2 = 0$, and so $\mu_1 = \mu_s$ and $\mu_2 = \mu_a$. □

Remark 7.B (Extended real-valued signed measures): Take two measures, $\eta\colon \mathcal{X} \to \overline{\mathbb{R}}$ and $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$, on the same σ-algebra $\mathcal{X}$. An extended real-valued signed measure $\nu = \eta - \mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ on $\mathcal{X}$ may be defined, according to the axioms of Definition 2.3, as $(\eta - \mu)(E) = \eta(E) - \mu(E)$ for every E in $\mathcal{X}$ if we declare that $(\eta - \mu)(E) = 0$ whenever E is such that $\eta(E) = \mu(E) = +\infty$. Thus the above uniqueness proof involving the signed measures $\mu_s - \mu_1$ and $\mu_a - \mu_2$ remains accurate even when allowing extended real-valued signed measures. Along this line, part of Problems 7.1 to 7.7 and 7.10 to 7.12 can be extended to extended real-valued signed measures.

Remark 7.C (Integral with respect to sums of measures): Consider a measurable partition $\{A, B\}$ of X as in Definition 7.9 and take the decomposition $\mu = \mu_a + \mu_s$ as in Theorem 7.10 such that $\mu_a(A) = \mu_s(B) = 0$. Since $\int_X f\,d\mu = \int_A f\,d\mu + \int_B f\,d\mu$, we get (cf. Problem 3.8(a) and proof of Theorem 7.10)

$$\mu_s \perp \mu_a \quad \text{ so that } \quad \int f\,d(\mu_a + \mu_s) = \int f\,d\mu_a + \int f\,d\mu_s$$

for every function f in $\mathcal{M}^+(X, \mathcal{X})$ (or in $\mathcal{L}(X, \mathcal{X}, \mu)$).

Theorem 7.10 decomposes every σ-finite measure μ into two parts: an absolutely continuous measure μ_a and a singular measure μ_s, both with respect to a σ-finite *reference measure* η. Next we refine this decomposition.

Definition 7.11. Take a measurable space $(X, \mathcal{X})$. A measure μ on $\mathcal{X}$ is *continuous* with respect to a measure η on $\mathcal{X}$ if, for $\{x\} \in \mathcal{X}$,

$$\eta(\{x\}) = 0 \quad \text{ implies } \quad \mu(\{x\}) = 0$$

(i.e., $\mu(\{x\}) = 0$ for every *measurable* singleton $\{x\}$ such that $\eta(\{x\}) = 0$).

Definition 7.12. Let $(X, \mathcal{X})$ be a measurable space. A measure μ on $\mathcal{X}$ is *discrete* with respect to a measure η on $\mathcal{X}$ if there exists a measurable partition $\{A, B\}$ of X such that (i) B is a countable set all of whose subsets are measurable (equivalently, $B = \{b_n\}_{n\in I}$ with each singleton $\{b_n\}$ lying in $\mathcal{X}$, where the index set I is either finite or $I = \mathbb{N}$), and (ii)

$$\mu(A) = \eta(B) = 0.$$

That is, μ is discrete with respect to η if μ is concentrated on a *countable* set of η-measure zero, and in this case μ and η are singular — see Problems 7.13 and 7.14. In particular, if both measures act on the Borel σ-algebra $\Re$ of subsets of the real line $\mathbb{R}$ and if the reference measure is the Lebesgue measure, then things get considerably simplified.

Proposition 7.13. *Consider a measurable space $(X, \mathcal{X})$. Suppose μ and η are measures on $\mathcal{X}$. If μ is σ-finite and measurable singletons of X have η-measure zero, then there is a unique pair of measures μ_c and μ_d on $\mathcal{X}$ such that μ_c is continuous and μ_d is discrete, both with respect to η, and*

$$\mu = \mu_c + \mu_d.$$

Proof. If μ is σ-finite, then there is a sequence $\{E_n\}$ of sets in $\mathcal{X}$ that covers X such that each E_n has finite μ-measure. For each integer $k \geq 1$, set

$$B_k(n) = \big\{x \in E_n\colon\ \{x\} \in \mathcal{X} \text{ and } \mu(\{x\}) \geq \tfrac{1}{k}\big\}.$$

Suppose $B_k(n)$ is an infinite set. Thus it has a countably infinite subset, say $C_k(n) = \bigcup_m \{b_m\} \subseteq B_k(n) \subseteq E_n$, consisting of distinct points b_m of $B_k(n)$. Since each singleton $\{b_m\}$ is $\mathcal{X}$-measurable, $C_k(n)$ also lies in $\mathcal{X}$, and hence $\mu(C_k(n)) = \sum_m \mu(\{b_m\}) = \infty$ because $\mu(\{b_m\}) \geq \frac{1}{k}$ for all m. But this contradicts the fact that $\mu(C_k(n)) \leq \mu(E_n) < \infty$. So each $B_k(n)$ is a finite subset of E_n, which lies in $\mathcal{X}$. Thus, since $X = \bigcup_n E_n$, for each $k \geq 1$, the subset

$$B_k = \bigcup\nolimits_n B_k(n) = \big\{x \in X\colon\ \{x\} \in \mathcal{X} \text{ and } \mu(\{x\}) \geq \tfrac{1}{k}\big\}$$

of X is countable (a countable union of finite sets), and so

$$B = \bigcup\nolimits_k B_k = \big\{x \in X\colon\ \{x\} \in \mathcal{X} \text{ and } \mu(\{x\}) \neq 0\big\}$$

is again a countable subset of X (recall: a countable union of countable sets is countable), which lies in $\mathcal{X}$. Indeed, B is measurable because, after all, it is a countable union of measurable singletons. Take the measurable partition $\{A, B\}$ of X so that $A = X\backslash B$, and set

$$\mu_c = \mu_A \quad \text{and} \quad \mu_d = \mu_B,$$

where the measures μ_A and μ_B on $\mathcal{X}$ are defined as in Problem 2.11, that is,

$$\mu_c(E) = \mu_A(E) = \mu(E \cap A) \quad \text{and} \quad \mu_d(E) = \mu_B(E) = \mu(E \cap B)$$

for every $E \in \mathcal{X}$. Since $B = \{b_n\}$ is a countable set consisting of measurable singletons, and since measurable singletons have η-measure zero, it follows that μ_d is discrete with respect to η, because

$$\mu_d(A) = \mu(A \cap B) = 0 \quad \text{and} \quad \eta(B) = \eta\Big(\bigcup\nolimits_n \{b_n\}\Big) = \sum\nolimits_n \eta(\{b_n\}) = 0.$$

If $\{x\}$ is an $\mathcal{X}$-measurable singleton, then either $\mu(\{x\}) = 0$ or $\mu(\{x\}) \neq 0$. In the former case, $\{x\} \subseteq A$ so that $\mu_c(\{x\}) = \mu(\{x\} \cap A) = \mu(\{x\}) = 0$. In the latter case, $\{x\} \subseteq B$, so $\mu_c(\{x\}) = \mu(\{x\} \cap A) \subseteq \mu(B \cap A) = \mu(\varnothing) = 0$. Thus $\mu_c(\{x\}) = 0$ for all $\{x\} \in \mathcal{X}$, and hence μ_c is continuous with respect to η because $\eta(\{x\}) = 0$ for all $\{x\} \in \mathcal{X}$. Therefore

$$\mu(E) = \mu\big((E \cap A) \cup (E \cap B)\big) = \mu(E \cap A) + \mu(E \cap B) = \mu_c(E) + \mu_d(E)$$

for every $E \in \mathcal{X}$, and hence

$$\mu = \mu_c + \mu_d.$$

To prove uniqueness, suppose $\mu = \mu_1 + \mu_2$, where μ_1 and μ_2 are measures on $\mathcal{X}$ such that μ_1 is continuous and μ_2 is discrete, both with respect to η. Take a singleton $\{x\}$ in $\mathcal{X}$. Since $\eta(\{x\}) = 0$, it follows that $\mu_1(\{x\}) = \mu_c(\{x\}) = 0$, and so $\mu_2(\{x\}) = \mu(\{x\}) = \mu_d(\{x\})$. Thus $\mu_2 = \mu_d$ by Problem 7.14. If $\mu_1 \neq \mu_c$, then there is a measurable set $E \subseteq A$ such that $\mu_1(E) \neq \mu_c(E)$. But $\mu_2(E) = \mu_d(E) = 0$ because $E \subseteq A$ and $\mu_2 = \mu_d$, and hence $\mu(E) = \mu_1(E) = \mu_c(E)$, which is a contradiction. Then $\mu_1 = \mu_c$. □

Corollary 7.14. *Consider a measurable space $(X, \mathcal{X})$. If η is a σ-finite measure on $\mathcal{X}$ such that measurable singletons of X have η-measure zero, then every σ-finite measure μ on $\mathcal{X}$ has a unique decomposition*

$$\mu = \mu_a + \mu_{sc} + \mu_{sd},$$

where the measures μ_a, μ_{sc}, μ_{sd} on $\mathcal{X}$ are absolutely continuous, singular and continuous, singular and discrete, respectively, all with respect to η.

Proof. By Theorem 7.10, $\mu = \mu_a + \mu_s$, where μ_a is absolutely continuous and μ_s is singular, with respect to η. If μ is σ-finite, then so is μ_s (the same countable covering of X that makes μ σ-finite, works for μ_s). Thus

$$\mu_s = \mu_{sc} + \mu_{sd}$$

by Proposition 7.13, where μ_{sc} is continuous and μ_{sd} is discrete, with respect to η. Since $\mu_s \perp \eta$, there exists a measurable partition $\{A, B\}$ of X such that $\mu_s(A) = \mu_{sc}(A) + \mu_{sd}(A) = 0 = \eta(B)$, and so $\mu_{sc}(A) = \mu_{sd}(A) = 0$. Hence μ_{sc} and μ_{sd} also are singular with respect to η. □

The identity in Corollary 7.14, viz., $\mu = \mu_a + \mu_{sc} + \mu_{sd}$, is the *canonical decomposition* of a measure μ with respect to a reference measure η. It splits the measure μ into three components, μ_a, μ_{sc}, and μ_{sd}, referred to as the

absolutely continuous, *singular-continuous*, and *singular-discrete*

parts of μ with respect to the reference measure η. For the particular case of $(X, \mathcal{X}) = (\mathbb{R}, \Re)$, such a reference measure is often the Lebesgue measure.

7.4 Problems

Problem 7.1. Consider a signed measure $\nu\colon \mathcal{X} \to \mathbb{R}$ on $\mathcal{X}$. If A and B are $\mathcal{X}$-measurable sets and $\{E_n\}$ is a sequence of sets in $\mathcal{X}$, then show that

(a) $\nu(B\backslash A) = \nu(B) - \nu(A)$ if $A \subseteq B$,

(b) $\nu\big(\bigcup_n E_n\big) = \lim_n \nu(E_n)$ if $\{E_n\}$ is increasing,

(c) $\nu\big(\bigcap_n E_n\big) = \lim_n \nu(E_n)$ if $\{E_n\}$ is decreasing.

Hint: Proof of Proposition 2.2 and Definition 2.3.

Problem 7.2. Consider Definition 7.1 and prove the following assertions.

(a) A measurable subset of a positive set is positive.

(b) A countable union of positive sets is a positive set.

Hint: (b) Let $\{A_n\}$ be a sequence of sets. Set $A'_{n+1} = A_{n+1}\backslash\big(\bigcup_{i=1}^n A_i\big)$ with $A'_1 = A_1$. This $\{A'_n\}$ is a sequence of disjoint sets such that $\bigcup_n A'_n = \bigcup_n A_n$, referred to as a *disjointification* of $\{A_n\}$. Now suppose each A_n is a positive set. Each A'_n is a measurable subset of A_n and is a positive set by (a). Then $\nu(E \cap \bigcup_n A_n) = \nu\big(E \cap \bigcup_n A'_n\big) = \nu\big(\bigcup_n (E \cap A'_n)\big) = \sum_n \nu(E \cap A'_n) \geq 0$.

Problem 7.3. Take a signed measure ν on a σ-algebra $\mathcal{X}$. If A and B lie in $\mathcal{X}$ and B is positive with respect to ν, then

$$A \subseteq B \quad \text{implies} \quad 0 \leq \nu(A) \leq \nu(B).$$

Hint: Problems 7.1(a) and 7.2(a).

Problem 7.4. This is the signed-measure version of Problem 2.8. Suppose $\nu\colon \mathcal{X} \to \mathbb{R}$ is a signed measure on a σ-algebra $\mathcal{X}$, and let $\{E_n\}$ be a sequence of $\mathcal{X}$-measurable sets. Show that

(a) $\nu\big(\bigcup_n E_n\big) = \lim_n \nu\big(\bigcup_{i=1}^n E_i\big)$.

If each E_n is positive with respect to ν, then

(b) $\nu\big(\bigcup_n E_n\big) \le \sum_n \nu(E_n)$ (*countable subadditivity*).

Hints: (a) Problem 7.1(b). (b) Disjointification and Problem 7.3.

Problem 7.5. Take a signed measure ν on $\mathcal{X}$. For an arbitrary $E \in \mathcal{X}$, set $\mathcal{E} = \wp(E) \cap \mathcal{X}$ (the σ-algebra of all measurable subsets of E). Show that

$$\nu^+(E) = \sup_{F \in \mathcal{E}} \nu(F) \quad \text{and} \quad \nu^-(E) = -\inf_{F \in \mathcal{E}} \nu(F).$$

Hint: Theorem 7.4: $\nu(F) = \nu^+(F) - \nu^-(F) \le \nu^+(F) \le \nu^+(E) = \nu(A^+ \cap E)$.

Problem 7.6. Again, take a signed measure ν on $\mathcal{X}$. Prove that

$$N \in \mathcal{X} \text{ is a null set with respect to } \nu \text{ if and only if } |\nu|(N) = 0.$$

Hint: Definition of $|\nu|$ on the one hand; Problem 7.5 on the other hand.

Problem 7.7. Consider a measurable space $(X, \mathcal{X})$. Suppose ν, ν_1, and ν_2 are signed measures on $\mathcal{X}$. Let α be any real number. Verify that $\nu_1 + \nu_2$ and $\alpha\nu$ are again signed measures on $\mathcal{X}$, where for each $E \in \mathcal{X}$,

$$(\alpha\nu)(E) = \alpha\nu(E) \quad \text{and} \quad (\nu_1 + \nu_2)(E) = \nu_1(E) + \nu_2(E).$$

Now let $\mathcal{S} = \mathcal{S}(X, \mathcal{X}, \mathbb{R})$ stand for the collection of all signed measures on $\mathcal{X}$. Since addition and scalar multiplication of signed measures are again signed measures, it follows that $\mathcal{S}$ is a (real) linear space (in fact, a linear manifold of the real linear space $\mathbb{R}^{\mathcal{X}}$ of all real-valued set functions on $\mathcal{X} \subseteq \mathcal{P}(X)$). Consider the total variation of signed measures in $\mathcal{S}$. Prove that

$$\text{(a)} \quad |\alpha\nu| = |\alpha||\nu| \qquad \text{and} \qquad \text{(b)} \quad |\nu_1 + \nu_2| \le |\nu_1| + |\nu_2|.$$

Hint: Take a Hahn decomposition $\{A_\nu^+, A_\nu^-\}$ of X with respect to ν. Show that $\{A_{\alpha\nu}^+, A_{\alpha\nu}^-\}$ is a Hahn decomposition of X with respect to $\alpha\nu$, where $A_{\alpha\nu}^+ = A_\nu^+$ and $A_{\alpha\nu}^- = A_\nu^-$ if $\alpha \ge 0$ or $A_{\alpha\nu}^+ = A_\nu^-$ and $A_{\alpha\nu}^- = A_\nu^+$ if $\alpha \le 0$. Then show that $(\alpha\nu)^+ = \alpha\nu^+$ and $(\alpha\nu)^- = \alpha\nu^-$ if $\alpha \ge 0$ or $(\alpha\nu)^+ = -\alpha\nu^-$ and $(\alpha\nu)^- = -\alpha\nu^+$ if $\alpha \le 0$. Thus conclude the first identity: $|\alpha\nu| = (\alpha\nu)^+ + (\alpha\nu)^- = |\alpha|(\nu^+ + \nu^-) = |\alpha||\nu|$. To verify the second inequality, note that $\nu_1 + \nu_2 = (\nu_1^+ + \nu_2^+) - (\nu_1^- + \nu_2^-)$, apply Theorem 7.4 again to show that $(\nu_1 + \nu_2)^+ \le \nu_1^+ + \nu_2^+$ and $(\nu_1 + \nu_2)^- \le \nu_1^- + \nu_2^-$, and hence $|\nu_1 + \nu_2| = (\nu_1 + \nu_2)^+ + (\nu_1 + \nu_2)^- \le (\nu_1^+ + \nu_1^-) + (\nu_2^+ + \nu_2^-) = |\nu_1| + |\nu_2|$.

Next consider the function $\|\ \| : \mathcal{S} \to \mathbb{R}$ defined by

$$\|\nu\| = |\nu|(X)$$

for every $\nu \in \mathcal{S}$. This is a norm on $\mathcal{S}$. In other words, show that

(c) $(\mathcal{S}, \|\ \|)$ is a normed space.

Hint: Use (a) and (b) to verify axioms (iii) and (iv) of Definition 5.1.

Also show that this normed space is complete. That is, show that

(d) $(\mathcal{S}, \| \ \|)$ is a Banach space.

Hint: Consider the following well-known result from functional analysis.

> A normed space is a Banach space if and only if every absolutely summable sequence is summable

(cf. Suggested Readings at the end of Chapter 5). Observe that if $\{\nu_n\}$ is a sequence of signed measures in $\mathcal{S}$, then

$$\max\{\nu_n^+(E), \nu_n^-(E)\} \le \nu_n^+(E) + \nu_n^-(E) \le \nu_n^+(X) + \nu_n^-(X) = |\nu_n|(X),$$

where $0 \le \min\{\nu_n^+(E), \nu_n^-(E)\}$, so that $\nu_n(E) = \nu_n^+(E) - \nu_n^-(E)$ makes a summable sequence for every E in $\mathcal{X}$ (i.e., $\{\nu_n\}$ is summable in $\mathcal{S}$) whenever $\{\|\nu_n\|\}$ is summable in $\mathbb{R}$ (i.e., whenever $\{\nu_n\}$ is absolutely summable).

Problem 7.8. (a) Take a measurable space $(X, \mathcal{X})$. Show that absolute continuity $\ll$ is a reflexive and transitive but not a symmetric relation on the collection of all measures on $\mathcal{X}$. That is, if μ, η, and ν are measures on $\mathcal{X}$, then show that $\eta \ll \eta$ (reflexivity), and that $\mu \ll \eta$ and $\eta \ll \nu$ imply $\mu \ll \nu$ (transitivity), but $\mu \ll \eta$ does not imply $\eta \ll \mu$. If $\mu \ll \eta$ and $\eta \ll \mu$, then μ and η are called *equivalent measures* (notation: $\mu \equiv \eta$ or $\mu \sim \eta$).

(b) Show that: $\mu \ll \eta$ and η finite does not imply μ finite.

Hint: $\eta = \lambda_{[0,1]}$ (Lebesgue), $\mu(E) = \int_E \frac{1}{x}\, dx$. Propositions 3.5(c) and 3.7(b).

Problem 7.9. Consider a measurable space $(X, \mathcal{X})$. Let μ, η and ν be σ-finite measures on $\mathcal{X}$. Prove the following propositions.

(a) If $\mu \ll \eta$ and $g \in \mathcal{M}(X, \mathcal{X})^+$, then $\int_E g\, d\mu = \int_E g\, \frac{d\mu}{d\eta}\, d\eta$ for every $E \in \mathcal{X}$.

Hint: Theorem 7.8 and Problem 3.13 (recall: $g\chi_E \in \mathcal{M}(X, \mathcal{X})^+$).

(b) If $\mu \ll \nu$ and $\eta \ll \nu$, then $\frac{d(\mu+\eta)}{d\nu} = \frac{d\mu}{d\nu} + \frac{d\eta}{d\nu}$ ν-almost everywhere.

Hint: Theorem 7.8, Proposition 3.5(b), and Problem 3.8(d).

(c) If $\mu \ll \eta \ll \nu$, then $\frac{d\mu}{d\nu} = \frac{d\mu}{d\eta}\frac{d\eta}{d\nu}$ ν-almost everywhere.

Hint: Recall that $\mu \ll \nu$. Apply Theorem 7.8 for each relation $\ll$ followed by Problem 3.13 as in part (a). Then use Problem 3.8(d).

(d) If $\mu \ll \eta$ and $\eta \ll \mu$, then $\frac{d\mu}{d\eta} = (\frac{d\eta}{d\mu})^{-1}$ almost everywhere.

Note: $\mu \ll \eta \ll \mu$ means $\mu \equiv \eta$ (i.e., μ and η are equivalent measures) so that η-almost everywhere is equivalent to μ-almost everywhere.

Hint: $\frac{d\mu}{d\mu}$ is the identity. Use part (a) with $\nu = \mu$. Swap μ and η.

Problem 7.10. Let ν and η be signed measures on a σ-algebra $\mathcal{X}$. The signed measure ν is *absolutely continuous* with respect to η if, for $E \in \mathcal{X}$,

$$|\eta|(E) = 0 \quad \text{implies} \quad \nu(E) = 0$$

(i.e., $\nu(E) = 0$ for every $E \in \mathcal{X}$ such that $|\eta|(E) = 0$). Same notation as for measures: $\nu \ll \eta$. Show that the following assertions are pairwise equivalent.

(a) $\nu \ll \eta$.

(b) $\nu^+ \ll \eta$ and $\nu^- \ll \eta$.

(c) $|\nu| \ll |\eta|$.

Hint: Take a Hahn decomposition $\{A_\nu^+, A_\nu^-\}$ of X with respect to ν. Verify that $|\eta|(E) = 0$ implies $|\eta|(A^+ \cap E) = |\eta|(A^- \cap E) = 0$ and, if (a) holds, this implies $\nu^+(E) = \nu^-(E) = 0$. Thus conclude that (a) implies (b). That (b) implies (a) follows from the fact that $\nu = \nu^+ - \nu^-$ (Theorem 7.4). Similarly, verify that (b) and (c) are equivalent because $|\nu| = \nu^+ + \nu^-$.

Consider a third signed measure μ on $\mathcal{X}$ and show that

(d) $\mu \ll \eta$ and $\nu \ll \eta$ imply $(\mu + \nu) \ll \eta$.

Hint: $|\mu + \nu| \le |\mu| + |\nu|$ according to Problem 7.7(b).

Problem 7.11. Let ν and η be signed measures on a σ-algebra $\mathcal{X}$. One is *singular* with respect to the other (or *mutually singular*, or simply *singular*) if their total variations $|\nu|$ and $|\eta|$ are singular measures on $\mathcal{X}$ (according to Definition 7.9). Same notation as for measures: $\nu \perp \eta$. That is,

$$\nu \perp \eta \quad \text{if and only if} \quad |\nu| \perp |\eta|.$$

Since $|\nu| = \nu^+ + \nu^-$, show that

(a) $$\nu \perp \eta \quad \text{implies} \quad \nu^+ \perp \eta \ \text{ and } \ \nu^- \perp \eta.$$

Consider a third signed measure μ on $\mathcal{X}$ and show that

(b) $$\mu \perp \eta \ \text{ and } \ \nu \perp \eta \quad \text{imply} \quad (\mu + \nu) \perp \eta.$$

Hint: If $|\mu|(A) = |\eta|(B) = 0$ and $|\nu|(C) = |\eta|(D)$, where $\{A, B\}$ and $\{C, D\}$ are measurable partitions of X, then $\{E, E'\}$ forms another measurable partition of X, with $E = (A \cap C)$ and $E' = (B \cap C) \cup (A \cap D) \cup (B \cap D)$, such that $|\mu|(E) = |\nu|(E) = |\eta|(E') = 0$. Now recall from Problem 7.7(b) that $|\mu + \nu| \le |\mu| + |\nu|$, and so $|\mu + \nu|(E) = \eta(E') = 0$.

Problem 7.12. Let $(X, \mathcal{X})$ be a measurable space. If μ and η are measures (or signed measures) on $\mathcal{X}$ such that $\mu \ll \eta$ and $\mu \perp \eta$, then $\mu = 0$. Prove.

Problem 7.13. Take $(X, \mathcal{X})$ and let μ and η be measures on $\mathcal{X}$. Also prove:

(a) If $\mu \ll \eta$, then μ is continuous with respect to η.

(b) If μ is discrete with respect to η, then $\mu \perp \eta$.

(c) If μ is continuous and discrete with respect to η, then $\mu = 0$.

Note that the converses are not true. In particular, although *discrete* implies *singular*, *singular* does not imply *discrete*. See examples in Problem 7.15.

Hint: Definitions 7.6, 7.9, and 7.11. (c) If μ is discrete and continuous with respect to η, then $\mu(X) = \mu(A) + \mu(B) = \mu(B) = \sum_n \mu(b_n) = 0$, since each $\{b_n\}$ is measurable and $\eta(\{b_n\}) \le \eta(B) = 0$, so $\mu(\{b_n\}) = 0$.

(d) μ is discrete if and only if it is singular-discrete (with respect to η).

Problem 7.14. Let μ and η be measures on a σ-algebra $\mathcal{X}$. For each $E \in \mathcal{X}$ take the σ-algebra $\mathcal{E} = \wp(E) \cap \mathcal{X}$. If μ is discrete with respect to η, then

$$\mu(E) = \sum\nolimits_{\{x\} \in \mathcal{E}} \mu(\{x\}) \quad \text{for every} \quad E \in \mathcal{X},$$

and $\eta(\{x\}) = 0$ whenever $\{x\}$ in $\mathcal{X}$ is such that $\mu(\{x\}) \ne 0$.

Hint: $\mu(E) = \mu(E \cap B)$, where B is a countable set of measurable singletons.

Problem 7.15. Let μ, η, and λ be measures on the Borel σ-algebra $\Re$ of subsets of $\mathbb{R}$, where λ is the Lebesgue measure (Example 2C, Problem 2.7). Consider the components of the canonical decomposition in Corollary 7.14.

(a) Set $F = \chi_{[\alpha,\infty)} \colon \mathbb{R} \to \{0,1\}$, where $\chi_{[\alpha,\infty)}$ is the characteristic function of $[\alpha, \infty)$ for an arbitrary $\alpha \in \mathbb{R}$ (Example 1B). Set $\mu = \lambda_F \colon \Re \to \{0,1\}$, the Borel–Stieltjes measure generated by F (Example 2D). This is the Dirac measure $\delta_\alpha \colon \Re \to \{0,1\}$ at α (Example 2A). Show that $\mu \perp \lambda$. That is, show that, with respect to Lebesgue measure λ on $\Re$, such a measure

$$\mu = \lambda \chi_{[\alpha,\infty)} = \delta_\alpha \text{ is singular-discrete}$$

(i.e., *discrete*), and so is the measure $\mu = \sum_{q \in \mathbb{Q}} \delta_q$ (see Problem 2.18).

(b) Let $f \colon \mathbb{R} \to \mathbb{R}$ and $F \colon \mathbb{R} \to \mathbb{R}$ be measurable functions; f is nonnegative, and F is nondecreasing and continuously differentiable. Set μ and η as

$$\mu(B) = \int_B f \, d\lambda \quad \text{and} \quad \eta(B) = \lambda_F(B) = \int_B \tfrac{dF}{d\alpha} \, d\lambda(\alpha),$$

for $B \in \Re$, where η is the Borel–Stieltjes measure generated by F (Proposition 3.5 and Example 2D). Show that these are σ-finite measures (if f is locally L^1) and, with respect to Lebesgue measure λ on $\Re$, both

$$\mu \text{ and } \eta \text{ are absolutely continuous.}$$

Hint: Example 2D, Propositions 3.5 and 3.7, Theorem 7.8.

(c) Consider the Cantor set $C \subset [0,1]$ obtained by successive removal of the central open third of $[0,1]$ (Problem 2.9). Writing each point of $[0,1]$ in its ternary (i.e., base 3) expansion, it can be shown that α lies in C if and only if it has only 0's and 2's, and no 1's, in its ternary expansion. Moreover, it can be verified that the map $\Phi\colon C \to [0,1]$ that changes the 2's into 1's, and interprets the result in its binary expansion (i.e., as a base 2 number), is a surjective map of C onto $[0,1]$ (showing, *en passant*, that C is uncountable). It can also be verified that Φ is increasing, uniformly continuous, and has the same value at the endpoints of every bounded open interval in the complement $[0,1]\backslash C$ of the Cantor set. Thus the function $\Phi\colon C \to [0,1]$ has a unique extension $F\colon [0,1] \to [0,1]$ over the closed interval $[0,1]$ (so that $F(\alpha) = \Phi(\alpha)$ for $\alpha \in C$) that is piecewise constant (constant on each successively removed open third remaining in $[0,1]\backslash C$ — e.g., $F(\alpha) = \frac{1}{2}$ for every α in the central open third $(\frac{1}{3}, \frac{2}{3})$ of $[0,1]$), continuous, nondecreasing ($F(0) = 0$ and $F(1) = 1$), surjective, differentiable almost everywhere ($F' = 0$ λ-a.e., since $\lambda(C) = 0$, where λ stands for the Lebesgue measure on $\Re$), not absolutely continuous (i.e., $F(\alpha) - F(0) = \int_0^\alpha F'd\lambda$ fails, and so not continuously differentiable), and $\int_0^1 F d\lambda = \frac{1}{2}$. This function $F\colon [0,1] \to [0,1]$ is the *Cantor function*.

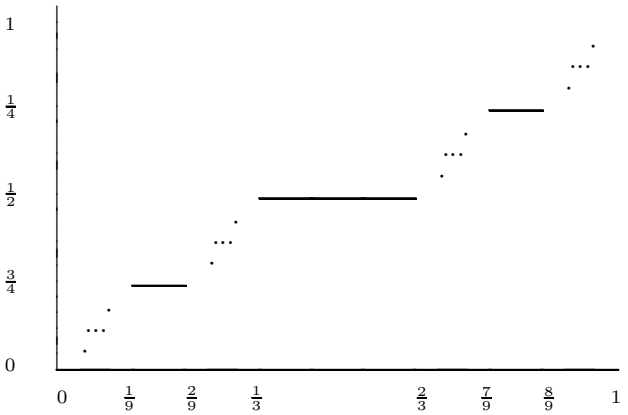


Fig. 7.1. A sketch of the Cantor function.

The Cantor function $F\colon [0,1] \to [0,1]$ is also referred to as the *Cantor–Lebesgue function*, or as the *Lebesgue singular function*, whose graph sketched above is sometimes called *Devil's staircase*.

(c_1) Take the Borel σ-algebra $\Re_{[0,1]} = \wp([0,1]) \cap \Re$ of Borel subsets of the unit interval $[0,1]$. Let $\mu_{[0,1]} = \lambda_F$ on $\Re_{[0,1]}$ be the Borel–Stieltjes measure generated by the Cantor function F via Lebesgue measure $\lambda = \lambda_{[0,1]}$ over $[0,1]$ (Example 2D and Problem 2.11). This $\mu_{[0,1]} = \lambda_F\colon \Re_{[0,1]} \to \mathbb{R}$ is the

Cantor–Lebesgue measure on $\mathfrak{R}_{[0,1]}$,

or the *Cantor–Lebesgue measure over the unit interval* $[0,1]$, which is also referred to as the *Borel–Stieltjes measure on* $\mathfrak{R}_{[0,1]}$ *generated by the Cantor function associated with the Cantor set over the unit interval* $[0,1]$. Recall that every singleton in $\mathbb{R}$ is $\mathfrak{R}$-measurable (Problem 2.7), and F is continuous. Show that the measure $\mu_{[0,1]} = \lambda_F$ is such that

$$\mu_{[0,1]}(\{\alpha\}) = 0 \quad \text{for every singleton } \{\alpha\} \subset [0,1],$$

so that $\mu_{[0,1]}$ is continuous with respect to Lebesgue measure $\lambda_{[0,1]}$ on $\mathfrak{R}_{[0,1]}$ (Definition 7.11), and $\mu_{[0,1]}$ is concentrated on the Cantor set C:

$$\mu_{[0,1]}([0,1]) = \mu_{[0,1]}(C) = 1.$$

Hint: First, $\mu_{[0,1]}(\{\alpha\}) = \lambda_F([\alpha,\alpha]) = 0$. Next, $\mu_{[0,1]}([0,1]) = \lambda_F([0,1]) = F(1) - F(0) = 1$. Let (α,β) be a generic open third that has been successively removed in the construction of the Cantor set. The closure of these open intervals are the parts where the Cantor function is constant. Thus $\mu_{[0,1]}((\alpha,\beta)) = \lambda_F((\alpha,\beta)) = F(\beta) - F(\alpha) = 0$. So $\mu_{[0,1]}([0,1]\backslash C) = 0$ because $[0,1]\backslash C$ is a countable disjoint union of such open intervals of $\mu_{[0,1]}$-measure zero (countable, because the uncountable Cantor set is a countable intersection of the complements of those open intervals).

Verify: $\mu_{[0,1]}$ and $\lambda_{[0,1]}$ on $\mathfrak{R}_{[0,1]}$ are singular (i.e., $\mu_{[0,1]} \perp \lambda_{[0,1]}$). So the

Cantor–Lebesgue measure $\mu_{[0,1]}$ over $[0,1]$ is singular-continuous

with respect to Lebesgue measure $\lambda_{[0,1]}$ (where $\lambda_{[0,1]}$ on $\mathfrak{R}_{[0,1]}$ is the restriction to $[0,1]$ of the Lebesgue measure λ on $\mathfrak{R}$ — Problem 2.11).

(c_2) By scaling the Cantor set, say βC for any $0 < \beta \in \mathbb{R}$, and removing the endpoints if necessary, the Cantor function retains all its essential properties when its domain (originally, the closed unit interval $[0,1]$) is replaced by any bounded interval I between 0 and β, so the Cantor function can be defined on half-open, or open, or closed bounded intervals as well. This leads to a Cantor function (same notation) $F\colon I \to I$ for each bounded interval $I \in \{[0,\beta), (0,\beta], (0,\beta), [0,\beta]\}$. Take the Borel σ-algebra $\mathfrak{R}_I = \wp(I) \cap \mathfrak{R}$ of Borel subsets of a bounded interval I, and let $\mu_I = \lambda_F$ on $\mathfrak{R}_I$ be the Borel–Stieltjes measure generated by the Cantor function F via Lebesgue measure $\lambda_I\colon \mathfrak{R}_I \to \mathbb{R}$ over I (cf. Example 2D and Problem 2.11 again). This measure $\mu_I = \lambda_F\colon \mathfrak{R}_I \to \mathbb{R}$ is the

Cantor–Lebesgue measure on $\mathfrak{R}_I$,

or the *Cantor–Lebesgue measure over a bounded interval* I. Use the hint to item (c_1), with C replaced by βC and with $F\colon I \to I$, to verify that

$$\mu_I(I) = \mu_I(\beta C) = \lambda_I(I) = \lambda(I) = \beta,$$

where λ denotes the Lebesgue measure on $\Re$, as usual. Again, using the same argument as in item (c_1), also verify that the

$$\text{Cantor–Lebesgue measure } \mu_I \text{ over } I \text{ is singular-continuous}$$

with respect to Lebesgue measure λ_I on $\Re_I$ (where $\lambda_I = \lambda|_{\Re_I}$ is the restriction to $\Re_I$ of the Lebesgue measure λ on $\Re$ — Problem 2.11).

Problem 7.16. Take the unit circle about the origin in the complex plane $\mathbb{C}$,

$$\mathbb{T} = \big\{z \in \mathbb{C} \colon |z| = 1\big\}.$$

A *nondegenerate interval* is an interval of the real line with more than one element (not empty and not a singleton). Take an arbitrary nondegenerate bounded half-open-on-the-right interval of the real line, translate and scale it to get the unit half-open-on-the-right interval, and set

$$\mathbb{I} = [0, 1).$$

Now consider the function

$$\gamma \colon \mathbb{I} \to \mathbb{T} \quad \text{given by} \quad \gamma(\alpha) = e^{2\pi i \alpha} \ \text{ for every } \ \alpha \in \mathbb{I} = [0, 1).$$

Recall that the Borel σ-algebra $\Re$ of subsets of $\mathbb{R}$ is generated by open (or half-open) intervals in $\mathbb{R}$, and so an arbitrary nondegenerate bounded interval lies in $\Re$. Thus take, in particular, the σ-algebra $\Re_{\mathbb{I}} = \wp(\mathbb{I}) \cap \Re$ of Borel subsets of $\mathbb{I}$. Next, let $\Re_{\mathbb{T}}$ be the Borel σ-algebra of subsets of $\mathbb{T}$ generated by half-open arcs, such as $[z_1, z_2) = \gamma([\alpha_1, \alpha_2)) \subseteq \mathbb{T}$, with $z_1 = \gamma(\alpha_1) = e^{2\pi i \alpha_1}$ and $z_2 = \gamma(\alpha_2) = e^{2\pi i \alpha_2}$ in $\mathbb{T}$ for α_1 and α_2 in $\mathbb{I} = [0, 1)$ with $\alpha_1 < \alpha_2$. So the σ-algebras $\Re_{\mathbb{I}}$ and $\Re_{\mathbb{T}}$ are generated by open (or half-open) intervals from $\mathbb{I}$, and by half-open arcs from $\mathbb{T}$, respectively; equivalently, they are generated by the usual topologies on $\mathbb{R}$ and on $\mathbb{T}$.

(a) Consider the usual metrics in $\mathbb{I}$ and in $\mathbb{T}$ (the restriction to $\mathbb{I}$ of the usual metric on $\mathbb{R}$ and the *arc-length metric* given by the angle of an arc in $\mathbb{T}$). Verify that the function $\gamma \colon \mathbb{I} \to \mathbb{T}$ defined above is continuous and invertible with a continuous inverse (i.e., a homeomorphism between $\mathbb{I} = [0, 1)$ and $\mathbb{T}$), with both γ and γ^{-1} being measurable, so that

$$\gamma \text{ and } \gamma^{-1} \text{ are measurable transformations.}$$

(Problems 1.8 to 1.13 — we will return to these notions in Section 11.1.)

(b) Set $\lambda_{\mathbb{I}} = \lambda|_{\Re_{\mathbb{I}}} \colon \Re_{\mathbb{I}} \to \mathbb{R}$, the restriction to $\Re_{\mathbb{I}}$ of the Lebesgue measure λ on $\Re$ (cf. Example 2C and Problem 2.11). This is the *Lebesgue measure over the interval* $\mathbb{I} = [0, 1)$, or the *Lebesgue measure on* $\Re_{\mathbb{I}}$. Associated with $\gamma \colon \mathbb{I} \to \mathbb{T}$, define a set function $\lambda_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$ induced by $\lambda_{\mathbb{I}}$ via γ,

$$\lambda_{\mathbb{T}}(B) = \lambda_{\mathbb{I}}\big(\gamma^{-1}(B)\big) \quad \text{for every} \quad B \in \Re_{\mathbb{T}}.$$

Verify that $\lambda_{\mathbb{T}} \colon \Re_{\mathbb{T}} \to \mathbb{R}$ is a measure on $\Re_{\mathbb{T}}$, and show that

$$\lambda_{\mathbb{T}}(\mathbb{T}) = \lambda_{\mathbb{I}}(\mathbb{I}) = 1.$$

Hint: Problems 2.19, 3.14. Check: $\lambda_{\mathbb{T}}(\mathbb{T}) = \lambda_{\mathbb{I}}(\gamma^{-1}(\mathbb{T})) = \lambda_{\mathbb{I}}(\mathbb{I}) = \lambda(\mathbb{I})$.

According to the above identity, the measure $\lambda_{\mathbb{T}}$ is referred to as the *normalized arc-length measure*, or the *normalized Lebesgue measure over the unit circle*, or still the *normalized Lebesgue measure on* $\Re_{\mathbb{T}}$. Also show that the arc-length measure is *rotation invariant*:

$$\lambda_{\mathbb{T}}(zB) = \lambda_{\mathbb{T}}(B) \quad \text{for all} \quad z \in \mathbb{T} \ \text{ and } \ B \in \Re_{\mathbb{T}}.$$

(We return to this point in Property P_6, Section 8.3, and in Chapter 13.)

(c) Consider the Cantor–Lebesgue measure $\mu_{\mathbb{I}} = \lambda_F$ on $\Re_{\mathbb{I}}$ (where F is the Cantor function as in Problem 7.15(c_2)). With $\gamma\colon \mathbb{I} \to \mathbb{T}$ as above, take the measure $\mu_{\mathbb{T}}\colon \Re_{\mathbb{T}} \to \mathbb{R}$ defined (as $\lambda_{\mathbb{T}}$ was defined in item (b)) by

$$\mu_{\mathbb{T}}(B) = \mu_{\mathbb{I}}\big(\gamma^{-1}(B)\big) \quad \text{for every} \quad B \in \Re_{\mathbb{T}}.$$

Recall that this defines the *measure* $\mu_{\mathbb{T}}$ *on* $\Re_{\mathbb{T}}$ *induced by the measure* $\mu_{\mathbb{I}}$ *on* $\Re_{\mathbb{I}}$ via γ (cf. Problems 2.19, 3.14, 4.19). Verify that

$$\mu_{\mathbb{T}}(\mathbb{T}) = \mu_{\mathbb{I}}(\mathbb{I}) = 1.$$

Hint: $\mu_{\mathbb{T}}(\mathbb{T}) = \mu_{\mathbb{I}}(\gamma^{-1}(\mathbb{T})) = \mu_{\mathbb{I}}(\mathbb{I}) = \lambda_F(\mathbb{I}) = \text{length}\,(\mathbb{I})$.

(d) Also show that if an arbitrary measure $\eta_{\mathbb{I}}$ on $\Re_{\mathbb{I}}$ is singular-discrete, absolutely continuous, or singular-continuous (with respect to Lebesgue measure $\lambda_{\mathbb{I}}$ on $\Re_{\mathbb{I}}$ — the restriction to $\mathbb{I}$ of the Lebesgue measure λ on $\Re$), then so is any induced measure $\eta_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$, induced by $\eta_{\mathbb{I}}$ via γ,

$$\eta_{\mathbb{T}}(B) = \eta_{\mathbb{I}}\big(\gamma^{-1}(B)\big) \quad \text{for every} \quad B \in \Re_{\mathbb{T}}.$$

That is, if a measure $\eta_{\mathbb{I}}$ on $\Re_{\mathbb{I}}$ is singular-discrete, absolutely continuous, or singular-continuous with respect to Lebesgue measure $\lambda_{\mathbb{I}}$ on $\Re_{\mathbb{I}}$, then so is the measure $\eta_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$ induced by $\eta_{\mathbb{I}}$ with respect to Lebesgue measure $\lambda_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$ induced by $\lambda_{\mathbb{I}}$ (i.e., with respect to arc-length measure $\lambda_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$).

For instance, $\quad \eta_{\mathbb{I}} \ll \lambda_{\mathbb{I}} \implies \eta_{\mathbb{T}} \ll \lambda_{\mathbb{T}} \quad$ or $\quad \eta_{\mathbb{I}} \perp \lambda_{\mathbb{I}} \implies \eta_{\mathbb{T}} \perp \lambda_{\mathbb{T}}$.

(e) Take the Cantor–Lebesgue measure $\mu_{\mathbb{I}} = \lambda_F\colon \Re_{\mathbb{I}} \to \mathbb{R}$ generated by a Cantor function $F\colon \mathbb{I} \to \mathbb{I}$ as in Problem 7.15(c_2). Consider the measure $\mu_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$ induced by the Cantor–Lebesgue measure $\mu_{\mathbb{I}}$ on $\Re_{\mathbb{I}}$, as in item (c), for the unit interval $\mathbb{I} = [0, 1)$. This $\mu_{\mathbb{T}}\colon \Re_{\mathbb{T}} \to \mathbb{R}$ is the

Cantor–Lebesgue measure on $\Re_{\mathbb{T}}$,

or the *Cantor–Lebesgue measure over the unit circle* $\mathbb{T}$, or still the *Borel–Stieltjes measure on* $\Re_{\mathbb{T}}$ *generated by the Cantor function*. Thus $\mu_{\mathbb{T}}$ is associated with the Cantor set over the unit circle (i.e., with $\gamma(C\backslash\{1\}) \subseteq \mathbb{T}$, where C is the Cantor set over $[0, 1]$). Also recall from item (c) that

$$\mu_{\mathbb{T}}(\mathbb{T}) = \mu_{\mathbb{I}}(\gamma^{-1}(\mathbb{T})) = \mu_{\mathbb{I}}(\mathbb{I}) = 1,$$

and so the Cantor–Lebesgue measure $\mu_{\mathbb{T}}$ is *normalized*. Show that the

Cantor–Lebesgue measure $\mu_{\mathbb{T}}$ over $\mathbb{T}$ is singular-continuous

with respect to normalized Lebesgue measure $\lambda_{\mathbb{T}}$ on $\mathfrak{R}_{\mathbb{T}}$ (induced by Lebesgue measure $\lambda_{\mathbb{I}}$ on $\mathfrak{R}_{\mathbb{I}}$ via measurable transformation γ).

Problem 7.17. Let z be in the unit circle $\mathbb{T}$ of the complex plane $\mathbb{C}$ and let k be an integer in $\mathbb{Z}$. Take the σ-algebra $\mathfrak{R}_{\mathbb{T}}$ of Borel subsets of $\mathbb{T}$ generated by half-open arcs. A *finite* measure $\eta_{\mathbb{T}}\colon \mathfrak{R}_{\mathbb{T}} \to \mathbb{R}$ on $\mathfrak{R}_{\mathbb{T}}$ is a *Rajchman measure* if

$$\int_{\mathbb{T}} z^k d\eta_{\mathbb{T}} \to 0 \quad \text{as} \quad |k| \to \infty.$$

Note 1: Take the measure $\eta_{\mathbb{I}} = \eta_{\mathbb{T}}(\gamma(\ \ ))$ on $\mathfrak{R}_{\mathbb{I}}$ with $\gamma\colon \mathbb{I} \to \mathbb{T}$ and $\mathbb{I} = [0,1)$ as in Problem 7.16. The above integral sets the Fourier transform $\widetilde{\eta}_{\mathbb{I}}$ of $\eta_{\mathbb{I}}$,

$$\widetilde{\eta}_{\mathbb{I}}(k) = \int_{\mathbb{T}} z^k d\eta_{\mathbb{T}}(z) = \int_{\mathbb{I}} e^{2\pi i k\alpha}\, d\eta_{\mathbb{I}}(\alpha) \quad \text{for every} \quad k \in \mathbb{Z}$$

(sometimes $\widetilde{\eta}_{\mathbb{I}}(k)$ is equivalently defined as $\widetilde{\eta}_{\mathbb{I}}(-k)$ for $k \in \mathbb{Z}$).

Absolute continuity, continuity, singularity, and singular continuity of measures on $\mathfrak{R}_{\mathbb{T}}$ will be with respect to normalized Lebesgue measure $\lambda_{\mathbb{T}}$ on $\mathfrak{R}_{\mathbb{T}}$.

(a) Show that every absolutely continuous finite measure on $\mathfrak{R}_{\mathbb{T}}$ is Rajchman.

Hint: Consider the Riemann–Lebesgue Lemma, which reads as follows: $\int_{\mathbb{T}} z^k \psi(z)\, d\lambda(z) \to 0$ *as* $|k| \to \infty$ *for every* $\psi \in L^2(\mathbb{T}, \lambda)$, where the integral is the Riemann integral or the Lebesgue integral with respect to Lebesgue measure $\lambda = \lambda_{\mathbb{T}}$ on $\mathfrak{R}_{\mathbb{T}}$ — cf. Problem 4.1(a). Since measures in this problem are all finite, take a measure η on $\mathfrak{R}_{\mathbb{T}}$ such that $\eta \ll \lambda$ and apply the Radon–Nikodym Theorem (Theorem 7.8) with $f = d\eta\,/d\lambda$.

Note 2: There are different proofs of the Riemann–Lebesgue Lemma. One of them that uses a bit of operator theory is this. Take the multiplication operator U from $L^2(\mathbb{T}, \lambda)$ to itself given by $(U\psi)(z) = z\psi(z)$, $z \in \mathbb{T}$, for every ψ in $L^2(\mathbb{T}, \lambda)$. So $\int_{\mathbb{T}} z^k \psi(z)\, d\lambda(z) = \langle U^k \psi\,; 1\rangle$, the inner product on the Hilbert space $L^2(\mathbb{T}, \lambda)$, which is separable as we will see in Proposition 12.L, where ψ and the unit function 1 are in $L^2(\mathbb{T}, \lambda)$. Since λ is the Lebesgue measure, $\{z^k\}_{k\in\mathbb{Z}}$ is an orthonormal basis for $L^2(\mathbb{T}, \lambda)$. Since U shifts the orthonormal basis $\{z^k\}_{k\in\mathbb{Z}}$, it is a bilateral shift. Since every bilateral shift on a Hilbert space is weakly stable, $\langle U^k\psi;\, \phi\rangle \to 0$ for every $\psi, \phi \in L^2(\mathbb{T}, \lambda)$.

Further basic properties of Rajchman measures are listed below. The survey on Rajchman measures mentioned in the Suggested Readings at the end of this chapter provides original references for the results stated below. The next one is Neder's answer to Riesz's question.

(b) Every Rajchman measure is continuous.

Let $\mu_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$ be the Cantor–Lebesgue measure over the unit circle $\mathbb{T}$. It can be shown that $\int_{\mathbb{T}} z^k d\mu_{\mathbb{T}} = \int_{\mathbb{T}} z^{3k} d\mu_{\mathbb{T}}$ (i.e., $\widetilde{\mu}_{\mathbb{I}}(k) = \widetilde{\mu}_{\mathbb{I}}(3k)$) for all k, and so the

(c) Cantor–Lebesgue singular-continuous measure over $\mathbb{T}$ is not Rajchman.

However, Menshov's example, which is a modification of the above Cantor–Lebesgue singular-continuous non-Rajchman measure $\mu_{\mathbb{T}}$ on $\Re_{\mathbb{T}}$, ensures:

(d) There exist singular Rajchman measures.

Therefore, according to items (b) and (d), the following assertion holds true.

(e) Every singular Rajchman measure is singular-continuous.

Problem 7.18. Let η be a finite measure on $\Re_{\mathbb{T}}$ and consider again the multiplication operator $U\colon L^2(\mathbb{T},\eta) \to L^2(\mathbb{T},\eta)$ given by

$$(U\psi)(z) = z\psi(z) \quad \eta\text{-a.e. for } z \in \mathbb{T} \quad \text{for every } \psi \in L^2(\mathbb{T},\eta),$$

now acting on $L^2(\mathbb{T},\eta)$. Show that

$$\eta \text{ is a Rajchman measure} \quad \text{if and only if} \quad \int_{\mathbb{T}} U^n \psi\, \overline{\phi}\, d\eta \to 0$$

as $n \to \infty$ for every $\psi, \phi \in L^2(\mathbb{T},\eta)$.

Hint: If η is Rajchman, then $\int_{\mathbb{T}} U^n z^k z^{-\ell}\, d\eta = \int_{\mathbb{T}} z^{n+k-\ell}\, d\eta \to 0$ as $n \to \infty$ for each $k, \ell \in \mathbb{Z}$. Since the set $\mathbb{T}$ is *compact* in $\mathbb{C}$ and the measure η is *finite*, the Stone–Weierstrass Theorem ensures that the set $P(\mathbb{T})$ of all polynomials $p(\ ,\ )\colon \mathbb{T}\times\mathbb{T} \to \mathbb{C}$ in z and $\overline{z}$ is dense in $(L^1(\mathbb{T},\eta), \|\ \|_1)$, as we will see in Proposition 12.D and in the proof of Proposition 12.K. Hence, by the above convergence, $\int_{\mathbb{T}} U^n |\psi|^2\, d\eta \to 0$ for every ψ in $L^2(\mathbb{T},\eta)$. As $L^2(\mathbb{T},\eta)$ is a *complex* Hilbert space, this implies $\int_{\mathbb{T}} U^n \psi\, \overline{\phi}\, d\eta \to 0$ for every $\psi, \phi \in L^2(\mathbb{T},\eta)$. The converse is immediate. In fact, if $\int_{\mathbb{T}} U^n \psi\, \overline{\phi}\, d\eta \to 0$, then with $\psi = \phi = 1$ in $L^2(\mathbb{T},\eta)$ we get $\int_{\mathbb{T}} z^n\, d\eta \to 0$ as $n \to \infty$, and so $\int_{\mathbb{T}} z^k\, d\eta \to 0$ as $|k| \to \infty$.

Note: Since $\int_{\mathbb{T}} U^n \psi\, \overline{\phi}\, d\eta = \langle U^n\psi\,;\phi\rangle$ for every ψ, ϕ in $L^2(\mathbb{T},\eta)$ (inner product on $L^2(\mathbb{T},\eta)$), in the jargon of operator theory the above equivalence reads:

η is a Rajchman measure if and only if U on $L^2(\mathbb{T},\eta)$ is weakly stable.

Suggested Readings

Bartle [5], Berberian [9], Halmos [31], Kelley and Srinivasan [38], Royden [62], Rudin [63], Shilov and Gurevich [66]. See also [51, Section 6.8]. For discussions on the Cantor function (Problem 7.15(c)) see [5, Chapter 17], [10, Example 6.23], [14, Example 7.Q], [58, Example 1.43], [64, Section 5.3], [65, Section 4.2]. For a survey on Rajchman measures, including the results mentioned in Problem 7.17, see [50] (along this line, also see [36, Chapter IX]).



8

Extension of Measures

8.1 Measure on an Algebra and Outer Measure

In Chapter 2 (Example 2C), we considered the Lebesgue measure λ on the Borel σ-algebra $\Re$ of subsets of the real line $\mathbb{R}$, which is generated by the collection of all open intervals; and we have been using the notion of Lebesgue measure since then, although it has not been properly constructed so far. Indeed, we promised in Example 2C to prove existence and uniqueness of the Lebesgue measure $\lambda\colon \Re \to \overline{\mathbb{R}}$ in Chapter 8. We will comply with that promise in Section 8.3 as a special case of the following program.

(1) First we introduce the concept of a measure μ on an algebra $\mathcal{A}$ (rather than on a σ-algebra) of subsets of a set X.

(2) Next we consider the notion of an outer measure μ^* generated by a measure μ on an algebra $\mathcal{A}$, which is a set function on the power set $\wp(X)$.

(3) Then we show that an outer measure μ^* (generated by a measure μ on an algebra $\mathcal{A}$) induces a σ-algebra $\mathcal{A}^*$ of subsets of X (such that $\mathcal{A} \subseteq \mathcal{A}^*$), upon which the restriction $\mu^*|_{\mathcal{A}^*}$ is a measure on the σ-algebra $\mathcal{A}^*$ (and the restriction $\mu^*|_{\mathcal{A}}$ is that measure μ on $\mathcal{A}$).

This is the Carathéodory Extension Theorem, the central result of this chapter, whose applications go as far as Chapters 9, 11, and 13.

The difference between an algebra and a σ-algebra of subsets of a set X is that in an algebra $\mathcal{A}$ any *finite union* of sets in $\mathcal{A}$ is required to remain in $\mathcal{A}$, while in a σ-algebra $\mathcal{X}$ it is imposed, in addition, that any *countable*

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_8

union of sets in $\mathcal{X}$ must remain in $\mathcal{X}$ (see Definition 1.1). We will now define the notion of a measure μ on an algebra $\mathcal{A}$. Since a countable union of sets in $\mathcal{A}$ is not necessarily in $\mathcal{A}$, countable additivity for μ will be restricted to countable families of sets in $\mathcal{A}$ whose union is assumed to be in $\mathcal{A}$.

Definition 8.1. A *measure on an algebra* is an extended real-valued set function μ on an algebra $\mathcal{A}$ of subsets of a set X,

$$\mu\colon \mathcal{A} \to \overline{\mathbb{R}},$$

that fulfills the following axioms.

(a) $\mu(\varnothing) = 0$,

(b) $\mu(E) \geq 0$ for every $E \in \mathcal{A}$ (*positiveness*),

(c) $\mu\big(\bigcup_n E_n\big) = \sum_n \mu(E_n)$ (*countable additivity*)
for every countable family $\{E_n\}$ of pairwise disjoint sets in $\mathcal{A}$ for which $\bigcup_n E_n$ lies in $\mathcal{A}$.

As observed after Definitions 2.1 and 2.3, if the countable set $\{\mu(E_n)\}$ of nonnegative extended real numbers in Definition 8.1(c) is infinite, then the infinite series $\sum_n \mu(E_n)$ of nonnegative extended real numbers either converges unconditionally to a real number (i.e., the real value of the sum does not depend on the order of the summands) or diverges to infinity.

Remark 8.A (Properties of measures on an algebra): The properties of measures on a σ-algebra in Proposition 2.2(a,b) are naturally transferred to measures on an algebra, and so are the properties in Proposition 2.2(c,d) under the additional assumption that $\bigcup_n E_n \in \mathcal{A}$.

Let $\mathcal{A}$ be any algebra of subsets of a set X. A measure μ on $\mathcal{A}$ generates a set function μ^* on the power set $\wp(X)$ as follows.

Definition 8.2. Suppose $\mu\colon \mathcal{A} \to \overline{\mathbb{R}}$ is a measure on an algebra $\mathcal{A}$ of subsets of a set X. For each subset S of X (i.e., for each $S \in \wp(X)$) consider the collection $\mathcal{C}_S$ of all countable families $\{E_n\}$ of sets in $\mathcal{A}$ that cover S,

$$\mathcal{C}_S = \big\{\{E_n\}\colon E_n \in \mathcal{A} \text{ and } S \subseteq \textstyle\bigcup_n E_n\big\}.$$

The extended real-valued set function μ^* on the power set $\wp(X)$,

$$\mu^*\colon \wp(X) \to \overline{\mathbb{R}},$$

defined for each $S \in \wp(X)$ by

$$\mu^*(S) = \inf_{\{E_n\} \in \mathcal{C}_S} \sum\nolimits_n \mu(E_n),$$

is the *outer measure* generated by the measure μ on the algebra $\mathcal{A}$.

Differently from measures, outer measures are not necessarily additive. We discuss this issue after the proof of the next proposition, which presents the basic properties of outer measures that will be required in the sequel.

Proposition 8.3. *Let $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ be the outer measure generated by a measure $\mu\colon \mathcal{A} \to \overline{\mathbb{R}}$ on an algebra $\mathcal{A}$ of subsets of a set X. Then*

(a) $\mu^*(\varnothing) = 0$,

(b) $\mu^*(S) \geq 0$ *for every* $S \in \wp(X)$ (*positiveness*),

(c) $\mu^*(S') \leq \mu^*(S)$ *whenever* $S' \subseteq S \subseteq X$,

(d) $\mu^*(E) = \mu(E)$ *for every* $E \in \mathcal{A}$ (*i.e.*, $\mu^*|_{\mathcal{A}} = \mu$),

(e) $\mu^*\big(\bigcup_n S_n\big) \leq \sum_n \mu^*(S_n)$ (*countable subadditivity*), *for every countable family* $\{S_n\}$ *of subsets of* X.

Proof. Properties (a), (b), and (c) are trivially verified by the definition of outer measure (see also the remark after Definition 8.2). To verify property (d), take an arbitrary $E \in \mathcal{A}$, and let $\{E'_n\}$ be a sequence of sets in $\mathcal{A}$ such that $E'_1 = E$ and $E'_n = \varnothing$ for all $n \neq 1$. Since $\{E'_n\} \in \mathcal{C}_E$, we get

$$\mu^*(E) = \inf_{\{E_n\} \in \mathcal{C}_E} \sum\nolimits_n \mu(E_n) \leq \sum\nolimits_n \mu(E'_n) = \mu(E).$$

To verify the reverse inequality proceed as follows. If $\{E_n\}$ is any countable family in $\mathcal{C}_E$, then $\{E \cap E_n\}$ is again a countable family in $\mathcal{C}_E$ such that $E = \bigcup_n (E \cap E_n)$, and so (cf. Problem 2.8(b) and Proposition 2.2(a))

$$\mu(E) \leq \sum\nolimits_n \mu(E \cap E_n) \leq \sum\nolimits_n \mu(E_n).$$

Since this holds for all $\{E_n\} \in \mathcal{C}_E$, we get

$$\mu(E) \leq \inf_{\{E_n\} \in \mathcal{C}_E} \sum\nolimits_n \mu(E_n) = \mu^*(E),$$

completing the proof of (d). Finally, we verify (e). Take an arbitrary $\varepsilon > 0$ and recall that, for each $S \in \wp(X)$,

$$\mu^*(S) = \inf_{\{E_k\} \in \mathcal{C}_S} \sum\nolimits_k \mu(E_k),$$

where

$$\mathcal{C}_S = \big\{\{E_k\}\colon E_k \in \mathcal{A} \text{ and } S \subseteq \textstyle\bigcup_k E_k\big\},$$

and hence, for each $n \geq 1$, there is a countable family $\{E_k\} \in \mathcal{C}_S$ for which

$$\sum\nolimits_k \mu(E_k) \leq \mu^*(S) + \tfrac{\varepsilon}{n^2}.$$

Now let $\{S_n\}$ be a countable family of subsets of X so that, for each $n \geq 1$,

$$\mu^*(S_n) = \inf_{\{E_{(n)k}\} \in \mathcal{C}_{S_n}} \sum\nolimits_k \mu(E_{(n)k}),$$

where each k-indexed family $\{E_{(n)k}\}$ in $\mathcal{C}_{S_n}$ covers each S_n. So the double-indexed (in k as well as n) countable families $\{E_{(n)k}\}$ cover $\bigcup_n S_n$. Then

$$\begin{aligned}\mu^*\Big(\bigcup\nolimits_n S_n\Big) &\leq \inf_{\{E_{(n)k}\} \in \mathcal{C}_{\cup_n S_n}} \sum\nolimits_{n,k} \mu\big(E_{(n)k}\big) \leq \sum\nolimits_n \sum\nolimits_k \mu\big(E_{(n)k}\big) \\ &\leq \sum\nolimits_n \big[\mu^*(S_n) + \tfrac{\varepsilon}{n^2}\big] = \sum\nolimits_n \mu^*(S_n) + \varepsilon \sum\nolimits_n \tfrac{1}{n^2}\end{aligned}$$

and therefore, since this holds for every $\varepsilon > 0$, and since $\sum_n \frac{1}{n^2} < \infty$,

$$\mu^*\Big(\bigcup\nolimits_n S_n\Big) \leq \inf_{\varepsilon > 0} \Big(\sum\nolimits_n \mu^*(S_n) + \varepsilon \sum\nolimits_n \tfrac{1}{n^2}\Big) = \sum\nolimits_n \mu^*(S_n). \qquad \square$$

Remark 8.B (Countable additivity and subadditivity):

(a) According to Definitions 2.1, 2.3, and 8.1, all measures (either on a σ-algebra or on an algebra and also signed measures) are *countably additive* (sometimes referred to as *σ-additive*) in the sense that the measure of the union of a countable *pairwise disjoint* collection of measurable sets is the sum (unconditional convergence) of the sequence of measures of those sets. *Countable subadditivity*, however, holds for any collection, not necessarily of pairwise disjoint sets, either for measures or signed measures (cf. Problems 2.8(b) and 7.4(b)), as well as for measures on an algebra.

(b) In spite of the terminology, an outer measure may be far from being a measure (in the sense of Definition 2.1 or 8.1) since it is not necessarily additive (not even finitely additive). Actually, if μ^* is an outer measure, then it may happen that $\mu^*(S \cup S') \lneqq \mu^*(S) + \mu^*(S')$ for some disjoint sets S and S' (Problems 8.16 and 8.22). For an outer measure, additivity (for pairwise disjoint sets) does not hold, but subadditivity (for not necessarily pairwise disjoint sets) remains valid. As in the case of measures and signed measures, outer measures are also countably subadditive (cf. Proposition 8.3(e)).

(c) On the other hand, an outer measure μ^* has the advantage of being defined for every subset of X. And it also inherits some properties of a measure, such as $\mu^*(\varnothing) = 0$, $\mu^*(S) \geq 0$ and $\mu^*(S') \leq \mu^*(S)$ for every $S' \subseteq S \subseteq X$ and, what is more important, $\mu^*(E) = \mu(E)$ for every $E \in \mathcal{A}$ (Proposition 8.3).

8.2 The Carathéodory Extension Theorem

Consider the outer measure μ^* generated by a measure μ on an algebra $\mathcal{A}$ of subsets of a set X. A set $E \in \wp(X)$ is said to be *μ^*-measurable* (or satisfies the *Carathéodory condition*) if

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \backslash E)$$

for every $S \in \wp(X)$. This suggests an additive behavior of μ^* at S. Let

$$\mathcal{A}^* = \big\{E \in \wp(X)\colon\ E \text{ is } \mu^*\text{-measurable}\big\}$$

be the collection of all μ^*-measurable subsets of X. The next result is a decisive one for constructing measures on a σ-algebra out of measures on an algebra. It says that

(i) *this collection* $\mathcal{A}^*$ *is a* σ*-algebra,* (ii) *it includes the algebra* $\mathcal{A}$ *(i.e.,* $\mathcal{A} \subseteq \mathcal{A}^*$*),* (iii) *the restriction* $\mu^*|_{\mathcal{A}^*}$ *of the outer measure* μ^* *to* $\mathcal{A}^*$ *is a measure on the* σ*-algebra* $\mathcal{A}^*$*, and* (iv) *this measure* $\mu^*|_{\mathcal{A}^*}$ *extends the measure* μ *on the algebra* $\mathcal{A}$ *over* $\mathcal{A}^*$ *(i.e.,* $(\mu^*|_{\mathcal{A}^*})|_{\mathcal{A}} = \mu^*|_{\mathcal{A}} = \mu$*).*

Theorem 8.4. (Carathéodory Extension Theorem). $\mathcal{A}^*$ *is a* σ*-algebra that includes the algebra* $\mathcal{A}$*, and the restriction of* μ^* *to* $\mathcal{A}^*$ *is a measure on* $\mathcal{A}^*$*.*

Proof. Note that the empty set $\varnothing$ and the whole set X clearly lie in $\mathcal{A}^*$ (i.e., they are μ^*-measurable). Also, since $S \cap E = S\backslash(X\backslash E)$ for every pair of sets E and S in $\wp(X)$, it follows that the complement of every set in $\mathcal{A}^*$ is again in $\mathcal{A}^*$. Next take an arbitrary $S \in \wp(X)$. If $E \in \mathcal{A}^*$, then

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S\backslash E),$$

and if $E' \in \mathcal{A}^*$, then

$$\mu^*(S \cap E) = \mu^*(S \cap E \cap E') + \mu^*((S \cap E)\backslash E').$$

Now observe that $[S\backslash(E' \cap E)] \cap E = (S \cap E)\backslash(E' \cap E) = (S \cap E)\backslash E'$, and also that $[S\backslash(E' \cap E)]\backslash E = (S\backslash E)\backslash(E' \cap E) = S\backslash E$. Thus, if $E \in \mathcal{A}^*$,

$$\begin{aligned}\mu^*(S\backslash(E' \cap E)) &= \mu^*([S\backslash(E' \cap E)] \cap E) + \mu^*([S\backslash(E' \cap E)]\backslash E)\\ &= \mu^*((S \cap E)\backslash E') + \mu^*(S\backslash E).\end{aligned}$$

The above three displayed identities ensure that if E and E' lie in $\mathcal{A}^*$, then

$$\begin{aligned}\mu^*(S) &= \mu^*(S \cap E \cap E') + \mu^*((S \cap E)\backslash E') + \mu^*(S\backslash E)\\ &= \mu^*(S \cap E' \cap E) + \mu^*(S\backslash(E' \cap E)),\end{aligned}$$

and so $E' \cap E$ lies in $\mathcal{A}^*$. Therefore, since intersections of sets in $\mathcal{A}^*$ and complements of sets in $\mathcal{A}^*$ are both again in $\mathcal{A}^*$, it follows that unions of sets in $\mathcal{A}^*$ also lie in $\mathcal{A}^*$ (because $E' \cup E = X\backslash[(X\backslash E') \cap (X\backslash E)]$ — De Morgan laws). Then a trivial induction ensures that any finite union of sets in $\mathcal{A}^*$ remains in $\mathcal{A}^*$, and hence $\mathcal{A}^*$ is an algebra. That is,

(a) $\bigcup_{i=1}^n E_i$ lies in $\mathcal{A}^*$ for every finite family $\{E_i\}_{i=1}^n$ of sets in $\mathcal{A}^*$.

Furthermore, if S is a set in $\mathcal{P}(X)$, if E and E' are sets in $\mathcal{A}^*$, and if E and E' are disjoint (so that $S \cap (E' \cup E) \backslash E' = S \cap E$), then

$$\begin{aligned}\mu^*(S \cap (E' \cup E)) &= \mu^*(S \cap (E' \cup E) \cap E') + \mu^*(S \cap (E' \cup E) \backslash E') \\ &= \mu^*(S \cap E') + \mu^*(S \cap E),\end{aligned}$$

and so μ^* acts additively on the intersection of any set in $\mathcal{P}(X)$ with every pair of disjoint sets in $\mathcal{A}^*$. Thus another trivial induction ensures that

(b) $\mu^*\big(S \cap \big(\bigcup_{i=1}^n E_i\big)\big) = \sum_{i=1}^n \mu^*(S \cap E_i)$ for every finite family $\{E_i\}_{i=1}^n$ of pairwise disjoint sets in $\mathcal{A}^*$ and every set S in $\mathcal{P}(X)$.

In particular, for $S = X$, this shows that μ^* is *finitely additive* on $\mathcal{A}^*$:

(c) $\mu^*\big(\bigcup_{i=1}^n E_i\big) = \sum_{i=1}^n \mu^*(E_i)$ for every finite family $\{E_i\}_{i=1}^n$ of pairwise disjoint sets in $\mathcal{A}^*$.

Next, these results on finite families are extended to countably infinite families, which shows that $\mathcal{A}^*$ is a σ-algebra and μ^* is countably additive on $\mathcal{A}^*$, and hence the restriction of μ^* to $\mathcal{A}^*$ is a measure (by Definition 2.1). Thus, take any (infinite) sequence $\{E''_n\}$ of sets in $\mathcal{A}^*$ and consider the disjointification $\{E'_n\}$ of $\{E''_n\}$ given by $E'_1 = E''_1$ and $E'_{n+1} = E''_{n+1} \backslash \big(\bigcup_{i=1}^n E''_i\big)$ (see hints to Problems 2.3 and 7.2). Clearly, $\{E'_n\}$ is a sequence of pairwise disjoint sets in $\mathcal{A}^*$ (each E'_n lies in $\mathcal{A}^*$ because finite union and difference of sets in an algebra remain in the algebra). For each integer $n \geq 1$, set $E_n = \bigcup_{i=1}^n E'_i = \bigcup_{i=1}^n E''_i$, which lies in $\mathcal{A}^*$ (a finite union of sets in $\mathcal{A}^*$), and set $E = \bigcup_{n=1}^\infty E'_n = \bigcup_{n=1}^\infty E''_n$ in $\mathcal{P}(X)$. Take an arbitrary S in $\mathcal{P}(X)$. Note by the countable subadditivity of Proposition 8.3(e) that

$$\mu^*(S \cap E) = \mu^*\Big(S \cap \bigcup_{i=1}^\infty E'_i\Big) = \mu^*\Big(\bigcup_{i=1}^\infty (S \cap E'_i)\Big) \leq \sum_{i=1}^\infty \mu^*(S \cap E'_i),$$

and hence, since $S = (S \cap E) \cup (S \backslash E)$, subadditivity ensures again that

$$\mu^*(S) \leq \mu^*(S \cap E) + \mu^*(S \backslash E) \leq \sum_{i=1}^\infty \mu^*(S \cap E'_i) + \mu^*(S \backslash E).$$

On the other hand, since each $E_n = \bigcup_{i=1}^n E'_i$ lies in $\mathcal{A}^*$, and since the sequence $\{E'_n\}$ consists of pairwise disjoint sets in $\mathcal{A}^*$, it follows by (b) that

$$\mu^*(S) = \mu^*(S \cap E_n) + \mu^*(S \backslash E_n) = \sum_{i=1}^n \mu^*(S \cap E'_i) + \mu^*(S \backslash E_n)$$

for all n. Since $E_n \subseteq E$, we get $S \backslash E \subseteq S \backslash E_n$, so $\mu^*(S \backslash E) \leq \mu^*(S \backslash E_n)$ by Proposition 8.3(c). Therefore,

$$\sum_{i=1}^{n} \mu^*(S \cap E'_i) + \mu^*(S \backslash E) \leq \mu^*(S)$$

for all n, which implies that

$$\sum_{i=1}^{\infty} \mu^*(S \cap E'_i) + \mu^*(S \backslash E) \leq \mu^*(S).$$

Hence, for every $S \in \wp(X)$,

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \backslash E) = \sum_{i=1}^{\infty} \mu^*(S \cap E'_i) + \mu^*(S \backslash E).$$

The first identity in the above expression says that E lies in $\mathcal{A}^*$, that is,

(a′) $\bigcup_n E''_n$ lies in $\mathcal{A}^*$ for every countable family $\{E''_n\}$ (not necessarily pairwise disjoint) of sets in $\mathcal{A}^*$,

proving that the algebra $\mathcal{A}^*$ is, in fact, a σ-algebra. Furthermore, taking $S = E$ (so that $\mu^*(S \backslash E) = 0$ and $S \cap E'_i = E'_i$ for every $i \geq 1$), we also get

$$\mu^*(E) = \sum_{i=1}^{\infty} \mu^*(E'_i).$$

If $\{E''_n\}$ is a sequence of pairwise disjoint sets, then $E'_n = E''_n$ for each n, and so the above identity ensures that μ^* is countably additive on $\mathcal{A}^*$:

(c′) $\mu^*\big(\bigcup_n E'_n\big) = \sum_n \mu^*(E'_n)$ for every countably infinite family $\{E'_n\}$ of pairwise disjoint sets in $\mathcal{A}^*$.

Properties (a′) and (c′) ensure that the restriction of μ^* to $\mathcal{A}^*$ is a measure on the σ-algebra $\mathcal{A}^*$. Finally we verify the inclusion $\mathcal{A} \subseteq \mathcal{A}^*$. Take an arbitrary $A \in \mathcal{A}$, an arbitrary $S \in \wp(X)$, and an arbitrary $\varepsilon > 0$. According to Definition 8.2, there exists a sequence $\{A_n\}$ of sets in $\mathcal{A}$ such that

$$S \subseteq \bigcup\nolimits_n A_n \quad \text{and} \quad \sum\nolimits_n \mu(A_n) \leq \mu^*(S) + \varepsilon.$$

Note that $\{(A_n \cap A), (A_n \backslash A)\}$ is a partition of A_n for each n, which is made up of sets in the algebra $\mathcal{A}$. Thus, according to Proposition 8.3(c,e,d),

$$\mu^*(S \cap A) \leq \mu^*\Big(\bigcup\nolimits_n (A_n \cap A)\Big) \leq \sum\nolimits_n \mu^*(A_n \cap A) = \sum\nolimits_n \mu(A_n \cap A),$$

$$\mu^*(S \backslash A) \leq \mu^*\Big(\bigcup\nolimits_n (A_n \backslash A)\Big) \leq \sum\nolimits_n \mu^*(A_n \backslash A) = \sum\nolimits_n \mu(A_n \backslash A),$$

and so, by using the additivity of Definition 8.1(c),

$$\mu^*(S\cap A)+\mu^*(S\backslash A)\le \sum\nolimits_n \big(\mu(A_n\cap A)+\mu(A_n\backslash A)\big)$$
$$=\sum\nolimits_n \mu\big((A_n\cap A)\cup(A_n\backslash A)\big)=\sum\nolimits_n \mu(A_n)\le \mu^*(S)+\varepsilon.$$

Since $\varepsilon>0$ is arbitrary, this implies

$$\mu^*(S\cap A)+\mu^*(S\backslash A)\le \mu^*(S).$$

On the other hand, since $S=(S\cap A)\cup(S\backslash A)$,

$$\mu^*(S)\le \mu^*(S\cap A)+\mu^*(S\backslash A)$$

by the subadditivity of Proposition 8.3(e). Hence,

$$\mu^*(S)=\mu^*(S\cap A)+\mu^*(S\backslash A)$$

for every $S\in\wp(X)$, which means that A lies in $\mathcal{A}^*$. Thus $\mathcal{A}\subseteq\mathcal{A}^*$. □

The Carathéodory Extension Theorem is sometimes referred to as the Carathéodory–Fréchet–Hahn–Kolmogorov Extension Theorem. Since the restriction of the outer measure μ^* to the algebra $\mathcal{A}$ coincides with the measure μ on $\mathcal{A}$ (i.e., $\mu^*|_{\mathcal{A}}=\mu$), since the restriction $\mu^*|_{\mathcal{A}^*}$ of the outer measure μ^* to the σ-algebra $\mathcal{A}^*$ is a measure, and since $\mathcal{A}\subseteq\mathcal{A}^*\subseteq\wp(X)$, it follows that the measure $\mu^*|_{\mathcal{A}^*}$ on the σ-algebra $\mathcal{A}^*$ extends the measure μ on the algebra $\mathcal{A}$ over $\mathcal{A}^*$. Moreover, in the sense of Section 2.3,

This measure $\mu^*|_{\mathcal{A}^*}$ is complete on $\mathcal{A}^*$.

Equivalently, the σ-algebra $\mathcal{A}^*$ is complete with respect to $\mu^*|_{\mathcal{A}^*}$. This means that if $N\in\mathcal{A}^*$ and $\mu^*(N)=0$, then every subset E of N lies in $\mathcal{A}^*$ (and so $\mu^*(E)=0$ according to Proposition 8.3(b,c)). More is true: *all sets with outer measure zero are μ^*-measurable* (which implies that $\mathcal{A}^*$ is, indeed, complete). We will prove this in the proposition below.

Proposition 8.5. *Consider the outer measure μ^* on $\wp(X)$ (induced by a measure on $\mathcal{A}$). Let $\mathcal{A}^*$ be the σ-algebra of all μ^*-measurable sets in $\wp(X)$.*

(a) *If $N\in\wp(X)$ is such that $\mu^*(N)=0$, then $N\in\mathcal{A}^*$.*

(b) *If $E\subseteq N\in\wp(X)$ and $\mu^*(N)=0$, then $E\in\mathcal{A}^*$ and $\mu(E)=0$.*

Proof. Assertion (b) follows from assertion (a) and Proposition 8.3(b,c). Assertion (a) says that every set with outer measure zero is μ^*-measurable. To verify this take arbitrary sets S and N in $\wp(X)$ such that $\mu^*(N)=0$. Since $S\cup N=(S\cap N)\cup(S\backslash N)\cup(N\backslash S)$, we get from Proposition 8.3(c,e) that

$$\mu^*(S)\le\mu^*(S\cup N)\le\mu^*(S\cap N)+\mu^*(S\backslash N)\le\mu^*(N)+\mu^*(S)=\mu^*(S),$$

and so $\mu^*(S)=\mu^*(S\cap N)+\mu^*(S\backslash N)$; that is, $N\in\mathcal{A}^*$. □

A measure on an algebra inherits most of the attributes of a measure on a σ-algebra. For instance, a measure μ on an algebra $\mathcal{A}$ of subsets of a set X is *finite* if $\mu(X) < \infty$. Similarly, μ is *σ-finite* if X is covered by a countable family of sets in $\mathcal{A}$ of finite measure, that is, if there exists a sequence $\{A_n\}$ of sets in $\mathcal{A}$ such that $\mu(A_n) < \infty$ for every n and $X = \bigcup_n A_n$. Finiteness and σ-finiteness are naturally extended from a measure μ on an algebra $\mathcal{A}$ to the associated outer measure μ^* on the power set $\wp(X)$ by Proposition 8.3(d) since $X \in \mathcal{A} \subseteq \wp(X)$ — just replace μ with μ^*. The next result says that if μ is σ-finite, then its extension over the σ-algebra $\mathcal{A}^*$ is unique.

Theorem 8.6. (Hahn Extension Theorem). *If a measure μ on the algebra $\mathcal{A}$ is σ-finite, then its extension to a measure on the σ-algebra $\mathcal{A}^*$ is unique.*

Proof. Consider the outer measure μ^* generated by a measure μ on an algebra $\mathcal{A}$ of subsets of a set X. The collection $\mathcal{A}^*$ of all μ^*-measurable subsets of X is a σ-algebra such that $\mathcal{A} \subseteq \mathcal{A}^*$, and the restriction of μ^* to $\mathcal{A}^*$ is a measure that extends μ over $\mathcal{A}^*$ (Proposition 8.5). Let η be a measure on $\mathcal{A}^*$ that extends μ over $\mathcal{A}^*$ (i.e., suppose η is a measure on $\mathcal{A}^*$ such that $\eta(E) = \mu(E)$ for every $E \in \mathcal{A}$). The proof will be split into two parts. The claimed result is proved supposing μ is a finite measure in part (a). This is extended to the case when μ is σ-finite in part (b).

(a) Assume that μ is a finite measure on the algebra $\mathcal{A}$. Then any extension η of μ to $\mathcal{A}^*$ is again finite (because $X \in \mathcal{A} \subseteq \mathcal{A}^* \subseteq \wp(X)$ implies $\mu^*(X) = \eta(X) = \mu(X) < \infty$). Take an arbitrary set E in $\mathcal{A}^*$, and an arbitrary sequence $\{A_n\}$ in $\mathcal{C}_E$. That is, let $\{A_n\}$ be any sequence of sets in $\mathcal{A}$ such that $E \subseteq \bigcup_n A_n$. (These sequences do exist; trivial example: $A_1 = X$.) Thus

$$\eta(E) \le \eta\Big(\bigcup\nolimits_n A_n\Big) \le \sum\nolimits_n \eta(A_n) = \sum\nolimits_n \mu(A_n).$$

(See Proposition 2.2(a) and Problem 2.8(b).) Since this holds for every $\{A_n\} \in \mathcal{C}_E$, it follows by the definition of the outer measure μ^* induced by the measure μ on the algebra $\mathcal{A}$ (Definition 8.2) that

$$\eta(E) \le \inf_{\{E_n\} \in \mathcal{C}_E} \sum\nolimits_n \mu(E_n) = \mu^*(E).$$

Recall that $\mu^*|_{\mathcal{A}^*}$ (μ^* restricted to $\mathcal{A}^*$) and η are measures on the σ-algebra $\mathcal{A}^*$, so that they are additive by Definition 2.1(c). Moreover, also recall that

$$\mu^*(X) = \eta(X) = \mu(X) < \infty.$$

Thus the above inequality leads to

$$\mu^*(E) = \mu^*(X) - \mu^*(X \backslash E) = \eta(X) - \mu^*(X \backslash E) \le \eta(X) - \eta(X \backslash E) = \eta(E).$$

Hence $\eta(E) = \mu^*(E)$ for all $E \in \mathcal{A}^*$, which proves uniqueness if μ is finite.

(b) Assume that μ is σ-finite. Thus any extension η of μ to $\mathcal{A}^*$ is again σ-finite. In fact, if μ is σ-finite, then there exists a sequence $\{A_n\}$ of sets in $\mathcal{A} \subseteq \mathcal{A}^* \subseteq \wp(X)$ for which $\mu^*(A_n) = \eta(A_n) = \mu(A_n) < \infty$ for every $n \geq 1$ and $X = \bigcup_n A_n$, and so μ^* and η are σ-finite as well. Set $A'_n = \bigcup_{i=1}^n A_i$ so that $\{A'_n\}$ is an *increasing* sequence of sets of finite measure that covers X. Take an arbitrary E in $\mathcal{A}^*$. Since $\{E \cap A'_n\}$ is a sequence of sets in $\mathcal{A}^*$ (intersection and finite union of sets in $\mathcal{A}^*$ remain in $\mathcal{A}^*$) of finite measure (since $\mu^*(E \cap A'_n) \leq \mu^*(A'_n) < \infty$ and $\eta(E \cap A'_n) \leq \eta(A'_n) < \infty$), it follows from part (a) that for every $n \geq 1$,

$$\mu^*(E \cap A'_n) = \eta(E \cap A'_n).$$

Thus, since $\{E \cap A'_n\}$ is an increasing sequence of sets in $\mathcal{A}^*$ such that $E = \bigcup_n (E \cap A'_n)$, and since both η and the restriction $\mu^*|_{\mathcal{A}^*}$ of μ^* to the σ-algebra $\mathcal{A}^*$ are measures on $\mathcal{A}^*$, we get by Proposition 2.2(c) that

$$\mu^*(E) = \lim_n \mu^*(E \cap A'_n) = \lim_n \eta(E \cap A'_n) = \eta(E).$$

Hence $\eta(E) = \mu^*(E)$ for all $E \in \mathcal{A}^*$, proving uniqueness if μ is σ-finite. □

Therefore, let η denote the measure on the σ-algebra $\mathcal{A}^*$ obtained by the restriction of the outer measure $\mu^* : \wp(X) \to \overline{\mathbb{R}}$ to $\mathcal{A}^*$. In other words, set

$$\eta = \mu^*|_{\mathcal{A}^*} : \mathcal{A}^* \to \overline{\mathbb{R}}.$$

The previous results are summarized as follows.

> *If $\mu : \mathcal{A} \to \overline{\mathbb{R}}$ is a σ-finite measure defined on an algebra $\mathcal{A}$, then there exist a σ-algebra $\mathcal{A}^*$ including $\mathcal{A}$ and a unique extension of μ to a measure over the σ-algebra $\mathcal{A}^*$. This measure on $\mathcal{A}^*$ is σ-finite and, by uniqueness, coincides with $\eta : \mathcal{A}^* \to \overline{\mathbb{R}}$.*

That is, there exists a unique measure η on the σ-algebra $\mathcal{A}^*$ such that

$$\mu = \eta|_{\mathcal{A}} : \mathcal{A} \to \overline{\mathbb{R}};$$

equivalently, such that $\eta(E) = \mu(E)$ for every $E \in \mathcal{A}$. Moreover, *the measure space $(X, \mathcal{A}^*, \eta)$ is complete*. This means that the σ-algebra $\mathcal{A}^*$ is complete with respect to the measure η or, equivalently, the measure η is complete on the σ-algebra $\mathcal{A}^*$.

8.3 Construction of Lebesgue Measure

We split this section into two parts. First, we construct the Lebesgue measure, as promised in Example 2C. Then, we discuss several properties of it, some of which are definitely nontrivial.

Construction. Consider the classes of (left-open) intervals of the real line.

Class $\mathcal{C}_1$: $\big\{(\alpha,\beta]\subseteq\mathbb{R}\colon\ \alpha,\beta\in\mathbb{R},\ \alpha\le\beta\big\}$.

Class $\mathcal{C}_2$: $\big\{(-\infty,\beta]\subseteq\mathbb{R}\colon\ \beta\in\mathbb{R}\big\}$.

Class $\mathcal{C}_3$: $\big\{(\alpha,+\infty)\subseteq\mathbb{R}\colon\ \alpha\in\mathbb{R}\big\}$.

Class $\mathcal{C}_4$: $\big\{(-\infty,+\infty)\big\}$.

These four classes are exhaustive. Let $\mathcal{I}$ be the family of all (left-open) intervals of the real line, each of them belonging to one of the above classes. The empty set $\varnothing$ is of class $\mathcal{C}_1$ (for the case of $\alpha=\beta$), and the only interval of class $\mathcal{C}_4$ is $\mathbb{R}$ itself. It is clear that the intersection of any two sets in $\mathcal{I}$ is again a set in $\mathcal{I}$ and that the complement of any set in $\mathcal{I}$ is a finite union of disjoint sets in $\mathcal{I}$. This means that $\mathcal{I}$ is a *semialgebra.* But the finite union of sets in $\mathcal{I}$ is not necessarily a set in $\mathcal{I}$, and so $\mathcal{I}$ is not an algebra. However, the collection $\mathfrak{S}$ of all finite unions of sets in $\mathcal{I}$ is an algebra (Problem 8.3).

Definition 8.7. Consider the collection $\mathfrak{S}\subseteq\wp(\mathbb{R})$ consisting of all *finite* unions of (left-open) intervals of the real line (i.e., of all *finite* unions of intervals from the semialgebra $\mathcal{I}$),

$$\mathfrak{S}=\big\{F\subseteq\mathbb{R}\colon\ F \text{ is a finite union of intervals from } \mathcal{I}\ \big\}.$$

Let $\ell\colon\mathfrak{S}\to\overline{\mathbb{R}}$ be an extended real-valued set function such that

(a) $\ell\big((\alpha,\beta]\big)=\beta-\alpha$ for any $\alpha,\beta\in\mathbb{R}$ such that $\alpha\le\beta$,

(b) $\ell\big((-\infty,\beta]\big)=\ell\big((\alpha,+\infty)\big)=\ell\big((-\infty,+\infty)\big)=\infty$,

(c) $\ell\big(\bigcup_i I_i\big)=\sum_i\ell(I_i)$ (*countable additivity*)
for every *finite* family $\{I_i\}$ of pairwise disjoint intervals in $\mathcal{I}$.

A function with these properties is referred to as the *length function* on $\mathfrak{S}$.

We show that properties (a), (b), (c) are enough to make the length function $\ell\colon\mathfrak{S}\to\overline{\mathbb{R}}$ well defined (and unique) on the collection $\mathfrak{S}$, which is an algebra, and on which ℓ is a measure. For the proof, we proceed as follows. First we check that ℓ is well defined at every set in $\mathfrak{S}$ (Problem 8.2). Next we show that $\mathfrak{S}$ is an algebra (Problem 8.3), and so we conclude that the set function ℓ is increasing (Problem 8.4). This is applied to prove the auxiliary result in Proposition 8.8, which is used to prove in Lemma 8.9 that the length function ℓ is countably additive, and so it is a measure on the algebra $\mathfrak{S}$.

Proposition 8.8. *Let $(a,b]$ be an interval of class $\mathcal{C}_1$. If $\{(\alpha_k,\beta_k]\}$ is a countable family of disjoint intervals of class $\mathcal{C}_1$, then*

(a) $\qquad (a,b]=\bigcup_k(\alpha_k,\beta_k] \quad\text{implies}\quad \ell\big((a,b]\big)=\sum_k\ell\big((\alpha_k,\beta_k]\big).$

If $\{(a_i, b_i]\}$ *is a finite set of disjoint intervals of class* $\mathcal{C}_1$*, then*

$$\text{(b)} \quad \bigcup\nolimits_i (a_i, b_i] = \bigcup\nolimits_k (\alpha_k, \beta_k] \quad \text{implies} \quad \ell\Big(\bigcup\nolimits_i (a_i, b_i]\Big) = \sum\nolimits_k \ell\big((\alpha_k, \beta_k]\big).$$

Proof. First note that if the countable family $\{(\alpha_k, \beta_k]\}$ of disjoint intervals is finite, then the results in (a) and (b) are trivially verified by Definition 8.7(c). Thus suppose the countable family $\{(\alpha_k, \beta_k]\}$ is infinite. To avoid trivialities, assume that all intervals $\{(\alpha_k, \beta_k]\}$ are *nonempty*, which means that $\alpha_k < \beta_k$ for every k.

(a) Suppose $(a, b] = \bigcup_k (\alpha_k, \beta_k]$. Take an arbitrary finite subfamily $\{(\alpha_i, \beta_i]\}$ of the infinite family $\{(\alpha_k, \beta_k]\}$. Since $\{(\alpha_i, \beta_i]\}$ has a finite number of subintervals of $(a, b]$, it follows that $a \le \min\{\alpha_i\}$ and $\max\{\beta_i\} \le b$, and so

$$\sum\nolimits_i \ell\big((\alpha_i, \beta_i]\big) = \sum\nolimits_i (\beta_i - \alpha_i) \le b - a = \ell\big((a, b]\big).$$

This holds for all finite subfamilies $\{(\alpha_i, \beta_i]\}$ of $\{(\alpha_k, \beta_k]\}$. By taking the supremum over all finite subsets of the countably infinite set $\{\ell((\alpha_k, \beta_k])\}$ of positive numbers we get

$$\text{(a}_1\text{)} \qquad \sum\nolimits_k \ell\big((\alpha_k, \beta_k]\big) = \sup \sum\nolimits_i \ell\big((\alpha_i, \beta_i]\big) \le \ell\big((a, b]\big).$$

To verify the reverse inequality, take an arbitrary $\varepsilon > 0$ and let $\{\varepsilon_k\}$ be any sequence of *positive* numbers such that $\sum_k \varepsilon_k \le \varepsilon$. Since $(a, b] = \bigcup_k (\alpha_k, \beta_k]$, $b \in (\alpha_{k_1}, \beta_{k_1}]$ for some index k_1, and so $b = \beta_{k_1}$. Furthermore, it also follows that $a = \inf_k\{\alpha_k\}$, which ensures the existence of an index $k_0 \ne k_1$ such that $\alpha_{k_0} < a + \varepsilon_0$. Hence the following infinite family of *open* intervals

$$\big\{(\alpha_{k_0} - \varepsilon_0, \beta_{k_0} + \varepsilon_0)\,,\ (\alpha_k, \beta_k + \varepsilon_k) \ \text{ for every } \ k \ne k_0 \ \text{ with } \ k_0 \ne k_1\big\}$$

covers the closed and bounded interval $[a, b]$ (i.e., it covers the compact interval $[a, b]$ by the Heine–Borel Theorem — as will be discussed in the remark after the proof of Lemma 11.2). The definition of compactness says that this family of open intervals has a *finite* subfamily of open intervals that still covers $[a, b]$ (cf. Definition 11.1(g) for compactness in topological spaces),

$$\big\{(\alpha_{k_0} - \varepsilon_0, \beta_{k_0} + \varepsilon_0)\,,\ (\alpha_j, \beta_j + \varepsilon_j) \ \text{ for every } \ j \in J\big\},$$

with J being a finite index set such that $k_0 \notin J$ and $k_1 \in J$. If the intervals in $\{(\alpha_k, \beta_k]\}$ are pairwise disjoint, then we may assume that $\alpha_{k_0} < \alpha_j < \alpha_{k_1}$ for all $j \in J \backslash \{k_1\}$. Note that J is not empty ($k_1 \in J$). Let $n \in \mathbb{N}$ be the cardinality of the finite set J (i.e., the number of elements of J), and relabel this finite family of open intervals with nonnegative integers $i \in [0, 1, \ldots, n]$. Since the intervals in $\{(\alpha_k, \beta_k]\}$ are disjoint, we can order the endpoints of

the intervals that appear in the above finite family as $\alpha_i < \beta_i \le \alpha_{i+1} < \beta_{i+1}$ for each $i \in [0, 1, \ldots, n-1]$, identifying k_0 with 0 and k_1 with n, so that

$$\alpha_0 - \varepsilon_0 < a \le \alpha_0 < \beta_0 \le \alpha_i < \beta_i \le \alpha_{i+1} < \beta_{i+1} < \alpha_n < \beta_n = b < \beta_n + \varepsilon_n$$

for $i \in [1, \ldots, n-2]$. This finite family of open intervals is then rewritten as

$$\big\{(\alpha_0 - \varepsilon_0, \beta_0 + \varepsilon_0)\,,\ (\alpha_i, \beta_i + \varepsilon_i) \ \text{ for every } \ i \in [1, \ldots, n]\big\},$$

which covers $[a, b]$, and so it covers the interval $(a, b]$ of class $\mathcal{C}_1$:

$$\begin{aligned}(a, b] \subseteq [a, b] \ &\subseteq \ (\alpha_0 - \varepsilon_0, \beta_0 + \varepsilon_0) \cup \bigcup_{i=1}^{n} (\alpha_i, \beta_i + \varepsilon_i) \\ &\subseteq (\alpha_0 - \varepsilon_0, \alpha_0] \cup (\alpha_0, \beta_0] \cup (\beta_0, \beta_0 + \varepsilon_0] \cup \bigcup_{i=1}^{n} (\alpha_i, \beta_i] \cup \bigcup_{i=1}^{n} (\beta_i, \beta_i + \varepsilon_i].\end{aligned}$$

Let i be an arbitrary index in $[0, \ldots, n-1]$. Since $\{(\alpha_k, \beta_k]\}$ consists of disjoint intervals, it follows that $\beta_i \le \alpha_{i+1}$. Also, since the above union of intervals of class $\mathcal{C}_1$ covers $[a, b]$, it follows that $\alpha_{i+1} \le \beta_i + \varepsilon_i$. Then consider a finite sequence $\{\delta_i\}$ of nonnegative numbers given by

$$0 \le \delta_i = \alpha_{i+1} - \beta_i \le \varepsilon_i \ \text{ for each } \ i \in [0, \ldots, n-1] \quad \text{ and } \quad 0 < \delta_n = \varepsilon_n.$$

Replacing $\{\varepsilon_i\}$ with $\{\delta_i\}$ in the intervals of the form $(\beta_i, \beta_i + \varepsilon_i]$ we get a *finite* covering for $(a, b]$ consisting of *disjoint* intervals of class $\mathcal{C}_1$, viz.,

$$(\alpha_0 - \varepsilon_0, \alpha_0] \cup (\alpha_0, \beta_0] \cup (\beta_0, \beta_0 + \delta_0] \cup \bigcup_{i=1}^{n} (\alpha_i, \beta_i] \cup \bigcup_{i=1}^{n} (\beta_i, \beta_i + \delta_i].$$

Hence, according to Definition 8.7(a,c) and Problem 8.4,

$$\ell\big((a, b]\big) \le \varepsilon_0 + \sum_{i=0}^{n} \ell\big((\alpha_i, \beta_i]\big) + \sum_{i=0}^{n} \delta_i \le \sum\nolimits_k \ell\big((\alpha_k, \beta_k]\big) + 2\varepsilon$$

for $\sum_{i=0}^{n} \delta_i < \sum_k \varepsilon_k \le \varepsilon$. Since this holds for an arbitrary $\varepsilon > 0$, we get

$$(\mathrm{a}_2) \qquad \ell\big((a, b]\big) \le \sum\nolimits_k \ell\big((\alpha_k, \beta_k]\big).$$

Therefore the result in (a) follows by (a_1) and (a_2).

(b) The disjointness assumption on both $\{(\alpha_k, \beta_k]\}$ and $\{(a_i, b_i]\}$ ensures that, if $\bigcup_i (a_i, b_i] = \bigcup_k (\alpha_k, \beta_k]$, then $(a_i, b_i] = \bigcup_j (\alpha_{i,j}, \beta_{i,j}]$ for each i, where $\{(\alpha_{i,j}, \beta_{i,j}]\} = \{(\alpha_k, \beta_k]\}$. Thus, applying Definition 8.7(c) and the result in item (a), we get

$$\ell\Big(\bigcup\nolimits_k (\alpha_k, \beta_k]\Big) = \ell\Big(\bigcup\nolimits_i (a_i, b_i]\Big) = \sum\nolimits_i \ell\big((a_i, b_i]\big) = \sum\nolimits_i \sum\nolimits_j \ell\big((\alpha_{i,j}, \beta_{i,j}]\big).$$

Since the summands $\{\ell((\alpha_{i,j}, \beta_{i,j}])\}$ are nonnegative real numbers, the doubly indexed sum is unconditionally convergent, and hence

$$\sum\nolimits_i \sum\nolimits_j \ell\big((\alpha_{i,j}, \beta_{i,j}]\big) = \sum\nolimits_k \ell\big((\alpha_k, \beta_k]\big). \qquad \square$$

Lemma 8.9. *The collection $\mathfrak{F}$ of all finite unions of (left-open) intervals of the real line is an algebra of subsets of $\mathbb{R}$, and the length function $\ell\colon \mathfrak{F} \to \overline{\mathbb{R}}$ is a measure defined on the algebra $\mathfrak{F}$.*

Proof. Take the collection $\mathfrak{F}$ of all finite unions of intervals from $\mathcal{I}$ as in Definition 8.7. It is readily verified that $\mathfrak{F}$ is an algebra of subsets of $\mathbb{R}$ (cf. Problem 8.3). We claim that the length function ℓ is a measure on the algebra $\mathfrak{F}$. Note that $\ell(\varnothing) = 0$ by Definition 8.7(a). Next note that every set in $\mathfrak{F}$ can be expressed as a finite union of *disjoint* intervals from $\mathcal{I}$ so that $\ell(F) \geq 0$ for every $F \in \mathfrak{F}$ by Definition 8.7(a,b,c). Indeed, every countable family of intervals in $\mathcal{I}$ admits a disjointification *consisting of intervals in* $\mathcal{I}$ (see hints to Problems 2.3 and 7.2). To complete the proof, it remains to verify axiom (c) — countable additivity — of Definition 8.1, viz.,

(c) $\ell\big(\bigcup_n F_n\big) = \sum_n \ell(F_n)$
for every countable family $\{F_n\}$ of pairwise disjoint sets in $\mathfrak{F}$ for which $\bigcup_n F_n$ lies in $\mathfrak{F}$.

Recall that every set in the algebra $\mathfrak{F}$ can be expressed as a finite union of *disjoint* intervals from $\mathcal{I}$. Hence the identity in property in (c) is readily verified by Definition 8.7(c) if $\{F_i\}$ is a *finite* family of pairwise disjoint sets in $\mathfrak{F}$. Thus suppose $\{F_n\}$ is a countably infinite family (equivalently, an infinite sequence) of pairwise disjoint sets in $\mathfrak{F}$ such that $\bigcup_n F_n$ lies in $\mathfrak{F}$. Take the extended nonnegative-valued sequence $\{\ell(F_n)\}$. If $\sum_n \ell(F_n) < \infty$, then $\ell\big(\bigcup_i F_i\big) = \sum_i \ell(F_i) < \sum_n \ell(F_n) < \infty$ for every finite subunion $\bigcup_i F_i$ of $\bigcup_n F_n$ because (c) holds for finite families of disjoint sets in $\mathfrak{F}$. So

$$\ell\Big(\bigcup\nolimits_n F_n\Big) = \sup \ell\Big(\bigcup\nolimits_i F_i\Big) \leq \sum\nolimits_n \ell(F_n) < \infty,$$

where the supremum is taken over all *finite subunions* of the countably infinite union $\bigcup_n F_n$. Hence, if $\ell\big(\bigcup_n F_n\big) = \infty$, then axiom (c) holds:

$$\ell\Big(\bigcup\nolimits_n F_n\Big) = \sum\nolimits_n \ell(F_n) = \infty.$$

Thus suppose $\ell\big(\bigcup_n F_n\big) < \infty$ so that $\ell(F_n) < \infty$ for each n by Problem 8.4, and recall that each F_n and also $\bigcup_n F_n$ are sets in $\mathfrak{F}$. Then each set F_n is a finite union of disjoint intervals of class $\mathcal{C}_1$,

$$F_n = \bigcup\nolimits_j (\alpha_{n,j}, \beta_{n,j}].$$

Since $\{F_n\}$ is a sequence of disjoint sets, this implies that its union

$$\bigcup_n F_n = \bigcup_{n,j} (\alpha_{n,j}, \beta_{n,j}] = \bigcup_k (\alpha_k, \beta_k]$$

(where $\{(\alpha_{n,j}, \beta_{n,j}]\} = \{(\alpha_k, \beta_k]\}$ is an infinite family of disjoint intervals of class $\mathcal{C}_1$) also is a finite union of disjoint intervals of class $\mathcal{C}_1$,

$$\bigcup_n F_n = \bigcup_i (a_i, b_i].$$

Thus $\bigcup_i (a_i, b_i] = \bigcup_k (\alpha_k, \beta_k]$. Using Proposition 8.8(b) and recalling the unconditional convergence argument that closed that proof, we get

$$\begin{aligned}\ell\Big(\bigcup_n F_n\Big) &= \ell\Big(\bigcup_i (a_i, b_i]\Big) = \sum\nolimits_k \ell\big((\alpha_k, \beta_k]\big) = \sum\nolimits_{n,j} \ell\big((\alpha_{n,j}, \beta_{n,j}]\big) \\ &= \sum\nolimits_n \sum\nolimits_j \ell\big((\alpha_{n,j}, \beta_{n,j}]\big) = \sum\nolimits_n \ell\Big(\bigcup_j (\alpha_{n,j}, \beta_{n,j}]\Big) = \sum\nolimits_n \ell(F_n)\end{aligned}$$

by Definition 8.7(c), so that (c) holds, thus completing the proof. □

Now we can to apply the Carathéodory Extension Theorem of Section 8.2 to build up the Lebesgue measure.

Theorem 8.10. *There exists a σ-algebra $\Im^*$ of subsets of $\mathbb{R}$ that includes the algebra $\Im$ and extends the length function $\ell\colon \Im \to \overline{\mathbb{R}}$ uniquely to a measure $\lambda^*\colon \Im^* \to \overline{\mathbb{R}}$, which is σ-finite and complete on the σ-algebra $\Im^*$.*

Proof. Lemma 8.9 says that $\ell\colon \Im \to \overline{\mathbb{R}}$ is a measure on the algebra $\Im$. Let $\ell^*\colon \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ be the outer measure generated by ℓ (as in Definition 8.2), and let $\Im^*$ be the collection of all sets in $\wp(\mathbb{R})$ that are ℓ^*-measurable:

$$\Im^* = \big\{E \in \wp(\mathbb{R})\colon\ \ell^*(S) = \ell^*(S \cap E) + \ell^*(S \backslash E) \ \text{ for every } \ S \in \wp(\mathbb{R})\big\}.$$

According to Theorem 8.4, $\Im^*$ is a σ-algebra of subsets of $\mathbb{R}$ that includes the algebra $\Im$, and the restriction of ℓ^* to $\Im^*$ is a measure. In other words, $\Im \subseteq \Im^* \subseteq \wp(\mathbb{R})$ and $\ell^*|_{\Im^*}$ is a measure on $\Im^*$. Thus this measure $\ell^*|_{\Im^*}$ extends ℓ over $\Im^*$ according to Proposition 8.3(d). Also, it is readily verified that the measure ℓ on the algebra $\Im$ is σ-finite. Indeed, the real line $\mathbb{R}$ is covered by the countably infinite family of intervals $\{(q_k - \varepsilon, q_k + \varepsilon]\}$ of class $\mathcal{C}_1$ of length 2ε for any $\varepsilon > 0$, where $\{q_k\}$ (with k running over all integers $\mathbb{Z}$) is an enumeration of the rational numbers $\mathbb{Q}$ (see Example 2C). Since ℓ is σ-finite, Theorem 8.6 ensures that there is a unique measure on $\Im^*$, say $\lambda^*\colon \Im^* \to \overline{\mathbb{R}}$, that extends ℓ over $\Im^*$, which is again σ-finite. By uniqueness, this extension of ℓ over $\Im^*$ coincides with the restriction of ℓ^* to $\Im^*$; that is, $\lambda^* = \ell^*|_{\Im^*}$. Thus Proposition 8.5 ensures that λ^* is a complete measure on the σ-algebra $\Im^*$ (i.e., the measure space $(\mathbb{R}, \Im^*, \lambda^*)$ is complete). □

Sets in $\Im^*$ are called *Lebesgue sets* (or $\Im^*$-measurable), and $\Im^*$ is referred to as the *Lebesgue σ-algebra*. The measure λ^* on $\Im^*$, as defined in Theorem 8.10, is also called *Lebesgue measure*. We will see below that $\Im^*$ is the completion of $\Re$, and λ^* is the completion of the Lebesgue measure λ on $\Re$.

Remark 8.C (The Lebesgue measure construction — a summary of steps):

- $\diamond$ $\mathcal{I} \subset \wp(\mathbb{R})$: the semialgebra of all (left-open) intervals in $\mathbb{R}$.
- $\diamond$ $\Im \subset \wp(\mathbb{R})$: the algebra of all *finite* unions of intervals from $\mathcal{I} \subset \Im$.
- $\diamond$ $\ell : \Im \to \overline{\mathbb{R}}$: the length function; a measure on the algebra $\Im$.
- $\diamond$ $\ell^* : \wp(\mathbb{R}) \to \overline{\mathbb{R}}$: the outer measure generated by ℓ (countably subadditive).
- $\diamond$ $\Im^* \subset \wp(\mathbb{R})$: the (complete) σ-algebra extending the algebra $\Im \subset \Im^*$; the Lebesgue σ-algebra of Lebesgue subsets of $\mathbb{R}$.
- $\diamond$ $\lambda^* : \Im^* \to \overline{\mathbb{R}}$: the (complete) Lebesgue measure on the σ-algebra $\Im^*$.
- $\diamond$ $\Re \subset \wp(\mathbb{R})$: the σ-algebra of all Borel subsets of $\mathbb{R}$, for which $\Im \subset \Re \subset \Im^*$.
- $\diamond$ $\lambda : \Re \to \overline{\mathbb{R}}$: the restriction of λ^* to $\Re$; the Lebesgue measure on $\Re$.

We close the section by considering a collection of basic properties of Lebesgue measure.

Properties. Consider the Borel σ-algebra $\Re$, consisting of the Borel sets (i.e., $\Re$-measurable sets), which is the σ-algebra generated by the open intervals of the real line $\mathbb{R}$ or, equivalently, by the left-open intervals in $\mathcal{I}$, which means that $\Re$ is the intersection of all σ-algebras of subsets of $\mathbb{R}$ that include the semialgebra $\mathcal{I}$. Then (see also the remark following Problem 1.14),

$$\Re \text{ is the smallest } \sigma\text{-algebra including the semialgebra } \mathcal{I}.$$

Since any σ-algebra that includes the semialgebra $\mathcal{I}$ necessarily includes the algebra $\Im$ (finite union of intervals from $\mathcal{I}$), it follows that

$$(\mathrm{P}_1) \qquad \Re \text{ is the smallest } \sigma\text{-algebra including the algebra } \Im.$$

Thus $\Re$ properly includes the algebra $\Im$ (because $\Im$ is not a σ-algebra — Problem 8.3). Moreover, $\Re$ is included in the σ-algebra $\Im^*$ (since $\Im^*$ is a σ-algebra that includes $\Im$). This leads to the following chain of inclusions:

$$\Im \subset \Re \subseteq \Im^* \subseteq \wp(\mathbb{R}).$$

It is clear that the restriction of the Lebesgue measure $\lambda : \Re \to \overline{\mathbb{R}}$ of Example 2C to the algebra $\Im$ is the length function $\ell : \Im \to \overline{\mathbb{R}}$ (Problem 2.7(c)). Recall that the restriction of the outer measure $\ell^* : \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ to the Lebesgue σ-algebra $\Im^*$ is the Lebesgue measure $\lambda^* : \Im^* \to \overline{\mathbb{R}}$ (Proof of Theorem 8.10). So

$$\lambda|_{\Im} = \ell \quad \text{and} \quad \ell^*|_{\Im^*} = \lambda^*.$$

Since the length function ℓ on $\Im$ extends to the measure λ^* on $\Im^*$ so that $\lambda^*_{\Im} = \ell = \lambda|_{\Im}$, and since this extension is unique (Theorem 8.10), we can infer that the restriction of the Lebesgue measure $\lambda^* : \Im^* \to \overline{\mathbb{R}}$ to the Borel σ-algebra $\Re$ is the Lebesgue measure $\lambda : \Re \to \overline{\mathbb{R}}$ of Example 2C. That is,

$$(\mathrm{P}_2) \qquad \lambda^*|_{\Re} = \lambda.$$

Summing up,

$$\ell = \lambda|_{\Im} = \lambda^*|_{\Im} = \ell^*|_{\Im}, \qquad \lambda = \lambda^*|_{\Re} = \ell^*|_{\Re}, \qquad \lambda^* = \ell^*|_{\Im^*}.$$

Recall that λ is a Borel measure in the sense of Problem 2.13 (which implies that all bounded sets in $\Re$ have finite measure). Then so is its extension λ^* over $\Im^*$. Since $\Re$-measurable sets are $\Im^*$-measurable, all the Borel sets of Problem 2.7 are Lebesgue sets, and their λ^*-measures (as Lebesgue sets) coincide with their λ-measures (as Borel sets), which in turn coincide with their ℓ^*-outer measures. In particular, the Cantor set in $\Re$ of Problem 2.9 is an uncountable set with Lebesgue measure zero. Also, the measure space $(\mathbb{R}, \Im^*, \lambda^*)$ *is the completion of the measure space* $(\mathbb{R}, \Re, \lambda)$. Equivalently, the σ-algebra $\Im^*$ *is the completion of the* σ*-algebra* $\Re$ with respect to λ, or the measure λ^* *is the completion of the measure* λ on $\Re$. Symbolically,

$$(\mathrm{P}_3) \qquad \Im^* = \widehat{\Re} \quad \text{and} \quad \lambda^* = \widehat{\lambda}.$$

(See the paragraph that closes Section 2.3.) In fact, since $(\mathbb{R}, \Im^*, \lambda^*)$ is a complete measure space, and according to Problem 8.6(b), it follows that

$$\Im^* = \big\{ \widehat{B} \subseteq \mathbb{R} : \ \widehat{B} = B \cup A, \ \text{with} \ B \in \Re, \ A \subseteq N \in \Re \ \text{and} \ \lambda(N) = 0 \big\}.$$

This is precisely the definition of $\widehat{\Re}$. So, according to Problem 2.15,

$\Im^*$-measurable functions are a.e. equal to $\Re$-measurable functions.

In other words, if $f : \mathbb{R} \to \overline{\mathbb{R}}$ is an $\Im^*$-measurable (or a *Lebesgue measurable*) function, then there is an $\Re$-measurable (or a Borel measurable) function $g : \mathbb{R} \to \overline{\mathbb{R}}$ such that $g = f$ λ^*-a.e. Observe that the Borel–Stieltjes measure on $\Re$ of Example 2D can be naturally extended to a complete *Lebesgue–Stieltjes measure* on $\Im^*$. The above properties are easily verified. However, $\Im^*$ is neither the smallest nor the largest σ-algebra including $\Im$, so the following inclusions are proper inclusions:

$$(\mathrm{P}_4) \qquad \Re \subset \Im^* \subset \wp(\mathbb{R}).$$

These *proper* inclusions are all but trivial. They have been quite critical in many aspects (including historical aspects) in the early days of measure theory, by the beginning of the twentieth century. Completeness of the measure

space $(\mathbb{R}, \Im^*, \lambda^*)$ can be used to give an existential proof for the first proper inclusion. Indeed, let $\mathcal{I}_{\mathbb{Q}}$ be the subfamily of $\mathcal{I}$ consisting of those intervals from $\mathcal{I}$ with rational endpoints. This is a countably infinite set (in fact, $\#\mathcal{I}_{\mathbb{Q}} = \#\mathbb{Q}^2 = \#\mathbb{Q} = \#\mathbb{N}$, where $\mathbb{Q}$ is the rational field and $\#$ stands for cardinality), and the smallest σ-algebra of subsets of $\mathbb{R}$ that includes $\mathcal{I}_{\mathbb{Q}}$ is the smallest σ-algebra that includes $\mathcal{I}$ (since $\mathbb{Q}$ is dense in $\mathbb{R}$, i.e., closure of $\mathbb{Q}$ coincides with $\mathbb{R}$), and so it is the smallest σ-algebra that includes the algebra $\Im$, which is precisely the Borel σ-algebra $\Re$. Then $\Re$ is the σ-algebra generated by $\mathcal{I}_{\mathbb{Q}}$. Since $\#\mathcal{I}_{\mathbb{Q}} = \#\mathbb{N} < \#\mathbb{R}$, we get $\#\Re \leq \#\mathbb{R}$ [31, Problem 9(c), p. 22]. But $\#\mathbb{R} \leq \#\Re$ trivially (for each $x \in \mathbb{R}$, $(x, x+1) \in \Re$). Hence,

$$\#\Re \;=\; \#\mathbb{R}.$$

Now recall that the Cantor set $C \in \Re \subseteq \Im^*$ of Problem 2.9 is uncountable (actually, $\#C = \#[0,1] = \#\mathbb{R}$) and has Lebesgue measure zero. Since λ^* is a complete measure on $\Im^*$ and $\lambda^*(C) = 0$, all subsets of C are measurable; that is, $\wp(C) \subseteq \Im^*$. Thus, recalling that $\#X < \#\wp(X)$ for every set X, we get

$$\#\mathbb{R} \;=\; \#C \;<\; \#\wp(C) \;\leq\; \#\Im^*.$$

Therefore $\#\Re < \#\Im^*$ and $\Re \subseteq \Im^*$ so that

$$\Re \;\subset\; \Im^*.$$

This proves the first proper inclusion in P_4 and, *en passant*, it also proves that $\Re$ is not complete (as anticipated in the paragraph that closes Section 2.3) since $\Im^*$ is the completion of $\Re$. As for the second proper inclusion,

$$\Im^* \subset \wp(\mathbb{R}),$$

a proof of it is worked out in Problem 8.14 based on translation invariance (an important property that we discuss below). It is also worth noticing that, since the σ-algebra $\Re$ is not complete with respect to λ (by Properties P_3 and P_4, as we saw above), there are Lebesgue sets of measure zero that are not Borel sets. However, *every Lebesgue set of measure zero is a subset of a Borel set of measure zero*. In symbols,

$$(\mathrm{P}_5) \qquad N \in \Im^* \text{ and } \lambda^*(N) = 0 \quad \text{imply} \quad N \subseteq B \in \Re \text{ and } \lambda(B) = 0.$$

In fact, since $N \in \Im^*$ and $\lambda^*(N) = 0$ imply $N \in \wp(\mathbb{R})$ and $\ell^*(N) = 0$, Property P_5 is an immediate consequence of Problem 8.8(c), which also says that the above Borel set B of measure zero is actually a G_δ (i.e., a countable intersection of open sets). Next we focus on the *translation invariance property*. For every $\alpha \in \mathbb{R}$ and every $S \subseteq \mathbb{R}$, set

$$S + \alpha = \big\{\varsigma + \alpha \in \mathbb{R} \colon\; \varsigma \in S\big\} \;\subseteq\; \mathbb{R}.$$

If $E \in \Im^*$, then $E + \alpha \in \Im^*$ and

(P_6) $$\lambda^*(E+\alpha) = \lambda^*(E).$$

(cf. Problem 8.12). Translation invariance plays a central role when we set about to build up a nonmeasurable set (i.e., a set in $\wp(\mathbb{R})\backslash\Im^*$, as we do in Problem 8.14). Another important consequence of it reads as follows.

(P_7) Sets with positive outer measure include nonmeasurable subsets.

(See Problem 8.18.) Special case:

Lebesgue sets of positive measure include nonmeasurable sets.

Translation invariance is the central topic of Chapter 13 (see, in particular, Proposition 13.F in Chapter 13).

Remark 8.D (Domain of a measure): In the first paragraph of Chapter 1 we observed that "the power set is too large a set to be the domain" of some measures. Now (and only now) we can offer a proper explanation to this.

(a) In fact, we have the following problem. It is not possible to construct a set function on a σ-algebra of sets from $\mathbb{R}$ with the following four attributes:

(1) defined on the whole $\wp(\mathbb{R})$,

(2) assigning to each interval the value of its length,

(3) countably additive, and

(4) translation invariant.

Recall from Property P_1 that an arbitrary σ-algebra $\mathcal{R} \subseteq \wp(\mathbb{R})$ of subsets of $\mathbb{R}$ containing all intervals (so that (2) makes sense) includes $\Re$.

(b) We can infer assertion (a) from Problem 8.14 below. It is shown there, by using the Axiom of Choice, that there is a set $V \subseteq [0,1]$ for which, if a set function $\mu\colon \mathcal{R} \to \overline{\mathbb{R}}$ satisfies (2), (3), and (4) (modulo 1) then $V \notin \mathcal{R}$.

(c) However, by weakening (1), as we did in Chapters 2 and 8, the Lebesgue measure (λ or λ^*) is defined on a σ-algebra ($\Re$ or $\Im^*$) properly included in the power set $\wp(\mathbb{R})$, and satisfies (2), (3), and (4) — cf. Problem 8.12 below.

(d) There are other approaches to face the above problem. For instance, the outer measure $\ell^*\colon \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ generated by the measure $\ell\colon \Im \to \overline{\mathbb{R}}$ on the algebra $\Im$ (cf. Definitions 8.2, 8.7, Proposition 8.3, and Lemma 8.9), satisfies properties (1), (2), and (4) but replaces (3) with countable subadditivity.

(e) Moreover, as we saw in Problem 1.20, after using the Axiom of Choice to show the existence of a set $V \in \wp(\mathbb{R})\backslash\Re$ as in item (b) above, the existence of an $\Re$-nonmeasurable function $f\colon \mathbb{R} \to \mathbb{R}$ (i.e., $f \notin \mathcal{M}(\mathbb{R}, \Re)$) may be confirmed by using the Continuum Hypothesis.

8.4 Problems

Problem 8.1. Recall that the axiom (c) of Definition 8.1 (for measures on algebras) and the axiom (c) of Definition 2.1 (for measures on σ-algebras) have been referred to as *countable additivity*. Let $\mu\colon \mathcal{A} \to \overline{\mathbb{R}}$ be a set function on an algebra $\mathcal{A}$, and consider the following property:

$\mu\big(\bigcup_i E_i\big) = \sum_i \mu(E_i)$ for every *finite* family $\{E_i\}$ of disjoint sets in $\mathcal{A}$.

This is referred to as *finite additivity*. It is clear (tautologically, actually) that countable additivity implies finite additivity. Verify that the converse fails.

Hint: Problem 2.5.

Next suppose a set function μ is such that

(i) $\mu(E) \geq 0$ for every $E \in \mathcal{A}$

and, if $\{A_n\}$ is a sequence of sets in $\mathcal{A}$,

(ii) $\lim_k \mu\big(\bigcup_{n=1}^k A_n\big) = \mu\big(\bigcup_{n=1}^\infty A_n\big)$

whenever $\bigcup_{n=1}^\infty A_n$ lies in $\mathcal{A}$. Prove the following proposition. *If such a μ is finitely additive, then it is countably additive.* In other words, take an arbitrary sequence $\{E_n\}$ of pairwise disjoint sets in $\mathcal{A}$. If

$$\mu\Big(\bigcup_{n=1}^m E_n\Big) = \sum_{n=1}^m \mu(E_n)$$

for every $m \geq 1$ (i.e., *finite additivity*) and (i) and (ii) hold, then show that

$$\mu\Big(\bigcup_{n=1}^\infty E_n\Big) = \sum_{n=1}^\infty \mu(E_n)$$

whenever $\bigcup_{n=1}^\infty E_n$ lies in $\mathcal{A}$ (i.e., *countable additivity*).

Hint: First use the proof of Proposition 2.2(a), recalling that now μ is finitely additive and acts on an algebra $\mathcal{A}$, to show that if (i) holds, then

(i′) $A', A \in \mathcal{A}$ and $A' \subseteq A$ imply $\mu(A') \leq \mu(A)$.

Next take an arbitrary sequence $\{E_n\}$ of pairwise disjoint sets in $\mathcal{A}$ such that $\bigcup_{n=1}^\infty E_n \in \mathcal{A}$. Use finite additivity, the fact that $\big\{\bigcup_{n=1}^m E_n\big\}$ is an increasing sequence of sets in $\mathcal{A}$, and the result in (i′) to check that

$$\sum_{n=1}^m \mu(E_n) \;=\; \mu\Big(\bigcup_{n=1}^m E_n\Big) \;\leq\; \mu\Big(\bigcup_{n=1}^\infty E_n\Big) \quad \text{for all} \quad m \geq 1.$$

Then apply the same argument, recalling from (i′) that $\big\{\mu\big(\bigcup_{n=1}^m E_n\big)\big\}$ is an increasing sequence of nonnegative elements from $\overline{\mathbb{R}}$, to show that

$$\mu\Big(\bigcup_{n=1}^{k} E_n\Big) \le \lim_m \mu\Big(\bigcup_{n=1}^{m} E_n\Big) = \lim_m \sum_{n=1}^{m} \mu(E_n) = \sum_{n=1}^{\infty} \mu(E_n) \quad \text{for all} \quad k \ge 1.$$

Problem 8.2. The notion of length function $\ell\colon \mathfrak{F} \to \overline{\mathbb{R}}$ was introduced in Definition 8.7. Show that properties (a), (b), and (c) in Definition 8.7 are enough to ensure that $\ell\colon \mathfrak{F} \to \overline{\mathbb{R}}$ is defined on the whole collection $\mathfrak{F}$.

Hint: Every countable family of intervals in $\mathcal{I}$ admits a disjointification *consisting of intervals in* $\mathcal{I}$ (use the hints to Problems 2.3 and 7.2). Use Definition 8.7(c) to verify that a length is defined for every set in $\mathfrak{F}$.

Problem 8.3. Consider again the setup of Definition 8.7.

(a) First show that $\mathfrak{F}$ is an algebra of subsets of the real line $\mathbb{R}$.

Hint: In order to show that axiom (b) of Definition 1.1 holds true; that is, to check that *the complement of a set in* $\mathfrak{F}$ *belongs to* $\mathfrak{F}$, proceed as follows. Verify that (i) if I is an interval in $\mathcal{I}$, then $\mathbb{R}\backslash I$ is the union of no more than two intervals in $\mathcal{I}$, and so it is a set in $\mathfrak{F}$, and (ii) if F_1 and F_2 are sets in $\mathfrak{F}$, then $F_1 \cap F_2$ is a set in $\mathfrak{F}$ (since the intersection of any pair of intervals in $\mathcal{I}$ is an interval in $\mathcal{I}$). According to the De Morgan laws, $\mathbb{R}\backslash\bigcup_i I_i = \bigcap_i(\mathbb{R}\backslash I_i) \in \mathfrak{F}$ for every *finite* union $\bigcup_i I_i$ of intervals I_i in $\mathcal{I}$.

(b) Now verify that $\mathfrak{F}$ is not a σ-algebra of subsets of $\mathbb{R}$.

Hint: $\bigcup_{n=1}^{\infty}(2n-1, 2n] = (1,2] \cup (3,4] \cup \ldots$ is not a set in $\mathfrak{F}$.

Problem 8.4. Let $\mathfrak{F}$ be the algebra of Problem 8.3(a). Verify that

$$F' \subseteq F \quad \text{with} \quad F', F \in \mathfrak{F} \quad \text{implies} \quad \ell(F') \le \ell(F).$$

Hint: $F = F' \cup (F\backslash F')$ and $F\backslash F' \in \mathfrak{F}$. Recall the hint to Problem 8.2.

Problem 8.5. Consider the classes $\mathcal{C}_1$ and $\mathcal{C}_3$ of Section 8.3. Consider the union in the hint to Problem 8.3(b). Observe that it suggests the following infinite union in $\mathfrak{F}$ (that lies in class $\mathcal{C}_3$) of disjoint intervals of class $\mathcal{C}_1$,

$$\bigcup_{k=1}^{\infty}(k, k+1] \;=\; (1,2] \cup (2,3] \cup (3,4] \cup \ldots \;=\; (1, +\infty),$$

which has an infinite subunion $\bigcup_{n=1}^{\infty}(2n-1, 2n]$ not in $\mathfrak{F}$. Show that this can happen even if the original infinite union is of class $\mathcal{C}_1$ (thus having finite length) by exhibiting a sequence $\{(\alpha_k, \beta_k]\}$ of intervals of class $\mathcal{C}_1$ such that

$$\bigcup_{k=1}^{\infty}(\alpha_k, \beta_k] = (0,1] \quad \text{but} \quad \bigcup_{n=1}^{\infty}(\alpha_n, \beta_n] \notin \mathfrak{F}$$

for some subsequence $\{(\alpha_n, \beta_n]\}$ of $\{(\alpha_k, \beta_k]\}$.

Hint: $\left\{\left(\frac{1}{2^k}, \frac{1}{2^{k-1}}\right]\right\}$.

Problem 8.6. Take the Lebesgue and Borel σ-algebras $\Im^*$ and $\Re \subseteq \Im^*$, the Lebesgue measures λ^* on $\Im^*$ and $\lambda = \lambda^*|_{\Re}$, and prove the assertions below.

(a) If $E \in \Im^*$, then there exists $B \in \Re$ such that $\lambda^*(E \backslash B) = 0$.

Hint: As $\lambda^*(E) = \lambda^*(E \cap B) + \lambda^*(E \backslash B)$, it follows that, if $\lambda^*(E \backslash B) > 0$ for every $B \in \Re$, then $\lambda^*(E \cap B) < \lambda^*(E)$ (whenever $\lambda^*(E) < \infty$) for every $B \in \Re$, which is a contradiction because $\mathbb{R} \in \Re$.

(b) If $E \in \Im^*$, then there exist $B \in \Re$ and $N \in \Im^*$ such that $E = B \cup N$ and $B \cap N = \varnothing$, where $\lambda^*(N) = 0$ and so $\lambda^*(E) = \lambda(B)$.

Problem 8.7. Let $\ell \colon \Im \to \overline{\mathbb{R}}$ be the length function, which is a measure on the algebra $\Im$ (Lemma 8.9). Take the outer measure $\ell^* \colon \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ generated by ℓ. Apply Definitions 8.2 and 8.7 to write ℓ^* as

$$\text{(a)} \qquad \ell^*(S) \;=\; \inf_{\{I_n\} \in \mathcal{I}_S} \sum\nolimits_n \ell(I_n)$$

for every $S \in \wp(\mathbb{R})$, where

$$\mathcal{I}_S = \big\{\{I_n\} \colon\; I_n \in \mathcal{I} \;\text{ and }\; S \subseteq \textstyle\bigcup_n I_n\big\}$$

is the collection of all countable families $\{I_n\}$ of intervals in $\mathcal{I}$ that cover S. Let $I \in \mathcal{I}$ be an arbitrary interval, and let I° and $\overline{I}$ denote interior and closure of I, respectively (with respect to the usual topology of $\mathbb{R}$). Let $\lambda \colon \Re \to \overline{\mathbb{R}}$ be the Lebesgue measure on the σ-algebra $\Re$. Recall that $\Re$ includes the algebra $\Im$ (i.e., $\Im \subset \Re$). Verify that I° and $\overline{I}$ lie in $\Re$, and

$$\text{(b)} \qquad \lambda(I^\circ) = \lambda(I) = \lambda(\overline{I})$$

for every $I \in \mathcal{I}$. (See Problem 2.7.) Recall that $I^\circ \subseteq I \subseteq \overline{I}$ and show that

$$\text{(c)} \qquad \ell^*(S) \;=\; \inf_{\{I_n\} \in \mathcal{I}_S^\circ} \sum\nolimits_n \lambda(I_n^\circ) \;=\; \inf_{\{I_n\} \in \overline{\mathcal{I}}_S} \sum\nolimits_n \lambda(\overline{I}_n),$$

where the infimum is taken over all countable coverings of S consisting either of *open* intervals (equivalently, of the interior of intervals in $\mathcal{I}$),

$$\mathcal{I}_S^\circ = \big\{\{I_n\} \colon\; I_n \in \mathcal{I} \;\text{ and }\; S \subseteq \textstyle\bigcup_n I_n^\circ\big\},$$

or of *closed* intervals (equivalently, of the closure of intervals in $\mathcal{I}$),

$$\overline{\mathcal{I}}_S = \big\{\{I_n\} \colon\; I_n \in \mathcal{I} \;\text{ and }\; S \subseteq \textstyle\bigcup_n \overline{I}_n\big\}.$$

Hint: The covering by closed intervals is a consequence of item (b) because $I \subseteq \overline{I}$ and $\ell(I) = \lambda(I)$. For the case of a covering by open intervals, take an

arbitrary $\varepsilon > 0$ and observe that, for each $I_n \in \mathcal{I}$, there is an open interval $J_n \in \Re$ for which $I_n \subseteq J_n$ and $\lambda(J_n) \leq \ell(I_n) + \frac{\varepsilon}{2^n}$. Thus $\bigcup_n I_n \subseteq \bigcup_n J_n$ and $\sum_n \lambda(J_n) \leq \sum_n \ell(I_n) + \varepsilon$ (with n running over $\mathbb{N}$).

Problem 8.8. Take the measure $\ell : \Im \to \overline{\mathbb{R}}$ on the algebra $\Im$, let $\ell^* : \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ be the outer measure generated by ℓ (Definition 8.2), let $\lambda : \Re \to \overline{\mathbb{R}}$ be the Lebesgue measure on the σ-algebra $\Re$, and let $\lambda^* : \Im^* \to \overline{\mathbb{R}}$ be the Lebesgue measure on the σ-algebra $\Im^*$. Recall that $\Re \subseteq \Im^*$ and $\lambda(B) = \lambda^*(B)$ for every $B \in \Re$, and also that open sets are measurable (open sets are Borel sets by Problem 2.7(d), and so they are Lebesgue sets). Take any $S \in \wp(\mathbb{R})$ and an arbitrary $\varepsilon > 0$. Show that there is an open set $U_\varepsilon \subseteq \mathbb{R}$ such that

$$\text{(a)} \qquad S \subseteq U_\varepsilon \quad \text{and} \quad \lambda(U_\varepsilon) \leq \ell^*(S) + \varepsilon.$$

So, since $\ell^*(S) \leq \lambda(U_\varepsilon)$ (and infimum is the maximum of all lower bounds),

$$\text{(b)} \qquad \ell^*(S) = \inf_{U \in \mathcal{T}} \big\{\lambda(U) \colon S \subseteq U\big\},$$

where $\mathcal{T}$ is the topology (i.e., the collection of all open sets) of $\mathbb{R}$.

Hint: Take the first identity in Problem 8.7(c) with the infimum taken over $\mathcal{I}_S^\circ$. Thus there is $\{J_n\} \in \mathcal{I}_S^\circ$ such that $\sum_n \lambda(J_n^\circ) \leq \ell^*(S) + \varepsilon$. Since every union of open sets is again an open set, set $U = \bigcup_n J_n^\circ$ in $\Re$ and verify that $\lambda(U) = \lambda^*(U) = \ell^*(U) \leq \sum_n \ell^*(J_n^\circ) = \sum_n \lambda(J_n^\circ)$ — Proposition 8.3(e).

Finite intersections of open sets are open sets, but a countable intersection of open sets, which is referred to as a G_δ, is not necessarily an open set, although always measurable (i.e., a G_δ is a Borel set). Show that there exists a G_δ, say $B \in \Re$, such that

$$\text{(c)} \qquad S \subseteq B \quad \text{and} \quad \ell^*(S) = \lambda(B).$$

Hint: Item (a) ensures the existence of a sequence $\{U_n\}$ of open sets such that $S \subseteq U_n$ and $\lambda(U_n) \leq \ell^*(S) + \frac{1}{n}$ for all n. Set $B = \bigcap_n U_n$ in $\Re$ and verify that $S \subseteq B \subseteq U_n$, and so $\ell^*(S) \leq \ell^*(B) = \lambda(B) \leq \lambda(U_n)$ for all n.

Problem 8.9. Apply Problem 8.8(c), and the fact that the measure λ^* is complete, to show that every set with outer measure zero is a Lebesgue set:

(a) $N \in \wp(\mathbb{R})$ and $\ell^*(N) = 0$ imply $N \in \Im^*$ and $\lambda^*(N) = 0$;

(b) $E, N \in \wp(\mathbb{R})$, $E \subseteq N$, and $\ell^*(N) = 0$ imply $E \in \Im^*$ and $\lambda^*(E) = 0$.

Take an arbitrary set $S \in \wp(\mathbb{R})$. According to Problem 8.8(c), $S \subseteq B$ with $\ell^*(S) = \lambda(B)$ for some $B \in \Re$. Since $B = S \cup (B \backslash S)$ and $S = B \backslash (B \backslash S)$, use item (a) to prove the following assertion.

(c) If $\ell^*(S) < \infty$, then $S \in \Im^*$ if and only if $\ell^*(B \backslash S) = 0$.

Problem 8.10. Consider the setup of Problem 8.8. Prove that

$$\lambda^*(E) = \inf_{U \in \Re} \{\lambda(U) \colon U \text{ is open and } E \subseteq U\} \quad \text{for every} \quad E \in \Im^*.$$

Since $\Re \subseteq \Im$ and $\lambda = \lambda^*|_{\Re}$, conclude that

$$\lambda(B) = \inf_{U \in \Re} \{\lambda(U) \colon U \text{ is open and } B \subseteq U\} \quad \text{for every} \quad B \in \Re.$$

This means that λ^* on $\Im$ and λ on $\Re$ are *outer regular* (see Section 11.2).

Hint: Problem 8.8(b) and Property P_2.

Problem 8.11. Consider again the setup of Problem 8.8. Prove that

$$\lambda^*(E) = \sup_{V \in \Re} \{\lambda(V) \colon V \text{ is closed and } V \subseteq E\} \quad \text{for every} \quad E \in \Im^*.$$

Again, since $\Re \subseteq \Im$ and $\lambda = \lambda^*|_{\Re}$, conclude that

$$\lambda(B) = \sup_{V \in \Re} \{\lambda(V) \colon V \text{ is closed and } V \subseteq B\} \quad \text{for every} \quad B \in \Re.$$

This implies that λ^* on $\Im$ and λ on $\Re$ are *inner regular* (see Section 11.2).

Hint: Take an arbitrary $E \in \Im^*$ and an arbitrary $\varepsilon > 0$. Set $E' = \mathbb{R} \backslash E \in \Im^*$ (the complement of E lies in $\Im^*$). According to Problem 8.8(a), there is an open set U_ε such that $E' \subseteq U_\varepsilon$ and $\lambda(U_\varepsilon) \leq \lambda^*(E') + \varepsilon$. First prove that

$$\lambda^*(U_\varepsilon \backslash E') \leq \varepsilon.$$

If $\lambda^*(E') < \infty$, then the above inequality is an immediate consequence of Proposition 2.2(b). If $\lambda^*(E') = \infty$, then proceed as follows. Since λ^* is a σ-finite measure on $\Im^*$, every set in $\Im^*$ is σ-finite, and so there is a countable covering of E', say $\{E'_n\}$, made up of measurable subsets of E' of finite measure. Since $\lambda^*(E'_n) < \infty$, there is an open set $U_{\varepsilon,n}$ such that $E'_n \subseteq U_{\varepsilon,n}$ and $\lambda^*(U_{\varepsilon,n} \backslash E'_n) \leq \frac{\varepsilon}{2^n}$. Take the open set $U_\varepsilon = \bigcup_n U_{\varepsilon,n} \supseteq \bigcup_n E'_n = E'$. Show that $\lambda^*(U_\varepsilon \backslash E') \leq \sum_n \lambda^*(U_{\varepsilon,n} \backslash E'_n) \leq \varepsilon$. This proves the claimed inequality. Now take the closed set $V_\varepsilon = \mathbb{R} \backslash U_\varepsilon \subseteq E$, verify that $E \backslash V_\varepsilon = U_\varepsilon \backslash E'$, and so $\lambda^*(E \backslash V_\varepsilon) \leq \varepsilon$. Thus $\lambda^*(E) = \lambda^*(E \cap V_\varepsilon) + \lambda^*(E \backslash V_\varepsilon) \leq \lambda^*(V_\varepsilon) + \varepsilon$. But $\lambda^*(V_\varepsilon) \leq \lambda^*(E)$ (recall: supremum is the minimum of all upper bounds).

Problem 8.12. Prove the translation invariance property (i.e., prove P_6).

Hint: Consider the length function $\ell \colon \Im \to \overline{\mathbb{R}}$ and let $\ell^* \colon \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ be the outer measure generated by the measure ℓ on the algebra $\Im$. Take an arbitrary real number $\alpha \in \mathbb{R}$. Verify that

$$\ell(I + \alpha) = \ell(I)$$

for every interval $I \in \mathcal{I} \subset \Im$. Apply Problem 8.7(a) to show that

$$\ell^*(S + \alpha) = \ell^*(S)$$

for every $S \in \wp(\mathbb{R})$. Recall that $E \in \Im^*$ if and only if

$$\ell^*(S) = \ell^*(S \cap E) + \ell^*(S \backslash E) \quad \text{for every} \quad S \in \wp(\mathbb{R}).$$

Take arbitrary $A, B \in \wp(\mathbb{R})$. Verify that $(A + \alpha) \cap B = (A \cap (B - \alpha)) + \alpha$ and $(\mathbb{R} \backslash B) + \alpha = \mathbb{R} \backslash (B + \alpha)$, and so $(A + \alpha) \backslash B = (A \backslash (B - \alpha)) + \alpha$. Thus show that if $E \in \Im^*$, then

$$\begin{aligned} \ell^*(S) = \ell^*(S - \alpha) &= \ell^*\big((S - \alpha) \cap E\big) + \ell^*\big((S - \alpha) \backslash E\big) \\ &= \ell^*\big((S \cap (E + \alpha)) - \alpha\big) + \ell^*\big((S \backslash (E + \alpha)) - \alpha\big) \\ &= \ell^*\big(S \cap (E + \alpha)\big) + \ell^*\big(S \backslash (E + \alpha)\big). \end{aligned}$$

Hence, $E + \alpha$ also lies in $\Im^*$. Since $\lambda^* = \ell^*|_{\Im^*}$, conclude the result in P_6:

$$E \in \Im^* \quad \text{implies} \quad E + \alpha \in \Im^* \ \text{ and } \ \lambda^*(E + \alpha) = \lambda^*(E).$$

Problem 8.13. Consider a binary operation $\dot{+}\colon [0,1) \times [0,1) \to [0,1)$ defined as follows. If α and β lie in the interval $[0,1)$, then

$$\alpha \dot{+} \beta = \begin{cases} \alpha + \beta, & \alpha + \beta < 1, \\ \alpha + \beta - 1, & \alpha + \beta \geq 1. \end{cases}$$

This is called *sum modulo* 1. For every $\alpha \in [0,1)$ and every $S \subseteq [0,1)$, set

$$S \dot{+} \alpha = \big\{\varsigma \dot{+} \alpha \in \mathbb{R} \colon \varsigma \in S\big\} \ \subseteq \ [0,1).$$

This is called *translation modulo* 1. Prove the *translation invariance modulo* 1 for the Lebesgue measure.

(a) First, prove translation invariance with respect to sum modulo 1 for the outer measure $\ell^*\colon \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ generated by the length function $\ell\colon \Im \to \overline{\mathbb{R}}$:

$$S \subseteq [0,1) \quad \Longrightarrow \quad \ell^*(S \dot{+} \alpha) = \ell^*(S).$$

(b) Then extend it to the Lebesgue measure $\lambda^*\colon \Im^* \to \overline{\mathbb{R}}$:

$$E \subseteq [0,1) \ \text{ and } \ E \in \Im^* \quad \Longrightarrow \quad E \dot{+} \alpha \in \Im^* \ \text{ and } \ \lambda^*(E \dot{+} \alpha) = \lambda^*(E).$$

This will be necessary for proving the next fundamental result.

Hint: Take an arbitrary real number $\alpha \in [0,1)$, and take an arbitrary set $E \subseteq [0,1)$ such that $E \in \Im^*$. Consider the sets $E_1 = E \cap [0, 1-\alpha)$ and $E_2 =$

$E \cap [1-\alpha, 1)$. Show that E_1 and E_2 lie in $\Im^*$. Since $\varsigma + \alpha < 1$ for every $\varsigma \in E_1$, and $\varsigma + \alpha \geq 1$ for every $\varsigma \in E_2$, verify that $E_1 \dot{+} \alpha = E_1 + \alpha$ and $E_2 \dot{+} \alpha = E_2 + (\alpha - 1)$. Thus conclude that the translated sets $E_1 \dot{+} \alpha$ and $E_2 \dot{+} \alpha$ lie in $\Im^*$. Finally, using translation invariance again (for ordinary sums, as in Problem 8.12), show that

$$\begin{aligned}\lambda^*(E \dot{+} \alpha) &= \lambda^*(E_1 \dot{+} \alpha) + \lambda^*(E_2 \dot{+} \alpha) \\ &= \lambda^*(E_1 + \alpha) + \lambda^*(E_2 + \alpha) = \lambda^*(E_1) + \lambda^*(E_2) = \lambda^*(E).\end{aligned}$$

Problem 8.14. Let $\sim$ be a relation on the interval $[0, 1)$ defined by

$$\alpha \sim \beta \quad \text{if and only if} \quad \alpha - \beta \in \mathbb{Q}.$$

In other words, α and β in $[0, 1)$ are related if their difference is a rational number. This is an equivalence relation. That is, $\sim$ is reflexive (i.e., $\alpha \sim \alpha$), transitive (i.e., $\alpha \sim \beta$ and $\beta \sim \gamma$ imply $\alpha \sim \gamma$), and symmetric (i.e., $\alpha \sim \beta$ implies $\beta \sim \alpha$). Thus $\sim$ induces a partition of $[0, 1)$ into equivalence classes

$$[\alpha] = \{\alpha' \in [0, 1) \colon \alpha' \sim \alpha\} = \{\alpha' \in [0, 1) \colon \alpha' - \alpha \in \mathbb{Q}\}.$$

The Axiom of Choice ensures the existence of sets consisting of elements from every equivalence class (one and just one from each equivalence class). These are the *Vitali sets*. Let $V \subseteq [0, 1)$ be a Vitali set and let $\{q_n\}$ be an enumeration of $\mathbb{Q} \cap [0, 1)$. For each n set $V_n = V \dot{+} q_n \subseteq [0, 1)$; rational translations modulo 1 of V. Show that $\{V_n\}$ forms a partition of $[0, 1)$.

(a) $V_m \cap V_n = \varnothing$ whenever $m \neq n$.

Hint: Take an arbitrary $\varsigma \in V_n \cap V_m$. Show that $\varsigma = \alpha \dot{+} q_n = \beta \dot{+} q_m$, with $\alpha, \beta \in V$, $\alpha \dot{+} q_n \in V_n$, and $\beta \dot{+} q_m \in V_m$. Then $\alpha - \beta \in \mathbb{Q}$ (i.e., $\alpha \sim \beta$), so α and β come from the same equivalence class. Since V has only one element from each equivalence class, $\alpha = \beta$. Hence $m = n$.

(b) $\bigcup_n V_n = [0, 1)$.

Hint: Take an arbitrary $\alpha \in [0, 1)$. Thus α is in some equivalence class (since these classes form a partition of $[0, 1)$), and so $\alpha \sim \beta$ for some $\beta \in V \subseteq [0, 1)$ (since V has one element from each class), which means $\alpha - \beta \in \mathbb{Q}$ so that $\alpha = \beta + q$ for some q in $\mathbb{Q}$. First show that if $\beta \leq \alpha$, then $\alpha = \beta + q_n$ for q_n in $\mathbb{Q} \cap [0, 1)$ so that $\alpha = \beta \dot{+} q_n$, and therefore $\alpha \in V_n$ for some n. Next verify that if $\alpha < \beta$, then $\alpha = \beta - q_m$ for q_m in $\mathbb{Q} \cap (0, 1)$ so that $\alpha = \beta \dot{+} p_m$ with $p_m = 1 - q_m$ in $\mathbb{Q} \cap (0, 1)$, and so $\alpha \in V_m$ for some m. Then conclude that $[0, 1) \subseteq \bigcup_n V_n$.

Now use Problem 8.13 (translation invariance modulo 1) to prove that

$$V \notin \Im^*.$$

Hint: If $V \in \Im^*$, then $\lambda^*([0,1)) = \sum_n \lambda^*(V_n) = \sum_n \lambda^*(V) \neq 1$.

Outcome: *Vitali sets are not measurable*, thus completing the proof of P_4. This was the first example of a nonmeasurable set, given by Giuseppe Vitali in 1905. The use of the Axiom of Choice is essential here.

Although not explicitly mentioned in Problems 8.15 to 8.23 below, $\ell^*: \wp(\mathbb{R}) \to \overline{\mathbb{R}}$ stands for the outer measure generated by the length function $\ell: \Im \to \overline{\mathbb{R}}$, and $\lambda^*: \Im^* \to \overline{\mathbb{R}}$ stands for the Lebesgue measure on the Lebesgue σ-algebra $\Im^*$, as always in this book.

Problem 8.15. Exhibit a sequence $\{V_n\}$ of disjoint sets in $\wp(\mathbb{R})$ such that

$$\ell^*\Big(\bigcup\nolimits_n V_n\Big) < \sum\nolimits_n \ell^*(V_n).$$

Problem 8.16. Exhibit a pair of disjoint sets S and S' in $\wp(\mathbb{R})$ such that

$$\ell^*(S \cup S') \neq \ell^*(S) + \ell^*(S').$$

Hint: According to Propositions 8.3(c) and 8.5(a), the outer measure of a Vitali set V is such that $0 < \ell^*(V) \leq 1$. Take the sequence $\{V_n\}$ of Problem 8.14. Set $q_0 = 0$ so that $V_0 = V$. Note that the equivalence class containing zero is $[0] = \mathbb{Q} \cap [0,1)$. Use the same argument as in the hint to Problem 8.14(b) and show that $\bigcup_n V_n \backslash V_0 = (0,1)$. Set $S = V_0$ and $S' = \bigcup_n V_n \backslash V_0$.

Problem 8.17. Show that measurable subsets of Vitali sets have measure zero. In other words, prove that

$$E \in \Im^* \text{ and } E \subset V \quad \text{imply} \quad \lambda^*(E) = 0.$$

Hint: Consider the setup of Problem 8.14. Set $E_n = E \dotplus q_n \subset V_n$. Show that $\{E_n\}$ is a sequence of disjoint sets in $\Im^*$ with $\lambda^*(E_n) = \lambda^*(E)$ and

$$\sum\nolimits_n \lambda^*(E_n) = \lambda^*\Big(\bigcup\nolimits_n E_n\Big) \leq \ell^*\Big(\bigcup\nolimits_n V_n\Big) = \ell^*\big([0,1)\big) = 1.$$

Problem 8.18. Prove Property P_7.

If $S \in \wp(\mathbb{R})$ and $\ell^*(S) > 0$, then there exists $S_0 \subseteq S$ such that $S_0 \notin \Im^*$.

Hint: Suppose $\ell^*(S) > 0$. Then use Problem 8.7(c) to show that there is a translation of S, say S', such that $\ell^*\big(S' \cap (0,1)\big) > 0$. Set $A = S' \cap (0,1)$ and $A_n = A \cap V_n$, with $\{V_n\}$ as in Problem 8.14. If all subsets of S are measurable, then show that A_n is measurable. Since $A_n \subseteq V$, Problem 8.17 ensures that $\lambda^*(A_n) = 0$. Thus verify the following contradiction:

$$0 < \ell^*(A) \le \ell^*\Big(A \cap \bigcup\nolimits_n V_n\Big) = \ell^*\Big(\bigcup\nolimits_n A_n\Big) \le \sum\nolimits_n \ell^*(A_n) = \sum\nolimits_n \lambda^*(A_n) = 0.$$

Problem 8.19. Take $E \in \Im^*$ and $S \in \wp(\mathbb{R})$ arbitrary. Show that

(a) $\ell^*(E \cup S) + \ell^*(E \cap S) = \lambda^*(E) + \ell^*(S)$,

(b) $E \subseteq S$ and $\lambda^*(E) < \infty$ imply $\ell^*(S\backslash E) = \ell^*(S) - \lambda^*(E)$.

Hint: $\ell^*(E \cup S) = \ell^*((E \cup S) \cap E) + \ell^*((E \cup S)\backslash E) = \lambda^*(E) + \ell^*(S\backslash E)$ and $\ell^*(E \cap S) + \ell^*(S\backslash E) = \ell^*(S)$, proving (a). Replace S with $S\backslash E$ in (a).

Problem 8.20. If $S \in \wp(\mathbb{R})$ is such that $\ell^*(S) < \infty$, then show that

(a) $S \in \Im^*$ if and only if $\lambda^*(E) = \ell^*(S)$ for some $E \in \Im^*$ such that $E \subseteq S$,

(b) $S \in \Im^*$ if and only if $\ell^*(S\backslash E) = 0$ for some $E \in \Im^*$ such that $E \subseteq S$.

Hint: For the nontrivial part of (a): $\ell^*(S\backslash E) = \ell^*(S) - \lambda(E) = 0$ by Problem 8.19(b); use Problem 8.9(a) to conclude that $S = (S\backslash E) \cup E \in \Im^*$.

Problem 8.21. Prove that if $E \in \Im^*$, $\lambda^*(E) < \infty$, and $S \subseteq E$, then

$$S \in \Im^* \quad \text{if and only if} \quad \lambda^*(E) = \ell^*(S) + \ell^*(E\backslash S).$$

This is a special application of the Carathéodory condition for measurability. It gives a necessary and sufficient condition for a subset of a Lebesgue measurable set of finite measure to be Lebesgue measurable.

Hint: If $S \in \Im^*$, then the claimed equation follows from the Carathéodory condition. Conversely, note that $E\backslash S \subseteq B$ with $\ell^*(E\backslash S) = \lambda(B)$ for some $B \in \Re$ by Problem 8.8(c). Thus show that $\ell^*(E\backslash S) \le \lambda^*(E \cap B) \le \lambda(B) = \ell^*(E\backslash S)$, and hence $\lambda^*(E) = \lambda^*(E \cap B) + \lambda^*(E\backslash B) = \ell^*(E\backslash S) + \lambda^*(E\backslash B)$. If the claimed equation holds, then $\lambda^*(E\backslash B) = \ell^*(S) < \infty$, since $\lambda^*(E) < \infty$. Verify that $E\backslash B \subseteq S$, apply Problem 8.20(a), and conclude that $S \in \Im^*$.

Problem 8.22. Prove the following proposition. Every measurable set E in $\Im^*$ with $0 < \lambda^*(E) < \infty$ has a nonmeasurable partition, S and S' in $\wp(\mathbb{R})\backslash\Im^*$ with $S \cup S' = E$ and $S \cap S' = \varnothing$, such that

$$\lambda^*(E) = \lambda^*(S \cup S') < \ell^*(S) + \ell^*(S').$$

Hint: According to Problem 8.18, there is a nonmeasurable set $S \subset E$. So $\{S, S'\}$, with $S' = E\backslash S$, is a nonmeasurable partition of E. Thus, by subadditivity, $\lambda^*(E) \le \ell^*(S) + \ell^*(S')$. The inequality is strict by Problem 8.21.

Problem 8.23. Now prove the converse. Take any nonmeasurable set with finite outer measure, say $S \in \wp(\mathbb{R})\backslash\Im^*$ with $\ell^*(S) < \infty$, and let $B \in \Re$

be such that $S \subset B$ and $\ell^*(S) = \lambda(B)$ as in Problem 8.8(c). Show that $S' = B \backslash S \in \wp(\Re) \backslash \Im^*$ is such that $\ell^*(S') > 0$ (by Problem 8.9), and hence

$$\lambda(B) = \lambda(S \cup S') < \ell^*(S) + \ell^*(S').$$

Problem 8.24. Consider the properties P_i that close Section 8.3. Note that Problem 8.14, which shows the existence of Vitali sets V, uses the Axiom of Choice and Problem 8.13 (besides countable additivity). Problem 8.13 uses Problem 8.12, and Problems 8.12 and 8.13 are about property P_6. That is, about translation invariance of the Lebesgue measure on $\Im^*$ (and on $\Re$, since $\Re \subset \Im^*$ and $\lambda = \lambda^*|_{\Re}$). So the ingredients to prove that $V \in \wp(\mathbb{R}) \backslash \Im^*$ were

- the Axiom of Choice,
- translation invariance of the Lebesgue measure, and
- countable additivity (of measures).

These were the only assumptions used to prove that, given the Lebesgue spaces, $(\mathbb{R}, \Re, \lambda)$ or its completion $(\mathbb{R}, \Im^*, \lambda^*)$, there are sets in $\wp(\mathbb{R}) \backslash \Im^*$, and so in $\wp(\mathbb{R}) \backslash \Re$. Recall that the Lebesgue measure is translation invariant and also that measures in general, as we have defined them, are all countably additive. Now verify the following result.

> Take a measure space $(\mathbb{R}, \mathcal{R}, \mu)$, where $\mathcal{R}$ is a σ-algebra of subsets of the real line $\mathbb{R}$, and μ is a measure (thus countably additive) on $\mathcal{R}$. Suppose the measure μ is translation invariant. If a Vitali set V lies in $\mathcal{R}$, then either $\mu([0,1)) = 0$ or $\mu([0,1)) = \infty$.

Hint: Let $V \subset [0,1) \subset \mathbb{R}$ be a Vitali set as in Problem 8.14. Recall from Problem 8.14 that $V_n = V \dotplus q_n \subseteq [0,1)$ is such that $\{V_n\}$ makes a partition of $[0,1)$. Use the hint to Problem 8.14 to show that if V lies in $\mathcal{R}$, then $\mu([0,1)) = \sum_n \mu(V_n) = \sum_n \mu(V)$.

Suggested Readings

Bartle [5], Cohn [16], Halmos [31], Royden [62]. See also [51, Part Two].



9

Product Measures

9.1 Construction of Product Measure

Let $X \times Y$ denote the *Cartesian product* of two sets X and Y, which is the set of all ordered pairs (x, y) where $x \in X$ and $y \in Y$. Consider two measure spaces $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \eta)$. In this section we construct a σ-algebra of subsets of the Cartesian product $X \times Y$, denoted by $\mathcal{X} \times \mathcal{Y}$, which is induced by the σ-algebras $\mathcal{X}$ and $\mathcal{Y}$, such that a measure π on $\mathcal{X} \times \mathcal{Y}$ is given by the product of the measures μ on $\mathcal{X}$ and η on $\mathcal{Y}$. Since we will be dealing with the product of measures, we must consider the problem of defining the product "zero times infinity" because these are possible values for extended real-valued measures. Thus we declare once again (see Sections 1.3 and 3.1) that $0 \cdot +\infty = +\infty \cdot 0 = 0$.

Consider an arbitrary pair of subsets of X and Y, $A \subseteq X$ and $B \subseteq Y$. The Cartesian product $A \times B \subseteq X \times Y$ is called a *rectangle* from $X \times Y$. If $(X, \mathcal{X})$ and $(Y, \mathcal{Y})$ are measurable spaces, and if E and F are measurable subsets of X and Y, respectively (i.e., if $E \in \mathcal{X}$ and $F \in \mathcal{Y}$), then $E \times F$ is referred to as a *measurable rectangle* from $X \times Y$ (i.e., a rectangle $E \times F$ from $X \times Y$ consisting of an $\mathcal{X}$-measurable set $E \subseteq X$ and a $\mathcal{Y}$-measurable set $F \subseteq Y$). Let $\mathcal{X} \times \mathcal{Y}$ denote the σ-algebra generated by the measurable rectangles from $X \times Y$ (i.e., the smallest σ-algebra of subsets of $X \times Y$ containing all measurable rectangles $E \times F$ with $E \in \mathcal{X}$ and $F \in \mathcal{Y}$), and consider the measurable space $(X \times Y, \mathcal{X} \times \mathcal{Y})$, which is referred to as the *Cartesian product* of the measurable spaces $(X, \mathcal{X})$ and $(Y, \mathcal{Y})$. In particular, it is clear that $E \times F$ is a measurable rectangle from $X \times Y$ if and only if $E \times F \in \mathcal{X} \times \mathcal{Y}$.

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_9

Example 9A. Take the Borel algebra $\mathfrak{R}$ of subsets of $\mathbb{R}$ and consider the Cartesian product $(\mathbb{R}\times\mathbb{R}, \mathfrak{R}\times\mathfrak{R})$ of two copies of the same measurable space $(\mathbb{R}, \mathfrak{R})$. Note that open subsets of $\mathbb{R}^2 = \mathbb{R}\times\mathbb{R}$ lie in $\mathfrak{R}\times\mathfrak{R}$. In fact, since $\mathbb{R}^2$ is a separable space, it follows by Problem 1.14 that open sets of $\mathbb{R}^2$ are countably covered by the topological base consisting of open rectangles made up of open intervals. Indeed, it can be shown that the σ-algebra $\mathfrak{R}\times\mathfrak{R}$ coincides with the σ-algebra generated by the open sets of $\mathbb{R}^2$, and so $\mathfrak{R}\times\mathfrak{R}$ is the *Borel σ-algebra* of subsets of $\mathbb{R}^2$ (see the remark that follows Problem 1.14).

Proposition 9.1. *The collection $\mathcal{P}$ of all finite unions of measurable rectangles is an algebra of subsets of $X\times Y$, included in $\mathcal{X}\times\mathcal{Y}$.*

Proof. Every finite union of measurable rectangles admits a disjointification (cf. hints to Problems 2.3 and 7.2) consisting of measurable rectangles. Thus finite unions of sets in $\mathcal{P}$ are in $\mathcal{P}$. It is readily verified that the complement of a measurable rectangle lies in $\mathcal{P}$, and that a finite intersection of sets in $\mathcal{P}$ is a set in $\mathcal{P}$. Applying the De Morgan laws we conclude that the complement of a set in $\mathcal{P}$ lies in $\mathcal{P}$. So $\mathcal{P}$ is an algebra. It is clear that $\mathcal{P}$ is included in the σ-algebra $\mathcal{X}\times\mathcal{Y}$ generated by measurable rectangles. □

As we have observed in the above proof, any set in $\mathcal{P}$ admits a disjointification consisting of measurable rectangles. This means that any set in $\mathcal{P}$ can be expressed as a *finite* union of *disjoint* measurable rectangles.

Definition 9.2. Take an arbitrary set $P = \bigcup_i E_i\times F_i$ in $\mathcal{P}$, where $\{E_i\times F_i\}$ is an arbitrary *finite* partition of P consisting of measurable rectangles from $X\times Y$. Let $\mu\colon \mathcal{X}\to\overline{\mathbb{R}}$ and $\eta\colon \mathcal{Y}\to\overline{\mathbb{R}}$ be measures on $\mathcal{X}$ and $\mathcal{Y}$, respectively, and define a set function $\varpi\colon \mathcal{P}\to\overline{\mathbb{R}}$ by the following *finite* sum: for $P\in\mathcal{P}$,

$$\text{(a)}\qquad \varpi(P) = \varpi\Big(\bigcup\nolimits_i E_i\times F_i\Big) = \sum\nolimits_i \mu(E_i)\,\eta(F_i) = \sum\nolimits_i \varpi(E_i\times F_i).$$

In particular, for each measurable rectangle $E\times F$ from $X\times Y$,

$$\text{(b)}\qquad \varpi(E\times F) = \mu(E)\,\eta(F).$$

In (a) we have three identities. The first identity just reminds us that $\{E_i\times F_i\}$ is a partition of P, the second identity is the definition of the set function ϖ on $\mathcal{P}$, and the third identity is a consequence of the second, according to the particular case in (b). By additivity of the measures μ and η, it is easy to verify that the sums in (a) are "partition invariant": they remain the same for every finite partition of P made up of measurable rectangles. We (i) show that the set function ϖ is a measure on the algebra $\mathcal{P}$, and then we (ii) extend this to a measure on the σ-algebra $\mathcal{X}\times\mathcal{Y}$. But first we need the following auxiliary result to prove that ϖ is countably additive.

Proposition 9.3. *Let $\{E_k \times F_k\}$ be a countable family of disjoint rectangles in $\mathcal{X} \times \mathcal{Y}$, and let $A \times B$ be a rectangle in $\mathcal{X} \times \mathcal{Y}$. Then*

(a) $\quad A \times B = \bigcup_k E_k \times F_k \quad$ implies $\quad \varpi(A \times B) = \sum_k \varpi(E_k \times F_k)$.

If $\{A_i \times B_i\}$ is a finite *set of disjoint rectangles in $\mathcal{X} \times \mathcal{Y}$, then*

(b) $\bigcup_i A_i \times B_i = \bigcup_k E_k \times F_k \quad$ implies $\quad \varpi\Big(\bigcup_i A_i \times B_i\Big) = \sum_k \varpi(E_k \times F_k)$.

Proof. Let $A \times B$ be any measurable rectangle with $A \in \mathcal{X}$ and $B \in \mathcal{Y}$, and let $\{E_k \times F_k\}$ be an arbitrary countable partition of $A \times B$ with E_k in $\mathcal{X}$ and F_k in $\mathcal{Y}$ so that $\{E_k\}$ and $\{F_k\}$ are countable coverings of A and B, respectively. If $\{E_k \times F_k\}$ is a finite partition, then the results in (a) and (b) are trivially verified by Definition 9.2(a). Thus suppose it is countably infinite.

(a) As usual, let χ_S denote the characteristic function of a set S. Then

$$\chi_A(x)\, \chi_B(y) = \chi_{A \times B}\big((x,y)\big) = \sum\nolimits_k \chi_{E_k \times F_k}\big((x,y)\big) = \sum\nolimits_k \chi_{E_k}(x)\, \chi_{F_k}(y),$$

for every pair $(x, y) \in X \times Y$. (The second identity holds since $\{E_k \times F_k\}$ is a family of disjoint sets that cover $A \times B$.) Fix x, integrate with respect to η, and apply the Monotone Convergence Theorem (cf. Problem 3.7) to get

$$\chi_A(x)\, \eta(B) = \sum\nolimits_k \chi_{E_k}(x)\, \eta(F_k).$$

Next, integrate with respect to μ, repeating the same argument, to get

$$\mu(A)\, \eta(B) = \sum\nolimits_k \mu(E_k)\, \eta(F_k).$$

Thus we get the result in (a) by Definition 9.2(b), namely,

$$\varpi(A \times B) = \sum\nolimits_k \varpi(E_k \times F_k).$$

(b) The disjointness assumption on both $\{E_k \times F_k\}$ and $\{A_i \times B_i\}$ ensures that if $\bigcup_i A_i \times B_i = \bigcup_k E_k \times F_k$, then $A_i \times B_i = \bigcup_j E_{i,j} \times F_{i,j}$ for each i, where $\{E_{i,j} \times F_{i,j}\} = \{E_k \times F_k\}$. Hence, using Definition 9.2(a) and the result in (a), we get

$$\varpi\Big(\bigcup\nolimits_k E_k \times F_k\Big) = \varpi\Big(\bigcup\nolimits_i A_i \times B_i\Big) = \sum\nolimits_i \varpi(A_i \times B_i) = \sum\nolimits_i \sum\nolimits_j \varpi(E_{i,j} \times F_{i,j}).$$

If $\varpi(A_i \times B_i) = \infty$ for some i, then $\varpi\big(\bigcup_i A_i \times B_i\big) = \sum_k \varpi(E_k \times F_k) = \infty$ and the result in (b) holds. Suppose $\varpi(A_i \times B_i) < \infty$ for every i. Then the summands $\{\varpi(E_{i,j} \times F_{i,j})\}$ are nonnegative real numbers by (a), and so the double-indexed sum is unconditionally convergent. Therefore,

$$\sum\nolimits_i \sum\nolimits_j \varpi(E_{i,j} \times F_{i,j}) = \sum\nolimits_k \varpi(E_k \times F_k),$$

and the result in (b) still holds. (Compare with Proposition 8.8.) $\square$

Lemma 9.4. *The set function ϖ is a measure on the algebra $\mathcal{P}$.*

Proof. Observe that $\varpi(\varnothing) = 0$ and $\varpi(P) \geq 0$ for every $P \in \mathcal{P}$, trivially. Consider an arbitrary countable family $\{P_n\}$ of pairwise disjoint sets in $\mathcal{P}$ for which $\bigcup_n P_n$ lies in $\mathcal{P}$. Then, as we have seen before, each set P_n is a finite union of disjoint measurable rectangles,

$$P_n = \bigcup\nolimits_j E_{n,j} \times F_{n,j}.$$

Since $\{P_n\}$ is a sequence of disjoint sets, this implies that its union

$$\bigcup\nolimits_n P_n = \bigcup\nolimits_{n,j} E_{n,j} \times F_{n,j} = \bigcup\nolimits_k E_k \times F_k$$

(where $\{E_{n,j} \times F_{n,j}\} = \{E_k \times F_k\}$ is a countable family of disjoint measurable rectangles) also is a finite union of disjoint measurable rectangles,

$$\bigcup\nolimits_n P_n = \bigcup\nolimits_i A_i \times B_i.$$

Therefore, $\bigcup_i A_i \times B_i = \bigcup_k E_k \times F_k$. Using Proposition 9.3(b), and recalling the unconditional convergence argument that closed that proof, we get

$$\begin{aligned}\varpi\Big(\bigcup\nolimits_n P_n\Big) &= \varpi\Big(\bigcup\nolimits_i A_i \times B_i\Big) = \sum\nolimits_k \varpi(E_k \times F_k) = \sum\nolimits_{n,j} \varpi(E_{n,j} \times F_{n,j}) \\ &= \sum\nolimits_n \sum\nolimits_j \varpi(E_{n,j} \times F_{n,j}) = \sum\nolimits_n \varpi\Big(\bigcup\nolimits_j E_{n,j} \times F_{n,j}\Big) = \sum\nolimits_n \varpi(P_n)\end{aligned}$$

by Definition 9.2(a). This shows that ϖ is countably additive. □

Theorem 9.5. (Product Measure Theorem). *Let $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \eta)$ be measure spaces. Then there exists a measure $\pi\colon \mathcal{X} \times \mathcal{Y} \to \overline{\mathbb{R}}$ such that*

$$\pi(E \times F) = \mu(E)\, \eta(F)$$

for every measurable rectangle $E \times F$ with $E \in \mathcal{X}$ and $F \in \mathcal{Y}$. Moreover, if μ and η are σ-finite, then the measure $\pi\colon \mathcal{X} \times \mathcal{Y} \to \overline{\mathbb{R}}$ satisfying the above identity is unique and σ-finite as well.

Proof. Take the measure ϖ on the algebra $\mathcal{P}$ such that $\varpi(E \times F) = \mu(E)\, \eta(F)$ for every measurable rectangle $E \times F$ from $X \times Y$ (Definition 9.2 and Lemma 9.4). The Carathéodory Theorem (Theorem 8.4) says that there exists an extension of ϖ to a measure π^* on a σ-algebra $\mathcal{P}^*$ that includes the algebra $\mathcal{P}$. Also, it is readily verified that if μ and η are σ-finite on $\mathcal{X}$ and $\mathcal{Y}$, then ϖ is σ-finite on $\mathcal{P}$, and so the Hahn Theorem (Theorem 8.6) says that π^* is unique on $\mathcal{P}^*$ and σ-finite. Let π be the restriction of π^* to $\mathcal{X} \times \mathcal{Y}$, the smallest σ-algebra including $\mathcal{P}$. Thus π is a σ-finite measure (as π^* is) which (by the uniqueness of π^*) is the extension of ϖ over $\mathcal{X} \times \mathcal{Y}$. Summing up:

$$\mathcal{P} \subseteq \mathcal{X}\times\mathcal{Y} \subseteq \mathcal{P}^*, \quad \pi = \pi^*|_{\mathcal{X}\times\mathcal{Y}}, \quad \text{and} \quad \varpi = \pi|_{\mathcal{P}} = \pi^*|_{\mathcal{P}}.$$

As $\varpi(E\times F) = \mu(E)\,\eta(F)$ for each measurable rectangle $E\times F$, and since $\varpi = \pi|_{\mathcal{P}}$ so that $\varpi(E\times F) = \pi(E\times F)$, it follows that $\pi\colon \mathcal{X}\times\mathcal{Y} \to \overline{\mathbb{R}}$ satisfies

$$\pi(E\times F) = \mu(E)\,\eta(F) \quad \text{for every} \quad E\times F \in \mathcal{X}\times\mathcal{Y}. \qquad \square$$

The value of π at each Cartesian product $E\times F$ in $\mathcal{X}\times\mathcal{Y}$ is the product of the values of μ at E in $\mathcal{X}$ and η at F in $\mathcal{Y}$. This motivates the notation.

Notation and Terminology:

$$\pi = \mu \times \eta,$$

which is referred to as the *product measure* (or as the *product* of the measures μ and η). Accordingly, $(X\times Y,\, \mathcal{X}\times\mathcal{Y},\, \mu\times\eta)$ (or $(X\times Y,\, \mathcal{X}\times\mathcal{Y},\, \pi)$) is the (Cartesian) *product space* of the measure spaces $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \eta)$.

9.2 Sections of Sets and Functions

Let S be an arbitrary subset of the Cartesian product $X\times Y$. Associated to a point $x \in X$ and a point $y \in Y$, consider its *sections*, which are the sets

$$S_x = \big\{y \in Y\colon\ (x,y) \in S\big\},$$

called the *x-section* of S, and

$$S^y = \big\{x \in X\colon\ (x,y) \in S\big\},$$

called the *y-section* of S. The reason for this notation with subscript and superscript is to distinguish x-sections (subsets of Y for each $x \in X$) from y-sections (subsets of X for each $y \in Y$). Observe that sections are not *slices*, which means that $S_x \subseteq Y$ (or $S^y \subseteq X$) and $\{x\}\times S_x \subseteq X\times Y$ (or $S^y\times\{y\} \subseteq X\times Y$) are, in general, different sets. Also note that sections of a rectangle $A\times B \subseteq X\times Y$ are either empty or *sides* of the rectangle:

$$(A\times B)_x = \begin{cases} B, & x \in A, \\ \varnothing, & x \notin A, \end{cases} \qquad (A\times B)^y = \begin{cases} A, & y \in B, \\ \varnothing, & y \notin B. \end{cases}$$

In particular,

$$(X\times Y)_x = Y \quad \text{and} \quad (X\times Y)^y = X.$$

Now let $f\colon X\times Y \to \overline{\mathbb{R}}$ be an arbitrary extended real-valued function on $X\times Y$. For each $x \in X$ consider the function $f_x\colon Y \to \overline{\mathbb{R}}$ defined by

$$f_x(y) = f(x,y) \quad \text{for every} \quad y \in (X\times Y)_x = Y,$$

which is called the *x-section* of f. Similarly, for each $y \in Y$ consider the function $f^y\colon X \to \overline{\mathbb{R}}$ defined by

$$f^y(x) = f(x,y) \quad \text{for every} \quad x \in (X\times Y)^y = X,$$

which is called the *y-section* of f. If $f\colon A\times B \to \overline{\mathbb{R}}$ is defined on a rectangle $A\times B \subseteq X\times Y$, then its x-sections and y-sections are defined on $B \subseteq Y$ and $A \subseteq X$, respectively: $f_x\colon B \to \overline{\mathbb{R}}$ and $f^y\colon A \to \overline{\mathbb{R}}$.

Proposition 9.6. *Every section of a measurable set is measurable*:

(a) $W \in \mathcal{X}\times\mathcal{Y}$ *implies* $W^y \in \mathcal{X}$ *and* $W_x \in \mathcal{Y}$ *for every* $x \in X$ *and* $y \in Y$ (*i.e., if* $W \subseteq X\times Y$ *is* $\mathcal{X}\times\mathcal{Y}$*-measurable, then* $W^y \subseteq X$ *is* $\mathcal{X}$*-measurable and* $W_x \subseteq Y$ *is* $\mathcal{Y}$*-measurable*).

Every section of a measurable function is measurable:

(b) *If* $f\colon X\times Y \to \overline{\mathbb{R}}$ *is* $\mathcal{X}\times\mathcal{Y}$*-measurable, then* $f^y\colon X \to \overline{\mathbb{R}}$ *is* $\mathcal{X}$*-measurable and* $f_x\colon Y \to \overline{\mathbb{R}}$ *is* $\mathcal{Y}$*-measurable for every* $x \in X$ *and* $y \in Y$.

Proof. Take a pair of measurable spaces $(X,\mathcal{X})$ and $(Y,\mathcal{Y})$, and consider their Cartesian product $(X\times Y,\, \mathcal{X}\times\mathcal{Y})$.

(a) The assertion in (a) is an immediate consequence of Problem 9.5. In fact, $\mathcal{X}\times\mathcal{Y}$ is included in the collection $(\mathcal{X}\times\mathcal{Y})_X$ of all subsets of $X\times Y$ for which all x-sections are $\mathcal{Y}$-measurable. Hence, if $W \in \mathcal{X}\times\mathcal{Y}$, then $W_x \in \mathcal{Y}$. Similarly, $W \in \mathcal{X}\times\mathcal{Y}$ also implies $W^y \in \mathcal{X}$. (See Problem 9.5.)

(b) To prove assertion (b) proceed as follows. Take any $\alpha \in \mathbb{R}$. If a function $f\colon X\times Y \to \overline{\mathbb{R}}$ is $\mathcal{X}\times\mathcal{Y}$-measurable, then the set $\{(x,y) \in X\times Y\colon f(x,y) > \alpha\}$ is $\mathcal{X}\times\mathcal{Y}$-measurable, and so every x-section $\{(x,y) \in X\times Y : f(x,y) > \alpha\}_x$ is $\mathcal{Y}$-measurable by item (a). Note that for each $x \in X$,

$$\big\{(x,y) \in X\times Y\colon f(x,y) > \alpha\big\}_x = \big\{y \in Y\colon f(x,y) > \alpha\big\} = \big\{y \in Y\colon f_x(y) > \alpha\big\}.$$

Therefore the set $\{y \in Y :\ f_x(y) > \alpha\}$ is $\mathcal{Y}$-measurable. This means that the function $f_x\colon Y \to \overline{\mathbb{R}}$ is measurable. Similarly, using the same argument, the function $f^y\colon X \to \overline{\mathbb{R}}$ is measurable. □

Next we apply the Monotone Class Lemma (cf. Problems 1.18 and 1.19) to prove an important result that will play a crucial role in Section 9.3.

Lemma 9.7. *Let* $(X,\mathcal{X},\mu)$ *and* $(Y,\mathcal{Y},\eta)$ *be measure spaces. For each measurable set* W *in* $\mathcal{X}\times\mathcal{Y}$ *consider the nonnegative functions* $f_W : X \to \overline{\mathbb{R}}$ *and* $g_W : Y \to \overline{\mathbb{R}}$ *defined by*

$$f_W(x) = \eta(W_x) \quad \textit{and} \quad g_W(y) = \mu(W^y)$$

for every $x \in X$ and every $y \in Y$, respectively. If μ and η are σ-finite measures, then $f_W\colon X \to \overline{\mathbb{R}}$ and $g_W\colon Y \to \overline{\mathbb{R}}$ are measurable functions such that

$$\int_X f_W \, d\mu \;=\; \pi(W) \;=\; \int_Y g_W \, d\eta.$$

Proof. Consider the collection $\mathcal{K}$ of all sets W in $\mathcal{X}\times\mathcal{Y}$ such that

$$f_W \in \mathcal{M}(X,\mathcal{X})^+, \;\; g_W \in \mathcal{M}(Y,\mathcal{Y})^+, \;\; \text{and} \;\; \textstyle\int_X f_W d\mu = \pi(W) = \int_Y g_W d\eta.$$

This $\mathcal{K}$ is the subcollection of $\mathcal{X}\times\mathcal{Y}$ for which the conclusion of the lemma holds true. We split the proof into two parts. In part (a) we apply the Monotone Class Lemma to prove that if μ and η are finite measures, then

$$\mathcal{K} = \mathcal{X}\times\mathcal{Y}$$

(i.e., the stated assertion holds true for finite measures). In part (b) we apply the Monotone Convergence Theorem to extend it to σ-finite measures.

(a) Let $A\times B$ in $\mathcal{X}\times\mathcal{Y}$ be any measurable rectangle. Recall that x-sections $(A\times B)_x$ are B if $x \in A$ and empty otherwise, and y-sections $(A\times B)^y$ are A if $y \in B$ and empty otherwise. Thus, for each $x \in X$ and each $y \in Y$ set

$$\begin{aligned} f_{A\times B}(x) &= \eta\big((A\times B)_x\big) = \eta(B)\,\chi_A(x), \\ g_{A\times B}(y) &= \mu\big((A\times B)^y\big) = \mu(A)\,\chi_B(y). \end{aligned}$$

If μ and η are finite measures, then these define *real-valued* functions on X and on Y such that (cf. Example 1B and Proposition 1.5) $f_{A\times B} = \eta(B)\,\chi_A$ is in $\mathcal{M}(X,\mathcal{X})^+$, $g_{A\times B} = \mu(A)\,\chi_B$ is in $\mathcal{M}(Y,\mathcal{Y})^+$, and (cf. Problem 3.3(a))

$$\int_X f_{A\times B}\, d\mu \;=\; \eta(B)\,\mu(A) \;=\; \pi(A\times B) \;=\; \mu(A)\,\eta(B) \;=\; \int_Y g_{A\times B}\, d\eta.$$

Consider the algebra $\mathcal{P}$ of Proposition 9.1. Let P be an arbitrary set in $\mathcal{P}$. Since $P = \bigcup_{i=1}^n A_i\times B_i$ is a finite union of *disjoint* measurable rectangles $\{A_i\times B_i\}$, the above results ensure that $P \in \mathcal{K}$, and so $\mathcal{P} \subseteq \mathcal{K}$. Indeed,

$$\begin{aligned} f_P(x) &= \eta(P_x) = \eta\big(\big(\textstyle\bigcup_{i=1}^n A_i\times B_i\big)_x\big) = \eta\big(\textstyle\bigcup_{i=1}^n (A_i\times B_i)_x\big) \\ &= \textstyle\sum_{i=1}^n \eta\big((A_i\times B_i)_x\big) = \sum_{i=1}^n \eta(B_i)\,\chi_{A_i}(x) = \sum_{i=1}^n f_{A_i\times B_i}(x), \end{aligned}$$

$$\begin{aligned} g_P(y) &= \mu(P^y) = \mu\big(\big(\textstyle\bigcup_{i=1}^n A_i\times B_i\big)^y\big) = \mu\big(\textstyle\bigcup_{i=1}^n (A_i\times B_i)^y\big) \\ &= \textstyle\sum_{i=1}^n \mu\big((A_i\times B_i)^y\big) = \sum_{i=1}^n \mu(A_i)\,\chi_{B_i}(y) = \sum_{i=1}^n g_{A_i\times B_i}(y), \end{aligned}$$

for every $x \in X$ and $y \in Y$. Hence (cf. Proposition 1.5) $f_P = \sum_{i=1}^n f_{A_i\times B_i}$ is in $\mathcal{M}(X,\mathcal{X})^+$, $g_P = \sum_{i=1}^n g_{A_i\times B_i}$ is in $\mathcal{M}(Y,\mathcal{Y})^+$, and (cf. Problem 3.7))

$$\begin{aligned}\int_X f_P\,d\mu &= \sum_{i=1}^n \int_X f_{A_i\times B_i}\,d\mu \\ &= \sum_{i=1}^n \mu(A_i)\,\eta(B_i) = \sum_{i=1}^n \pi(A_i\times B_i) = \pi\Big(\bigcup_{i=1}^n A_i\times B_i\Big) = \pi(P) \\ &= \sum_{i=1}^n \int_Y g_{A_i\times B_i}\,d\eta = \int_Y g_P\,d\eta.\end{aligned}$$

Therefore, $P \in \mathcal{K}$ and so

(i) $$\mathcal{P} \subseteq \mathcal{K}.$$

First suppose $\{C_n\}$ is an increasing sequence of sets in $\mathcal{K}$. For each n set

$$f_{C_n}(x) = \eta\big((C_n)_x\big) \quad \text{and} \quad g_{C_n}(y) = \mu\big((C_n)^y\big)$$

for every $x \in X$ and every $y \in Y$. Since $C_n \in \mathcal{K}$, this defines two sequences $\{f_{C_n}\}$ and $\{g_{C_n}\}$ of functions in $\mathcal{M}(X,\mathcal{X})^+$ and in $\mathcal{M}(Y,\mathcal{Y})^+$ such that

$$\int_X f_{C_n}\,d\mu = \pi(C_n) = \int_Y g_{C_n}\,d\eta.$$

But $\{f_{C_n}\}$ and $\{g_{C_n}\}$ are increasing sequences of extended real-valued functions because $\{C_n\}$ is an increasing sequence of sets. Thus these sequences of functions converge pointwise. Set $C = \bigcup_n C_n$, which lies in the σ-algebra $\mathcal{X}\times\mathcal{Y}$. Since the sequences $\{(C_n)_x\}$ and $\{(C_n)^y\}$ of sections in $\mathcal{Y}$ and $\mathcal{X}$ are increasing with $\bigcup_n (C_n)_x = C_x$ in $\mathcal{Y}$ and $\bigcup_n (C_n)^y = C^y$ in $\mathcal{X}$ for each $x \in X$ and each $y \in Y$, it follows by Proposition 2.2(c) that the functions $f_C\colon X \to \overline{\mathbb{R}}$ and $g_C\colon Y \to \overline{\mathbb{R}}$ are the pointwise limits of $\{f_{C_n}\}$ and $\{g_{C_n}\}$:

$$\lim_n f_{C_n}(x) = \lim_n \eta\big((C_n)_x\big) = \eta\Big(\bigcup\nolimits_n (C_n)_x\Big) = \eta(C_x) = f_C(x),$$

$$\lim_n g_{C_n}(y) = \lim_n \mu\big((C_n)^y\big) = \mu\Big(\bigcup\nolimits_n (C_n)^y\Big) = \mu(C^y) = g_C(y),$$

for every $x \in X$ and $y \in Y$. Still by Proposition 2.2(c) we get

$$\lim_n \pi(C_n) = \pi\Big(\bigcup\nolimits_n C_n\Big) = \pi(C).$$

Note from Proposition 1.8 that $f_C \in \mathcal{M}(X,\mathcal{X})^+$ and $g_C \in \mathcal{M}(Y,\mathcal{Y})^+$. Use the Monotone Convergence Theorem (Theorem 3.4) to conclude:

$$\int_X f_C\,d\mu = \lim_n \int_X f_{C_n}\,d\mu = \pi(C) = \lim_n \int_Y g_{C_n}\,d\eta = \int_Y g_C\,d\eta.$$

This implies that

$$C \in \mathcal{K}.$$

Next suppose $\{D_n\}$ is a decreasing sequence of sets in $\mathcal{K}$ and set $D = \bigcap_n D_n$ in $\mathcal{X}\times\mathcal{Y}$. Proceeding as before, consider the similarly defined functions f_{D_n} in $\mathcal{M}(X,\mathcal{X})^+$ and g_{D_n} in $\mathcal{M}(Y,\mathcal{Y})^+$ such that

$$\int_X f_{D_n} d\mu \,=\, \pi(D_n) \,=\, \int_Y g_{D_n} d\eta$$

for each n. Still under the assumption that μ and η are finite measures, which implies that the product measure $\pi = \mu\times\eta$ is finite as well, we may apply Proposition 2.2(d) for the finite measure π (instead of Proposition 2.2(c) for general measures) to verify that $\{f_{D_n}\}$ and $\{g_{D_n}\}$ are both decreasing sequences of real-valued functions so that they converge pointwise, and

$$\lim_n f_{D_n}(x) = f_D(x) \quad \text{and} \quad \lim_n g_{D_n}(y) = g_D(y)$$

for every $x \in X$ and $y \in Y$, which define the real-valued limit functions $f_D \in \mathcal{M}(X,\mathcal{X})^+$ and $g_D \in \mathcal{M}(Y,\mathcal{Y})^+$. Proposition 2.2(d) also ensures that

$$\lim_n \pi(D_n) = \pi\Big(\bigcap\nolimits_n D_n\Big) = \pi(D).$$

Recall: $\{f_{D_n}\}$ and $\{g_{D_n}\}$ are decreasing sequences of nonnegative real-valued measurable functions, and $\int f_{D_1} d\mu$ and $\int g_{D_1} d\eta$ are bounded by $\pi(X\times Y) = \mu(X)\,\eta(Y) < \infty$ (since $\int_X f_{D_1} d\mu = \pi(D_1) = \int_Y g_{D_1} d\eta$, $\mu(X) < \infty$, $\eta(Y) < \infty$). Thus, by the Dominated Convergence Theorem (Theorem 4.7), we get

$$\int_X f_D \, d\mu \,=\, \pi(D) \,=\, \int_Y g_D \, d\eta.$$

This implies that

$$D \in \mathcal{K}.$$

Hence (cf. Problem 1.15), under the finite measure assumption,

(ii) $\mathcal{K}$ is a monotone class.

According to (i) and (ii), the Monotone Class Lemma (see Problems 1.18 and 1.19) ensures that if μ and η are finite measures, then

$$\mathcal{K} = \mathcal{X}\times\mathcal{Y}.$$

(b) If the measures μ and η are σ-finite, then there is a pair of *increasing* sequences $\{E_n\}$ and $\{F_n\}$ of $\mathcal{X}$-measurable and $\mathcal{Y}$-measurable sets covering X and Y, respectively, such that $\mu(E_n) < \infty$ and $\eta(F_n) < \infty$. For each n consider the σ-algebras $\mathcal{E}_n = \wp(E_n) \cap \mathcal{X}$ and $\mathcal{F}_n = \wp(F_n) \cap \mathcal{Y}$ so that μ and η are finite measures when restricted to them. Let W be an arbitrary set in $\mathcal{X}\times\mathcal{Y}$. For each n set $W_n = W \cap (E_n\times F_n)$ in $\mathcal{E}_n\times\mathcal{F}_n$. Hence

$$(W_n)_x = \big(W \cap (E_n \times F_n)\big)_x = W_x \cap (E_n \times F_n)_x = W_x \cap F_n \in \mathcal{F}_n,$$
$$(W_n)^y = \big(W \cap (E_n \times F_n)\big)^y = W^y \cap (E_n \times F_n)^y = W^y \cap E_n \in \mathcal{E}_n.$$

Take the functions $f_{W_n}\colon X \to \overline{\mathbb{R}}$ and $g_{W_n}\colon Y \to \overline{\mathbb{R}}$ defined for each n by

$$f_{W_n}(x) = \eta\big((W_n)_x\big) = \eta\big((W_n)_x\big)\,\chi_{E_n},$$
$$g_{W_n}(x) = \mu\big((W_n)^y\big) = \mu\big((W_n)^y\big)\,\chi_{F_n},$$

for every $x \in X$ and $y \in Y$. Let $\mu_n = \mu|_{\mathcal{E}_n}$ and $\eta_n = \eta|_{\mathcal{F}_n}$ be the restrictions of μ and η to $\mathcal{E}_n$ and $\mathcal{F}_n$ so that $(E_n, \mathcal{E}_n, \mu_n)$ and $(F_n, \mathcal{F}_n, \eta_n)$ are finite measure spaces. Since the stated assertion holds for finite measures (as we saw in item (a)), it follows that $f_{W_n} \in \mathcal{M}(X, \mathcal{X})^+$, $g_{W_n} \in \mathcal{M}(Y, \mathcal{Y})^+$, and

$$\int_X f_{W_n}\, d\mu = \int_{E_n} f_{W_n}\, d\mu_n = \pi(W_n) = \int_{F_n} g_{W_n}\, d\eta_n = \int_Y g_{W_n}\, d\eta.$$

Since $\{E_n\}$ and $\{F_n\}$ are increasing sequences of $\mathcal{X}$-measurable and $\mathcal{Y}$-measurable sets covering X and Y, conclude that $\{W_n\}$, $\{(W_n)_x\}$, and $\{(W_n)^y\}$ are increasing sequences of $\mathcal{X}\times\mathcal{Y}$-measurable, $\mathcal{Y}$-measurable, and $\mathcal{X}$-measurable sets that cover W, W_x, and W^y, respectively. So $W = \bigcup_n W_n$, $W_x = \bigcup_n (W_n)_x$, and $W^y = \bigcup_n (W_n)^y$. Therefore, Proposition 2.2(c) ensures that $\lim_n \pi\big(W_n\big) = \pi\big(\bigcup_n W_n\big) = \pi(W)$ and

$$\lim_n f_{W_n}(x) = \lim_n \eta\big((W_n)_x\big) = \eta\Big(\bigcup\nolimits_n (W_n)_x\Big) = \eta(W_x) = f_W(x),$$
$$\lim_n g_{W_n}(y) = \lim_n \mu\big((W_n)^y\big) = \mu\Big(\bigcup\nolimits_n (W_n)^y\Big) = \mu(W^y) = g_W(y),$$

for every $x \in X$ and $y \in Y$. But $\{f_{W_n}\}$ and $\{g_{W_n}\}$ are increasing sequences (since $\{E_n\}$ and $\{F_n\}$ are increasing) of nonnegative measurable functions (according to item (a)). Hence the Monotone Convergence Theorem (Theorem 3.4) ensures that $f_W \in \mathcal{M}(X, \mathcal{X})^+$, $g_W \in \mathcal{M}(Y, \mathcal{Y})^+$, and

$$\int_X f_W\, d\mu = \lim_n \int_X f_{W_n}\, d\mu = \pi(W) = \lim_n \int_Y g_{W_n}\, d\eta = \int_Y g_W\, d\eta. \quad \square$$

9.3 Fubini and Tonelli Theorems

The following two theorems are central in this chapter, and crucial in integration theory, in general. They give sufficient conditions for interchanging the order of integration and, as one would expect, are based on Lemma 9.7. Theorem 9.8 (Tonelli's) deals with extended real-valued nonnegative functions, while Theorem 9.9 (Fubini's) dismisses nonnegativeness but assumes integrability (i.e., finite integrals).

Theorem 9.8. (Tonelli Theorem). *Consider two measure spaces* $(X, \mathcal{X}, \mu)$ *and* $(Y, \mathcal{Y}, \eta)$. *If* μ *and* η *are* σ*-finite and* $h \in \mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$, *then the extended real-valued functions* f_h *and* g_h *defined by*

$$f_h(x) = \int_Y h_x \, d\eta \quad \text{and} \quad g_h(y) = \int_X h^y \, d\mu$$

are in $\mathcal{M}(X, \mathcal{X})^+$ *and in* $\mathcal{M}(Y, \mathcal{Y})^+$, *respectively, and*

$$\int_X f_h \, d\mu = \int_{X\times Y} h \, d\pi = \int_Y g_h \, d\eta.$$

Proof. Let $\top = \top(X{\times}Y, \mathcal{X}{\times}\mathcal{Y}, \mu{\times}\eta) \subseteq \mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$ be the set of all nonnegative functions $h \in \mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$ such that

$$f_h \in \mathcal{M}(X, \mathcal{X})^+, \ g_h \in \mathcal{M}(Y, \mathcal{Y})^+, \text{ and } \textstyle\int_X f_h \, d\mu = \int_{X\times Y} h \, d\pi = \int_Y g_h \, d\eta.$$

That is, $\top$ is the subcollection of $\mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$ for which the conclusion of the Tonelli Theorem holds true. The program here is to prove that

$$\top = \mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+.$$

Equivalently, to prove that $\mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+ \subseteq \top$. Let χ_W be the characteristic function of W in $\mathcal{X}{\times}\mathcal{Y}$. Recall that $\chi_W \in \mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$ (Example 1B). Note that $(\chi_W)_x = \chi_{W_x}$ and $(\chi_W)^y = \chi_{W^y}$ (Problem 9.8). Set

$$f_{\chi_W}(x) = \int_Y (\chi_W)_x \, d\eta = \int_Y \chi_{W_x} \, d\eta = \eta(W_x),$$

$$g_{\chi_W}(y) = \int_X (\chi_W)^y \, d\mu = \int_X \chi_{W^y} \, d\mu = \mu(W^y),$$

for every $x \in X$ and $y \in Y$. Since μ and η are σ-finite measures, Lemma 9.7 ensures that $f_{\chi_W} \in \mathcal{M}(X, \mathcal{X})^+$, $g_{\chi_W} \in \mathcal{M}(Y, \mathcal{Y})^+$, and

$$\int_X f_{\chi_W} \, d\mu = \int_{X\times Y} \chi_W \, d\pi = \int_Y g_{\chi_W} \, d\eta.$$

Hence every characteristic function of sets in $\mathcal{X}{\times}\mathcal{Y}$ lies in $\top$. Therefore, by additivity and positive homogeneity of the set of positive measurable functions (Proposition 1.9), and also by the integral itself (Proposition 3.5(a,b)), we can conclude that every simple function (Definition 3.1) lies in $\top$. Now let h be an arbitrary function in $\mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$. Problem 1.6 says that there exists an increasing sequence $\{\varphi_n\}$ of simple functions in $\mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$ converging pointwise to h. For each n and for every $x \in X$ and $y \in Y$, set

$$f_{\varphi_n}(x) = \int_Y (\varphi_n)_x \, d\eta \quad \text{and} \quad g_{\varphi_n}(y) = \int_X (\varphi_n)^y \, d\mu. \qquad (*)$$

Since $\varphi_n \in \top$ (simple functions lie in $\top$), the above identities define a pair of functions f_{φ_n} and g_{φ_n} such that $f_{\varphi_n} \in \mathcal{M}(X, \mathcal{X})^+$, $g_{\varphi_n} \in \mathcal{M}(Y, \mathcal{Y})^+$, and

$$\int_X f_{\varphi_n} d\mu = \int_{X\times Y} \varphi_n \, d\pi = \int_Y g_{\varphi_n} d\eta. \qquad (**)$$

Since $\varphi_n \to h$, it follows that $(\varphi_n)_x \to h_x$ and $(\varphi_n)^y \to h^y$ for every $x \in X$ and $y \in Y$, where these convergences are pointwise. Since $\{\varphi_n\}$ is an increasing sequence of nonnegative functions, it is clear that the sequences $\{(\varphi_n)_x\}$ and $\{(\varphi_n)^y\}$ are also increasing and consist of nonnegative functions for each $x \in X$ and $y \in Y$. Furthermore, since φ_n is measurable, then so are all sections $(\varphi_n)_x$ and $(\varphi_n)^y$, for each n, by Proposition 9.6. Thus, applying the Monotone Convergence Theorem (Theorem 3.4) we get from $(*)$ that

$$\lim_n f_{\varphi n}(x) = \lim_n \int_Y (\varphi_n)_x \, d\eta = \int_Y \lim_n (\varphi_n)_x \, d\eta = \int_Y h_x \, d\eta = f_h(x),$$

$$\lim_n g_{\varphi n}(y) = \lim_n \int_X (\varphi_n)^y \, d\mu = \int_X \lim_n (\varphi_n)^y \, d\mu = \int_X h^y \, d\mu = g_h(y),$$

for every $x \in X$ and every $y \in Y$, so that $f_{\varphi_n} \to f_h$ and $g_{\varphi_n} \to g_h$ pointwise. Recall that $\{f_{\varphi_n}\}$ and $\{g_{\varphi_n}\}$ are increasing sequences (because $\{\varphi_n\}$ is increasing). Using the Monotone Convergence Theorem once again we get from $(**)$ that $f_h \in \mathcal{M}(X, \mathcal{X})^+$, $g_h \in \mathcal{M}(Y, \mathcal{Y})^+$, and

$$\int_X f_h \, d\mu = \int_{X\times Y} h \, d\pi = \int_Y g_h \, d\eta,$$

and hence $h \in \top$. Then $\mathcal{M}(X \times Y, \mathcal{X} \times \mathcal{Y})^+ \subseteq \top$. □

The conclusion of the Tonelli Theorem can be rewritten as

$$\int_X \left(\int_Y h \, d\eta \right) d\mu = \int_{X\times Y} h \, d\pi = \int_Y \left(\int_X h \, d\mu \right) d\eta.$$

This is a significant result. It says that *the order of the integrals can be interchanged*. The same applies to the next theorem, which has exactly the same conclusion but a different hypothesis, where the function h is not necessarily nonnegative but should be integrable with respect to the product measure.

Theorem 9.9. (Fubini Theorem). *Consider two measure spaces $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \eta)$. If μ and η are σ-finite measures and $h \in \mathcal{L}(X \times Y, \mathcal{X} \times \mathcal{Y}, \pi)$, then there are real-valued functions f_h and g_h defined a.e. on X and Y by*

$$f_h(x) = \int_Y h_x \, d\eta \quad \text{and} \quad g_h(y) = \int_X h^y \, d\mu,$$

which are in $\mathcal{L}(X, \mathcal{X}, \mu)$ and in $\mathcal{L}(Y, \mathcal{Y}, \eta)$, respectively, and

$$\int_X f_h \, d\mu = \int_{X\times Y} h \, d\pi = \int_Y g_h \, d\eta.$$

Proof. Take an arbitrary h in $\mathcal{M}(X\times Y, \mathcal{X}\times\mathcal{Y})$, and consider its positive and negative parts h^+ and h^- in $\mathcal{M}(X\times Y, \mathcal{X}\times\mathcal{Y})^+$ such that $h = h^+ - h^-$ (Proposition 1.6), and also its x-section h_x in $\mathcal{M}(Y,\mathcal{Y})$ and y-section h^y in $\mathcal{M}(X,\mathcal{X})$ (Proposition 9.6). Now consider the parts of the sections $(h_x)^\pm$ in $\mathcal{M}(Y,\mathcal{Y})^+$ and $(h^y)^\pm$ in $\mathcal{M}(X,\mathcal{X})^+$, which coincide with the sections of the parts $(h^\pm)_x$ and $(h^\pm)^y$ (Problem 9.9). As the positive and negative parts h^+ and h^- lie in $\mathcal{M}(X\times Y, \mathcal{X}\times\mathcal{Y})^+$, because $(h^\pm)_x = (h_x)^\pm$ and $(h^\pm)^y = (h^y)^\pm$, and since the measures μ and η are σ-finite, the Tonelli Theorem ensures that the functions $f_{h^\pm}$ and $g_{h^\pm}$ defined on X and Y by

$$f_{h^\pm}(x) = \int_Y (h_x)^\pm \, d\eta \quad \text{and} \quad g_{h^\pm}(y) = \int_X (h^y)^\pm \, d\mu \qquad (*)$$

are in $\mathcal{M}(X,\mathcal{X})^+$ and in $\mathcal{M}(Y,\mathcal{Y})^+$, respectively, and

$$\int_X f_{h^\pm}\, d\mu = \int_{X\times Y} h^\pm \, d\pi = \int_Y g_{h^\pm}\, d\eta. \qquad (**)$$

In addition, if $h \in \mathcal{L}(X\times Y, \mathcal{X}\times\mathcal{Y}, \pi)$, then the parts h^+ and h^- are nonnegative functions in $\mathcal{L}(X\times Y, \mathcal{X}\times\mathcal{Y}, \pi)$ by Definition 4.1. Even though their sections are real-valued, the nonnegative measurable functions $f_{h^\pm}$ and $g_{h^\pm}$ are not necessarily real-valued (cf. Problem 9.11) but have finite integrals according to $(**)$. Then they are real-valued almost everywhere by Problem 3.9(b), and the differences $f_{h^+}(x) - f_{h^-}(x)$ and $g_{h^+}(y) - g_{h^-}(y)$ are defined almost everywhere with respect to μ and η. Thus take real-valued functions f_h and g_h defined almost everywhere on X and Y, respectively, as follows:

$$f_h = f_{h^+} - f_{h^-} \quad \text{and} \quad g_h = g_{h^+} - g_{h^-}$$

and zero otherwise, which lie in $\mathcal{L}(X,\mathcal{X},\mu)$ and $\mathcal{L}(Y,\mathcal{Y},\eta)$. In fact (cf. Problems 3.8 and 3.9), there is an $\mathcal{X}$-measurable set N with $\mu(N) = 0$ such that $\int_X f_{h^\pm}|_{X\setminus N}\, d\mu = \int_X f_{h^\pm}\, \chi_{X\setminus N}\, d\mu = \int_{X\setminus N} f_{h^\pm}\, d\mu = \int_X f_{h^\pm}\, d\mu < \infty$ for which $f_{h^\pm}|_{X\setminus N}$ is *real-valued*, and hence $f_{h^\pm}|_{X\setminus N}$ lies in $\mathcal{L}(X,\mathcal{X},\mu)$. Thus the function f_h defined by $f_{h^+}|_{X\setminus N} - f_{h^-}|_{X\setminus N}$ on $X\setminus N$ and zero on N is real-valued and lies in $\mathcal{L}(X,\mathcal{X},\mu)$ — see Lemma 4.5. Similarly, the function g_h is real-valued and lies in $\mathcal{L}(Y,\mathcal{Y},\eta)$. Then, by $(*)$ and Definition 4.1,

$$f_h(x) = \int_Y (h_x)^+ \, d\eta - \int_Y (h_x)^- \, d\eta = \int_Y h_x \, d\eta,$$

$$g_h(y) = \int_X (h^y)^+ \, d\mu - \int_X (h^y)^- \, d\mu = \int_X h^y \, d\mu,$$

for μ-almost every x in X and η-almost every y in Y. Also, by $(**)$ and applying Definition 4.1 again, and since $f_h = f_{h^+} - f_{h^-}$ on $X\setminus N$ so that $\int_X f_h\, d\mu = \int_{X\setminus N} f_{h^+}\, d\mu - \int_{X\setminus N} f_{h^-}\, d\mu$ (see Proposition 4.3), it follows that

$$\int_{X\times Y} h\,d\pi = \int_{X\times Y} h^+\,d\pi - \int_{X\times Y} h^-\,d\pi = \int_X f_{h+}\,d\mu - \int_X f_{h-}\,d\mu = \int_X f_h\,d\mu,$$

and similarly

$$\int_{X\times Y} h\,d\pi \;=\; \int_Y g_{h+}\,d\eta - \int_Y g_{h-}\,d\eta \;=\; \int_Y g_h\,d\eta. \qquad \square$$

Suppose N is an $\mathcal{X}$-measurable set such that $\mu(N)=0$ for which $f_h(x) = \int_Y h_x(y)d\eta$ on $X\backslash N$ and zero on N (as in the preceding proof). Observe that $\int_X f_h(x)d\mu = \int_{X\backslash N} f_h(x)d\mu = \int_{X\backslash N}\big(\int_Y h_x(y)d\eta\big)d\mu = \int_X\big(\int_Y h(x,y)d\eta\big)d\mu$. In a similar way, $\int_Y g_h(y)d\eta = \int_Y\big(\int_X h(x,y)d\mu\big)d\eta$. So, as we have commented before, the conclusion of the Fubini Theorem can also be rewritten as

$$\int_X \left(\int_Y h\,d\eta\right) d\mu \;=\; \int_{X\times Y} h\,d\pi \;=\; \int_Y \left(\int_X h\,d\mu\right) d\eta.$$

The middle integral, $\int_{X\times Y} h\,d\pi$, is the *double integral* of h. The left and right integrals, $\int_X\big(\int_Y h\,d\eta\big)d\mu$ and $\int_Y\big(\int_X h\,d\mu\big)d\eta$, are the *iterated integrals* of h.

9.4 Problems

Problem 9.1. Let $\Re\times\Re$ be the Borel σ-algebra of subsets of $\mathbb{R}^2$ (Example 9A). If $E\in\Re$, show that the *difference set* D_E is $\Re\times\Re$-measurable; that is,

$$D_E = \big\{(x,y)\in\mathbb{R}^2\colon\ y-x\in E\big\} \;\in\; \Re\times\Re.$$

Hint: (a) The "smart" way: $f(x,y)=y-x$ defines a continuous function f from $\mathbb{R}^2$ to $\mathbb{R}$, thus a measurable function (Section 11.1), and $D_E = f^{-1}(E)$.

(b) The "tour de force" way: Take $E\in\Re$ so that $E\cup\{\beta\}+\alpha\in\Re$ for every $\alpha,\beta\in\mathbb{R}$. If E is unbounded, then consider a countable partition of E made up of bounded measurable sets. If E is bounded, write D_E as a countable union of *measurable triangles* as follows. Let $\operatorname{diam} E = \sup_{x,y\in E}|x-y| < \infty$ be the *diameter* of the bounded set $E\subseteq\mathbb{R}$. For each integer $k\in\mathbb{Z}$ set

$$E_k = (E\cup\{\sup E\}) + k\operatorname{diam} E,$$

$$F_k = E_k + \inf E = \big[(E\cup\{\sup E\}) + \inf E\big] + k\operatorname{diam} E,$$

which are subsets of $\mathbb{R}$. Now consider the following subsets of $\mathbb{R}^2$:

$$L = \big\{(x,y)\in\mathbb{R}^2\colon\ y\le \sup E + x\big\} \quad \text{or} \quad L = \big\{(x,y)\in\mathbb{R}^2\colon\ y < \sup E + x\big\}$$

accordingly as $\sup E$ lies or does not lie in E (either open or closed in $\mathbb{R}^2$),

$$U = \big\{(x,y) \in \mathbb{R}^2 \colon y \geq \inf E + x\big\} \quad \text{or} \quad U = \big\{(x,y) \in \mathbb{R}^2 \colon y > \inf E + x\big\}$$

accordingly as $\inf E$ lies or does not lie in E (either open or closed in $\mathbb{R}^2$), and

$$\Delta_k = (E_k \times F_k) \cap U \quad \text{and} \quad \nabla_k = (E_k \times F_{k+1}) \cap L$$

for each $k \in \mathbb{Z}$. Verify that the above *triangle sets* cover the *strip-shape set* D_E (as suggested in the sketch below):

$$D_E = \big\{(x,y) \in \mathbb{R}^2 \colon y - x \in E\big\} = \bigcup\nolimits_k (\Delta_k \cup \nabla_k).$$

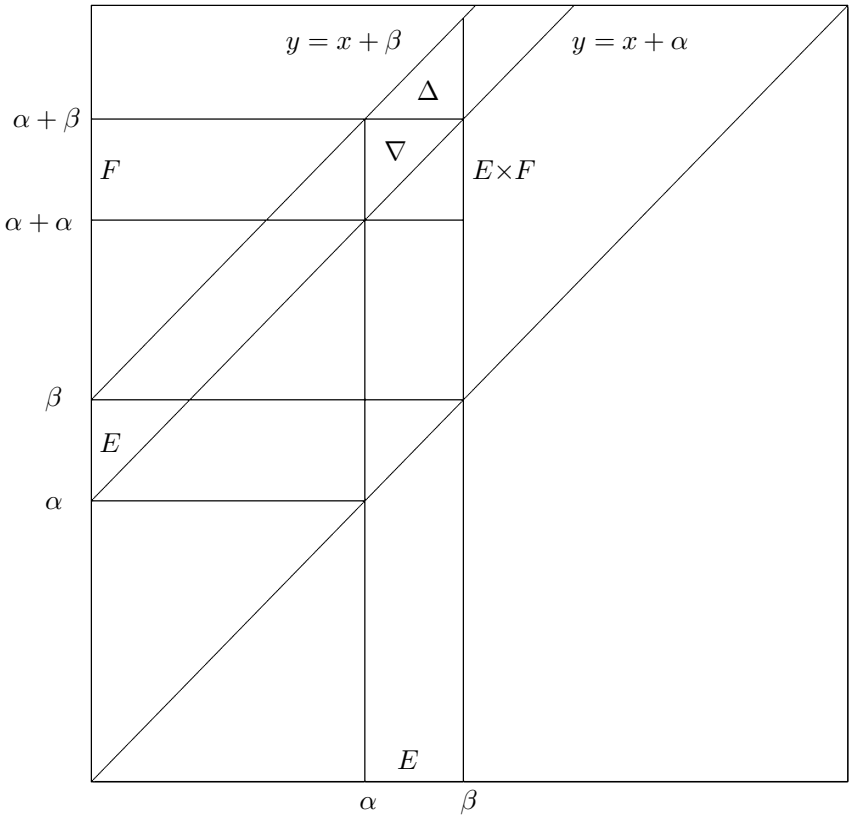


Fig. 9.1. Problem 9.1(b)

Problem 9.2. Show that if $g \colon \mathbb{R} \to \mathbb{R}$ is $\mathfrak{R}$-measurable, then the function $h \colon \mathbb{R}^2 \to \mathbb{R}$ given by $h(x,y) = g(y-x)$ for (x,y) in $\mathbb{R}^2$ is $\mathfrak{R}\times\mathfrak{R}$-measurable.

Hint: Use Problem 9.1 to verify that $f \colon \mathbb{R}^2 \to \mathbb{R}$ such that $f(x,y) = y - x$ for (x,y) in $\mathbb{R}^2$ is $\mathfrak{R}\times\mathfrak{R}$-measurable. Recall Problems 1.8 and 1.10.

Problem 9.3. Consider the Cartesian product $(X\times Y,\, \mathcal{X}\times\mathcal{Y})$ of the measurable spaces $(X,\mathcal{X})$ and $(Y,\mathcal{Y})$. Let $f\in\mathcal{M}(X,\mathcal{X})$ and $g\in\mathcal{M}(Y,\mathcal{Y})$ be real-valued functions. Show that the real-valued function $h=f\,g$, given by

$$h(x,y)=f(x)\,g(y) \quad \text{for every } (x,y)\in X\times Y, \quad \text{lies in } \mathcal{M}(X\times Y,\,\mathcal{X}\times\mathcal{Y}).$$

Problem 9.4. Take an arbitrary subset S of the Cartesian product $X\times Y$ of two sets X and Y. Let $\{S_\alpha\}$ be an arbitrary collection of subsets of $X\times Y$, and take any x in X. Use the definition of x-section to show that

(a) $\big((X\times Y)\backslash S\big)_x=(X\times Y)_x\backslash S_x=Y\backslash S_x$,

(b) $\big(\bigcup_\alpha S_\alpha\big)_x=\bigcup_\alpha (S_\alpha)_x$.

Problem 9.5. Let $(X,\mathcal{X})$ and $(Y,\mathcal{Y})$ be measurable spaces, and take the measurable space $(X\times Y,\,\mathcal{X}\times\mathcal{Y})$ of their Cartesian product. Consider the collections of all subsets of $X\times Y$ for which all x-sections are $\mathcal{Y}$-measurable,

$$(\mathcal{X}\times\mathcal{Y})_X=\big\{W\subseteq X\times Y\colon\, W_x\in\mathcal{Y} \text{ for every } \ x\in X\big\},$$

and of all subsets of $X\times Y$ for which all y-sections are $\mathcal{X}$-measurable,

$$(\mathcal{X}\times\mathcal{Y})_Y=\big\{W\subseteq X\times Y\colon\, W^y\in\mathcal{X} \text{ for every } \ y\in Y\big\}.$$

Show that $(\mathcal{X}\times\mathcal{Y})_X$ and $(\mathcal{X}\times\mathcal{Y})_Y$ are σ-algebras of subsets of $X\times Y$ (use Problem 9.4), and that they contain all measurable rectangles (recall that sections of rectangles are either empty or sides of them), and conclude that

$$\mathcal{X}\times\mathcal{Y}\subseteq(\mathcal{X}\times\mathcal{Y})_X \quad \text{ and } \quad \mathcal{X}\times\mathcal{Y}\subseteq(\mathcal{X}\times\mathcal{Y})_Y.$$

Problem 9.6. Consider the unit interval $X=[0,1]\subset\mathbb{R}$, and let $\mathcal{X}$ be the σ-algebra of all subsets of X that either are countable or are the complement of a countable set (see Problem 2.6). Take the measurable space $(X\times X,\,\mathcal{X}\times\mathcal{X})$ of the Cartesian product of two copies of the measurable space $(X,\mathcal{X})$. For $\alpha\ge 0$ and an arbitrary $\varepsilon>0$, consider the *sector*

$$S_{\alpha,\varepsilon}=\big\{(x,y)\in X\times X\colon\, \alpha x\le y\le(\alpha+\varepsilon)x\big\}.$$

(a) Show that $S_{\alpha,\varepsilon}$ is nonmeasurable $\ (S_{\alpha,\varepsilon}\notin\mathcal{X}\times\mathcal{X})$.

Hint: Proposition 9.6(a).

For each $\alpha\ge 0$, consider the *line segments*

$$I_\alpha=\big\{(x,y)\in X\times X\colon\, y=\alpha x\big\}.$$

(b) Show that the horizontal line segment I_0 is measurable $\ (I_0\in\mathcal{X}\times\mathcal{X})$.

(c) Show that if $\alpha>0$, then I_α is nonmeasurable $\ (I_\alpha\notin\mathcal{X}\times\mathcal{X})$.

Hint: Set $L_\alpha=\{(x,y)\in X\times X\colon \alpha x<y\}$ and $U_\alpha=\{(x,y)\in X\times X\colon y<\alpha x\}$. L_α and U_α are not measurable. I_α is measurable if and only if $L_\alpha\cup U_\alpha$ is.

Problem 9.7. Use Proposition 9.6(a) to prove the failure of its own converse: *There exist nonmeasurable sets for which all sections are measurable.*

Hint: See Problem 9.6.

Problem 9.8. Consider the characteristic function χ_S of a subset S of the Cartesian product $X \times Y$ of two sets X and Y. Show that for each $x \in X$ and each $y \in Y$, the sections $(\chi_S)_x$ and $(\chi_S)^y$ of the function χ_S are the characteristic functions of the sections S_x and S^y of the set S. That is,

$$(\chi_S)_x = \chi_{S_x} \quad \text{and} \quad (\chi_S)^y = \chi_{S^y}.$$

Thus conclude that if $A \times B$ is any rectangle from $X \times Y$, then

$$\chi_{A\times B}(x, y) = \chi_A(x)\chi_B(y) \quad \text{for every} \quad (x, y) \in X \times Y.$$

Problem 9.9. Let $f : X \times Y \to \overline{\mathbb{R}}$ be an arbitrary extended real-valued function on the Cartesian product $X \times Y$. Take its positive and negative parts, f^+ and f^-, and its x-sections and y-sections, f_x and f^y. First show that

$$(f^+)_x = (f_x)^+, \ \ (f^-)_x = (f_x)^- \quad \text{and} \quad (f^+)^y = (f^y)^+, \ \ (f^-)^y = (f^y)^-.$$

Hint: Recall that $f^+ = f\chi_{F^+}$, where $F^+ = \{(x, y) \in X \times Y : f(x, y) \geq 0\}$, and verify that $(f^+)_x = (f\chi_{F^+})_x = f_x(\chi_{F^+})_x = f_x\chi_{F^+} = (f_x)^+$.

Next show that the x-sections and y-sections of $f = f^+ - f^-$ are given by

$$f_x = (f^+)_x - (f^-)_x \quad \text{and} \quad f^y = (f^+)^y - (f^-)^y.$$

Problem 9.10. Consider the Borel σ-algebra $\mathfrak{R}$ (or the Lebesgue σ-algebra $\mathfrak{L}^*$) and take the Lebesgue measure λ on $\mathfrak{R}$ (or on $\mathfrak{L}^*$), which is referred to as the *linear Lebesgue measure* or as the *length* on $\mathbb{R}$. Consider the product of λ with itself; that is, the measure

$$\pi = \lambda \times \lambda$$

on the σ-algebra $\mathfrak{R} \times \mathfrak{R}$ (or on $\mathfrak{L}^* \times \mathfrak{L}^*$) of subsets of the plane $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. This measure is referred to as the *planar Lebesgue measure* or as the *area* on $\mathbb{R}^2$. Note that $\mathfrak{R} \times \mathfrak{R} \subset \mathfrak{L}^* \times \mathfrak{L}^*$, so $\mathfrak{R} \times \mathfrak{R}$-measurable sets are $\mathfrak{L}^* \times \mathfrak{L}^*$-measurable. Thus the area of a measurable rectangle $E \times F$ in $\mathbb{R}^2$ is $\lambda(E)\lambda(F)$. Give an example of an uncountable measurable subset of $[0, 1] \times [0, 1]$ with area zero such that all sections of it (x-section and y-section) are either empty or uncountable with length zero.

Problem 9.11. Consider the Lebesgue measure space $(\mathbb{R}, \mathfrak{R}, \lambda)$. Take the product space $(\mathbb{R}^2, \mathfrak{R} \times \mathfrak{R}, \pi)$, where $\pi = \lambda \times \lambda$ is the area on $\mathbb{R}^2$ (cf. Problem 9.10). Exhibit a function $f : \mathbb{R}^2 \to \mathbb{R}$ satisfying the following properties.

(1) $f = 0$ π-almost everywhere.

That is, $f = 0$ up to a measurable set of area zero. Thus verify that this function f is $\Re\times\Re$-measurable, and apply Proposition 4.2(a) to infer that this f is also integrable with $\int_{\mathbb{R}^2} f\,d\pi = 0$. Moreover,

(2) $\int_{\mathbb{R}} f_x\,d\lambda = \infty$ for some x-section $f_x\colon \mathbb{R} \to \mathbb{R}$ of f.

Problem 9.12. Set $E = [0,1] \subset \mathbb{R}$. Let $\mathcal{E} = \wp(E) \cap \Re$ be the σ-algebra of all Borel subsets of $[0,1]$. Take the Lebesgue measure λ on $\mathcal{E}$ (i.e., the restriction of the Lebesgue measure to $\mathcal{E}$ as in Problem 2.11). Let μ be the counting measure on $\mathcal{E}$ as in Problem 2.4(b), which is not σ-finite. Consider the measure spaces $(E,\mathcal{E},\lambda)$ and $(E,\mathcal{E},\mu)$. Use Example 9A and verify that the *identity segment*, or the *diagonal set*, of the rectangle $[0,1]\times[0,1]$, viz., the set $I = \{(x,y) \in E\times E\colon\ x = y\}$, is $\mathcal{E}\times\mathcal{E}$-measurable, and show that

$$\int \mu(I_x)\,d\lambda \neq \int \lambda(I^y)\,d\mu.$$

Hence the assumption of σ-finiteness cannot be omitted from Lemma 9.7.

Problem 9.13. Consider the setup of the previous problem. Use the characteristic function $\chi_I \in \mathcal{M}(E\times E,\ \mathcal{E}\times\mathcal{E},\ \lambda\times\mu)^+$ of the set I above to show that the assumption of σ-finiteness cannot be omitted from the Tonelli Theorem.

Problem 9.14. Let $(X,\mathcal{X},\mu)$ and $(Y,\mathcal{Y},\eta)$ be σ-finite measure spaces. Take their product space $(X\times Y,\ \mathcal{X}\times\mathcal{Y},\ \pi)$. If W and W' are sets in $\mathcal{X}\times\mathcal{Y}$ such that $\eta(W_x) = \eta(W'_x)$ for (almost) every $x \in X$ (or $\mu(W^y) = \mu(W'^y)$ for (almost) every $y \in Y$), then show that $\pi(W) = \pi(W')$.

Hint: Apply the Tonelli Theorem to the characteristic functions χ_W and $\chi_{W'}$ of the $\mathcal{X}\times\mathcal{Y}$-measurable sets W and W'. Use the results of Problem 9.8.

Problem 9.15. Take a pair $f, g \in \mathcal{L}(\mathbb{R},\Re,\lambda)$ of Lebesgue integrable functions. Define a function $h\colon \mathbb{R}^2 \to \mathbb{R}$ by $h(x,y) = f(x-y)$ for every (x,y) in $\mathbb{R}^2$. According to Problem 9.2, h is $\Re\times\Re$-measurable. Verify that the function $hg\colon \mathbb{R}^2 \to \mathbb{R}$ mapping each $(x,y) \in \mathbb{R}^2$ into $f(x-y)g(y) \in \mathbb{R}$ is $\Re\times\Re$-measurable as well. Now apply the Fubini Theorem to show that there is a real-valued function $f * g \in \mathcal{L}(\mathbb{R},\Re,\lambda)$ defined almost everywhere on $\mathbb{R}$ by

$$(f*g)(x) = \int_{\mathbb{R}} h_x\,g\,dy = \int_{\mathbb{R}} f(x-y)g(y)\,dy.$$

This function $f*g\colon \mathbb{R} \to \mathbb{R}$ is the *convolution* of f and g. Also show that

$$\int |f*g|\,dx \leq \left(\int |f|\,dx\right)\left(\int |g|\,dy\right).$$

Problem 9.16. Let $\{\alpha_{m,n}\}$ be a countable family of *nonnegative* real numbers, double-indexed by $(m,n) \in \mathbb{N}\times\mathbb{N}$.

(a) Use elementary analysis to show that

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\alpha_{m,n} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\alpha_{m,n}.$$

Hint: Recall that a family of real numbers is absolutely summable if and only if it is unconditionally summable.

Take the product space $(\mathbb{N}\times\mathbb{N},\ \wp(\mathbb{N}\times\mathbb{N}),\ \mu\times\mu)$ of two copies of the measure space $(\mathbb{N}, \wp(\mathbb{N}), \mu)$, where μ is the counting measure of Example 2B. Consider the integral of the function $\alpha_{m,n}\colon \mathbb{N}\times\mathbb{N} \to \mathbb{R}$ with respect to the product measure $\pi = \mu\times\mu$ (see Problem 3.4). That is, for every $E \in \wp(\mathbb{N}\times\mathbb{N})$,

$$\int_E \alpha_{m,n}\, d\pi = \sum\nolimits_E \alpha_{m,n}.$$

(b) If $\sum_{\mathbb{N}\times\mathbb{N}} \alpha_{m,n} < \infty$ (i.e., under the assumption of finite integral), then show that the result in (a) can be proved by using the Fubini Theorem.

Problem 9.17. Consider the setup of the previous problem, but now suppose $\{\alpha_{m,n}\}$ is an arbitrary countable family of integers (not necessarily nonnegative). Suppose $\alpha_{m,m} = 1$ and $\alpha_{m,m+1} = -1$ for every integer $m \in \mathbb{N}$, and $\alpha_{m,n} = 0$ otherwise. Show that

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\alpha_{m,n} = 0 \quad \text{and} \quad \sum_{n=1}^{\infty}\sum_{m=1}^{\infty}\alpha_{m,n} = 1.$$

Conclude that the nonnegativeness assumption (i.e., $\alpha_{m,n} \geq 0$ for all (m,n) in $\mathbb{N}\times\mathbb{N}$) cannot be dropped in the Tonelli Theorem, and integrability (i.e., $\sum_{\mathbb{N}\times\mathbb{N}} |\alpha_{m,n}| < \infty$) cannot be omitted from the Fubini Theorem.

Hint:

$$\{\alpha_{m,n}\} = \begin{pmatrix} 1 & -1 & 0 & & \\ 0 & 1 & -1 & 0 & \\ & 0 & 1 & -1 & \\ & & 0 & 1 & \\ & & & & \ddots \end{pmatrix}.$$

Problem 9.18. Let $\big([0,1]\times[0,1],\ \wp([0,1]\times[0,1]) \cap (\Re\times\Re),\ \lambda\times\lambda\big)$ be the product space made from two copies of the finite Lebesgue measure space $\big([0,1],\ \wp([0,1]) \cap \Re,\ \lambda\big)$. Take the functions $f \in \mathcal{L}\big([0,1],\ \wp([0,1]) \cap \Re,\ \lambda\big)$ and the real-valued $g \in \mathcal{M}\big([0,1],\ \wp([0,1]) \cap \Re\big)^+$ given by

$$f(x) = \begin{cases} 1, & 0 < x \leq \frac{1}{2}, \\ -1, & \frac{1}{2} < x \leq 1, \end{cases} \qquad g(y) = \begin{cases} 0, & y = 0, \\ \frac{1}{y}, & y \in (0,1]. \end{cases}$$

Let the function $h \in \mathcal{M}\big([0,1]\times[0,1],\ \wp([0,1]\times[0,1]) \cap (\Re\times\Re)\big)$ be given by

$$h(x,y) = f(x)\,g(y)$$

for every $(x,y) \in [0,1]\times[0,1]$, which is real-valued (Problem 9.3). Show that

$$\int_0^1 \left(\int_0^1 h(x,y)dx \right) dy = 0.$$

What is the value of the other iterated integral of h? Is h integrable?

Problem 9.19. Recall that an integrable function f (i.e., f in $\mathcal{L}(X,\mathcal{X},\mu)$) is a *real-valued* measurable function with a *finite integral* (i.e., $\int f^{\pm}\,d\mu < \infty$; equivalently, $\int |f|\,d\mu < \infty$ — see Lemma 4.4). Theorem 9.9 assumes that h is integrable and concludes that there are integrable functions f_h and g_h such that $\int_X f_h\,d\mu = \int_{X\times Y} h\,d\pi = \int_Y g_h\,d\eta$, where μ and η are σ-finite measures. As integrable functions, these h, f_h, and g_h are real-valued. Show that if μ and η are σ-finite and if h is an extended real-valued $\mathcal{X}\times\mathcal{Y}$-measurable function with a finite integral, then there are extended real-valued $\mathcal{X}$-measurable and $\mathcal{Y}$-measurable functions f_h and g_h (defined as in Theorem 9.9) with finite integrals such that

$$\int_X f_h\,d\mu \;=\; \int_{X\times Y} h\,d\pi \;=\; \int_Y g_h\,d\eta.$$

The Fubini Theorem holds if we allow extended real-valued functions but retain the assumption of finite integrals (i.e., $\int_{X\times Y} |h|\,d\pi < \infty$).

Problem 9.20. Take measure spaces $(X,\mathcal{X},\mu)$, $(Y,\mathcal{Y},\eta)$, $(X\times Y,\,\mathcal{X}\times\mathcal{Y},\pi)$, where $\pi = \mu\times\eta$. Let h be a real-valued function on $X\times Y$. Show that if $f \in \mathcal{L}(X,\mathcal{X},\mu)$ and $g \in \mathcal{L}(Y,\mathcal{Y},\eta)$, then

$$h = fg \quad \text{implies} \quad \int_{X\times Y} h\,d\pi \;=\; \left(\int_X f\,d\mu\right)\left(\int_Y g\,d\eta\right)$$

where $h \in \mathcal{L}(X\times Y,\,\mathcal{X}\times\mathcal{Y},\,\pi)$. That is, if $h(x,y) = f(x)\,g(y)$ for every (x,y) in $X\times Y$, and $f \in \mathcal{L}(X,\mathcal{X},\mu)$ and $g \in \mathcal{L}(Y,\mathcal{Y},\eta)$, then $h \in \mathcal{L}(X\times Y,\,\mathcal{X}\times\mathcal{Y},\pi)$ and $\int_{X\times Y} h\,d\pi = \big(\int_X f\,d\mu\big)\big(\int_Y g\,d\eta\big)$ even if the μ and η are not σ-finite.

Hint: Take the *kernels* of f and g, namely, $\mathcal{N}(f) = \{x \in X\colon f(x) = 0\}$ in $\mathcal{X}$ and $\mathcal{N}(g) = \{y \in Y\colon g(y) = 0\}$ in $\mathcal{Y}$ so that $X' = X\backslash\mathcal{N}(f) \in \mathcal{X}$ and $Y' = Y\backslash\mathcal{N}(g) \in \mathcal{Y}$ are σ-finite sets with respect to μ and η (Problem 3.9(c)). Set $\mathcal{X}' = \wp(X')\cap\mathcal{X}$ and $\mathcal{Y}' = \wp(Y')\cap\mathcal{Y}$, the sub-$\sigma$-algebras of $\mathcal{X}$ and $\mathcal{Y}$ made up of subsets of X' and Y'. Verify that μ and η (in fact, their restrictions to $\mathcal{X}'$ and $\mathcal{Y}'$) are σ-finite measures on $\mathcal{X}'$ and $\mathcal{Y}'$. Since $h = fg$, get

$$\int_{X\times Y} h(x,y)\,d\pi \;=\; \int_{X'\times Y'} f(x)\,g(y)\,d\pi.$$

Apply the Fubini Theorem for σ-finite measures on $\mathcal{X}'$ and $\mathcal{Y}'$.

Problem 9.21. Consider two measure spaces $(X, \mathcal{X}, \mu)$ and $(Y, \mathcal{Y}, \eta)$. Take their Cartesian product $(X\times Y,\ \mathcal{X}\times\mathcal{Y},\ \pi)$, where $\pi = \mu\times\eta$, and let h be a function in $\mathcal{M}(X\times Y,\ \mathcal{X}\times\mathcal{Y})$. Prove the following two assertions. If

(a) h is nonnegative, or

(b) h is π-integrable and essentially bounded, then

$$\int_{E\times F} h\,d\pi = 0 \ \text{ for every } \ E\times F \in \mathcal{X}\times\mathcal{Y} \quad \text{ if and only if } \quad h = 0 \ \pi\text{-a.e.}$$

Hint: Note that there are sets in $\mathcal{X}\times\mathcal{Y}$ that are not rectangles. Also, the essentially bounded assumption can be dismissed if π is finite (Problem 5.19).

(a) Straightforward by Proposition 3.7(a) since, for $h \in \mathcal{M}(X\times Y, \mathcal{X}\times\mathcal{Y})^+$, $0 \le \int_{E\times F} h\,d\pi \le \int_{X\times Y} h\,d\pi$ and $\int_{X\times Y} h\,d\pi = 0$ if and only if $h = 0$ π-a.e.

(b) Take the $\mathcal{X}\times\mathcal{Y}$-measurable sets

$$H^+ = \big\{(x,y) \in X\times Y\colon h(x,y) \ge 0\big\} = h^{-1}([0,+\infty)),$$
$$H^- = \big\{(x,y) \in X\times Y\colon h(x,y) \le 0\big\} = h^{-1}((-\infty,0]).$$

As $h \in \mathcal{M}(X\times Y, \mathcal{X}\times\mathcal{Y})$, set $h^+ = h\chi_{H^+}$ and $h^- = h\chi_{H^-}$ in $\mathcal{M}(X\times Y, \mathcal{X}\times\mathcal{Y})^+$. Take an increasing sequence $\{G_n\} = \{\bigcup_{i=1}^n E_i\times F_i\}$ of disjoint rectangles from $\mathcal{X}\times\mathcal{Y}$ such that $G_n \subseteq H^+$ for all n, converging in measure to H^+ in the following sense: for every $\varepsilon > 0$ there exists a positive integer n_ε such that

$$n \ge n_\varepsilon \quad \text{ implies } \quad \pi(H^+ \setminus G_n) < \varepsilon.$$

Set h_n^+ in $\mathcal{M}(X, \mathcal{X})^+$ such that $0 \le h_n^+ = h\chi_{G_n} \le h\chi_{H^+} = h^+$. Therefore, $0 \le h^+ - h_n^+ = h\chi_{H^+\setminus G_n}$ is integrable for every n because h is. Now suppose $\int_{E\times F} h\,d\pi = 0$ for all $E\times F \in \mathcal{X}\times\mathcal{Y}$. Then

(i) $\int h_n^+\,d\pi = \int_{G_n} h\,d\pi = 0$, so that $\int h^+ d\pi = \int (h^+ - h_n^+)\,d\pi$, for all n, and

(ii) if $\operatorname{ess\,sup}_{(x,y)\in X\times Y} |h| = K < \infty$ (h bounded π-a.e.), then, for every $\varepsilon > 0$,

$$\begin{aligned} 0 \le \int h^+ d\pi &= \int (h^+ - h_n^+)\,d\pi = \int_{H^+\setminus G_n} h\,d\pi \\ &\le K\,\pi(H^+\setminus G_n) < K\,\varepsilon \end{aligned}$$

whenever $n \ge n_\varepsilon$. Thus $0 = \int h_n^+ d\pi \to \int h^+\,d\pi$, so that $\int h^+\,d\pi = 0$, and hence $h^+ = 0$ π-a.e. by Proposition 3.7(a). A symmetric argument shows that $h^- = 0$ π-a.e. Thus $h = h^+ - h^- = 0$ π-a.e. The converse follows by Proposition 4.2(a) since $h\chi_{E\times F} \in \mathcal{L}(X\times Y, \mathcal{X}\times\mathcal{Y}, \pi)$ by Proposition 4.2(d).

Problem 9.22. Prove the following converse of Problem 9.20. Suppose the measures μ and η are σ-finite, the functions f, g, and h are integrable, $f \in \mathcal{L}(X, \mathcal{X}, \mu)$, $g \in \mathcal{L}(Y, \mathcal{Y}, \eta)$, $h \in \mathcal{L}(X\times Y,\ \mathcal{X}\times\mathcal{Y},\ \pi)$ and the difference $h - fg$ is π-essentially bounded ($\operatorname{ess\,sup}_{(x,y)\in X\times Y} |h - fg| < \infty$). Then

$$\int_{E\times F} h\,d\pi = \left(\int_E f\,d\mu\right)\left(\int_F g\,d\eta\right) \text{ for every } E\times F\in\mathcal{X}\times\mathcal{Y} \text{ implies } h=fg \ \pi\text{-a.e.}$$

Hint: By Fubini, $\left(\int_E f\,d\mu\right)\left(\int_F g\,d\eta\right)=\int_E\left(\int_F fg\,d\eta\right)d\mu)=\int_{E\times F} fg\,d\pi$, so that $\int_{E\times F} h\,d\pi=\left(\int_E f\,d\mu\right)\left(\int_F g\,d\eta\right)$ means $\int_{E\times F} h\,d\pi=\int_{E\times F} fg\,d\pi$; that is, $\int_{E\times F}(h-fg)\,d\pi=0$, for every $E\times F\in\mathcal{X}\times\mathcal{Y}$. Apply Problem 9.21(b).

Problem 9.23. Let $(X,\mathcal{X},\wp)$ be a probability space and let $f\colon X\to\mathbb{R}$ and $g\colon X\to\mathbb{R}$ be random variables (cf. Problem 2.20). Let $\alpha\in\mathbb{R}$ and $\beta\in\mathbb{R}$ be real numbers and let $A\in\Re$ and $B\in\Re$ be Borel subsets of the real line $\mathbb{R}$. Take the following subsets of X, which are $\mathcal{X}$-measurable (cf. Problem 1.8):

$$\begin{aligned} E_f(\alpha) &= E_f((-\infty,\alpha]) = \{x\in X\colon f(x)\le\alpha\} = f^{-1}((-\infty,\alpha]),\\ E_g(\beta) &= E_g((-\infty,\beta]) = \{x\in X\colon g(x)\le\beta\} = g^{-1}((-\infty,\beta]),\\ E_f(A) &= \{x\in X\colon f(x)\in A\} = f^{-1}(A),\\ E_g(B) &= \{x\in X\colon g(x)\in B\} = g^{-1}(B). \end{aligned}$$

Given two random variables f and g, let $(f,g)\colon X\to\mathbb{R}\times\mathbb{R}$ be a map assigning to each x in X a pair of real numbers, forming the vector $(f(x),g(x))$ in $\mathbb{R}\times\mathbb{R}$.

(a) Verify that the inverse image under (f,g) of a rectangle $A\times B$ in $\Re\times\Re$ is

$$(f,g)^{-1}(A\times B)=f^{-1}(A)\cap g^{-1}(B)=E_f(A)\cap E_g(B),$$

and conclude that $(f,g)\colon X\to\mathbb{R}\times\mathbb{R}$ is a measurable transformation with respect to the σ-algebras $\mathcal{X}$ and $\Re\times\Re$ as in Problem 1.8.

Hint: A set in the range of the map (f,g) is of the form $A\times B$ in $\mathbb{R}\times\mathbb{R}$.

The pair (f,g) is called a *random vector* or a *vector of random variables*.

(b) Prove that there exists a unique measure $\mu_{f,g}\colon\Re\times\Re\to\mathbb{R}$ such that

$$\mu_{f,g}(A\times B)=\wp\big((f,g)^{-1}(A\times B)\big)=\wp\big(E_f(A)\cap E_g(B)\big). \qquad (\S')$$

Hint: If $\{A_k\times B_k\}$ is a countable family of rectangles in $\Re\times\Re$, then the definition of the map (f,g) ensures that

$$(f,g)^{-1}\big(\textstyle\bigcup_k A_k\times B_k\big)=\bigcup_k\big((f,g)^{-1}(A_k\times B_k)\big)=\bigcup_k\big(E_f(A_k)\cap E_g(B_k)\big).$$

Now consider a real-valued set function $\overline{\mu_{f,g}}$ defined on the collection of all countable unions of measurable rectangles in $\Re\times\Re$ (in principle, we should begin with the collection of all finite unions as in Section 9.1), such that

$$\overline{\mu_{f,g}}\big(\textstyle\bigcup_k A_k\times B_k\big)=\wp\big((f,g)^{-1}\big(\bigcup_k A_k\times B_k\big)\big)=\wp\big(\bigcup_k\big(E_f(A_k)\cap E_g(B_k)\big)\big)$$

for countable families of rectangles $\{A_k \times B_k\} \in \Re \times \Re$. In particular, (§′) holds for $\overline{\mu_{f,g}}$. Let $\{\bigcup_j A'_j \times B'_j\}$ be a disjointification of $\{\bigcup_k A_k \times B_k\}$ such that $\{A'_j\}$ and $\{B'_j\}$ are families of disjoint measurable sets, so that $\bigcup_k A_k \times B_k = \bigcup_j A'_j \times B'_j$, and $\{E_f(A'_j)\}$ and $\{E_g(B'_j)\}$ are families of disjoint measurable sets. Thus the collection of measurable sets $\{\bigcup_k (E_f(A'_j) \cap E_g(B'_j))\}$ is disjoint (since $E_f(A'_j) \cap E_g(B'_j) \subseteq E_f(A'_j)$). Therefore, by (§′),

$$\overline{\mu_{f,g}}\big(\bigcup_k A'_k \times B'_k\big) = \sum_k \wp\big(E_f(A'_k) \cap E_g(B'_k)\big) = \sum_k \overline{\mu_{f,g}}(A'_k \times B'_k).$$

So $\overline{\mu_{f,g}}$ is countably additive for disjointified rectangles. Then, by the argument in Section 9.1, culminating with Theorem 9.5, the above identity can be extended to a unique finite measure $\mu_{f,g}$ on $\Re \times \Re$ satisfying (§′).

Thus, following the terminology of Problem 2.19, this unique measure $\mu_{f,g} \colon \Re \times \Re \to \mathbb{R}$ for which (§′) holds is

the measure $\mu_{f,g}$ on $\Re \times \Re$ induced by the measure $\wp$ on $\mathcal{X}$ via f and g.

(c) Show that $\mu_{f,g} \colon \Re \times \Re \to \mathbb{R}$ is a probability measure, and therefore $(\mathbb{R} \times \mathbb{R}, \Re \times \Re, \mu_{f,g})$ is a probability space.

Such a probability measure $\mu_{f,g}$ on $\Re \times \Re$ induced by the probability measure $\wp$ on $\mathcal{X}$ via random variables f and g is the *joint probability distribution of the random variables f and g*, or the *joint distribution law of f and g*.

Now define a function $F_{f,g} \colon \mathbb{R} \times \mathbb{R} \to [0,1]$ as follows: for every $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$,

$$\begin{aligned} F_{f,g}(\alpha,\beta) &= \mu_{f,g}\big((-\infty,\, \alpha] \times (-\infty,\, \beta]\big) = \wp\big(E_f(\alpha) \cap E_g(\beta)\big) \\ &= \wp\big(\{x \in X\colon\ f(x) \le \alpha \ \text{ and } \ g(x) \le \beta\}\big). \end{aligned}$$

This $F_{f,g}$ is the *joint distribution function of the random variables f and g*. By a slight abuse of notation, similar to that observed in Problem 2.20, the above pointwise definition of $F_{f,g}$ is also written as

$$F_{f,g}(\alpha,\beta) = \mu_{f,g}\big((-\infty,\, \alpha] \times (-\infty,\, \beta]\big) = \wp(f \le \alpha,\, g \le \beta),$$

and read: "$F_{f,g}(\alpha,\beta)$ is the probability of f being less than or equal to α and g being less than or equal to β".

(d) Show that, in general, $F_{f,g}(\alpha,\beta) \ne \wp\big(E_f(\alpha)\big)\, \wp\big(E_g(\beta)\big)$. In other words,

$$\wp\big(E_f(\alpha) \cap E_g(\beta)\big) \ne \wp\big(E_f(\alpha)\big)\, \wp\big(E_g(\beta)\big).$$

Hint: If $f = g$, $F_{f,g}(\alpha,\alpha) = \wp\big(E_f(\alpha)\big) \ne \wp\big(E_f(\alpha)\big)^2$ for $0 \ne \wp\big(E_f(\alpha)\big) \ne 1$.

Note: It is worth trying to verify that the framework of this Problem 9.23 can be extended to an n-tuple $(f_1, \ldots, f_n)$ of random variables on the same

set X and measurable with respect to the same σ-algebra $\mathcal{X}$, and such an n-tuple can be further extended beyond finiteness. However, as happens in essentially all parts of mathematics, "when a notion is primarily defined for an n-tuple, it may be wise to reduce the multiplicity down to a pair to see clearly what is really happening" [45, Remark 4.11].

Problem 9.24. Let $(X, \mathcal{X}, \wp)$ be a probability space. Set $\pi = \wp \times \wp$, the product measure of $\wp$ with itself on the σ-algebra $\mathcal{X} \times \mathcal{X}$. Let E_1 and E_2 be $\mathcal{X}$-measurable subsets of X. So $E_1 \times E_2$ is an $\mathcal{X} \times \mathcal{X}$-measurable rectangle in $X \times X$. By Theorem 9.5 (and the notation and terminology that follow it),

$$\pi(E_1 \times E_2) = \wp(E_1)\, \wp(E_2).$$

(a) Show that $\pi(X \times X) = 1$ and so $(X \times X, \mathcal{X} \times \mathcal{X}, \pi)$ is a probability space.

Definition: $\mathcal{X}$-measurable subsets E_1 and E_2 of X are *independent sets* if

$$\wp(E_1 \cap E_2) = \wp(E_1)\, \wp(E_2).$$

Equivalently, if

$$\wp(E_1 \cap E_2) = \pi(E_1 \times E_2).$$

(b) Verify: if $\wp(E_1 \cap E_2) = \wp(E_1)$ (e.g., if $E_1 \subseteq E_2$), then E_1 and E_2 are independent if and only if either $\wp(E_1) = 0$ or, if $\wp(E_1) \neq 0$, $\wp(E_2) = 1$.

Concrete examples: Set $X = [0,1]$ and $\mathcal{X} = \Re_{[0,1]} = \wp([0,1]) \cap \Re$. Let $\lambda_{[0,1]}$ and $\mu_{[0,1]}$ be the Lebesgue measure and the Cantor–Lebesgue measure over $[0,1]$, respectively. Both are probability measures (cf. Problem 7.15). Suppose $E_1 = C$, the Cantor set, and $E_2 = [0,1]$. If either $\wp = \lambda_{[0,1]}$ or $\wp = \mu_{[0,1]}$, then verify that E_1 and E_2 are independent.

For a given pair of random variables f and g, consider the sets $E_f(A)$, $E_g(B)$, $E_f(\alpha)$, and $E_g(\beta)$ in $\mathcal{X}$, for $A, B \in \Re$ and $\alpha, \beta \in \mathbb{R}$, defined in Problem 9.23. Moreover, consider the distribution functions $F_f \colon \mathbb{R} \to [0,1]$ and $F_g \colon \mathbb{R} \to [0,1]$ of f and g, respectively, as defined in Problem 2.20. Thus

$$F_f(\alpha) = \wp\big(E_f(\alpha)\big) \quad \text{and} \quad F_g(\beta) = \wp\big(E_g(\beta)\big)$$

for every $\alpha, \beta \in \mathbb{R}$. Therefore,

$$F_f(\alpha)\, F_g(\beta) = \wp\big(E_f(\alpha)\big)\, \wp\big(E_g(\beta)\big).$$

Now consider their joint distribution function $F_{f,g} \colon \mathbb{R} \times \mathbb{R} \to [0,1]$ defined in Problem 9.23: for every $(\alpha, \beta) \in \mathbb{R} \times \mathbb{R}$,

$$F_{f,g}(\alpha, \beta) = \wp\big(E_f(\alpha) \cap E_g(\beta)\big).$$

As we saw in Problem 9.23(c), in general,

$$F_{f,g}(\alpha,\beta) \neq F_f(\alpha)\,F_g(\beta).$$

Definition: f and g are *independent random variables* if

$$F_{f,g}(\alpha,\beta) = F_f(\alpha)\,F_g(\beta)$$

for every $(\alpha,\beta) \in \mathbb{R}\times\mathbb{R}$. Equivalently, f and g are independent random variables if $E_f(\alpha)$ and $E_g(\beta)$ are independent sets for every $\alpha,\beta \in \mathbb{R}$:

$$\wp\big(E_f(\alpha) \cap E_g(\beta)\big) = \wp\big(E_f(\alpha)\big)\,\wp\big(E_g(\beta)\big).$$

(c) Give an example of a pair of independent random variables.

Hint: (i) Set $X = [0,1]$, $\mathcal{X} = \wp([0,1]) \cap \Re$, $\wp = \lambda$, the Lebesgue measure over $[0,1]$, $f = \chi_{[0,1]}$, and $g = -f$. We know from Problem 2.20(b) that

(ii) $f^{-1}(A) = [0,1]$ if $1 \in A$ and $\varnothing$ otherwise, so that $E_f(\alpha) = [0,1]$ if $1 \leq \alpha$ and $\varnothing$ otherwise, $\mu_f = \delta_1$, and $F = \chi_{[1,\infty)}$. Verify that

(iii) $g^{-1}(B) = [0,1]$ if $-1 \in B$ and $\varnothing$ otherwise, so that $E_g(\beta) = [0,1]$ if $-1 \leq \beta$ and $\varnothing$ otherwise, $\mu_g = \delta_{-1}$, and $F = \chi_{[-1,\infty)}$. Also verify that

(vi) $E_f(\alpha) \cap E_f(\beta) = [0,1]$ if $1 \leq \alpha$ and $-1 \leq \beta$ and $\varnothing$ otherwise, so that $\lambda\big(E_f(\alpha) \cap E_f(\beta)\big) = 1$ if $1 \leq \alpha$ and $-1 \leq \beta$ and 0 otherwise, and hence $\lambda\big(E_f(\alpha) \cap E_f(\beta)\big) = \lambda\big(E_f(\alpha)\big)\lambda\big(E_f(\beta)\big)$ for every $\alpha,\beta \in \mathbb{R}$.

(v) Thus conclude that f and g are independent random variables.

(d) Verify the converse. Two random variables f and g are independent if and only if $E_f(A)$ and $E_g(B)$ are independent sets for every $A, B \in \Re$:

$$\wp\big(E_f(A) \cap E_g(B)\big) = \wp\big(E_f(A)\big)\,\wp\big(E_g(B)\big).$$

Hint: Last paragraph of Section 1.1 and the remark following Problem 1.14.

Since $\wp\big(E_f(A)\big)\,\wp\big(E_g(B)\big) = \pi\big(E_f(A)\times E_g(B)\big)$, item (d) above says that f and g are independent if and only if, for every $A, B \in \Re$,

$$\wp\big(E_f(A) \cap E_g(B)\big) = \pi\big(E_f(A)\times E_g(B)\big).$$

For two random variables f and g, take their probability distributions; that is, the probability measures μ_f and μ_g on $\Re$ induced by the probability measure $\wp$ on $\mathcal{X}$ via f and g as in (§) of Problems 2.19 and 2.20: for $A, B \in \Re$,

$$\mu_f(A) = \wp\big(f^{-1}(A)\big) = \wp\big(E_f(A)\big),$$

$$\mu_g(B) = \wp\big(g^{-1}(B)\big) = \wp\big(E_g(B)\big).$$

Now take their joint probability distribution; that is, the probability measure $\mu_{f,g}$ on $\Re\times\Re$ induced by the probability measure $\wp$ on $\mathcal{X}$ via f and g, as in Problem 9.23, which is the unique probability measure on $\Re\times\Re$ for which (§′) in Problem 9.23 holds: for every $A\times B \in \Re\times\Re$,

$$\mu_{f,g}(A\times B) = \wp\big((f,g)^{-1}(A\times B)\big) = \wp\big(E_f(A)\cap E_g(B)\big).$$

(e) Show that two random variables f and g are independent if and only if

$$\mu_{f,g} = \mu_f\times\mu_g$$

(i.e., $\mu_{f,g}$ is the product measure of μ_f and μ_g, which is the unique measure on $\Re\times\Re$ such that $\mu_{f,g}(A\times B)=\mu_f(A)\,\mu_g(B)$ for every $A\times B\in\Re\times\Re$).

Hint: By (d), *independence* means $\wp\big(E_f(A)\cap E_f(B)\big)=\wp\big(E_f(A)\big)\,\wp\big(E_f(B)\big)$ for every $A\times B\in\Re\times\Re$. Thus f and g are independent if and only if

$$\mu_{f,g}(A\times B) = \mu_f(A)\,\mu_g(B)$$

for every $A\times B\subseteq\Re\times\Re$. As μ_f and μ_g are finite, Theorem 9.5 ensures that $\mu_f\times\mu_g$ is the unique measure on $\Re\times\Re$ such that, for every $A\times B\in\Re\times\Re$,

$$(\mu_f\times\mu_g)(A\times B) = \mu_f(A)\,\mu_g(B).$$

Problem 9.25. Let $(X,\mathcal{X},\wp)$ be a probability space. Let $f,g\colon X\to\mathbb{R}$ be random variables. Set $\pi=\wp\times\wp$, the product measure of $\wp$ with itself on the σ-algebra $\mathcal{X}\times\mathcal{X}$. Take the map $(f,g)\colon X\to\mathbb{R}\times\mathbb{R}$ (as in Problem 9.23) and the measure $\mu_{f,g}\colon\Re\times\Re\to\mathbb{R}$ induced by $\wp$ satisfying (§′) in Problem 9.23(b):

$$\mu_{f,g}(A\times B) = \wp\big((f,g)^{-1}(A\times B)\big), \qquad (\S')$$

which extends to two variables the expression (§) in Problems 3.14 and 4.19. This problem extends Problems 3.14 and 4.19 to a pair of random variables.

Prove the *joint change of variables* for functions $\phi\colon\mathbb{R}\times\mathbb{R}\to\mathbb{R}$ of two variables, for $\phi\in\mathcal{M}(\mathbb{R}\times\mathbb{R},\Re\times\Re)^+$ and for $\phi\in\mathcal{L}(\mathbb{R}\times\mathbb{R},\Re\times\Re,\mu_{f,g})$:

$$\int_X \phi\circ(f,g)\,d\wp = \int_{\mathbb{R}\times\mathbb{R}} \phi\,d\mu_{f,g}. \qquad (\dagger')$$

Hint: This (†′) extends to two variables the expression (†) in Problems 3.14 and 4.19. If $\phi=\chi_{A\times B}$ is the characteristic function of a rectangle $A\times B$ in $\Re\times\Re$ of Borel sets A,B in $\Re$, then verify the following auxiliary identity:

$$\chi_{A\times B}\circ(f,g) = \chi_{(f,g)^{-1}(A\times B)}, \qquad \text{(i)}$$

extending a similar equation used to prove (†) in Problem 3.14. So, by (§′),

$$\int_{(f,g)^{-1}(A\times B)} d\wp = \int_{A\times B} d\mu_{f,g}.$$

Apply (i) to check that (†′) holds for characteristic functions $\chi_{A\times B}$:

$$\int_X \chi_{A\times B} \circ (f,g)\, d\wp = \int_X \chi_{(f,g)^{-1}(A\times B)}\, d\wp = \int_{(f,g)^{-1}(A\times B)} d\wp$$
$$= \int_{A\times B} d\mu_{f,g} = \int_{\mathbb{R}\times\mathbb{R}} \chi_{A\times B}\, d\mu_{f,g}.$$

Thus (†′) holds for simple functions. Use the argument of Problem 3.14(b) if $\phi \in \mathcal{M}(\mathbb{R}\times\mathbb{R}, \Re\times\Re)^+$, and then the argument of Problem 4.19(a) if $\phi \in \mathcal{L}(\mathbb{R}\times\mathbb{R}, \Re\times\Re, \mu_{f,g})$ to conclude the proof of (†′).

Problem 9.26. Consider the setup of Problem 9.25. Take a pair of random variables f and g. Their product $fg\colon X \to \mathbb{R}$ is, as we know, a random variable. Thus consider its expectation as defined in Problem 4.20.

(a) Show that

$$\mathrm{Exp}(fg) = \int_X fg\, d\wp = \int_{\mathbb{R}\times\mathbb{R}} \alpha\,\beta\, d\mu_{f,g}(\alpha,\beta).$$

Hint: Since the product fg is measurable (cf. Proposition 1.5), suppose $fg \in \mathcal{M}(X,\mathcal{X})^+$ or $fg \in \mathcal{L}(X,\mathcal{X},\wp)$ and, according to Problem 4.20, set

$$\mathrm{Exp}(fg) = \int_X fg\, d\wp.$$

Take the function $\phi\colon \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ given by $\phi(\alpha,\beta) = \alpha\,\beta$ for every (α,β) in $\mathbb{R}\times\mathbb{R}$, which is measurable in the sense that $\phi^{-1}(C) \in \Re\times\Re$ for $C \in \Re$.

Indeed, if $C \neq \varnothing$ is in the range of ϕ, then it is of the form $C = A\cdot B$ for $A, B \in \Re$, where $A\cdot B = \{\alpha\,\beta \in \mathbb{R} : \alpha \in A \text{ and } \beta \in B\}$, so that $\phi^{-1}(C) = A\times B$. Thus, ϕ is measurable. (Actually, ϕ is continuous — see Problem 9.1.)

Consider assumptions A_1 and A_2 of Problem 4.20, now according to whether fg lies in $\mathcal{M}(X,\mathcal{X})^+$ or in $\mathcal{L}(X,\mathcal{X},\wp)$, and apply (†′) in Problem 9.25:

$$\int_X fg\, d\wp = \int_X \phi\circ(f,g)\, d\wp = \int_{\mathbb{R}\times\mathbb{R}} \phi\, d\mu_{f,g} = \int_{\mathbb{R}\times\mathbb{R}} \alpha\,\beta\, d\mu_{f,g}(\alpha,\beta).$$

(b) Verify that, in general,

$$\mathrm{Exp}(fg) \neq \mathrm{Exp}(f)\,\mathrm{Exp}(g).$$

Hint: Problem 5.20 says that $\mathrm{Exp}(f)^2 \leq \mathrm{Exp}(f^2)$, but it may happen that $\mathrm{Exp}(f)^2 < \mathrm{Exp}(f^2)$. For instance, set $X = [0,1]$, $\mathcal{X} = \wp([0,1]) \cap \Re$, $\wp = \lambda$, the Lebesgue measure over $[0,1]$, and let f be the identity function, $f(x) = x$ for all $x \in X = [0,1]$, so that $\mathrm{Exp}(f)^2 = \frac{1}{4}$ and $\mathrm{Exp}(f^2) = \frac{1}{3}$.

Definition: f and g are *uncorrelated random variables* if

$$\mathrm{Exp}(fg) = \mathrm{Exp}(f)\,\mathrm{Exp}(g).$$

(c) Show that if f and g are independent, then f and g are uncorrelated:

$$\textit{Independence} \quad \Longrightarrow \quad \textit{Uncorrelatedness}.$$

Hint: By 4.20 and according to item (a) we get (see also Problem 5.20)

$$\mathrm{Exp}(f) = \int_X f\,d\wp = \int_{\mathbb{R}} \alpha\,d\mu_f(\alpha) \quad \text{and} \quad \mathrm{Exp}(g) = \int_X g\,d\wp = \int_{\mathbb{R}} \beta\,d\mu_f(\beta),$$

$$\mathrm{Exp}(fg) = \int_X fg\,d\wp = \int_{\mathbb{R}\times\mathbb{R}} \alpha\,\beta\,d\mu_{f,g}(\alpha,\beta).$$

If f, g are positive and $\mathrm{Exp}(fg)$ is not finite, then one of $\mathrm{Exp}(f)$ or $\mathrm{Exp}(g)$ is not finite (Problem 5.1(b)) and the claimed identity is trivial.Thus suppose $f, g \in \mathcal{L}(X, \mathcal{X}, \wp)$, so the integrals for $\mathrm{Exp}(f)$ and $\mathrm{Exp}(g)$ are all finite, and the problem follows by the Fubini Theorem. Indeed, by Problem 9.24(d), independence means $\mu_{f,g} = \mu_f \times \mu_g$. Use Problem 9.20 to get uncorrelatedness:

$$\int_{\mathbb{R}\times\mathbb{R}} \alpha\,\beta\,d\mu_{f,g} = \left(\int_{\mathbb{R}} \alpha\,d\mu_f\right)\left(\int_{\mathbb{R}} \beta\,d\mu_g\right).$$

(d) Give an example of a pair of uncorrelated random variables that are not independent (i.e., show that the converse to (c) fails).

Hint: Suppose f and h are independent (e.g., Problem 9.24(c)). Then they are uncorrelated by (c): $\mathrm{Exp}(fh) = \mathrm{Exp}(f)\,\mathrm{Exp}(h)$. Set

$$g = h + f.$$

Thus, by linearity of the integral, $\mathrm{Exp}(g) = \mathrm{Exp}(h) + \mathrm{Exp}(f)$, and $\mathrm{Exp}(fg) = \mathrm{Exp}(fh) + \mathrm{Exp}(f^2) = \mathrm{Exp}(f)\,\mathrm{Exp}(h) + \mathrm{Exp}(f^2)$. Suppose $\mathrm{Exp}(f^2) = \mathrm{Exp}(f)^2$ (e.g., the same f in Problem 9.24(c): $f = 1$ on $[0,1]$ with $\wp = \lambda$ over $[0,1]$). So

$$\mathrm{Exp}(fg) = \mathrm{Exp}(f)\,\big(\mathrm{Exp}(h) + \mathrm{Exp}(f)\big) = \mathrm{Exp}(f)\,\mathrm{Exp}(g),$$

and f and g are uncorrelated. However f and $g = h + f$ are not independent. Indeed, $E_f(\alpha) = \{x \in X\colon f(x) \le \alpha\}$, $E_g(\alpha) = \{x \in X\colon h(x) + f(x) \le \alpha\}$. If $f \ge 0$ and $h \ge 0$, $E_f(\alpha) \cap E_g(\alpha) = \{x \in X\colon h(x) + f(x) \le \alpha \text{ and } f(x) \le \alpha\} = E_g(\alpha)$. Hence $\wp\big(E_f(\alpha) \cap E_g(\alpha)\big) = \wp\big(E_g(\alpha)\big) \ne \wp\big(E_f(\alpha)\big)\wp\big(E_g(\alpha)\big)$ whenever $\wp\big(E_g(\alpha)\big) \ne 0$ and $\wp\big(E_f(\alpha)\big) \ne 1$ for some $\alpha \in \mathbb{R}$ (i.e., if there exists $\alpha \in \mathbb{R}$ such that $F_g(\alpha) \ne 0$ and $F_f(\alpha) \ne 1$).

Suggested Readings

Bartle [5], Bauer [8], Berberian [9], Halmos [31], Lang [46], Royden [62].

Appendix to Part I

A

A Glimpse at Probability Measures

A.1 Introduction

This is a short essay on basic concepts arising when a measure is normalized.

The introductory probability saga in this book began with Problem 2.19, which brought an arbitrary measure space $(X, \mathcal{X}, \eta)$ down to a concrete real measure space $(\mathbb{R}, \Re, \mu)$ such that $\mu(\mathbb{R}) = \eta(X)$. This was followed by Problems 3.14 and 4.19 that established a change of variable rule transforming integrals over X, with respect to a measure η on $\mathcal{X}$, into integrals over $\mathbb{R}$, with respect to a measure μ on $\Re$. Basic probability notions have been considered in subsequent problems from Chapter 2 to 9, as follows.

Problem 2.19: definition of a measure μ on $\Re$, induced by a measure on $\mathcal{X}$, that will give rise to the probability distribution of a random variable;

Problem 2.20: probability measure, probability space, random variable, probability distribution, and distribution function of a random variable;

Problem 2.21: basic properties of distribution functions;

Problem 3.14: change of variable rule for positive functions, supporting the change of variable rule for integrable functions;

Problem 3.15: density function of a random variable;

Problem 4.19: change of variable rule for integrable functions, supporting the notion of expectation in terms of the probability distribution;

Problem 4.20: expectation of a random variable;

Problem 5.20: variance of a random variable;

Problem 9.23: joint probability distribution and joint distribution function;

Problem 9.24: the notion of independence;

Problem 9.25: joint change of variable rule for random variables, supporting the notion of uncorrelatedness;

Problem 9.26: the notion of uncorrelatedness.

As we saw in Chapter 2, probability is a normalized measure (i.e., a positive measure such that the value of it at the whole space is equal to one), and random variables are real-valued measurable functions. In probability theory, the main role is played by random variables. For each random variable f, the original probability-measure space $(X, \mathcal{X}, \wp)$ acts as a starting point for generating another probability-measure space $(\mathbb{R}, \Re, \mu_f)$ upon which the probability plot takes place, where the main characters are the random variables themselves. Thus, differently from the general measure theory framework, probability properties are to be expressed in terms of the probability distribution μ_f of each random variable f. Probability problems are centered on the concepts of independence and conditioning, expressed in terms of sub-σ-algebras of the original σ-algebra $\mathcal{X}$.

Probability theory has its own notation and terminology. We show below a concise list containing a correspondence between some probability terminology and notation and their equivalent in measure theory. (Table A.1 below was partially borrowed from [48, Chapter III].)

Probability	***Measure***
probability measure P	normalized measure $\wp$
random variable X	measurable function f
elementary random variable	simple function
event A	measurable set E
sure event Ω	whole set, underlying set X
impossible event	empty set
elementary event $\omega \in \Omega$	point $x \in X$
σ-field of events $\mathcal{A}$	σ-algebra $\mathcal{X}$
probability space $(\Omega, \mathcal{A}, P)$	probability space $(X, \mathcal{X}, \wp)$
expectation $E \quad - \quad E(X)$	integral $\int \quad - \quad \mathrm{Exp}(f)$
second order random variable $E(X^2) < \infty$	function $f \in L^2(X, \mathcal{X}, \wp)$
almost sure (a.s.), with probability one (w.p.1)	almost everywhere (a.e.)
mean square (m.s.) convergence	convergence in L^2 (L^2)
convergence in probability $(\wp)$	convergence in measure (μ)

Tab. A.1. Correspondence between probability and measure terms.

According to the above table, the middle convergence diagram of Problem 6.6 on almost everywhere convergence, convergence in measure, and convergence in L^p, in particular, in L^2, can be redrawn in terms of probability terminology, leading to the relationship among convergence with probability one, mean square convergence, and convergence in probability, as follows.

$$\begin{array}{ccc} \text{(w.p.1)} & \Longrightarrow & (\wp) \\ \\ \text{(m.s.)} & \Longrightarrow & (\wp) \\ & & \begin{array}{c}\textbf{\textit{Probability}}\\ \textbf{\textit{measure}}\end{array} \end{array}$$

The diagram is exhaustive: no arrows are missing.

This appendix presents an overview of some fundamental notions of probability measures. It supplies and explains crucial definitions, and invites the reader to proceed to probability theory properly through the bibliography listed in the Suggested Readings at the end of the appendix. As measurable functions, random variables can be defined between any pair of measurable spaces, but we will stick to real-valued random variables.

A.2 Random Vectors, Sequences, and Functions

All measurable functions mentioned in this section are defined on the same underlying set X, measurable with respect to the same σ-algebra $\mathcal{X}$, where all probability properties refer to the same probability measure $\wp$.

Fix a probability space $(X, \mathcal{X}, \wp)$. A real random variable is a real-valued measurable function $f \in \mathcal{M}(X, \mathcal{X})$, meaning an $\mathcal{X}$-measurable function

$$f\colon X \to \mathbb{R}$$

assigning to each $x \in X$ a number $f(x)$ in $\mathbb{R}$. Now we consider the following extensions of the notion of a (real) random variable. *Random vectors* (or *vectors of random variables*) are ordered finite families of measurable functions. *Random sequences* (or *sequences of random variables*) are well ordered, countably infinite families of measurable functions. *Random functions* (or *functions of random variables*) are arbitrary families of measurable functions. These are symbolically defined as follows.

(1) 2-DIMENSIONAL RANDOM VECTOR. For a pair of real-valued measurable functions

$$(f_1, f_2) \in \mathcal{M}(X, \mathcal{X}) \times \mathcal{M}(X, \mathcal{X}),$$

a *2-dimensional random vector* is the map

$$(f_1, f_2)\colon X \to \mathbb{R}^2$$

assigning to each $x \in X$ a pair of real numbers $(f_1(x), f_2(x))$ in $\mathbb{R}^2$.

(2) n-DIMENSIONAL RANDOM VECTOR. Given an n-tuple of real-valued measurable functions

$$(f_1, \dots, f_n) \in \prod\nolimits_{i=1}^{n} \mathcal{M}(X, \mathcal{X})$$

in the Cartesian product of n copies of $\mathcal{M}(X, \mathcal{X})$, an *$n$-dimensional random vector* is a map

$$(f_1, \dots, f_n)\colon X \to \mathbb{R}^n$$

assigning to each $x \in X$ an n-tuple of real numbers $(f_1(x), \dots, f_n(x))$ in $\mathbb{R}^n$.

(3) RANDOM SEQUENCE. Consider a sequence of real-valued measurable functions

$$\{f_k\} = (f_1, f_2, \dots) \in \prod\nolimits_{k\in\mathbb{N}} \mathcal{M}(X, \mathcal{X}) = \prod\nolimits_{k=1}^{\infty} \mathcal{M}(X, \mathcal{X})$$

in the Cartesian product of countably infinite copies of $\mathcal{M}(X, \mathcal{X})$, indexed by a well ordered countably infinite set, for instance, by the set of all positive integers $\mathbb{N}$, as assumed above for simplicity. A *random sequence* is the map

$$\{f_k\} = (f_1, f_2, \dots)\colon X \to s$$

(whose codomain s is a sequence space, e.g., $s = \ell^p$ or $s = \ell^\infty$) assigning to each x in X a sequence of real numbers $\{f_k(x)\} = (f_1(x), f_2(x), \dots)$ in the real sequence space s (i.e., a function from X to s). Note that, by fixing an arbitrary $x \in X$, this map generates the map

$$f_{(\ )}(x)\colon \mathbb{N} \to \mathbb{R} \quad \in \quad s,$$

which is a sequence in the sequence space s taking each $k \in \mathbb{N}$ to a real number $f_k(x)$. Fixing $k \in \mathbb{N}$, we get the random variable $f_k \in \mathcal{M}(X, \mathcal{X})$. Fixing a pair $(k, x) \in \mathbb{N}\times X$ we get the real number $f_k(x) \in \mathbb{R}$.

(4) RANDOM FUNCTION. Take an arbitrary family of real-valued measurable functions

$$\{f_\gamma\} \in \prod\nolimits_{\gamma\in\Gamma} \mathcal{M}(X, \mathcal{X})$$

in the Cartesian product of copies of $\mathcal{M}(X, \mathcal{X})$, indexed by an arbitrary index set Γ. A *random function* is a map

$$\{f_\gamma\}\colon X \to \mathcal{F}$$

(whose codomain $\mathcal{F}$ is a function space) assigning to each x in X a real-valued function $\{f_\gamma(x)\}$ in the real function space $\mathcal{F}$. That is, assigning to each $x \in X$ a function from Γ to $\mathbb{R}$ in $\mathcal{F}$:

$$f_{(\ )}(x) \colon \Gamma \to \mathbb{R} \quad \in \quad \mathcal{F}.$$

The notions in the previous items, all of them, are particular cases of (real) random functions. In particular, for the last item, it is possible to have an uncountable index set Γ. For instance, if $\Gamma = \mathbb{R}$ (or $\Gamma = [0,1]$), then the function space $\mathcal{F}$ may be the Banach spaces $\mathcal{F} = L^p(\Gamma, \mathcal{F}, \eta)$ or $\mathcal{F} = L^\infty(\Gamma, \mathcal{F}, \eta)$, with $\mathcal{F}$ standing for $\Re$ (or $\Re \cap \wp([0,1])$) and $\eta \colon \mathcal{F} \to \mathbb{R}$ being a measure on it. For another example, still with $\Gamma = \mathbb{R}$ (or $\Gamma = [0,1]$), the function space $\mathcal{F}$ may be $\mathcal{F} = (C(\Gamma), \| \ \|_p)$ or $\mathcal{F} = (C(\Gamma), \| \ \|_\infty)$, the Banach spaces of real-valued continuous functions equipped with the p-norm or with the sup-norm. Recall again that by fixing $\gamma \in \Gamma$ we get the random variable $f_\gamma \in \mathcal{M}(X, \mathcal{X})$, and by fixing a pair $(\gamma, x) \in \Gamma \times X$ we get $f_\gamma(x) \in \mathbb{R}$.

(5) STOCHASTIC PROCESS. A *stochastic process* is a family $\{f_t\}_{t \in T}$ of random variables. In other words, it is a random function as defined in (4) above:

$$\{f_t\} \colon X \to \mathcal{F},$$

assigning to each x in X a real-valued function $\{f_t(x)\}$ in the function space $\mathcal{F}$. That is, assigning to each $x \in X$ the function (from T to $\mathbb{R}$ in $\mathcal{F}$)

$$f_{(\ )}(x) \colon T \to \mathbb{R} \quad \in \quad \mathcal{F}.$$

The special notation for the index set T is deliberate to suggest "time" (a term we will not dare to define). The codomain $\mathcal{F}$ is a space of real-valued functions on T. If T is any class of intervals of the real line as in Section 8.3, in particular, if $T = [0, \infty)$, then the stochastic process is referred to as a *continuous process*, and $\mathcal{F}$ may be any of the function spaces considered in item (4), among others. If T is an interval from $\mathbb{Z}$, for instance, if $T = \mathbb{N}$, then it is referred to as a *discrete process*, and $\mathcal{F}$ may be any of the sequence spaces considered in item (3), among others. As before, all random variables f_t are real-valued and defined in terms of a common probability space $(X, \mathcal{X}, \wp)$, all acting on the same set X, all being $\mathcal{X}$-measurable. For each $t \in T$ we get the real-valued random variable $f_t \in \mathcal{M}(X, \mathcal{X})$, and for each pair $(t, x) \in T \times X$ we get the real number $f_t(x) \in \mathbb{R}$.

Remark A.1 (Function of a random variable): A different notion, not related to the above items, is that of a *function of a random variable* $f \colon X \to \mathbb{R}$, meaning a composition $\phi \circ f$ of a real-valued function $\phi \colon \mathbb{R} \to \mathbb{R}$ on $\mathbb{R}$ with a real random variable f on X, also written as

$$\phi(f) \colon X \to \mathbb{R},$$

assigning to each $x \in X$ a number $\phi(f(x))$ in $\mathbb{R}$, which is again a random variable whenever ϕ is measurable; that is, if it is $\Re$-measurable.

A.3 Distributions, Characteristic, and Density Functions

Most probability properties are attributes of a random variable described in terms of distributions, either probability distributions or distribution functions. Attached to these, their transformations, such as density functions and characteristic functions, also play a central role in probability theory. Take a probability space $(X, \mathcal{X}, \wp)$ and fix a random variable $f\colon X \to \mathbb{R}$.

(1) PROBABILITY DISTRIBUTION. The *probability distribution*, or *distribution law*, of a random variable f is the probability measure $\mu_f\colon \Re \to \mathbb{R}$ induced by the probability measure $\wp$ via f (cf. Problem 2.20) given by

$$\mu_f(B) = \wp\big(f^{-1}(B)\big) = \wp\big(\{x \in X\colon\ f(x) \in B\}\big) \quad \text{for every} \quad B \in \Re.$$

(2) DISTRIBUTION FUNCTION. The *distribution function* of a random variable f is the function $F\colon \mathbb{R} \to [0,1]$ defined in terms of the probability distribution μ_f as follows (cf. Problems 2.20 and 2.21). For every $\alpha < \beta$ in $\mathbb{R}$,

$$F(\alpha) = \mu_f\big((-\infty, \alpha]\big) = \wp\big(\{x \in X\colon\ f(x) \le \alpha\}\big),$$

$$F(\beta) - F(\alpha) = \mu_f\big((\alpha, \beta]\big) = \wp\big(\{x \in X\colon\ \alpha < f(x) \le \beta\}\big).$$

(3) CHARACTERISTIC FUNCTION. The *characteristic function* $\Phi\colon \mathbb{R} \to \mathbb{C}$, either of a probability distribution μ_f or of a distribution function F of a random variable f, is the Fourier transform of μ_f, also called the Fourier–Stieltjes transform of F, described as follows. For every $\xi \in \mathbb{R}$,

$$\Phi(\xi) \,=\, \mathrm{Exp}(e^{i\xi f}) \,=\, \int_X e^{i\xi f}\, d\wp \,=\, \int_{\mathbb{R}} e^{i\xi\alpha}\, d\mu_f(\alpha) \,=\, \int_{-\infty}^{+\infty} e^{i\xi\alpha}\, dF(\alpha).$$

The first two identities are notation and definition. The third identity is a consequence of the linearity of the integral and the change of variables rule of Problem 4.19, $\int_X \phi \circ f\, d\wp = \int_{\mathbb{R}} \phi\, d\mu_f$, where $\phi(\ ) = e^{i\xi(\ )}\colon \mathbb{R} \to \mathbb{C}$ is written as $\phi(\ ) = \phi_1(\ ) + i\,\phi_2(\ ) = \cos(\xi(\ )) + i\,\sin(\xi(\ ))$, with $\phi_1, \phi_2\colon \mathbb{R} \to \mathbb{R}$ being integrable with respect to μ_f (because $|\phi_1| \le 1$, $|\phi_2| \le 1$ and $\mu_f(\mathbb{R}) = 1$). The fourth (last) identity holds because the probability distribution μ_f is the Borel–Stieltjes measure λ_F generated by the distribution function F, as we saw in Problem 2.20(d); that is, $\mu_f = \lambda_F$. In such a Borel–Stieltjes context, λ_F is identified with F, and it is usual to write dF for $d\lambda_F$. Hence, dF for $d\mu_f$.

(4) DENSITY FUNCTION. The *density function* $p\colon \mathbb{R} \to \mathbb{R}$ of a random variable f, if it exists, is a nonnegative $\Re$-measurable function ($p \in \mathcal{M}(\mathbb{R}, \Re)^+$) defined in Problem 3.15 in terms of the probability distribution μ_f (along the lines of Problem 3.13) as the nonnegative function p for which

$$\int_B d\mu_f = \int_B p\, d\lambda\,, \quad \text{equivalently,} \quad \mu_f(B) = \int_B d\mu_f = \int_B p\, d\lambda = \int_B p(\alpha)\, d\alpha,$$

for every $B \in \Re$, where $\lambda\colon \Re \to \overline{\mathbb{R}}$ is the Lebesgue measure on $\Re$ (as usual, we sometimes write $d\alpha$ for $d\lambda(\alpha)$ or $d\lambda$). The notation discussed in Problem 3.13 applies, so it is also usual to write $p = \frac{d\mu_f}{d\lambda}$ or $d\mu_f = p\, d\lambda$. If p exists, then for every $\beta \in \mathbb{R}$ it is related to the distribution function F by the integral

$$F(\beta) \,=\, \int_{-\infty}^{\beta} p(\alpha)\, d\alpha \quad \text{ so that } \quad \int_{\mathbb{R}} p(\alpha)\, d\alpha \,=\, 1.$$

Conversely, as we saw in Problem 3.15(b),

> if the distribution function F is absolutely continuous, then the density function p exists and is the derivative of F: $p = \frac{dF}{d\alpha}$.

(5) CHARACTERISTIC FUNCTION AGAIN. The characteristic function Φ is the Fourier transform of the density function p if it exists. In this case, $\mu_f(B) = \int_B p\, d\lambda$, and so $\int g\, d\mu_f = \int gp\, d\lambda$ for $g \in \mathcal{M}(\mathbb{R}, \Re)^+$ by Problem 3.13, which is enough to get (by the same argument used in (3) to get $\Phi(\xi)$ for $\xi \in \mathbb{R}$),

$$\Phi(\xi) \,=\, \int_{\mathbb{R}} e^{i\xi\alpha}\, p(\alpha)\, d\lambda(\alpha) \,=\, \int_{-\infty}^{+\infty} e^{i\xi\alpha}\, p(\alpha)\, d\alpha.$$

A.4 Absolute Continuities and Normal Distribution

(1) ABSOLUTE CONTINUITIES. Another circumstance where the density function p exists and is unique is when the probability distribution μ_f of a random variable f is absolutely continuous with respect to Lebesgue measure λ (i.e., if $\mu_f \ll \lambda$). Since measures in this appendix are probability measures, they are all finite, and so the Radon–Nikodym Theorem ensures that

> if μ_f is absolutely continuous with respect to Lebesgue measure λ, then there is a unique $p \in \mathcal{M}(\mathbb{R}, \Re)^+$ such that $\mu_f(B) = \int_B p\, d\lambda$ for $B \in \Re$, so that the density function of f exists and is precisely this p.

Actually, these two sufficient conditions for the existence of a density function p for a given random variable f are the same:

> *the probability distribution μ_f of a random variable f is absolutely continuous with respect to Lebesgue measure λ if and only if the distribution function F of f is absolutely continuous.*

Indeed, as we saw above, $\mu_f \ll \lambda$ implies $\mu_f(B) = \int_B p\, d\lambda$ (cf. Theorem 7.8, the Radon–Nikodym Theorem), so that $\int_{\mathbb{R}} p\, d\lambda < \infty$ and $\mu_f(\alpha, \beta) = F(\beta) - F(\alpha) = \int_{[\alpha,\beta]} p\, d\lambda$ (by Problem 3.15(a)), so that F is absolutely continuous (by the definition of absolute continuity in Problem 3.15). Conversely, if F is absolutely continuous (cf. Problem 3.15(b)), then there exists $p = \frac{dF}{d\alpha}$ so that

p is continuous and $\int_{\mathbb{R}} p(\alpha)\,d\alpha = 1$, and hence p is bounded. Thus, for an arbitrary $B \in \Re$, $\mu_f(B) = \int_B p\,d\lambda \le \sup_{\alpha\in B} p(\alpha) \int_B d\lambda \le \sup_{\alpha\in\mathbb{R}} p(\alpha)\,\lambda(B)$ (cf. Problem 3.15), so that $\lambda(B) = 0 \Longrightarrow \mu_f(B) = 0$, which means $\mu_f \ll \lambda$.

(2) NORMAL DISTRIBUTION. An important example of a probability distribution for a random variable that is absolutely continuous with respect to Lebesgue measure, thus ensuring the existence of a density function, is the *normal distribution*. A way to construct the normal probability distribution for a random variable (for a random variable that yields the normal probability distribution) goes as follows. Let m and σ be real constants. These will be the parameters of the normal probability distribution. The next well-known integral (with a lot of history behind it) is a suitable starting point:

$$\int_{-\infty}^{+\infty} e^{-\xi^2} d\xi = \sqrt{\pi} \quad \text{so that} \quad \tfrac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{+\infty} e^{-(\alpha-m)^2/2\sigma^2} d\alpha = 1$$

by a change of variable ($\xi = (\alpha - m)/\sqrt{2}\,\sigma$, for $\sigma > 0$). So consider the nonnegative (clearly continuous) function $p\colon \mathbb{R} \to \mathbb{R}$ given, for every $\alpha \in \mathbb{R}$, by

$$p(\alpha) = \tfrac{1}{\sqrt{2\pi}\,\sigma} e^{-(\alpha-m)^2/2\sigma^2} \quad \text{so that} \quad \int_{-\infty}^{+\infty} p(\alpha)\,d\alpha = 1.$$

Take a set function $\mu\colon \Re \to \mathbb{R}$ given by (where λ means Lebesgue measure)

$$\mu(B) = \int_B p\,d\lambda = \tfrac{1}{\sqrt{2\pi}\,\sigma} \int_B e^{-(\alpha-m)^2/2\sigma^2} d\lambda(\alpha)$$

for every $B \in \Re$. First, recall from Proposition 3.5(c) that this μ on $\Re$ is a measure, a probability measure, actually, since $\mu(\mathbb{R}) = \int_{\mathbb{R}} p\,d\lambda(\alpha) = 1$. Next, define a function $F\colon \mathbb{R} \to \mathbb{R}$ as follows: for every $\beta \in \mathbb{R}$,

$$F(\beta) = \mu\big((-\infty, \beta]\big) = \tfrac{1}{\sqrt{2\pi}\,\sigma} \int_{-\infty}^{\beta} e^{-(\alpha-m)^2/2\sigma^2} d\alpha.$$

(No elementary primitive describing F, I'm afraid, just the usual description in terms of the so-called erf (error function) that essentially means the above integral.) It is readily verified that the range of F is $[0, 1]$, F is nondecreasing, is right-continuous, vanishes at $-\infty$, and approaches 1 at ∞. Thus

(a) F is a distribution function of some random variable f with respect to some probability space $(X, \mathcal{X}, \wp)$, according to Problem 2.21. Hence,

(b) the probability measure μ on $\Re$ such that $F(\beta) = \mu\big((-\infty, \beta]\big)$ for every $\beta \in \mathbb{R}$ is the probability distribution μ_f of the same random variable f by the uniqueness in Problem 2.20(d). Therefore, $\mu = \mu_f$ of such an f, and

(c) the function p for which $\mu_f(B) = \int_B p\,d\lambda$ is the density function of f, as defined in Problem 3.15.

This probability measure μ_f on $\Re$ is the *normal probability distribution* of the random variable f, and f is referred to as being *normally distributed*. Accordingly, F is called the *normal distribution function* of f, and p is the *normal density function* of f, and independently of any probabilistic or measure-theoretical context, the function p itself is referred to as one of the forms of a *Gaussian function* (the simplest form of a Gaussian function is $e^{-\xi^2}$ for $\xi \in \mathbb{R}$). There are two ways to verify that

> the normal probability distribution μ_f of a random variable f is absolutely continuous with respect to Lebesgue measure λ.

Directly: since $\mu_f(B) = \int_B p\, d\lambda$, and p is bounded (because it is defined everywhere in $\mathbb{R}$, is positive, is continuous, and is integrable), it follows that $\mu_f(B) = \int_B p\, d\lambda \leq \sup_{\alpha\in B} p(\alpha) \int_B d\lambda \leq \sup_{\alpha\in\mathbb{R}} p(\alpha)\, \lambda(B)$, and therefore $\lambda(B) = 0 \Longrightarrow \mu_f(B) = 0$, which means $\mu_f \ll \lambda$ by Definition 7.6.

Indirectly: since, as defined above, the function F is such that

$$F(\beta) - F(\alpha) = \int_{[\alpha,\beta]} p(\alpha)\, d\alpha \quad \text{with} \quad \int_{[\alpha,\beta]} p(\alpha)\, d\alpha \leq 1 < \infty$$

for every closed interval $[\alpha, \beta]$, it follows by the definition of absolutely continuous functions in Problem 3.15 that F is an absolutely continuous function, and so, as we proved in item (1) above, $\mu_f \ll \lambda$.

Still by the definition of absolutely continuous functions in Problem 3.15, the absolute continuity of the continuous function F ensures that

$$p = F' = \frac{dF}{d\alpha}.$$

The parameters m and σ in the above distribution and density function are:

$$m = \mathrm{Exp}(f) = \int_X f\, d\wp = \int_{\mathbb{R}} \alpha\, \mu_f(\alpha) = \int_{-\infty}^{+\infty} \alpha\, p(\alpha)\, d\alpha,$$

$$\sigma^2 = \mathrm{Var}(f) = \mathrm{Exp}[(f-m)^2] = \mathrm{Exp}(f^2) - m^2 = \int_{-\infty}^{+\infty} \alpha^2 p(\alpha)\, d\alpha - m^2,$$

as given in Problems 4.20 and 5.20 (after some omitted calculations).

A.5 Joint Distributions and Density Functions

(1) JOINT PROBABILITY DISTRIBUTION AND INDEPENDENCE. The *joint probability distribution* of a pair of random variables $f\colon X \to \mathbb{R}$ and $g\colon X \to \mathbb{R}$ is the unique probability measure $\mu_{f,g}\colon \Re\times\Re \to \mathbb{R}$ such that

$$\mu_{f,g}(A\times B) = \wp\big((f,g)^{-1}(A\times B)\big)$$

for every rectangle $A\times B \in \Re\times\Re$, where the map $(f,g)\colon X \to \mathbb{R}\times\mathbb{R}$ assigns to each $x \in X$ the vector $(f(x), g(x)) \in \mathbb{R}\times\mathbb{R}$, as in Problem 9.23(b). The random variables f and g are *independent* if and only if (cf. Problem 9.24(e))

$$\mu_{f,g} = \mu_f \times \mu_g,$$

where μ_f and μ_g are the probability distributions of f and g, respectively.

(2) JOINT DISTRIBUTION FUNCTION AND INDEPENDENCE. The *joint distribution function* of a pair of random variables f and g is $F_{f,g}\colon \mathbb{R}\times\mathbb{R} \to [0,1]$, a function defined in terms of the joint probability distribution $\mu_{f,g}$ by

$$F_{f,g}(\gamma,\delta) = \mu_{f,g}\big((-\infty,\,\gamma]\times(-\infty,\,\delta]\big)$$

for every $(\gamma,\delta) \in \mathbb{R}\times\mathbb{R}$. Recall that the random variables f and g are *independent* if (by definition)

$$F_{f,g}(\gamma,\delta) = F_f(\gamma)\, F_g(\delta)$$

for every $(\gamma,\delta) \in \mathbb{R}\times\mathbb{R}$, where $F_f\colon \mathbb{R} \to [0,1]$ and $F_g\colon \mathbb{R} \to [0,1]$ are the distribution functions of f and g, respectively (cf. Problems 9.23 and 9.24).

(3) JOINT DENSITY FUNCTION: INDEPENDENCE AND UNCORRELATEDNESS. The *joint density function* of a pair of random variables f and g, if it exists, is the nonnegative function $p_{f,g}\colon \mathbb{R}\times\mathbb{R} \to \mathbb{R}$ also defined in terms of the joint probability distribution $\mu_{f,g}$ as follows. For every measurable set $W \in \Re\times\Re$ (in particular, for every measurable rectangle $A\times B \in \Re\times\Re$),

$$\mu_{f,g}(W) = \int_W p_{f,g}\, d(\lambda\times\lambda)\,; \quad \text{in particular} \quad \mu_{f,g}(A\times B) = \int_{A\times B} p_{f,g}\, d(\lambda\times\lambda),$$

where $\lambda\times\lambda$ is the planar Lebesgue measure on $\Re\times\Re$ (as in Problem 9.10), and $d(\lambda\times\lambda)$ is also denoted by $d\lambda\, d\lambda$ or $d\lambda(\alpha)\, d\lambda(\beta)$, or simply by $d\alpha\, d\beta$. Hence, if $p_{f,g}$ exists, then the joint distribution $F_{f,g}$ is such that

$$F_{f,g}(\gamma,\delta) = \mu_{f,g}\big((-\infty,\,\gamma]\times(-\infty,\,\delta]\big) = \int_{(-\infty,\gamma]\times(-\infty,\delta]} p_{f,g}(\alpha,\beta)\, d\alpha\, d\beta$$

for all $(\delta,\gamma) \in \mathbb{R}\times\mathbb{R}$. If the density functions p_f, p_g of f and g also exist, then f and g are independent if (and only if) $F_{f,g}(\gamma,\delta) = F_f(\gamma)\, F_g(\delta)$, which means

$$\int_{(-\infty,\gamma]\times(-\infty,\delta]} p_{f,g}(\alpha,\beta)\, d\alpha\, d\beta = \left(\int_{-\infty}^{\gamma} p_f(\alpha)\, d\alpha\right)\left(\int_{-\infty}^{\delta} p_g(\beta)\, d\beta\right),$$

for every $\gamma,\delta \in \mathbb{R}$. More is true. As f and g are independent if and only if $\mu_{f,g} = \mu_f\times\mu_g$, we get $\mu_{f,g}(A\times B) = \mu_f(A)\,\mu_g(B)$ that, in fact, is equivalent to independence (see Problem 9.24(e)) and its hint), which means

$$\int_{A\times B} p_{f,g}(\alpha,\beta)\,d\alpha\,d\beta = \left(\int_A p_f(\alpha)\,d\alpha\right)\left(\int_B p_g(\beta)\,d\beta\right)$$

for every $A\times B \subseteq \Re\times\Re$. Since, when they exist, $p_{f,g}$, p_f and p_g are nonnegative (actually, they are integrable and bounded as well), Problems 9.20 and 9.22 ensure that the above integral equation holds; equivalently, f and g are *independent* if and only if

$$p_{f,g} = p_f\, p_g.$$

Recall from Problem 9.26: two random variables f and g are *uncorrelated* if

$$\mathrm{Exp}(fg) = \mathrm{Exp}(f)\,\mathrm{Exp}(g),$$

where $\mathrm{Exp}(f) = \int_X f\,d\wp = \int_{\mathbb{R}} \alpha\,d\mu_f(\alpha)$, $\mathrm{Exp}(g) = \int_X g\,d\wp = \int_{\mathbb{R}} \beta\,d\mu_g(\beta)$, and $\mathrm{Exp}(fg) = \int_X fg\,d\wp = \int_{\mathbb{R}\times\mathbb{R}} \alpha\,\beta\,d\mu_{f,g}(\alpha,\beta)$. So, if the identity function is integrable with respect to both probability distributions, and if the density functions exist, then by the definitions of p_f, p_g, and $p_{f,g}$, the random variables f and g are *uncorrelated* if and only if (cf. Problem 4.20)

$$\int_{\mathbb{R}\times\mathbb{R}} a\,\beta\,p_{f,g}(\alpha,\beta)\,d\alpha\,d\beta = \left(\int_{\mathbb{R}} \alpha\,p_f(\alpha)\,d\alpha\right)\left(\int_{\mathbb{R}} \beta\,p_g(\beta)\,d\beta\right).$$

A.6 Conditioning

In this context, the term "conditioning" essentially refers to "restricting to a sub-σ-algebra". Starting with two versions of conditional probability, there follow the notions of conditional probability distribution and conditional expectation of a random variable, which are themselves random variables. All this evolves around a sub-σ-algebra $\mathcal{J}$ of a σ-algebra $\mathcal{X}$ that contains X.

(1) SUB-σ-ALGEBRAS. A σ-algebra $\mathcal{X}$ is a σ-algebra of subsets of a set X, by definition. Thus a σ-algebra $\mathcal{X}$ has a maximum (in the inclusion ordering) element $X = \max \mathcal{X}$ (i.e., X is an upper bound of $\mathcal{X}$ and belongs to $\mathcal{X}$).

A sub-σ-algebra $\mathcal{J}$ of an arbitrary σ-algebra $\mathcal{X}$ is a subcollection of $\mathcal{X}$ (i.e., $\mathcal{J}\subseteq\mathcal{X}$) satisfying the axioms (a), (b), and $(\widehat{\mathrm{c}})$ of Definition 1.1.The whole set (i.e., the underlying set) of $\mathcal{J}$ exists and is $J = \max \mathcal{J} \in \mathcal{J}$. So it may not be necessary to specify the underlying sets J and X of $\mathcal{J}$ and $\mathcal{X}$ in some cases, since they are uniquely determined as the maximum of all sets in $\mathcal{J}$ and in $\mathcal{X}$:

$$J = \max \mathcal{J} \subseteq \max \mathcal{X} = X.$$

There are two special kinds of sub-σ-algebras $\mathcal{J}$ of a σ-algebra $\mathcal{X}$. One is that made up of all $\mathcal{X}$-measurable subsets of the set $J = \max \mathcal{J}$ in $\mathcal{X}$,

which is the intersection of two σ-algebras of subsets of the set $X = \max \mathcal{X}$, namely, the power set $\wp(J)$ of J and $\mathcal{X}$. This is the particular case discussed in Section 1.1, that is, $\mathcal{J} = \wp(J) \cap \mathcal{X}$. However, an arbitrary sub-σ-algebra of a σ-algebra $\mathcal{X}$ may not be made up uniquely of $\mathcal{X}$-measurable subsets of a proper subset of X. Thus, the other special particular case is that where the underlying set X of $\mathcal{X}$ is contained in $\mathcal{J}$, and $\mathcal{J}$ is properly included in $\mathcal{X}$.

Example A.1: If $\{A, B\}$ is a partition of X consisting of nonempty sets, then $\mathcal{J} = \{\varnothing, X\}$ is a sub-σ-algebra of the σ-algebra $\mathcal{X} = \{\varnothing, A, B, X\}$ for which not all $\mathcal{X}$-measurable subsets of $J = \max \mathcal{J} = X$ belong to $\mathcal{J}$.

Example A.2: If $\mathcal{X}$ is a σ-algebra containing more than four elements (perhaps with uncountable cardinality), and $\{A, B\}$ is a nontrivial $\mathcal{X}$-measurable partition of $X = \max \mathcal{X}$, then the sub-σ-algebra $\mathcal{J} = \{\varnothing, A, B, X\}$ of $\mathcal{X}$ (properly included in $\mathcal{X}$) is such that $J = \max \mathcal{J} = X$.

Example A.3: Take the probability space $(X, \mathcal{X}, \lambda)$, where $X = [0, 1]$, $\mathcal{X} = \Re \cap \wp([0, 1])$, and λ is the Lebesgue measure on $\mathcal{X}$. Let $\mathcal{J}$ be the collection of all sets in $\mathcal{X}$ that are either countable or are the complement of a countable set (see Problem 2.6). This $\mathcal{J}$ is a sub-σ-algebra of $\mathcal{X}$ including X, so $J = X$.

Sub-σ-algebras $\mathcal{J}$ of $\mathcal{X}$ that include X will take an important part in the forthcoming items. Since all sets in $\mathcal{J}$ are sets in $\mathcal{X}$, it follows at once that

every $\mathcal{J}$-measurable set is $\mathcal{X}$-measurable, and so if $X \in \mathcal{J}$,
every $\mathcal{J}$-measurable function $X \mapsto \mathbb{R}$ is $\mathcal{X}$-measurable.

(2) CONDITIONAL PROBABILITY. Consider a probability space $(X, \mathcal{X}, \wp)$.

(i) Fix a set $Q \in \mathcal{X}$ such that $\wp(Q) > 0$. Define a function $\wp(\ \ |Q) \colon \mathcal{X} \to \mathbb{R}$ by

$$\wp(E|Q) = \frac{\wp(E \cap Q)}{\wp(Q)} \quad \text{for every} \quad E \in \mathcal{X}.$$

For each $E \in \mathcal{X}$, the number $\wp(E|Q)$ is the *conditional probability of a set E with respect to a set Q*. The set function $\wp(\ \ |Q) : \mathcal{X} \to \mathbb{R}$ is readily verified to be a probability measure, as we will check below. Particular cases (recall that E and Q are independent if $\wp(E \cap Q) = \wp(E)\,\wp(Q)$):

$$\wp(E|Q) = \wp(E) \quad \text{if and only if} \quad E \text{ and } Q \text{ are independent,}$$

$$\wp(E|Q) = \wp(E)\,/\,\wp(Q) \quad \text{if} \quad E \subseteq Q \quad (\text{and so} \quad \wp(\varnothing|Q) = 0),$$

$$\text{and} \quad \wp(E|Q) = 1 \quad \text{whenever} \quad Q \subseteq E.$$

Now take the measure $\wp_Q(\ \ ) \colon \mathcal{X} \to \mathbb{R}$ defined in Problem 2.11 by $\wp_Q(E) = \wp(E \cap Q)$ for every $E \in \mathcal{X}$, and consider the multiple of it given by $\wp(\ \ |Q) =$

$\frac{\wp_Q(\ )}{\wp(Q)}$, which is a probability measure (for $\wp(X\,|\,Q)=1$). Let $\mathcal{Q}=\wp(Q)\cap\mathcal{X}$ be the sub-σ-algebra of the σ-algebra $\mathcal{X}$ generated by the $\mathcal{X}$-measurable set $Q=\max\mathcal{Q}$. Also take the restriction $\wp|_{\mathcal{Q}}\colon\mathcal{Q}\to\mathbb{R}$ of the probability measure $\wp\colon\mathcal{X}\to\mathbb{R}$ to $\mathcal{Q}$ defined in Problem 2.11 as well. Thus

$$\wp_Q(E)=\wp(E\cap Q)=\wp(E\,|\,Q)\,\wp(Q)\quad\text{ for every }\quad E\in\mathcal{X},$$
$$\wp|_{\mathcal{Q}}(E)=\wp(E)\quad\text{ for every }\quad E\in\mathcal{Q}.$$

As $\wp_Q(X)=\wp(Q)=\wp|_{\mathcal{Q}}(Q)$, these are not probability measures if $\wp(Q)<1$, but their normalized versions are. (Normalized measures are obtained when they are multiplied by $\frac{1}{\wp(Q)}$.) These two measures will be required below.

(ii) Next we define a different notion. In what was said above, we have fixed the second element in $\wp(\ \ |Q)$; that is, we had fixed the set Q in $\mathcal{X}$. In what will be said below, we will do the opposite: the first element will be fixed. So fix a set $E\in\mathcal{X}$. Let $\mathcal{J}$ be an arbitrary sub-σ-algebra of the σ-algebra $\mathcal{X}$ that contains X, so that $J=\max\mathcal{J}=\max\mathcal{X}=X$. Consider the finite measure $\wp_E\colon\mathcal{J}\to\mathbb{R}$ discussed above (cf. Problem 2.11 again),

$$\wp_E(G)=\wp(E\cap G)\quad\text{ for every }\quad G\in\mathcal{J},$$

which is absolutely continuous with respect to the measure $\wp|_{\mathcal{J}}\colon\mathcal{J}\to\mathbb{R}$, the restriction of the probability measure $\wp\colon\mathcal{X}\to\mathbb{R}$ to $\mathcal{J}$ (because, for $G\in\mathcal{J}$, $\wp|_{\mathcal{J}}(G)=0\iff\wp(G)=0\Longrightarrow\wp(E\cap G)=0$, since $E\cap G\subseteq G$). Thus the Radon–Nikodym Theorem (Theorem 7.8) ensures that there exists a nonnegative $\mathcal{J}$-measurable function $f_E\colon X\to\mathbb{R}$ (i.e., a random variable on X) that is integrable with respect to $\wp|_{\mathcal{J}}$ (and so with respect to $\wp$) such that

$$\wp(E\cap G)\;=\;\wp_E(G)\;=\int_G f_E\,d\wp\quad\text{ for every }\quad G\in\mathcal{J},$$

where the second identity holds for f_E $\wp$-a.e. (i.e., it holds for the equivalence class of f_E). Such a nonnegative random variable $f_E\colon X\to\mathbb{R}$ is denoted by

$$f_E=\wp(E\,|\,\mathcal{J}),\tag{A.6.2.0}$$

and is referred to as the *conditional probability of a set $E\in\mathcal{X}$ given a sub-σ-algebra $\mathcal{J}$ of $\mathcal{X}$* (that contains X). Summing up:

Conditional Probability of a Set Given a Sub-σ-algebra. The conditional probability of E in $\mathcal{X}$ given $\mathcal{J}$ is the random variable $\wp(E\,|\,\mathcal{J})$ on X satisfying the following two conditions:

$$\wp(E\,|\,\mathcal{J})\text{ is }\mathcal{J}\text{-measurable and }\wp|_{\mathcal{J}}\text{-integrable,}\tag{A.6.2.I}$$

$$\int_G\wp(E\,|\,\mathcal{J})\,d\wp\;=\;\wp(E\cap G)\quad\wp\text{-a.e. for every}\quad G\in\mathcal{J}.\tag{A.6.2.II}$$

Properties of the conditional probability $\wp(E\,|\,\mathcal{J})$ of a set $E \in \mathcal{X}$ given a sub-σ-algebra $\mathcal{J}$ of $\mathcal{X}$ that will be required ahead are presented below. All of them hold $\wp$-almost everywhere (i.e., $\wp$-a.e.). By Proposition 3.7, Problem 4.5, and (A.6.2.0), the random variable $f_E = \wp(E\,|\,\mathcal{J})\colon X \to \mathbb{R}$ is such that

(a) $0 \le \wp(E\,|\,\mathcal{J}) \le 1$, $\quad \wp(\varnothing\,|\,\mathcal{J}) = 0$, $\quad \wp(X\,|\,\mathcal{J}) = 1 \quad$ (since $X \in \mathcal{J}$),

(b) $\wp(E) = 0 \;$ implies $\; \wp(E\,|\,\mathcal{J}) = 0$, $\quad \wp(E) = 1 \;$ implies $\; \wp(E\,|\,\mathcal{J}) = 1$.

Moreover, by (A.6.2.0) and (A.6.2.II), for sets $D, E \in \mathcal{X}$ such that $D \subseteq E$, we get $\int_G f_{E\setminus D}\, d\wp \le \int_G f_E\, d\wp - \int_G f_D\, d\wp$, and hence (Problem 4.5 again) $f_{E\setminus D} \le f_E - f_D$ $\wp$-a.e. So if D and E are $\mathcal{X}$-measurable subsets of X, then

(c) $D \subseteq E \;$ implies $\; \wp(E\setminus D\,|\,\mathcal{J}) \le \wp(E\,|\,\mathcal{J}) - \wp(D\,|\,\mathcal{J})$,

(d) $D \subseteq E \;$ implies $\; \wp(D\,|\,\mathcal{J}) \le \wp(E\,|\,\mathcal{J})$.

For simplicity, use the term "increasing" for "nondecreasing" and "decreasing" for "nonincreasing", as we have been doing sometimes since Section 1.3. Let $\{E_n\}$ be a sequence of sets from $\mathcal{X}$. If $\{E_n\}$ is increasing or decreasing, then so is the integral $\int_G f_{E_n} d\wp$ in (A.6.2.II) and, by Proposition 2.2(c,d), $\lim_n \wp(E_n) = \wp\big(\bigcup_n E_n\big)$ for the increasing case and $\lim_n \wp(E_n) = \wp\big(\bigcap_n E_n\big)$ for the decreasing case. So, by (A.6.2.II), $\int_G \lim_n f_{E_n} d\wp = \int_G f_{\cup E_n} d\wp$, convergence from below for the increasing case, and $\int_G \lim_n f_{E_n} d\wp = \int_G f_{\cap E_n} d\wp$, convergence from above for the decreasing case. Thus, by calling Problem 4.5 again, we get $\lim_n f_{E_n} \nearrow f_{\cup E_n}$ (convergence from below $\wp$-a.e.) for the increasing case and $\lim_n f_{E_n} \searrow f_{\cap E_n}$ (convergence from above $\wp$-a.e.) for the decreasing case. Therefore, by using the notation (A.6.2.0) once again,

(e) $E_n \nearrow E \;$ implies $\; \wp(E_n\,|\,\mathcal{J}) \nearrow \wp(E\,|\,\mathcal{J})$,

(f) $E_n \searrow E \;$ implies $\; \wp(E_n\,|\,\mathcal{J}) \searrow \wp(E\,|\,\mathcal{J})$.

Notation: Set $E = \bigcup E_n$ and write $E_n \nearrow E$ if $\{E_n\}$ is increasing and, using the same notation, set $E = \bigcap E_n$ and write $E_n \searrow E$ if $\{E_n\}$ is decreasing.

(3) CONDITIONAL PROBABILITY DISTRIBUTION. Take a probability space $(X, \mathcal{X}, \wp)$ and let $\mathcal{J}$ be a sub-σ-algebra of $\mathcal{X}$ containing X. Take a random variable $f\colon X \to \mathbb{R}$. For each $\alpha \in \mathbb{R}$, set

$$E_\alpha = \{x \in X\colon\ f(x) \le \alpha\} = f^{-1}\big((-\infty, \alpha]\big).$$

Now take the ($\mathcal{J}$-measurable) random variable defined in (A.6.2.0):

$$f_{E_\alpha} = \wp(E_\alpha\,|\,\mathcal{J})\colon X \to \mathbb{R} \quad (\wp\text{-a.e}).$$

The arguments below use properties (a) to (f) and the notation (A.6.2.0) of item (2), and all assertions involving f_{E_α} hold $\wp$-a.e. for $x \in X$. As in item (2), we use increasing for nondecreasing and decreasing for nonincreasing.

(i) First, fix an arbitrary $x \in X$. Set

$$F(\alpha, x) = f_{E_\alpha}(x)$$

for every $\alpha \in \mathbb{R}$, and consider the function

$$F(\ \ , x)\colon \mathbb{R} \to \mathbb{R}.$$

To begin with, note that $E_\alpha \subseteq E_\beta$ whenever $\alpha \le \beta$ for $\alpha, \beta \in \mathbb{R}$. Thus, by property (d) and notation (A.6.2.0) of item (2) above, we get

$$\alpha \le \beta \quad \text{implies} \quad f_{E_\alpha} = \wp(E_\alpha \,|\, \mathcal{J}) \le f_{E_\beta} = \wp(E_\beta \,|\, \mathcal{J}),$$

$\wp$-a.e. for $x \in X$, which means

$$\alpha \le \beta \quad \text{implies} \quad F(\alpha, x) = f_{E_\alpha}(x) \le f_{E_\beta}(x) = F(\beta, x),$$

for $\alpha, \beta \in \mathbb{R}$, and therefore

(a) $F(\ \ , x)\colon \mathbb{R} \to \mathbb{R}$ is increasing.

Next, since $\{E_n\}$ is an increasing sequence and $\{E_{-n}\}$ is a decreasing sequence of $\mathcal{X}$-measurable sets (for n running over the positive integers $\mathbb{N}$), we get by Proposition 2.2(c,d) that $\lim_{-n\to-\infty} \wp(E_{-n}) = \wp\big(\bigcap_{-n} E_{-n}\big) = 0$ and $\lim_{n\to+\infty} \wp(E_n) = \wp\big(\bigcup_n E_n\big) = 1$ (as we have seen before when verifying properties (e) and (f) in item (2)). Thus, by properties (b,d,e,f) in item (2), it follows that $\lim_{-n\to-\infty} \wp(E_{-n} \,|\, \mathcal{J}) = 0$ and $\lim_{n\to+\infty} \wp(E_n \,|\, \mathcal{J}) = 1$. Hence $\lim_{-n\to-\infty} f_{E_{-n}} = 0$ and $\lim_{n\to+\infty} f_{E_n} = 1$, which implies

$$\lim_{-n\to-\infty} F(-n, x) = \lim_{-n\to-\infty} f_{E_{-n}}(x) = 0 \quad \text{and} \quad \lim_{n\to+\infty} F(n, x) = \lim_{n\to+\infty} f_{E_n}(x) = 1.$$

Recall from property (a) and (A.6.2.0) in (2) that $0 \le f_{E_\alpha} = \wp(E_\alpha \,|\, \mathcal{J}) \le 1$. So, as $F(\alpha, x) = f_{E_\alpha}(x)$, we get $0 \le F(\alpha, x) \le 1$ for all $\alpha \in \mathbb{R}$. Thus, since $F(\ \ , x)\colon \mathbb{R} \to \mathbb{R}$ is increasing by (a), the above limits imply, for $\alpha \in \mathbb{R}$, that

(b) $\lim_{\alpha\to-\infty} F(\alpha, x) = 0 \quad$ and $\quad \lim_{\alpha\to+\infty} F(\alpha, x) = 1$.

Finally, take an arbitrary $\alpha \in \mathbb{R}$ and suppose a sequence of real numbers $\{\alpha_n\}$ is such that $\alpha \le \alpha_n$ for all positive integers n and $\alpha_n \to \alpha$. With no loss of generality, let $\{\alpha_n\}$ be decreasing (we can always reorder it or take a subsequence to get a decreasing sequence). Recall that $\{E_{\alpha_n}\}$ is decreasing as well. Consider the following string of implications.

$$\begin{aligned} \alpha_n \searrow \alpha \implies & E_{\alpha_n} \searrow \textstyle\bigcap_n E_{\alpha_n} = E_\alpha \implies \wp(E_{\alpha_n}) \searrow \wp(E_\alpha) \\ \implies & \wp(E_{\alpha_n} \,|\, \mathcal{J}) \searrow \wp(E_\alpha \,|\, \mathcal{J}) \iff f_{E_{\alpha_n}} \searrow f_{E_\alpha} \\ \implies & f_{E_{\alpha_n}}(x) \searrow f_{E_\alpha}(x) \iff F(\alpha_n, x) \searrow F(\alpha, x), \end{aligned}$$

where the second implication comes from Proposition 2.2(d) and the third implication comes from property (f) in item (2) above; the other relations are trivial, notation, or definition. Therefore,

(c) $F(\ ,x)\colon \mathbb{R} \to \mathbb{R}$ is right-continuous.

According to (a), (b), and (c), it follows from Problem 2.21 that $F(\ ,x)$ is a distribution function of some random variable. Actually, by Problem 2.21(b),

> given a measurable space $(X, \mathcal{X})$ and an $\mathcal{X}$-measurable real-valued function $f\colon X \to \mathbb{R}$, there is a probability measure $\wp_x\colon \mathcal{X}_f \to \mathbb{R}$ on the sub-σ-algebra $\mathcal{X}_f$ of $\mathcal{X}$ induced by f, so that f is a random variable for which the function $F(\ ,x)\colon \mathbb{R} \to \mathbb{R}$ is a distribution of f. Hence,
>
> $$\wp_x(E_\alpha|\mathcal{X}_f) = \wp_x(E_\alpha) = F(\alpha, x) = \wp(E_\alpha|\,\mathcal{J})(x) \quad \text{for every} \quad \alpha \in \mathbb{R}.$$

Associated with f set, for such an $x \in X$,

$$\mu_f\big((-\infty, \alpha], x\big) = F(\alpha, x) = \wp_x\big(\{x \in X\colon f(x) \le \alpha\}\big) \quad \text{for every} \quad \alpha \in \mathbb{R},$$

so that $\mu_f\big((\alpha, \beta], x\big) = F(\beta, x) - F(\alpha, x) = \lambda_F\big((\alpha, \beta], x\big)$, with $\alpha \le \beta$ in $\mathbb{R}$, is the Borel–Stieltjes measure of Example 2D as in Problem 2.20(d). This leads to a probability measure on $\Re$ (by extending μ_f from intervals to $\Re$, via Theorem 8.4 with help from Problem 8.3). The outcome is:

$$\mu_f(\ ,x)\colon \Re \to \mathbb{R} \text{ is a probability measure for each } x \in X.$$

(ii) Now, fix an arbitrary left-open interval $(\alpha, \beta] \subseteq \mathbb{R}$ $(\alpha, \beta \in \mathbb{R}$ and $\alpha \le \beta)$. Consider the function $\mu_f\big((\alpha, \beta],\ \big)\colon X \to \mathbb{R}$ given by the above measure $\mu_f(\ ,x)\colon \Re \to \mathbb{R}$ so that, as we have seen before, for every $x \in X$ we get

$$\mu_f\big((\alpha, \beta], x\big) = F(\beta, x) - F(\alpha, x) = f_{E_\beta} - f_{E_\alpha}.$$

Then the function $\mu_f\big((\alpha, \beta],\ \big)\colon X \to \mathbb{R}$ is $\mathcal{J}$-measurable, since f_{E_α} and f_{E_β} are. This holds for every $\alpha, \beta \in \overline{\mathbb{R}}$ with $\alpha \le \beta$ (because the limit arguments in the verification of item (b) ensure the cases of $\pm\infty$). Thus, $\mathcal{J}$-measurability holds for the classes $\mathcal{C}_1$ to $\mathcal{C}_4$ of Section 8.3, and hence for the algebra $\mathfrak{S}$ of all finite unions of intervals in these classes (cf. Definition 8.7). Therefore, $\mu_f(B,\ )\colon X \to \mathbb{R}$ is $\mathcal{J}$-measurable for every set B in the algebra $\mathfrak{S}$. So

$$\mathfrak{S} \subseteq \mathcal{B},$$

where $\mathcal{B}$ is the class of all $B \subseteq \mathbb{R}$ such that $\mu_f(B,\ )$ is $\mathcal{J}$-measurable. That is,

$$B \in \mathcal{B} \quad \Longleftrightarrow \quad M_B(\alpha) = \{x \in X\colon \mu_f(B, x) \le \alpha\} \in \mathcal{J}$$

for every $\alpha \in \mathbb{R}$. (Note: if $\alpha < 0$, then $M_B(\alpha) = \varnothing \in \mathcal{J}$, and if $\alpha \ge 1$, then $M_B(\alpha) = X \in \mathcal{J}$, for all $B \subseteq \mathbb{R}$). Thus, for arbitrary $\alpha \in [0, 1)$,

$$B_n \nearrow \textstyle\bigcup_n B_n \Longrightarrow \mu_f(B_n, x) \nearrow \mu_f\big(\bigcup_n B_n, x\big) \Longrightarrow M_{B_n}(\alpha) \searrow \bigcap_n M_{B_n}(\alpha) \in \mathcal{J},$$

$$B_n \searrow \textstyle\bigcap_n B_n \Longrightarrow \mu_f(B_n, x) \searrow \mu_f\big(\bigcap_n B_n, x\big) \Longrightarrow M_{B_n}(\alpha) \nearrow \bigcup_n M_{B_n}(\alpha) \in \mathcal{J},$$

because $\mathcal{J}$ is a σ-algebra. (And the implications can be reversed up to sets of measure zero.) Then $\mathcal{B}$ is a monotone class (cf. Problem 1.15). Since $\Im$ is an algebra, Problem 1.19 ensures that

$$\Im \subseteq \mathcal{B} \quad \Longrightarrow \quad \mathcal{R}_{\Im} \subseteq \mathcal{B},$$

where $\mathcal{R}_{\Im}$ stands for the σ-algebra generated by the algebra $\Im$, that is, the smallest σ-algebra including $\Im$. But $\Re$ is the smallest σ-algebra including the algebra $\Im$ (cf. Property P_1 in Section 8.3). Hence

$$\Re \subseteq \mathcal{B}.$$

Therefore,

$$\mu_f(B,\ \,) \colon X \to \mathbb{R} \text{ is a } \mathcal{J}\text{-measurable function for every } B \in \Re.$$

Moreover, since, for every $\alpha \in \mathbb{R}$,

$$\mu_f\big((-\infty, \alpha],\ \,\big) = F(\alpha,\ \,) = f_{E_\alpha} = \wp(E_\alpha \,|\, \mathcal{J}),$$

where $\wp(E_\alpha \,|\, \mathcal{J})$ is the random variable in (A.6.2.0) of item (2), and so

$$\int_G \mu_f\big((-\infty, \alpha],\ \,\big)\, d\wp = \wp(E_\alpha \cap G) = \wp\big(f^{-1}\big((-\infty, \alpha]\big) \cap G\big)$$

$\wp$-a.e. for every $G \in \mathcal{J}$. The same argument used above (based on Section 8.2) allows us to extend from intervals $(-\infty, \alpha] \in \Re$ to arbitrary $B \in \Re$:

$$\int_G \mu_f(B,\ \,)\, d\wp = \wp\big(f^{-1}(B) \cap G\big) \tag{A.6.3.0}$$

$\wp$-a.e. for every $G \in \mathcal{J}$ and $B \in \Re$, and so the function $\mu_f(B,\ \,)$ is an integrable random variable, thus satisfying (A.6.2.I) and (A.6.2.II). Then, $\wp$-a.e.,

$$\mu_f(B,\ \,) = \wp\big(f^{-1}(B) \,\big|\, \mathcal{J}\big).$$

That is, the $\mathcal{J}$-measurable function $\mu_f(B,\ \,) \colon X \to \mathbb{R}$ coincides $\wp$-a.e. with the random variable $\wp\big(f^{-1}(B) \,\big|\, \mathcal{J}\big) \colon X \to \mathbb{R}$, which is the conditional probability of the inverse image $f^{-1}(B)$ in $\mathcal{X}$ of B in $\Re$ under the real random variable f on X, given the sub-σ-algebra $\mathcal{J}$ of $\mathcal{X}$ (that contains X).

(iii) The function $\mu_f(\ ,\ \,) \colon \Re \times X \to \mathbb{R}$ is the *conditional probability measure of the random variable* $f \colon X \to \mathbb{R}$ *given the sub-*σ*-algebra* $\mathcal{J}$. Summing up:

Conditional Probability Distribution of a Random Variable Given a Sub-σ-algebra. The conditional probability distribution of f given $\mathcal{J}$ is the function $\mu_f(\ ,\ \,) \colon \Re \times X \to \mathbb{R}$ satisfying the following two conditions:

$$\mu_f(\ \,, x) \colon \Re \to \mathbb{R} \text{ is a probability measure for every } x \in X, \tag{A.6.3.I}$$

$$\mu_f(B,\ \,) = \wp\big(f^{-1}(B) \,\big|\, \mathcal{J}\big) \colon X \to \mathbb{R} \ \ \wp\text{-a.e for every } B \in \Re. \tag{A.6.3.II}$$

(4) CONDITIONAL EXPECTATION. Consider a probability space $(X, \mathcal{X}, \wp)$ and let $\mathcal{J}$ be a sub-σ-algebra of $\mathcal{X}$ containing X.

Note A.1: Take the linear space $\mathcal{L}(X, \mathcal{X}, \wp)$ of $\wp$-integrable functions on $\mathcal{X}$ (Lemma 4.5), which are necessarily $\mathcal{X}$-measurable. Let $\wp|_{\mathcal{J}} \colon \mathcal{J} \to \mathbb{R}$ be the restriction of $\wp \colon \mathcal{X} \to \mathbb{R}$ to $\mathcal{J} \subseteq \mathcal{X}$. If $g \colon X \to \mathbb{R}$ lies in $\mathcal{L}(X, \mathcal{X}, \wp)$ and is $\mathcal{J}$-measurable, then $g \in \mathcal{L}(X, \mathcal{J}, \wp|_{\mathcal{J}})$ and $\int_G g \, d\wp = \int_G g \, d\wp|_{\mathcal{J}}$ for every $G \in \mathcal{J}$. We will use this identification throughout this item (writing $\wp$ for $\wp|_{\mathcal{J}}$).

Note A.2: Let $f \colon X \to \mathbb{R}$ and $g \colon X \to \mathbb{R}$ be $\wp$-integrable functions. Thus, besides being $\mathcal{X}$-measurable, they lie in $\mathcal{L}(X, \mathcal{X}, \wp)$. So Problem 4.5(c) says that $\int_E f \, d\wp = \int_E g \, d\wp$ for every $E \in \mathcal{X}$ if and only if $f = g$ $\wp$-a.e. Now suppose, in addition, that g is $\mathcal{J}$-measurable. Consider the identity $\int_G g \, d\wp = \int_G g \, d\wp|_{\mathcal{J}}$ for every $G \in \mathcal{J}$ as discussed above. Note that $\int_G f \, d\wp = \int_G g \, d\wp$ for every $G \in \mathcal{J}$ does not imply $f = g$ $\wp$-a.e. (f does not need to be $\mathcal{J}$-measurable).

Note A.3: Let again $f \colon X \to \mathbb{R}$ and $g \colon X \to \mathbb{R}$ be in $\mathcal{L}(X, \mathcal{X}, \wp)$ (so they are $\mathcal{X}$-measurable) and let g be $\mathcal{J}$-measurable. In general, f is not supposed to be $\mathcal{J}$-measurable. However, if it is, then both f and g are $\mathcal{J}$-measurable and lie in $\mathcal{L}(X, \mathcal{J}, \wp)$ (because they lie in $\mathcal{L}(X, \mathcal{X}, \wp)$ and $\mathcal{J} \subseteq \mathcal{X}$), so that Problem 4.5(c) ensures, in this particular case, that $\int_G f \, d\wp = \int_G g \, d\wp$ for every $G \in \mathcal{J}$ if and only if $f = g$ $\wp$-a.e. for $\wp = \wp|_{\mathcal{J}} \colon \mathcal{J} \to \mathbb{R}$ (but not $f = g$ $\wp$-a.e. for $\wp \colon \mathcal{X} \to \mathbb{R}$; they might differ in a set $E \in \mathcal{X} \backslash \mathcal{J}$ for which $\wp(E) \neq 0$).

Let f be an integrable random variable with respect to the probability measure $\wp \colon \mathcal{X} \to \mathbb{R}$, so that $f \colon X \to \mathbb{R}$ is $\mathcal{X}$-measurable and $\int_X f \, d\wp < \infty$, that is, $f \in \mathcal{L}(X, \mathcal{X}, \wp)$. Suppose for a while that f is nonnegative, so that (cf. Proposition 3.5(c)) the set function $\eta \colon \mathcal{J} \to \mathbb{R}$ given by $\eta(G) = \int_G f \, d\wp$ for every $G \in \mathcal{J}$ is also a finite measure, now on $\mathcal{J}$, and absolutely continuous with respect to $\wp$. Thus the Radon–Nikodym Theorem (Theorem 7.8) ensures that there exists a $\mathcal{J}$-measurable function $g \colon X \to \mathbb{R}$ such that $\eta(G) = \int_G g \, d\wp$ for every $G \in \mathcal{J}$. Next, consider an arbitrary $\mathcal{X}$-measurable random variable $f \colon X \to \mathbb{R}$ in $f \in \mathcal{L}(X, \mathcal{X}, \wp)$ (not necessarily nonnegative) so that the above argument holds for the nonnegative functions f^+ and f^- in the decomposition $f = f^+ - f^-$. Thus, by linearity of the integral, the argument holds for f, so that for every integrable random variable $f \colon X \to \mathbb{R}$, there exists a $\mathcal{J}$-measurable integrable function $g \colon X \to \mathbb{R}$ for which

$$\int_G g \, d\wp = \int_G f \, d\wp \quad \text{for every} \quad G \in \mathcal{J}.$$

This ($\mathcal{J}$-measurable) random variable $g \colon X \to \mathbb{R}$ is denoted by

$$g = \mathrm{Exp}(f \,|\, \mathcal{J}) \tag{A.6.4.0}$$

and is referred to as the *conditional expected value of the* ($\mathcal{X}$-measurable) *random variable* $f \colon X \to \mathbb{R}$ *given the sub-σ-algebra* $\mathcal{J}$. Summing up:

Conditional Expectation of an Integrable Random Variable Given a Sub-σ-algebra. The conditional expectation of an integrable f given $\mathcal{J}$ is the random variable $\mathrm{Exp}(f\,|\,\mathcal{J})$ on X satisfying the following two conditions:

$$\mathrm{Exp}(f\,|\,\mathcal{J}) \text{ is } \mathcal{J}\text{-measurable and } \wp|_{\mathcal{J}}\text{-integrable,} \tag{A.6.4.I}$$

$$\int_G \mathrm{Exp}(f\,|\,\mathcal{J})\,d\wp = \int_G f\,d\wp \quad \text{for every} \quad G \in \mathcal{J}. \tag{A.6.4.II}$$

Some basic properties of the conditional expectation $\mathrm{Exp}(f\,|\,\mathcal{J})$ of an integrable random variable f on X given a sub-σ-algebra $\mathcal{J}$ of $\mathcal{X}$ are pulled together below.

(a) $\mathrm{Exp}\big(\mathrm{Exp}(f\,|\,\mathcal{J})\big) = \mathrm{Exp}(f)$ $\wp$-a.e.

By the definition of expectation (Problem 4.20) and by (A.6.4.II) we get: $\mathrm{Exp}\big(\mathrm{Exp}(f\,|\,\mathcal{J})\big) = \int_X \mathrm{Exp}(f\,|\,\mathcal{J})\,d\wp = \int_X f\,d\wp = \mathrm{Exp}(f)$, since $X \in \mathcal{J}$.

(b) If f is $\mathcal{J}$-measurable, then $\mathrm{Exp}(f\,|\,\mathcal{J}) = f$ $\wp$-a.e.

This follows from Problem 4.5(c), as observed in Note A.3 above.

(c) If f is a constant function $\wp$-a.e., then $\mathrm{Exp}(f\,|\,\mathcal{J}) = f$ $\wp$-a.e.

(d) If $\mathcal{J} = \mathcal{X}$, then $\mathrm{Exp}(f\,|\,\mathcal{J}) = f$ $\wp$-a.e.

These are particular cases of (b).

(e) If $\mathcal{J} = \{\varnothing, X\}$, then $\mathrm{Exp}(f\,|\,\mathcal{J}) = \mathrm{Exp}(f)$ $\wp$-a.e.

(f) $\mathrm{Exp}(\ \ |\,\mathcal{J})\colon \mathcal{L}(X, \mathcal{X}, \wp) \to \mathcal{L}(X, \mathcal{X}, \wp)$ is a linear transformation.

Item (e) follows by the definition of expected value (Problem 4.20) and Note A.3. Item (f) follows by the linearity of the integral functional (Lemma 4.5).

(g) If $f \le g$ $\wp$-a.e., then $\mathrm{Exp}(f\,|\,\mathcal{J}) \le \mathrm{Exp}(g\,|\,\mathcal{J})$ $\wp$-a.e.

(h) $\mathrm{Exp}(f\,|\,\mathcal{J}) \le \mathrm{Exp}(|f|\,|\,\mathcal{J})$ $\wp$-a.e.

In fact, if $f\colon X \to \mathbb{R}$ and $g\colon X \to \mathbb{R}$ are random variables and $f \le g$ $\wp$-a.e., then so is $g - f$ with $0 \le g - f$ $\wp$-a.e., and $0 \le \int_G \mathrm{Exp}(g - f)\,|\,\mathcal{J})\,d\wp$ for all $G \in \mathcal{J}$, so that $\mathrm{Exp}(f\,|\,\mathcal{J}) \le \mathrm{Exp}(g\,|\,\mathcal{J})$, according to (A.6.4.II), item (f), and Problem 4.5(b), both $\wp$-a.e., leading to (g) and, consequently, to (h).

(i) If $f_n \to f$ and $|f_n| \le g \in \mathcal{L}(X, \mathcal{X}, \wp)$, then $\mathrm{Exp}(f_n\,|\,\mathcal{J}) \to \mathrm{Exp}(f\,|\,\mathcal{J})$,

where convergence and inequalities are $\wp$-a.e. This is an application of the Dominated Convergence Theorem (Theorem 4.7) to (A.6.4.II).

(j) $\mathrm{Exp}(fg\,|\,\mathcal{J}) = f\,\mathrm{Exp}(g\,|\,\mathcal{J})$ if f is $\mathcal{J}$-measurable and fg, g are integrable.

Compare with (b). Recall that conditional expectation only makes sense for integrable random variables. Now suppose $f = \chi_{G'}$ for any $G' \in \mathcal{J}$ so that the claimed identity holds naturally. Indeed, applying (A.6.4.II) twice we get $\int_G \mathrm{Exp}(\chi_{G'} g \,|\, \mathcal{J})\, d\wp = \int_G \chi_{G'} g\, d\wp = \int_{G\cap G'} g\, d\wp = \int_{G\cap G'} \mathrm{Exp}(g \,|\, \mathcal{J})\, d\wp = \int_G \chi_{G'} \mathrm{Exp}(g \,|\, \mathcal{J})\, d\wp$ for every $G \in \mathcal{J}$. As $\mathrm{Exp}(\chi_{G'} g \,|\, \mathcal{J})$ and $\chi_{G'} \mathrm{Exp}(g \,|\, \mathcal{J})$ are $\mathcal{J}$-measurable, we obtain the claimed identity $\wp$-a.e. by (A.6.4.I) and by Note A.3. If the identity holds for characteristic functions of $\mathcal{J}$-measurable sets, then it holds for every $\mathcal{J}$-measurable simple function $f = \varphi\colon X \to \mathbb{R}$ by (A.6.4.II) and by linearity of the integral and of the conditional expectation in (f). Recalling from (g) that the conditional expectation is positive whenever the function is, the identity holds for positive functions f and g by (A.6.4.II) with the help of the Monotone Convergence Theorem (Corollary 3.10). So the decompositions $f = f^+ - f^-,\ g = g^+ - g^-,\ fg = (fg)^+ - (fg)^- = (f^+g^+ + f^-g^-) - (f^+g^- + f^-g^+)$ lead to the general case stated in (j) by linearity again (of the integral, and so of the conditional expectation).

(k) $\mathrm{Exp}\big(\mathrm{Exp}(f \,|\, \mathcal{J}) \,|\, \mathcal{J}'\big) = \mathrm{Exp}(f \,|\, \mathcal{J}')$ $\wp$-a.e. if $\mathcal{J}'$ is a sub-σ-algebra of $\mathcal{J}$.

Indeed, by (A.6.4.I), $\mathrm{Exp}\big(\mathrm{Exp}(f \,|\, \mathcal{J}) \,|\, \mathcal{J}'\big)$ is $\mathcal{J}'$-measurable and integrable, and $\mathrm{Exp}(f \,|\, \mathcal{J})$ is integrable. By (A.6.4.II), $\int_G \mathrm{Exp}\big(\mathrm{Exp}(f \,|\, \mathcal{J}) \,|\, \mathcal{J}'\big)\, d\wp = \int_G \mathrm{Exp}(f \,|\, \mathcal{J})\, d\wp$ for every G in $\mathcal{J}'$. As $\int_G \mathrm{Exp}(f \,|\, \mathcal{J})\, d\wp = \int_G f\, d\wp$ for every G in $\mathcal{J}$ by (A.6.4.II), we get $\int_G \mathrm{Exp}(f \,|\, \mathcal{J})\, d\wp = \int_G f\, d\wp$ for every G in $\mathcal{J}'$ since $\mathcal{J}' \subseteq \mathcal{J}$. Hence $\int_G \mathrm{Exp}\big(\mathrm{Exp}(f \,|\, \mathcal{J}) \,|\, \mathcal{J}'\big)\, d\wp = \int_G f\, d\wp = \int_G \mathrm{Exp}(f \,|\, \mathcal{J}')\, d\wp$ for every $G \in \mathcal{J}'$ by (A.6.4.II) again. Thus the claimed identity by definition.

The connection between conditional probability in item (2) and conditional probability distribution in item (3) is clear by (A.6.3.II) (definition of conditional probability distribution). The next property shows how conditional expectation $\mathrm{Exp}(f \,|\, \mathcal{J})\colon X \to \mathbb{R}$ in item (4) and conditional probability distribution $\mu_f(B, \ ) = \wp\big(f^{-1}(B) \,|\, \mathcal{J}\big)\colon X \to \mathbb{R}$ in item (3) are connected.

(ℓ) If $\phi\colon \mathbb{R} \to \mathbb{R}$ is $\Re$-measurable and the composition $\phi \circ f\colon X \to \mathbb{R}$ is $\wp$-integrable, then $\int_{\mathbb{R}} \phi\, d\mu_f(\ , x) = \mathrm{Exp}(\phi \circ f \,|\, \mathcal{J})(x)$ $\wp$-a.e. for every $x \in X$.

To verify this result, first suppose $\phi = \chi_B$ for $B \in \Re$. Thus, according to the identity $\chi_B \circ f = \chi_{f^{-1}(B)}$ (cf. Problem 3.14), with (A.6.4.II) and (A.6.3.0), we get $\int_G \mathrm{Exp}(\chi_B \circ f \,|\, \mathcal{J})\, d\wp = \int_G \mathrm{Exp}(\chi_{f^{-1}(B)} \,|\, \mathcal{J})\, d\wp = \int_G \chi_{f^{-1}(B)}\, d\wp = \int_{f^{-1}(B)\cap G} d\wp = \wp\big(f^{-1}(B) \cap G\big) = \int_G \mu_f(B, \ )\, d\wp$ for every $G \in \mathcal{J}$. Since the functions $\mathrm{Exp}(\chi_B \circ f \,|\, \mathcal{J})\colon X \to \mathbb{R}$ and $\mu_f(B, \ )\colon X \to \mathbb{R}$ are $\mathcal{J}$-measurable (cf. (2.I), (3.II), and (4.II)), lie in $\mathcal{L}(X, \mathcal{J}, \wp)$, and $\int_G \mathrm{Exp}(\chi_B \circ f \,|\, \mathcal{J})\, d\wp = \int_G \mu_f(B, \ )\, d\wp$ for $G \in \mathcal{J}$, Problem 4.5(c) says: $\mathrm{Exp}(\chi_B \circ f \,|\, \mathcal{J}) = \mu_f(B, \ )$ $\wp$-a.e. for $\wp = \wp|_{\mathcal{J}}\colon \mathcal{J} \to \mathbb{R}$. But $\mu_f(B, x) = \int_{\mathbb{R}} \chi_B\, d\mu_f(\ , x)$ for every $x \in X$. This proves the identity in (ℓ) for $\phi = \chi_B$ for any $B \in \Re$, and so (by linearity) the identity holds for every $\mathcal{J}$-measurable simple function, and the same argument used to verify (j) applies here, leading to the general case in (ℓ).

(m) $\mathrm{Exp}(f\,|\,\mathcal{J})(x) = \int_{\mathbb{R}} \alpha\, d\mu_f(\alpha, x)$ $\wp$-a.e. for every $x \in X$.

(n) $\mu_f(\mathbb{R}, x) = \int_{\mathbb{R}} d\mu_f(\alpha, x) = \mathrm{Exp}(1\,|\,\mathcal{J})(x) = 1$ $\wp$-a.e. for every $x \in X$.

These are particular cases of (ℓ). For item (m), ϕ is the identity function ($\phi(\alpha) = \alpha$); for (n), the expected result (i.e., $\mu_f(\ \ , x)\colon \Re \to \mathbb{R}$ is a probability measure for every $x \in X$) holds for ϕ being the unit function $\phi = \chi_{\mathbb{R}}$.

A.7 Markov and Martingale Processes

Stochastic processes were discussed in item (5) of Section A.2: a family of random variables or, equivalently, a random function. Among the myriad of important, useful, common, relevant, and applicable stochastic processes, we will define next only a pair of them that are given in terms of conditioning, as we saw in Section A.6. The Markov process is given in terms of conditional probability and the martingale process in terms of conditional expectation. (For discussions on a large number of specific stochastic processes, see, e.g., the books cited in the Suggested Readings at the end of this Appendix.)

(1) FILTRATIONS. Let $(X, \mathcal{X})$ be a measurable space and $(X, \mathcal{X}, \wp)$ a probability space. Consider the natural partial order in $\mathcal{X}$ given by the inclusion relation $\subseteq$, so that $(\mathcal{X}, \subseteq)$ is a partially ordered set in the inclusion ordering. Denote by $\boldsymbol{J}$ the collection of all sub-σ-algebras of $\mathcal{X}$ (which includes the collection of all sub-σ-algebras of $\mathcal{X}$ that include X). Let $\leq$ be a simple order on a set T, so that $(T, \leq)$ is a simply ordered set. For each $t \in T$, let $\mathcal{J}_t \subseteq \mathcal{X}$ be a sub-σ-algebra of the σ-algebra $\mathcal{X}$ (i.e., $\mathcal{J}_t \in \boldsymbol{J}$ for every $t \in T$).

(a) A *filtration* $\mathcal{F}\colon T \to \boldsymbol{J}$ is an increasing order-preserving map into the collection of sub-σ-algebras of a fixed σ-algebra. In other words, a filtration is a mapping $t \mapsto \mathcal{J}_t$ assigning to each $t \in T$ a sub-σ-algebra $\mathcal{F}(t) = \mathcal{J}_t$ of $\mathcal{X}$, which is increasing in the sense that, if $s, t \in T$, then $s \leq t \Longrightarrow \mathcal{F}(s) \subseteq \mathcal{F}(t)$.

(b) The triple $(X, \mathcal{X}, \mathcal{F})$ is referred to as a *filtered measure space*, and the quadruple $(X, \mathcal{X}, \mathcal{F}, \wp)$ as a *filtered probability space*.

(c) If a family $\{\mathcal{J}_t\}$ of sub-σ-algebras of a σ-algebra $\mathcal{X}$, indexed by a simply ordered set $(T, \leq)$, is such that $\mathcal{J}_s \subseteq \mathcal{J}_t$ whenever $s \leq t$, then $\{\mathcal{J}_t\}$ is said *to lead to a filtration* $\mathcal{F}\colon T \to \boldsymbol{J}$ for which $\mathcal{F}(t) = \mathcal{J}_t$ for every $t \in T$.

(d) If $\{f_t\}$ is a family of $\mathcal{J}_t$-measurable random variables, where $\{\mathcal{J}_t\}$ is a family of sub-σ-algebras of $\mathcal{X}$ leading to a filtration $\mathcal{F}$, then the family $\{f_t\}$ is said to be *adapted to the filtration* $\mathcal{F}$.

(e) A *filtered stochastic process* is a family $\{f_t\}$ of random variables adapted to a filtration $\mathcal{F}$. Sometimes a filtered stochastic process is denoted by the quintuple $(X, \mathcal{X}, \mathcal{F}, \{f_t\}, \wp)$.

(f) If, in addition, each f_t is $\wp$-integrable, then we get an *integrable filtered stochastic process*.

Example A.4: Let $\{f_k\}$, for $k \in \mathbb{N}$, be a random sequence (as in (3) of Section A.2), so a discrete stochastic process indexed by the positive integers. Let

$$F_j = \big\{f_j^{-1}(B) \in \mathcal{X} \colon B \in \Re\big\} \subset \mathcal{X} \quad \text{for each} \quad j \in \mathbb{N}$$

be the $\mathcal{X}$-measurable set consisting of all inverse images of each real random variable $f_j \colon X \to \mathbb{R}$ in the discrete process $\{f_k\}$, and consider the collection of accumulated inverse images of these real random variables,

$$\mathcal{C}_k = \textstyle\bigcup_{j=1}^{k} F_j = \bigcup_{j=1}^{k} \big\{f_j^{-1}(B) \in \mathcal{X} \colon B \in \Re\big\} \subset \mathcal{X} \quad \text{for each} \quad k \in \mathbb{N},$$

which is a subcollection of $\mathcal{X} \subseteq \wp(X)$. Set

$$\mathcal{J}_k = \mathcal{X}_{\mathcal{C}_k} \quad \text{for each} \quad k \in \mathbb{N},$$

the sub-σ-algebra of $\mathcal{X}$ generated by $\mathcal{C}_k = \bigcup_{j=1}^{k} \{f_j^{-1}(B) \colon B \in \Re\}$ (i.e., $\mathcal{J}_k$ is the smallest sub-σ-algebra of $\mathcal{X}$ that includes the collection $\mathcal{C}_k$). This $\mathcal{J}_k$ is also referred to as *the sub-σ-algebra generated by the set of random variables* $\{f_1, \ldots, f_k\}$. In this case the map

$$\mathcal{F} \colon \mathbb{N} \to \boldsymbol{J} \quad \text{such that} \quad \mathcal{F}(k) = \mathcal{J}_k \quad \text{for each} \quad k \in \mathbb{N},$$

is a filtration to which the sequence $\{f_k\}$ is adapted, which means

$$j \le k \implies \mathcal{J}_j \subseteq \mathcal{J}_k \quad \text{and} \quad f_k \text{ is } \mathcal{J}_k\text{-measurable} \quad \text{for every} \quad k \in \mathbb{N}.$$

Thus the random sequence $\{f_k\}$ is a real discrete filtered stochastic process. We will also need to consider the sub-σ-algebra $\mathcal{X}_{\{F_j\}}$ generated by a collection $\{F_j\}$ consisting of a unique set F_j (i.e., a singleton in $\mathcal{X}$) with respect to a single random variable f_j, which will be denoted by

$$[f_j] = \mathcal{X}_{\{F_j\}} \quad \text{for each} \quad j \in \mathbb{N}.$$

This $[f_j]$ is *the sub-σ-algebra generated by the random variable f_j*.

Next we define Markov and martingale processes. The definitions will be restricted to the particular case of *real* and *discrete* stochastic processes.

(2) Markov process. A real discrete filtered stochastic process $\{f_k\}$, with k running over $\mathbb{N}$, is a *Markov process* if, for every $B \in \Re$,

$$j < k \quad \Longrightarrow \quad \wp\big(f_k^{-1}(B) \,\big|\, \mathcal{J}_j\big) = \wp\big(f_k^{-1}(B) \,\big|\, [f_j]\big) \quad \wp\text{-a.e.},$$

where $\mathcal{J}_j$ is the sub-σ-algebra of $\mathcal{X}$ generated by $\{f_1, \ldots, f_j\}$ and $[f_j]$ is the sub-σ-algebra of $\mathcal{X}$ generated by f_j (as defined in Example A.4 above). Recall from Example A.4 that, for every $j \in \mathbb{N}$, this means

$$\mathcal{J}_j = \mathcal{X}_{\cup_{i\in[1,\,j]}\{f_i^{-1}(B):\,B\in\Re\}} \quad \text{and} \quad [f_j] = \mathcal{X}_{\{f_j^{-1}(B):\,B\in\Re\}}.$$

So the above implication can be rewritten as: if $B_0 \in \Re$ and $j < k$, then $\wp$-a.e.

$$\wp\big(f_k^{-1}(B_0) \,\big|\, \mathcal{X}_{\cup_{i\in[1,\,j]}\{f_i^{-1}(B):\,B\in\Re\}}\big) = \wp\big(f_k^{-1}(B_0) \,\big|\, \mathcal{X}_{\{f_j^{-1}(B):\,B\in\Re\}}\big).$$

By the same abuse of notation as in Problem 2.20, let $f_k^{-1}(B_0)$ be written as $\{f_k \in B_0\}$. Thus, roughly speaking, the above identity characterizing a discrete Markov process may be read as: "in a time-sequence of random variables, the probability of a random variable being in a Borel set given the sub-σ-algebra generated by all past random variables in the time-sequence is the same as the probability given the sub-σ-algebra generated only by the most recent previous random variable in the time-sequence".

(3) MARTINGALE. An integrable real discrete filtered stochastic process $\{f_k\}$ is a *martingale* if

$$j < k \quad \Longrightarrow \quad f_j = \mathrm{Exp}(f_k \,|\, \mathcal{J}_j) \quad \wp\text{-a.e.}$$

Replacing the equality sign $=$ in the above relation with $\leq$ or $\geq$, we get the definitions of *submartingales* and *supermartingales*, respectively. Such a definition of a real discrete martingale can be rewritten as follows. Let $\{f_k\}$ be a real-valued sequence of random variables with respect to a probability space $(X, \mathcal{X}, \wp)$ and let $\{\mathcal{J}_k\}$ be a sequence of sub-σ-algebras of $\mathcal{X}$ (with k running over the positive integers $\mathbb{N}$). If

$$\mathcal{J}_k \subset \mathcal{J}_{k+1} \quad \text{(i.e., } \{\mathcal{J}_k\} \text{ leads to a filtration } \mathcal{F}\text{)},$$

$$f_k \text{ is } \mathcal{J}_k\text{-measurable} \quad \text{(i.e., } \{f_k\} \text{ is adapted to } \mathcal{F}\text{)},$$

$$f_k \text{ is } \wp\text{-integrable} \quad \text{(i.e., } \mathrm{Exp}(|f_k|) < \infty\text{)},$$

and

$$f_k = \mathrm{Exp}(f_{k+1} \,|\, \mathcal{J}_k) \quad \wp\text{-a.e.},$$

then the discrete stochastic process $\{f_k\}$ is a *martingale*. Sometimes the sequence of pairs $(f_k, \mathcal{J}_k)$ is also called a *martingale*, meaning that $\{f_k\}$ is an integrable filtered stochastic process, which is adapted to the filtration led to by the chain $\{\mathcal{J}_k\}$, and the identity

$$\mathrm{Exp}(f_{k+1} \,|\, \mathcal{J}_k) = f_k \quad \wp\text{-a.e.}$$

holds. Roughly speaking again, the above identity characterizing a discrete martingale process may be read as: "the expectation of a random variable in a time-sequence of random variables given the sub-σ-algebra generated by all past random variables in the time-sequence is precisely the most recent previous random variable in the time-sequence".

A.8 Why Does Normalization Change Everything?

As we have seen through Chapters 2 to 9, σ-finite measures are an essential ingredient in many of the more relevant theorems in measure theory. Finite measures, however, make things simpler and, quite often, make things possible. Sometimes, being finite is not enough, and normalized measures become necessary. What makes being equal-to-one special is, after all, that 1 is the neutral element under multiplication in a ring with identity, in particular in any field, in our case, in $\mathbb{R}$. Roughly speaking, the contrast between finite and normalized measures is somewhat analogous to that between a bounded Banach-space operator and an isometry.

There are many examples where being a probability measure ($\wp(X) = 1$ rather than $\wp(X) < \infty$; equivalently, $\mu_f(\mathbb{R}) = 1$ rather than $\mu_f(\mathbb{R}) < \infty$; see Problems 2.19 and 2.20) makes the difference. We focus on four examples.

(a) The first time we met this issue was in Problem 5.20, when variance was defined. There is no concept such as variance on an arbitrary finite measure space. It works for probability measures since the expression

$$\mathrm{Exp}(f)^2 \leq \mathrm{Exp}(f^2)$$

requires $\wp(X) \leq 1$, in particular, $\wp(X) = 1$, as discussed in the remark following Proposition 5.6. In fact, the unity function χ_X lies in $L^p(X, \mathcal{X}, \wp)$ with $\|\chi_X\|_p = 1$ for every $p \geq 1$, for every measurable space $(X, \mathcal{X})$, if and only if $\wp(X) = 1$. For the special case of $p = 2$, the Schwarz inequality implies the above displayed inequality as $(\int f d\wp)^2 \leq \int f^2 d\wp$ for every $f \in \mathcal{L}(X, \mathcal{X}, \wp)$, for every $(X, \mathcal{X})$, if $\wp(X) = 1$.

(b) Another well-known and interesting result along this line is the Borel–Cantelli Lemma, which reads as follows.

> Borel–Cantelli Lemma. *Let $(X, \mathcal{X}, \eta)$ be a measure space and let $\{E_n\}$ be a sequence of $\mathcal{X}$-measurable sets.*
>
> (1) *If $\sum_n \eta(E_n) < \infty$, then $\eta(\limsup_n E_n) = 0$.*
>
> (2) *The converse holds for independent sets if $\eta(X) = 1$. That is, if $\{E_n\}$ is a sequence of $\mathcal{X}$-measurable independent sets, if $\eta(X) = 1$, and if $\eta(\limsup_n E_n) \neq 1$, then $\sum_n \eta(E_n) < \infty$.*
>
> *(And $\sum_n \eta(E_n) < \infty$ implies $\eta(\limsup_n E_n) = 0$ by (1), which in turn implies $\eta(\limsup_n E_n) \neq 1$.)*

Note A.4: For an arbitrary sequence $\{E_n\}$ of subsets of X, define:

$$\liminf_n E_n = \bigcup_{n\geq 1} \bigcap_{k\geq n} E_k \quad \text{and} \quad \limsup_n E_n = \bigcap_{n\geq 1} \bigcup_{k\geq n} E_k.$$

If $\liminf_n E_n = \limsup_n E_n$, then the common set is called $\lim_n E_n$. If $\{E_n\}$ is monotone, then $\lim_n E_n$ exists in the above sense. If $\{E_n\}$ is decreasing (or nonincreasing), then $\lim_n E_n = \bigcap_n E_n$ (as in the construction of the Cantor set in Problem 2.9). If $\{E_n\}$ is increasing (or nondecreasing), then $\lim_n E_n = \bigcup_n E_n$. Compare with the numerical counterpart in Section 1.3.

(c) There are several limit theorems for sequences of random variables referred to as the law of large numbers. Perhaps those that serve better as samples of results restricted to probability measures are the so-called Weak and Strong Laws of Large Numbers. One of their forms is stated below.

LAW OF LARGE NUMBERS. *Let $(X, \mathcal{X}, \wp)$ be a probability space and let $\{f_n\}$ be a sequence of independent random variables with finite variance and zero mean.*

WEAK LAW.

$$\lim_n \frac{1}{n^2} \sum_{k=1}^{n} \mathrm{Var}(f_k) = 0 \quad \Longrightarrow \quad \frac{1}{n} \sum_{k=1}^{n} f_k \to 0$$

in measure (i.e., in probability).

STRONG LAW.

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \mathrm{Var}(f_k) < \infty \quad \Longrightarrow \quad \frac{1}{n} \sum_{k=1}^{n} f_k \to 0$$

almost everywhere (i.e., with probability one).

It seems clear that the above assumption of zero mean (i.e., $\mathrm{Exp}(f_n) = 0$ for all n) can be adapted to an arbitrary (finite) constant mean. For instance, let i.i.d. stand for *independent and identically distributed* random variables. So $\{f_n\}$ is i.i.d. if it is a sequence of independent random variables for which all f_n have the same distribution (i.e., $\mu_{f_n} = \mu_f$ for all n). Being i.i.d. implies, in particular, that the sequence of means $\{\mathrm{Exp}(f_n)\}$ and the sequence of variances $\{\mathrm{Var}(f_n)\}$ are constant (Problem 5.20); more particularly, $\mathrm{Exp}(f_n) = m \in \mathbb{R}$ and/or $\mathrm{Var}(f_n) = \sigma^2 \in \mathbb{R}$, if an i.i.d. sequence $\{f_n\}$ has *finite* mean and/or *finite* variance. The Kolmogorov strong law of large numbers reads as follows.

KOLMOGOROV STRONG LAW.

$$\{f_n\} \ \textit{is i.i.d. with finite mean } m \quad \Longrightarrow \quad \frac{1}{n} \sum_{k=1}^{n} f_k \to m$$

almost everywhere (i.e., with probability one).

The strong laws have stronger assumptions and stronger conclusions than the weak law. As the reader may have noticed, these results are somewhat

related to mean-ergodic-like theorems of elementary analysis on convergence of Cesàro means, such as, for instance, *if* $\{\alpha_n\}$ *is a real-valued sequence, then* $\big\{\alpha_n \to \alpha \in \mathbb{R} \Longrightarrow \frac{1}{n}\sum_{k=1}^{n} \alpha_k \to \alpha\big\}$, but this would be another story.

(d) Our last example of a result holding only when measures, more than just being finite, are normalized as well, is the Central Limit Theorem. In its simplest and classical form it asserts that, under reasonable conditions, sums of random variables tend to be normally distributed (as in item (2) of Section A.4). This is a conspicuous example of a result that is meaningful in a probability-space setup. But first we need the following notion.

Note A.5: Let $(X, \mathcal{X}, \wp)$ be a fixed probability space. A sequence of random variables $\{f_n\}$ *converges in distribution* to a random variable f with probability distribution μ_f (notation: $f_n \to f$ in distribution) if the sequence $\{\mu_{f_n}\}$ of probability distributions μ_{f_n} of f_n is such that

$$\mu_{f_n}\big((-\infty, \alpha]\big) \;\to\; \mu_f((-\infty, \alpha]\big)$$

(convergence in $\mathbb{R}$) for every $\alpha \in \mathbb{R}$ for which $\mu_f(\{\alpha\}) = 0$. Equivalently (Problem 2.20), if the sequence $\{F_n\}$ of distribution functions F_n of f_n converges pointwise to the distribution function F of f at every point of continuity of F.

> Central Limit Theorem. *Let* $(X, \mathcal{X}, \wp)$ *be a probability space. If* $\{f_n\}$ *is a sequence of independent and identically distributed random variables with finite mean* $m = \mathrm{Exp}(f_n)$ *and variance* $0 \neq \sigma^2 = \mathrm{Var}(f_n)$, *then*
>
> $$\tfrac{1}{\sqrt{n\sigma^2}} \sum\nolimits_{i=1}^{n} (f_i - m) \to f$$
>
> *in distribution to a normally distributed random variable* f *with zero mean and unit variance* (*in the probability jargon, to* $f \in N(0, 1)$).

For proofs of the Borel–Cantelli Lemma, the Laws of Large Numbers, and the Central Limit Theorem, see the references in the Suggested Readings. It may be an ambitious wish, but a wish anyway, to expect that the aperitif presented in this appendix may motivate a proper reading of probability measure, *comme il faut*, as addressed in the references listed below.

Suggested Readings

Billingsley [11], Breiman [12], Chung [15], Doob [19, 20], Gikhman and Skorokhod [26, 27, 28], Halmos [31], Kingman and Taylor [39], Kolmogorov [40], Loève [48, 49], Neveu [57], Wong [74]; see also recent texts in [4], [30], [47], [61], [67].

Part II

Measures on Topological Spaces



10

Remarks on Integrals

10.1 Positive Measures

This is an introductory chapter to Part II, dealing with basic properties, needed in later chapters, of integrals with respect to positive, signed, and complex measures. The main focus of Part II, the investigation of measures on topological spaces, is left to the next three chapters. While we will not yet equip the underlying set X with a topology in this chapter, we do assume here, to avoid trivialities, that the underlying set X is nonempty. By a *positive measure* we simply mean a measure $\mu\colon \mathcal{X} \to \mathbb{R}$ on a σ-algebra $\mathcal{X}$ of subsets of a nonempty set X so that $\mu(X) \geq 0$. Sometimes this is used to specify a nonzero measure, that is, a measure μ such that $\mu(X) > 0$, but we allow the zero measure here. The term *positive measure* is employed just to distinguish it from *signed measure* (also called *real measure*) and *complex measure*. In this section we summarize the basic properties of integrals with respect to a positive measure, as discussed in Chapters 3, 4, and 5. These basic properties will be extended to integrals with respect to a signed measure and with respect to a complex measure in the forthcoming sections.

Remark 10.1. REAL-VALUED FUNCTIONS. Take $f\colon X \to \mathbb{R}$. If $\mu\colon \mathcal{X} \to \mathbb{R}$ is a *finite* positive measure on a σ-algebra $\mathcal{X}$ of subsets of X, then the characteristic function $\chi_E\colon X \to \mathbb{R}$ is integrable for every measurable set $E \in \mathcal{X}$, and so is the unit function $1 = \chi_X\colon X \to \mathbb{R}$ such that $1(x) = 1$ for every $x \in X$ (reason: $\int d\mu = \mu(X) < \infty$). If $f\colon X \to \mathbb{R}$ is an integrable function, then

$$\left| \int_E f \, d\mu \right| \leq \int_E |f| \, d\mu \leq \sup |f| \, \mu(E)$$

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_10

for every $E \in \mathcal{X}$ by Lemma 4.4 and Problem 5.1 (where $\sup|f| \in \overline{\mathbb{R}}$ is defined as usual: $\sup|f| = \sup_{x\in X}|f(x)|$). Therefore,

$$\sup_{|f|\le 1}\left|\int f\,d\mu\right| \le \mu(X) = \int d\mu = \int 1\,d\mu = \left|\int d\mu\right| = \left|\int 1\,d\mu\right|,$$

and so (since $\sup|1| = 1$, trivially),

$$\sup_{|f|\le 1}\left|\int f\,d\mu\right| = \max_{|f|\le 1}\left|\int f\,d\mu\right| = \left|\int 1\,d\mu\right| = \int 1\,d\mu = \mu(X),$$

where the supremum is taken over all real-valued integrable functions f on X such that $|f| \le 1$. The set $\mathcal{L} = \mathcal{L}(X, \mathcal{X}, \mu)$ of all real-valued integrable functions on X with respect to the measure μ is a linear space, and the integral $\int(\ )\,d\mu\colon \mathcal{L} \to \mathbb{R}$ is a linear functional (cf. Lemma 4.5). For every positive measure μ, the integral is a *positive functional* in the sense that it takes positive functions to positive numbers:

$$0 \le \int f\,d\mu \quad \text{whenever} \quad 0 \le f \in \mathcal{L}$$

(i.e., if $f\colon X \to \mathbb{R}$ in $\mathcal{L}(X, \mathcal{X}, \mu)$ is such that $0 \le f$ μ-a.e., then $0 \le \int f\,d\mu$ — cf. Proposition 4.2(a) and Problem 4.4(b)). That $\sup_{|f|\le 1}\left|\int f\,d\mu\right| = \left|\int 1\,d\mu\right|$ (i.e., the supremum is actually attained by the function 1, which lies in $\mathcal{L}$ whenever μ is finite) is a consequence of the positivity of the linear functional $\int(\ )\,d\mu\colon \mathcal{L} \to \mathbb{R}$ with respect to a positive measure μ. Another consequence of the positivity of the integral with respect to a positive measure is this (see Problem 4.4(a)). If f and g are functions in $\mathcal{L}$, then

$$f \le g \quad \text{implies} \quad \int f\,d\mu \le \int g\,d\mu.$$

(In fact, since the integral is a positive functional, $0 \le \int(g - f)\,d\mu$ whenever $f \le g$ μ-a.e., and since it is linear, $\int f\,d\mu \le \int g\,d\mu$.)

Remark 10.2. COMPLEX-VALUED FUNCTIONS. Take $f\colon X \to \mathbb{C}$ and let $f = f_1 + i\,f_2$ be its Cartesian decomposition, where $f_1 = \operatorname{Re} f\colon X \to \mathbb{R}$ and $f_2 = \operatorname{Im} f\colon X \to \mathbb{R}$ are real-valued functions on X (the *real* and *imaginary parts* of f). A complex-valued function f is measurable if f_1 and f_2 are measurable (Problem 1.7); and f is integrable with respect to a positive measure $\mu\colon \mathcal{X} \to \mathbb{R}$ if f_1 and f_2 are integrable with respect to μ, and its integral is

$$\int f\,d\mu = \int f_1\,d\mu + i\int f_2\,d\mu,$$

where f is integrable if and only if $|f|$ is (Problem 4.7), and for every $E \in \mathcal{X}$,

$$\left|\int_E f\,d\mu\right| \le \int_E |f|\,d\mu \le \sup|f|\,\mu(E).$$

The proof of the first inequality in the above expression requires a little care (cf. Hint to Problem 4.7). However, as a consequence, we get the same result obtained for real-valued functions, namely,

$$\sup_{|f|\le 1}\left|\int f\,d\mu\right| = \left|\int 1\,d\mu\right| = \int 1\,d\mu \;=\; \mu(X),$$

where the supremum is taken over all complex-valued integrable functions f on X such that $|f| \le 1$. The supremum is actually a maximum, attained by the real-valued function $1\colon X \to \mathbb{R}$, which is integrable whenever μ is finite, and this is a consequence of the positivity of the integral with respect to a positive measure μ (when applied to a real-valued positive function).

10.2 Real Measures

If $\mu\colon \mathcal{X} \to \mathbb{R}$ and $\eta\colon \mathcal{X} \to \mathbb{R}$ are *finite positive measures* on a σ-algebra $\mathcal{X}$ of subsets of a set X, then consider the (finite real-valued) signed measure $\nu\colon \mathcal{X} \to \mathbb{R}$ on $\mathcal{X}$ defined by $\nu = \mu - \eta$, and recall that $\nu = \nu^+ - \nu^-$, where $\nu^+\colon \mathcal{X} \to \mathbb{R}$ and $\nu^-\colon \mathcal{X} \to \mathbb{R}$ are finite positive measures on $\mathcal{X}$ (singular to each other), referred to as the positive and negative variation of ν, which are such that $\nu^+ \le \mu$ and $\nu^- \le \eta$ (Theorem 7.4). The total variation of ν is the finite (positive) measure $|\nu|\colon \mathcal{X} \to \mathbb{R}$ given by $|\nu| = \nu^+ + \nu^-$ (Example 7A).

Remark 10.3. REAL-VALUED FUNCTIONS. A function $f\colon X \to \mathbb{R}$ is integrable with respect to a signed measure $\nu = \mu - \eta$, for arbitrary finite positive measures μ and η, if it is integrable with respect to the positive measures μ and η. And so if it is integrable with respect to the positive measures ν^+ and ν^-, as $\nu^+ \le \mu$ and $\nu^- \le \eta$, so that $\int |f|\,d\nu^+ \le \int |f|\,d\mu$ and $\int |f|\,d\nu^- \le \int |f|\,d\eta$; cf. Problem 3.3(d) and Lemma 4.4(a). The integral of $f\colon X \to \mathbb{R}$ with respect to a signed measure ν is unambiguously defined by

$$\int f\,d\nu \;=\; \int f\,d\mu - \int f\,d\eta \;=\; \int f\,d\nu^+ - \int f\,d\nu^-.$$

Note that $\chi_E\colon X \to \mathbb{R}$ is integrable for every $E \in \mathcal{X}$ since ν is finite, and so

$$\nu(E) \;=\; \nu^+(E) - \nu^-(E) \;=\; \int_E d\nu^+ - \int_E d\nu^- \;=\; \int_E d\nu.$$

Equivalently,

$$\nu(E) \;=\; \mu(E) - \eta(E) \;=\; \int_E d\mu - \int_E d\eta \;=\; \int_E d\nu.$$

This ensures that the integral with respect to a signed measure ν is equivalently defined as the difference of the integrals with respect to the positive

measures ν^+ and ν^-, or with respect to the positive measures μ and η. Similarly, the integral with respect to the positive measure $|\nu|$ is defined by

$$\int f\, d|\nu| \,=\, \int f\, d\nu^+ + \int f\, d\nu^-.$$

Again, set $\sup|f| = \sup_{x\in X}|f(x)|$. Thus (cf. Lemma 4.4), for every $E \in \mathcal{X}$,

$$\left|\int_E f\, d\nu\right| \,\le\, \int_E |f|\, d\nu^+ + \int_E |f|\, d\nu^- = \int_E |f|\, d|\nu| \,\le\, \sup|f|\,|\nu|(E),$$

so that

$$\sup_{|f|\le 1}\left|\int f\, d\nu\right| \,\le\, |\nu|(X),$$

where the supremum is taken over all real-valued integrable functions f on X such that $|f| \le 1$. Since ν^+ and ν^- are singular, there is a measurable partition $\{A^+, A^-\}$ of X such that $\nu^+(A^-) = \nu^-(A^+) = 0,\ \nu^+(A^+) = \nu^+(X)$, and $\nu^-(A^-) = \nu^-(X)$ (e.g., any Hahn decomposition of X with respect to ν — see Section 7.1). Consider the function $\chi_{A^+} - \chi_{A^-} : X \to \mathbb{R}$ for which

$$\begin{aligned}\int (\chi_{A^+} - \chi_{A^-})\, d\nu \,&=\, \int \chi_{A^+}\, d\nu^+ + \int \chi_{A^-}\, d\nu^- = \int_{A^+} d\nu^+ + \int_{A^-} d\nu^- \\ &=\, \nu^+(A^+) + \nu^-(A^-) \,=\, \nu^+(X) + \nu^-(X) \,=\, |\nu|(X).\end{aligned}$$

Therefore, since $|\chi_{A^+} - \chi_{A^-}| = 1$,

$$|\nu|(X) \,=\, \int d|\nu| \,=\, \int 1\, d\nu^+ + \int 1\, d\nu^- = \int (\chi_{A^+} - \chi_{A^-})\, d\nu \,\le\, \sup_{|f|\le 1}\left|\int f\, d\nu\right|,$$

and so the supremum is actually attained by the function $\chi_{A^+} - \chi_{A^-}$, which is a consequence of the positivity of the integrals $\int(\ )\, d\nu^+$ and $\int(\ )\, d\nu^-$ with respect to the positive measures ν^+ and ν^-. Summing up:

$$\sup_{|f|\le 1}\left|\int f\, d\nu\right| \,=\, |\nu|(X).$$

Next we show another proof of the above identity. This alternate proof will be required in the sequel. Suppose, in particular, that f is a real-valued, nonnegative, integrable function on X with $|f| \le 1$ such that $\chi_B \le f \le \chi_C$, where χ_B and χ_C are characteristic functions of measurable sets $B, C \in \mathcal{X}$ with $B \subseteq C$. As $B \cap A^- \subseteq C \cap A^-$ and $B \cap A^+ \subseteq C \cap A^+$, there is no loss of generality in assuming that $B \subseteq C \subseteq A^+$ or $B \subseteq C \subseteq A^-$. If $B \subseteq C \subseteq A^+$, then $0 \le \nu(B) = \int \chi_B\, d\nu \le \int f\, d\nu \le \int \chi_C\, d\nu = \nu(C)$; on the other hand, if $B \subseteq C \subseteq A^-$, then $\nu(C) = \int \chi_C\, d\nu \le \int f\, d\nu \le \int \chi_B\, d\nu = \nu(B) \le 0$. Outcome: If either $B \subseteq C \subseteq A^+$ or $B \subseteq C \subseteq A^-$, and $\chi_B \le f \le \chi_C$, then

$$|\nu(B)| \,\le\, \left|\int f\, d\nu\right| \le |\,\nu(C)|\,.$$

Note that this fails without the assumption that $B \subseteq C$ are subsets of either A^+ or A^- (e.g., set $B = A^+$, $C = X$, $0 < -\nu(A^-) = \nu(A^+)$, and $f = 1$). Now, if $\{A_i\}$ is any measurable covering of X, then consider a measurable partition $\{E_k\}$ of X such that $\{E_k\} = \{A_i^+\} \cup \{A_i^-\}$, where $A_i^+ = A_i \cap A^+$ and $A_i^- = A_i \cap A^-$ for each i. So $\{E_k\}$ consists of subsets of either A^+ or A^-. If B_k and C_k are measurable sets with $B_k \subseteq C_k$ in $\{E_k\}$ and $\chi_{B_k} \leq f_k \leq \chi_{C_k}$, then

$$|\nu(B_k)| \leq \left| \int f_k \, d\nu \right| \leq |\nu(C_k)|$$

for each k. If $\{A_i\}$ is any finite measurable covering of X, then $\{E_k\}$ is a finite measurable partition of X. Take $C_k = E_k$ and set $f_k = \chi_{E_k}$. In this case,

$$\sup \sum\nolimits_k \left| \int \chi_{E_k} \, d\nu \right| = \sup \sum\nolimits_k |\nu(E_k)| = |\nu|(X)$$

according to Examples 2I and 7A, where the supremum is taken over all finite measurable partitions of X. Recalling that $\sum_k \chi_{E_k} = \chi_X = 1$ (and so $\left|\sum_k \chi_{E_k}\right| = 1$), and also that $0 \leq \chi_{E_k} \leq 1$, we get

$$\sup \sum\nolimits_k \left| \int \chi_{E_k} \, d\nu \right| \leq \sup_{|f| \leq 1} \left| \int f \, d\nu \right|,$$

where the supremum on the right-hand side is taken over all real-valued integrable functions f on X for which $|f| \leq 1$. Hence,

$$|\nu|(X) = \sup \sum\nolimits_k \left| \int \chi_{E_k} \, d\nu \right| \leq \sup_{|f| \leq 1} \left| \int f \, d\nu \right| \leq |\nu|(X).$$

This leads to another proof that

$$\sup_{|f| \leq 1} \left| \int f \, d\nu \right| = |\nu|(X).$$

Remark 10.4. COMPLEX-VALUED FUNCTIONS. A function $f : X \to \mathbb{C}$ is integrable with respect to a signed measure $\nu : \mathcal{X} \to \mathbb{R}$ if $f_1 = \operatorname{Re} f$ and $f_2 = \operatorname{Im} f$ are integrable with respect to ν, and the integral of $f = f_1 + i f_2 : X \to \mathbb{C}$ with respect to a signed measure $\nu = \nu^+ - \nu^- : \mathcal{X} \to \mathbb{R}$ is defined by

$$\begin{aligned}
\int f \, d\nu &= \int f_1 \, d\nu + i \int f_2 \, d\nu \\
&= \int f_1 \, d\nu^+ - \int f_1 \, d\nu^- + i \left[\int f_2 \, d\nu^+ - \int f_2 \, d\nu^- \right] \\
&= \int f_1 \, d\nu^+ + i \int f_2 \, d\nu^+ - \int f_1 \, d\nu^- - i \int f_2 \, d\nu^- \\
&= \int f \, d\nu^+ - \int f \, d\nu^-,
\end{aligned}$$

so that, by the inequalities in Remark 10.2 and according to the definition of the integral with respect to $|\nu|$ in Remark 10.3 we get, for every $E \in \mathcal{X}$,

$$\left|\int_E f\,d\nu\right| \le \left|\int_E f\,d\nu^+\right| + \left|\int_E f\,d\nu^-\right|$$
$$\le \int_E |f|\,d\nu^+ + \int_E |f|\,d\nu^- = \int_E |f|\,d|\nu| \le \sup|f|\,|\nu|(E).$$

By the above inequality we get $\sup_{|f|\le 1}\left|\int f\,d\nu\right| \le |\nu|(X)$. However, as we saw in Remark 10.3, such a supremum is attained by the real-valued (as a particular case of a complex-valued) function $\chi_{A^+} - \chi_{A^-} : X \to \mathbb{R} \subseteq \mathbb{C}$ for any Hahn decomposition $\{A^+, A^-\}$ of X with respect to ν. Therefore,

$$\sup_{|f|\le 1}\left|\int f\,d\nu\right| = |\nu|(X),$$

where the supremum is taken over all complex-valued integrable functions f on X such that $|f| \le 1$.

10.3 Complex Measures

Now recall the definition of complex measure (cf. Problem 2.16): a complex measure $\zeta : \mathcal{X} \to \mathbb{C}$ is a complex-valued set function on a σ-algebra $\mathcal{X}$ of subsets of a set X such that $\zeta = \nu_1 + i\,\nu_2$, where $\nu_1 = \operatorname{Re}\zeta : \mathcal{X} \to \mathbb{R}$ and $\nu_2 = \operatorname{Im}\zeta : \mathcal{X} \to \mathbb{R}$, the real and imaginary parts of ζ, are (finite real-valued) signed measures on $\mathcal{X}$. We say that a positive measure $\mu : \mathcal{X} \to \mathbb{R}$ *dominates* a complex measure $\zeta : \mathcal{X} \to \mathbb{C}$ if $|\zeta(E)| \le \mu(E)$ for every $E \in \mathcal{X}$ (warning: see Example 2G). Set $\mu = |\nu_1| + |\nu_2|$. It is clear that $\mu : \mathcal{X} \to \mathbb{R}$ is a finite positive measure (since it is the sum of finite positive measures), and it is readily verified that μ dominates ζ. Indeed, $|\zeta(E)| \le |\nu_1(E)| + |\nu_2(E)| \le |\nu_1|(E) + |\nu_2|(E) = (|\nu_1| + |\nu_2|)(E) < \infty$ for every $E \in \mathcal{X}$ (cf. Examples 2I and 7A). Consider the *total variation* $|\zeta| : \mathcal{X} \to \mathbb{R}$ of $\zeta : \mathcal{X} \to \mathbb{C}$ defined along the same line as in Example 2I: for each $n \in \mathbb{N}$ let $\boldsymbol{E}(n)$ be the collection of all measurable partitions of $E \in \mathcal{X}$ containing n sets so that $\bigcup \boldsymbol{E}(n)$ is the collection of all *finite* measurable partitions of E. For each $E \in \mathcal{X}$ set

$$|\zeta|(E) = \sup_{\{E_j\} \in \bigcup \boldsymbol{E}(n)} \sum\nolimits_j |\zeta(E_j)|\,.$$

Again, following the same steps of Examples 2I and 7A, we can show that the total variation $|\zeta| : \mathcal{X} \to \mathbb{R}$ of the complex measure $\zeta : \mathcal{X} \to \mathbb{C}$ is a finite positive measure, which is the least positive measure that dominates ζ (i.e., if μ dominates ζ, then $|\zeta| \le \mu$).

Remark 10.5. REAL-VALUED FUNCTIONS. A real-valued function $f\colon X \to \mathbb{R}$ on X is integrable with respect to a complex measure $\zeta = \nu_1 + i\,\nu_2\colon \mathcal{X} \to \mathbb{C}$ if it is integrable with respect to both signed measures ν_1 and ν_2, and the integral of an integrable real-valued function $f\colon X \to \mathbb{R}$ with respect to a complex measure $\zeta = \nu_1 + i\,\nu_2$ is defined by

$$\int f\,d\zeta \;= \int f\,d\nu_1 \,+\, i\int f\,d\nu_2.$$

Observe that since $\chi_E\colon X \to \mathbb{R}$ is integrable for each $E \in \mathcal{X}$ because ν_1 and ν_2 are finite, it follows that (see Remark 10.3), for every $E \in \mathcal{X}$,

$$\zeta(E) \;=\; \nu_1(E) + i\,\nu_2(E) \;=\; \int_E d\nu_1 + i\int_E d\nu_2 \;=\; \int_E d\zeta.$$

Also note that $\left|\int_E f\,d\zeta\right| \le \left|\int_E f\,d\nu_1\right| + \left|\int_E f\,d\nu_2\right| \le \sup|f|\,(|\nu_1|+|\nu_2|)(E)$ for every $E \in \mathcal{X}$. But we can get a tighter inequality as follows. Take an arbitrary $E \in \mathcal{X}$. Let $g\colon X \to \mathbb{R}$ be a nonnegative measurable function, and consider the collection $\Phi_g^+(E)$ of all positive simple functions $\varphi = \sum_j \alpha_j \chi_{E_j}$, for all finite measurable partitions $\{E_j\} \in \bigcup \boldsymbol{E}(n)$, such that $0 \le \varphi \le g$. Note that $\left|\int_E \varphi\,d\zeta\right| = \left|\sum_j \alpha_j\,\zeta(E_j)\right| \le \sum_j \alpha_j\,|\zeta(E_j)| \le \int_E \varphi\,d|\zeta|$, and therefore $\left|\int_E g\,d\zeta\right| = \sup_{\varphi\in\Phi_g^+(E)}\left|\int_E \varphi\,d\zeta\right| \le \sup_{\varphi\in\Phi_g^+(E)}\int_E \varphi\,d|\zeta| = \int_E g\,d|\zeta|$. By linearity of the integral, and recalling that $f = f^+ - f^-$ and $|f| = f^+ + f^-$, where f^+ and f^- are nonnegative and measurable, we get, for every $E \in \mathcal{X}$,

$$\left|\int_E f\,d\zeta\right| = \left|\int_E f^+ d\zeta - \int_E f^- d\zeta\right| \le \left|\int_E f^+ d\zeta\right| + \left|\int_E f^- d\zeta\right|$$

$$\le \int_E f^+ d|\zeta| + \int_E f^- d|\zeta| \;=\; \int_E |f|\,d|\zeta| \;\le\; \sup|f|\,|\zeta|(E),$$

so that

$$\left|\int f\,d\zeta\right| \;\le\; \sup|f|\,|\zeta|(X).$$

Let $\{A_1^+, A_1^-\}$ and $\{A_2^+, A_2^-\}$ be Hahn decompositions of X with respect to the signed measures ν_1 and ν_2, respectively. Consider the definition of disjointification as in the hint to Problem 7.2(b). Let $\mathcal{A}_{1,2}^{+-}$ be a disjointification of the collection $\{A_1^+ \cap A_2^+,\ A_1^+ \cap A_2^-,\ A_1^- \cap A_2^+,\ A_1^- \cap A_2^-\}$ so that $\mathcal{A}_{1,2}^{+-}$ is a measurable partition of X. Regarding ν_1 and ν_2, the disjoint sets in $\mathcal{A}_{1,2}^{+-}$ bear the same properties as the sets $A_i{}^{\pm} \cap A_j{}^{\pm}$. Let f be a real-valued, nonnegative, integrable function on X such that $|f| \le 1$ and $\chi_B \le f \le \chi_C$ for sets $B \subseteq C$ in $\mathcal{X}$ (as in Remark 10.3). If B and C are subsets of sets in $\mathcal{A}_{1,2}^{+-}$, then we get $|\nu_1(B)|^2 + |\nu_2(B)|^2 \le \left|\int f\,d\nu_1\right|^2 + \left|\int f\,d\nu_2\right|^2 \le |\nu_1(C)|^2 + |\nu_2(C)|^2$ by Remark 10.3. Outcome: If $B \subseteq C$ are subsets of sets in $\mathcal{A}_{1,2}^{+-}$ and $\chi_B \le f \le \chi_C$, then

$$|\zeta(B)| \;\le\; \left|\int f\,d\zeta\right| \;\le\; |\zeta(C)|\,.$$

An argument similar to that in Remark 10.3 shows that, for an arbitrary measurable covering $\{A_k\}$ of X, there is a measurable partition $\{E_k\}$ of X whose sets E_k are the intersection of each A_k with sets from the partition $\mathcal{A}_{1,2}^{+-}$. Thus, if B_k and C_k lie in $\{E_k\}$, with $B_k \subseteq C_k$ and $\chi_{B_k} \le f_k \le \chi_{C_k}$, then

$$|\zeta(B_k)| \le \left| \int f_k \, d\zeta \right| \le |\zeta(C_k)|$$

for every k. Once again, as in Remark 10.3, for every finite measurable partition of X there exists such a finite measurable partition $\{E_k\}$ of X, so that

$$\sup \sum\nolimits_k \left| \int \chi_{E_k} \, d\zeta \right| = \sup \sum\nolimits_k |\zeta(E_k)| = |\zeta|(X),$$

with the supremum taken over all finite measurable partitions of X. Thus, following the same argument as in Remark 10.3, since $\sum_k \chi_{E_k} = \chi_X = 1$ (so that $\left|\sum_k \chi_{E_k}\right| = 1$) and $0 \le \chi_{E_k} \le 1$, we get

$$\sup \sum\nolimits_k \left| \int \chi_{E_k} \, d\zeta \right| \le \sup_{|f| \le 1} \left| \int f \, d\zeta \right|,$$

with the supremum on the right-hand side taken over all real-valued integrable functions f on X such that $|f| \le 1$. Hence, $|\zeta|(X) = \sup \sum_k |\int \chi_{E_k} d\zeta| \le \sup_{|f|\le 1} |\int f \, d\zeta|$. Since $\sup_{|f|\le 1} \left|\int f \, d\zeta\right| \le |\zeta|(X)$, it follows that

$$\sup_{|f| \le 1} \left| \int f \, d\zeta \right| = |\zeta|(X).$$

Remark 10.6. COMPLEX-VALUED FUNCTIONS. $f = f_1 + i\, f_2 \colon X \to \mathbb{C}$ is integrable with respect to a complex measure $\zeta = \nu_1 + i\, \nu_2 \colon \mathcal{X} \to \mathbb{C}$ if the real-valued functions f_1 and f_2 (the real and imaginary parts of f) are integrable with respect to ζ, and the integral of an integrable function $f \colon X \to \mathbb{C}$ with respect to a complex measure $\zeta \colon \mathcal{X} \to \mathbb{C}$ is given by

$$\begin{aligned} \int f \, d\zeta &= \int f_1 \, d\zeta + i \int f_2 \, d\zeta \\ &= \int f_1 \, d\nu_1 + i \int f_1 \, d\nu_2 + i \left[\int f_2 \, d\nu_1 + i \int f_2 \, d\nu_2 \right] \\ &= \int f_1 \, d\nu_1 + i \int f_2 \, d\nu_1 + i \left[\int f_1 \, d\nu_2 + i \int f_2 \, d\nu_2 \right] \\ &= \int f \, d\nu_1 + i \int f \, d\nu_2 . \end{aligned}$$

Observe that $\left|\int_E f \, d\zeta\right| \le \left|\int_E f_1 \, d\zeta\right| + \left|\int_E f_2 \, d\zeta\right| \le (\sup |f_1| + \sup |f_2|)|\zeta|(E)$ for every $E \in \mathcal{X}$, by Remark 10.5. However, we can again get a tighter inequality, with $\sup |f_1| + \sup |f_2|$ replaced by $\sup |f| = \big(\sup (|f_1|^2 + |f_2|^2)\big)^{1/2}$.

Indeed, write $\int f\,d\zeta = \rho\, e^{i\theta}$, which implies that $e^{-i\theta}\int f\,d\zeta = \rho \geq 0$. Now set $g = e^{-i\theta} f : X \to \mathbb{C}$. With $g_1 = \operatorname{Re} g : X \to \mathbb{R}$, we get $|g_1| \leq |g|$ and $|g| = |f|$. So $0 \leq \left|\int f\,d\zeta\right| = \rho = \int e^{-i\theta} f\,d\zeta = \int g\,d\zeta = \operatorname{Re}\int g\,d\zeta = \left|\operatorname{Re}\int g\,d\zeta\right| \leq \left|\int \operatorname{Re} g\,d\zeta\right|$.

(To verify the inequality: as $\int g\,d\zeta = \int g_1\,d\nu_1 - \int g_2\,d\nu_2 + i\left[\int g_1\,d\nu_2 + \int g_2\,d\nu_1\right]$, $0 \leq \operatorname{Re}\int g\,d\zeta = \int g_1\,d\nu_1 - \int g_2\,d\nu_2$. Suppose $0 \leq \int g_2\,d\nu_2$ (no loss of generality; multiply f by -1 if necessary). Thus $\left|\operatorname{Re}\int g\,d\zeta\right| = \operatorname{Re}\int g\,d\zeta \leq \int g_1\,d\nu_1$. Since $\int \operatorname{Re} g\,d\zeta = \int g_1\,d\zeta = \int g_1\,d\nu_1 + i\int g_1\,d\nu_2$, we get $\operatorname{Re}\int \operatorname{Re} g\,d\zeta = \int g_1\,d\nu_1$. So $0 \leq \left|\operatorname{Re}\int g\,d\zeta\right| \leq \int g_1\,d\nu_1 = \operatorname{Re}\int \operatorname{Re} g\,d\zeta = \left|\operatorname{Re}\int \operatorname{Re} g\,d\zeta\right| \leq \left|\int \operatorname{Re} g\,d\zeta\right|$.)

Then, by Remark 10.5, $\left|\int f\,d\zeta\right| \leq \left|\int g_1\,d\zeta\right| \leq \int |g_1|\,d|\zeta|$. Since $|g_1| \leq |f|$,

$$\left|\int f\,d\zeta\right| \leq \int |f|\,d|\zeta| \leq \sup|f|\,|\zeta|(X)$$

by using Remark 10.5 again. Therefore, as $\left|\int_E f\,d\zeta\right| = \left|\int f\chi_E\,d\zeta\right|$,

$$\left|\int_E f\,d\zeta\right| \leq \int_E |f|\,d|\zeta| \leq \sup|f|\,|\zeta|(E)$$

for every $E \in \mathcal{X}$. Hence $\sup_{|f|\leq 1}\left|\int f\,d\zeta\right| \leq |\zeta|(X)$, where the supremum is taken over all complex-valued integrable functions f on X such that $|f| \leq 1$. However, as we saw in Remark 10.5, this inequality becomes an identity even for the particular case where the supremum is taken over the real-valued functions, and so it holds for the general case. That is,

$$\sup_{|f|\leq 1}\left|\int f\,d\zeta\right| = |\zeta|(X).$$

10.4 Additional Propositions

The expression $\left|\int f\right| \leq \int |f|$ was central in this chapter, but $\left|\int fg\right| \not\leq \left|\int f|g|\right|$ even if $|f| \leq 1$ (e.g., $f = g = 1$ for $x \in [0,1)$ and $f = g = -1$ for $x \in [1,2]$).

A finite positive measure (whose range is in $[0,+\infty)$) is a particular case of a signed measure (whose range is in $\mathbb{R}$), and a signed measure is a particular case of a complex measure (whose range is in $\mathbb{C}$). However, if a positive measure is not finite, then it is not a particular case of any of the above measures since its range is in $[0,+\infty]$. Thus a signed measure $\nu = \nu^+ - \nu^-$ and a complex measure $\zeta = \nu_1 + i\,\nu_2 = \nu_1^+ - \nu_1^- + i\,(\nu_2^+ - \nu_2^-)$ are always *finite* in the sense that $|\nu(E)| < \infty$ and $|\zeta(E)| < \infty$ for every measurable set E. So they are, in particular, σ-finite in the sense that ν^+, ν^-, ν_1^+, ν_1^-, ν_2^+, ν_2^-, being all finite positive measures, are naturally σ-finite. Let μ and η be arbitrary measures (positive, or finite positive, or signed, or complex) on the same σ-algebra. Exactly as in the case of positive measures (cf. Definition 7.6),

μ is *absolutely continuous* with respect to η (notation: $\mu \ll \eta$) if, for an arbitrary measurable set E, $\eta(E) = 0$ implies $\mu(E) = 0$. Similarly, as in the case of positive measures (cf. Definition 7.9), μ and η are *singular* (notation: $\mu \perp \eta$) if $\mu(A) = \eta(B) = 0$ for some measurable partition $\{A, B\}$ of X.

With the above extended definitions in mind, the Radon–Nikodym Theorem (Theorem 7.8) has an immediate extension to integrable functions (instead of nonnegative measurable functions), where a σ-finite positive measure μ is replaced by a (finite) signed measure ν, as follows: *Let $(X, \mathcal{X})$ be a measurable space. If ν is a signed measure and η is a positive σ-finite measure, both on $\mathcal{X}$, and if ν is absolutely continuous with respect to η, then there is a unique (η-almost everywhere unique) real-valued function f in $\mathcal{L}(X, \mathcal{X}, \eta)$ such that $\nu(E) = \int_E f\, d\eta$ for each $E \in \mathcal{X}$.* The proof follows by Theorem 7.8 for $f = f^+ - f^-$ with $\nu^+(E) = \int_E f^+ d\eta$ and $\nu^-(E) = \int_E f^- d\eta$ (cf. Proposition 7.5). Theorem 7.8 is not a particular case of the above version. But the above version is a particular case of the next result.

Proposition 10.A. *Let $(X, \mathcal{X})$ be a measurable space. If ζ is a complex measure and η is a positive σ-finite measure, both on $\mathcal{X}$, and if ζ is absolutely continuous with respect to η, then there exists a unique (η-almost everywhere unique) complex-valued η-integrable function f on X such that*

$$\zeta(E) = \int_E f\, d\eta \quad \text{for each} \quad E \in \mathcal{X}.$$

A similar approach extends the Lebesgue Decomposition Theorem (Theorem 7.10), which in fact is a corollary of the preceding Radon–Nikodym Theorem, as follows. Let $\mathbb{F}$ denote either $\mathbb{R}$ or $\mathbb{C}$. So an *$\mathbb{F}$-valued measure* means either a finite positive measure, or a signed measure, or a complex measure.

Proposition 10.B. *Let $(X, \mathcal{X})$ be a measurable space. If η and μ are both measures on $\mathcal{X}$, η positive σ-finite and μ $\mathbb{F}$-valued, then there is a unique pair of $\mathbb{F}$-valued measures μ_a and μ_s on $\mathcal{X}$ such that $\mu_a \ll \eta$, $\mu_s \perp \eta$, and*

$$\mu = \mu_a + \mu_s.$$

The Fundamental Theorem of Calculus of elementary calculus asserts that integration and differentiation are the inverse of each other as follows. Take, for instance, a real-valued function $F\colon [\alpha, \beta] \to \mathbb{R}$ (not assumed *a priori* to be measurable with respect to any σ-algebra). Suppose it is continuously differentiable on the interval $[\alpha, \beta] \subset \mathbb{R}$. The derivative of the indefinite (Riemann) integral coincides with the function, which is often written as

$$F(x) = \left(\int_\alpha^x F(t)\, dt \right)'$$

for every $x \in [\alpha, \beta]$. Conversely, the function is recovered by integrating its derivative F' (which, being continuous on $[\alpha, \beta]$, is integrable) by the formula

$$F(x) - F(\alpha) = \int_{\alpha}^{x} F'(t)\, dt$$

for every $x \in [\alpha, \beta]$. Perhaps it might come as no big surprise that such an inverse relationship between differentiation and integration may be properly stated in a measure-theoretical framework; in particular, in the context of the Lebesgue measure. As expected, the Radon–Nikodym Theorem (Theorem 7.8 — Proposition 10.A) plays a crucial role in establishing this with the help of the Lebesgue Decomposition Theorem (Theorem 7.10 — Proposition 10.B), where the reference measure λ is the Lebesgue measure.

We now proceed along the lines of the following propositions towards the Lebesgue version of the Fundamental Theorem of Calculus. In what follows, $\lambda\colon \Re \to \overline{\mathbb{R}}$ will denote the Lebesgue measure on the Borel σ-algebra $\Re$ generated by the open sets of $\mathbb{R}$, equipped with its usual topology, as in Section 8.3 (see the remark that follows Problem 1.14).

Recall that a *compact* subset of $\mathbb{R}$ is precisely a closed and bounded subset of $\mathbb{R}$, and that a Borel measure on $\Re$ is a positive measure $\mu\colon \Re \to \overline{\mathbb{R}}$ assigning a finite value to every compact set of $\mathbb{R}$ (i.e., $\mu(K) < \infty$ for every compact $K \subset \mathbb{R}$; cf. Problem 2.13). It is readily verified that the Lebesgue measure (which assigns its finite length to every bounded interval) is a Borel measure, and also that every Borel measure is σ-finite (cf. Problem 2.13).

Also recall that a *finite* positive measure on $\Re$ (which is tautologically a Borel measure) is a particular case of a signed measure on $\Re$, which in turn is a particular case of a complex measure on $\Re$. Thus we actually have essentially two distinct cases: a positive nonfinite (possibly σ-finite) measure, or an $\mathbb{F}$-*valued measure* — meaning a complex (complex-valued), or a signed (real-valued), or a positive finite measure, such as

$$\zeta = \nu_1 + i\,\nu_2 = (\mu_1 - \eta_1) + i\,(\mu_2 - \eta_2)\colon \Re \to \mathbb{C},$$

or

$$\nu = \mu - \eta\colon \Re \to \mathbb{R}, \qquad \text{or} \qquad \mu\colon \Re \to [0, +\infty),$$

respectively, where μ, η, μ_1, η_1, μ_2, η_2 are (finite) positive measures on $\Re$.

Similarly, a real-valued function on $\mathbb{R}$ can be viewed as a particular case of a complex-valued function on $\mathbb{R}$, and so we will refer to an $\mathbb{F}$-*valued function*, meaning either a real-valued or a complex-valued function. Some of the next results can be stated for measures on the σ-algebra generated by the open sets of $\mathbb{R}^n$, or for functions on $\mathbb{R}^n$. We will, however, consider only measures on $\Re$ and functions on $\mathbb{R}$ in Propositions 10.C to 10.F.

Let X be a nonempty open subset of $\mathbb{F}$ (particular case, $X = \mathbb{R}$). Take a function $F\colon X \to \mathbb{F}$, and an arbitrary point $x \in X$. Let $F'(x)$ be a number in $\mathbb{F}$ with the following property. For every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$0 < |y - x| < \delta \quad \text{implies} \quad \left|\tfrac{F(y)-F(x)}{y-x} - F'(x)\right| < \varepsilon.$$

If there exists such a number $F'(x) \in \mathbb{F}$, then it is called the *derivative of* F *at* x, and the function F is said to be *differentiable at* x.

Proposition 10.C. *Let μ be an $\mathbb{F}$-valued measure and let λ be the Lebesgue measure, both on $\Re$, and consider the function $F\colon \mathbb{R} \to \mathbb{F}$ defined by*

$$F(x) = \mu\big((-\infty, x)\big) \quad \textit{for every} \quad x \in \mathbb{R}.$$

The function F is differentiable at $x \in X$ (so that there exists $F'(x) \in \mathbb{F}$) and $F'(x) = \alpha$ if and only if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\lambda\big(I(x)\big) < \delta \quad \textit{implies} \quad \left|\tfrac{\mu(I(x))}{\lambda(I(x))} - \alpha\right| < \varepsilon$$

for every open interval $I(x)$ that contains x.

Given an arbitrary $x \in I(x) = (\alpha, \beta)$, we say that the open interval $I(x)$ *shrinks to* x (notation: $I(x) \to \{x\}$) if for every $\delta > 0$ there exists an interval $I_\delta(x) = (\alpha_\delta, \beta_\delta) \subseteq I(x) = (\alpha, \beta)$ containing x (i.e., $\alpha \le \alpha_\delta < x < \beta_\delta \le \beta$) such that $|\beta_\delta - \alpha_\delta| < \delta$. Proposition 10.C suggests the following definition. Take any point $x \in I(x) = (\alpha, \beta)$. Let $(D\mu)(x)$ be a number in $\mathbb{F}$ with the following property. For every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$|\beta_\delta - \alpha_\delta| < \delta \quad \text{implies} \quad \left|\tfrac{\mu(I_\delta(x))}{\lambda(I_\delta(x))} - (D\mu)(x)\right| < \varepsilon$$

whenever the interval $I_\delta(x) = (\alpha_\delta, \beta_\delta) \subseteq I(x) = (\alpha, \beta)$ contains x. If this number $(D\mu)(x) \in \mathbb{F}$ exists, then it is called the *derivative of the measure μ at a point x with respect to Lebesgue measure λ*. In other words, $(D\mu)(x)$ is the limit of $\frac{\mu(I(x))}{\lambda(I(x))}$ as the open interval $I(x)$ containing x shrinks to x:

$$(D\mu)(x) = \lim_{I(x) \to \{x\}} \tfrac{\mu(I(x))}{\lambda(I(x))}$$

for every $x \in \mathbb{R}$ at which this limit exists. If the limit exists for every $x \in \mathbb{R}$, then the function $D\mu\colon \mathbb{R} \to \mathbb{F}$ defined by the preceding limit is referred to as the *derivative of the measure μ with respect to Lebesgue measure λ*. Recall that μ is an $\mathbb{F}$-valued measure on $\Re$, which may be a finite positive measure, or a signed measure, or a complex measure.

The next result says that the Radon–Nikodym derivative $\frac{d\mu}{d\lambda}$ of a measure μ with respect to Lebesgue measure λ coincides with the derivative $D\mu$ of

the measure μ with respect to Lebesgue measure λ. (Warning: all measures act on $\Re$ and the reference measure must be the Lebesgue measure λ.)

Proposition 10.D. *Let μ be an $\mathbb{F}$-valued measure and let λ be the Lebesgue measure, both on $\Re$. Suppose μ is absolutely continuous with respect to λ ($\mu \ll \lambda$). Let $\frac{d\mu}{d\lambda}$ be the Radon–Nikodym derivative of μ with respect to λ.*

Claim: $D\mu = \frac{d\mu}{d\lambda}$ *(λ-almost everywhere), so that*

$$\mu(E) = \int_E (D\mu)\, d\lambda \quad \textit{for every} \quad E \in \Re.$$

A further form of Proposition 10.D, representing the first part of the Fundamental Theorem of Calculus extended to Lebesgue integrals, is considered in Proposition 10.E. It requires the notion of Lebesgue points of an integrable function. Let $\lambda\colon \Re \to \mathbb{R}$ be the Lebesgue measure. A function $f\colon \mathbb{R} \to \mathbb{F}$ is *integrable in the Lebesgue sense* if $f \in \mathcal{L}(\mathbb{R}, \Re, \lambda)$, that is, if it is λ-integrable. For $\mathbb{F} = \mathbb{C}$, this means that $f = f_1 + i\, f_2\colon \mathbb{R} \to \mathbb{C}$, with $f_1, f_2\colon \mathbb{R} \to \mathbb{R}$ such that $f_1, f_2 \in \mathcal{L}(\mathbb{R}, \Re, \lambda)$. In this case, every $x \in \mathbb{R}$ for which

$$\lim_{I(x) \to \{x\}} \tfrac{1}{\lambda(I(x))} \int_{I(x)} |f - f(x)|\, d\lambda = 0$$

is called a *Lebesgue point of f*.

Proposition 10.E. *Let $f\colon \mathbb{R} \to \mathbb{F}$ be integrable in the Lebesgue sense. For each x in $\mathbb{R}$, set*

$$F(x) = \int_{-\infty}^{x} f\, d\lambda$$

in $\mathbb{F}$. This expression defines a function $F\colon \mathbb{R} \to \mathbb{F}$, which is differentiable at every Lebesgue point x, whose derivative at these points x is $F'(x) = f(x)$.

A function $f\colon [\alpha, \beta] \to \mathbb{F}$ is *absolutely continuous* on the interval $[\alpha, \beta]$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\sum\nolimits_i (\beta_i - \alpha_i) < \delta \quad \text{implies} \quad \sum\nolimits_i |f(\beta_i) - f(\alpha_i)| < \varepsilon$$

for every *finite* disjoint collection of open intervals $(\alpha_i, \beta_i) \subseteq [\alpha, \beta]$. As a particular case, if the above holds, then it holds for every collection containing just one interval (i.e., $\beta_i - \alpha_i < \delta$ implies $|f(\beta_i) - f(\alpha_i)| < \varepsilon$ for every open subinterval (α_i, β_i) of $[\alpha, \beta]$), leading to the standard notion of continuity. Thus absolute continuity trivially implies continuity.

Note: The above definition of absolutely continuous functions coincides with the definition of absolutely continuous functions in Problem 3.15 for $\mathbb{F} = \mathbb{R}$ — see reference in the Notes that close this chapter.

The first part of the Fundamental Theorem of Calculus extended to Lebesgue integrals in Proposition 10.E asserted that the derivative of the indefinite integral coincides with the function. Conversely, the second part of the Fundamental Theorem of Calculus extended to Lebesgue integrals asserts that the function is recovered by integrating its derivative, as follows.

Proposition 10.F. *Take a function $f\colon[\alpha,\beta]\to\mathbb{F}$ and let λ be the Lebesgue measure on $\Re$. If either f is absolutely continuous (so that it is differentiable λ-almost everywhere on $[\alpha,\beta]$ and f' is λ-integrable), or f is differentiable at every point of the interval $[\alpha,\beta]$ and f' is λ-integrable, then*

$$f(x)-f(\alpha)=\int_\alpha^x f'\,d\lambda \quad \textit{for every} \quad x\in[\alpha,\beta]\,.$$

Recall again that by an $\mathbb{F}$-valued function $f\colon X\to\mathbb{F}$ on a nonempty set X we mean either a complex-valued function or the particular case of a real-valued function. Similarly, by an $\mathbb{F}$-valued measure $\mu\colon\mathcal{X}\to\mathbb{F}$ on a σ-algebra $\mathcal{X}$ of subsets of X we mean either a complex measure or the particular cases of a signed measure or a finite positive measure. Since the notion of integral has been extended in this chapter to $\mathbb{F}$-valued functions and $\mathbb{F}$-valued measures, the notion of the Banach space $L^1=L^1(X,\mathcal{X},\mu)$ is naturally extended from the case of real-valued functions and positive measures, considered in Chapter 5, to $\mathbb{F}$-valued functions and measures. In particular, exactly the same expression for the norm in L^1 holds for $\mathbb{F}$-valued functions and measures due to the upper bounds obtained in Remarks 10.1 to 10.6,

$$\left|\int f\,d\mu\right|\le\int|f|\,d|\mu| \;=\; \|f\|_1<\infty \quad \text{for every} \quad f\in L^1(X,\mathcal{X},\mu),$$

so the results in Chapter 5 (also Section 4.2) still hold for $\mathbb{F}$-valued functions and measures. The above inequality says that the integral, as a transformation from the Banach space $(L^1(X,\mathcal{X},\mu),\|\ \|_1)$ of integrable functions to the Banach space $(\mathbb{F},|\ |)$ equipped with its usual norm $|\ |$, is bounded (in fact, a contraction). Since for linear transformations boundedness is equivalent to continuity, we get the next result for $\mathbb{F}$-valued measures and functions.

Proposition 10.G. *The integral $\int(\ )\,d\mu\colon L^1(X,\mathcal{X},\mu)\to\mathbb{F}$, as a functional between the Banach spaces $L^1(X,\mathcal{X},\mu)$ and $\mathbb{F}$, for an $\mathbb{F}$-valued measure μ, is linear and continuous (i.e., the integral is a continuous linear functional).*

For the particular case of $\mathbb{F}=\mathbb{R}$ and μ being a positive measure, see the remark that precedes Problem 5.16.

Extending the notion of integral of scalar-valued (either real or complex) functions to vector-valued functions on a finite-dimensional (real or complex) normed space seems quite natural and, in fact, it is natural. Things

become more delicate and more interesting if we consider vector-valued functions on not necessarily finite-dimensional spaces. We use capital letters also to denote functions into normed spaces, and call them 'mappings'.

Let Y be a normed space and let $\|\ \|$ denote the norm $\|\ \|_Y$ on Y (cf. Definition 5.1), either as a finite-dimensional or an infinite-dimensional, real or complex, normed space. Let $\mathcal{Y}_\mathcal{T}$ be the σ-algebra generated by the collection $\mathcal{T}$ of all open subsets of Y (i.e., the σ-algebra generated by a topology $\mathcal{T}$ on Y — cf. Problem 1.12 — which is called a Borel σ-algebra of subsets of Y — see the remark following Problem 1.14). Now let X be a nonempty set, and let $\mathcal{X}$ be a σ-algebra of subsets of X. Note that, in accordance with the first paragraph of this chapter, no topology is being assigned to the set X.

A Y-valued mapping $F\colon X \to Y$ on X is said to be measurable if the inverse images of sets in $\mathcal{Y}_\mathcal{T}$ are sets in $\mathcal{X}$ (i.e., $F^{-1}(E) \in \mathcal{X}$ for every $E \in \mathcal{Y}_\mathcal{T}$ — cf. Problem 1.8). Consider the real-valued function $f\colon X \to \mathbb{R}$ defined by $f(x) = \|F(x)\|$ for every $x \in X$. When we say that $\|F(\ )\|$ (norm on Y) is measurable, we mean in the sense of Definition 1.2, which coincides with the above definition (inverse images of sets in the Borel algebra $\Re$ lie in $\mathcal{X}$ — cf. Problem 1.8 again). Let $\mu\colon \mathcal{X} \to \overline{\mathbb{R}}$ be a positive measure on the σ-algebra $\mathcal{X}$ of subsets of the nonempty set X, and consider a measure space $(X, \mathcal{X}, \mu)$. Let $\mathcal{L}(X,Y) = \mathcal{L}(X, \mathcal{X}, \mu; Y)$ denote the collection of all those measurable mappings $F\colon X \to Y$ for which the function $\|F(\ )\|\colon X \to \mathbb{R}$ is integrable; that is, for which $\|F(\ )\|$ is measurable and $\int \|F(\ )\|\, d\mu < \infty$.

Proposition 10.H. *Let $(X, \mathcal{X}, \mu)$ be a measure space and Y a normed space.*

(a) *If a Y-valued mapping $F\colon X \to Y$ on X is measurable, then the real-valued function $\|F(\ )\|\colon X \to \mathbb{R}$ on X is also measurable.*

(b) *$\mathcal{L}(X,Y)$ is a linear space and the function $\|\ \|_\mathcal{L}\colon \mathcal{L}(X,Y) \to \mathbb{R}$ defined by*

$$\|F\|_\mathcal{L} = \int \|F(\ )\|\, d\mu \quad \text{for every} \quad F \in \mathcal{L}(X,Y),$$

for a positive measure μ, is a seminorm on the linear space $\mathcal{L}(X,Y)$.

Compare with Lemma 4.5 and Proposition 5.4. Note that we are using the notation $\|F(x)\|$ for the norm $\|F(x)\|_Y$ of $F(x)$ on Y for each $x \in X$, and the notation $\|F\|_\mathcal{L}$ for the seminorm $\|F\|_{\mathcal{L}(X,Y)}$ of F on the linear space $\mathcal{L} = \mathcal{L}(X,Y)$. Now suppose Y is a Banach space (i.e., a complete normed space — see the paragraph that precedes Proposition 5.4 and Theorem 5.9).

A Banach-space-valued mapping $F\colon X \to Y$ on X is said to be *Bochner integrable* if it lies in $\mathcal{L}(X,Y)$ (i.e., F is measurable and $\int \|F(\ )\|\, d\mu < \infty$) and if its range $F(X)$ is separable (i.e., $F(X) \subseteq Y$ has a countable dense subset, which means that $F(X)$ has a countable subset whose closure coincides

with the closure of $F(X)$ in Y — the notion of denseness and separability will be discussed in the first section of the next chapter). Since $F(X)$ is separable whenever Y is (i.e., whenever Y has a countable dense subset; equivalently, if Y has a countable subset whose closure coincides with Y), it follows that *if Y is a separable Banach space, then $F: X \to Y$ is Bochner integrable if and only if $F \in \mathcal{L}(X,Y)$*. It is worth recalling that $\mathbb{F}$ equipped with its usual norm (as well as $\mathbb{F}^n$ equipped with any norm) is a separable Banach space. Just as in Sections 5.1 and 5.2, consider the equivalence class $[F]$ of all mappings in $\mathcal{L}(X,Y)$ that are equal to F μ-a.e., and denote the set of these equivalence classes by $L^1(X,Y) = L^1(X,\mathcal{X},\mu;Y)$. As in Lemma 5.8, the seminorm of Proposition 10.H becomes a norm $\| \ \|_1$ on $L^1(X,Y)$,

$$\|F\|_1 = \|[F]\|_1 = \int \|F(\ )\|\, d\mu,$$

which does not depend on the exemplar F in $[F]$, so that $(L^1(X,Y), \| \ \|_1)$, or simply $L^1(X,Y)$ when the norm $\| \ \|_1$ is clear in the context, is a normed space. Again, note that the norm $\| \ \|_{L^1(X,Y)}$ on $L^1(X,Y)$ is denoted by $\| \ \|_1$ as usual. A mapping $\Phi: X \to Y$ is a *simple mapping* if its range is finite, and it is an *integrable simple mapping* if it is simple and lies in $\mathcal{L}(X,Y)$ (i.e., $\#\Phi(X) < \infty$, Φ is measurable, and $\int \|\Phi(\ )\|\, d\mu < \infty$).

Proposition 10.I. *Take $(X,\mathcal{X},\mu)$, μ positive, and let Y be a Banach space.*

(a) *Then $L^1(X,Y)$ is a Banach space.*

(b) *If Y is separable, then the collection of all Y-valued integrable simple mappings on X is a dense linear manifold of the Banach space $L^1(X,Y)$.*

The first part of Proposition 10.I is the counterpart of Theorem 5.9; the second part (which extends Problem 5.4) allows us to define an integral for mappings in $L^1(X,Y)$ as follows. For a separable Banach space Y, write a measurable simple mapping Φ as a weighted sum of characteristic functions $\chi_{E_i}: X \to \{0,1\} \subset \mathbb{F}$ of a finite measurable partition $\{E_i\}$ of X with a finite set of distinct elements $\{a_i\}$ from Y (the finite set $\{a_i\}$ is the range of Φ), say

$$\Phi = \sum_{i=1}^{n} a_i \chi_{E_i},$$

and define the *integral of $\Phi: X \to Y$* with respect to μ as in Definition 3.1:

$$\int \Phi\, d\mu = \sum_{i=1}^{n} a_i\, \mu(E_i),$$

which is a vector in Y whenever $\Phi \in L^1(X,Y)$. Since the collection of all Y-valued integrable simple mappings on X is a dense linear manifold of $L^1(X,Y)$, every mapping $F \in L^1(X,Y)$ is the limit, in the norm $\| \ \|_1$ on

$L^1(X,Y)$, of a sequence $\{\Phi_k\}$ of integrable simple mappings in $L^1(X,Y)$. That is, for every $F \in L^1(X,Y)$, there exists a sequence $\{\Phi_k\}$ such that

$$\|\Phi_k - F\|_1 \to 0 \quad \text{as} \quad k \to \infty.$$

In other words, a Bochner integrable mapping $F\colon X \to Y$ of X into a separable Banach space Y (i.e., $F \in L^1(X,Y)$ where Y is a separable Banach space) is, according to Proposition 10.I, the limit in $L^1(X,Y)$ of a sequence of integrable simple mappings in $L^1(X,Y)$; that is,

$$F = \lim_k \Phi_k.$$

Summing up: *every $F \in L^1(X,Y)$ is the limit in $L^1(X,Y)$ of a sequence $\{\Phi_k\}$ of integrable simple mappings in $L^1(X,Y)$.*

This leads to the next result, which defines the integral of F as the limit in Y of the integrals of Φ_k: for each F in $L^1(X,Y)$ there exists a unique vector $s(F)$ in Y, denoted by $s(F) = \int F\,d\mu$, such that $\| \int \Phi_k\,d\mu - s(F)\| \to 0$ as $k \to \infty$, where $\|\ \|$ is the norm on Y, which is written as

$$\int F\,d\mu = \int \lim_k \Phi_k\,d\mu = \lim_k \int \Phi_k\,d\mu\,.$$

The existence of such a limit in Y is ensured by the next proposition. Indeed, since *the integral transformation $\int(\ )\,d\mu\colon L^1(X,Y) \to Y$ is linear and continuous*, as in Proposition 10.J below, and since $F = \lim_k \Phi_k$ in $L^1(X,Y)$, we get the above limit exchange, namely, $\int \lim_k \Phi_k\,d\mu = \lim_k \int \Phi_k\,d\mu$.

Proposition 10.J. *If $(X,\mathcal{X},\mu)$ is a measure space with a positive measure μ, and if Y is a separable Banach space, then*

(a) *there is a unique bounded linear transformation (i.e., a continuous linear transformation) $\int(\ )\,d\mu\colon L^1(X,Y) \to Y$ such that $\int \Phi\,d\mu = \sum_{i=1}^n a_i\,\mu(E_i)$ for every integrable simple mapping $\Phi = \sum_{i=1}^n a_i\,\chi_{E_i} \in L^1(X,Y)$. Also,*

(b) *for every $F \in L^1(X,Y)$, $F = \lim_k \Phi_k$ and $\int F d\mu = \lim_k \int \Phi_k d\mu$ for some sequence $\{\Phi_k\}$, and this bounded linear transformation $\int(\ )\,d\mu$ is such that*

$$\left\| \int F\,d\mu \right\| \le \int \|F(\ )\|\,d\mu = \|F\|_1.$$

The inequality $\left\|\int F\,d\mu\right\| \le \|F\|_1$ for $F \in L^1(X,Y)$, where $\|\ \|$ is the norm on Y, means that the transformation $\int(\ )\,d\mu\colon L^1(X,Y) \to Y$ is a contraction.

The value at $F \in L^1(X,Y)$ of the bounded linear transformation $\int(\ )\,d\mu$ from the Banach space $L^1(X,Y)$ to a separable Banach space Y, viz., $\int F\,d\mu$, is called the *Bochner integral* of F with respect the a positive measure μ.

Positive measures have been assumed since Proposition 10.H. This will also be the case for the next proposition on the composition of transformations T and F, ensuring the identity $T\big(\int_X F(x)\,d\mu(x)\big) = \int_X T\big(F(x)\big)\,d\mu(x)$.

Proposition 10.K. *If $(X, \mathcal{X}, \mu)$ is a measure space with a positive measure μ, if Y and Z are separable Banach spaces, and if T is a bounded linear transformation of Y to Z, then for every mapping $F \in L^1(X, Y)$,*

$$T\int F\,d\mu = \int (T \circ F)\,d\mu.$$

For an exposition on further notions of integral, extended in a different direction from what has been done here (also referred to as generalized Riemann integral, Kurzweil–Henstock, or gauge integral), restricted to functions on the real line, that corrects some defects in the classical Riemann theory, simplifying and extending the Lebesgue theory, see, for instance, [7].

Notes: The whole chapter, in particular the propositions in Section 10.4, comprise a set of basic results on integration that will be required in the sequel. The $\mathbb{F}$-valued versions of the Radon–Nikodym Theorem and of the Lebesgue Decomposition Theorem in Propositions 10.A and 10.B are natural and immediate consequences of their positive versions in Theorems 7.8 and 7.10 (e.g., also see [63, Theorem 6.10]). For the discussion on the Fundamental Theorem of Calculus extended to Lebesgue integrals, along the lines we have approached here, the reader is referred to [63, Chapter 7]. In particular, see [63, Theorems 7.1, 7.8, and 7.11] for Propositions 10.C, 10.D, and 10.E; and [63, Theorems 7.20, 7.21] for Proposition 10.F. For the equivalence between the definitions of absolutely continuous functions (as those in Problem 3.15 and Proposition 10.F), see, e.g., [56, Theorem D.13.3]. Since the integral remains linear for $\mathbb{F}$-valued functions and measures (naturally extending the result for real-valued functions and positive measures of Lemma 4.5 and Section 5.1), it follows that Proposition 10.G is an immediate consequence of the fact that for linear transformations boundedness coincides with continuity (see, e.g., [43, Theorem 4.14]). Propositions 10.H, 10.I, 10.J, and 10.K lead to the notion of Bochner integral of Banach-space-valued mappings, and some of its basic results. For their proofs, see, e.g., [13, Theorems 17.8, 17.9, 17.11, 17.13, 17.14, 17.15]. (For the paragraph before Proposition 10.J, see, e.g., [43, Corollary 3.8 and Theorem 4.14].)

Suggested Readings

Berberian [9], Brown and Pearcy [13], Cohn [16], Halmos [31], Kingman and Taylor [39], Royden [62], Rudin [63], Weir [73]. See also [7], [69].



11

Borel Measure

11.1 Topological Spaces

A *topological space* $(X, \mathcal{T})$ was defined in Problem 1.12 as a set X equipped with a topology $\mathcal{T}$. We drop the "$\mathcal{T}$" from the notation $(X, \mathcal{T})$ and write simply X for a topological space when the topology is implicit. A *topology* $\mathcal{T}$ on a set X is a collection of subsets of X satisfying the following axioms:

(i) the whole set X and the empty set $\varnothing$ lie in $\mathcal{T}$,

(ii) finite intersections of sets in $\mathcal{T}$ lie in $\mathcal{T}$,

(iii) arbitrary unions of sets in $\mathcal{T}$ lie in $\mathcal{T}$.

The sets in $\mathcal{T}$ are called the *open* sets of X (with respect to $\mathcal{T}$).

A *metric space* (X, d) is a set X equipped with a metric d. Again, we drop the "d" from the notation (X, d) and write simply X for a metric space when the metric is clear or immaterial. A *metric* in a set X is a function $d\colon X \times X \to \mathbb{R}$ such that for every $x, y, z \in X$,

(i) $d(x, y) \geq 0$ and $d(x, x) = 0$ (*nonnegativeness*),

(ii) $d(x, y) = 0$ if and only if $x = y$ (*positiveness*),

(iii) $d(x, y) = d(y, x)$ (*symmetry*),

(iv) $d(x, y) \leq d(x, z) + d(z, y)$ (*triangle inequality*).

These are the *metric axioms*. If $d\colon X \times X \to \mathbb{R}$ satisfies axioms (i), (iii), and (iv), but not necessarily axiom (ii), then it is called a *pseudometric*. An *open ball*, centered at $x_0 \in X$ with radius ε, in a metric space X is the set

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_11

$B_\varepsilon(x_0) = \{x \in X\colon\ d(x, x_0) < \varepsilon\}$. A set U is *open* in a metric space X if U includes a nonempty open ball centered at each one of its points. Open balls are open sets in a metric space. The collection $\mathcal{T}$ of all open sets in a metric space X satisfies the three axioms of topology; this is referred to as the *metric topology* on X, or the *topology induced* (or generated, or determined) *by a metric* d. When we refer to the topology of a metric space, it will be understood that this is the metric topology, unless otherwise stated. So every metric space is a topological space, where the (metric) topology is that induced by the metric. This topology $\mathcal{T}$ induced by the metric d, and the topological space obtained by equipping X with $\mathcal{T}$, are said to be metrized by d.

In general, if X is a topological space with topology $\mathcal{T}$, and if there exists a metric d on $X \times X$ (so that X also is a metric space with respect to the metric d), such that the collection of all open sets in X with respect to d coincides with $\mathcal{T}$, then the metric d is said to metrize the topology $\mathcal{T}$, and the topological space X and the topology $\mathcal{T}$ are called *metrizable*. The notion of topological space is broader than the notion of metric space. Every metric space is a topological space, but the converse fails: there are topological spaces that are not metrizable.

Definition 11.1. Let X be a topological space (i.e., a nonempty set X equipped with a topology $\mathcal{T}$, whose elements are the open subsets of X).

(a) A set $V \subseteq X$ is *closed* if its complement $X \backslash V$ is open.

(b) The *closure* $\overline{A}$ of $A \subseteq X$ is the smallest closed subset of X that includes A. (The intersection of all closed subsets of X that include A.)

(c) A set $A \subseteq X$ is *dense* if $\overline{A} = X$.

(d) A topological space is *separable* if it has a countable dense subset.

(e) The *interior* A° of $A \subseteq X$ is the largest open subset of X included in A. (The union of all open subsets of X included in A.)

(f) A *covering* of a set $A \subseteq X$ is a collection of subsets of X whose union includes A. An *open covering* of A is a covering of A consisting entirely of open subsets of X.

(g) A set $K \subseteq X$ is *compact* if every open covering of K includes a finite subcovering. If X is a compact set itself, then X is a *compact space*.

(h) A set in X is *σ-compact* if it is a countable union of compact sets.

(i) A set $A \subseteq X$ is *relatively compact* (or *conditionally compact*) if its closure $\overline{A}$ is compact.

(j) A *base* (or a *topological base*) for X is a subcollection of $\mathcal{T}$ that covers each open subset of X (i.e., covers each set in $\mathcal{T}$). Equivalently, $\mathcal{B} \subseteq \mathcal{T}$

is a base for a topology $\mathcal{T}$ of subsets of X if, for each $U \in \mathcal{T}$ and each $x \in U$, there is a $G \in \mathcal{B}$ with $x \in G \subseteq U$.

(k) A *neighborhood* N_x of a point x in X is any subset of X that includes an open set that contains x. An *open neighborhood* of $x \in X$ is any open subset of X that contains x.

(ℓ) A *Hausdorff* space is a topological space X such that, for every pair of distinct points x and y in X, there exist neighborhoods N_x and N_y of x and y, respectively, such that $N_x \cap N_y = \varnothing$.

(m) A *locally compact space* is a topological space X such that every point of X has a compact neighborhood.

The concepts of closed and open are duals of each other in the sense that a set is open if and only if its complement is closed, and vice versa. The three axioms of topology say that $\varnothing$ and X are open, the intersection of a finite collection of open sets is open, and the union of an arbitrary collection of open sets is open. It is readily verified that the dual result for closed sets reads as follows: $\varnothing$ and X are closed, the union of a finite collection of closed sets is closed, and the intersection of an arbitrary collection of closed sets is closed. If a set is both open and closed, then it is said to be a *clopen* set. The standard properties stated in the next lemma will be required in the sequel.

Lemma 11.2. *Let A, B, G, U, V, K be subsets of a topological space X.*

(a) *A closed subset of a compact set is compact.*

(b) *If $A \subseteq B$ and $\overline{B}$ is compact, then $\overline{A}$ is compact.*

(c) *A nonempty subset of a topological space is open if and only if it is (or includes) a neighborhood of each one of its points.*

(d) *Every metric space is Hausdorff.*

(e) *Every singleton in a Hausdorff space is closed.*

(f) *If K is compact, X is Hausdorff, and $x \in X \backslash K$, then there is an open set U with $K \subseteq U$ and a neighborhood N_x of x such that $U \cap N_x = \varnothing$.*

(g) *Every compact subset of a Hausdorff space is closed.*

(h) *If X is Hausdorff, V is closed, and K is compact, then $V \cap K$ is compact.*

(i) *If the intersection of an infinite (not necessarily countable) collection of compact subsets of a Hausdorff space is empty, then there is a finite subcollection whose intersection is also empty.*

Proof. Let X be topological space.

(a) *If $V \subseteq K$, where K is compact and V is closed, then V is compact.* In fact, suppose V is a closed subset of a compact set $K \subseteq X$. Take an arbitrary

covering of V, say $\mathcal{U}$, consisting of open subsets of X. So $\mathcal{U} \cup \{X \backslash V\}$ is an open covering of K. Since K is compact, this covering includes a finite subcovering, say $\mathcal{U}'$, so that $\mathcal{U}' \backslash \{X \backslash V\} \subseteq \mathcal{U}$ is a finite subcovering of V. Therefore, every open covering of V has a finite subcovering.

(b) This is a direct consequence of (a), since $A \subseteq B$ implies $\overline{A} \subseteq \overline{B}$.

(c) If $A \subseteq X$ is open and nonempty, then A trivially is (and so A trivially includes) an open neighborhood of each one of its points. Conversely, take $A \subseteq X$. If A includes (or if A is) a neighborhood N_a of every $a \in A$, then there is an open set $U_a \subseteq N_a \subseteq A$ such that $a \in U_a$ for every $a \in A$. Since $A = \bigcup_{a \in A} U_a$, it follows that A is open (cf. axiom (iii) of a topological space).

(d) It is readily verified (by the triangle inequality) that, for every pair of distinct points x and y in a metric space X, there are nonempty open balls $B_\varepsilon(x)$ and $B_\rho(y)$ centered at x and y (which clearly are neighborhoods of x and y), respectively, such that $B_\varepsilon(x) \cap B_\rho(y) = \varnothing$.

(e) Take an arbitrary $x \in X$, consider the singleton $\{x\} \subseteq X$ and its complement $X \backslash \{x\} \subseteq X$, and take an arbitrary $y \in X \backslash \{x\}$. If X is Hausdorff, then there exists a neighborhood N_y of y such that $N_y \subseteq X \backslash \{x\}$. Hence there exists an open $U_y \subseteq X \backslash \{x\}$ such that $y \in U_y$. Thus $X \backslash \{x\}$ is itself a neighborhood of each one of its points, and so $X \backslash \{x\}$ is open by item (c).

(f) Let K be a compact proper subset of X. If $K = \varnothing$, then the result is trivially verified for $U = \varnothing$. Thus suppose $K \neq \varnothing$ and take an arbitrary point x in $X \backslash K$. Since x is distinct from every point in K, it follows that for every $y \in K$ there exists an open neighborhood K_y of y and an open neighborhood X_y of x such that $K_y \cap X_y = \varnothing$ (reason: X is a Hausdorff space). But $K \subseteq \bigcup_{y \in K} K_y$ so that $\{K_y\}_{y \in K}$ is a covering of K consisting of nonempty open subsets of X. If K is compact, then there is a finite subset of K, say $\{y_i\}_{i=1}^n$, such that $K \subseteq U = \bigcup_{i=1}^n K_{y_i}$, which is open (each K_{y_i} is open). Set $N_x = \bigcap_{i=1}^n X_{y_i}$, which is a neighborhood of x (in fact, an open neighborhood of x, since it is a finite intersection of open neighborhoods X_{y_i} of x). Since $K_{y_i} \cap X_{y_i} = \varnothing$, it follows that $K_{y_i} \cap N_x = \varnothing$, for each i, and therefore $(\bigcup_{i=1}^n K_{y_i}) \cap N_x = \varnothing$. That is, $U \cap N_x = \varnothing$.

(g) Let K be a compact subset of a Hausdorff space X. If either $K = \varnothing$ or $K = X$, then K is trivially closed in X. Thus suppose $\varnothing \neq K \neq X$. According to (f), for every $x \in X \backslash K$ there exists an open set U with $K \subseteq U$ and a (open) neighborhood N_x of x such that $U \cap N_x = \varnothing$. Hence $N_x \subseteq X \backslash K$. Thus $X \backslash K$ includes a (open) neighborhood of each one of its points, which means by (c) that $X \backslash K$ is open, and so K is closed.

(h) *If X is Hausdorff, then $V \cap K$ is compact whenever K is compact and V is closed.* Indeed, let K be a compact subset and V a closed subset of a

Hausdorff space. By item (g), K is closed. Thus $V \cap K$ is a closed subset of the compact K, and therefore compact itself according to (a).

(i) *If an infinite collection $\{K_\gamma\}$ of compact sets in a Hausdorff space X is such that $\bigcap_\gamma K_\gamma = \varnothing$, then there exists a finite subcollection $\{K_i\}$ of $\{K_\gamma\}$ such that $\bigcap_i K_i = \varnothing$.* Indeed, set $U_\gamma = X \backslash K_\gamma$, which is open if X is Hausdorff according to (g). Take an arbitrary K from the collection $\{K_\gamma\}$. If $\bigcap_\gamma K_\gamma = \varnothing$, then there is no point of K that belongs to every K_γ. Hence $\{U_\gamma\}$ is an open covering of K. Since K is compact, there is a finite subcollection $\{U_i\}$ of $\{U_\gamma\}$ such that $K \subseteq \bigcup_i U_i$. Thus $X \backslash \bigcup_i U_i \subseteq X \backslash K$, and so $\bigcap_i K_i = \bigcap_i X \backslash U_i = X \backslash \bigcup_i U_i \subseteq X \backslash K$. Then $K \cap \bigcap_i K_i = \varnothing$. □

Remark 11.A (Boundedness in metric and locally compact spaces):

(a) Boundedness in metric spaces. A set B in a metric space (X, d) is *bounded* if $\sup_{x,y \in B} d(x, y) < \infty$ (i.e., if it has a finite diameter); and B is *totally bounded* if for every $\varepsilon > 0$ there exists a finite ε-net B_ε for B. (A subset B_ε of B is an *ε-net* for B if for every point x of B there exists a point y in B_ε such that $d(x, y) < \varepsilon$.) It is readily verified that total boundedness implies boundedness, but the converse fails: the unit ball in ℓ^p equipped with the usual metric is bounded but not totally bounded. The Compactness Theorem says that *in a metric space, a set is compact if and only if it is complete and totally bounded*. (See, e.g., [43, Corollary 3.81] — a sequence $\{x_n\}$ in a metric space is Cauchy if $\lim_{n,m} d(x_m, x_n) = 0$, a set in a metric space is complete if every Cauchy sequence in it converges to a point in it — compare with Section 5.1.) Since metric spaces are Hausdorff, compact sets in a metric space are closed (Lemma 11.2(d,g)). Therefore, *compact sets in a metric space are closed and bounded*. The Heine–Borel Theorem states the converse for the metric space $\mathbb{F}^n$ (where $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$) equipped with any of its usual metrics: *in $\mathbb{F}^n$, compact means closed and bounded* (see, e.g., [43, Theorem 3.83 and Corollary 4.32]). Since $\mathbb{F}^n$ is a complete metric space, it follows by the Compactness Theorem that what the Heine–Borel Theorem says is that in $\mathbb{F}^n$ boundedness implies total boundedness (i.e., in $\mathbb{F}^n$, boundedness coincides with total boundedness). Note that the metric (thus Hausdorff) space $\mathbb{F}^n$ is locally compact, but the metric space ℓ^p is not. In fact, $\mathbb{F}^n$ *is a prototype of a locally compact Hausdorff space*.

(b) Boundedness in locally compact spaces. In light of these facts, a set in a locally compact space is said to be *bounded* (or *topologically bounded*) if it is included in a compact set. So, also in this case, *every compact set is bounded* (i.e., topologically bounded); in fact, every relatively compact set is bounded. Thus, in this case, a closed and bounded set is a closed set included in a compact set, and hence (by Lemma 11.2(a)) *a closed and bounded set is compact*. The converse holds in a locally compact Hausdorff space, where compact sets are closed (Lemma 11.2(g)): *in a locally compact Hausdorff*

space, compact means closed and bounded (where bounded means relatively compact — Lemma 11.2(b)). A set in a locally compact space is *σ-bounded* if it is included in a σ-compact set. Finally, note that *in a locally compact complete metric space, topologically bounded means totally bounded*. Indeed, B totally bounded $\Longrightarrow \overline{B}$ closed and totally bounded $\Longleftrightarrow B$ compact (reason: in a complete metric space a set is complete if and only if it closed) $\Longrightarrow B$ topologically bounded $\Longrightarrow \overline{B}$ closed and topologically bounded $\Longleftrightarrow$ $\overline{B}$ compact $\Longrightarrow \overline{B}$ totally bounded $\Longrightarrow B$ totally bounded.

Remark 11.B (Denseness): It is readily verified (cf. Proposition 11.B) that *if a topological space has a countable base, then it is separable*. The converse is not true in general. Example: if X is uncountable with topology consisting of the empty set and the complements of the finite sets, then X is separable but has no countable base (see, e.g., [37, p. 49]). However, the converse holds in a metric space: *a metric space is separable if and only if it has a countable base* (see, e.g., [43, Theorem 3.35]).

Theorem 11.3. *In a locally compact Hausdorff space X, if K is compact, U is open, and $K \subseteq U$, then there exists G open with compact closure such that*

$$K \subseteq G \subseteq \overline{G} \subseteq U.$$

In particular, for each $x \in U$ there exists an open neighborhood N of x and a compact C, such that $N \subseteq C \subseteq U$.

Proof. Since X is locally compact, every point of X, in particular, every point x of $K \subseteq X$ has a compact neighborhood, and therefore an open neighborhood B_x with a compact closure. Thus, $K \subseteq \bigcup_x B_x$. Since K is compact, this open covering has a finite subcovering $\{B_i\}$ such that $K \subseteq \bigcup_i B_i$. Set $B = \bigcup_i B_i$, so that B is open and $K \subseteq B$, and hence $\overline{B} = \bigcup_i \overline{B}_i$ (recall: the closure of a finite union of sets coincides with the union of their closures). Since each $\overline{B}_i$ is compact, and since a finite union of compact sets is clearly compact, it follows that $\overline{B}$ is compact. If the open set U is such that $U = X$, then the result is verified with $G = B$ (i.e., $K \subseteq B \subseteq \overline{B} \subseteq X$). Thus suppose $U \subset X$, and take an arbitrary y in the closed set $V = X \backslash U$. Since $K \subseteq U$, it follows that $y \in X \backslash K$. Thus, since X is Hausdorff, Lemma 11.2(f) ensures that there exists an open set U'_y with $K \subseteq U'_y$ and $y \notin U'_y$. Set $K_y = V \cap \overline{B} \cap \overline{U'_y}$ which is compact by Lemma 11.2(h) — $\overline{B}$ is compact and X is Hausdorff. Since $\{K_y\}_{y \in X \backslash U}$ is a collection of compact sets such that $\bigcap_y K_y = \varnothing$ (because $\bigcap_y U'_y = \varnothing$ — if there exists $z \in X \backslash U$ such that $z \in \bigcap_y U'_y$, then $z \in U'_z$, which is a contradiction), it follows by Lemma 11.2(i) that there exists a finite subcollection, say $\{K_i\}$, such that $\bigcap_i K_i = \varnothing$. Hence $(X \backslash U) \cap \overline{B} \cap \bigcap_i \overline{U'_i} = \varnothing$ so that $\overline{B} \cap \bigcap_i \overline{U'_i} \subseteq U$. Recall that $K \subseteq B \cap \bigcap_i U'_i$. Thus, by setting $G = B \cap \bigcap_i U'_i$, which is open (finite intersection of open sets), we get $\overline{G} \subseteq \overline{B} \cap \bigcap_i \overline{U'_i}$, and so

$$K \subseteq G \subseteq \overline{G} \subseteq U.$$

In particular, since a singleton is clearly compact, the above inclusion ensures that $\{x\} \subseteq N \subseteq C \subseteq U$ for every $x \in U$, where $\{x\} = K$ is compact, $N = G$ is an open neighborhood of x, and $C = \overline{G}$ is compact. □

Remark 11.C (Borel sets, Borel σ-algebras, and Borel functions):

(i) Let X be a topological space with topology $\mathcal{T}$. Define, as in Problem 1.13:

$$\mathcal{X}_{\mathcal{T}} \text{ is the } \sigma\text{-algebra generated by the topology } \mathcal{T}.$$

That is, $\mathcal{X}_{\mathcal{T}}$ is the smallest σ-algebra of subsets of X that includes $\mathcal{T}$. The elements of $\mathcal{X}_{\mathcal{T}}$ (i.e., the $\mathcal{X}_{\mathcal{T}}$-measurable sets) are referred to as the *Borel sets* of X — see the remark that follows Problem 1.14.

> A *Borel σ-algebra* of subsets of X is any σ-algebra $\mathcal{A}$ of subsets of X that includes $\mathcal{X}_{\mathcal{T}}$; in particular, $\mathcal{X}_{\mathcal{T}}$ itself.

Recall that all open and all closed subsets of X are Borel sets (as they lie in $\mathcal{X}_{\mathcal{T}}$). A subset of X is a G_δ (read: *G-delta*) if it is a countable intersection of open subsets of X, and an F_σ (read: *F-sigma*) if it is a countable union of closed subsets of X. These are also Borel sets. In a Hausdorff space, all compact sets are Borel sets, since they are closed by Lemma 11.2(g).

(ii) Let $f\colon X \to Y$ be a function between topological spaces X and Y.

> (a) f is *continuous* if the inverse images of open sets are open, equivalently, if the inverse images of closed sets are closed, and
>
> (b) f is *$\mathcal{X}$-measurable* if the inverse images of open sets (or of closed sets) are measurable with respect to a σ-algebra $\mathcal{X}$ of subsets of X.

In other words, if X and Y are topological spaces with topologies $\mathcal{T}_X$ and $\mathcal{T}_Y$, and if $f\colon X \to Y$ is a function of X into Y, then

$$f\colon X \to Y \text{ is continuous} \iff f^{-1}(U) \in \mathcal{T}_X \text{ for every } U \in \mathcal{T}_Y$$

and, if $\mathcal{X}$ is an arbitrary σ-algebra of subsets of X, then

$$f\colon X \to Y \text{ is } \mathcal{X}\text{-measurable} \iff f^{-1}(U) \in \mathcal{X} \text{ for every } U \in \mathcal{T}_Y.$$

Theorem 11.4(b) below says that *a measurable function is precisely a measurable transformation* (in the sense of Problem 1.8) if $\mathcal{Y} = \mathcal{Y}_{\mathcal{T}}$, that is, if $\mathcal{Y}$ is the Borel σ-algebra $\mathcal{Y}_{\mathcal{T}}$ of subsets of Y generated by the topology $\mathcal{T}_Y$ on Y:

$$f^{-1}(U) \in \mathcal{X} \text{ for every } U \in \mathcal{T}_Y \iff f^{-1}(E) \in \mathcal{X} \text{ for every } E \in \mathcal{Y}_{\mathcal{T}}.$$

In particular, if $\mathcal{A}$ is an arbitrary Borel σ-algebra of subsets of X (i.e., if $\mathcal{A}$ is a σ-algebra of subsets of X such that $\mathcal{X}_{\mathcal{T}} \subseteq \mathcal{A}$, where $\mathcal{X}_{\mathcal{T}}$ is the σ-algebra generated by a topology $\mathcal{T}_X$ on X), and if $\mathcal{T}_Y$ is a topology on Y, then

(c) $f\colon X \to Y$ is *Borel measurable* if it is $\mathcal{A}$-measurable,

that is, if $f^{-1}(U) \in \mathcal{A}$ for every $U \in \mathcal{T}_Y$. Moreover,

(d) a *Borel function* is simply a Borel measurable function.

According to (a), (b), (c), and (d), since $\mathcal{T}_X \subseteq \mathcal{X}_\mathcal{T}$ and $f^{-1}(U) \in \mathcal{T}_X$ for every $U \in \mathcal{T}_Y$ implies $f^{-1}(U) \in \mathcal{X}_\mathcal{T} \subseteq \mathcal{A}$ for every $U \in \mathcal{T}_Y$, it is clear that

continuous functions are measurable functions

in the sense that

continuous functions are Borel functions.

Theorem 11.4. *Let $\mathcal{X}$ be a σ-algebra of subsets of a set X, let Y and Z be topological spaces, and consider the functions $f\colon X \to Y$ and $g\colon Y \to Z$.*

(a) *The collection $\mathcal{Y} = \{E \subseteq Y\colon f^{-1}(E) \in \mathcal{X}\}$ is a σ-algebra of subsets of Y.*

(b) *If f is $\mathcal{X}$-measurable, then $\mathcal{Y}_\mathcal{T} \subseteq \mathcal{Y}$ (i.e., $\mathcal{Y}$ is a Borel σ-algebra). In other words, if f is $\mathcal{X}$-measurable, then $f^{-1}(E) \in \mathcal{X}$ for every $E \in \mathcal{Y}_\mathcal{T}$.*

(c) *If $Y = \mathbb{R}$ (equipped with the usual topology), then f is $\mathcal{X}$-measurable if and only if $f^{-1}((\alpha, \infty)) \in \mathcal{X}$ for every $\alpha \in \mathbb{R}$.*

(d) *If g is a Borel function and f is $\mathcal{X}$-measurable, then the composition $h = g \circ f\colon X \to Z$ is $\mathcal{X}$-measurable.*

(e) *If g is continuous and f is $\mathcal{X}$-measurable, then the composition $h = g \circ f\colon X \to Z$ is $\mathcal{X}$-measurable.*

Proof. Let $\mathcal{X}$ be a σ-algebra of subsets of X.

(a) This goes in the opposite direction of Problem 1.9. Indeed, $f^{-1}(\varnothing) = \varnothing$ and $f^{-1}(Y) = X$. Moreover, $f^{-1}(Y \backslash E) = X \backslash f^{-1}(E)$ for every $E \in \mathcal{X}$ and $f^{-1}\big(\bigcup_n E_n\big) = \bigcup_n f^{-1}(E_n)$ for every countable collection $\{E_n\}$ of $\mathcal{X}$-measurable sets. Thus $\mathcal{Y}$ is a σ-algebra according to Definition 1.1.

(b) If f is $\mathcal{X}$-measurable (inverse images of open sets are measurable), then the σ-algebra $\mathcal{Y}$ of item (a) includes all open sets of Y. That is, it includes the topology $\mathcal{T}_Y$ on Y: $\mathcal{T}_Y \subseteq \mathcal{Y}$. Thus, since $\mathcal{Y}$ is a σ-algebra that includes $\mathcal{T}_Y$, and $\mathcal{Y}_\mathcal{T}$ is the smallest σ-algebra that includes $\mathcal{T}_Y$ (i.e., the σ-algebra generated by the topology $\mathcal{T}_Y$), it follows that $\mathcal{Y}_\mathcal{T} \subseteq \mathcal{Y}$.

(c) Let $Y = \mathbb{R}$ equipped with the usual topology, take $f\colon X \to Y$, and set $\mathcal{Y} = \{E \subseteq Y\colon f^{-1}(E) \in \mathcal{X}\}$. Take an arbitrary $\alpha \in \mathbb{R}$. Suppose $f^{-1}(\alpha, \infty)$ lies in $\mathcal{X}$. Thus (α, ∞) lies in $\mathcal{Y}$. Let $\{\alpha_n\}$ be a real-valued sequence converging to α and such that $\alpha_n < \alpha$ for each n. Since (i) each $(\alpha_n, \infty) \in \mathcal{Y}$, (ii) $(-\infty, \alpha) = \bigcup_n (-\infty, \alpha_n] = \bigcup_n Y \backslash (\alpha_n, \infty)$, and (iii) $\mathcal{Y}$ is a σ-algebra by (a), it follows that $(-\infty, \alpha) \in \mathcal{Y}$. So $(\alpha, \beta) = (-\infty, \beta) \cap (\alpha, \infty) \in \mathcal{Y}$ for every

$\beta \in \mathbb{R}$; that is, intervals of the real line are $\mathcal{Y}$-measurable. Since every open set of $\mathbb{R}$ is a countable union of open intervals — every separable metric space has a countable topological base of open balls (see, e.g., [43, Corollary 3.16 and Theorem 3.35]), and since $\mathcal{Y}$ is a σ-algebra, it follows that every open set in $\mathbb{R}$ lies in $\mathcal{Y}$, and hence, according to the definition of $\mathcal{Y}$, $f^{-1}(U) \in \mathcal{X}$ for every open set $U \subseteq \mathbb{R}$, which means that f is $\mathcal{X}$-measurable. Outcome: if $f^{-1}(\alpha, \infty) \in \mathcal{X}$, then f is $\mathcal{X}$-measurable. The converse is trivial since (α, ∞) is open in Y: if f is $\mathcal{X}$-measurable, then $f^{-1}(\alpha, \infty) \in \mathcal{X}$.

(d) Take $U \in \mathcal{T}_Z$, an arbitrary open subset of Z. If $g\colon Y \to Z$ is a Borel function, then $g^{-1}(U) \in \mathcal{Y}_\mathcal{T}$, where $\mathcal{Y}_\mathcal{T}$ is the Borel σ-algebra generated by the topology $\mathcal{T}_Y$. So, if f is $\mathcal{X}$-measurable, then $f^{-1}(g^{-1}(U)) \in \mathcal{X}$ by (b). If $h = g \circ f\colon X \to Z$, then $h^{-1}(C) = f^{-1}(g^{-1}(C))$ for every $C \subseteq Z$. (Indeed, $x \in h^{-1}(C) \iff h(x) \in C \iff g(f(x)) \in C \iff f(x) \in g^{-1}(C) \iff x \in f^{-1}(g^{-1}(C))$.) Thus $h^{-1}(U) \in \mathcal{X}$, and hence h is $\mathcal{X}$-measurable.

(e) This is a particular case of (d), since every continuous function is a Borel function (i.e., continuous functions are Borel measurable). □

11.2 Regular Measures

Set $X = \mathbb{R}$, the real line equipped with its usual (metric) topology, and take $\mathcal{X}_\mathcal{T} = \Re$, the Borel σ-algebra. That is, the σ-algebra of subsets of $\mathbb{R}$ generated by the open intervals, which coincides with the σ-algebra generated by the (usual) topology of $\mathbb{R}$ — i.e., by the open sets of $\mathbb{R}$ (cf. remark below Problem 1.14). Now take the extended real line $\overline{\mathbb{R}}$ (cf. warning above Problem 5.18). If a measure $\mu\colon \Re \to \overline{\mathbb{R}}$ is such that $\mu(K) < \infty$ for every compact subset K of $\mathbb{R}$ (i.e., every closed and bounded set), then it was called a Borel measure in Problem 2.13. (Note: $K \in \Re$ because closed subsets of $\mathbb{R}$ lie in $\Re$.) We generalize the notion of Borel measure from the concrete space $\mathbb{R}$ to general topological spaces and, in particular, to Hausdorff spaces.

Let $\mathcal{X}_\mathcal{T}$ be the Borel σ-algebra of subsets of a topological space X generated by a topology $\mathcal{T}$ on X. Every closed subset of X is a Borel set. Recall that if X is Hausdorff, then every compact set is closed (Lemma 11.2(g)), and hence a Borel set. Take any Borel σ-algebra $\mathcal{A}$ of subsets of X (i.e., any σ-algebra $\mathcal{A}$ including $\mathcal{X}_\mathcal{T}$; in particular, $\mathcal{X}_\mathcal{T}$ itself or its completion).

> A *Borel measure* is a (positive) measure $\mu\colon \mathcal{A} \to \overline{\mathbb{R}}$ on $\mathcal{A}$ (with $\mathcal{X}_\mathcal{T} \subseteq \mathcal{A}$) such that $\mu(K) < \infty$ for every compact set $K \in \mathcal{A}$.

Remark 11.D (Borel measures): Therefore (compare with Problem 2.13),

(a) every finite measure on a Borel σ-algebra $\mathcal{A}$ is a Borel measure,

(b) if X is σ-compact, then every Borel measure is σ-finite.

Let $\mathcal{A}$ be a Borel σ-algebra of subsets of a Hausdorff space X (where compact sets are Borel sets, so that if U is an open subset of X and K is a compact subset of X, then U and K lie in $\mathcal{A}$). An arbitrary set E in $\mathcal{A}$ is *outer regular* with respect to a measure $\mu\colon \mathcal{A} \to \overline{\mathbb{R}}$ on $\mathcal{A}$ if

$$\mu(E) = \inf\big\{\mu(U)\colon\ E \subseteq U,\ U \text{ open in } X\big\},$$

and *inner regular* with respect to a measure $\mu\colon \mathcal{A} \to \overline{\mathbb{R}}$ on $\mathcal{A}$ if

$$\mu(E) = \sup\big\{\mu(K)\colon\ K \subseteq E,\ K \text{ compact in } X\big\}.$$

A set E in $\mathcal{A}$ is *regular* with respect to μ if it is outer and inner regular with respect to μ. A measure μ on a Borel σ-algebra $\mathcal{A}$ is *regular, outer regular, or inner regular* (or *semiregular*) if every set E in $\mathcal{A}$ is regular, outer regular, or inner regular with respect to it. A set E in $\mathcal{A}$ is *quasiregular* with respect to μ if it is outer regular, and, if it is open, then it is inner regular; that is,

$$\mu(E) = \sup\big\{\mu(K)\colon\ K \subseteq E,\ K \text{ compact in } X\big\} \quad \text{if } E \text{ is open in } X.$$

The difference between regular and quasiregular is that a nonopen quasiregular set may not be inner regular. A measure μ on a Borel σ-algebra $\mathcal{A}$ is *quasiregular* if every E set in $\mathcal{A}$ is quasiregular with respect to it. Clearly, regularity implies quasiregularity: a regular measure is quasiregular.

Lemma 11.5. *Take the Borel σ-algebra $\mathcal{X}_\mathcal{T}$ of subsets of a Hausdorff space X. All sets below are Borel sets, and measures are Borel measures on $\mathcal{X}_\mathcal{T}$.*

(a) *A set of infinite measure is outer regular.*

(b) *A set of measure zero is inner regular.*

(c) *An open set is outer regular.*

(d) *A compact set is inner regular.*

(e) *A countable union of open sets is outer regular.*

(f) *A finite intersection of compact sets is inner regular.*

(g) *A finite intersection of open sets is outer regular.*

The analogous result for inner regular sets reads: A finite union of compact sets is inner regular. However, more is true, as stated in (i).

(h) *A countable intersection of open sets of finite measure is outer regular.*

(i) *A countable union of compact sets is inner regular.*

(j) *A finite disjoint union of inner regular sets of finite measure is inner regular.*

The analogous result for outer regular sets reads: A finite disjoint union of outer regular sets of finite measure is outer regular. However, more is true.

(k) *A countable union of outer regular sets is outer regular.*

(ℓ) *A countable intersection of inner regular sets of finite measure is inner regular.*

(m) *An increasing countable union of inner regular sets is inner regular.*

(n) *A decreasing countable intersection of outer regular sets of finite measure is outer regular.*

Proof. Consider the definitions of outer and inner regular Borel sets with respect to a Borel measure μ on $\mathcal{X}_{\mathcal{T}}$.

(a) This is immediate by the definition of outer regular sets: X is an open Borel set, and $\mu(E) = \infty$ implies $\mu(X) = \infty$ for every Borel set $E \subseteq X$.

(b) This is the dual of item (a), since $\varnothing$ is a compact Borel set with $\mu(\varnothing) = 0$, and $\varnothing \subseteq E$ for every Borel set E.

(c) This is trivial by the definition of outer regular sets (set $E = U$).

(d) This is the dual of item (c). Also trivial by the definition of inner regular sets (set $E = K$).

(e) A countable union of Borel sets is a Borel set, an arbitrary union of open sets is an open set, and open sets are outer regular by (c).

(f) This is the dual of item (e). A countable intersection of Borel sets is a Borel set, a finite intersection of compact sets is a compact set, and compact sets are inner regular by (d).

(g) A countable intersection of Borel sets is a Borel set, a finite intersection of open sets is an open set, and open sets are outer regular by (c).

(h) Let $\{E_k\}_{k=1}^{\infty}$ be a sequence of open sets with $\mu(E_k) < \infty$. Take a sequence $\{U_n\}_{n=1}^{\infty}$ of open sets with $U_n = \bigcap_{i=1}^{n} E_i$. Set $E = \bigcap_{k=1}^{\infty} E_k = \bigcap_{n=1}^{\infty} U_n$, which is a Borel set. Since $\{U_n\}_{n=1}^{\infty}$is a decreasing sequence with $\mu(U_n) \leq \min_{1 \leq i \leq n} \mu(E_i) < \infty$ for each n, it follows by Proposition 2.2(d) that

$$\mu(E) = \lim_{n} \mu(U_n),$$

where $\{U_n\}_{n=1}^{\infty}$ is a decreasing sequence of open sets such that $E \subseteq U_n$, which implies that E is outer regular.

(i) This is the dual of item (h). Let $\{E_k\}_{k=1}^{\infty}$ be a sequence of compact sets. Take a sequence $\{K_n\}_{n=1}^{\infty}$ of compact sets with $K_n = \bigcup_{i=1}^{n} E_i$. Set $E = \bigcup_{k=1}^{\infty} E_k = \bigcup_{n=1}^{\infty} K_n$, which is a Borel set. Since $\{K_n\}_{n=1}^{\infty}$is an increasing sequence, it follows by Proposition 2.2(c) that

$$\mu(E) = \lim_n \mu(K_n),$$

where $\{K_n\}_{n=1}^{\infty}$ is an increasing sequence of compact sets such that $K_n \subseteq E$. This implies that E is inner regular.

(j) Take an arbitrary $\varepsilon > 0$. If $\{E_i\}_{i=1}^n$ is a finite collection of disjoint inner regular Borel sets of finite measure, then there is a compact set $K_i \subseteq E_i$ such that $\mu(E_i) \le \mu(K_i) + \frac{\varepsilon}{n}$ for each i. Set $K = \bigcup_{i=1}^n K_i$, which is compact (finite union of compact sets), and $E = \bigcup_{i=1}^n E_i$. These are Borel sets such that $K \subseteq E$. Since $\{E_i\}_{i=1}^n$ is a disjoint collection,

$$\mu(E) = \sum_{i=1}^n \mu(E_i) \le \sum_{i=1}^n \mu(K_i) + \varepsilon = \mu(K) + \varepsilon,$$

which implies that E is inner regular.

(k) Again take an arbitrary $\varepsilon > 0$. If $\{E_k\}_{k=1}^{\infty}$ is a sequence of outer regular Borel sets, then there exists an open set U_k such that $E_k \subseteq U_k$ and $\mu(U_k) \le \mu(E_k) + \frac{\varepsilon}{2^k}$ for each k. Set $U = \bigcup_{k=1}^{\infty} U_k$, which is open, and $E = \bigcup_{k=1}^{\infty} E_k$. These are Borel sets (countable unions of Borel sets) such that $E \subseteq U$. If $\mu(E) = \infty$, then E is outer regular by (c). If $\mu(E) < \infty$, then

$$\begin{aligned}\mu(U) - \mu(E) = \mu(U\backslash E) &\le \mu\Big(\bigcup_{k=1}^{\infty} U_k\backslash E_k\Big) \\ &\le \sum_{k=1}^{\infty} \mu(U_k\backslash E_k) = \sum_{k=1}^{\infty} \big(\mu(U_k) - \mu(E_k)\big) \le \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon.\end{aligned}$$

Thus $\mu(U) \le \mu(E) + \varepsilon$, which implies that E is outer regular.

(ℓ) This is the dual of item (k). Take an arbitrary $\varepsilon > 0$. If $\{E_k\}_{k=1}^{\infty}$ is a sequence of inner regular Borel sets such that $\mu(E_k) < \infty$ for every k, then there exists a compact set K_k such that $K_k \subseteq E_k$ and $\mu(E_k) \le \mu(K_k) + \frac{\varepsilon}{2^k}$ for each k. Set $K = \bigcap_{k=1}^{\infty} K_k$, which is compact, and $E = \bigcap_{k=1}^{\infty} E_k$. These are Borel sets (countable intersections of Borel sets) such that $K \subseteq E$. Moreover, since $E\backslash K = \bigcap_{k=1}^{\infty} E_k\backslash(\bigcap_{k=1}^{\infty} K_k) \subseteq \bigcup_{k=1}^{\infty}(E_k\backslash K_k)$, we get

$$\begin{aligned}\mu(E) - \mu(K) = \mu(E\backslash K) &\le \mu\Big(\bigcup_{k=1}^{\infty} E_k\backslash K_k\Big) \\ &\le \sum_{k=1}^{\infty} \mu(E_k\backslash K_k) = \sum_{k=1}^{\infty} \big(\mu(E_k) - \mu(K_k)\big) \le \sum_{i=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon.\end{aligned}$$

Thus $\mu(E) \le \mu(K) + \varepsilon$, which implies that E is inner regular.

(m) Suppose $\{E_k\}_{k=1}^{\infty}$ is a increasing sequence of inner regular Borel sets. Set $E = \bigcup_{k=1}^{\infty} E_k$, which is a Borel set. If $\mu(E) = 0$, then E is inner regular

by item (b). Thus suppose $\mu(E) > 0$. Take any real number δ such that $0 < \delta < \mu(E)$. Since $\{E_k\}_{k=1}^{\infty}$ is increasing, it follows that $\delta < \sup_k \mu(E_k) = \lim_k \mu(E_k) = \mu(\bigcup_{k=1}^{\infty} E_k) = \mu(E)$ — cf. Proposition 2.2(c). Since each E_k is inner regular (and recalling again that $\{E_k\}_{k=1}^{\infty}$ is increasing), there exists a compact set K such that $K \subseteq \sup_k E_k = \bigcup_{k=1}^{\infty} E_k = E$ and $\delta < \mu(K) \leq \sup_k \mu(E_k) = \mu(E)$. Summing up: for every $0 < \delta < \mu(E)$ there exists a compact set K such that $K \subseteq E$ and

$$\delta < \mu(K) \leq \mu(E),$$

which implies that E is inner regular.

(n) This is the dual of item (m). Suppose $\{E_k\}_{k=1}^{\infty}$ is a decreasing sequence of outer regular Borel sets with $\mu(E_k) < \infty$. Set $E = \bigcap_{k=1}^{\infty} E_k$, which is a Borel set with $\mu(E) < \infty$. Take any $\delta > 0$ such that $\mu(E) < \delta$. Since $\{E_k\}_{k=1}^{\infty}$ is decreasing, $\mu(E) = \mu(\bigcap_{k=1}^{\infty} E_k) = \lim_k \mu(E_k) = \inf_k \mu(E_k) < \delta$ — cf. Proposition 2.2(d). Since each E_k is outer regular (and since $\{E_k\}_{k=1}^{\infty}$ is decreasing), there is an open set U such that $E = \bigcap_{k=1}^{\infty} E_k = \inf_k E_k \subseteq U$ and $\mu(E) = \inf_k \mu(E_k) \leq \mu(U) < \delta$. Summing up: for every $\mu(E) < \delta$ there is an open set U such that $E \subseteq U$ and

$$\mu(E) \leq \mu(U) < \delta.$$

This implies that E is outer regular. □

Theorem 11.6. *Let μ be a Borel measure on the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ of subsets of a locally compact Hausdorff space X. The following assertions are pairwise equivalent.*

(a) *Every compact set is outer regular.*

(b) *Every bounded open set is inner regular.*

(c) *Every bounded set is regular.*

(d) *Every σ-bounded set is regular.*

If X is σ-compact, then the above equivalent assertions are also equivalent to the following assertion.

(e) *The Borel measure $\mu\colon \mathcal{X}_{\mathcal{T}} \to \overline{\mathbb{R}}$ is regular.*

Proof. Sets are subsets of a locally compact Hausdorff space (where compact means closed and bounded, and bounded means relatively compact), and μ is a Borel measure, thus finite at every Borel set included in a compact set.

(a)⇒(b) Take an arbitrary $\varepsilon > 0$. Let U be an arbitrary bounded open set. So $U \subseteq K$ for some compact set K. Note that $K \backslash U$ is compact (intersection

of a compact set with a closed set in a Hausdorff space — Lemma 11.2(h)). If every compact set is outer regular, then $K\backslash U$ is outer regular, so that there exists an open set G such that $K\backslash U \subseteq G$ and $\mu(G) \le \mu(K\backslash U) + \varepsilon$. Note that $K\backslash G$ is compact and that $K\backslash G \subseteq K\backslash(K\backslash U) = U$. Thus

$$\begin{aligned}\mu(U) - \mu(K\backslash G) &= \mu(U\backslash(K\backslash G)) = \mu(U \cap G) = \mu(G \cap (K\backslash[K\backslash U])) \\ &\le \mu(G\backslash[K\backslash U]) = \mu(G) - \mu(K\backslash U) \le \varepsilon.\end{aligned}$$

Hence $\mu(U) \le \mu(K\backslash G) + \varepsilon$, which implies that U is inner regular.

(b) $\Rightarrow$(a) Dually, again take an arbitrary $\varepsilon > 0$. Let K be an arbitrary compact set. Let U be a bounded open set such that $K \subseteq U$. Note that $U\backslash K$ is a bounded open set (intersection of a bounded open set with an open set). If every bounded open set is inner regular, then $U\backslash K$ is inner regular, so that there exists a compact set C such that $C \subseteq U\backslash K$ and $\mu(U\backslash K) \le \mu(C) + \varepsilon$. Note that $U\backslash C$ is open and that $K = U\backslash(U\backslash K) \subseteq U\backslash C$. Thus

$$\begin{aligned}\mu(U\backslash C) - \mu(K) &= \mu((U\backslash C)\backslash K) = \mu((U\backslash K)\backslash C) \\ &= \mu(U\backslash K) - \mu(C) \le \varepsilon.\end{aligned}$$

Hence $\mu(U\backslash C) \le \mu(K) + \varepsilon$, which implies that K is outer regular.

(d) $\Rightarrow$(c) $\Rightarrow$(a,b) Trivial.

(c) $\Rightarrow$(d) Let $E = \bigcup_k E_k$ be an arbitrary σ-bounded Borel set, where each E_k is a Borel bounded set. If (c) holds, then each E_k is, in particular, outer regular. Thus Lemma 11.5(k) ensures that E is outer regular. Now set $F_n = \bigcup_{i=1}^{n} E_i$, so that $\{F_n\}_{n=1}^{\infty}$ is an increasing sequence of bounded Borel sets, which are, in particular, inner regular according to (c). Since $E = \bigcup_n F_n$, it follows by Lemma 11.5(k) that E is inner regular. Therefore, E is regular.

(a,b) $\Rightarrow$(c) Take an arbitrary $\varepsilon > 0$. Let E be any bounded Borel set, so that $E^\circ \subseteq E \subseteq \overline{E}$, where the interior E^o is open and bounded, and the closure $\overline{E}$ is compact. If (a,b) holds, then E° and $\overline{E}$ are regular (cf. Lemma 11.5(c,d)). Therefore, since $\mu(\overline{E}) < \infty$, Proposition 11.F(a) ensures that there exist compact sets C and K and open sets G and U such that $C \subseteq E^\circ \subseteq G$ and $K \subseteq \overline{E} \subseteq U$, for which $\mu(G\backslash C) = \mu(G) - \mu(C) < \varepsilon$ and $\mu(U\backslash K) = \mu(U) - \mu(K) < \varepsilon$. In fact, we may take K and G such that $\mu(K) \le \mu(G)$ (e.g., set $G = E^\circ$ and $K = \overline{E}$). Thus $C \subseteq E \subseteq K$ and

$$\begin{aligned}\mu(U) - \mu(C) &= \mu(U) - \mu(K) + \mu(K) - \mu(C) \\ &\le \mu(U) - \mu(K) + \mu(G) - \mu(C) \le 2\varepsilon,\end{aligned}$$

which means by Proposition 11.F(a) that E is regular.

(d) $\Leftrightarrow$(e) Assertion (e) implies (d) trivially and, if X is σ-compact, then every Borel set is σ-bounded, so that (d) implies (e). □

11.3 Construction of Borel Measures

An *outer measure* (or a *plain outer measure*) is an extended real-valued set function $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ on the power set $\wp(X)$ of a given set X such that

(a) $\mu^*(\varnothing) = 0$,

(b) $\mu^*(S) \geq 0$ for every $S \in \wp(X)$ (*positiveness*),

(c) $\mu^*(S') \leq \mu^*(S)$ whenever $S' \subseteq S \subseteq X$,

(d) $\mu^*\big(\bigcup_n S_n\big) \leq \sum_n \mu^*(S_n)$ (*countable subadditivity*), for countable families $\{S_n\}$ of sets in $\wp(X)$.

Observe that the *outer measure generated by a measure on an algebra*, as in Definition 8.2, is a (plain) outer measure according to Proposition 8.3. By analogy with Section 8.2, given an outer measure $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$, we say that a set $E \in \wp(X)$ is μ^*-*measurable* if

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \backslash E)$$

for every $S \in \wp(X)$. Also note that $S\backslash E = S \cap (X\backslash E) = S\backslash(S \cap E)$, and so $S = (S \cap E) \cup (S\backslash E)$ and $(S \cap E) \cap (S\backslash E) = \varnothing$. Since μ^* is subadditive (i.e., since (d) holds for the sets $S \cap E$ and $S\backslash E$), it follows that $E \in \wp(X)$ is μ^*-measurable if and only if

$$\mu^*(S \cap E) + \mu^*(S \backslash E) \leq \mu^*(S)$$

for every $S \in \wp(X)$. A *quasiregular outer measure* (or a *topologically regular outer measure*) on the power set $\wp(X)$ of a locally compact Hausdorff space X is an outer measure μ^* such that

(i) $\mu^*(S) = \inf\{\mu^*(U)\colon S \subseteq U,\ U \in \wp(X) \text{ open}\}$ for every $S \in \wp(X)$,

(ii) $\mu^*(G \cup U) = \mu^*(G) + \mu^*(U)$ if G and U are disjoint open in $\wp(X)$,

(iii) $\mu^*(U) = \sup\{\mu^*(K)\colon K \subseteq U,\ K \in \wp(X) \text{ compact}\}$ for U open in $\wp(X)$.

Property (iii) justifies the terminology *quasiregular*. It is readily verified that item (iii) is equivalent to the following property: for each open set $U \in \wp(X)$,

(iii′) $\mu^*(U) = \sup\{\mu^*(G)\colon \overline{G} \subseteq U,\ G \in \wp(X) \text{ open with } \overline{G} \text{ compact}\}$.

The next theorem is the counterpart of Theorem 8.4, building a Borel measure from an outer measure (instead of building a measure on a σ-algebra from an outer measure generated by a measure on an algebra).

Theorem 11.7. *If $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ is a quasiregular outer measure on the power set of a locally compact Hausdorff space X, then the collection $\mathcal{A}^*$ of all μ^*-measurable subsets of X,*

$$\mathcal{A}^* = \big\{E \in \wp(X)\colon\ E \text{ is } \mu^*\text{-measurable}\big\},$$

is a Borel σ-algebra (i.e., $\mathcal{X}_\mathcal{T} \subseteq \mathcal{A}^*$, meaning that $\mathcal{A}^*$ is a σ-algebra of subsets of X that includes the σ-algebra $\mathcal{X}_\mathcal{T}$). Also, the restriction

$$\eta = \mu^*|_{\mathcal{A}^*} : \mathcal{A}^* \to \overline{\mathbb{R}}$$

of μ^* to $\mathcal{A}^*$ is a quasiregular complete measure. Moreover, if $\mu^*(S) < \infty$ for every bounded set $S \subseteq X$, then $\eta\colon \mathcal{A}^* \supseteq \mathcal{X}_\mathcal{T} \to \overline{\mathbb{R}}$ is a Borel measure. Furthermore, the restriction of μ^* to the Borel σ-algebra $\mathcal{X}_\mathcal{T}$,

$$\mu = \eta|_{\mathcal{X}_\mathcal{T}} = \mu^*|_{\mathcal{X}_\mathcal{T}} : \mathcal{X}_\mathcal{T} \to \overline{\mathbb{R}},$$

is a quasiregular measure on $\mathcal{X}_\mathcal{T}$, which is a Borel measure if $\mu^*(S) < \infty$ for every bounded set $S \subseteq X$. (However, μ is not necessarily complete.)

Proof. Let $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ be a quasiregular outer measure on $\wp(X)$, where X is a locally compact Hausdorff space.

(a) That the class $\mathcal{A}^* = \{E \in \wp(X)\colon E \text{ is } \mu^*\text{-measurable}\}$ of all μ^*-measurable sets forms a σ-algebra has already been verified in the proof of Theorem 8.4.

(b) Now we show that every Borel set in the Borel σ-algebra $\mathcal{X}_\mathcal{T}$ is μ^*-measurable (i.e., $\mathcal{X}_\mathcal{T} \subseteq \mathcal{A}^*$). Let X be a locally compact Hausdorff space. It can be verified by property (i) of quasiregular outer measures (previous page) that $E \in \wp(X)$ is μ^*-measurable if and only if, for every open $U \in \wp(X)$,

$$\mu^*(U \cap E) + \mu^*(U \backslash E) \le \mu^*(U).$$

Since this holds trivially if $\mu^*(U) = \infty$, it follows that *a set E in $\wp(X)$ is μ^*-measurable if and only if the above inequality holds for every open set $U \in \wp(X)$ with $\mu^*(U) < \infty$*. Since $\mathcal{X}_\mathcal{T}$ is a σ-algebra generated by $\mathcal{T}$, and since $\mathcal{A}^*$ is a σ-algebra, in order to show that $\mathcal{X}_\mathcal{T} \subseteq \mathcal{A}^*$ it is enough to verify that the above inequality holds for every open set E in $\mathcal{T} \subseteq \mathcal{X}_\mathcal{T}$ (instead of for every set $E \in \mathcal{X}_\mathcal{T}$). That is, it suffices to show that each open set E is μ^*-measurable. So take arbitrary open sets E and U in $\mathcal{T} \subseteq \mathcal{X}_\mathcal{T} \subseteq \wp(X)$ with $\mu^*(U) < \infty$, and take an arbitrary $\varepsilon > 0$. Property (iii′) of quasiregular outer measures (previous page) says that for each open set U',

$$\mu^*(U') = \sup\big\{\mu^*(G)\colon\ \overline{G} \subseteq U',\ G \text{ open with } \overline{G} \text{ compact}\big\}.$$

As $U \cap E$ is open and $\mu^*(U \cap E) < \infty$, there is an open set G for which $G \subseteq U \cap E$ and $\mu^*(U \cap E) < \mu^*(G) + \varepsilon$. Note: $U \backslash E = U \cap (X \backslash E) \subseteq U \backslash G$ and $(U \backslash G) \cap G = \varnothing$ (so $\mu^*(G) + \mu^*(U \backslash G) = \mu^*(G \cup (U \backslash G))$ by property (ii) of quasiregular outer measures on the previous page. Therefore,

$$\begin{aligned}\mu^*(U \cap E) + \mu^*(U \backslash E) &< \mu^*(G) + \mu^*(U \backslash G) + \varepsilon = \mu^*(G \cup (U \backslash G)) + \varepsilon \\ &= \mu^*(U \cup G) + \varepsilon = \mu^*(U) + \varepsilon.\end{aligned}$$

Thus $\mu^*(U \cap E) + \mu^*(U \backslash E) \le \mu^*(U)$, and so E is μ^*-measurable.

(c) That the restriction $\eta = \mu^*|_{\mathcal{A}^*} : \mathcal{A}^* \to \overline{\mathbb{R}}$ of μ^* to $\mathcal{A}^*$ is a complete measure has been verified in the proofs of Theorem 8.4 and Proposition 8.5.

(d) Next we show that $\eta = \mu^*|_{\mathcal{A}^*} : \mathcal{A}^* \supseteq \mathcal{X}_{\mathcal{T}} \to \overline{\mathbb{R}}$ is a quasiregular measure. Take any set $E \in \mathcal{A}^*$. Since every open set U lies in $\mathcal{X}_{\mathcal{T}} \subseteq \mathcal{A}^*$, we get by property (i) of quasiregular outer measures at the beginning of this section that

$$\begin{aligned} \eta(E) = \mu^*(E) &= \inf\big\{\mu^*(U)\colon\ E \subseteq U,\ U \text{ open}\big\} \\ &= \inf\big\{\eta(U)\colon\ E \subseteq U,\ U \text{ open}\big\}. \end{aligned}$$

Also, since every compact set K lies in $\mathcal{X}_{\mathcal{T}} \subseteq \mathcal{A}^*$, it follows by property (iii) of quasiregular outer measures, again at the beginning of this section, that

$$\begin{aligned} \eta(E) = \mu^*(E) &= \sup\big\{\mu^*(K)\colon\ K \subseteq E,\ K \text{ compact}\big\} \\ &= \sup\big\{\eta(K)\colon\ K \subseteq E,\ K \text{ compact}\big\} \end{aligned}$$

whenever E is open (and so E lies in $\mathcal{X}_{\mathcal{T}} \subseteq \mathcal{A}^*$).

(e) That $\eta\colon \mathcal{A}^* \supseteq \mathcal{X}_{\mathcal{T}} \to \overline{\mathbb{R}}$ is a Borel measure goes as follows. If $\mu^*(S) < \infty$ for every bounded subset S of X (i.e., for every $S \in \wp(X)$ with compact closure), and since every compact set lies in $\mathcal{X}_{\mathcal{T}}$ and is closed because X is Hausdorff, we get $\eta(K) = \mu^*(K) < \infty$ for every compact set K.

(f) Finally, observe that all properties of the measure η except for completeness are readily transferred to the measure μ. □

A *content on a topology* (or simply a *content*) is an extended real-valued set function $\mu^\#\colon \mathcal{T} \to \overline{\mathbb{R}}$ on the topology $\mathcal{T}$ of a topological space X such that for arbitrary open sets U, G, and U_n in $\mathcal{T}$,

(1) $\mu^\#(\varnothing) = 0$,

(2) $\mu^\#(U) \ge 0$, (*positiveness*),

(3) $\mu^\#(U) < \infty$ whenever $\overline{U}$ is compact,

(4) $\mu^\#(G) \le \mu^\#(U)$ whenever $G \subseteq U$,

(5) $\mu^\#(G \cup U) = \mu^\#(G) + \mu^\#(U)$ whenever $G \cap U = \varnothing$,

(6) $\mu^\#\big(\bigcup_n U_n\big) \le \sum_n \mu^\#(U_n)$ (*countable subadditivity*), for countable families $\{U_n\}$ of sets in $\mathcal{T}$.

It is an *inner content* if, in addition,

(7) $\mu^\#(U) = \sup\{\mu^\#(G)\colon\ \overline{G} \subseteq U,\ \overline{G}\,\text{compact}\}$.

Lemma 11.8. *Let $\mu^\#\colon \mathcal{T} \to \overline{\mathbb{R}}$ be an inner content. If X is a locally compact Hausdorff space, then the set function $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ given by*

$$\mu^*(S) = \inf\{\mu^\#(U)\colon S \subseteq U,\ U \text{ open}\} \quad \text{for every} \quad S \in \wp(X)$$

is a quasiregular outer measure such that $\mu^*(S) < \infty$ *for every bounded* S.

Proof. Properties (a), (b), (c), and (d) in the definition of an outer measure μ^* follow at once by properties (1) — null measure of the empty set, (2) — nonnegativeness, (4) — monotonicity, and (6) — countable subadditivity in the definition of the content $\mu^\#$, respectively.

Thus μ^* is an outer measure.

Property (4) in the definition of the content $\mu^\#$ ensures that

$$\mu^*(U) = \mu^\#(U) \quad \text{for every open set } U.$$

Recall properties (i), (ii), and (iii′) in the definition of quasiregular outer measure at the beginning of this section. Thus property (i) holds by the definition of μ^*. Moreover, the above identity also shows that property (7) in the definition of an inner content ensures that property (iii′) of a quasiregular outer measure holds. Observe that property (5) in the definition of an inner content trivially implies property (ii) of a quasiregular outer measure.

Thus the outer measure μ^* is quasiregular.

Finally, property (3) in the definition of the content $\mu^\#$ and property (c) in the definition of an outer measure μ^* imply that

$$\mu^*(S) < \infty \text{ for every bounded set } S \in \wp(X)$$

(i.e., for every $S \in \wp(X)$ with compact closure). □

The combination of Lemma 11.8 and Theorem 11.7 concludes the program of constructing a Borel measure on an arbitrary topological compact Hausdorff space out of an inner content in two steps:

- ◇ an inner content $\mu^\#$ generates a quasiregular outer measure μ^* which is finite at bounded sets (by Lemma 11.8),
- ◇ and this generates a quasiregular *Borel measure* μ (by Theorem 11.7).

11.4 Additional Propositions

A collection $\mathcal{C}$ of sets has the *finite intersection property* if every finite subcollection of $\mathcal{C}$ has a nonempty intersection.

Proposition 11.A. *A space is compact if and only if every family of closed sets that has the finite intersection property has a nonempty intersection.*

A *Lindelöf* space is a topological space X such that every open covering of X includes a countable subcovering.

Proposition 11.B. *If a topological space has a countable base, then it is a Lindelöf space and separable. Every σ-compact space is a Lindelöf space. Every locally compact space with a countable base is σ-compact. A locally compact Hausdorff space is Lindelöf if and only if it is σ-compact.*

Along this line, recall that a *Polish space* is a separable topological space that is metrizable by a metric that makes it complete.

Proposition 11.C. *Take the Borel σ-algebra $\mathcal{X}_\mathcal{T}$ of subsets of a locally compact Hausdorff space. If $(E \cap K) \in \mathcal{X}_\mathcal{T}$ for every compact K, then $E \in \mathcal{X}_\mathcal{T}$.*

Proposition 11.D. *Let K_1 and K_2 be distinct compact sets in a Borel σ-algebra of subsets of a locally compact Hausdorff space.*

(a) *If every compact set is outer regular, then so is $K_1 \backslash K_2$.*

(b) *If every bounded open set is inner regular, then so is $K_1 \backslash K_2$.*

Proposition 11.E. *Let μ be a finite Borel measure on a Borel σ-algebra of subsets of a locally compact Hausdorff space. If every Borel set is inner regular, then μ is regular.*

Proposition 11.F. *Suppose μ is a Borel measure on a Borel σ-algebra of subsets of a locally compact Hausdorff space.*

(a) *A Borel set E of finite measure is regular if and only if for every $\varepsilon > 0$ there exists a compact set K and an open set U such that $K \subseteq E \subseteq U$ and $\mu(U \backslash K) < \varepsilon$.*

(b) *If E and F are regular Borel sets of finite measure, then $E \backslash F$ is also regular of finite measure.*

Proposition 11.G. *If μ is a Borel measure on a Borel σ-algebra of subsets of a locally compact Hausdorff space X, and if every open set is σ-compact, then μ is regular.* (Example: the Euclidean space $X = \mathbb{R}^n$ is a locally compact Hausdorff space where open sets are countable unions of closed balls.)

Proposition 11.H. *Lebesgue is a regular Borel measure. Either as a measure on the Borel σ-algebra $\Re$ generated by the topology of $\mathbb{R}$ or as a measure on its completion $\Im^*$, the Lebesgue σ-algebra, the Lebesgue measure is a regular Borel measure.* (See Section 8.3; compare with Problems 8.10 and 8.11.)

Recall from Problem 2.13 the definition of support of a measure on the Borel algebra $\Re$ generated by the open sets of $\mathbb{R}$, and extend it to the σ-algebra $\mathcal{X}_\mathcal{T}$ of Borel sets of a locally compact Hausdorff space X. The *support* $[\mu]$ of a measure $\mu\colon \mathcal{X}_\mathcal{T} \to \overline{\mathbb{R}}$ is the (unique) set $[\mu] = X \backslash U$, where U is the

union of all open sets of measure zero, so that $[\mu]$ is a closed set in $\mathcal{X}_\mathcal{T}$ such that $X\backslash[\mu]$ is the largest open set of measure zero:

$$X\backslash[\mu] = \sup\big\{U \in \mathcal{X}_\mathcal{T} \colon U \text{ is open in } X \text{ and } \mu(U) = 0\big\}.$$

Proposition 11.I. *If μ is a regular Borel probability measure (i.e., $\mu(X) = 1$) on a compact Hausdorff space X, then its support $[\mu]$ is the smallest compact set such that $\mu([\mu]) = 1$ and $\mu(K) < 1$ for every compact set $K \subset [\mu]$.*

Let X be a locally compact Hausdorff space, and define: $\mathcal{X}_G$ is the σ-algebra of subsets of X generated by the compact G_δ's in X. Sets in $\mathcal{X}_G$ are called *Baire sets*. Since every $G_\delta \subseteq X$ is a Borel set, we get

$$\mathcal{X}_G \subseteq \mathcal{X}_\mathcal{T}\colon \quad \text{every Baire set is a Borel set.}$$

The notions of outer, inner, regular, and quasiregular extend naturally to Baire sets (with respect to a measure on $\mathcal{X}_G$). Take a measure $\mu\colon \mathcal{X}_G \to \overline{\mathbb{R}}$.

If $\mu(K) < \infty$ for every compact $K \in \mathcal{X}_G$, then μ is a *Baire measure*.

Proposition 11.J. *If $E \in \mathcal{X}_G$, then E or $X\backslash E$ is σ-bounded. Every Baire set is σ-bounded if and only if X is σ-compact.*

Proposition 11.K. *A σ-bounded Baire set is a countable disjoint union of bounded Baire sets.*

Proposition 11.L. *Compact Baire sets are G_δ; open Baire sets are F_σ.*

Proposition 11.M. *Let μ be a measure on a σ-algebra including $\mathcal{X}_G$. If μ is inner regular or quasiregular, then for each measurable set E with finite measure there is a Baire set B such that $\mu(E \triangledown B) = 0$.* (See Problem 6.5.)

Proposition 11.N. *If X is a separable locally compact metric space, then $\mathcal{X}_G = \mathcal{X}_\mathcal{T}$* (i.e., *every Borel set is a Baire set*).

Proposition 11.O. *If X is a σ-compact locally compact Hausdorff space, then every Baire measure on $\mathcal{X}_G$ is regular.*

Let $\mu\colon \mathcal{X}_G \to \overline{\mathbb{R}}$ be a Baire measure on the σ-algebra $\mathcal{X}_G$ of all Baire sets in a locally compact Hausdorff space X, and let $[\mu] \in \mathcal{X}_G$ be the support of μ, now with respect to $\mathcal{X}_G$; that is, the set $[\mu] = X\backslash U$, where U is the union of all open Baire sets of measure zero.

Proposition 11.P. *Consider the support $[\mu]$ of a Baire measure μ.*

(a) *If U is an open Baire set such that $U \cap [\mu] \neq \varnothing$, then $\mu(U) > 0$.*

(b) *If K is a compact Baire set such that $K \cap [\mu] = \varnothing$, then $\mu(K) = 0$.*

(c) *If E is a σ-bounded Baire set such that $E \cap [\mu] = \varnothing$, then $\mu(E) = 0$.*

A measure on a Borel σ-algebra of subsets of a Hausdorff space X is *locally finite* if every $x \in X$ has an open neighborhood of finite measure. That is, a measure $\mu : \mathcal{A} \to \overline{\mathbb{R}}$, where $\mathcal{A}$ is a Borel σ-algebra of subsets of a Hausdorff space X, is *locally finite* if for every $x \in X$ there exists an open set $U \subseteq X$ such that $x \in U$ and $\mu(U) < \infty$.

Proposition 11.Q. *Every locally finite measure is a Borel measure.*

A *neighborhood base* of a point x in a topological space (or a *local base* at x) is a family of neighborhoods of x such that every neighborhood of x includes a member of the family (cf., e.g., [13, Problem 3.A(ii)] — in particular, the family of all open neighborhoods of x is trivially a neighborhood base of x). Note that *if a topological space has a countable base, then it is a topological space for which every point has a countable neighborhood base* (see Definition 11.1 and Proposition 11.B).

A measure μ on a Borel σ-algebra of subsets of a Hausdorff space is a *Radon measure* if it is both locally finite and inner regular.

Proposition 11.R. *Consider a Borel σ-algebra of subsets of a Hausdorff space X. Suppose every point in X has a countable neighborhood base. (Observe that such an assumption holds, in particular, if X has a countable base; more particularly, if X is separable.) Under the above assumption,*

(a) *every inner regular Borel measure is locally finite, and so*

(b) *every inner regular Borel measure is a Radon measure.*

Thus, by Proposition 11.Q, still under the above assumption,

(c) *a Radon measure is precisely an inner regular Borel measure.*

Proposition 11.S. *Suppose X is a separable σ-compact locally compact metric space. Then*

(a) *every Baire measure on $\mathcal{X}_G$ is regular,*

(b) *and consequently, in this case, every Borel measure on $\mathcal{X}_T$ is regular.*

We will see in the next chapter (Section 12.4, Proposition 12.H) that the above σ-compactness assumption can be dismissed, since it is a consequence of separability and local compactness in a metric space.

Consider a metric space (X, d). The *diameter* $\operatorname{diam}(S)$ of a nonempty subset S of X is given by $\operatorname{diam}(S) = \sup_{x,y \in S} d(x, y)$. So a nonempty set S is bounded in (X, d) if and only if $\operatorname{diam}(S) < \infty$. By convention the empty set $\varnothing$ is bounded and $\operatorname{diam}(\varnothing) = 0$. Take an arbitrary $\varepsilon > 0$. An ε-covering of a set $S \subseteq X$ is a covering of S made up of subsets of X of diameter not greater than ε. This notion is applied below to define *Hausdorff measures*.

Proposition 11.T. *Let X be a locally compact metric space. For every pair of real numbers $p \geq 0$ and $\varepsilon > 0$, consider a nonnegative extended real-valued set function defined by*

$$\mu^*_{\varepsilon,p}(S) \,=\, \inf \sum\nolimits_i \operatorname{diam}(S_i)^p \quad \textit{for every} \quad S \in \wp(X).$$

The infimum is taken over all finite open ε-coverings $\{S_i\}$ of S. This defines an outer measure $\mu^*_{\varepsilon,p} \colon \wp(X) \to \overline{\mathbb{R}}$. Take the limit $\lim_{\varepsilon \to 0} \mu^*_{\varepsilon,p}(S)$. The limit exists in $\overline{\mathbb{R}}$ and coincides with $\sup_{\varepsilon > 0} \mu^*_{\varepsilon,p}(S)$ for every $S \subseteq X$. Thus set

$$\mu^*_p(S) \,=\, \lim_{\varepsilon \to 0} \mu^*_{\varepsilon,p}(S) \,=\, \sup_{\varepsilon > 0} \mu^*_{\varepsilon,p}(S) \quad \textit{for every} \quad S \in \wp(X).$$

Such a set function $\mu^*_p \colon \wp(X) \to \overline{\mathbb{R}}$ is another outer measure. Following the setup of Theorem 11.7, let $\mathcal{A}^*_p$ be the Borel σ-algebra of all μ^*_p-measurable sets (so that $\mathcal{A}^*_p$ includes the Borel σ-algebra $\mathcal{X}_\mathcal{T}$). The restrictions of μ^*_p,

$$\eta_p \,=\, \mu^*_p|_{\mathcal{A}^*_p} \colon \mathcal{A}^*_p \to \overline{\mathbb{R}},$$

$$\mu_p \,=\, \eta_p|_{\mathcal{X}_\mathcal{T}} \,=\, \mu^*_p|_{\mathcal{X}_\mathcal{T}} \colon \mathcal{X}_\mathcal{T} \to \overline{\mathbb{R}},$$

are measures on $\mathcal{A}^*_p$ and on $\mathcal{X}_\mathcal{T}$, respectively.

The outer measure μ^*_p and the measures η_p and μ_p, are called p-dimensional Hausdorff measures.

Both η_p and μ_p are Borel measures if $\mu^*_p(S) < \infty$ for every bounded set S in $\wp(X)$, and η_p is complete. For each $S \in \wp(X)$ there is a unique real number

$$\dim_H(S) \,=\, \inf \big\{p \geq 0 \colon\, \mu^*_p(S) = 0\big\} \,=\, \sup \big\{p > 0 \colon\, \mu^*_p(S) = \infty\big\}$$

called the Hausdorff dimension of S, *which is such that*

$$\mu^*_p(S) = \infty \quad \textit{if} \quad p < \dim_H(S),$$

$$\mu^{*}{}_{\dim_H(S)}(S) \,\in\, [0, \infty],$$

$$\mu^*_p(S) = 0 \quad \textit{if} \quad \dim_H(S) < p.$$

Examples: (i) *If $X = \mathbb{R}$, then μ^*_1 coincides with the outer measure ℓ^* generated by the length function ℓ (so μ_1 coincides with the Lebesgue measure λ); if C is the Cantor set, then $\dim_H(C) = \log 2 / \log 3$ and $\mu^{*}{}_{\dim_H(C)}(C) = 1$.* (ii) *If U is an open set in $X = \mathbb{R}^n$, then $\dim_H(U) = n$.*

If $X = \mathbb{R}^n$ or $X = \mathbb{C}^n$, equipped with the usual (Euclidean) topology (or any equivalent topology), then X is a σ-compact locally compact metric (thus Hausdorff) space with a countable base (thus separable), and so it satisfies all the assumptions in every one of the preceding propositions.

Notes: Besides $\mathbb{R}^n$ and $\mathbb{C}^n$, compact Hausdorff spaces, and open subspaces of a compact Hausdorff space, are examples of locally compact Hausdorff spaces (e.g., [13, Problem 3.V]). Recall from Lemma 11.2(g) that a compact subset of a Hausdorff space is closed. Actually (e.g., [13, Proposition 3.2]),

> *a subset of a compact Hausdorff space is a compact Hausdorff space if and only if it is closed* — as in metric spaces, so also in a Hausdorff space, "inside" a compact set, compact is a synonym of closed.

For the general topology summarized in Section 11.1 see, e.g., [21] and [37]. This Section 11.4 contains standard results on measures on topological spaces whose proofs can be found in many texts. For instance, Proposition 11.A can be found in [37, Theorem 5.1], and Proposition 11.B in [37, Theorems 1.14, 1.15, Problem 5.Y(b)], [62, Problem 8.16, Theorem 9.21], and [8, Example IV.29-2(c)]. For Proposition 11.C see, e.g., [62, Lemma 13.9], and for Proposition 11.D, [31, Theorem 52.A]. Propositions 11.E and 11.F are related to regular Borel measures (e.g., [62, Proposition 13.10] and [13, Problem 10.F]). Proposition 11.G (e.g., [63, Theorem 2.18]) is a first step towards Proposition 11.H, which says that Lebesgue measure is a prototype of a regular Borel measure (see, e.g., [62, Proposition 3.15]). Proposition 11.I deals with support of probability measures; see, e.g., [63, Exercise 2.9]. Propositions 11.J, 11.K, 11.L, and 11.M introduce Baire sets and measures (e.g., [62, Lemmas 13.6, 13.7, Problem 13.12(e,f), Proposition 13.15]). Proposition 11.N shows when Baire and Borel coincide, and Proposition 11.O gives a necessary condition for regularity (see, e.g., [62, Problem 13.1, Corollary 13.12]). Proposition 11.P deals with supports of Baire measures (e.g., [62, Problem 13.24]). Propositions 11.Q and 11.R on locally finite and Radon measures establish the connection between Borel and Radon measure (e.g., [8, Definition 25.2, Eq. (25.7) and Lemma 25.4]).

Proof of Proposition 11.S. Recall that a metric space is Hausdorff (Lemma 11.2(d)). Thus part (a) of Proposition 11.S is a particular case of Proposition 11.O, and part (b) is an immediate consequence of part(a) since, in this case, $\mathcal{X}_G = \mathcal{X}_\mathcal{T}$ by Proposition 11.N, and therefore, in this case again, Baire measures are Borel measures. □

For the Hausdorff measures of Proposition 11.T see, e.g., [23, Chapter 1], [24, Chapter 3], and also [60].

Suggested Readings

Bauer [8], Brown and Pearcy [13], Bercovici, Brown, and Pearcy [10], Halmos [31], Kechris [35], Royden [62], Rudin [63].



12

Representation Theorems

12.1 Continuous Functions and Compact Support

Let $\mathbb{F}$ be either the real field $\mathbb{R}$ or the complex field $\mathbb{C}$. Recall that a *scalar-valued* function $f : X \to \mathbb{F}$ on a nonempty set X is called *real-valued* if $\mathbb{F} = \mathbb{R}$ and *complex-valued* if $\mathbb{F} = \mathbb{C}$. If it is not relevant whether a scalar-valued function is real-valued or complex-valued, then we also refer to it as $\mathbb{F}$-valued. The *kernel* $\mathcal{N}(f)$ and *range* $\mathcal{R}(f)$ of a function $f : X \to \mathbb{F}$ are the sets

$$\mathcal{N}(f) = f^{-1}(\{0\}) = \big\{x \in X : \ f(x) = 0\big\},$$

$$\mathcal{R}(f) = f(X) = \big\{\gamma \in \mathbb{F} : \ \gamma = f(x) \ \text{ for some } \ x \in X\big\}.$$

The *support* $\overline{[f]}$ of an $\mathbb{F}$-valued function $f : X \to \mathbb{F}$ on a topological space X is the closure of the complement of its kernel:

$$\overline{[f]} = \overline{(X \backslash \mathcal{N}(f))} = \overline{\big\{x \in X : \ f(x) \neq 0\big\}}.$$

From now on suppose X is a topological space and consider the usual (metric) topology of $\mathbb{F}$. Let $C_c(X)$ denote the collection of all $\mathbb{F}$-valued continuous functions on X with *compact support*; that is,

$$C_c(X) = \big\{f : X \to \mathbb{F} : \ f \text{ is continuous and } \overline{[f]} \text{ is compact in } X\big\}.$$

If it is necessary to make it clear whether a collection of scalar-valued continuous functions on X consists of real-valued functions only, or whether it may contain complex-valued functions as well, then we may use the notation

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_12

$$C_c(X,\mathbb{R}) = \big\{f\colon X\to\mathbb{R}\colon\ f \text{ is continuous and } \overline{[f]} \text{ is compact in } X\big\},$$

or

$$C_c(X,\mathbb{C}) = \big\{f\colon X\to\mathbb{C}\colon\ f \text{ is continuous and } \overline{[f]} \text{ is compact in } X\big\}.$$

It is worth noticing, *en passant*, that if X is a locally compact Hausdorff space, then the class of Baire sets coincides with the smallest σ-algebra of subsets of X on which all functions in $C_c(X,\mathbb{R})$ are measurable (see, e.g., [62, p. 331]), which in turn is included in the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ (Section 11.4).

Lemma 12.1. *The range of a function $f\in C_c(X)$ is a compact subset of $\mathbb{F}$ (the complex plane $\mathbb{C}$ or the real line $\mathbb{R}$, either equipped with its usual topology).*

Proof. Take $f\in C_c(X)$ with compact support K. As f is continuous and K is closed, $f(X)=f(K)$. If we show that $f(K)$ is compact in $\mathbb{F}$, then we are done.

Claim. Let $f\colon X\to Y$ be a continuous mapping of a topological space X into a topological space Y. If K is compact in X, then $f(K)$ is compact in Y (i.e., a continuous image of a compact set is compact).

Proof of Claim. Let $\mathcal{U}$ be a covering of $f(K)$ (i.e., $f(K)\subseteq\bigcup_{U\in\mathcal{U}}U$) consisting of open subsets U of Y. If f is continuous, then $f^{-1}(U)$ is an open subset of X for every $U\in\mathcal{U}$. Set $f^{-1}(\mathcal{U})=\{f^{-1}(U)\colon\ U\in\mathcal{U}\}$, a collection of open subsets of X. Since $K\subseteq f^{-1}(f(K))\subseteq f^{-1}\big(\bigcup_{U\in\mathcal{U}}U\big)=\bigcup_{U\in\mathcal{U}}f^{-1}(U)$, it follows that $f^{-1}(\mathcal{U})$ is a covering of K made up of open subsets of X. If K is compact, then there exists a finite subcollection of $f^{-1}(\mathcal{U})$ covering K; that is, there exists $\{U_i\}_{i=1}^n\subseteq\mathcal{U}$ such that $K\subseteq\bigcup_{i=1}^n f^{-1}(U_i)\subseteq X$. So $f(K)\subseteq f\big(\bigcup_{i=1}^n f^{-1}(U_i)\big)\subseteq\bigcup_{i=1}^n U_i\subseteq Y$, and hence $f(K)$ is compact. □

Take the collection $C_c(X)$ of all continuous $\mathbb{F}$-valued functions f on X with compact support $\overline{[f]}$. Let K be a compact subset, and U an open subset, of the topological space X (with topology $\mathcal{T}$). For each K and U set

$$C_c(X)_K = \big\{f\in C_c(X)\colon\ f(X)\subseteq[0,1] \text{ and } f(K)=\{1\}\big\},$$

$$C_c(X)^U = \big\{f\in C_c(X)\colon\ f(X)\subseteq[0,1] \text{ and } \overline{[f]}\subseteq U\big\}.$$

Observe that when dealing with $C_c(X)_K$ or $C_c(X)^U$, we work only with functions in $C_c(X,\mathbb{R})$, since $f(X)\subseteq[0,1]\subset\mathbb{R}$. A real-valued (or extended real-valued) function $f\colon X\to\mathbb{R}$ (or $f\colon X\to\overline{\mathbb{R}}$) on a topological space X is said to be *lower semicontinuous* if the inverse image of the open interval (α,∞) under f is an open subset of X for every real number α,

$$f^{-1}\big((\alpha,\infty)\big)=\big\{x\in X\colon\ f(x)>\alpha\big\}\in\mathcal{T}\quad\text{ for every }\quad\alpha\in\mathbb{R}$$

(see Definition 1.2), and *upper semicontinuous* if the inverse image of the open interval $(-\infty,\alpha)$ under f is an open subset of X for every real α,

$$f^{-1}\big((-\infty,\alpha)\big) = \big\{x \in X\colon\ f(x) < \alpha\big\} \in \mathcal{T} \quad \text{for every} \quad \alpha \in \mathbb{R}.$$

Equipping $\mathbb{R}$ with its usual topology, it is readily verified that *f is continuous if and only if it is both lower and upper semicontinuous*. Examples of functions that are either lower or upper semicontinuous (but possibly not both): a characteristic function $\chi_U\colon X \to \mathbb{R}$ of an open set $U \subseteq X$ is lower semicontinuous (compare with Example 1B) and, dually, a characteristic function $\chi_V\colon X \to \mathbb{R}$ of a closed set $V \subseteq X$ is upper semicontinuous (neither of which is continuous if $X = \mathbb{R}$ is equipped with its usual topology). Consider the definitions of $\sup_\gamma f_\gamma$ and $\inf_\gamma f_\gamma$ for a family $\{f_\gamma\}$ of functions $f_\gamma\colon X \to \mathbb{R}$ or $f_\gamma\colon X \to \overline{\mathbb{R}}$, as we did in Section 1.3 for countable families. It is readily verified that *$\sup_\gamma f_\gamma\colon X \to \overline{\mathbb{R}}$ is lower semicontinuous whenever each f_γ is*, and *$\inf_\gamma f_\gamma\colon X \to \overline{\mathbb{R}}$ is upper semicontinuous whenever each f_γ is*.

So we can interpret continuity of functions in $C_c(X)_K$ and in $C_c(X)^U$ as functions being both lower and upper semicontinuous.

Lemma 12.2. (Urysohn Lemma). *If U and K are, respectively, open and compact sets in a locally compact Hausdorff space X such that $K \subseteq U$, then $C_c(X)_K \cap C_c(X)^U \neq \varnothing$. That is, there is a function f in $C_c(X)_K \cap C_c(X)^U$, which means that there exists a function $f \in C_c(X)$ such that $\chi_K \leq f \leq \chi_U$.*

Proof. Take $K \subseteq U \subseteq X$, where U is an open set, K is a compact set, and X is a locally compact Hausdorff space.

Claim. There exists a countable family of open subsets of X with compact closure, say $\{G_q\}_{q\in\mathbb{Q}\cap[0,1]}$ indexed by the rationals in $[0,1]$, such that

$$K \subseteq \overline{G}_1, \quad G_0 \subseteq U, \quad \text{and} \quad \overline{G_q} \subseteq G_p \ \text{ whenever } \ p < q.$$

Proof of Claim. Let the set $\{q_k\}_{k=1}^{\infty}$ be an enumeration of the countable set $\mathbb{Q} \cap (0,1) \subset [0,1] \subset \mathbb{R}$ of all rational numbers in the open interval $(0,1)$. Set $q_0 = 0$ and $q_\infty = 1$. According to Theorem 11.3, there exist open subsets G_0 and G_1 of X with compact closure such that

$$K \subseteq G_1 \subseteq \overline{G_1} \subseteq G_0 \subseteq \overline{G_0} \subseteq U.$$

Take any $q \in \mathbb{Q} \cap (0,1)$. Again by Theorem 11.3, there is an open set with compact closure G_q associated with q such that

$$\overline{G_1} \subseteq G_q \subseteq \overline{G_q} \subseteq G_0.$$

Take any $q' \in \mathbb{Q} \cap (0,1)$ such that $q' \neq q$. Use Theorem 11.3 and get an open set with compact closure $G_{q'}$ such that if $q' < q$, then $G_q \subseteq G_{q'}$ and

$$\overline{G_1} \subseteq G_q \subseteq \overline{G_q} \subseteq G_{q'} \subseteq \overline{G_{q'}} \subseteq G_0,$$

while if $q < q'$, then $G_{q'} \subseteq G_q$ and

$$\overline{G_1} \subseteq G_{q'} \subseteq \overline{G_{q'}} \subseteq G_q \subseteq \overline{G_q} \subseteq G_0.$$

Proceeding along this line, if we already have a finite set $\{q_i\}_{i=1}^n$ of rational numbers from $\mathbb{Q} \cap (0,1)$, and an associated set of open sets with compact closure $\{G_{q_i}\}_{i=1}^n$ such that $G_{q_{i+1}} \subseteq G_{q_i}$ for $q_i < q_{i+1}$ and

$$G_{q_{i+1}} \subseteq \overline{G_{q_{i+1}}} \subseteq G_{q_i} \subseteq \overline{G_{q_i}},$$

we can then take any $q_{n+1} \in \mathbb{Q} \cap (0,1)$ such that $q_{n+1} \neq q_i$ for every integer i in $[1,n]$. Using Theorem 11.3 once again, we get an open set with compact closure $G_{q_{n+1}}$ such that

$$\overline{G_{q_j}} \subseteq G_{q_{n+1}} \subseteq \overline{G_{q_{n+1}}} \subseteq G_{q_\ell},$$

where q_ℓ is the largest number from $\{q_i\}_{i=1}^n$ that is smaller than q_{n+1}, and q_j is the smallest number from $\{q_i\}_{i=1}^n$ that is greater than q_{n+1}. Continuing this way, we obtain, for each $q \in \{q_k\}_{k=1}^\infty \cup \{q_0, q_\infty\} = \mathbb{Q} \cap [0,1]$, an open set with compact closure G_q, such that $K \subseteq G_1$, $\overline{G_0} \subseteq U$, and $\overline{G_q} \subseteq G_p$ whenever $p < q$. This concludes the proof of the claimed statement. □

For each $q \in \{q_k\}_{k=1}^\infty \cup \{q_0, q_\infty\} = \mathbb{Q} \cap [0,1]$, take the characteristic functions of G_q and $X \backslash \overline{G_q}$, and consider the following functions of X into $\mathbb{R}$:

$$f_q = q\,\chi_{G_q}, \qquad g_q = q\,\chi_{X\backslash\overline{G_q}} + \chi_{\overline{G_q}},$$

$$f = \sup_k f_{q_k}, \qquad g = \inf_k g_{q_k}.$$

As we saw before, characteristic functions of open sets are lower semicontinuous, as are the suprema of families of lower semicontinuous functions; dually, characteristic functions of closed sets are upper semicontinuous, as are the infima of families of upper semicontinuous functions. Therefore, it is readily verified that each f_q is a lower semicontinuous function, as is f, while each g_q is an upper semicontinuous function, as is g. Now we show that

$$f \leq g \quad \text{and} \quad g \leq f, \quad \text{and hence} \quad f = g.$$

Indeed, if $g_q(x) < f_p(x)$, then $q < p$, $x \in G_p$ and $x \notin \overline{G_q}$ (i.e., $x \in (G_p \backslash \overline{G_q})$). But $q < p$ implies $\overline{G_p} \subseteq G_q$ so that $G_p \backslash \overline{G_q} = \varnothing$. So $f_p < g_q$ for every p and q in $\mathbb{Q} \cap [0,1]$, which implies that $f \leq g$. On the other hand, if $f(x) < g(x)$ for some $x \in X$, then there are $p, q \in \mathbb{Q} \cap [0,1]$ such that $f(x) < p < q < g(x)$. But $f(x) < p$ implies that $x \notin G_p$, and $q < g(x)$ implies that $x \in \overline{G_q}$, so that $x \in \overline{G_q} \backslash G_p$. Again, $p < q$ implies $\overline{G_q} \subseteq G_p$, so that $\overline{G_q} \backslash G_p = \varnothing$. Then we get $g \leq f$. Hence $f = g$. Therefore, since f is lower semicontinuous, g is upper semicontinuous, and $f = g$, it follows that

$$f\colon X \to \mathbb{R} \text{ is continuous.}$$

Clearly, f_q has compact support $\overline{Gq} \subseteq [0,1]$ and $f_q(X) \subseteq [0,1]$ for every q in $\mathbb{Q} \cap [0,1]$, and so (i) f has compact support and (ii) $f(X) \subseteq [0,1]$. Since $K \subseteq \overline{[f_q]}$ for all $q \in \mathbb{Q} \cap [0,1]$ and $\sup(\mathbb{Q} \cap [0,1]) = 1$, we get (iii) $f(K) = \{1\}$. Since $\bigcup_q G_q \subseteq G_0 \subseteq U$, is also follows that (iv) $\overline{[f]} \subseteq U$. Outcome:

$$f \in C_c(X)_K \cap C_c(X)^U. \qquad \square$$

Corollary 12.3. *Let U_i and K be subsets of a locally compact Hausdorff space X. If $\{U_i\}_{i=1}^n$ is an open covering of the compact set K, then there is an $f_i \in C_c(X)^{U_i}$ for each $i \in [1,n]$ such that $\sum_{i=1}^n f_i(x) = 1$ for every $x \in K$.*

Proof. Since $K \subseteq U = \bigcup_{i=1}^n U_i$, where K is compact and U is open, there exists an open set G with compact closure (cf. Theorem 11.3) such that

$$K \subseteq G \subseteq \overline{G} \subseteq U.$$

Take an arbitrary integer $i \in [1,n]$. Set $G_i = G \cap U_i \subseteq U_i$, which is an open set (intersection of two open sets) with compact closure (since $\overline{(G \cap U_i)} \subseteq \overline{G}$, which is compact — see Lemma 11.2). Let K_i be the finite union of all sets $\overline{G_j}$ such that $G_j \subseteq U_i$. Thus K_i is compact (because each $\overline{G_j}$ is compact and K_i is a finite union of them) and $K_i \subseteq U_i$, where U_i is open. Therefore, by Lemma 12.2, there exists a function $g_i\colon X \to \mathbb{R}$ such that

$$g_i \in C_c(X)_{K_i} \cap C_c(X)^{U_i}.$$

Now set

$$f_1 = g_1 \quad \text{and} \quad f_i = \prod_{j=1}^{i-1}(1 - g_j)g_i \quad \text{if} \quad i \geq 2.$$

Since $g_i \in C_c(X)^{U_i}$, it follows by the definition of $C_c(X)^{U_i}$ that

$$f_i \in C_c(X)^{U_i}.$$

It is readily verified by induction that

$$\sum_{i=1}^n f_i = 1 - \prod_{i=1}^n (1 - g_i).$$

Next note that $K \subseteq \bigcup_{i=1}^n K_i$. Indeed, since $\bigcup_{i=1}^n \overline{G_i} = \bigcup_{i=1}^n K_i$, it follows that $K \subseteq G \subseteq \bigcup_{i=1}^n (G \cap U_i) = \bigcup_{i=1}^n G_i \subseteq \bigcup_{i=1}^n \overline{G_i} = \bigcup_{i=1}^n K_i$. If $g_i(x) < 1$ for every $i \in [1,n]$ for some $x \in K \subseteq \bigcup_{j=1}^n \overline{K_j}$, then $g_i \notin C_c(X)_{K_i}$, which is a contradiction. Thus for every $x \in K$ there exists an integer $i' \in [1,n]$ such that $g_{i'}(x) = 1$. Hence $1 - g_{i'}(x) = 0$, and so $\prod_{i=1}^n (1 - g_i(x)) = 0$, for every $x \in K$. Therefore,

$$\sum_{i=1}^n f_i(x) = 1 \quad \text{for every} \quad x \in K. \qquad \square$$

The preceding consequence of the Urysohn Lemma (i.e., Corollary 12.3), together with the Urysohn Lemma itself (i.e., Lemma 12.2), will play an important role for proving the first version of the Riesz Representation Theorem, namely, Theorem 12.5 in Section 12.3.

12.2 Bounded Linear Functionals

Let S be a nonempty set and consider the set $\mathbb{F}^S$ of all functions $f\colon S \to \mathbb{F}$ of S into $\mathbb{F}$ (i.e., of all $\mathbb{F}$-valued functions on S). It is readily verified that $\mathbb{F}^S$ is a *linear space* over the field $\mathbb{F}$, where addition and scalar multiplication in $\mathbb{F}^S$ are defined pointwise: the linear structure of $\mathbb{F}^S$ is inherited from the linear structure of $\mathbb{F}$. Let $L(S) \subseteq \mathbb{F}^S$ be any *linear manifold* of the linear space $\mathbb{F}^S$. That is, $L(S) \subseteq \mathbb{F}^S$ is such that $f+g$ and αf lie in $L(S)$ if $f, g \in L(S)$ and $\alpha \in \mathbb{F}$ — defined by $(f+g)(s) = f(s) + g(s)$ and $(\alpha f)(s) = \alpha f(s)$ for every $s \in S$. So $L(S)$ is itself a linear space over $\mathbb{F}$. If $\mathbb{F} = \mathbb{C}$ or $\mathbb{F} = \mathbb{R}$, then $L(S)$ is called a *complex linear space* or a *real linear space* and, if necessary, it will be denoted by $L(S, \mathbb{C})$ or $L(S, \mathbb{R})$ — see the remark that follows Lemma 4.5.

Take a *linear functional* $\Phi\colon L(S) \to \mathbb{F}$ on the linear space $L(S)$, which means an *additive* and *homogeneous* functional (i.e., $\Phi(f+g) = \Phi(f) + \Phi(g)$ and $\Phi(\alpha f) = \alpha\,\Phi(f)$ for every f, g in $L(S)$ and every α in $\mathbb{F}$). A functional Φ acting on $L(S)$ is said to be *positive* if $\Phi(f) \geq 0$ for every $f \geq 0$ (i.e., for every function $f \in L(S)$ whose range $f(S)$ is included in $[0, \infty) \subset \mathbb{R}$); that is, if

$$\Phi(f) \geq 0 \text{ for every } f \in L(S) \text{ such that } f(s) \geq 0 \text{ for every } s \in S.$$

Let $\Phi\colon L(S) \to \mathbb{F}$ be a positive linear functional on the linear space $L(S)$. If $f, g \in L(S)$ are such that $f \leq g$, then $0 \leq \Phi(g - f) = \Phi(g) - \Phi(f)$. So

$$\Phi(f) \leq \Phi(g) \quad \text{whenever} \quad f \leq g.$$

Moreover, if $f\colon S \to \mathbb{R}$ is a real-valued function in $L(S)$, then (considering the decomposition $f = f^+ - f^-$ of Section 1.2 where $f^+ \geq 0$ and $f^- \geq 0$)

$$\begin{aligned} |\Phi(f)| &= |\Phi(f^+ - f^-)| = |\Phi(f^+) - \Phi(f^-)| \\ &\leq |\Phi(f^+)| + |\Phi(f^-)| = \Phi(f^+) + \Phi(f^-) = \Phi(f^+ + f^-) = \Phi(|f|). \end{aligned}$$

Also, if the unit function $1\colon S \to \mathbb{F}$ (i.e., $1(s) = 1$ for all $s \in S$) lies in $L(S)$, and $f\colon S \to \mathbb{F}$ is such that $|f| \leq 1$, then $0 \leq \Phi(1 - |f|) = \Phi(1) - \Phi(|f|)$. So

$$\Phi(|f|) \leq \Phi(1) \quad \text{whenever} \quad |f| \leq 1.$$

Summing up: *If $L(S)$ is a linear space of $\mathbb{F}$-valued functions on a nonempty set S, if $\Phi\colon L(S) \to \mathbb{F}$ is a positive linear functional, if $f\colon S \to \mathbb{R}$ is a real-valued function in $L(S)$, and if the unit function 1 lies in $L(S)$, then*

$$|\Phi(f)| \le \Phi(|f|) \le \Phi(1) \quad \text{whenever} \quad |f| \le 1.$$

An important particular case: A *bounded function* is a function $f\colon S \to \mathbb{F}$ for which the range $f(S)$ is a bounded set in the metric space $\mathbb{F}$ equipped with the usual metric (i.e., a function f such that $\sup_{s\in S} |f(s)| < \infty$ — cf. the remark on boundedness after the proof of Lemma 11.2). Now consider the collection $B(S)$ of all $\mathbb{F}$-valued bounded functions on S,

$$B(S) = \big\{f\colon S \to \mathbb{F}\colon\ f \text{ is bounded}\big\},$$

which is a linear space (sum and scalar multiplication of bounded functions are again bounded functions); in fact, it is a normed space with *sup-norm*

$$\|f\|_\infty = \sup_{s\in S} |f(s)|\,.$$

Note that the unit function $1\colon S \to \{1\} \subset \mathbb{R} \subseteq \mathbb{F}$ lies in $B(S)$. If a function f lies in $B(S)$, then $|f| \le 1$ means $|f(s)| \le 1$ for every $s \in S$, which is equivalent to saying that $\sup_{s\in S} |f(s)| \le 1$. Therefore,

$$\|f\|_\infty \le 1 \quad \text{if and only if} \quad |f| \le 1.$$

If $L(S) \subseteq B(S)$ is a linear manifold of $B(S)$, then $L(S)$ inherits the norm of $B(S)$ and $(L(S), \|\ \|_\infty)$ is a normed space.

A *bounded linear functional* on the normed space $L(S)$ is a linear functional $\Phi\colon L(S) \to \mathbb{F}$ such that $\sup_{f\ne 0} \frac{|\Phi(f)|}{\|f\|_\infty} < \infty$. (It is worth recalling that, if $\Phi \ne 0$, then the image of Φ, viz., $\Phi(L(S))$, is not bounded in $\mathbb{F}$ — indeed, $\Phi(L(S)) = \mathbb{F}$: nonzero *linear* functionals are surjective.) The *induced uniform norm* of a bounded linear functional (in fact, the induced uniform norm on the normed space of all $\mathbb{F}$-valued bounded linear functionals on $L(S)$ — the *dual* of $L(S)$) is

$$\|\Phi\| = \sup_{f\ne 0} \frac{|\Phi(f)|}{\|f\|_\infty} = \sup_{\|f\|_\infty \le 1} |\Phi(f)| = \sup_{|f|\le 1} |\Phi(f)|\,.$$

Hence, if the bounded linear functional Φ is positive and if $1 \in L(S)$, then $\Phi(1) \le \|\Phi\| = \sup_{|f|\le 1} |\Phi(f)|$, and also $\sup_{|f|\le 1} \Phi(|f|) \le \Phi(1)$. Moreover, if $f \in L(S)$ is real-valued, then $|\Phi(f)| \le \Phi(|f|)$. Therefore, if $\mathbb{F} = \mathbb{R}$, then

$$\Phi(1) \le \|\Phi\| = \sup_{|f|\le 1} |\Phi(f)| \le \sup_{|f|\le 1} \Phi(|f|) \le \Phi(1).$$

Summing up: *If $L(S)$ is a real linear space of real-valued bounded functions on a nonempty set S, if Φ is a real-valued positive bounded linear functional on $L(S)$, and if $1 \in L(S)$, then*

$$\|\Phi\| = \Phi(1).$$

Again, the complex linear space of all *complex-valued bounded functions* on S, and the real linear space of all *real-valued bounded functions* on S will, when necessary, be denoted respectively by

$$B(S,\mathbb{C}) = \big\{f\colon S \to \mathbb{C}\colon\ f \text{ is bounded}\big\},$$

and

$$B(S,\mathbb{R}) = \big\{f\colon S \to \mathbb{R}\colon\ f \text{ is bounded}\big\}.$$

A *linear lattice* is a linear space equipped with a partial ordering such that every pair of elements has an infimum and a supremum in it. That is,

$$f \wedge g = \inf\{f,g\} \quad \text{and} \quad f \vee g = \sup\{f,g\}$$

lie in the linear space for every f and g in the linear space. Set $\mathbb{F} = \mathbb{R}$ and let $L(S,\mathbb{R})$ be a real linear space of real-valued functions on S. For f, g in $L(S,\mathbb{R})$ let $f \wedge g$ and $f \vee g$ be defined by $(f \wedge g)(s) = \min\{f(s), g(s)\}$ and $(f \vee g)(s) = \max\{f(s), g(s)\}$ for each $s \in S$. If $L(S,\mathbb{R})$ has (in addition to the linear properties) the property that $f \wedge g$ and $f \vee g$ lie in $L(S,\mathbb{R})$ whenever f, g lie in $L(S,\mathbb{R})$, then $L(S,\mathbb{R})$ is a *linear lattice* of real-valued functions on S. Example: the linear space $B(S,\mathbb{R})$ of all *real-valued bounded functions* on S equipped with the partial ordering $\leq$ (defined by $f \leq g$ if $f(s) \leq g(s)$ for every $s \in S$), which induces the above binary operations $\wedge$ and $\vee$, is a linear lattice that contains the unit function 1 (i.e., $1(s) = 1$ for all $s \in S$).

Now endow S with a topology. Let $S = X$ be a topological space and consider the set $C(X)$ of all $\mathbb{F}$-valued continuous functions on X:

$$C(X) = \big\{f\colon X \to \mathbb{F}\colon\ f \text{ is continuous}\big\}.$$

Since sum and scalar multiples of continuous functions are again continuous functions, $C(X)$ is a linear manifold of the linear space $\mathbb{F}^X$ of all $\mathbb{F}$-valued functions on X, and so $C(X)$ is itself a linear space. Clearly, $C_c(X) \subseteq C(X)$. Since the support of a sum of two functions in $C_c(X)$ is the union of their supports (because a finite union of compact sets is compact), we can infer that $C_c(X)$ is a linear manifold of $C(X)$, and hence $C_c(X)$ is itself a linear space. If X is a compact space, then closed subsets of X are compact sets (Lemma 11.2(a)) so that $C_c(X) = C(X)$ by the definition of $C_c(X)$; that is,

$$C_c(X) = C(X) \ \text{ whenever } X \text{ is a compact topological space.}$$

By Lemma 12.1 (recalling that in $\mathbb{F}$ compact means closed and bounded),

$$C_c(X) \subseteq B(X),$$

and so the linear space $C_c(X)$, and also $C(X)$ if X is compact, is again a normed space equipped with the sup-norm $\|\ \|_\infty$.

Once again, the complex linear space of all *complex-valued continuous functions* on X, and the real linear space of all *real-valued continuous functions* on X will, when necessary, be denoted respectively by

$$C(X,\mathbb{C}) = \big\{f\colon X\to\mathbb{C}\colon\ f \text{ is continuous}\big\},$$

and

$$C(X,\mathbb{R}) = \big\{f\colon X\to\mathbb{R}\colon\ f \text{ is continuous}\big\}.$$

Note that $C(X,\mathbb{R})$ has the additional property that if f,g lie in $C(X,\mathbb{R})$, then so do the functions $f\wedge g$ and $f\vee g$. This means that $C(X,\mathbb{R})$ is a linear lattice of real-valued functions on X. Also, the unit function, $1(x)=1$ for all $x\in X$, lies in $C(X)$. Moreover, if X is compact, then $C(X)\subseteq B(X)$ (i.e., $\mathbb{F}$-valued continuous functions on a compact space are bounded — cf. Claim in the proof of Lemma 12.1 plus remark on boundedness in Section 11.1).

Lemma 12.4. *Let $L(S,\mathbb{R})$ be a linear lattice of real-valued bounded functions on a nonempty set S. If $\Phi\colon L(S,\mathbb{R})\to\mathbb{R}$ is a real-valued bounded linear functional on $L(S,\mathbb{R})$, then there are two positive real-valued bounded linear functionals $\Phi^+\colon L(S,\mathbb{R})\to\mathbb{R}$ and $\Phi^-\colon L(S,\mathbb{R})\to\mathbb{R}$ on $L(S,\mathbb{R})$ such that*

$$\Phi = \Phi^+ - \Phi^-.$$

If $L(S,\mathbb{R})$ contains the unit function 1, then $\|\Phi\| = \Phi^+(1) + \Phi^-(1)$.

Proof. Let $\Phi\colon L(S,\mathbb{R})\to\mathbb{R}$ be a bounded linear functional. Set

$$\Psi^+(f) = \sup_{0\le\phi\le f} \Phi(\phi) \quad \text{for each} \quad 0\le f\in L(S,\mathbb{R}).$$

Thus $\Psi^+(0)=0$ (since $\Phi(0)=0$ by linearity of Φ) and hence $\Psi^+(f)\ge 0$ for every $0\le f\in L(S,\mathbb{R})$, and $\sup_{0\le f}\Psi^+(f)<\infty$ (since Φ is bounded). Note that, since Φ is linear, for every $0\le\alpha\in\mathbb{R}$ and every $0\le f\in L(S,\mathbb{R})$,

(i) $$\Psi^+(\alpha f) = \alpha\,\Psi^+(f).$$

Now take an arbitrary pair of functions $0\le f,g$ in $L(S,\mathbb{R})$. Let ϕ,ψ be functions in $L(S,\mathbb{R})$ such that $0\le\phi\le f$ and $0\le\psi\le g$. Thus, by the definition of Ψ^+, since Φ is additive, $\Phi(\phi)+\Phi(\psi) = \Phi(\phi+\psi)\le\Psi^+(f+g)$, and so (definition of Ψ^+ again — taking the sup for all such φ and ψ)

$$\Psi^+(f)+\Psi^+(g)\le\Psi^+(f+g).$$

Conversely, take any function $h\in L(S,\mathbb{R})$ such that $0\le h\le f+g$. Since $0\le h\wedge f = \inf\{h,f\}\le f$, and since $0\le h\wedge f\le h\le f+g$, it follows that $0\le h-(h\wedge f)\le g$. (Indeed, $(h-(h\wedge f))(s)=0$ if $h(s)\le f(s)$ and $(h-(h\wedge f))(s) = (h-f)(s)\le g(s)$ if $f(s)\le h(s)$.) Again, by the definition

of Ψ^+, since Φ is linear, $\Phi(h) = \Phi(h \wedge f) + \Phi(h - (h \wedge f)) \leq \Psi^+(f) + \Psi^+(g)$, and so (definition of Ψ^+ once again — taking the sup for all such h)

$$\Psi^+(f+g) \leq \Psi^+(f) + \Psi^+(g).$$

Therefore, for every $0 \leq f, g \in L(S, \mathbb{R})$,

$$\Psi^+(f+g) = \Psi^+(f) + \Psi^+(g). \tag{ii}$$

Since every real-valued function $f \in L(S, \mathbb{R})$ can be written as $f = f^+ - f^-$, where $f^+ = f \vee 0$ and $f^- = -f \wedge 0$ (the positive and negative parts of f) are nonnegative functions in $L(S, \mathbb{R})$, the values $\Psi^+(f^+)$ and $\Psi^+(f^-)$ are well defined. So consider the functional $\Phi^+ \colon L(S, \mathbb{R}) \to \mathbb{R}$ defined on the whole $L(S, \mathbb{R})$ as follows. For every $f \in L(S, \mathbb{R})$,

$$\Phi^+(f) = \Psi^+(f^+) - \Psi^+(f^-).$$

First we show that Φ^+ is homogeneous. Take any $f \in L(S, \mathbb{R})$. Since $\alpha \geq 0$ implies $(\alpha f)^+ = \alpha f^+$ and $(\alpha f)^- = \alpha f^-$, we get by (i) that $\Psi^+((\alpha f)^+) = \alpha \Psi^+(f^+)$ and $\Psi^+((\alpha f)^-) = \alpha \Psi^+(f^-)$, and hence $\Phi^+(\alpha f) = \alpha \Phi^+(f)$ whenever $\alpha \geq 0$. Next observe that since $f^- = (-f)^+$ and $f^+ = (-f)^-$,

$$\Phi^+(-f) = \Psi^+((-f)^+) - \Psi^+((-f)^-) = \Psi^+(f^-) - \Psi^+(f^+) = -\Phi^+(f).$$

So $\Phi^+(\alpha f) = \Phi^+(-|\alpha| f) = -\Phi^+(|\alpha| f) = -|\alpha| \Phi^+(f) = \alpha \Phi^+(f)$ if $\alpha \leq 0$. Therefore, for every $\alpha \in \mathbb{R}$ and every $f \in L(S, \mathbb{R})$,

$$\Phi^+(\alpha f) = \alpha \Phi^+(f). \tag{i$'$}$$

In order to show additivity, take any f in $L(S, \mathbb{R})$, and let ϕ, ψ in $L(S, \mathbb{R})$ be such that $0 \leq \phi$, $0 \leq \psi$, $0 \leq f + \phi$ and $0 \leq f + \psi$. Note that by (ii),

$$\Psi^+(f + \phi) + \Psi^+(\psi) = \Psi^+(f + \phi + \psi) = \Psi^+(f + \psi) + \Psi^+(\phi),$$

and so

$$\Psi^+(f + \phi) - \Psi^+(\phi) = \Psi^+(f + \psi) - \Psi^+(\psi).$$

In particular, with $\phi = f^-$ (so that $f + \phi = f^+$), we get

$$\Phi^+(f) = \Psi^+(f^+) - \Psi^+(f^-) = \Psi^+(f + \psi) - \Psi^+(\psi).$$

Summing up: *For every* $f \in L(S, \mathbb{R})$,

$$\Phi^+(f) = \Psi^+(f + \psi) - \Psi^+(\psi)$$

whenever $0 \leq \psi$ *is such that* $0 \leq f + \psi$. Thus take any $g \in L(S, \mathbb{R})$, and let ψ be such that, in addition, $0 \leq g + \psi$. Then

$$\Phi^+(g) = \Psi^+(g+\psi) - \Psi^+(\psi).$$

Therefore, by using (i) and (ii), we get

$$\begin{aligned}\Phi^+(f) + \Phi^+(g) &= \Psi^+(f+\psi) + \Psi^+(g+\psi) - 2\,\Psi^+(\psi)\\ &= \Psi^+(f+g+2\,\psi) - \Psi^+(2\,\psi).\end{aligned}$$

But $0 \le f+\psi$ and $0 \le g+\psi$ imply that $0 \le f+g+2\,\psi$. Hence

$$\Phi^+(f+g) = \Psi^+(f+g+2\,\psi) - \Psi^+(2\,\psi).$$

Outcome: For every $f, g \in L(S,\mathbb{R})$,

$$\text{(ii}') \qquad \Phi^+(f+g) = \Phi^+(f) + \Phi^+(g).$$

Properties (i′) and (ii′) mean that the functional $\Phi^+\colon L(S,\mathbb{R}) \to \mathbb{R}$ is linear. It is also bounded, since $\sup_{f\ge 0} \Psi^+(f) < \infty$. Indeed,

$$\sup_{f\in L(S,\mathbb{R})} \Phi^+(f) = \sup_{f\in L(S,\mathbb{R})} \big(\Psi^+(f^+) - \Psi^+(f^-)\big) \le 2 \sup_{f\ge 0} \Psi^+(f) < \infty.$$

Moreover, observe that Φ^+ is positive as well:

$$\Phi^+(f) = \Psi^+(f) \ge 0 \quad \text{for every} \quad f \ge 0.$$

Next consider the functional $\Phi^-\colon L(S,\mathbb{R}) \to \mathbb{R}$ defined on $L(S,\mathbb{R})$ by

$$\Phi^- = \Phi^+ - \Phi,$$

which is trivially linear and bounded (since Φ and Φ^+ are). Since $\Phi(f) \le \Psi^+(f)$ for every $f \ge 0$ by the definition of Ψ^+, it follows that Φ^- is positive:

$$\Phi^-(f) = \Phi^+(f) - \Phi(f) = \Psi^+(f) - \Phi(f) \ge 0 \quad \text{for every} \quad f \ge 0.$$

Finally, suppose $L(S,\mathbb{R})$ contains the unit function 1. Since $L(S,\mathbb{R})$ is a real linear space of real-valued bounded functions, since Φ^+ and Φ^- are real-valued positive bounded linear functionals on $L(S,\mathbb{R})$, and since $1 \in L(S,\mathbb{R})$,

$$\|\Phi^+\| = \Phi^+(1) \quad \text{and} \quad \|\Phi^-\| = \Phi^-(1),$$

so that

$$\|\Phi\| = \|\Phi^+ - \Phi^-\| \le \|\Phi^+\| + \|\Phi^-\| = \Phi^+(1) + \Phi^-(1).$$

On the other hand, take any $\phi \in L(S,\mathbb{R})$ such that $0 \le \phi \le 1$, equivalently, such that $|2\,\phi - 1| \le 1$. First recall by the definition of Φ^- that

$$\Phi^+(1) + \Phi^-(1) = 2\,\Phi^+(1) - \Phi(1).$$

Observe that, according to the definitions of Ψ^+ and Φ^+,

$$\Phi^+(1) = \Psi^+(1) = \sup_{0\le\phi\le 1} \Phi(\phi).$$

Also, since $|2\,\phi - 1| \le 1$, and since Φ is linear,

$$2\,\Phi(\phi) - \Phi(1) = \Phi(2\,\phi - 1) \le \sup_{|f|\le 1} |\Phi(f)| = \|\Phi\|\,.$$

Hence,

$$\Phi^+(1) + \Phi^-(1) = \sup_{0\le\phi\le 1} 2\,\Phi(\phi) - \Phi(1) \le \|\Phi\|\,.$$

Therefore, $\Phi^+(1) + \Phi^-(1) = \|\Phi\|$. □

12.3 The Riesz Representation Theorem

As we saw in Section 11.1, *continuous functions are Borel measurable*. So *every $f \in C_c(X)$ is $\mathcal{X}_\mathcal{T}$-measurable*. Actually, *$f \in C_c(X)$ is integrable with respect to a Borel measure* (and so is $f \in C(X) = C_c(X)$ if X is compact), since these functions are bounded ($C_c(X) \subseteq B(X)$), and since Borel measures are finite on compact sets (cf. Lemma 4.4(b) and Problem 3.3(a,b)).

This section contains the central theme of the chapter. The next theorem, Theorem 12.5, is one of the forms of the celebrated Riesz Representation Theorem, which might be called Riesz–Markov–Radon–Banach–Kakutani–Halmos Theorem, since many versions of it have traveled a long way from Riesz's original 1909 paper to the general form on locally compact Hausdorff space developed in Halmos's book [31]. For a short account of such a long story see [62, p. 354]. We will see further versions up to Section 12.4.

Theorem 12.5. *Let X be a locally compact Hausdorff space. If Φ is a positive linear functional on $C_c(X)$, then there exists a unique quasiregular Borel measure μ on the Borel σ-algebra $\mathcal{X}_\mathcal{T}$ of subsets of X such that*

$$\Phi(f) = \int f\,d\mu \quad \textit{for every} \quad f \in C_c(X).$$

Note: Proposition 12.E in Section 12.4 shows that the quasiregular Borel measure μ can be taken to be regular.

Proof. Let X be a locally compact Hausdorff space. Take the linear space of all $\mathbb{F}$-valued continuous functions on X with compact support,

$$C_c(X) = \big\{f\colon X \to \mathbb{F}\colon\ f \text{ is continuous and } \overline{[f]} \text{ is compact in } X\big\},$$

let $\Phi\colon C_c(X) \to \mathbb{F}$ be a positive linear functional,

$\Phi(f) \geq 0$ for every $f \in C_c(X)$ such that $f(x) \geq 0$ for every $x \in X$,

and consider the set function $\mu^{\#} : \mathcal{T} \to \overline{\mathbb{R}}$ on the topology $\mathcal{T}$ on X defined by

$$\mu^{\#}(U) \;=\; \sup_{f \in C_c(X)^U} \Phi(f),$$

where $C_c(X)^U$ is a collection of *positive* functions for every open set $U \in \mathcal{T}$:

$$C_c(X)^U = \big\{ f \in C_c(X) \colon\; f(X) \subseteq [0,1] \text{ and } \overline{[f]} \subseteq U \big\} \;\subseteq\; C_c(X, \mathbb{R}).$$

First we show that $\mu^{\#}$ satisfies the properties (1) to (7) in the definition of an inner content (Section 11.3). Let G and U be arbitrary sets in $\mathcal{T}$. Clearly,

$$\mu^{\#}(\varnothing) = 0, \tag{1}$$

$$\mu^{\#}(U) \geq 0, \tag{2}$$

since Φ positive, and it is readily verified that

$$\mu^{\#}(U) < \infty \quad \text{whenever} \quad \overline{U} \text{ is compact}, \tag{3}$$

$$\mu^{\#}(G) \leq \mu^{\#}(U) \quad \text{whenever} \quad G \subseteq U. \tag{4}$$

Before verifying property (5) we need to show property (6). Let $\{U_k\}$ be a countable family of open sets and set $U = \bigcup_k U_k$, which is open. Take an arbitrary $f \in C_c(X)^U$. Thus $\{U_k\}$ is an open covering of the compact set $\overline{[f]}$, and hence there exists a finite open subcovering $\{U_i\}_{i=1}^n \subseteq \{U_k\}$ of $\overline{[f]}$. In this case, Corollary 12.3 ensures that there exists $f_i \in C_c(X)^{U_i}$ for each $i \in [1, n]$ such that $\sum_{i=1}^n f_i(x) = 1$ for every $x \in \overline{[f]}$. So

$$f = \sum_{i=1}^{n} f_i f \quad \text{where each } f_i f \text{ lies in } C_c(X)^{U_i}.$$

Hence, by the definition of $\mu^{\#}$, and since Φ is linear,

$$\Phi(f) \;= \sum_{i=1}^{n} \Phi(f_i f) \;\leq \sum_{i=1}^{n} \mu^{\#}(U_i) \;\leq \sum_{i=1}^{\infty} \mu^{\#}(U_i).$$

Then

$$\mu^{\#}(U) \;=\; \sup_{f \in C_c(X)^U} \Phi(f) \;\leq\; \sum_{i=1}^{\infty} \mu^{\#}(U_i).$$

Therefore,

$$\mu^{\#}\Big(\bigcup_k U_k\Big) \leq \sum\nolimits_k \mu^{\#}(U_k) \quad \text{for every countable family } \{U_k\} \subseteq \mathcal{T}. \tag{6}$$

Now suppose that $G \cap U = \varnothing$. If $f \in C_c(X)^U$ and if $g \in C_c(X)^G$, then $f + g \in C_c(X)^{U \cup G}$. Thus (cf. definition of $\mu^{\#}$),

$$\Phi(f) + \Phi(g) \;=\; \Phi(f + g) \;\leq\; \mu^{\#}(U \cup G)$$

for every $f \in C_c(X)^U$ and every $g \in C_c(X)^G$. This implies that

$$\mu^{\#}(U) + \mu^{\#}(G) = \sup_{f \in C_c(X)^U} \Phi(f) + \sup_{g \in C_c(X)^G} \Phi(g)$$
$$\leq \sup_{f \in C_c(X)^U,\, g \in C_c(X)^G} \big(\Phi(f) + \Phi(g)\big) \leq \mu^{\#}(U \cup G).$$

Since we have already verified that $\mu^{\#}(U \cup G) \leq \mu^{\#}(U) + \mu^{\#}(G)$ (independently of the disjointness assumption), it follows that

$$\mu^{\#}(G \cup U) = \mu^{\#}(G) + \mu^{\#}(U) \quad \text{whenever} \quad G \cap U = \varnothing. \tag{5}$$

Thus $\mu^{\#}$ is a content, which in fact is an inner content, since

$$\sup_{G \in \mathcal{G}(U)} \mu^{\#}(G) = \sup_{G \in \mathcal{G}(U)} \sup_{f \in C_c(X)^G} \Phi(f) = \sup_{f \in C_c(X)^U} \Phi(f) = \mu^{\#}(U),$$

with $\mathcal{G}(U) = \{G \in \mathcal{T}\colon \overline{G} \subseteq U,\ \overline{G} \text{ compact}\}$ for each $U \in \mathcal{T}$, so that

$$\mu^{\#}(U) = \sup\big\{\mu^{\#}(G)\colon \overline{G} \subseteq U,\ \overline{G} \text{ compact}\big\}. \tag{7}$$

Therefore, by Lemma 11.8, the set function $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ defined by

$$\mu^*(S) = \inf\big\{\mu^{\#}(U)\colon S \subseteq U\big\} \quad \text{for every} \quad S \in \wp(X)$$

is a quasiregular outer measure such that $\mu^*(S) < \infty$ if S is bounded. So

$$\mu = \mu^*|_{\mathcal{X}_{\mathcal{T}}}\colon \mathcal{X}_{\mathcal{T}} \to \overline{\mathbb{R}}$$

is a quasiregular Borel measure on the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ generated by the topology $\mathcal{T}$ according to Theorem 11.7. Next we show that

$$\Phi(f) = \int f\, d\mu \quad \text{for every} \quad f \in C_c(X).$$

Any complex-valued function f in $C_c(X)$ can be written as $f = f_1 + i\, f_2$, where $f_1 = \operatorname{Re} f$ and $f_2 = \operatorname{Im} f$ (the real and imaginary parts of f) are real-valued functions in $C_c(X)$, and any real-valued function f in $C_c(X)$ can be written as $f = f^+ - f^-$, where f^+ and f^- (the positive and negative parts of f) are nonnegative functions in $C_c(X)$. In fact, such a decomposition is possible since every function in $C_c(X)$ is bounded (reason: $f(X)$ is compact by Lemma 12.1 and so it is bounded — cf. remarks on boundedness in Section 11.1). Therefore, since Φ and $\int(\ )\, d\mu$ are linear functionals, it is enough to consider nonnegative functions. That is, if the above identity holds for every nonnegative function f in $C_c(X)$, then it holds for every $\mathbb{F}$-valued function f in $C_c(X)$. So take an arbitrary nonnegative function f in $C_c(X)$. Since Φ and $\int(\ )\, d\mu$ are linear functionals and $f(X)$ is bounded, we may assume without loss of generality that $f \leq 1$. Thus suppose $f(X) \subseteq [0, 1]$. Take an

arbitrary bounded open set $U \in \mathcal{T}$ such that $\overline{[f]} \subseteq U$. In other words, consider an arbitrary bounded open set U such that

$$f \in C_c(X)^U.$$

Also take an arbitrary positive integer n and set, for each integer $k \geq 0$,

$$U_0 = U, \qquad U_k = \left\{x \in X\colon \tfrac{k-1}{n} < f(x)\right\} = f^{-1}((\tfrac{k-1}{n}, \infty)) \quad \text{for} \quad k \geq 1,$$

which are open (inverse images of open sets under a continuous function). Note that $U_1 = X \backslash f^{-1}(\{0\}) = X \backslash \mathcal{N}(f)$ so that $\overline{U_1} = \overline{[f]}$, $\overline{U_{k+1}} \subseteq U_k$, and $U_k = \varnothing$ for every $k \geq n+1$. Set, for each integer $k \in [1, n]$,

$$f_k(x) = \begin{cases} 1, & x \in U_{k+1} \qquad (\text{i.e., if } \tfrac{k}{n} < f(x)), \\ nf(x) - (k-1), & x \in U_k \backslash U_{k+1} \qquad (\text{i.e., if } \tfrac{k-1}{n} < f(x) \leq \tfrac{k}{n}), \\ 0, & x \in X \backslash U_k \qquad (\text{i.e., if } f(x) \leq \tfrac{k-1}{n}), \end{cases}$$

which defines a collection of n real-valued (since f is real-valued) functions f_k on X with compact support. In fact, $\overline{[f_k]} \subseteq \overline{U_k} \subseteq U_{k-1}$. Each f_k is continuous (reason: $nf(x) - (k-1) = 0$ if $f(x) = \frac{k-1}{n}$, $nf(x) - (k-1) = 1$ if $f(x) = \frac{k}{n}$, and f is continuous). Thus $f_k(X) \subseteq [0, 1]$, and so $f_k \in C_c(X)^{G_k}$ for each $k \in [1, n]$ and each open set G_k such that $\overline{[f_k]} \subseteq G_k$. In particular,

$$f_k \in C_c(X)^{U_{k-1}}.$$

Thus, according to the definitions of the inner content $\mu^{\#}\colon \mathcal{T} \to \overline{\mathbb{R}}$ and of the quasiregular outer measure $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ induced by it, and recalling that the quasiregular Borel measure $\mu\colon \mathcal{X}_{\mathcal{T}} \to \overline{\mathbb{R}}$ generated by them is such that $\mu|_{\mathcal{T}} = \mu^*|_{\mathcal{T}} = \mu^{\#}$, we get, for each integer $k \in [1, n]$,

$$\mu^{\#}(U_{k+1}) = \mu^*(U_{k+1}) = \mu(U_{k+1}) \leq \int f_k \, d\mu \leq \mu(U_k) = \mu^*(U_k) = \mu^{\#}(U_k).$$

Since $f_k = 1$ on $U_{k+1} \subseteq U_k$, and $\overline{[f_k]} \subseteq \overline{U_k} \subseteq U_{k=1}$, it also follows that

$$\mu^{\#}(U_{k+1}) = \sup_{g \in C_c(X)^{U_{k+1}}} \Phi(g) \leq \Phi(f_k) \leq \sup_{g \in C_c(X)^{U_{k-1}}} \Phi(g) = \mu^{\#}(U_{k-1}).$$

Hence,

$$\int f_{k+1} \, d\mu \leq \Phi(f_k) \qquad \text{and} \qquad \Phi(f_{k+2}) \leq \int f_k \, d\mu$$

for every integer $k \in [1, n-1]$ if $n > 1$, and for every integer $k \in [1, n-2]$ if $n > 2$, respectively.

Claim. $$nf(x) = \sum_{k=1}^{n} f_k(x) \quad \text{for every} \quad x \in X.$$

Proof of Claim. If $n = 1$, then $f = f_1$. Thus suppose $n > 1$. Take an arbitrary $x \in X$. If $\frac{i-1}{n} < f(x) \le \frac{i}{n}$ for some integer i in $[1, n]$, then

$$f_k(x) = \begin{cases} 1, & \text{if } \ 1 \le k \le i-1, \\ 0, & \text{if } \ i+1 \le k \le n, \end{cases}$$

for every integer $k \neq i$ in $[1, n]$. Hence,

$$\sum_{k=1}^{n} f_k(x) = \sum_{k=1}^{i-1} f_k(x) + f_i(x) + \sum_{k=i+1}^{n} f_k(x) = (i-1) + nf(x) - (i-1) = nf(x). \ \square$$

Now suppose $n > 1$ so that $\int f_{k+1}\, d\mu \le \Phi(f_k)$ for each integer $k \in [1, n-1]$. Since Φ and $\int(\ )\, d\mu$ are linear functionals, we get

$$\begin{aligned} n\int f\, d\mu - \int f_1\, d\mu &= \int (nf - f_1)\, d\mu = \int \Big(\sum_{k=1}^{n} f_k - f_1 \Big)\, d\mu \\ = \int \sum_{k=1}^{n-1} f_{k+1}\, d\mu &= \sum_{k=1}^{n-1} \int f_{k+1}\, d\mu \le \sum_{k=1}^{n-1} \Phi(f_k) = \Phi\Big(\sum_{k=1}^{n-1} f_k \Big) \\ = \Phi\Big(\sum_{k=1}^{n} f_k - f_n \Big) &= \Phi(nf - f_n) = n\Phi(f) - \Phi(f_n). \end{aligned}$$

Recall that $\int f_1\, d\mu \le \mu(U_0)$, with $U_0 = U$, where U is bounded, and so $\mu(U) < \infty$, and also that $\Phi(f_n) \ge 0$ (because Φ is positive). Then

$$\int f\, d\mu \le \Phi(f) + \tfrac{1}{n} \int f_1\, d\mu - \tfrac{1}{n}\Phi(f_n) \le \Phi(f) + \tfrac{1}{n}\mu(U_0).$$

Since the preceding inequality holds for every integer $n > 1$, it follows that

$$\int f\, d\mu \le \Phi(f).$$

Next suppose $n > 2$ so that $\Phi(f_{k+2}) \le \int f_k\, d\mu$ for each integer $k \in [1, n-2]$. Again, since Φ and $\int(\ )\, d\mu$ are linear functionals, we get

$$\begin{aligned} n\Phi(f) - \Phi(f_1) - \Phi(f_2) &= \Phi(nf - f_1 - f_2) = \Phi\Big(\sum_{k=1}^{n} f_k - f_1 - f_2 \Big) \\ = \sum_{k=1}^{n-2} \Phi(f_{k+2}) &\le \sum_{k=1}^{n-2} \int f_k\, d\mu = \int \Big(\sum_{k=1}^{n} f_k - f_{n-1} - f_n \Big)\, d\mu \\ = \int (nf - f_{n-1} - f_n)\, d\mu &= n\int f\, d\mu - \int f_{n-1}\, d\mu - \int f_n\, d\mu. \end{aligned}$$

Recall that $\Phi(f_1) + \Phi(f_2) \le \mu(U_0) + \mu(U_1) \le 2\mu(U) < \infty$, and also that each $\int f_k\, d\mu \ge 0$. Then

$$\Phi(f) \le \int f\,d\mu + \tfrac{1}{n}\Big(\Phi(f_1)+\Phi(f_2) - \int f_{n-1}\,d\mu - \int f_n\,d\mu\Big) \le \int f\,d\mu + \tfrac{2}{n}\mu(U).$$

Since the preceding inequality holds for every integer $n > 2$, it follows that

$$\Phi(f) \le \int f\,d\mu.$$

Outcome:

$$\int f\,d\mu = \Phi(f).$$

Finally, we show that the quasiregular measure $\mu\colon \mathcal{X}_{\mathcal{T}} \to \overline{\mathbb{R}}$ on the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ of subsets of the locally compact Hausdorff space X generated by the topology $\mathcal{T}$ on X, which satisfies the above identity, is unique. Suppose μ and μ' on $\mathcal{X}_{\mathcal{T}}$ are quasiregular measures that satisfy the above identity. Observe by the definition of quasiregular measures on $\mathcal{X}_{\mathcal{T}}$ that they are determined by their values on open sets, which in turn are determined by their values on compact sets. So, in order to show that $\mu' = \mu$ on $\mathcal{X}_{\mathcal{T}}$, it suffices to verify that $\mu'(K) = \mu(K)$ for every compact set K. Thus take an arbitrary compact set $K \subseteq X$ and an arbitrary $\varepsilon > 0$. Since μ' is outer regular, there exists an open set U such that $K \subseteq U$ and $\mu'(U) \le \mu'(K) + \varepsilon$. By Lemma 12.2, let f be any function in $C_c(X)_K \cap C_c(X)^U$. Therefore, since $f(X) \subseteq [0,1]$, $f(K) = \{1\}$, and $\overline{[f]} \subseteq U$, it follows that

$$\mu(K) \le \int f\,d\mu = \Phi(f) = \int f\,d\mu' \le \mu'(U) \le \mu'(K) + \varepsilon,$$

and hence $\mu(K) \le \mu'(K)$. Swapping μ with μ' we get $\mu'(K) \le \mu(K)$. Thus $\mu'(K) = \mu(K)$, so that $\mu' = \mu$. □

A major application of Theorem 12.5 occurs in the proof of the Spectral Theorem for normal operators (see, e.g., [44, Proof of Theorem 3.11]). Comparing with the above version of the Riesz Representation Theorem, the next version in Theorem 12.6 exchanges positiveness with boundedness for the linear functional Φ, and the positive Borel measure is replaced with a (finite) signed measure. The price for it is that the domain on which such a functional acts is the real linear space $C(X,\mathbb{R})$ for a compact X, rather than the linear space $C_c(X)$ on a locally compact X.

Theorem 12.6. *Let X be a compact Hausdorff space. If Φ is a real-valued bounded linear functional on the real linear space $C(X,\mathbb{R})$, then there exists a unique quasiregular signed measure ν on the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ of subsets of X such that*

$$\Phi(f) = \int f\,d\nu \quad \textit{for every} \quad f \in C(X,\mathbb{R}).$$

Moreover, $\|\Phi\| = |\nu|(X)$. (*See also Proposition 12.E.*)

Proof. Let X be a compact Hausdorff space. Recall that the real linear space

$$C(X,\mathbb{R}) = \big\{f\colon X \to \mathbb{R}\colon\ f \text{ is continuous}\big\}$$

of all real-valued continuous functions on X is a linear lattice, and let $\Phi\colon C(X,\mathbb{R}) \to \mathbb{R}$ be a real-valued bounded linear functional on $C(X,\mathbb{R})$. Lemma 12.4 ensures that there exist two positive real-valued bounded linear functionals $\Phi^+\colon C(X,\mathbb{R}) \to \mathbb{R}$ and $\Phi^-\colon C(X,\mathbb{R}) \to \mathbb{R}$ on $C(X,\mathbb{R})$ such that

$$\Phi \,=\, \Phi^+ - \Phi^-.$$

Take an arbitrary $f \in C(X,\mathbb{R})$. Since X is compact, $C(X) = C_c(X)$, and so Theorem 12.5 says that there is a Borel measure μ, and a Borel measure η, both unique on $\mathcal{X}_\mathcal{T}$ (which are finite because X is compact), for which

$$\Phi^+(f) \,=\, \int f\,d\mu \quad \text{and} \quad \Phi^-(f) \,=\, \int f\,d\eta.$$

So the real-valued signed measure $\nu = \mu - \eta$ on $\mathcal{X}_\mathcal{T}$, which is quasiregular by Theorem 12.5 (the difference of finite quasiregular measures), is such that

$$\Phi(f) \,=\, \Phi^+(f) - \Phi^-(f) \,=\, \int f\,d\mu - \int f\,d\eta \,=\, \int f\,d\nu,$$

proving the existence of ν. To verify uniqueness, suppose ν and ν' are (finite) real-valued signed measures on $\mathcal{X}_\mathcal{T}$ such that for an arbitrary $f \in C(X,\mathbb{R})$,

$$\int f\,d\nu \,=\, \Phi(f) \,=\, \int f\,d\nu'.$$

Set $\widetilde{\nu} = \nu - \nu'$. This is a (finite) real-valued signed measure on $\mathcal{X}_\mathcal{T}$ for which

$$\int f\,d\widetilde{\nu} \,=\, \int f\,d\widetilde{\nu}^+ - \int f\,d\widetilde{\nu}^- \,=\, 0,$$

where $\widetilde{\nu}^+$ and $\widetilde{\nu}^-$ are finite positive measures on $\mathcal{X}_\mathcal{T}$ associated with the decomposition $\widetilde{\nu} = \widetilde{\nu}^+ - \widetilde{\nu}^-$: the positive and negative variations of $\widetilde{\nu}$. Now consider the (real-valued) positive linear functional Ψ on $C(X,\mathbb{R})$ such that

$$\Psi(f) \,=\, \int f\,d\widetilde{\nu}^+ = \int f\,d\widetilde{\nu}^-.$$

By uniqueness of the Borel measure in Theorem 12.5 we get $\widetilde{\nu}^+ = \widetilde{\nu}^-$, so that $\widetilde{\nu} = 0$, and hence $\nu = \nu'$, which proves uniqueness. Finally, recall that $C(X) \subseteq B(X)$ whenever X is compact, and so $\sup|f| < \infty$, and also that

$$|\Phi(f)| \,=\, \left|\int f\,d\nu\right| \,\le\, \int |f|\,d|\nu| \,\le\, \sup|f|\,|\nu|(X)$$

for every $f \in C(X,\mathbb{R})$ (as we saw in Remark 10.3). Hence,

$$\|\Phi\| \,=\, \sup_{|f|\le 1} |\Phi(f)| \le |\nu|(X).$$

On the other hand, according to Example 7A and Theorem 7.4,

$$|\nu|(X) = \nu^+(X) + \nu^-(X) \le \mu(X) + \eta(X) = \Phi^+(1) + \Phi^-(1) = \|\Phi\|$$

by Lemma 12.4 since $C(X,\mathbb{R})$ contains the function 1. Therefore,

$$\|\Phi\| = |\nu|(X). \qquad \square$$

The next result extends Theorem 12.6 to bounded linear functionals on $C(X,\mathbb{C})$, where (finite) signed measures are replaced with complex measures.

Theorem 12.7. *Let X be a compact Hausdorff space. If Φ is a complex-valued bounded linear functional on the complex linear space $C(X,\mathbb{C})$, then there exists a unique quasiregular complex measure ζ on the Borel σ-algebra $\mathcal{X}_T$ of subsets of X such that*

$$\Phi(f) = \int f \, d\zeta \quad \textit{for every} \quad f \in C(X,\mathbb{C}).$$

Moreover, $\|\Phi\| = |\zeta|(X)$. (*See also Proposition 12.E.*)

Proof. Let X be a compact Hausdorff space, let

$$C(X,\mathbb{C}) = \big\{f\colon X \to \mathbb{C}\colon\ f \text{ is continuous}\big\}$$

be the complex linear space of all complex-valued continuous functions on X, and let $\Phi\colon C(X,\mathbb{C}) \to \mathbb{C}$ be a complex-valued bounded linear functional on $C(X,\mathbb{C})$. A function f in $C(X,\mathbb{C})$ can be written as $f = f_1 + i\, f_2$, where f_1 and f_2 (the real and imaginary parts of f) are real-valued continuous functions in the real linear space $C(X,\mathbb{R}) = \{\phi\colon X \to \mathbb{R}\colon\ \phi \text{ is continuous}\}$. Also, write the complex numbers $\Phi(f_1)$ and $\Phi(f_2)$ in their Cartesian representation, $\Phi(f_1) = \operatorname{Re}\Phi(f_1) + i \operatorname{Im}\Phi(f_1)$ and $\Phi(f_2) = \operatorname{Re}\Phi(f_2) + i \operatorname{Im}\Phi(f_2)$, where $\operatorname{Re}\Phi(f_i)$ and $\operatorname{Im}\Phi(f_i)$ are real numbers. Thus, to each complex-valued functional $\Phi\colon C(X,\mathbb{C}) \to \mathbb{C}$ on the complex linear space $C(X,\mathbb{C})$ we can associate two real-valued functionals, say, $\Phi_1\colon C(X,\mathbb{R}) \to \mathbb{R}$ and $\Phi_2\colon C(X,\mathbb{R}) \to \mathbb{R}$ on the real linear space $C(X,\mathbb{R})$, defined by

$$\Phi_1(\phi) = \operatorname{Re}\Phi(\phi) \quad \text{and} \quad \Phi_2(\phi) = \operatorname{Im}\Phi(\phi),$$

so that

$$\Phi(\phi) = \Phi_1(\phi) + i\,\Phi_2(\phi)$$

for every $\phi \in C(X,\mathbb{R})$. These real-valued functionals Φ_1 and Φ_2 on $C(X,\mathbb{R})$ are linear and bounded. Indeed, since Φ is linear, it follows that for every $\alpha \in \mathbb{R}$ and every $\phi, \psi \in C(X,\mathbb{R})$,

$$\alpha\,\Phi_1(\phi) + i\,\alpha\,\Phi_2(\phi) = \alpha\,\Phi(\phi) = \Phi(\alpha\,\phi) = \Phi_1(\alpha\,\phi) + i\,\Phi_2(\alpha\,\phi),$$

$$\begin{aligned}\Phi_1(\phi+\psi) + i\,\Phi_2(\phi+\psi) &= \Phi(\phi+\psi) = \Phi(\phi) + \Phi(\psi)\\ &= \big(\Phi_1(\phi) + \Phi_1(\psi)\big) + i\big(\Phi_2(\phi) + \Phi_2(\psi)\big),\end{aligned}$$

and so Φ_1 and Φ_2 are linear. They are also bounded, since Φ is bounded:

$$\begin{aligned}\max\{\|\Phi_1\|^2, \|\Phi_2\|^2\} &= \max\Big\{\sup_{|\phi|\le 1} |\Phi_1(\phi)|^2, \sup_{|\psi|\le 1} |\Phi_2(\psi)|^2\Big\}\\ &\le \sup_{|\phi|\le 1} \big(|\Phi_1(\phi)|^2 + |\Phi_2(\phi)|^2\big)\\ &= \sup_{|\phi|\le 1} |\Phi_1(\phi) + i\,\Phi_2(\phi)|^2\\ &\le \sup_{|\phi|\le 1} |\Phi(\phi)|^2 \le \sup_{|f|\le 1} |\Phi(f)|^2 = \|\Phi\|^2.\end{aligned}$$

Therefore, according to Theorem 12.6, there exist unique quasiregular signed measures ν_1 and ν_2 on $\mathcal{X}_{\mathcal{T}}$ such that

$$\Phi_1(f_1) = \int f_1\, d\nu_1, \qquad \Phi_1(f_2) = \int f_2\, d\nu_1,$$

$$\Phi_2(f_1) = \int f_1\, d\nu_2, \qquad \Phi_2(f_2) = \int f_2\, d\nu_2.$$

Observe that by the linearity of Φ we get, for each $f \in C(X, \mathbb{C})$,

$$\begin{aligned}\Phi(f) &= \Phi(f_1 + i\, f_2) = \Phi(f_1) + i\,\Phi(\ f_2)\\ &= \Phi_1(f_1) + i\,\Phi_2(f_1) + i\,\big(\Phi_1(f_2) + i\,\Phi_2(f_2)\big)\\ &= \Phi_1(f_1) + i\,\Phi_1(f_2) - \Phi_2(f_2) + i\,\Phi_2(f_1).\end{aligned}$$

Hence, since $\int(\ )\,d\mu$ is a linear functional, for every $f \in C(X, \mathbb{C})$,

$$\begin{aligned}\Phi(f) &= \int f_1\, d\nu_1 + i\int f_2\, d\nu_1 - \int f_2\, d\nu_2 + i\int f_1\, d\nu_2\\ &= \int (f_1 + i\, f_2)\, d\nu_1 + i\int (f_1 + i\, f_2)\, d\nu_2\\ &= \int f\, d\nu_1 + i\int f\, d\nu_2 = \int f\, d\zeta,\end{aligned}$$

where $\zeta = \nu_1 + i\,\nu_2$ is a complex measure on $\mathcal{X}_{\mathcal{T}}$, which is quasiregular and unique since the signed measures ν_1 and ν_2 on $\mathcal{X}_{\mathcal{T}}$ are quasiregular and unique (cf. proof of Theorem 12.6). Finally, to verify that

$$\|\Phi\| = |\zeta|(X),$$

we proceed as follows. First recall that, since $C(X) \subseteq B(X)$ whenever X is compact, we get $\sup|f| < \infty$ for every $f \in C(X) = C(X, \mathbb{C})$, and also

$$|\Phi(f)| \;=\; \left|\int f\, d\zeta\right| \;\le\; \int |f|\, d|\zeta| \;\le\; \sup|f|\, |\zeta|(X)$$

for every $f \in C(X) = C(X, \mathbb{C})$ (as we saw in Remark 10.6), which implies

$$\|\Phi\| \;=\; \sup_{f\in C(X),\, |f|\le 1} |\Phi(f)| \;=\; \sup_{f\in C(X),\, |f|\le 1} \left|\int f\, d\zeta\right| \;\le\; |\zeta|(X).$$

Next we show the reverse inequality, so that

$$\sup_{f\in C(X),\ |f|\le 1}\left|\int f\,d\zeta\right| = |\zeta|(X).$$

In fact, we have already seen in Section 10.3 that the above identity holds,

$$\sup_{|f|\le 1}\left|\int f\,d\zeta\right| = |\zeta|(X),$$

when the supremum is taken over all complex-valued *integrable* functions f on X such that $|f|\le 1$. Now we show that this still holds if the supremum is taken over all complex-valued *continuous* functions f on X with $|f|\le 1$. As X is a compact Hausdorff space, we can apply Lemma 12.2 and Corollary 12.3.

(a) By Lemma 12.2 we get: *if* $\{U_k\}$ *is a finite open covering of* X*, then there is a finite compact covering* $\{K_k\}$ *of* X *such that* $K_k\subseteq U_k$ *and, for each index* k*, there is a function* $f_k\in C_c(X)=C(X)$ *such that* $\chi_{K_k}\le f_k\le\chi_{U_k}$. Let $\{A_1^+, A_1^-\}$ and $\{A_2^+, A_2^-\}$ be Hahn decompositions of X with respect to the signed measures ν_1 and ν_2. Take a disjointification $\mathcal{A}_{1,2}^{+-}$ of the collection $\{A_1^+\cap A_2^+,\ A_1^+\cap A_2^-,\ A_1^-\cap A_2^+,\ A_1^-\cap A_2^-\}$. So $\mathcal{A}_{1,2}^{+-}$ is a measurable partition of X. As we saw in Remark 10.5 via Remark 10.3, for every measurable covering of X there is a measurable partition of X consisting of subsets of the sets in the partition $\mathcal{A}_{1,2}^{+-}$. Then, by Lemma 12.2 (see also Theorem 11.3), take finite partitions $\{U_k'\}$ and $\{K_k'\}$ of X consisting of the intersection of U_k and K_k with sets from $\mathcal{A}_{1,2}^{+-}$ such that $K_k'\subset U_k'$ and $\chi_{K'_k}\le f_k\le\chi_{U'_k}$ with f_k continuous, for each k. Thus, according to the argument of Remark 10.5,

$$|\zeta(K_k')| \le \left|\int f_k\,d\zeta\right| \le |\zeta(U_k')|,$$

and so,

$$\sup\sum\nolimits_k|\zeta(K_k')| \le \sup\sum\nolimits_k\left|\int f_k\,d\zeta\right| \le \sup\sum\nolimits_k|\zeta(U_k')|,$$

with the supremum taken over all finite measurable partitions $\{U_k'\}$ of X of the form $U_k'=U_k\cap A$, for sets A in $\mathcal{A}_{1,2}^{+-}$, which generate finite measurable partitions $\{K_k'\}$, with the collection $\{f_k\}$ of continuous functions such that $\chi_{K'_k}\le f_k\le\chi_{U'_k}$. Recall that ζ is quasiregular. This means that $\zeta=\nu_1+i\,\nu_2$, where ν_1 and ν_2 are quasiregular signed measures (by Theorem 12.6); that is, ν_1 and ν_2 are differences of quasiregular finite positive measures, which are Borel measures μ on $\mathcal{X}_\mathcal{T}$ such that $\mu(U)=\sup\{\mu(K)\ K\subseteq U,\ K \text{ compact}\}$ for open sets U). Since $\sup\sum_k|\zeta(E_k)|=|\zeta|(X)$, as we saw in Remark 10.5, following the argument of Remark 10.3, where the supremum is taken over all finite measurable partitions $\{E_k\}$ of X, we may infer that

$$\sup\sum\nolimits_k\left|\int f_k\,d\zeta\right| = |\zeta|(X). \qquad (*)$$

(b) Arguing as in Corollary 12.3, we may take $\{f_k\}$ such that $\sum_k f_k=1$. Indeed, set $f=\sum_k f_k$, a real-valued continuous function on X for which $f\ge 1$

(since $\chi_{K_k} \le f_k$ and $\{K_k\}$ covers X). By setting $f'_k = f_k/f$ we get a finite collection $\{f'_k\}$ of functions in $C(X)$ with $0 \le f'_k \le \chi_{U_k}$ and $\sum_k f'_k = 1$. Such a system of functions is referred to as a *partition of unity* on X corresponding to an open covering $\{U_k\}$ of X. Thus, with $\sum_k f_k = 1 \in C(X)$ (so $\left|\sum_k f_k\right| = 1$) and $0 \le f_k \le \chi_{U_k}$ for each k, it is readily verified (taking again the supremum over all finite open coverings $\{U_k\}$ of X) that

$$\sup \sum\nolimits_k \left| \int f_k \, d\zeta \right| = \sup \sum\nolimits_k \left| \int_{U_k} f_k \, d\zeta \right| \le \sup_{f \in C(X),\, |f| \le 1} \left| \int f \, d\zeta \right|. \quad (**)$$

(c) Thus, by $(*)$ and $(**)$, we get the reverse inequality,

$$|\zeta|(X) = \sup \sum\nolimits_k \left| \int f_k \, d\zeta \right| \le \sup_{f \in C(X),\, |f| \le 1} \left| \int f \, d\zeta \right|.$$

Therefore,

$$\|\Phi\| = \sup_{f \in C(X),\, |f| \le 1} |\Phi(f)| = \sup_{f \in C(X),\, |f| \le 1} \left| \int f \, d\zeta \right| = |\zeta|(X). \qquad \square$$

12.4 Additional Propositions

We will split this section into two parts. Part 1 deals with several aspects of Riesz representations. Part 2 deals with regular Borel measures and their connection with the results of Part 1.

Riesz Representations

Take the Banach space $L^p = L^p(\mu) = L^p(X, \mu) = L^p(X, \mathcal{X}, \mu)$ of all (equivalence classes of) $\mathbb{F}$-valued functions $f : X \to \mathbb{F}$ on a nonempty set X that are p-integrable if $p \ge 1$, or essentially bounded if $p = \infty$, where $\mu : \mathcal{X} \to \mathbb{R}$ is a positive measure on a σ-algebra $\mathcal{X}$ of subsets of X (see Chapter 5). If $p > 1$, then the Hölder conjugate $q > 1$ of p is the solution to the equation $\frac{1}{p} + \frac{1}{q} = 1$; for $p = 1$, set $q = \infty$ (and vice versa). In Section 7.2, after proving the Radon–Nikodym Theorem, we remarked that "a major application of the Radon–Nikodym Theorem is the Riesz Representation Theorem." Such a version of the Riesz Representation Theorem reads as follows.

Proposition 12.A. *If $p \ge 1$ and $\Phi : L^p \to \mathbb{F}$ is a bounded linear functional, then there is a unique $g \in L^q$, where q is the Hölder conjugate of p, such that*

$$\Phi(f) = \int f \overline{g} \, d\mu \quad \text{for every} \quad f \in L^p.$$

Moreover, $\|\Phi\| = \|g\|_q$. If $p = 1$ (so that $q = \infty$), then the (positive) measure μ is supposed to be σ-finite.

A particularly important special case is that of $p = q = 2$, where L^2 is a Hilbert space (see Section 5.1). As we have noticed in Section 7.2 (see the remark after the proof of Theorem 7.8), this particular case can be proved without using the Radon–Nikodym Theorem. The Riesz Representation Theorem in an abstract Hilbert space reads as follows.

Proposition 12.B. *Let $\mathcal{H}$ be an arbitrary Hilbert space. For every bounded linear functional $\Phi\colon \mathcal{H} \to \mathbb{F}$, there exists a unique vector $g \in \mathcal{H}$ such that*

$$\Phi(f) = \langle f\,;g\rangle \quad \textit{for every} \quad f \in \mathcal{H}.$$

Moreover, $\|\Phi\| = \|g\|$. Such a unique vector g in $\mathcal{H}$ is called the Riesz representation *of the functional Φ.*

The space L^2 is a concrete Hilbert space. Among the Banach spaces L^p, for every $p \geq 1$ and for $p = \infty$, the only one which is a Hilbert space is L^2; only $\|\ \|_2$ is induced by an inner product. The inner product in L^2 is

$$\langle f\,;g\rangle = \int f\overline{g}\,d\mu \quad \text{for every} \quad f, g \in \mathcal{H}.$$

So the Riesz Representation Theorem in L^2 can be independently gotten by Proposition 12.A or Proposition 12.B. We can get it by Proposition 12.B, and use it to yield another proof of the Radon–Nikodym Theorem (different from our proof of Theorem 7.8), and use the Radon–Nikodym Theorem to prove Proposition 12.A (see, e.g., [63, Remark 6.17] or [5, Exercise 8.V]).

The next result extends Theorem 12.5 from the linear space $C_c(X)$ of all continuous functions with compact support to the linear space

$$C_0(X) \text{ of all continuous functions that vanish at infinity.}$$

A function $f\colon X \to \mathbb{F}$ on a topological space X *vanishes at infinity* if for every $\varepsilon > 0$ there is a compact set $K_\varepsilon \subset X$ such that $|f(x)| < \varepsilon$ whenever $x \in X \backslash K_\varepsilon$.

Remark 12.A ($\mathbb{F}$-valued regular measures): Recall that by an $\mathbb{F}$-valued measure ζ on $\mathcal{X}_\mathcal{T}$ we mean a *finite positive*, or *signed*, or *complex* measure. A *regular* (*quasiregular*) *signed measure* $\nu = \nu^+ - \nu^-$ is one such that the finite positive measures ν^+ and ν^- are regular (quasiregular); a *regular* (*quasiregular*) *complex measure* $\zeta = \nu_1 + i\,\nu_2$ is one such that the signed measures ν_1 and ν_2 are regular (quasiregular) — cf. proofs of Theorems 12.6 and 12.7.

Proposition 12.C. *Let X be a locally compact Hausdorff space. If Φ is an $\mathbb{F}$-valued bounded linear functional on $C_0(X)$, then there is a unique quasiregular $\mathbb{F}$-valued measure ζ on the Borel σ-algebra $\mathcal{X}_\mathcal{T}$ of subsets of X such that*

$$\Phi(f) = \int f\,d\zeta \quad \textit{for every} \quad f \in C_0(X).$$

Moreover, $\|\Phi\| = |\zeta|(X)$. (*See also Proposition 12.E.*)

Let X be a topological space, let μ be a positive Borel measure on $\mathcal{X}_{\mathcal{T}}$, and consider the following plain set inclusions.

$$C_c(X) \subseteq C_0(X) \subseteq B(X) \subseteq L^\infty(X,\mu) \quad \text{and} \quad C_c(X) \subseteq L^p(X,\mu)$$

for every real number $p \geq 1$ (see the opening of Section 12.3). Compare the above inclusions in $L^p(X,\mu)$ or $L^\infty(X,\mu)$ with the inclusions in Problems 5.8 and 5.12. The above inclusions are interpreted in terms of the equivalence classes of each representative in $C_c(X)$, $C_0(X)$, or $B(X)$.

The next proposition closes the representation theorems saga with a gold key, giving a denseness result of fundamental importance for the linear spaces on which the Riesz Representation Theorems were built, viz., $C_c(X)$, $C_0(X)$, $C(X)$, $L^p(X,\mu)$. Recall: $C_c(X) = C_0(X) = C(X)$ if X is compact.

Proposition 12.D. *Let X be a locally compact Hausdorff space. If μ is a positive quasiregular Borel measure on the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ of subsets of the set X, then the linear space $C_c(X)$ is dense in the Banach space $(L^p(X,\mathcal{X}_{\mathcal{T}},\mu),\| \ \|_p)$ for every real $p \geq 1$.*

Regular Borel Measures

Regular Borel measures play a crucial role in many aspects of measure theory, with an important influence in many parts of functional analysis. So far, we have been dealing with this topic in Theorem 11.6, and from Proposition 11.E to Proposition 11.I, inclusive, and also in Propositions 11.O, 11.R, and 11.S. Moreover, regular Borel measures have already played a significant part in the Riesz Representation Theorems (Theorems 12.5, 12.6, 12.7, and Proposition 12.C) and also in the central result stated in Proposition 12.D. The propositions below deal again with regular Borel measures; in particular, they are concerned with special situations where regularity comes as a consequence, and not as an imposed additional condition.

Proposition 12.E. *If X is a locally compact and σ-compact (in particular, a compact) Hausdorff space, then there exists a Borel σ-algebra of subsets of X (i.e., a σ-algebra including the σ-algebra $\mathcal{X}_{\mathcal{T}}$ — see the remark before Theorem 11.4) such that the Borel measure of Theorem 12.5 (and so of Theorems 12.6 and 12.7 and Propositions 12.C and 12.D) is regular.*

Proposition 12.F. *Let μ be an inner regular Borel measure on the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ of subsets of a locally compact Hausdorff space X. If X is σ-compact (in particular, compact), then μ is regular.*

Compare with Propositions 11.G, 11.R, and 11.S. By Proposition 12.F we can get more along those lines, namely, *every inner regular Borel measure on the Borel σ-algebra of a compact Hausdorff space is regular.*

First recall the following standard notions.

(i) A topological space satisfies the *second axiom of countability* if it possesses a countable (topological) base; it satisfies the *first axiom of countability* if there is a countable neighborhood base at every point of X (cf., e.g., [14, p. 235]). It is clear that the second axiom of countability implies the first (we have already met them in Propositions 11.B and 11.R).

Next consider the following well-known results.

(ii) *Every topological space that satisfies the second axiom of countability is separable* (see, e.g., [13, p. 33]).

(iii) *If a locally compact Hausdorff space satisfies the second axiom of countability, then it is σ-compact* (see, e.g., [13, Problem 10.A]).

(iv) (a) *A metric space satisfies the second axiom of countability if and only if it is separable.* (b) *Every compact metric space satisfies the second axiom of countability, thus is separable.* (c) *A compact Hausdorff space is metrizable if and only if it satisfies the second axiom of countability* (see, e.g., [13, Proposition 4.1 and Corollary 4.4]).

Proposition 12.G. *Over a locally compact Hausdorff space satisfying the second axiom of countability, a Borel measure is regular.*

Proposition 12.H. *Over a separable locally compact metric space, a Borel measure is regular.*

Proposition 12.I. *Over a compact metric space, a Borel measure is regular.*

Proposition 12.J. *A Borel measure on the σ-algebra of Borel subsets of an arbitrary Borel subset of $\mathbb{R}^n$ or $\mathbb{C}^n$ is regular.*

Proposition 12.K. *The Borel–Stieltjes measures in Problem 7.15(a,b,c), and the arc-length measure over the unit circle in Problem 7.16, are regular.*

Compare Propositions 12.G to 12.K with Propositions 11.G and 11.H, and also with Theorem 11.6. See Proposition 12.E as well.

We close the chapter with an application of Propositions 12.D and 12.I. Let X be a metric space and consider the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ of subsets of X. In particular, for $X = \mathbb{C}$, let $\mathcal{C}_{\mathcal{T}}$ be the σ-algebra of subsets of $\mathbb{C}$ generated by the open subsets of $\mathbb{C}$.

Proposition 12.L. *If X is a compact subset of $\mathbb{C}$ and μ is a finite measure on $\mathcal{X}_{\mathcal{T}}$ (or, if μ is a Borel measure on $\mathcal{C}_{\mathcal{T}}$), then $(L^p(X, \mathcal{X}_{\mathcal{T}}, \mu), \| \ \|_p)$ is separable for every $p \geq 1$.*

Notes on Riesz Representation: Propositions 12.A to 12.D are classical and crucial results in functional analysis, which extend the original form of the Riesz Representation Theorem (Theorem 12.5) in many directions. Proposition 12.A is the usual form of it in the concrete Banach spaces L^p. Proofs using somewhat different approaches can be found, for instance, in [5, Theorems 8.14 and 8.15], [62, Theorems 11.29 and 11.30], or [63, Theorem 6.16]). Proposition 12.B is the standard version for abstract Hilbert spaces, which does not depend on any measure-theoretical concept (see, e.g., [43, Theorem 5.62].) For the extension to the concrete linear space $C_0(X)$ in Proposition 12.C see [63, Theorem 6.19], and for the pivotal denseness result of Proposition 12.D see [8, Theorem 29.14].

Proposition 12.D does not hold for $p = \infty$: the Lebesgue measure λ on the real line is a locally compact regular Borel measure, and the unit function 1 on $\mathbb{R}$ lies in $L^\infty(\mathbb{R}, \lambda)$ but is not approached in the sup-norm by any continuous function with compact support (i.e., by any function in $C_c(\mathbb{R})$).

Notes on Regular Borel Measures: Proposition 12.E rescues regular measures in Theorems 12.5 to 12.7 (see [63, Theorem 2.17]). For Propositions 12.F and 12.G, see [13, Problem 10.M and Corollary 10.6], respectively. Proposition 12.I is an immediate consequence of Proposition 12.G (cf. (iv.b)), and so are Propositions 12.J and 12.K (cf. [13, Example 10.B]). According to items (iii) and (iv) above Proposition 12.G, a separable locally compact metric space is σ-compact, and so Proposition 12.H is a consequence of Proposition 11.S(b).

Proof of Proposition 12.L. Take a compact subset X of $\mathbb{C}$. Let $C(X)$ be the set of all complex-valued continuous functions on X, and let $P(X) \subset C(X)$ be the set of all polynomials $q(\ ,\ ) : X \times X \to \mathbb{C}$ in z and $\overline{z}$ (equivalently, $q(\ ,\overline{\ }) : X \to \mathbb{C}$ such that $z \mapsto q(z, \overline{z})$). The Stone–Weierstrass Theorem says:

if X is compact, then $P(X)$ is dense in the Banach space $(C(X), \|\ \|_\infty)$

(see, e.g., [18, Theorems 7.3.1, and 7.3.2]). That is,

$$X \text{ compact} \quad \Longrightarrow \quad \overline{P(X)} = C(X) \quad \text{in the sup-norm topology.}$$

If μ is a finite measure on $\mathcal{X}_\mathcal{T}$, and since $P(X)$ is dense in $(C(X), \|\ \|_\infty)$, it follows that $P(X)$ is dense in the normed space $(C(X), \|\ \|_p)$ as well:

$$\overline{P(X)} = C(X) \quad \text{in the } p\text{-norm topology for every } p \geq 1.$$

(Indeed, with $0 < \mu(X) < \infty$ and X compact, for every $\varepsilon > 0$ and an arbitrary $f \in C(X)$ there exists $q \in P(X)$ such that $\|f - q\|_\infty = \sup_X |f - q| < \frac{\varepsilon}{\mu(X)^{1/p}}$ because $P(X)$ is dense in $C(X)$ in the sup-norm topology, and hence $\|f - q\|_p^p = \int_X |f - q|^p \, d\mu \leq \sup_X |f - q|^p \, \mu(X) < \varepsilon^p$ by Remark 10.1, so that $\|f - q\|_p < \varepsilon$, and $P(X)$ is dense in $C(X)$ in the p-norm topology.)

Moreover, regard $(C(X), \|\ \|_p)$ as a linear manifold of $(L^p(X, \mathcal{X}_\mathcal{T}, \mu), \|\ \|_p)$ (in the equivalence-class sense), and recall from Proposition 12.I that, as X is compact and μ is finite, μ is regular. Thus Proposition 12.D ensures that

$$\overline{C(X)} = L^p(X, \mathcal{X}_\mathcal{T}, \mu) \quad \text{in the } p\text{-norm topology for every } p \geq 1.$$

Therefore, by transitivity, $P(X)$ is dense in $L^p(X, \mathcal{X}_\mathcal{T}, \mu)$; that is,

$$\overline{P(X)} = L^p(X, \mathcal{X}_\mathcal{T}, \mu) \quad \text{in the } p\text{-norm topology for every } p \geq 1.$$

Since the set $P(X)$ of all polynomials is a linear span of the countable set $\{e_k, \overline{e_\ell}\}_{k,\ell \in \mathbb{Z}}$ (with $e_k \colon X \to \mathbb{C}$ given by $e_k(z) = z^k$), the above density result ensures that $L^p(X, \mathcal{X}_\mathcal{T}, \mu)$ is spanned by a countable set, and so it is separable (see, e.g., [43, Proposition 4.9(b)]). □

Remark 12.B (Separability and nonseparability of L^p spaces):

(a) As we saw in the proof of Proposition 12.L, the Stone–Weierstrass Theorem ensures that if X is compact in $\mathbb{C}$, then $P(X)$ is dense in $C(X)$ in the sup-norm topology, and therefore, if X is compact,

$$(C(X), \|\ \|_\infty) \text{ is separable.}$$

Consequently, by Propositions 12.I and 12.D, if X is compact in $\mathbb{C}$ and if μ is a finite measure on $\mathcal{X}_\mathcal{T}$, or a Borel measure on $\mathcal{C}_\mathcal{T}$, then

$$(L^p(X, \mathcal{X}_\mathcal{T}, \mu), \|\ \|_p) \text{ is separable for every } p \geq 1.$$

On the other hand, according to Problem 5.18, we can infer that

$$(\ell^p, \|\ \|_p) \text{ is separable for every } p \geq 1$$

(as ℓ^0 is separable; see, e.g., [43, Example 3.P]). It is also well known that

$$(\ell^\infty, \|\ \|_\infty) \text{ is not separable}$$

(see, e.g., [43, Example 3.Q]), where the same argument can be applied to show that by setting $\mu = \lambda$, the Lebesgue measure on $\Re = \Re_{[0,1]}$, which is a Borel measure, or its restriction $\lambda = \lambda_{[0,1]}$, which is such that $\lambda\big([0,1]\big) = 1$,

$$(L^\infty([0,1], \Re, \lambda), \|\ \|_\infty) \text{ is not separable.}$$

(b) In fact, separability of $C(X)$ does not extend to L^∞ as it does for L^p, and this is illustrated as follows. As before, let $B(X)$ stand for the linear space of all complex-valued bounded functions on a set X. Equipping $B(X)$ with the plain sup-norm $\|\ \|_\infty$ (i.e., $\|f\|_\infty = \sup_{x \in X} |f(x)|$ for $f \in B(X)$), we get the Banach space $(B(X), \|\ \|_\infty)$. It is clear that $B(X) \subseteq L^\infty(X, \mu)$ for every measure μ on any σ-algebra $\mathcal{X}$ of subsets of X. The Weierstrass Theorem says: $C(X) \subseteq B(X)$ if X is a compact topological space (cf. Lemma 12.1). So

$$\text{if } X \text{ is compact,} \quad \text{then} \quad C(X) \subseteq B(X) \subseteq L^\infty(X, \mathcal{X}, \mu)$$

for every measure μ on $\mathcal{X}$. Equip $C(X)$ with the plain sup-norm so that $(C(X), \| \ \|_\infty)$ is a Banach space. Note that, even if X is compact in $\mathbb{C}$,

$$C(X) \text{ is not dense in } (B(X), \| \ \|_\infty),$$

that is, $\overline{C(X)} \neq B(X)$ because $C(X)$ is closed in $(B(X), \| \ \|_\infty)$. (Indeed, as is also well known, a subset of a complete metric space — in particular, of a Banach space — is complete if and only if it is closed — see, e.g., [43, Examples 3.S, 3.T and Corollary 3.41].) Therefore, even if μ is Borel on $\mathcal{C}_\mathcal{T}$ (equivalently, even if $\mu(X) < \infty$ on $\mathcal{X}_\mathcal{T}$ for X compact),

$$C(X) \text{ is not dense in } (L^\infty(X, \mathcal{X}_\mathcal{T}, \mu,), \| \ \|_\infty),$$

where now the same symbol $\| \ \|_\infty$ stands for the essential sup-norm (i.e., $\|f\|_\infty = \operatorname{ess\,sup}_{x \in X} |f(x)|$ for $f \in L^\infty(X, \mathcal{X}_\mathcal{T}, \mu)$ with respect to the measure μ); that is, $\overline{C(X)} \neq L^\infty(X, \mathcal{X}_\mathcal{T}, \mu)$ in the essential sup-norm topology. (In fact, if $\overline{C(X)} = L^\infty(X, \mathcal{X}_\mathcal{T}, \mu)$ in the essential sup-norm topology with respect to μ, then (by the above displayed string of inclusions) $\overline{C(X)} = B(X)$ in the essential sup-norm topology, and so in the plain sup-norm topology (in this case the measure μ is immaterial), which is false as we saw above.)

(c) However, compactness of X is not a *sine qua non* condition for separability of L^p. Indeed, let a σ-algebra $\mathcal{X}$ of subsets of a set X be *countably generated* if it is generated (i.e., $\mathcal{X} = \mathcal{X}_\mathcal{C}$) by a countable collection $\mathcal{C} \subseteq \mathcal{X}$.

If μ is a σ-finite measure on a countably generated $\mathcal{X}$, then
$L^p(X, \mathcal{X}, \mu)$ is separable for every $p \geq 1$.

(See, e.g., [16, Proposition 3.4.5].) Note that this item offers an alternate proof of Proposition 12.L. In particular, since $\Re$ is easily verified to be countably generated (apply the same argument used in Example 2C), we get

$$L^p(\mathbb{R}, \Re, \lambda) \text{ is separable for every } p \geq 1,$$

where $\lambda \colon \Re \to \overline{\mathbb{R}}$ is the Lebesgue measure, which is Borel, regular, and σ-finite on the Borel σ-algebra $\Re$ of subsets of the noncompact $\mathbb{R}$.

(d) On the other hand, in spite of the results in item (c) and, in particular, of Proposition 12.L, separability of L^p spaces does not hold for finite measures in general. Actually, there exist nonseparable L^2 spaces with finite measures, and also non-σ-finite measures for which L^2 is separable (see, e.g., [34, pp. 2, 3] and [13, pp. 174, 192, 376]).

Suggested Readings

Bauer [8], Brown and Pearcy [13], Cohn [16], Conway [17], Halmos [31], Kingman and Taylor [39], Royden [62], Rudin [63].



13

Invariant Measures

13.1 Topological Groups

A *binary operation* on a nonempty set X is a mapping of the Cartesian product of X with itself into X, say, $\star\colon X\times X\to X$. It is usual to write $x\star y$ instead of $\star(x,y)$ to indicate the value in X of the mapping $\star$ at the point (x,y) in $X\times X$. In this context it is convenient to interpret the binary operation $\star$ multiplicatively, so that $x\star y$ is interpreted as the product of x and y, and it is written in a simplified form as $x\,y$. If a binary operation on X has the property that

$$x\,(y\,z)\;=\;(x\,y)\,z$$

for every x, y, and z in X, then it is said to be *associative*. So we can drop the parentheses and write $x\,y\,z$. If there exists an element e in X such that

$$x\,e\;=\;e\,x\;=\;x$$

for every $x\in X$, then e is said to be the *identity element* (or the *neutral element*) with respect to the binary operation on X. It is easy to show that if a binary operation has a neutral element e, then e is unique. If an associative binary operation on X has an identity element e in X, and if for some $x\in X$ there exists $x^{-1}\in X$ such that

$$x\,x^{-1}\;=\;x^{-1}\,x\;=\;e,$$

then x^{-1} is called the *inverse* of x with respect to the underlying binary operation. It is also easy to show that if the inverse of x exists with respect

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4_13

to an associative binary operation, then it is unique. Note that $(x\,y)^{-1} = y^{-1}x^{-1}$ (indeed, $x\,y\,y^{-1}x^{-1} = e$).

A *group* is a nonempty set X on which is defined a binary operation that

$$\text{is associative, has an identity element } e \in X,$$
$$\text{and every } x \text{ in } X \text{ has an inverse } x^{-1} \text{ in } X.$$

In general, the binary operation is not commutative in the sense that there may exist a pair of elements $x, y \in X$ for which $x\,y \neq y\,x$. However, if the binary operation on a group X is such that, for every $x, y \in X$,

$$x\,y = y\,x,$$

then X is said to be a *commutative group* (or an *Abelian group*). A nonempty subset M of a group X is a *subgroup* of X if M is itself a group; that is, if

$$e \in M, \text{ and } x^{-1} \text{ and } x\,y \text{ lie in } M \text{ for every } x \text{ and } y \text{ in } M$$

(equivalently, if $x\,y^{-1} \in M$ whenever x and y are in M). For arbitrary subsets A and B of a group X, let A^{-1} and AB stand for the subsets of X consisting of all elements of the forms x^{-1} and $x\,y$, respectively, with $x \in A$ and $y \in B$. Thus a nonempty subset M of a group X is a subgroup of X if and only if $MM^{-1} \subseteq M$. Note that $A \cap A^{-1} = (A \cap A^{-1})^{-1}$ for every $A \subseteq X$.

We write A^2 for AA. Observe that $(AA)A = A(AA)$, and so we write A^3 for $AAA = (AA)A = A(AA)$. Generalizing, we write A^n for the multiplication of A with itself n times, $A \ldots A \subseteq X$, for every positive integer n. If $e \in A$, then $\{A^n\}$ is an increasing sequence (with respect to the inclusion ordering) since, in this case, $A^n = A^n\{e\} \subseteq A^n A = A^{n+1}$ for every positive integer n. For each $x \in X$ and each $A \subseteq X$, let xA and Ax denote $\{x\}A$ and $A\{x\}$, which are referred to as the *left translation* and *right translation* of the set $A \subseteq X$, respectively. If M is a subgroup of X, then the sets xM and Mx are referred to as the *left coset* and *right coset* of M. An *invariant subgroup* M of X is one for which $xM = Mx$ for every $x \in X$. A *homomorphism* is a map $\Phi\colon X \to Y$ of a group X into a group Y such that

$$\Phi(x\,y) = \Phi(x)\,\Phi(y) \quad \text{for every} \quad x, y \in X,$$

where $x\,y$ is the result of the binary operation on the group X and $\Phi(x)\Phi(y)$ is the result of the binary operation on the group Y.

A *topological group* is a group X that is also a topological space for which the map $(x, y) \mapsto x\,y^{-1}$ is continuous from $X \times X$ (equipped with the usual product topology) to X or, equivalently, for which

$$\text{the } \textit{multiplication map } (x, y) \mapsto x\,y \text{ from } X \times X \text{ to } X \text{ and}$$
$$\text{the } \textit{inversion map } x \mapsto x^{-1} \text{ from } X \text{ to } X \text{ are both continuous.}$$

Take an arbitrary $x_0 \in X$. The maps $y \mapsto x_0\, y$ and $y \mapsto y\, x_0$ from X to X are referred to as the *left multiplication* by x_0 and *right multiplication* by x_0, respectively. These are sections of the multiplication map $(x, y) \mapsto x\, y$ (see Section 9.2). Since a restriction of a continuous map is continuous, it follows that a section of a continuous function is continuous, and so the right and left multiplication maps are both continuous. It is readily verified that the map $y \mapsto x_0^{-1} y$ from X to X is the inverse of the left multiplication $y \mapsto x_0\, y$, and the map $y \mapsto y\, x_0^{-1}$ from X to X is the inverse of the right multiplication $y \mapsto y\, x_0$, which are again continuous (since these inverses are precisely the left and right multiplications by x_0^{-1}).

A *symmetric neighborhood* in a topological group is a neighborhood N of the identity that coincides with its inverse, that is,

$$e \in N \quad \text{and} \quad N = N^{-1}.$$

Topological properties are naturally attributed to topological groups. We refer to *compact*, *locally compact*, or *Hausdorff groups* if, as topological spaces, they are compact, locally compact, or Hausdorff. A map between topological spaces is *continuous* if the inverse images of open (closed) sets are open (closed). A *homeomorphism* Ψ of a topological space X onto a topological space Y is an invertible map that is continuous with a continuous inverse:

$$\Psi\colon X \to Y \text{ is continuous and there exists } \Psi^{-1}\colon Y \to X \text{ continuous.}$$

So a homeomorphism between topological spaces maps open (closed) sets into open (closed) sets, and the inverse image of an open (closed) set is an open (closed) set. That is, if $\Psi\colon X \to Y$ is a homeomorphism, then $\Psi(A)$ is open (closed) in Y if and only if $A = \Psi^{-1}(\Psi(A))$ is open (closed) in X. Observe that the left and right multiplication maps are homeomorphisms of a topological group X onto itself (and so is the inversion map).

Remark 13.A (Four classical examples of topological groups):

(a) *Multiplicative and additive scalar groups*. Let $\mathbb{F}$ denote the set of either all real or all complex numbers, and consider the set $\mathbb{F}\backslash\{0\}$ of all nonzero real or complex numbers. Both $\mathbb{F}$ and $\mathbb{F}\backslash\{0\}$ are commutative locally compact Hausdorff groups when equipped with the usual metric topology, where $\mathbb{F}\backslash\{0\}$ is equipped with the binary operation defined by the usual multiplication in $\mathbb{F}$ with identity $e = 1$ (a multiplicative group), and $\mathbb{F}$ is equipped with the binary operation defined by the usual addition in $\mathbb{F}$ with identity $e = 0$ (an additive group).

(b) The *circle group*. The unit circle about the origin $\mathbb{T} = \{z \in \mathbb{C}\colon |z| = 1\}$ in the complex plane $\mathbb{C}$, equipped with the usual metric topology, is a commutative compact Hausdorff (multiplicative) group with identity $e = 1$.

(c) The *general linear group*. Let $GL(n)$ be the collection of all (real or complex) $n \times n$ invertible matrices equipped with the ordinary matrix operations and the uniform topology induced by the usual topology of the normed space $\mathbb{F}^n$. This is a noncommutative locally compact Hausdorff (multiplicative) group with identity $e = I$, where I denotes the identity $n \times n$ matrix.

(d) The *unitary operator group*. Let $\mathcal{B}[\mathcal{H}]$ be the algebra of all *operators* on a Hilbert space $\mathcal{H}$ (i.e., of all bounded linear transformations of $\mathcal{H}$ into itself) equipped with the uniform topology. Take the set of all *unitary operators*; that is, of all operators T from $\mathcal{B}[\mathcal{H}]$ for which $T^*T = TT^* = I$, where I is the identity operator and T^* is the *adjoint* of T. This is a noncommutative Hausdorff (multiplicative) group with identity $e = I$ that is not locally compact if $\mathcal{H}$ is infinite-dimensional. (Here multiplication means composition.)

The following lemma collects standard results on topological groups that will be required in the sequel. The result in item (c) is central.

Lemma 13.1. *Let X be a topological group.*

(a) *If K and C are compact sets in X, then K^{-1} and CK are compact.*

(b) *Take $x \in X$ and $U \subseteq X$ arbitrary. The set U is open if and only if xU is open, if and only if Ux is open, if and only if U^{-1} is open.*

(c) *If G is an open (bounded) neighborhood of the identity e, then there is a decreasing sequence $\{U_n\}$ of open (bounded) symmetric neighborhoods U_n of e such that, for each integer $n \geq 1$, $U_n^k \subseteq G$ for every $1 \leq k \leq n+1$.*

(d) *If K is compact and G is open (bounded) in X such that $K \subseteq G$, then there is an open (bounded) symmetric neighborhood U of e such that $KU \subseteq G$, $UK \subseteq G$. If X is locally compact and Hausdorff, $UKU \subseteq G$.*

(e) *If X is locally compact and Hausdorff, and if C and K are disjoint compact subsets of X, then there exists an open bounded symmetric neighborhood U of the identity e such that each of UC, CU and UCU is disjoint with each of UK, KU and UKU.*

(f) *X is Hausdorff if and only if $\{e\}$ is a closed set.*

Proof. Let X be a topological group and let K, C, U, U_n, G be subsets of X. Suppose they are nonempty to avoid trivialities.

(a) Recall: continuous images of compact sets are compact (cf. Claim in the proof of Lemma 12.1). Since inversion is continuous, it follows that K^{-1} is compact if K is. Since the Cartesian product of compact sets is compact, it follows that $C \times K$ is compact in $X \times X$ whenever C and K are compact in X; since multiplication is continuous, it follows that CK is compact.

(b) Since the left and right multiplication maps are homeomorphisms, xU is open if and only if U is, and Ux is open if and only if U is. Since inversion also is a homeomorphism, U^{-1} is open if and only if U is.

(c) Since $G = eG$, it follows that G lies in the range of the multiplication map. Since G is open in X, and since the multiplication map from $X \times X$ to X is continuous (in particular, it is continuous at (e, e)), the inverse image of G under this map is open in $X \times X$, and contains the point (e, e) because G contains e. So there exist $A \subseteq X$ and $B \subseteq X$ such that $A \times B$ is open in $X \times X$, $(e, e) \in A \times B$, and $AB = G$. Since $e \in A \cap B$ and $AB = G$, it follows that $A \subseteq G$ and $B \subseteq G$. Therefore, since $A \times B$ is open in $X \times X$, there exist open neighborhoods of e, say, $W_1 \subseteq A \subseteq G$ and $W_2 \subseteq B \subseteq G$, such that $W_1 W_2 \subseteq AB = G$. Hence, the set $W = W_1 \cap W_2 \subseteq G$ is open in X (intersection of open sets), contains e, and $W^2 = WW \subseteq W_1 W_2 \subseteq G$. Since W^{-1} is open (according to (b)) and contains e, it follows that $U_1 = W \cap W^{-1} \subseteq W \subseteq G$ is a symmetric ($U_1 = U_1^{-1}$) open neighborhood of e (intersection of open sets containing e), and $U_1^2 = U_1 U_1 \subseteq WW = W^2 \subseteq G$.

Outcome 1: *For every open neighborhood G of the identity e there is a symmetric open neighborhood U_1 of e such that $U_1 \subseteq G$ and $U_1^2 \subseteq G$.*

Thus, applying the above result to U_1 itself (which is an open neighborhood of e), it follows that there exists a symmetric open neighborhood U_2 of e such that $U_2 \subseteq U_1 \subseteq G$ and $U_2^2 \subseteq U_1 \subseteq G$. Therefore, since $e \in U_2$,

$$U_2^3 = U_2 U_2 U_2 \subseteq U_2 U_2 U_2 U_2 = U_2^4 = (U_2^2)(U_2^2) \subseteq U_1 U_1 = U_1^2 \subseteq G.$$

Outcome 2: *For every open neighborhood G of the identity e, there is a symmetric open neighborhood U_2 of e such that $U_2 \subseteq G$, $U_2^2 \subseteq G$, and $U_2^3 \subseteq G$.*

Again, applying the above result to U_2 (which is an open neighborhood of e), it follows that there exists a symmetric open neighborhood U_3 of e such that $U_3 \subseteq U_2 \subseteq G$, $U_3^2 \subseteq U_2 \subseteq G$, and $U_3^3 \subseteq U_2 \subseteq G$. Since $e \in U_3$, we get

$$U_3^4 \subseteq U_3^5 = (U_3^2)(U_3^3) \subseteq U_2 U_2 = U_2^2 \subseteq G.$$

Outcome 3: *For every open neighborhood G of the identity e, there is a symmetric open neighborhood U_3 of e such that $U_3^k \subseteq G$ for every $1 \leq k \leq 4$.*

Proceeding this way, if G is an open neighborhood of e, we eventually get down to a pair of symmetric open neighborhoods U_n and U_{n+1} of e such that (i) $U_{n+1} \subseteq U_n \subseteq G$, (ii) $U_n^k \subseteq G$ for $1 \leq k \leq n+1$, and (iii) $U_{n+1}^j \subseteq G$ for $1 \leq j \leq n+2$, for each integer $n \geq 1$. Thus, $\{U_n\}$ is a decreasing sequence of open symmetric neighborhoods of e such that $U_n^k \subseteq G$ for every k such that $1 \leq k \leq n+1$. Note that $U \subseteq G$ is bounded whenever G is.

(d) Take an arbitrary $x \in K \subseteq G \subseteq X$, where K is compact and G is open (bounded). Thus Gx^{-1} and $x^{-1}G$ are open (bounded) neighborhoods of e.

Then, by (c), there are open (bounded) symmetric neighborhoods V_x and W_x of e such that $V_x^2 \subseteq Gx^{-1}$ and $W_x^2 \subseteq x^{-1}G$. Since $e \in V_x \cap W_x$, we get that $K \subseteq \bigcup_{x\in K} V_x x$ and $K \subseteq \bigcup_{x\in K} x\, W_x$. Since K is compact, there exist finite open subcoverings such that $K \subseteq \bigcup_{i=1}^n V_{x_i} x_i$ and $K \subseteq \bigcup_{i=1}^m x_i W_{x_i}$ with x_i in K. Set $V = \bigcap_{i=1}^n V_{x_i}$ and $W = \bigcap_{i=1}^m W_{x_i}$, which are again open (bounded) symmetric neighborhoods of e (because each V_{x_i} and W_{x_i} are). Since $x \in K$, it follows that $x \in V_{x_i} x_i$ and $x \in x_i W_{x_i}$ for some x_i, and so

$$Vx \subseteq V_{x_i} x \subseteq V_{x_i} V_{x_i} x_i = V_{x_i}^2 x_i \subseteq G x_i^{-1} x_i = G \quad \text{for every} \quad x \in K,$$

$$x\, W \subseteq x\, W_{x_i} \subseteq x_i W_{x_i} W_{x_i} = x_i W_{x_i}^2 \subseteq x_i^{-1} x_i G = G \quad \text{for every} \quad x \in K.$$

Thus, $VK \subseteq G$ and $KW \subseteq G$. If X is locally compact and Hausdorff, then (by Theorem 11.3) there is an open neighborhood N of e and a compact C such that $\{e\} \subseteq N \subseteq C \subseteq V$. Set $V' = N \cap N^{-1}$, an open (cf. (b)) symmetric neighborhood of e such that $V' \subseteq C \subseteq V$. Hence, $CK \subseteq VK \subseteq G$. Recall that CK is compact by (a). Then the preceding result ensures that there is an open (bounded) symmetric neighborhood W' of e such that $CKW' \subseteq G$. Set $U = V' \cap V \cap W \cap W'$, an open (bounded) symmetric neighborhood of e. So

$$UK \subseteq VK \subseteq G, \quad KU \subseteq KW \subseteq G, \quad \text{and} \quad UKU \subseteq V'KW' \subseteq CKW' \subseteq G.$$

(e) Suppose X is Hausdorff. If C and K are compact, then C and K are closed (Lemma 11.2(g)), and so $X\backslash C$ and $X\backslash K$ are open. If $C \cap K = \varnothing$, then $C \subseteq X\backslash K$ and $K \subseteq X\backslash C$. Since X is Hausdorff, it follows by Lemma 11.2(f) that for every $x \in X\backslash C$ there is an open set G_1 and an open neighborhood N_x of x such that $C \subseteq G_1$ and $G_1 \cap N_x = \varnothing$. In particular, this holds for every $x \in K \subset X\backslash C$, and so there is an open set $G_2 = \bigcup_{x\in K} N_x$ so that $K \subseteq G_2$ and $G_1 \cap G_2 = \varnothing$. If the Hausdorff X is also locally compact, then we can take G_1 and G_2 bounded by Theorem 11.3. Thus, by item (d), there are open bounded symmetric neighborhoods U_1 and U_2 of e such that

$$U_1 C \subseteq G_1,\ CU_1 \subseteq G_1,\ U_1 C U_1 \subseteq G_1,\ U_2 K \subseteq G_2,\ KU_2 \subseteq G_2,\ U_2 K U_2 \subseteq G_2.$$

Set $U = U_1 \cap U_2$, an open bounded symmetric neighborhood of e such that

$$UC \subseteq U_1 C \subseteq G_1, \quad CU \subseteq CU_1 \subseteq G_1, \quad UK \subseteq U_2 K \subseteq G_2, \quad KU \subseteq KU_2 \subseteq G_2,$$

$$UCU \subseteq U_1 C U_1 \subseteq G_1, \quad \text{and} \quad UKU \subseteq U_2 K U_2 \subseteq G_2.$$

Therefore, since $G_1 \cap G_2 = \varnothing$, it follows that each of UC, CU, and UCU is disjoint with each of UK, KU and UKU.

(f) Lemma 11.2(e) ensures that if a topological space X is Hausdorff, then $\{e\}$ is closed. Conversely, suppose X is a topological group, and suppose $\{e\}$ is closed. Consider the left multiplication map, so that $\{x\} = x\{e\}$ is closed

for every $x \in X$ (reason: the left multiplication map is a homeomorphism so that it maps open sets into open sets and, dually, closed sets into closed sets). Suppose $x \neq e$. Since $\{x\}$ is closed, its complement $X \backslash \{x\}$ is open, so that there exists an open neighborhood G of e included in $X \backslash \{x\}$, and so $x \notin G$. By item (c) there exists an open symmetric neighborhood U of e such that $U^2 \subseteq G$. This ensures that

$$U \cap x\,U = \varnothing.$$

Indeed, if $y \in U \cap x\,U$, then $y = x\,z$ for some $z \in U$, which implies that $x = y\,z^{-1} \in UU = U^2 \subseteq G$ (because U is symmetric; i.e., $U = U^{-1}$), which is a contradiction ($x \notin G$). Hence $U \cap x\,U$ is empty. Thus, since $e \in U$ and $x \in x\,U$, it follows that there exist neighborhoods N_x of x and N_e of e such that $N_x \cap N_e = \varnothing$. Take any distinct points x and y in X. Since $x \neq y$, we get $x^{-1}y \neq e$, and so there exist neighborhoods $N_{x^{-1}y}$ of $x^{-1}y$ and N_e of e such that $N_{x^{-1}y} \cap N_e = \varnothing$, which implies that $xN_{x^{-1}y} \cap xN_e = \varnothing$. Since $y \in xN_{x^{-1}y}$ and $x \in xN_e$, it follows that X is Hausdorff. □

13.2 Haar Measure

Let X be a locally compact Hausdorff group, and let $\mathcal{A}$ be a Borel σ-algebra of subsets of X (i.e., a σ-algebra that includes $\mathcal{X}_\mathcal{T}$ — cf. remark after Theorem 11.3). A measure μ on $\mathcal{A}$ is *left invariant* or *right invariant* if

$$\mu(xE) = \mu(E) \quad \text{or} \quad \mu(Ex) = \mu(E) \quad \text{for every } E \text{ in } \mathcal{A} \text{ and every } x \in X,$$

respectively. It is called an *invariant measure* if it is left invariant. In other words, a measure on a Borel σ-algebra of subsets of a topological group is left invariant or right invariant if it is invariant under left or right translations. These are the *invariance properties*. For commutative groups, a measure is left invariant if and only if it is right invariant (called a *translation invariant measure*, or a measure satisfying the *translation invariance* property).

A left (*right*) *Haar measure* is a left (right) invariant nonzero positive Borel measure on a Borel σ-algebra of subsets of a locally compact Hausdorff group. A *Haar measure* is a left Haar measure.

Remark 13.B (Left-right symmetry): There is an asymmetry in our definition of invariant and Haar measures. Left and right translations are naturally distinct but symmetrical properties in a (noncommutative) group. So why favor "left" when defining Haar measure? There is a good reason. In this context, as we might have already guessed when considering the proofs of items (d) and (e) in Lemma 13.1, the left-right symmetry of the invariance property preserves the pertinent algebraic (group) and topological

properties. The reason is that the inversion map $x \mapsto x^{-1}$ on X interchanges left and right (i.e., $(xA)^{-1} = A^{-1}x^{-1}$) and, being a homeomorphism, also preserves topological properties. In other words, a result on left translation implies and is implied by a corresponding result on right translation. In particular, if μ is a left Haar measure on a Borel σ-algebra, then the set function η on the same Borel σ-algebra obtained by the composition of inversion and μ (i.e., $\eta(E) = \mu(E^{-1})$) is a right Haar measure (and vice versa).

The next result shows that, under the inner regularity assumption, the "nonzero" condition in the definition of a Haar Borel measure is equivalent to saying that it assigns a positive value to every nonempty open set.

Lemma 13.2. *Let $\mu\colon \mathcal{A} \to \overline{\mathbb{R}}$ be a positive measure on a Borel σ-algebra of subsets of a locally compact Hausdorff group. Suppose μ is left invariant.*

(a) *If $\mu(U) = 0$ for some open $U \neq \varnothing$, then $\mu(K) = 0$ for every compact K.*

(b) *If $\mu(K) > 0$ for some compact K, then $\mu(U) > 0$ for every open $U \neq \varnothing$.*

If, in addition, μ is an inner regular Borel measure, then

(c) *$\mu \neq 0$ if and only if $\mu(U) > 0$ for every nonempty open set $U \in \mathcal{A}$,*

and, in this case (i.e., *if $\mu \neq 0$*), *then*

(d) *$0 < \int f d\mu < \infty$ for every $f \in C_c(X)$ such that $0 \leq f \neq 0$ (for every nonzero nonnegative continuous function on X with compact support).*

Proof. (a,b) If K and U are nonempty subsets of X, then for every $k \in K$ and every $u_0 \in U$ there exists $x = ku_0^{-1} \in X$ such that $k = x\,u_0$, and so $\bigcup_{x\in X} xU$ is a covering of K. (If U contains the identity e, then we can take $u_0 = e$ and, in this case, $\bigcup_{x\in K} xU$ is a covering of K.) Thus, if $\mu(U) = 0$ for some open set $U \neq \varnothing$, and if $K \neq \varnothing$ is any compact set, then $\bigcup_{x\in X} xU$ is an open covering of K, and therefore there exists a finite set of points in X, $\{x_i\}_{i=1}^n$, such that $K \subseteq \bigcup_{i=1}^n x_i U$. Since μ is left invariant,

$$\mu(K) \leq \sum_{i=1}^{n} \mu(x_i U) = \sum_{i=1}^{n} \mu(U) = n\mu(U) = 0.$$

This proves (a), which is equivalently stated as in (b): if $\mu(K) > 0$ for some compact K, then $\mu(U) > 0$ for every open $U \neq \varnothing$.

(c,d) Suppose, in addition, that μ is an inner regular Borel measure. Since $\mu(U) = 0$ for some open $U \neq \varnothing$ implies $\mu(K) = 0$ for every compact K, and since $\mu(E) = \sup\{\mu(K)\colon K \subseteq E,\ K \text{ compact}\}$ for every Borel set E (as μ is an inner regular measure), it follows that $\mu(U) = 0$ for some nonempty open set U implies $\mu(E) = 0$ for every $E \in \mathcal{A}$. That is, $\mu = 0$. In other words, *if*

every compact set has measure zero, and if the measure is inner regular, then $\mu = 0$. So $\mu \neq 0$ implies $\mu(U) > 0$ for every open $U \neq \varnothing$, proving (c) — the converse is trivial. For (d) note that if $0 \leq f \neq 0$ in $C_c(X)$, then there is an $\varepsilon > 0$ and an open set $\varnothing \neq U \subseteq [f]^\circ$ such that $\varepsilon \chi_U \leq f$ and so, by (c), $0 < \varepsilon\mu(U) = \varepsilon \int \chi_U \, d\mu \leq \int f \, d\mu \leq (\sup f)\mu([f]) < \infty$ because μ is Borel. □

The standard program to build a Haar measure on an arbitrary locally compact Hausdorff group is similar to that of building a Borel measure in an arbitrary locally compact Hausdorff space as in Section 11.3, where it was shown that an inner content on a topology generates a quasiregular outer measure (Lemma 11.8) that is finite at bounded sets, which in turn generates a Borel measure (Theorem 11.7). Now we define the notion of outer content on the collection of all compact sets, which is dual to the notion of inner content on a topology as in Section 11.3, and show in Lemma 13.3 below how an outer content on the compact sets generates a quasiregular outer measure, which is the dual of Lemma 11.8.

Let X be a topological space, and let $\mathcal{K}$ denote the *class of all compact subsets* of X. A *content on the compact sets* is a real-valued set function $\mu^{\#}\colon \mathcal{K} \to \mathbb{R}$ on $\mathcal{K}$ such that for arbitrary compact sets K, C, and K_i in $\mathcal{K}$,

$(1')$ $\mu^{\#}(\varnothing) = 0$,

$(2')$ $\mu^{\#}(K) \geq 0$, (*positiveness*),

$(3')$ $\mu^{\#}(K) < \infty$,

$(4')$ $\mu^{\#}(C) \leq \mu^{\#}(K)$ whenever $C \subseteq K$,

$(5')$ $\mu^{\#}(C \cup K) = \mu^{\#}(C) + \mu^{\#}(K)$ whenever $C \cap K = \varnothing$,

$(6')$ $\mu^{\#}\big(\bigcup_i K_i\big) \leq \sum_i \mu^{\#}(K_i)$ (*finite subadditivity*), for finite families $\{K_i\}$ of sets in $\mathcal{K}$.

It is an *outer content* if, in addition,

$(7')$ $\mu^{\#}(K) = \inf\{\mu^{\#}(C)\colon K \subseteq C^\circ,\ C \in \mathcal{K}\}$.

Lemma 13.3. *Let* $\mu^{\#}\colon \mathcal{K} \to \mathbb{R}$ *be an outer content. If* X *is a locally compact Hausdorff space, then the set function* $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$ *given by*

$$\mu^*(S) = \sup\big\{\mu^{\#}(K)\colon K \subseteq S,\ K \text{ compact}\big\} \quad \text{for every} \quad S \in \wp(X)$$

is a quasiregular outer measure such that $\mu^*(S) < \infty$ *for every bounded* S.

Proof. Properties $(1')$, $(2')$, and $(4')$ in the definition of the content $\mu^{\#}$ imply the properties (a), (b), and (c) in the definition of an outer measure μ^* (see Section 11.3). Property $(6')$ in the definition of the content $\mu^{\#}$ implies that

$$\mu^*\big(\bigcup\nolimits_i S_i\big) \leq \sup_{K_i \subseteq S_i} \mu^{\#}\big(\bigcup\nolimits_i K_i\big) \leq \sup_{K_i \subseteq S_i} \sum\nolimits_i \mu^{\#}(K_i) \leq \sum\nolimits_i \mu^*(S_i),$$

for every finite family $\{S_i\}$ of sets in $\wp(X)$, which leads to property (d) in the definition of an outer measure μ^* (see Section 11.3, again).

Thus μ^* is an outer measure.

Property $(4')$ in the definition of the content $\mu\#$ ensures that

$$\mu^*(K) = \mu\#(K) \quad \text{for every compact set } K.$$

Now recall properties (i), (ii), (iii) in the definition of a quasiregular outer measure at the beginning of Section 11.3. According to the above identity, property (iii) holds by the definition of μ^*. Moreover, the above identity also shows that property $(7')$ in the definition of an outer content ensures that

$$\mu^*(K) = \inf\{\mu^*(C)\colon\ K \subseteq C^\circ,\ C \in \mathcal{K}\}$$

for every compact set K, which, by the definition of μ^*, implies that

$$\mu^*(S) = \sup_{K\subseteq S} \mu^*(K) = \sup_{K\subseteq S} \inf_{K\subseteq C^\circ} \mu^*(C^\circ) = \inf_{S\subseteq C^\circ} \mu^*(C^\circ)$$

for every bounded set S, which leads to property (i) in the definition of a quasiregular outer measure. Property $(5')$ in the definition of an outer content implies property (ii) in the definition of a quasiregular outer measure, according to the definition of μ^*:

$$\mu^*(G\cup U) = \sup_{C\cup K\subseteq G\cup U} \mu\#(C\cup K) = \sup_{C\subseteq G,\ K\subseteq U} \mu\#(C)+\mu\#(K) = \mu^*(G)+\mu^*(U)$$

whenever $G \cap U = \varnothing$. (Indeed, according to the definition of μ^*, property (ii) of a quasiregular outer measure becomes, in this case, a consequence of the condition $\mu^*(C \cup K) = \mu^*(C) + \mu^*(K)$ whenever $C \cap K = \varnothing$ — property $(5')$, since this μ^* coincides with $\mu\#$ on $\mathcal{K}$ so that $\mu^*(S) = \sup_{K\subseteq S} \mu^*(K)$.)

Thus the outer measure μ^* is quasiregular.

Finally, property $(3')$ in the definition of the content $\mu\#$ and property (c) in the definition of an outer measure μ^* imply that

$$\mu^*(S) < \infty \text{ for every bounded set } S \in \wp(X)$$

(i.e., for every $S \subseteq K$ for some $K \in \mathcal{K}$). □

13.3 Construction of Haar Measures

In a locally compact Hausdorff group X, a bounded set means a relatively compact set (i.e., with compact closure) and so a compact set is precisely

a closed and bounded set (see remarks on boundedness after the proof of Lemma 11.2). Let $B \subseteq X$ be bounded (i.e., $\overline{B}$ is compact), and let $A \subseteq X$ have a nonempty interior (i.e., the open set A° is nonempty). As $\bigcup_{x \in X} xA^\circ$ is an open covering of $\overline{B}$, there is a finite set $\{x_i\}_{i=1}^n \subseteq X$ such that

$$B \subseteq \overline{B} \subseteq \bigcup_{i=1}^n x_i A^\circ \subseteq \bigcup_{i=1}^n x_i A.$$

The *covering number* $[B:A]$ of a bounded set B by a set A with nonempty interior is the least number of translations of A required to cover B. That is,

$$[B:A] \;=\; \min\left\{n \in \mathbb{N}\colon\; B \subseteq \textstyle\bigcup_{i=1}^n x_i A\right\} > 0$$

whenever $B \neq \varnothing$. As it should be, $[\varnothing:A] = 0$. Observe that

$$[B:A] \;=\; \min\left\{[C:A]\colon\; B \subseteq C^\circ,\; C \text{ bounded}\right\} \in \mathbb{N}.$$

Also, the covering number is translation invariant: for every $x \in X$,

$$[xB:A] = [B:A] = [B:xA]\,.$$

If $D \subseteq X$ is bounded with nonempty interior ($\overline{D}$ compact and $D^\circ \neq \varnothing$), then

$$[B:A] \leq [B:D]\,[D:A]$$

(reason: cover B with $[B:D]$ translations of D and cover each translation of D with $[D:A]$ translations of A), and

$$[D:D] = 1.$$

If B is a bounded set and U is a nonempty open set, then the covering number $[B:U]$ can be thought of as a comparison between the sizes of B and U. This is our starting point for constructing a Haar measure. For a given B, the covering number $[B:U]$ gets larger as U gets smaller. To control this growing process we consider instead the *covering ratio* $[B:U]/[D:U]$ for some reference bounded set D with nonempty interior, and define somehow a limit of this covering ratio as U gets smaller and smaller. This will lead us to a left invariant nonzero positive inner regular Borel measure.

From now on we assume the following setup. Let X be a locally compact Hausdorff group and consider the inclusions

$$\mathcal{T} \subseteq \mathcal{X}_\mathcal{T} \subseteq \mathcal{A} \subseteq \wp(X) \quad \text{ and } \quad \mathcal{K} \subseteq (\mathcal{B} \cap \mathcal{X}_\mathcal{T}),$$

where $\mathcal{T}$ is a topology on X, $\mathcal{X}_\mathcal{T}$ is the Borel σ-algebra of subsets of X generated by $\mathcal{T}$, $\mathcal{A}$ is any Borel σ-algebra of subsets of X, $\wp(X)$ is the power set

of X, $\mathcal{K} \subseteq \mathcal{X}_{\mathcal{T}}$ is the class of all compact subsets of X, and $\mathcal{B} \subseteq \wp(X)$ is the *class of all bounded subsets* of X. Let D be a bounded set with nonempty interior (i.e., $D \in \mathcal{B}$ and $D^\circ \neq \varnothing$). Take an arbitrary nonempty open set $U \in \mathcal{T}$ and consider the rational-valued set function $\vartheta_U \colon \mathcal{B} \to \mathbb{Q}$ defined by

$$\vartheta_U(B) = \tfrac{[B:U]}{[D:U]} \quad \text{for every} \quad B \in \mathcal{B}.$$

Lemma 13.4. *The covering ratio function $\vartheta_U \colon \mathcal{B} \to \mathbb{Q}$ has the following properties. For arbitrary bounded sets B, B_1, B_2, and B_i in $\mathcal{B}$,*

(a) $\vartheta_U(\varnothing) = 0$,

(b) $0 < \vartheta_U(B)$ *whenever* $B \neq \varnothing$, (*positiveness*),

(c) $0 \leq \vartheta_U(B) \leq [B\colon D] < \infty$,

(d) $\vartheta_U(B_1) \leq \vartheta_U(B_2)$ *whenever* $B_1 \subseteq B_2$,

(e) $\vartheta_U(B_1 \cup B_2) = \vartheta_U(B_1) + \vartheta_U(B_2)$ *whenever* $B_1 U^{-1} \cap B_2 U^{-1} = \varnothing$,

(f) $\vartheta_U\big(\bigcup_i B_i\big) \leq \sum_i \vartheta_U(B_i)$ (*finite subadditivity*), *for finite families* $\{B_i\}$ of sets in $\mathcal{B}$,

(g) $\vartheta_U(B) = \min\big\{\vartheta_U(C)\colon\ B \subseteq C^\circ,\ C \in \mathcal{B}\big\} \in \mathbb{Q}$,

(h) $\vartheta_U(xB) = \vartheta_U(B)$ *for every* $x \in X$,

(i) $\vartheta_U(D) = 1$,

(j) $U \mapsto \vartheta_U(B)$ *is a bounded map for each* B.

(k) *If K_1 and K_2 are disjoint compact sets, then there is an open bounded symmetric neighborhood U of the identity e such that*

$$\vartheta_G(K_1 \cup K_2) = \vartheta_G(K_1) + \vartheta_G(K_2)$$

for every open neighborhood G of e such that $G \subseteq U$.

Proof. Properties (a) to (f) are readily verified except, perhaps, for (e). To show (e), take $x \in X$ and observe that if $B_1 \cap xU \neq \varnothing$, then $x \in B_1 U^{-1}$. (Indeed, if $b = xu$, then $bu^{-1} = x$, so that $x \in B_1 U^{-1}$.) Similarly, if $B_2 \cap xU \neq \varnothing$, then $x \in B_2 U^{-1}$. Thus, if $B_1 U^{-1} \cap B_2 U^{-1} = \varnothing$, then there is no x such that $B_1 \cap xU \neq \varnothing$ and $B_2 \cap xU \neq \varnothing$; that is, no left translation of U meets both B_1 and B_2, and so $[B_1 \cup B_1 : U] = [B_1 : U] + [B_2 : U]$, proving property (e). Property (e) leads to property (k). Indeed, if K_1 and K_2 are disjoint compact sets, then there exists an open bounded symmetric neighborhood N of the identity e such that $K_1 N \cap K_2 N = \varnothing$ by Lemma 13.1(e). Set $U = N^{-1}$, again an open bounded symmetric neighborhood of e. Let G be an arbitrary open (bounded) neighborhood of e such that $G \subseteq U$. Set $N' = G^{-1}$, an open (bounded) neighborhood of e. Since $N'^{-1} = G \subseteq U = N^{-1}$, it follows that $N' \subseteq N$, and so $K_1 N' \cap K_2 N' = \varnothing$, equivalently, $K_1 G^{-1} \cap K_2 G^{-1} = \varnothing$. Thus $\vartheta_G(K_1 \cup K_2) = \vartheta_G(K_1) + \vartheta_G(K_2)$ by

(e), proving property (k). Properties (g) and (h) follow from the facts that $[B:U] = \min\{[C:U]\colon B \subseteq C^\circ,\ C \in \mathcal{B}\}$ and $[xB:U] = [B:U]$, (i) is trivial, and (j) follows from (c) as a consequence of $[B:U] \le [B:D]\,[D:U]$. □

Observe from properties (a) to (f) in the preceding lemma that ϑ_U (actually, the restriction of it to $\mathcal{K}$) is nearly a content; it fails to be a content on $\mathcal{K}$ just because additivity in property (e) may not hold for every pair of disjoint compact sets — property (k) is an attempt to establish additivity that will actually succeed in the next lemma.

The next lemma plays a central role in proving the existence of Haar measures. There are distinct ways of approaching its proof. Some authors use the Hahn–Banach Theorem (on extension of bounded linear functionals), others use the Arzelà–Ascoli Theorem (on compact subsets of continuous functions), and some use the Tychonoff Theorem, which says that *the Cartesian product* $\prod_{\gamma\in\Gamma} K_\gamma$ *of compact sets* K_γ *is compact* (in the product topology — see, e.g., [37, Theorem 5.13]).

Lemma 13.5. *The covering ratio function* $\vartheta_U\colon \mathcal{B} \to \mathbb{Q}$ *generates a nonzero left invariant outer content* $\mu^\#\colon \mathcal{K} \to \mathbb{R}$ *on* $\mathcal{K}$.

Proof. Let $\mathcal{U}$ denote the collection of all open neighborhoods of the identity $e \in X$. Fix an arbitrary compact set $D \in \mathcal{K} \subseteq \mathcal{B}$ with nonempty interior, and consider the covering ratio function $\vartheta_U\colon \mathcal{B} \to \mathbb{R}$ for an arbitrary $U \in \mathcal{U}$. Take the restriction of each ϑ_U to $\mathcal{K}$, denoted again by $\vartheta_U\colon \mathcal{K} \to \mathbb{R}$, so that $\vartheta_U(K) = [K:U]/[D:U]$ for every $K \in \mathcal{K}$. Let

$$\Theta = \bigcup_{U\in\mathcal{U}} \{\vartheta_U\}$$

be the collection of the covering ratio functions $\vartheta_U\colon \mathcal{K} \to \mathbb{R}$ restricted to $\mathcal{K}$ for all $U \in \mathcal{U}$. Take the Cartesian product

$$\Pi = \prod_{K\in\mathcal{K}} [0, [K:D]]$$

of the closed intervals $[0, [K:D]]$. Since closed (and bounded) intervals are compact sets in $\mathbb{R}$, the Tychonoff Theorem says that the Cartesian product $\prod_{K\in\mathcal{K}}[0,[K:D]]$ is compact in the product topology. Interpret the elements of the Cartesian product $\prod_{K\in\mathcal{K}}[0,[K:D]]$ as real-valued functions on $\mathcal{K}$. Indeed, by the definition of Cartesian product, $\prod_{K\in\mathcal{K}}[0,[K:D]]$ consists of all indexed families $\{\phi(K)\}_{K\in\mathcal{K}}$ such that $\phi(K) \in [0,[K:D]]$ for each $K \in \mathcal{K}$; equivalently, of all real-valued set functions $\phi\colon \mathcal{K} \to \mathbb{R}$ on $\mathcal{K}$ such that $0 \le \phi(K) \le [K:D]$ for every $K \in \mathcal{K}$. Since $0 \le \vartheta_U(K) \le [K:D] < \infty$ by Lemma 13.4(c), the function $\vartheta_U\colon \mathcal{K} \to \mathbb{R}$ lies in $\prod_{K\in\mathcal{K}}[0,[K:D]]$ for each $U \in \mathcal{U}$. That is, $\Theta \subseteq \Pi$. For each $U \in \mathcal{U}$ consider the set of functions

$$\Theta(U) = \big\{\vartheta_G \in \Theta\colon\, G \subseteq U\big\} \subseteq \Pi.$$

Let $\{U_i\}_{i=1}^n$ be an arbitrary finite family of open neighborhoods of e (i.e., each U_i lies in $\mathcal{U}$). Observe that $\bigcap_{i=1}^n U_i$ is again a set in $\mathcal{U}$, and also that

$$\varnothing \neq \Theta\Big(\bigcap_{i=1}^n U_i\Big) \subseteq \bigcap_{i=1}^n \Theta(U_i).$$

(Reason: $\bigcap_{i=1}^n U_i \subseteq U_j$ so that $\Theta(\bigcap_{i=1}^n U_i) \subseteq \Theta(U_j)$ for all $U_j \in \{U_i\}_{i=1}^n$ — cf. Lemma 13.4(d) — and $\Theta(U) \neq \varnothing$ for every $U \in \mathcal{U}$ because $\vartheta_U \in \Theta(U)$.) Hence, the family $\{\Theta(U) \subseteq \Pi\colon\, U \in \mathcal{U}\}$ of subsets of $\Theta \subseteq \Pi$ has the finite intersection property (i.e., every finite subcollection of it has a nonempty finite intersection). In particular, the family $\{\overline{\Theta(U)} \subseteq \Pi\colon\, U \in \mathcal{U}\}$ of closed subsets of the compact space Π has the finite intersection property (since $\varnothing \neq \bigcap_{i=1}^n \Theta(U_i) \subseteq \bigcap_{i=1}^n \overline{\Theta(U_i)}$). Therefore, Proposition 11.A ensures that $\{\overline{\Theta(U)} \subseteq \Pi\colon\, U \in \mathcal{U}\}$ has a nonempty intersection. So there exists

$$\mu^{\#} \in \bigcap_{U \in \mathcal{U}} \overline{\Theta(U)} = \bigcap_{U \in \mathcal{U}} \overline{\big\{\vartheta_G \in \Theta\colon\, G \subseteq U\big\}} \subseteq \Pi,$$

where $\mu^{\#}$ is a real-valued set function on $\mathcal{K}$, as is every element of the set $\Pi = \prod_{K \in \mathcal{K}}[0, [K\colon D]]$. Thus, for an arbitrary $K \in \mathcal{K}$, the value $\mu^{\#}(K) \in \mathbb{R}$ is approached by taking a $\vartheta_U(K) = [K\colon U]/[D\colon U] \in \mathbb{Q}$ for some $U \in \mathcal{U}$, and letting U get smaller and smaller in the above sense. We show that such a function $\mu^{\#}\colon \mathcal{K} \to \mathbb{R}$ is a nonzero left invariant outer content on $\mathcal{K}$.

(i) First note that properties (a) to (c) in Lemma 13.4 trivially imply properties $(1')$ to $(3')$ in the definition of a content on $\mathcal{K}$.

(ii) To verify property $(4')$ in the definition of a content on $\mathcal{K}$ proceed as follows. For each $K \in \mathcal{K}$ take the projection $\Phi_K\colon \Pi \to [0, [K\colon D]]$ defined by $\Phi_K(\phi) = \phi(K)$ for every function $\phi\colon \mathcal{K} \to [0, [K\colon D]$ in $\Pi = \prod_{K \in \mathcal{K}}[0, [K\colon D]]$. The product topology on $\prod_{K \in \mathcal{K}}[0, [K\colon D]]$ makes the projections Φ_K continuous (see, e.g., [37, p. 90]). Take arbitrary sets $K_1, K_2 \in \mathcal{K}$. So the difference $\Phi_{K_2 - K_1} = \Phi_{K_2} - \Phi_{K_1}\colon \Pi \to \mathbb{R}$ is continuous. Consider the set

$$\begin{aligned}\Sigma_{K_2 - K_1} &= \big\{\phi \in \Pi\colon\, \phi(K_1) \leq \phi(K_2)\big\} \\ &= \big\{\phi \in \Pi\colon\, 0 \leq \Phi_{K_2 - K_1}(\phi)\big\} = \Phi^{-1}_{K_2 - K_1}([0, \infty)) \subseteq \Pi,\end{aligned}$$

which is closed (since $\Phi_{K_2 - K_1}$ is continuous, so that the inverse image of a closed set is closed). If $K_1 \subseteq K_2$, then $\vartheta_U(K_1) \leq \vartheta_U(K_2)$ for every $U \in \mathcal{U}$ by Lemma 13.4(d), so that $\vartheta_U \in \Sigma_{K_2 - K_1}$ for every $U \in \mathcal{U}$, and hence we get $\Theta(U) \subseteq \Sigma_{K_2 - K_1}$ for every $U \in \mathcal{U}$. Thus $\mu^{\#} \in \bigcap_{U \in \mathcal{U}} \overline{\Theta(U)} \subseteq \Sigma_{K_2 - K_1}$ (since $\Sigma_{K_2 - K_1}$ is closed). Then $\mu^{\#}(K_1) \leq \mu^{\#}(K_2)$. That is, property (d) in Lemma 13.4 implies property $(4')$ in the definition of a content on $\mathcal{K}$.

(iii) Take $K_1, K_2 \in \mathcal{K}$. Since Φ_K is continuous for each $K \in \mathcal{K}$, it follows that $\Phi_{K_2+K_1} = \Phi_{K_2} + \Phi_{K_1} - \Phi_{K_1 \cup K_2} : \Pi \to \mathbb{R}$ is continuous. Consider the set

$$\begin{aligned}\Sigma_{K_2+K_1} &= \big\{\phi \in \Pi\colon\ \phi(K_1 \cup K_2) = \phi(K_1) + \phi(K_2)\big\} \\ &= \big\{\phi \in \Pi\colon\ \Phi_{K_2+K_1}(\phi) = 0\big\} = \Phi^{-1}_{K_2+K_1}(\{0\}) \ \subseteq\ \Pi,\end{aligned}$$

which is closed (since $\Phi_{K_2+K_1}$ is continuous and every singleton is closed in $\mathbb{R}$). If $K_1 \cap K_2 = \varnothing$, then there exists $U_0 \in \mathcal{U}$ such that $\vartheta_G(K_1 \cup K_2) = \vartheta_G(K_1) + \vartheta_G(K_2)$ for every $G \in \mathcal{U}$ such that $G \subseteq U_0$ by Lemma 13.4(k). That is, there exists $U_0 \in \mathcal{U}$ such that $\Theta(U_0) \subseteq \Sigma_{K_2+K_1}$. Therefore, since $\Sigma_{K_2+K_1}$ is closed, it follows that $\mu^{\#} \in \bigcap_{U\in\mathcal{U}} \overline{\Theta(U)} \subseteq \overline{\Theta(U_0)} \subseteq \Sigma_{K_2+K_1}$. Then $\mu^{\#}(K_1 \cup K_2) = \mu^{\#}(K_1) + \mu^{\#}(K_2)$. That is, property (k) in Lemma 13.4 implies property $(5')$ in the definition of a content on $\mathcal{K}$.

(iv) Property $(6')$ of a content on $\mathcal{K}$ results from property (f) in Lemma 13.4 by the same argument as in (ii) with the map $\Phi_{K_2+K_1}$ of (iii). Indeed, the set

$$\begin{aligned}\Sigma'_{K_2+K_1} &= \big\{\phi \in \Pi\colon\ \phi(K_1 \cup K_2) \le \phi(K_1) + \phi(K_2)\big\} \\ &= \big\{\phi \in \Pi\colon\ 0 \le \Phi_{K_2+K_1}(\phi)\big\} = \Phi^{-1}_{K_2+K_1}([0,\infty)) \ \subseteq\ \Pi\end{aligned}$$

is closed. As $\vartheta_U(K_1 \cup K_2) \le \vartheta_U(K_1) + \vartheta_U(K_2)$ for every $U \in \mathcal{U}$ by Lemma 13.4(f), $\vartheta_U \in \Sigma'_{K_2+K_1}$, and hence $\Theta(U) \subseteq \Sigma'_{K_2+K_1}$, for every $U \in \mathcal{U}$. Thus $\mu^{\#} \in \bigcap_{U\in\mathcal{U}} \overline{\Theta(U)} \subseteq \Sigma'_{K_2+K_1}$, and so $\mu^{\#}(K_1 \cup K_2) \le \mu^{\#}(K_1) + \mu^{\#}(K_2)$, which extends by induction to finite unions of compact sets. Then property (f) in Lemma 13.4 implies property $(6')$ in the definition of a content on $\mathcal{K}$.

(v) Property (g) in Lemma 13.4 leads to property $(7')$ of an outer content on $\mathcal{K}$. In fact, take $K \in \mathcal{K}$ arbitrary. Let C be any set in $\mathcal{K}$ such that $K \subseteq C^\circ$. Recall that Φ_K and Φ_C are continuous, and so $\{\Phi_{C-K} = \Phi_C - \Phi_K\}_{K\subseteq C^\circ}$ is a family of continuous maps. Take the set $\Sigma_{C-K} = \{\phi \in \Pi\colon\ 0 \le \Phi_{C-K}(\phi)\}$ of item (ii), which is closed, and consider the set

$$\begin{aligned}\Sigma_K &= \big\{\phi \in \Pi\colon\ \phi(K) = \inf_{K\subseteq C^\circ} \phi(C)\big\} = \big\{\phi \in \Pi\colon\ \Phi_K(\phi) = \inf_{K\subseteq C^\circ} \Phi_C(\phi)\big\} \\ &= \big\{\phi \in \Pi\colon\ \inf_{K\subseteq C^\circ} \Phi_{C-K}(\phi) = 0\big\} = \inf_{K\subseteq C^\circ} \Sigma_{C-K} = \bigcap_{K\subseteq C^\circ} \Sigma_{C-K} \ \subseteq\ \Pi,\end{aligned}$$

which is closed as well (intersection of closed sets). Thus, as before, since $\vartheta_U(K) = \inf_{K\subseteq C^\circ} \vartheta_U(C)$ by Lemma 13.4(g), so that $\vartheta_U \in \Sigma_K$, for every $U \in \mathcal{U}$, we get $\mu^{\#} \in \bigcap_{U\in\mathcal{U}} \overline{\Theta(U)} \subseteq \Sigma_K$ (because Σ_K is closed). Therefore, $\mu^{\#}(K) = \inf_{K\subseteq C^\circ} \mu^{\#}(C)$. Then property (g) in Lemma 13.4 ensures property $(7')$ in the definition of an outer content on $\mathcal{K}$.

Hence, according to (i)–(v), the function $\mu^{\#} : \mathcal{K} \to \mathbb{R}$ is an outer content on $\mathcal{K}$. That $\mu^{\#}$ is nonzero follows from the fact that $\mu^{\#}(D) = 1$. Indeed, the same continuity argument ensures that since Φ_D is continuous, the set

$$\Sigma_D = \big\{\phi \in \Pi\colon\ \phi(D) = 1\big\} = \big\{\phi \in \Pi\colon\ \Phi_D(\phi) = 1\big\} = \Phi_D^{-1}(\{1\}) \ \subseteq\ \Pi$$

is closed. Since $\vartheta_U \in \Sigma_D$ by Lemma 13.4(i), we get $\overline{\Theta(U)} \subseteq \Sigma_D$, for every $U \in \mathcal{U}$, and so $\mu^{\#} \in \Sigma_D$. Finally, take $x \in X,\ K \in \mathcal{K}$, and consider the continuous map $\Phi_{xK-K} = \Phi_{xK} - \Phi_K\colon \Pi \to \mathbb{R}$. Then the set

$$\Sigma_{xK-K} = \big\{\phi \in \Pi\colon \phi(xK) = \phi(K)\big\} = \big\{\phi \in \Pi\colon \Phi_{xK-K} = 0\big\} = \Phi_{xK-K}^{-1}(\{0\}) \subseteq \Pi$$

is closed, and so, according to Lemma 13.4(h), the same argument ensures that $\mu^{\#} \in \Sigma_{xK-K}$, which means that $\mu^{\#}$ is left invariant. □

The preceding lemma allows us to apply Theorem 11.7 to ensure the existence of a Haar measure on a Borel σ-algebra of subsets of an arbitrary locally compact Hausdorff group.

Theorem 13.6. *There exists a quasiregular Haar measure on a Borel σ-algebra of subsets of every locally compact Hausdorff group.*

Proof. We will prove the following statement that leads to the above one.

A left invariant outer content $\mu^{\#}\colon \mathcal{K} \to \mathbb{R}$, generated by a covering ratio function $\vartheta_U\colon \mathcal{B} \to \mathbb{Q}$, generates a quasiregular outer measure $\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$, finite on $\mathcal{B}$, which in turn generates a quasiregular complete Haar measure (i.e., a left invariant nonzero positive quasiregular complete Borel measure) *$\eta\colon \mathcal{A}^* \to \overline{\mathbb{R}}$ on a Borel σ-algebra $\mathcal{A}^*$ of subsets of a locally compact Hausdorff group X. Its restriction $\mu\colon \mathcal{X}_{\mathcal{T}} \to \overline{\mathbb{R}}$ to the Borel σ-algebra $\mathcal{X}_{\mathcal{T}}$ generated by $\mathcal{T}$ is again a quasiregular Haar measure, although μ may not be complete.*

Indeed, consider the covering ratio function

$$\vartheta_U\colon \mathcal{B} \to \mathbb{Q}$$

defined by $\vartheta_U(B) = [B\colon U]/[D\colon U]$ for every $B \in \mathcal{B}$, which is nonzero and left invariant by Lemma 13.4(h,i). This ϑ_U generates a nonzero left invariant outer content on $\mathcal{K}$ (whose existence was proved in Lemma 13.5),

$$\mu^{\#}\colon \mathcal{K} \to \mathbb{R},$$

which in turn generates a quasiregular outer measure

$$\mu^*\colon \wp(X) \to \overline{\mathbb{R}}$$

defined by $\mu^*(S) = \sup\{\mu^{\#}(K)\colon\ K \subseteq S,\ K \in \mathcal{K}\}$ for every $S \in \wp(X)$, which is finite on $\mathcal{B}$, according to Lemma 13.3. Therefore, by Theorem 11.7, there exists a quasiregular complete Borel measure

$$\eta = \mu^*|_{\mathcal{A}^*}\colon \mathcal{A}^* \to \overline{\mathbb{R}}$$

on a σ-algebra $\mathcal{A}^*$ that includes the Borel σ-algebra $\mathcal{X}_\mathcal{T}$. It remains to verify that the (positive) Borel measure η is nonzero and left invariant.

(a) The measure η is nonzero. In fact, $\mu^\#$ is nonzero by Lemma 13.5, and hence μ^* is nonzero according to its very definition in Lemma 13.3. Thus $\mu^*(X) > 0$. Then, since $X \in \mathcal{A}^*$, we get $\eta(X) = \mu^*|_{\mathcal{A}^*}(X) = \mu^*(X) > 0$.

(b) The measure η is left invariant. Indeed, since $\mu^\#$ is left invariant by Lemma 13.5 (i.e., $\mu^\#(xK) = \mu^\#(K)$ for every $x \in X$ and every $K \in \mathcal{K}$), it follows by the definition of μ^* in Lemma 13.3 that

$$\begin{aligned}\mu^*(x^{-1}S) &= \sup_{K \subseteq x^{-1}S} \mu^\#(K) = \sup_{xK \subseteq S} \mu^\#(K) \\ &= \sup_{xK \subseteq S} \mu^\#(xK) = \sup_{K' \subseteq S} \mu^\#(K') = \mu^*(S)\end{aligned}$$

for every $x \in X$ and $S \in \wp(X)$, so that μ^* is left invariant, and so is the restriction $\eta = \mu^*|_{\mathcal{A}^*}$ of it to $\mathcal{A}^*$ so that for every $x \in X$ and $E \in \mathcal{A}^*$,

$$\eta(xE) = \mu^*(xE) = \mu^*(E) = \eta(E).$$

Again, as in the proof of Theorem 11.7, the above properties of the quasiregular complete Haar measure η, except for completeness, are transferred to all restrictions of it to any Borel σ-algebra of subsets of X included in $\mathcal{A}^*$; in particular, to its restriction

$$\mu = \eta|_{\mathcal{X}_\mathcal{T}} \colon \mathcal{X}_\mathcal{T} \to \overline{\mathbb{R}}$$

to the smallest Borel σ-algebra $\mathcal{X}_\mathcal{T}$ of subsets of X, so that μ is a quasiregular Haar measure on $\mathcal{X}_\mathcal{T}$. □

It is clear that a Haar measure on a Borel σ-algebra of subsets of a given locally compact Hausdorff group X is not unique. Reason: μ is Haar if and only if $\gamma\,\mu$ is Haar for every positive number γ. However, this is essentially the only way they can differ.

Theorem 13.7. *If η and μ are quasiregular Haar measures on the same Borel σ-algebra of subsets of a given locally compact Hausdorff group, then there exists a positive constant γ such that*

$$\eta = \gamma\,\mu.$$

Proof. Let μ and ν be quasiregular Haar measures on a Borel σ-algebra $\mathcal{A}$ of subsets of a locally compact Hausdorff group X. Take the product $\mu \times \nu$ on $\mathcal{A} \times \mathcal{A}$ as in Theorem 9.5. Recall that continuous functions with compact support are integrable with respect to Borel measures. Consider the set

$$C_c(X)^+ = \big\{g \in C_c(X,\mathbb{R})\colon\ g(X) > 0\big\}$$

consisting of all nonzero and nonnegative continuous real-valued functions $g\colon X \to \mathbb{R}$ on X with compact support (see Section 12.1). Take an arbitrary μ-integrable Borel function $f \in \mathcal{L}(X,\mathcal{A},\mu)$, and an arbitrary function $g \in C_c(X)^+$, so that $\int f\,d\mu < \infty$ and $0 < \int g\,d\nu < \infty$ by Lemma 13.2(d). Consider the function $h\colon X\times X \to \mathbb{R}$ defined by

$$h(x,y) = \tfrac{f(x)g(yx)}{\int g(zx)\,d\nu(z)} \quad \text{for every} \quad x,y \in X,$$

which is again a $\mu\times\nu$-integrable Borel function: $h \in \mathcal{L}(X\times X, \mathcal{A}\times\mathcal{A}, \mu\times\nu)$ (cf. Problem 9.20). Now recall that the Fubini Theorem (Theorem 9.9) allows us to interchange the order of the iterated integrals (cf. Problem 9.20 again). The translation invariance property in Proposition 13.E says that we can swap x with zx in the argument of integrable functions when integrating them. (In particular, we can replace x with $y^{-1}x$ if the argument is x, and y with xy if the argument is y, and so we can replace $y^{-1}x = (y)^{-1}x$ with $(xy)^{-1}x = y^{-1}x^{-1}x = y^{-1}$ if the argument is y.) Therefore,

$$\begin{aligned}\int\left(\int h(x,y)\,d\nu(y)\right)d\mu(x) &= \int\left(\int h(x,y)\,d\mu(x)\right)d\nu(y)\\ &= \int\left(\int h(y^{-1}x,y)\,d\mu(x)\right)d\nu(y) = \int\left(\int h(y^{-1}x,y)\,d\nu(y)\right)d\mu(x)\\ &= \int\left(\int h(y^{-1},xy)\,d\nu(y)\right)d\mu(x)\end{aligned}$$

(where we reverse the integration order, then we replace x with $y^{-1}x$, reverse the integration order again, and replace $y^{-1}x$ with y^{-1} and y with xy). Since

$$h(y^{-1},xy) = \tfrac{f(y^{-1})g(x)}{\int g(zy^{-1})\,d\nu(z)} \quad \text{for every} \quad x,y \in X,$$

it then follows that

$$\begin{aligned}\int f(x)\,d\mu(x) &= \int f(x)\left(\tfrac{\int g(yx)\,d\nu(y)}{\int g(zx)\,d\nu(z)}\right)d\mu(x) = \int\left(\tfrac{\int f(x)g(yx)}{\int g(zx)\,d\nu(z)}\,d\nu(y)\right)d\mu(x)\\ &= \int\left(\int h(x,y)\,d\nu(y)\right)d\mu(x) = \int\left(\int h(y^{-1},xy)\,d\nu(y)\right)d\mu(x)\\ &= \int\left(\tfrac{\int f(y^{-1})g(x)}{\int g(zy^{-1})\,d\nu(z)}\,d\nu(y)\right)d\mu(x) = \int\left(\tfrac{\int f(y^{-1})\,d\nu(y)}{\int g(zy^{-1})\,d\nu(z)}\right)g(x)\,d\mu(x).\end{aligned}$$

Set $\alpha = \alpha(f,g,\nu) = \big(\int f(y^{-1})\,d\nu(y)\big)\big/\big(\int g(zy^{-1})\,d\nu(z)\big)$, which is a real number that does not depend on μ such that $\int f\,d\mu = \alpha\int g\,d\mu$. Thus, if η also is a quasiregular Haar measure on the same Borel σ-algebra $\mathcal{A}$ of subsets of X, then $\int f\,d\eta = \alpha\int g\,d\eta$ whenever $f \in \mathcal{L}(X,\mathcal{A},\eta)$. Hence,

$$\frac{\int f\,d\eta}{\int g\,d\eta} = \frac{\int f\,d\mu}{\int g\,d\mu}, \quad \text{which implies} \quad \int f\,d\eta = \left(\frac{\int g\,d\eta}{\int g\,d\mu}\right)\int f\,d\mu,$$

for every $f \in \mathcal{L}(X,\mathcal{A},\mu) \cap \mathcal{L}(X,\mathcal{A},\eta)$, for an arbitrary $g \in C_c(X)^+$. Then there is a positive constant $\gamma = \int g\,d\eta / \int g\,d\mu$ such that $\int f\,d\eta = \gamma \int f\,d\mu$ for every $f \in \mathcal{L}(X,\mathcal{A},\mu) \cap \mathcal{L}(X,\mathcal{A},\eta)$. Therefore, since μ and η are Borel measures on the Borel σ-algebra $\mathcal{A}$, it follows that for every compact set K, its characteristic function χ_K lies in $\mathcal{L}(X,\mathcal{A},\mu) \cap \mathcal{L}(X,\mathcal{A},\eta)$, and so

$$\eta(K) = \int_K d\eta = \int \chi_K\,d\eta = \gamma \int \chi_K\,d\mu = \gamma \int_K d\mu = \gamma\,\mu(K)$$

for every compact set K (all of them lie in $\mathcal{A}$). Since the Borel measures η and μ on the Borel σ-algebra $\mathcal{A}$ are quasiregular, the argument that closes the proof of Theorem 12.5 (which uses Lemma 12.2) ensures that if the above identity holds for every compact set K, then $\eta = \gamma\,\mu$. □

13.4 Additional Propositions

Recall that a *discrete space* is a topological space X whose topology $\mathcal{T}$ is the largest topology on X (i.e., $\mathcal{T} = \wp(X)$), so that every subset of X is open (and closed). Such a topology is called the *discrete topology*.

Proposition 13.A. *If μ is a Haar measure on a Borel σ-algebra of subsets of a locally compact Hausdorff group X, then X is discrete if and only if $\mu(\{x\}) \neq 0$ for at least one point $x \in X$.*

Proposition 13.B. *Let $\mathcal{A}$ be a Borel σ-algebra of subsets of a locally compact Hausdorff group X. If a Haar measure on $\mathcal{A}$ is finite, then X is compact.*

Proposition 13.C. *If μ is a Haar measure on a Borel σ-algebra of subsets of a locally compact Hausdorff group X, then the following assertions are pairwise equivalent.*

(a) *The measure μ is σ-finite.*

(b) *The space X is σ-compact.*

(c) *Every disjoint family of nonempty open sets is countable.*

(d) *For every nonempty open set U, there exists a sequence $\{x_n\}$ of points in X such that the family $\{x_n U\}$ covers X* (i.e., $X = \bigcup_n x_n U$).

Proposition 13.D. *Let μ be a Haar measure on a Borel σ-algebra of subsets of a locally compact Hausdorff group X, and let f and g be real-valued continuous functions on X. If $f = g$ μ-a.e*, then $f = g$ everywhere.

Proposition 13.E. *Let X be a locally compact Hausdorff group. For each real-valued function $f\colon X\to\mathbb{R}$ on X and each $y\in X$ let $f_y\colon X\to\mathbb{R}$ be defined by $f_y(x)=f(y^{-1}x)$ for every $x\in X$. Let $\mu\colon\mathcal{A}\to\overline{\mathbb{R}}$ be a nonzero positive quasiregular Borel measure on a Borel σ-algebra $\mathcal{A}$ of subsets of X. The measure μ is Haar* (i.e., *it is left invariant*) *if and only if*

$$\int f_y\,d\mu=\int f\,d\mu,$$

where $f_y\in\mathcal{L}(X,\mathcal{A},\mu)$, for every $y\in X$ and every $f\in\mathcal{L}(X,\mathcal{A},\mu)$; that is, for every μ-integrable Borel function $f\colon X\to\mathbb{R}$.

A *Haar integral* is an integral with respect to a Haar measure. The above property, namely, $f_y\in\mathcal{L}(X,\mathcal{A},\mu)$ *whenever* $f\in\mathcal{L}(X,\mathcal{A},\mu)$ *and the integrals coincide*, is referred to as the *translation invariance property* for the Haar integral (which, according to Proposition 13.E, characterizes the Haar measure). Recall again that continuous functions with compact support are integrable with respect to Borel measures. A positive linear functional Φ on the linear space $C_c(X,\mathbb{R})$ (where X is a locally compact Hausdorff group) with the above translation invariance property is also referred to as a *Haar integral*. Example: if $X=\mathbb{R}$, which is a locally compact Hausdorff additive group, then Proposition 13.E ensures that one can make the substitution $x\to x-y$ under the integral sign.

Proposition 13.F. *The Lebesgue measure is invariant under addition on $\mathbb{R}$ — either viewed as a measure on the Borel algebra $\mathfrak{R}$ generated by the usual topology of $\mathbb{R}$, or as a measure on its completion $\mathfrak{I}^*$, the Lebesgue algebra. Since it is a regular Borel measure* (see Proposition 11.H), *it follows that it is a regular Haar measure* (see the basic properties of Lebesgue measure that close Section 8.3).

Proposition 13.G. *Consider the Lebesgue measure $\lambda\colon\mathfrak{R}\to\overline{\mathbb{R}}$ on the Borel σ-algebra $\mathfrak{R}$ of subsets of $\mathbb{R}$ generated by the usual topology of $\mathbb{R}$. Take the set $\mathbb{R}^+=(0,\infty)$ of all the positive real numbers, which is a locally compact Hausdorff multiplicative group. The set function $\mu\colon\mathfrak{R}^+\to\overline{\mathbb{R}}$ on the Borel σ-algebra $\mathfrak{R}^+=\mathfrak{R}\cap\wp(\mathbb{R}^+)$ of subsets of $\mathbb{R}^+$ defined by*

$$\mu(E)=\int_E\tfrac{1}{x}\,d\lambda(x)\quad\text{ for every }\quad E\in\mathfrak{R}^+$$

is a Haar measure. (Reason: $\int\frac{dx}{x}=\log(x)$ and $\log(\beta)-\log(\alpha)=\log(\frac{\beta}{\alpha})$.)

The notion of absolute continuity on the same σ-algebra (cf. Definition 7.6) is naturally extended to a pair of measures, one of them acting on a sub-σ-algebra of the σ-algebra upon which the other measure acts. Indeed, if

$\eta\colon \mathcal{X} \to \overline{\mathbb{R}}$ is a measure on a σ-algebra $\mathcal{X}$, and $\mu\colon \mathcal{E} \to \overline{\mathbb{R}}$ is a measure on the σ-algebra $\mathcal{E} = \mathcal{X} \cap \wp(E)$, for some $E \in \mathcal{X}$, then μ is *absolutely continuous* with respect to η (same notation: $\mu \ll \eta$) if, for an arbitrary $E' \in \mathcal{E} \subseteq \mathcal{X}$,

$$\eta(E') = 0 \quad \text{implies} \quad \mu(E') = 0$$

(i.e., $\mu(E') = 0$ for every $E' \in \mathcal{E} \subseteq \mathcal{X}$ such that $\eta(E') = 0$).

Consider the setup of the previous proposition and note that $\Re^+ = \Re \cap \wp(\mathbb{R}^+)$ is a sub-σ-algebra of the σ-algebra $\Re$. From Propositions 3.5(c) and 3.7(b) we know that the Haar measure μ on $\Re^+$ of Proposition 13.G is absolutely continuous with respect to Lebesgue measure λ on $\Re$, and so (cf. the Radon–Nikodym Theorem — Theorem 7.8) the function $f\colon \mathbb{R}^+ \to \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ for every $x \in \mathbb{R}^+$ is the Radon–Nikodym derivative of μ with respect to λ. Now consider the sub-σ-algebra $\Re_0 = \Re \cap \wp(\mathbb{R}_0)$ of the σ-algebra $\Re$, where $\mathbb{R}_0 = \mathbb{R}\backslash\{0\}$ is the set of all nonzero real numbers, which is again a locally compact Hausdorff multiplicative group.

Proposition 13.H. *The same Haar measure μ as in Proposition 13.G, (i.e., $\mu(E) = \int_E \frac{1}{x}\, d\lambda(x)$), now acting on the σ-algebra $\Re_0 = \Re \cap \wp(\mathbb{R}\backslash\{0\})$ of subsets of the multiplicative group $\mathbb{R}\backslash\{0\}$ is absolutely continuous with respect to Lebesgue measure λ on the σ-algebra $\Re$ of subsets of the additive group $\mathbb{R}$.*

Observe that the Radon–Nikodym derivative of μ with respect to λ is the function $f\colon \mathbb{R}_0 \to \mathbb{R}$ given by $f(x) = \frac{1}{|x|}$ for every $x \in \mathbb{R}_0$ (Proposition 13.G — note that $\mathbb{R}_0 = \mathbb{R}^+ \cup (\mathbb{R}^+)^{-1}$ and so $\int_{\mathbb{R}_0} \frac{1}{|x|}\, d\lambda = \int_{\mathbb{R}^+} \frac{1}{x}\, d\lambda - \int_{(\mathbb{R}^+)^{-1}} \frac{1}{x}\, d\lambda$).

There are locally compact Hausdorff groups upon which the left and right Haar measures are intrinsically different.

Proposition 13.I. *Let X be the set of all 2×2 matrices of the form $\left(\begin{smallmatrix} \alpha & \beta \\ 0 & 1 \end{smallmatrix}\right)$ with real entries where $\alpha > 0$. Equip X with the matrix multiplication binary operation and with the half-plane Euclidean topology inherited from $\mathbb{R}^2$. So it becomes a locally compact Hausdorff multiplicative group. Take the sub-σ-algebra $\mathcal{X}_\mathcal{T}$ of $\Re\times\Re$ generated by the topology on X, take the planar Lebesgue measure $\pi = \lambda\times\lambda$ on $\Re\times\Re$, take its restriction to $\mathcal{X}_\mathcal{T}$ and, as before, write $d\alpha d\beta$ for $d\pi$. The set functions $\mu\colon \mathcal{X}_\mathcal{T} \to \overline{\mathbb{R}}$ and $\eta\colon \mathcal{X}_\mathcal{T} \to \overline{\mathbb{R}}$, given by*

$$\mu(E) = \int_E \tfrac{1}{\alpha^2}\, d\pi = \int_E \tfrac{1}{\alpha^2}\, d\alpha d\beta \quad \text{and} \quad \eta(E) = \int_E \tfrac{1}{\alpha}\, d\pi = \int_E \tfrac{1}{\alpha}\, d\alpha d\beta$$

for every $E \in \mathcal{X}_\mathcal{T} \subset \Re\times\Re$, are a left and a right Haar measure on $\mathcal{X}_\mathcal{T}$, respectively (left but not right, and vice versa). Since $\mu(E^{-1}) = \eta(E)$, there are Borel sets $E \in \mathcal{X}_\mathcal{T}$ such that $\mu(E) < \infty$ and $\mu(E^{-1}) = \eta(E) = \infty$, and X is a group upon which the left and right Haar measures behave quite differently.

In general, on separable locally compact metric spaces, Borel measures are regular according to Proposition 12.H. Thus, on separable metric locally compact groups, Haar measures that are simultaneously Borel measures are essentially unique in the sense of Theorem 13.7.

Hence, the Haar measures in Proposition 13.I are essentially unique.

In particular, on separable metric compact groups, Borel measures are regular (Proposition 12.I), and so essentially unique if they are Haar as well.

This is the case of the normalized Lebesgue measure $\lambda_{\mathbb{T}}\colon \Re_{\mathbb{T}} \to \mathbb{T}$ on the circle group $\mathbb{T}$, which is Haar, where $\Re_{\mathbb{T}}$ is the Borel σ-algebra of subsets of $\mathbb{T}$ generated by half-open arcs as in Problem 7.15. But, in this specific case, essentially unique actually means plain unique.

Proposition 13.J. *The normalized Lebesgue measure $\lambda_{\mathbb{T}}\colon \Re_{\mathbb{T}} \to \mathbb{T}$ over the unit circle $\mathbb{T}$ is Haar, and so it is the unique normalized Haar measure on $\Re_{\mathbb{T}}$.*

Notes: For continuity of the inversion and multiplication maps in the unitary operator group of Remark 13.A(d), see, e.g., [33, Problems 100 and 111].

Translation invariance was first mentioned at the end of Chapter 8, and it was fundamental to the building of nonmeasurable sets there. The propositions in this section complement the results on invariant measures discussed earlier in the chapter. For Propositions 13.A, 13.B and 13.C, see, e.g., [31, Problems 5.6, 5.8, 5.9], and for Proposition 13.D, see, e.g., [16, Exercise 9.2.2]. Proposition 13.E plays an important role in the proof of Theorem 13.7 (see, for instance, [9, Theorem 79.1]). We have already met those properties in previous propositions when we looked at Lebesgue measure. In fact, Proposition 13.F says that Lebesgue measure is a prototype of a regular Haar measure (see, e.g., [62, Proposition 14.24] or [8, Corollary 8.2]). Propositions 13.G and 13.H show that Haar measures on the Borel sets generated by the multiplicative groups $\mathbb{R}^+$ and $\mathbb{R}_0$ are absolutely continuous with respect to Lebesgue measure on the Borel sets generated by the additive group $\mathbb{R}$ (see, e.g., [16, Exercise 9.2.3] and [31, Problem 60.1]). For Proposition 13.I, see [31, Problem 58.4]. That the measure in Proposition 13.J is Haar was proved in Problem 7.15, and uniqueness comes from the fact that it is normalized, thus a probability measure, so that $\lambda_{\mathbb{T}}(\mathbb{T}) = 1$, and therefore, according to Theorem 13.7, it is unique (i.e., $1 = \eta(\mathbb{T}) = \gamma\lambda_{\mathbb{T}}(\mathbb{T}) = \gamma$ implies $\gamma = 1$).

Suggested Readings

Berberian [9], Cohn [16], Halmos [31], Nachbin [55], Royden [62].

References

1. S. Abbott, *Understanding Analysis*, 2nd ed., Springer, New York, 2016.
2. E. Asplund and L. Bungart, *A First Course in Integration*, Holt, Rinehart and Winston, New York, 1966.
3. S. Axler, *Measure, Integration & Real Analysis*, Springer-Switzerland, Cham, 2020.
4. P. Baldi, *Probability*, Springer-Switzerland, Cham, 2023.
5. R.G. Bartle, *The Elements of Integration and Lebesgue Measure*, Wiley, New York, 1995; enlarged 2nd ed. of *The Elements of Integration*, Wiley, New York, 1966.
6. R.G. Bartle, *The Elements of Real Analysis*, 2nd ed., Wiley, New York, 1976.
7. R.G. Bartle, *A Modern Theory of Integration*, Graduate Studies in Mathematics, Vol. 32, Amer. Math. Soc., Providence, 2001.
8. H. Bauer, *Measure and Integration Theory*, Walter de Gruyter, Berlin, 2001.
9. S.K. Berberian, *Measure and Integration*, Chelsea, New York, 1965.
10. H. Bercovici, A. Brown, and C. Pearcy, *Measure and Integration*, Springer-Switzerland, Cham, 2016.
11. P. Billingsley, *Probability and Measure*, 3rd ed., Wiley, New York, 1995.
12. L. Breiman, *Probability*, Addison-Wesley, Reading, 1968; reprinted: SIAM, Philadelphia, 1992.
13. A. Brown and C. Pearcy, *Introduction to Operator Theory I: Elements of Functional Analysis*, Springer, New York, 1977.
14. A. Brown and C. Pearcy, *An Introduction to Analysis*, Springer, New York, 1995.
15. K.L. Chung, *A Course in Probability Theory*, 3rd ed, Academic Press, New York, 2001.

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4

16. D.L. Cohn, *Measure Theory*, 2nd ed., Birkhäuser-Springer, New York, 2013.
17. J.B. Conway, *A Course in Abstract Analysis*, Graduate Studies in Mathematics, Vol. 141, Amer. Math. Soc., Providence, 2012.
18. J. Dieudonné, *Foundations of Modern Analysis*, Academic Press, New York, 1969.
19. J.L. Doob, *Stochastic Processes*, Wiley, New York, 1953; reprinted: Wiley, New York, 1990.
20. J.L. Doob, *Measure Theory*, Springer, New York, 1994.
21. J. Dugundji, *Topology*, Allyn & Bacon, Boston, 1966.
22. N. Dunford and J.T. Schwartz, *Linear Operators – Part I: General Theory*, Interscience, New York, 1958.
23. K.J. Falconer, *The Geometry of Fractal Sets*, Cambridge University Press, Cambridge, 1986.
24. K.J. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 3rd ed., Wiley, Chichester, 2014.
25. G.B. Folland, *Real Analysis: Modern Techniques and their Applications*, 2nd ed., Wiley, New York, 1999.
26. I.I. Gikhman and A.V. Skorokhod, *The Theory of Stochastic Processes I*, Springer, Berlin, 1974; reprinted: Springer, Berlin, 2004.
27. I.I. Gikhman and A.V. Skorokhod, *The Theory of Stochastic Processes II*, Springer, Berlin, 1975; reprinted: Springer, Berlin, 2004.
28. I.I. Gikhman and A.V. Skorokhod, *The Theory of Stochastic Processes III*, Springer, Berlin, 1979; reprinted: Springer, Berlin, 2007.
29. C. Goffman and G. Pedrick, *First Course in Functional Analysis*, Prentice-Hall, Englewood Cliffs, 1965.
30. A. Goswami and B.V. Rao, *Measure Theory for Analysis and Probability* Springer-Switzerland, Cham, 2025.
31. P.R. Halmos, *Measure Theory*, Van Nostrand, New York, 1950; reprinted: Springer, New York, 1974.
32. P.R. Halmos, *Naive Set Theory*, Van Nostrand, New York, 1960; reprinted: Springer, New York, 1974.
33. P.R. Halmos, *A Hilbert Space Problem Book*, Van Nostrand, New York, 1967; 2nd ed., Springer, New York, 1982.
34. P.R. Halmos and V.S. Sunder, *Bounded Integral Operators on L^2 Spaces*, Springer, Berlin, 1978.
35. A.S. Kechris, *Classical Descriptive Set Theory*, Springer, New York, 1995.
36. A.S. Kechris and A. Louveau, *Descriptive Set Theory and the Structure of Sets of Uniqueness*, Cambridge University Press, Cambridge, 1987.
37. J.L. Kelley, *General Topology*, Van Nostrand, New York, 1955; reprinted: Springer, New York, 1975.

38. J.L. Kelley and T.P. Srinivasan, *Measure and Integral – Volume 1*, Springer, New York, 1988.

39. J.F.C. Kingman and S.J. Taylor, *Introduction to Measure and Probability*, Cambridge University Press, Cambridge, 1966.

40. A.N. Kolmogorov, *Foundation of the Theory of Probability*, Chelsea, New York, 1956.

41. A.N. Kolmogorov and S.V. Fomin, *Introductory Real Analysis*, Prentice-Hall, Englewood Cliffs, 1970.

42. C.S. Kubrusly, *Measure Theory: A First Course*, Academic Press-Elsevier, San Diego, 2007.

43. C.S. Kubrusly, *The Elements of Operator Theory*, Birkhäuser-Springer, New York, 2011; enlarged 2nd ed. of *Elements of Operator Theory*, Birkhäuser, Boston, 2001.

44. C.S. Kubrusly, *Spectral Theory of Bounded Linear Operators*, Birkhäuser-Springer-Switzerland, Cham, 2020.

45. C.S. Kubrusly, *Bilinear Maps and Tensor Products in Operator Theory*, Springer-Switzerland, Cham, 2023.

46. S. Lang, *Real and Functional Analysis*, 3rd ed., Springer, New York, 1993.

47. J.-F. Le Gal, *Measure Theory, Probability, and Stochastic Processes*, Springer-Switzerland, Cham, 2022.

48. M. Loève, *Probability Theory I*, Springer, New York, 1977, enlarged 4th ed. of the first part of *Probability Theory*, 3rd ed., Van Nostrand, New York, 1963.

49. M. Loève, *Probability Theory II*, Springer, New York, 1978, enlarged 4th ed. of the second part of *Probability Theory*, 3rd ed., Van Nostrand, New York, 1963.

50. R. Lyons, *Seventy years of Rajchman measures*, J. Fourier Anal. Appl. (Kahane Special Issue) (1995), 363–377.

51. J.N. McDonald and N.A. Weiss, *A Course in Real Analysis*, Academic Press, San Diego, 1999.

52. R.E. Megginson, *An Introduction to Banach Space Theory*, Springer, New York, 1998.

53. S.N. Mukhopadhyay and S. Ray, *Measure and Integration*, Springer-Switzerland, Cham, 2025.

54. M.E. Munroe, *Measure and Integration*, 2nd ed., Addison-Wesley, Reading, 1971.

55. L. Nachbin, *The Haar Integral*, Van Nostrand, New York, 1965.

56. A.W. Naylor and G.R. Sell, *Linear Operator Theory in Engineering and Science*, Holt, Rinehart & Winston, New York, 1971; reprinted: Springer, New York, 1982.

57. J. Neveu, *Mathematical Foundations of the Calculus of Probability*, Holden-Day, San Francisco, 1965.

58. M. Reed and B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, 2nd ed., Academic Press, New York, 1980.

59. F. Riesz and B. Sz.-Nagy, *Functional Analysis*, Frederick Ungar, New York, 1955.

60. C.A. Rogers, *The Hausdorff Measure*, Cambridge University Press, Cambridge, 1970.

61. G.G. Roussas, *An Introduction to Measure-Theoretic Probability*, 2nd. ed, Academic Press-Elsevier, Amsterdam, 2014.

62. H.L. Royden, *Real Analysis*, 3rd ed., Macmillan, New York, 1988.

63. W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw-Hill, New York, 1987.

64. H. Sagan, *Space-Filling Curves*, Springer, New York, 1994.

65. B. Simon, *Real Analysis – A Comprehensive Course in Analysis, Part 1*, Amer. Math. Soc., Providence, 2015.

66. G.E. Shilov and B.L. Gurevich, *Integral, Measure and Derivative: A Unified Approach*, Prentice-Hall, Englewood Cliffs, 1966.

67. K. Shum, *Measure-Theoretical Probability*, Birkhäuser-Springer-Switzerland, Cham, 2023.

68. P. Suppes, *Axiomatic Set Theory*, Van Nostrand, New York, 1960.

69. D.W. Strook, *Essentials of Integration Theory for Analysis*, 2nd ed., Springer-Switzerland, Cham, 2011.

70. T. Tao, *An Introduction to Measure Theory*, Graduate Studies in Mathematics, Vol. 126, Amer. Math. Soc., Providence, 2011.

71. R.L. Vaught, *Set Theory: An Introduction*, 2nd ed., Birkhäuser, Boston, 1995.

72. A.J. Weir, *Lebesgue Integration and Measure*, Cambridge University Press, Cambridge, 1973.

73. A.J. Weir, *General Integration and Measure*, Cambridge University Press, Cambridge, 1974.

74. E. Wong, *Stochastic Processes in Information and Dynamical Systems*, McGraw-Hill, New York, 1971.

List of Symbols

$\#$, vii
$\mathbb{N}$, vii, 6
$\mathbb{I}_n$, vii
$\mathbb{R}$, viii, 5
$\mathbb{Q}$, viii, 8
$2^{\mathbb{N}}$, viii
$\aleph_0$, viii
$2^{\aleph_0}$, viii
$\wp(X)$, 3
$\mathcal{A}$, 3
$\varnothing$, 3, 18, 253
$\setminus$, 3
$\mathcal{X}$, 4
$(X, \mathcal{X})$, 4
$\bigcup$, 4
$\bigcap$, 4
$\subseteq$, 4
$\boldsymbol{X}_{\mathcal{C}}$, 4
$\in$, 5
$\mathcal{X}_{\mathcal{C}}$, 5
$\sigma(\mathcal{C})$, 5
$\Re$, 5
$\mathcal{T}$, 5, 18, 253
$\mathcal{R}_{\mathcal{T}}$, 5, 19
f^{-1}, 5
∞, 5
$>$, 5
$\geq$, 5
$<$, 5
$\leq$, 5
$\rightarrow$, 5
$\chi_E(\ )$, 6
$\{0, 1\}$, 6
$\subset$, 6
$1(\ )$, 6
$\mathbb{N}_e$, 6
$\mathbb{N}_o$, 6
$\mathcal{N}$, 6
$(\mathbb{N}, \mathcal{N})$, 6
$f|$, 6
$p(f, g)$, 7
F^+, 8, 63, 125

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4

F^-, 8, 63, 125
f^+, 8
f^-, 8
$|f|$, 9
$\overline{\mathbb{R}}$, 10
$-\infty$, 10
$+\infty$, 10
$f_{\mathbb{R}}$, 10
$\mathbb{N}_0$, 11
lim inf, 11
lim sup, 11
sup, 11
inf, 11
lim, 11
$\underline{f}$, 11
$\overline{f}$, 11
$\wedge$, 13, 284
$\vee$, 13, 284
min, 13
max, 13
$\mathcal{M}$, 14
$\mathcal{M}(X, \mathcal{X})$, 14
$\mathbb{R}^X$, 14
$F_{+\infty}$, 15
$F_{+\infty}$, 15
$\mathbb{C}$, 16
Re f, 16, 236
Im f, 16, 236
f^{-1}, 17, 18, 259, 260
$\mathcal{X}_f$, 17
$\circ$, 18, 57, 75, 202, 260
$(X, \mathcal{T})$, 18, 253
$\mathcal{X}_{\mathcal{T}}$, 18, 19, 259
$\Longrightarrow$, 21
$\wp(\mathbb{R})$, 21, 26, 157
$\{x\}$, 22
μ, 23, 148
$(X, \mathcal{X}, \mu)$, 24
N, 25, 28, 32, 121
δ_x, 25, 140
$(\mathbb{N}, \wp(\mathbb{N}), \mu)$, 25
λ, 25, 163
$\mathbb{Z}$, 26
λ_F, 26
a.e., 28
μ-a.e., 28
$(\mathbb{R}, \Re, \lambda)$, 28, 56, 163
ν, 29
E^+, 32, 124
E^-, 32, 124
$|\nu|$, 32, 124
$\mathcal{N}$, 32
$\widehat{\mathcal{X}}$, 32
$\widehat{\mu}$, 32
$(X, \widehat{\mathcal{X}}, \widehat{\mu})$, 33
$\widehat{\mathcal{N}}$, 34
$\mu|$, 38
$[\mu]$, 39, 271
$\widehat{f}$, 40
ζ, 40
$\wp$, 41
$(X, \mathcal{X}, \wp)$, 41
μ_f, 41, 212
$F\colon \mathbb{R} \to [0,1]$, 41, 212
$(\mathbb{R}, \Re, \mu_f)$, 42
$\int$, 44, 61
$\int_E$, 44
$\mathcal{M}^+$, 45
$\mathcal{M}(X, \mathcal{X})^+$, 45
Φ_f^+, 45
$\Longleftrightarrow$, 50
$\frac{d\mu}{d\eta}$, 57, 131
$p\colon \mathbb{R} \to \mathbb{R}$, 58, 212
$\mathcal{L}$, 61
$\mathcal{L}(X, \mathcal{X}, \mu)$, 61
F_0, 63, 125

F_-, 63, 125
F_+, 63, 125
ν^+, 68, 124
ν^-, 68, 124
$\mathbb{F}$, 75
Exp( ), 76
$\mathcal{L}(\mathbb{R}, \Re, \mu_f)$, 76
$\| \ \|$, 79
$\langle \ ; \ \rangle$, 80, 87
$\times$, 80, 117, 177
$\overline{\gamma}$, 80
$(\mathcal{L}, \| \ \|)$, 81
$\sim$, 83, 117, 138, 172
$[f]$, 83
L^1, 83
$L^1(\mu)$, 83
$L^1(X, \mu)$, 83
$L^1(X, \mathcal{X}, \mu)$, 83
$/$, 84, 118
$\| \ \|_1$, 84
L^p, 85
$L^p(\mu)$, 85
$L^p(X, \mu)$, 85
$L^p(X, \mathcal{X}, \mu)$, 85
$\| \ \|_p$, 85
L^∞, 85
$L^\infty(\mu)$, 85
$L^\infty(X, \mu)$, 85
$L^\infty(X, \mathcal{X}, \mu)$, 85
ess sup , 85
$\| \ \|_\infty$, 85, 283
L^2, 86
$\| \ \|_2$, 87
$(L^p, \| \ \|_p)$, 88
$(L^\infty, \| \ \|_\infty)$, 88
$(L^2, \| \ \|_2)$, 91
$\mathbb{R}^{\mathbb{N}}$, 92
ℓ^p, 93
ℓ^∞, 93
$(\ell^p, \| \ \|_p)$, 93, 303
$(\ell^\infty, \| \ \|_\infty)$, 93, 303
ℓ^1, 94
$L^\infty[0,1]$, 95
$L^p[0,1]$, 95
$L^1[0,1]$, 95
$L^\infty[1,\infty)$, 95
$L^p[1,\infty)$, 95
$L^1[1,\infty)$, 95
$L^2[1,\infty)$, 95
$C[0,1]$, 95
$(C[0,1], \| \ \|_p)$, 96
$(C[0,1], \| \ \|_\infty)$, 96
$R^p[0,1]$, 96
$(R^p[0,1], \| \ \|_p)$, 96
$(L^p[0,1], \| \ \|_p)$, 97
$\overline{A}$, 99, 254
ℓ^0, 99
Var( ), 100
$L^2(X, \mathcal{X}, \wp)$, 100
$\not\Longleftarrow$, 102
$\not\Longrightarrow$, 104
a.u., 111
μ-a.u. , 111
∇, 117
(a.e.), 118, 208
(a.u.), 118
(L^p), 118
(μ), 118, 208
(u.a.e.), 118
A^+, 121
A^-, 121
$|\nu|$, 124
$\ll$, 125
$\perp$, 132
μ_a , 132
μ_s , 132

μ_c , 134
μ_d , 134
μ_{sc} , 135
μ_{sd} , 135
$\|\nu\|$, 137
$\mathcal{S}$, 137
$\mathcal{S}(X, \mathcal{X}, \mathbb{R})$, 137
$\equiv$, 138
$\Re_{[0,1]}$, 141
$\lambda_{[0,1]}$, 141
$\mu_{[0,1]}$, 141
$\mathbb{I}$, 142
$\Re_{\mathbb{I}}$, 142
$\lambda_{\mathbb{I}}$, 143
$\mu_{\mathbb{I}}$, 143
$\mathbb{T}$, 143
$\Re_{\mathbb{T}}$, 143
$\lambda_{\mathbb{T}}$, 143
$\mu_{\mathbb{T}}$, 144
$\widetilde{\mu}$, 145
$\mathcal{C}_S$, 148
μ^*, 148, 267, 290
$\mathcal{A}^*$, 151
$\mathcal{C}_i$, $_{i=1,2,3,4}$, 157
$\mathcal{I}$, 157
$\Im$, 157
ℓ , 157
$\Im^*$, 161
ℓ^*, 161
λ^*, 161
$(\mathbb{R}, \Im^*, \lambda^*)$, 161, 163
$\widehat{\Re}$, 163
$\widehat{\lambda}$, 163
$\mathcal{I}_{\mathbb{Q}}$, 164
G_δ, 164, 169
I°, 168
$\overline{I}$, 168
$\mathcal{I}_S$, 168
$\mathcal{I}_S^\circ$, 168
$\overline{\mathcal{I}}_S$, 168
$\dot{+}$, 171
$(\mathbb{R}, \mathcal{X}, \mu)$, 175
$X{\times}Y$, 177
$\mathcal{X}{\times}\mathcal{Y}$, 177
$(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})$, 177
$(\mathbb{R}{\times}\mathbb{R}, \Re{\times}\Re)$, 178
$\mathbb{R}^2$, 178
$\mathcal{P}$, 178
ϖ, 178
π, 180
$\mu{\times}\eta$, 181
$(X{\times}Y, \mathcal{X}{\times}\mathcal{Y}, \mu{\times}\eta)$, 181
$(X{\times}Y, \mathcal{X}{\times}\mathcal{Y}, \pi)$, 181
S_x , 181
S^y, 181
$(A{\times}B)_x$, 181
$(A{\times}B)^y$, 181
$(A{\times}X)_x$, 181
$(X{\times}Y)^y$, 181
f_x , 181
f^y, 182
$\mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})^+$, 187
$\mathcal{L}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y}, \pi)$, 188
$\mathcal{M}(X{\times}Y, \mathcal{X}{\times}\mathcal{Y})$, 189
$(\mathcal{X}{\times}\mathcal{Y})_X$, 192
$(\mathcal{X}{\times}\mathcal{Y})_Y$, 192
$(\mathbb{R}^2, \Re{\times}\Re, \pi)$, 193
$*$, 194
$(\mathbb{N}{\times}\mathbb{N}, \wp(\mathbb{N}{\times}\mathbb{N}), \mu{\times}\mu)$, 195
$\mu_{f,g}$, 198
$(\mathbb{R}{\times}\mathbb{R}, \Re{\times}\Re, \mu_{f,g})$, 199
F_f , 200, 216
$F_{f,g}$, 200, 216
$\mathcal{L}(\mathbb{R}{\times}\mathbb{R}, \Re{\times}\Re, \mu_{f,g})$, 202
$(\Omega, \mathcal{A}, P)$, 208
(a.s.), 208

(w.p.1), 208
(m.s.), 208
(L^2), 208
$(\wp)$, 208
$\Phi\colon \mathbb{R} \to \mathbb{C}$, 212
p_f, 216
$p_{f,g}$, 216
$\wp(\ \ |Q)$, 218
$\wp(E\,|\,\mathcal{J})$, 219
$\mu_f(\ \ , x)$, 222
$\mu_f(B,\ \ )$, 223
$\mu_f(\ \ ,\ \ )$, 223
$\mathrm{Exp}(f\,|\,\mathcal{J})$, 224
$\mathcal{F}$, 227
$(X, \mathcal{X}, \mathcal{F})$, 227
$(X, \mathcal{X}, \mathcal{F}, \wp)$, 227
$(X, \mathcal{X}, \mathcal{F}, \{f_t\}, \wp)$, 227
$[f_j]$, 228
i.i.d. , 231
$N(0,1)$, 232
$\mathcal{L}(X,Y)$, 249
$\mathcal{L}(X, \mathcal{X}, \mu, Y)$, 249
$\|\ \ \|_Y$, 249
$\|\ \ \|_{\mathcal{L}(X,Y)}$, 249
$L^1(X,Y)$, 250
$L^1(X, \mathcal{X}, \mu; Y)$, 250
$[F]$, 250
$(L^1(X,Y), \|\ \ \|_1)$, 250
(X,d), 253
$d\colon \mathcal{X}\times\mathcal{X} \to \mathbb{R}$, 253
$B_\varepsilon(x)$, 254
A°, 254
N_x , 255
$\mathbb{F}^n$, 257
$\mu\#$, 269, 313
$\mathbb{R}^n$, 271
$\mathcal{X}_G$, 272
$\mu^*_{\varepsilon,p}$, 274
μ^*_p , 274
μ_p , 274
η_p , 274
$\dim_H$, 274
$\mathbb{C}^n$, 274
$\mathcal{N}(f)$, 277
$\mathcal{R}(f)$, 277
$\overline{[f]}$, 277
$C_c(X)$, 277
$C_c(X, \mathbb{R})$, 278
$C_c(X, \mathbb{C})$, 278
$C_c(X)_K$, 278
$C_c(X)^U$, 278
$\mathbb{F}^S$, 282
$L(S, \mathbb{C})$, 282
$L(S, \mathbb{R})$, 282
$B(S)$, 283
$B(S, \mathbb{C})$, 284
$B(S, \mathbb{R})$, 284
$\sup\{f, g\}$, 284
$\inf\{f, g\}$, 284
$C(X)$, 284
$\mathbb{F}^X$, 284
$B(X)$, 284
$C(X, \mathbb{C})$, 285
$C(X, \mathbb{R})$, 285
$C_0(X)$, 299
$(L^p(X, \mathcal{X}_\mathcal{T}, \mu), \|\ \ \|_p)$, 300
$\mathcal{C}_\mathcal{T}$, 301
$L^\infty(\mathbb{R}, \lambda)$, 302
$C_c(\mathbb{R})$, 302
$P(X)$, 302
$q(\ \ ,\ \ )$, 302
$q(\ \ , \overline{\ \ })$, 302
$(C(X), \|\ \ \|_\infty)$, 302
$(C(X), \|\ \ \|_p)$, 302
$(L^\infty([0,1], \Re, \lambda), \|\ \ \|_\infty)$, 303
$(B(X), \|\ \ \|_\infty)$, 304

$(L^\infty(X, \mathcal{X}_\mathcal{T}, \mu), \| \ \|_\infty)$, 304
$L^p(\mathbb{R}, \Re, \lambda)$, 304
$\star$, 305
x^{-1}, 305
A^{-1}, 306
AB, 306
A^n, 306
xA, 306
Ax, 306
$GL(n)$, 308
$\mathcal{B}[\mathcal{H}]$, 308
T^*, 308
$\mathcal{K}$, 313
$[B : A]$, 315
$\mathcal{B}$, 315
ϑ_U, 316
$\mathbb{R}^+$, 324
$\Re^+$, 324
$\mathbb{R}_0$, 325
$\Re_0$, 325

Index

A
a.e. convergence, 29, 103
Abelian group, 306
absolute continuity, 51, 126, 213
absolute homogeneity, 79
absolute integrability, 64
absolute value, 9
absolutely continuous function, 59, 213, 247
absolutely continuous measure, 51, 125, 140, 213, 215, 243, 325
absolutely continuous signed measure, 139
absolutely convergent series, 30
absolutely summable sequence, 138
additive functional, 66, 282
additive scalar group, 307
additivity, 23, 24, 29, 80, 148, 152, 157, 166, 175, 267, 282
adjoint, 308
algebra, 3
almost everywhere, 28
almost everywhere convergence, 29, 103
almost everywhere equality, 28
almost sure, 208
almost uniform Cauchy sequence, 112
almost uniform convergence, 111
arc-length metric, 143
area on $\mathbb{R}^2$, 193
Arzelà–Ascoli Theorem, 317
associative binary operation, 305
atom, 24
atom-free measure, 25
atom-free set, 25
atomic measure, 25
atomic set, 24
Axiom of Choice, viii, 22, 165, 172, 175

B
β-truncation, 16
Baire measure, 272
Baire set, 272
Banach space, 82, 88
base, 19, 254
Beppo Levi Theorem, 55
bilateral shift, 145
binary operation, 305
Bochner integrable function, 249
Bochner integral, 251
Boolean algebra, 3
Boolean ring, 4

© The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

C. S. Kubrusly, *The Essentials of Measure Theory*,
Universitext, https://doi.org/10.1007/978-3-032-12663-4

Borel algebra, 5
Borel–Cantelli Lemma, 230
Borel field, 4
Borel function, 15, 259
Borel measurable function, 15, 18, 260, 288
Borel measure, 39, 261, 270
Borel σ-algebra, 5, 19, 178, 259
Borel set, 5, 259
Borel–Stieltjes measure, 26, 42
Borel–Stieltjes measure on $\Re_{[0,1]}$, 142
Borel–Stieltjes measure on $\Re_{\mathbb{I}}$, 142
Borel–Stieltjes measure on $\Re_{\mathbb{T}}$, 144
bounded almost everywhere, 69
Bounded Convergence Theorem, 69
bounded function, 16, 283
bounded linear functional, 211, 283
bounded linear transformation, 251, 252, 308
bounded sequence of scalars, 11, 93
bounded sequence of vectors, 81
bounded set, 257
boundedness, 257

C

canonical decomposition, 136
canonical representation, 43
Cantor function, 141
Cantor–Lebesgue function, 141
Cantor–Lebesgue measure, 142–145
Cantor–Lebesgue measure on $\Re_{[0,1]}$, 142
Cantor–Lebesgue measure on $\Re_{\mathbb{I}}$, 142
Cantor–Lebesgue measure on $\Re_{\mathbb{T}}$, 144
Cantor set, 37
Cantor-like set, 38
Carathéodory condition, 150
Carathéodory Extension Theorem, 151, 161, 180
cardinality, vii
Cartesian product, 177
Cartesian product space, 181
Cauchy criterion, 81
Cauchy in measure, 106
Cauchy–Schwarz inequality, 80
Cauchy sequence, 81, 102, 257
Central Limit Theorem, 232
Cesàro means, 232
change of variable, 57, 75
characteristic function of a distribution, 212, 213
characteristic function of a set, 6
circle group, 307
class of bounded sets, 316
class of compact sets, 313
clopen set, 255
closed set, 254
closure, 99, 254
codomain, 210
commutative group, 306
compact group, 307
compact set, 245, 254
compact space, 254
compact support, 277
Compactness Theorem, 257
complete measure, 32, 268
complete measure space, 32
complete metric space, 257
complete normed space, 82
complete σ-algebra, 32
Completeness Theorem, 88
completion of a measure, 34
completion of a measure space, 34, 132
completion of a σ-algebra, 34
complex field $\mathbb{C}$, 16
complex inner product space, 80
complex linear space, 79, 282
complex measure, 40, 240
complex normed space, 79
complex-valued bounded function, 214
complex-valued continuous function, 285
complex-valued function, 16, 90, 236, 239, 242, 277
composition, 18, 260
concentrated measure, 24
concentrated on set of measure zero, 132
conditional expectation, 224, 225
conditional expected value, 224
conditional probability, 218, 219
conditional probability distribution, 220, 223

conditional probability measure, 223
conditionally compact set, 254
conditioning, 217
constant function, 6
construction of Borel measures, 267
construction of Haar measures, 314
construction of Lebesgue measure, 156
content, 269, 290
content on a topology, 269
content on the compact sets, 313
continuity, 97
continuous function,15, 18, 95, 259, 279, 284
continuous functional, 98
continuous linear functional, 97, 248
continuous linear transformation, 251
continuous mapping, 18, 98, 259, 307
continuous measure, 133
continuous process, 211
continuously differentiable, 27
Continuum Hypothesis, viii, 22, 165
convergence almost everywhere, 29, 52, 103
convergence almost uniform, 111
convergence everywhere, 29
convergence from below, 52
convergence in distribution, 232
convergence in L^2, 209
convergence in L^p, 103
convergence in measure, 106
convergence in probability, 209
convergence in the mean square sense, 209
convergence in the norm topology, 81
convergence of functions, 101
convergence pointwise, 12, 28, 101
convergence uniform, 16, 54, 101
convergence uniform almost everywhere, 111
convergence with probability one, 209
convergent scalar sequence, 11
convergent vector sequence, 81
convolution, 194
coset, 306
countable additivity, 23, 24, 29, 148, 157, 166, 175
countable set, viii
countable subadditivity, 37, 149, 267, 269
countably additive measure, 24, 166
countably infinite set, viii
counting measure, 25
covering, 31, 254
covering number, 315
covering ratio, 315

D

De Morgan laws, 4
decreasing scalar sequence, 11
decreasing sequence of functions, 45
decreasing sequence of sets, 27
dense set, 99, 254
denseness, 258
density function, 58, 77, 212
derivative of a function, 246
derivative of a measure, 246, 247
derivative of the distribution, 213
Devil's staircase, 141
diagonal set, 194
diameter, 190, 273
difference set, 190
differentiable function, 246
Dirac measure, 25, 54
Dirichlet function, 15
discrete measure, 134
discrete process, 211
discrete space, 323
discrete topology, 323
disjoint sequence, 31
disjoint sets, 24
disjoint supports, 132
disjointification, 35, 136
distribution function, 41, 212
distribution law, 41, 212
divergent series, 24
dominated convergence, 105
Dominated Convergence Theorem, 68, 72
dominating measure, 240
double integral, 190
dual, 283

E

ε-net, 257

$\mathcal{E}$-measurable function, 6
Egoroff Theorem, 114
equality almost everywhere, 28
equality everywhere, 28
equivalence class, 83
equivalence relation, 83
equivalent functions, 83
equivalent measures, 138
equivalent sets, vii, 117
essentially bounded function, 85
even positive integers, 6
expectation, 76
expected value, 76
extended real line, 10
extended real number system, 10
extended real-valued function, 9
extension of measure, 33, 147
extension of measure over a set, 38
extension of measure over $\widehat{\mathcal{X}}$, 33
extension theorems, 150, 124

F
F_σ set, 259
$\mathbb{F}$-valued function, 245
$\mathbb{F}$-valued measure, 244, 299
family adapted to a filtration, 227
Fatou's Lemma, 49
field, 3
filtration, 227
filtered measure space, 227
filtered probability space, 227
filtered stochastic process, 227
finite additivity, 152, 166
finite integral, 61, 163
finite intersection property, 270
finite measurable partition, 31, 239
finite measure, 24, 155
finite measure space, 69
finite positive measure, 237
finite set, vii, 25
finite subadditivity, 313
finitely additive measure, 166
first axiom of countability, 301
Fourier–Stieltjes transform, 212
Fourier transform of measure, 145, 212
Fourier transform of function, 213
Fubini Theorem, 188
function of a random variable, 211
function of random variables, 209
functional, 66
Fundamental Theorem of Calculus, 244
Fundamental Theorem of Calculus for Lebesgue integral, 247, 248

G
G_δ set, 164, 169, 259, 272
gauge integral, 252
Gaussian function, 214
general linear group, 308
generalized Hölder inequality, 94
generated measure, 47, 126
group, 306

H
Haar integral, 324
Haar measure, 311
Hahn–Banach Theorem, 317
Hahn decomposition of a set, 123
Hahn Decomposition Theorem, 121
Hahn Extension Theorem, 155
Hausdorff dimension, 274
Hausdorff group, 307
Hausdorff measure, 273
Hausdorff space, 255
Heine–Borel Theorem, 257
Hermitian symmetry, 80
Hilbert space, 82, 88
Hölder conjugates, 86
Hölder inequality, 86, 91
Hölder inequality generalized, 94
homeomorphism, 307
homogeneity, 80, 282
homogeneous functional, 66, 282
homomorphism, 306

I
$\Im^*$-measurable function, 163
$\Im^*$-measurable set, 163
identically distributed random variables, 231
identity element, 305
identity function, 6
identity segment, 194

imaginary part a function, 16, 366
inclusion ordering, 4
increasing scalar sequence, 11
increasing sequence of functions, 45
increasing sequence of sets, 27
indefinite integral, 68
independence, 202, 216, 217
independent and identically distributed random variables, 231, 232
independent random variables, 201, 211, 217
independent sets, 200
induced measure, 41, 144
induced uniform norm, 283
infimum, 284
infinite set, vii
inner content, 269, 290
inner product, 80, 288
inner product axioms, 80
inner product space, 80
inner regular measure, 170, 262
inner regular set, 262
integers $\mathbb{Z}$, 26
integrable filtered stochastic process, 228
integrable in the Lebesgue sense, 247
integrable function, 61
integrable simple mapping, 250
integral, 45
integral functional, 97, 248
integral of Banach-space-valued transformations, 249
integral of complex functions, 71, 236
integral of functions over a set, 44
integral of nonnegative functions, 45
integral of real functions, 61
integral of simple functions, 44, 250
integral of transformations, 251
integral transformation, 251
interior, 254
interval, 157, 246
interval shrinking to a point, 246
invariance property, 311
invariant measure, 311
invariant subgroup, 306
inverse element, 305
inverse image, 5, 17, 43
inversely induced σ-algebra, 17
inversion map, 306
iterated integral, 190

J

Jensen inequality, 93
joint change of variables, 202
joint density function, 216
joint distribution function, 199, 216
joint distribution law, 199
joint probability distribution, 199, 215
Jordan Decomposition Theorem, 124

K

kernel, 196, 277
Kolmogorov Strong Law of Large Numbers, 231
Kurzweil–Henstock integral, 252

L

Laws of Large Numbers, 231
Lebesgue Decomposition Theorem, 132, 244
Lebesgue dominated convergence, 68
Lebesgue integrable function, 62, 64
Lebesgue integral, 62, 64, 70
Lebesgue measurable function, 163
Lebesgue measure, 26, 144, 162, 271
Lebesgue measure on $\Im^*$, 162, 271
Lebesgue measure on $\Re$, 26, 62, 162, 271
Lebesgue measure on $\Re_{\mathbb{I}}$,, 143
Lebesgue measure on $\Re_{\mathbb{T}}$, 144
Lebesgue measure space, 56
Lebesgue point, 247
Lebesgue set, 162
Lebesgue σ-algebra, 34, 162
Lebesgue singular function, 141
Lebesgue spaces, 95
Lebesgue–Stieltjes measure, 163
Lebesgue Theorem, 70
left coset, 306
left Haar measure, 311
left invariant measure, 311
left multiplication, 307
left translation, 306
length function, 157

length on $\mathbb{R}$, 193
limit function, 12
limit inferior, 11
limit of a sequence, 81
limit superior, 11
Lindelöf space, 270
line segment, 192
linear continuous transformation, 251
linear functional, 66, 97, 282
linear lattice, 284
linear Lebesgue measure, 193
linear manifold, 14, 66, 282
linear space, 14, 66, 79, 282
linear space $\mathcal{L}$, 66
Littlewood second principle, 92
local base, 273
locally compact group, 307
locally compact Hausdorff space, 257
locally compact space, 255
locally finite measure, 273
locally L^p function, 95
lower semicontinuous, 278

M

μ-almost everywhere, 28
μ-almost everywhere convergence, 29, 103
μ-almost everywhere equality, 28
μ-equivalent functions, 83
μ-equivalent sets, 117
μ-integrable function, 61
μ^*-measurable set, 150, 267
Markov Process, 227, 228
Martingale Process, 227, 229
mean-ergodic-like theorems, 232
mean of a random variable, 76
mean square convergence, 209
measurability, 44
measurable covering, 31
measurable function, 5, 10, 259
measurable mapping, 17, 259
measurable partition, 31, 123
measurable rectangle, 177
measurable section, 182
measurable set, 4
measurable simple function, 44
measurable singleton, 36
measurable space, 4
measurable transformation, 17, 259
measurable triangle, 191
measure, 23
— absolutely continuous, 51, 125, 243
— atomic, 25
— atom-free, 25
— Baire, 272
— Borel, 39, 161
— Borel–Stieltjes, 26, 42
— Cantor–Lebesgue, 142–145
— complete, 32, 268
— complex, 40, 240
— concentrated, 24
— continuous, 133
— countably additive, 24, 166
— counting, 25
— Dirac, 25, 54
— discrete, 134
— dominating, 240
— equivalent, 138
— $\mathbb{F}$-valued, 244
— finite, 24, 155
— finite positive, 237
— finitely additive, 166
— generated, 47, 126
— Haar, 311
— Hausdorff, 273
— induced, 41, 144
— inner regular, 170, 262
— invariant, 311
— Lebesgue, 26, 144, 162, 271
— Lebesgue–Stieltjes, 163
— left Haar, 311
— left invariant, 311
— linear Lebesgue, 193
— locally finite, 273
— mutually singular, 132
— normalized, 145, 229
— normalized Lebesgue, 144, 326
— outer, 148, 262
— outer regular, 170, 262
— p-dimensional Hausdorff, 274
— plain outer, 267
— planar Lebesgue, 193
— point, 25
— positive, 235

— probability, 24, 41
— product, 181
— purely atomic, 25
— quasiregular, 262, 299
— quasiregular outer, 267, 290
— Rajchman, 145
— Radon, 273
— real, 237
— reference, 133
— regular, 262, 271
— regular Borel, 271, 300
— right Haar, 311
— right invariant, 311
— σ-additive, 150
— σ-finite, 24, 155
— semifinite, 35
— semiregular, 262
— signed, 29, 237
— singular, 132, 244
— singular-continuous, 136, 142
— singular-discrete, 136, 140
— topologically regular outer, 267
— translation invariant, 311
— uniformly σ-finite, 26, 53
— unit point, 25
measure axioms, 23
measure concentrated on a set, 24, 25 134, 142
measure induced by a measure, 41, 75, 143, 144, 145, 199
measure on a σ-algebra, 24
measure on an algebra, 148
measure over a set, 24
measure space, 24
measure zero, 24, 28, 32
Menshov example, 146
metric, 118, 253
metric axioms, 253
metric space, 253
metric topology, 254
metrizable topology, 254
Minkowski inequality, 87, 92
modus ponens, 118
moments, 77
monotone algebra, 19
monotone class, 19
monotone class generated, 20
Monotone Class Lemma, 20
Monotone Convergence Theorem, 46, 52
monotone scalar sequence, 11
monotone sequence of functions, 45
monotone sequence of sets, 27
multiplication map, 306
multiplication operator, 145
multiplicative scalar group, 307
mutually singular measures, 132
mutually singular signed measures, 139

N

n-dimensional random vector, 209
$\mathcal{N}$-measurable set, 6,
nth moments, 77
n-tuple, 199
natural ordering, 10
Neder's answer to Riesz's question, 145
neighborhood, 355
neighborhood base, 273
negative part of a function, 8
negative set, 121
negative variation, 124
neutral element, 305
nondegenerate interval, 39, 143
nonmeasurable set, 165, 173
nonnegative integers, 11
nonnegativeness, 79, 80, 253
nonzero linear functional, 283
norm, 79
norm axioms, 79
norm convergence, 81
normal distribution, 214
normal density function, 215
normal distribution function, 215
normal probability distribution, 215
normalized arc-length measure, 144
normalized Cantor–Lebesgue measure, 145
normalized Lebesgue measure, 144, 326
normalized measure, 144, 145, 208, 219, 229, 326
normally distributed random variable, 215, 232
normed linear space, 79
normed space, 79, 283

normed vector space, 79
null set, 121

O
odd positive integers, 6
open ball, 253
open covering, 254
open neighborhood, 255
open set, 18, 253
operator, 308
outer content, 313
outer measure, 148, 267
outer regular measure, 170, 262
outer regular set, 262

P
p-dimensional Hausdorff measure, 274
p-integrable function, 85
p-summable sequence, 93
partial ordering, 284
partition, 4, 31, 43
partition of unity, 298
plain outer measure, 267
planar Lebesgue measure, 193
point measure, 25
pointwise convergence, 12, 28, 101
Polish space, 271
polynomial, 7, 302
positive functional, 236, 282
positive integers $\mathbb{N}$, viii, 6
positive measure, 235
positive part of a function, 8
positive set, 121
positive variation, 124
positiveness, 24, 79, 80, 148, 149, 253, 267, 269, 313
power set, 3
probability distribution, 41, 212
probability measure, 24, 41
probability space, 24, 41, 58
product measure, 181
Product Measure Theorem, 180
product space, 181
properties of the Lebesgue measure, 162
pseudometric, 118, 253
pseudonorm, 81
purely atomic measure, 25
purely atomic set, 24

Q
quasiregular complex measure, 297, 299
quasiregular measure, 262, 299
quasiregular outer measure, 267, 290
quasiregular set, 262
quasiregular signed measure, 294, 299
quotient space, 84, 118

R
$\mathbb{R}$-valued sequence, 11
$\overline{\mathbb{R}}$-valued sequence, 11
$\mathfrak{R}$-measurable function, 15
$\mathfrak{R}$-measurable set, 5, 163
Radon measure, 273
Radon–Nikodym derivative, 131, 247
Radon–Nikodym Theorem, 127, 244
Rajchman measure, 145
random function, 209, 210
random sequence, 209, 210
random variable, 41
random vector, 198, 209
range, 277
rational field $\mathbb{Q}$, viii, 8
real field $\mathbb{R}$, viii, 5
real inner product space, 80
real line, 9
real linear space, 79, 282
real measure, 237
real normed space, 79
real numbers, viii
real part a function, 16, 236
real sequences, 92
real-valued bounded function, 284
real-valued continuous function, 285
real-valued function, 5, 235, 237, 241, 277
real-valued random variable, 41, 209
real-valued set function, 29
rectangle, 177
reference measure, 133
regular complex measure, 299
regular Borel measure, 271, 300
regular measure, 262, 271, 300, 301
regular set, 262

regular signed measure, 299
relatively compact set, 254
representation of a simple function, 43
restriction of function, 6
restriction of measure, 33, 38
Riemann–Lebesgue Lemma, 145
Riemann integrable function, 64, 96
Riemann integral, 64, 70
Riesz representation, 298
Riesz Representation Theorem, 131, 288, 298
Riesz Theorem, 88, 98
Riesz–Fischer Theorem, 88
Riesz–Weyl Theorem, 110
right continuity, 42
right coset, 306
right Haar measure, 311
right invariant measure, 311
right multiplication, 307
right translation, 306
ring, 4
rotation invariance, 144

S

S-valued sequence, 11
σ-additive measure, 150
σ-algebra, 4
σ-algebra generated, 5, 17, 42, 228
σ-algebra induced, 17
σ-algebra inversely induced, 17
σ-bounded set, 258
σ-compact set, 254
σ-field, 4
σ-finite measure, 24, 155
σ-finite set, 24, 55
σ-ring, 4
scalar, 81
scalar groups, 307
scalar-valued function, 277
Schwarz inequality, 80, 87
second axiom of countability, 301
second central moment, 100
second moment, 100
second-order random variable, 100
sections, 181
sector, 192
semialgebra, 157
semifinite measure, 35
seminorm, 81, 249
seminormed space, 81
semiregular measure, 262
separable Banach space, 250
separable space, 19, 254
sequence of complex functions, 75
sequence of elements in S, 11
sequence of elements in $\mathbb{R}$, 11
sequence of elements in $\overline{\mathbb{R}}$, 11
sequence of independent random variables, 231, 232
sequence of integrable functions, 73
sequence of measurable functions, 12
sequence of nonnegative numbers, 94
sequence of real functions, 12
sequence of real numbers, 29
sequence of random variables, 209
sequence of sets, 19
sequence of vectors,
set function, 23
sides, 181
signed measure, 29, 237
signed measure axioms, 29
simple function, 43
simple mapping, 250
singleton, 22, 25, 255, 36
singular measure, 132, 244
singular-continuous measure, 136, 142
singular-discrete measure, 136, 140
singular signed measure, 139
slice, 181
space $C[0,1]$, 96
spaces L^1, $L^1[0,1]$, $L^1[1,\infty)$, 67, 95
spaces L^2, $L[1,\infty)^2$, 86, 95, 299
spaces L^∞, $L^\infty[0,1]$, $L^\infty[1,\infty)$, 85, 95
spaces L^p, $L^p[0,1]$, 85, 95
space ℓ^0, 99
space ℓ^∞, 93
space ℓ^p, 93
space $R[0,1]$, 97
standard representation, 43
stochastic process, 211
Stone–Weierstrass Theorem, 302
strip-shape set, 191
Strong Law of Large Numbers, 231

subadditivity, 37, 79, 149
subcovering. 254
subgroup, 306
submartingale, 229
sub-σ-algebra, 4, 30, 217
sub-σ-algebra generated by one set, 5
sub-σ-algebra generated by sets, 5
sub-σ-algebra generated by one random variable, 42, 228
sub-σ-algebra generated by random variables, 228
sub-σ-algebras leading to a filtration, 227
subunion, 160
sum modulo one, 171
summable sequence, 93, 138
supermartingale, 229
sup-norm, 93, 283
support of a function, 277
support of a measure, 39, 271
supremum, 284
symmetric difference, 117
symmetric neighborhood, 307
symmetry, 253

T
Tonelli Theorem, 187
topological base, 18, 254
topological group, 306
topological space, 18, 253
topologically bounded set, 257
topologically regular outer measure, 267
topology, 18, 253
topology axioms, 18, 253
topology induced by a metric, 254
total variation, 124, 243
totally bounded set, 257
translation, 306
translation invariance, 164, 170, 175, 311, 324
translation invariance property, 311
translation invariance modulo one, 171
translation invariant measure, 311
translation modulo one, 171
triangle inequality, 79, 253
triangle set, 191
truncation, 16
two-dimensional random vector, 209
Tychonoff Theorem, 317

U
unconditionally convergent series, 24
uncorrelated random variables, 204, 217
uncorrelatedness, 204, 217
uncountable set, viii
uniform Cauchy sequence, 102
uniform convergence, 16, 54, 101
uniform convergence almost everywhere, 111
uniform topology, 308
uniformly bounded convergence, 105
uniformly σ-finite, 35
uniformly σ-finite measure, 26, 35
unit function, 6
unit point measure, 25
unitary operator, 308
unitary operator group, 308
upper semicontinuous, 278
Urysohn Lemma, 279

V
vanish at infinity, 299
variance, 100
variation of signed measure, 32
vector, 81
vector of random variables, 198, 209
vector space, 79
vector-valued function, 248
Vitali Convergence Theorem, 119
Vitali sets, 172

W
Weak Law of Large Numbers, 231
weak stability, 145, 146
with probability one, 208

X
$\mathcal{X}$-measurable function, 5, 10
$\mathcal{X}$-measurable set, 4
$\mathcal{X}$-valued sequence, 24
x-section of a function, 182
x-section of a set, 181

Y
y-section of a function, 182
y-section of a set, 181
Y-valued mapping, 249
Young inequality, 91

Z
Zorn's Lemma, viii